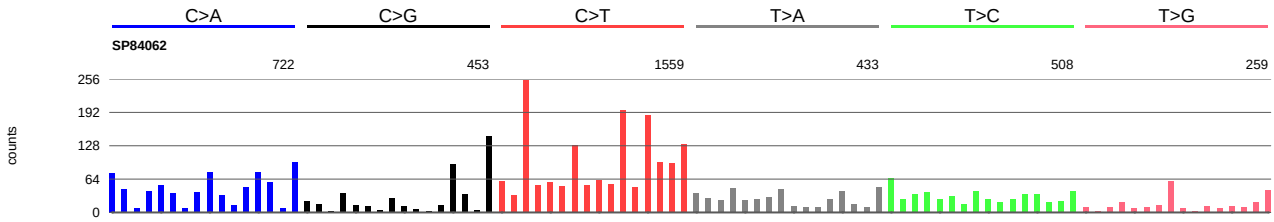
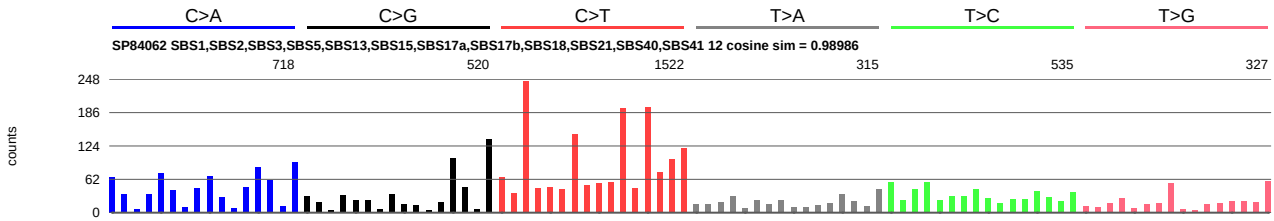
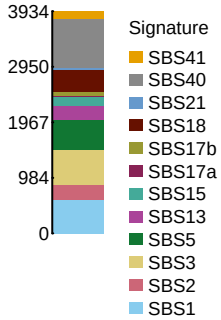
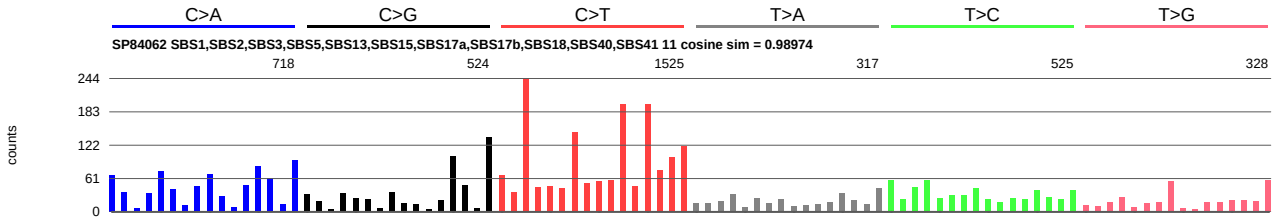
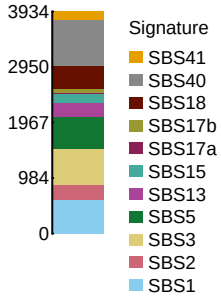
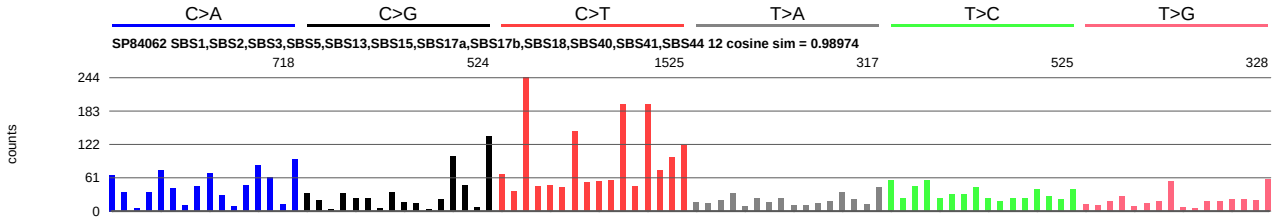

C>A
C>G
C>T
T>A
T>C
T>G
SP84062
722
453
1559
433
508
259
counts

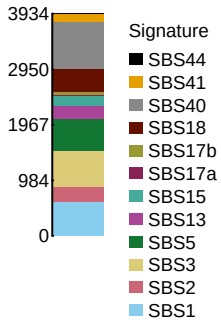

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 12 cosine sim = 0.98986
718
520
1522
315
535
327
counts

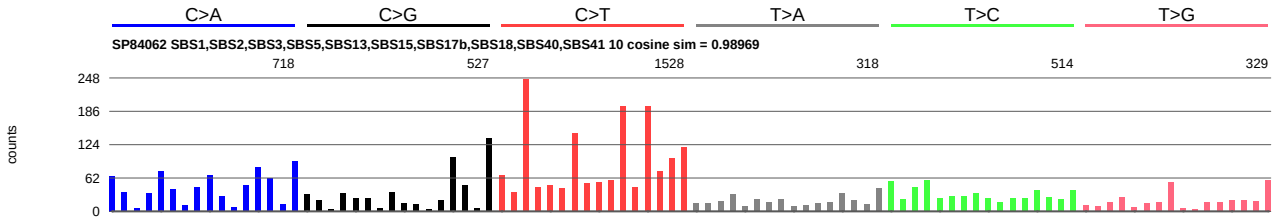

Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

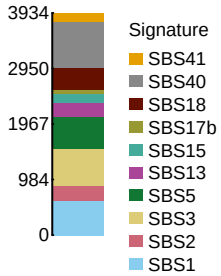

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.98974
718
524
1525
317
525
328
counts

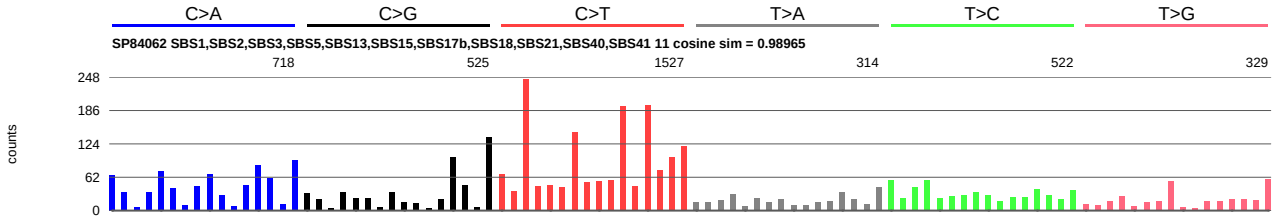

Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

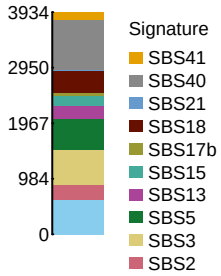

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.98974
718
524
1525
317
525
328
counts

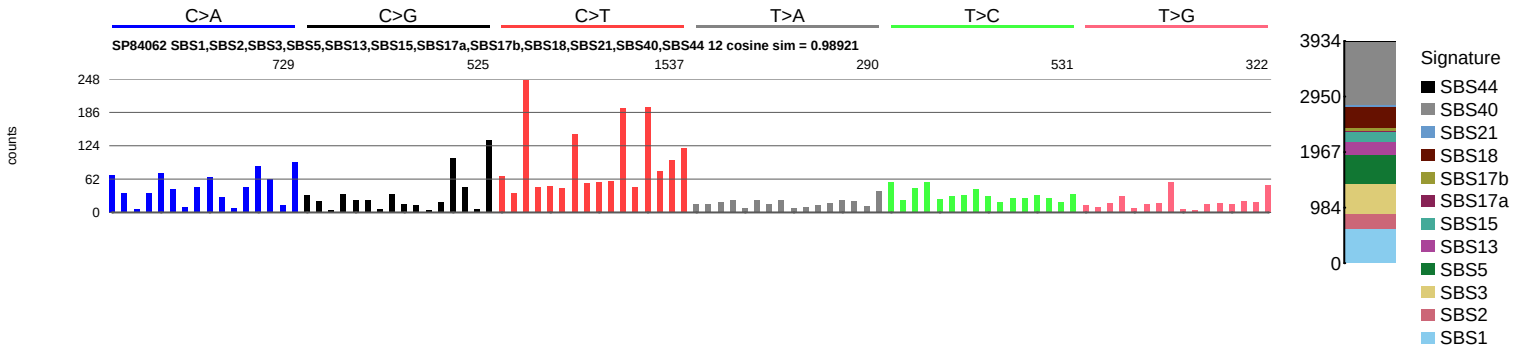

Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

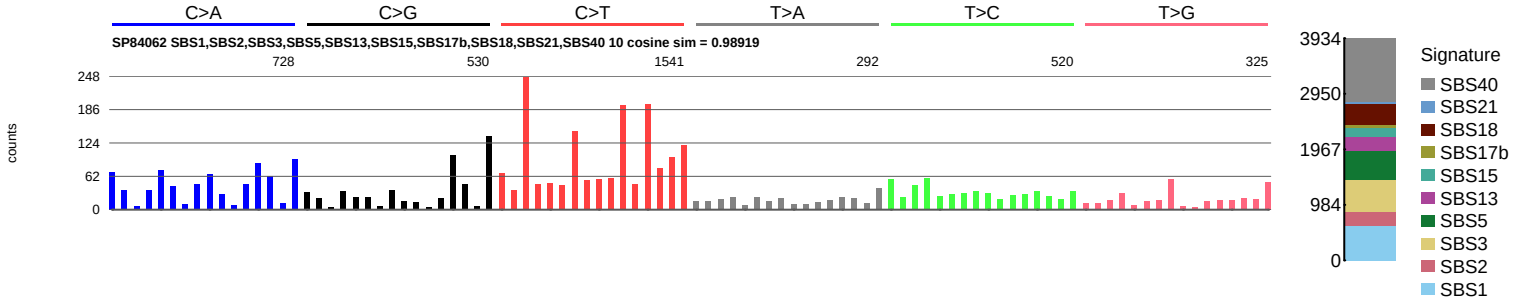

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98969
718
527
1528
318
514
329
counts

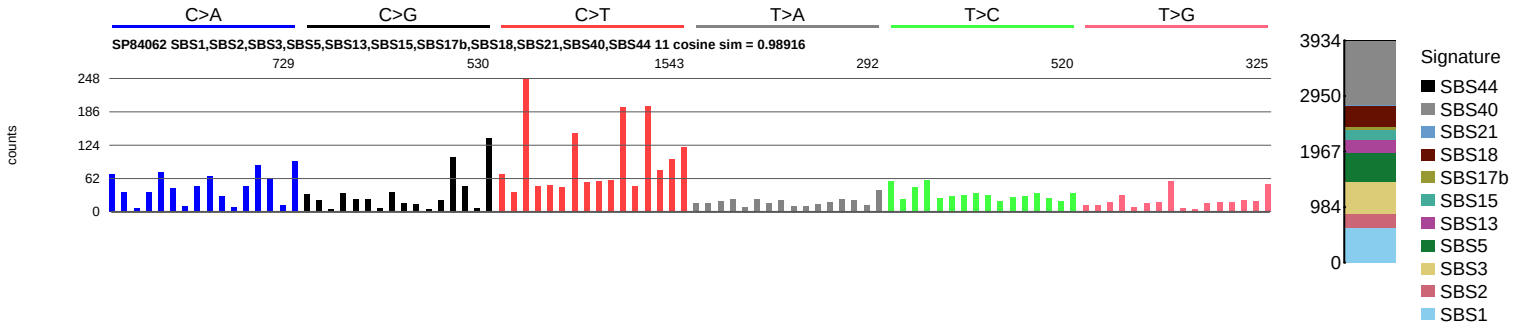

Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

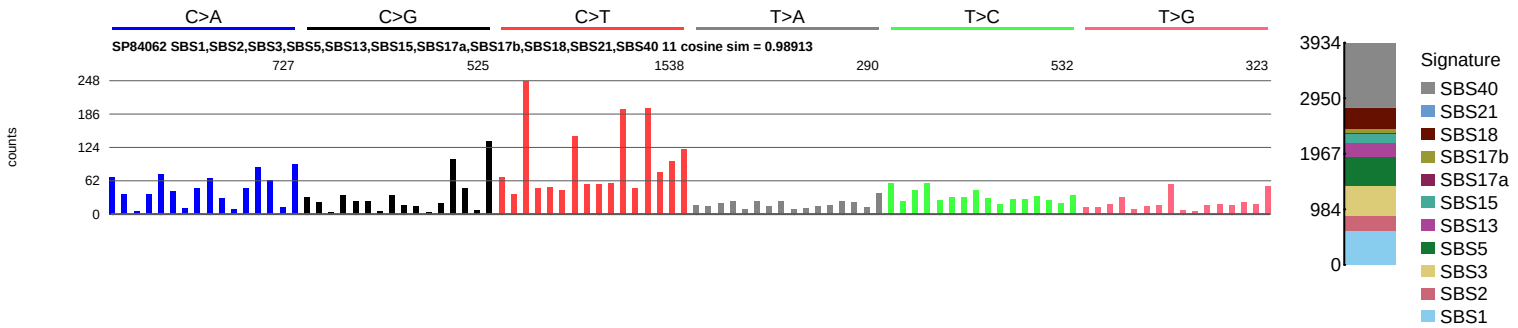

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98965
718
525
1527
314
522
329
counts

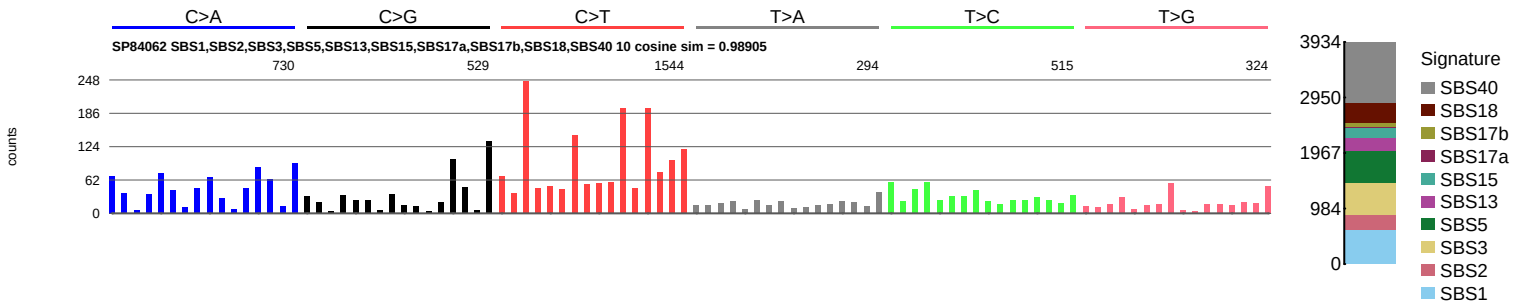

Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2

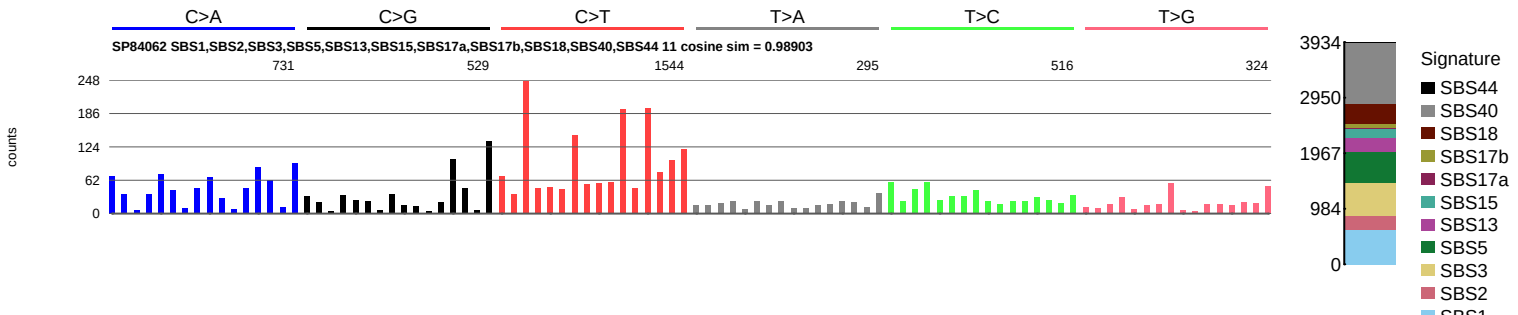

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.98921
729
525
1537
290
531
322
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

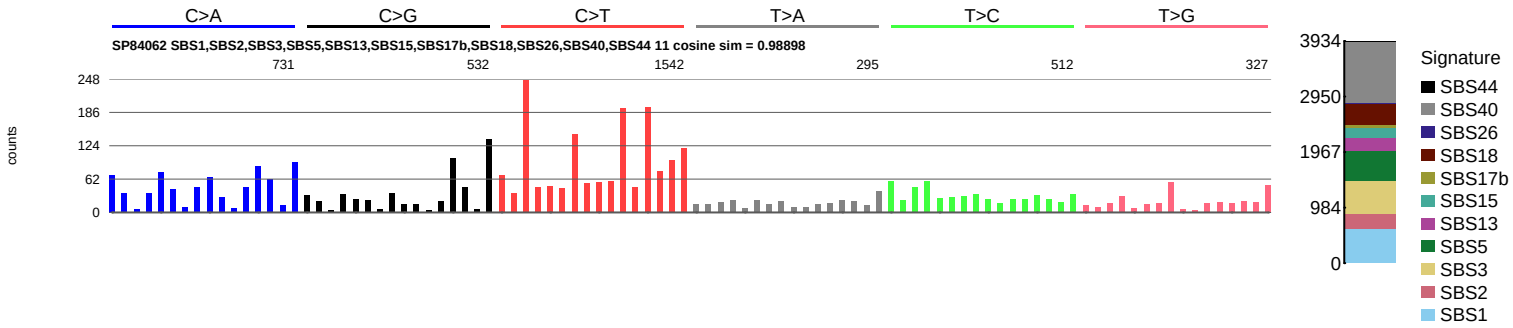

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98919
728
530
1541
292
520
325
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

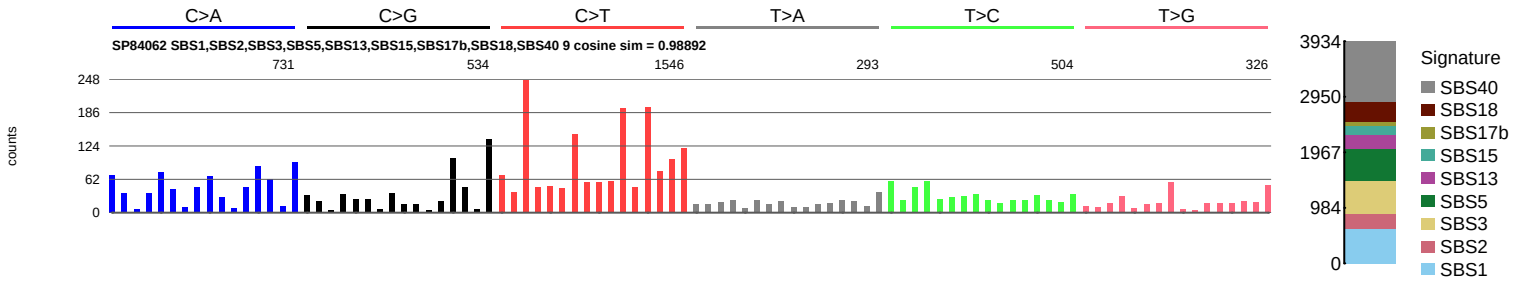

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98916
729
530
1543
292
520
325
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

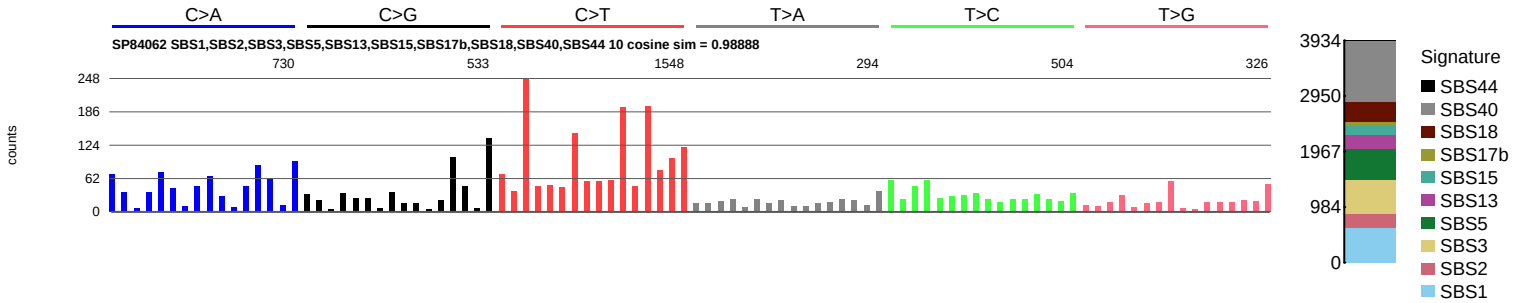

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.98913
727
525
1538
290
532
323
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

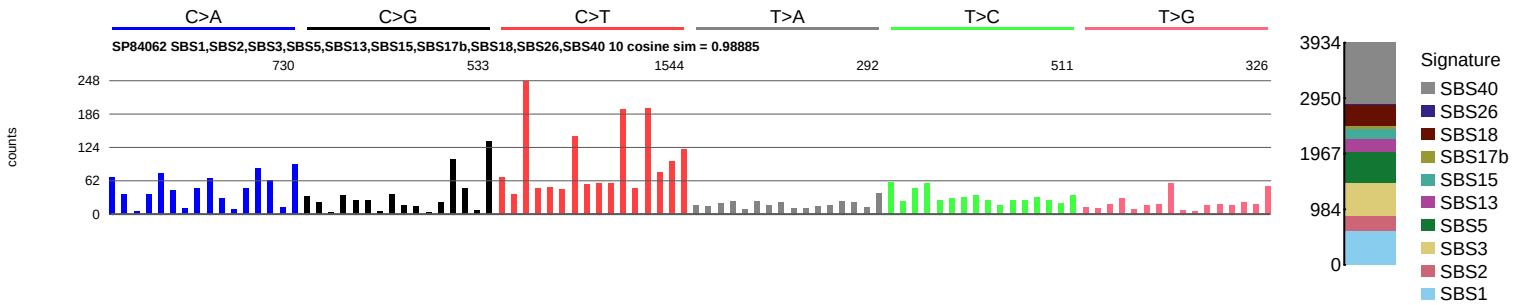

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98905
730
529
1544
294
515
324
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

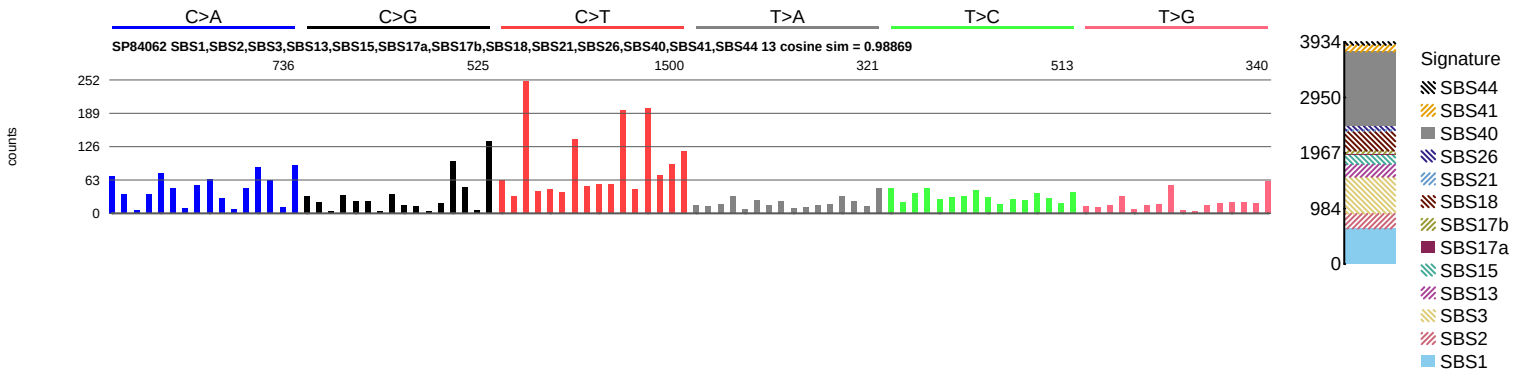

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98903
731
529
1544
295
516
324
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

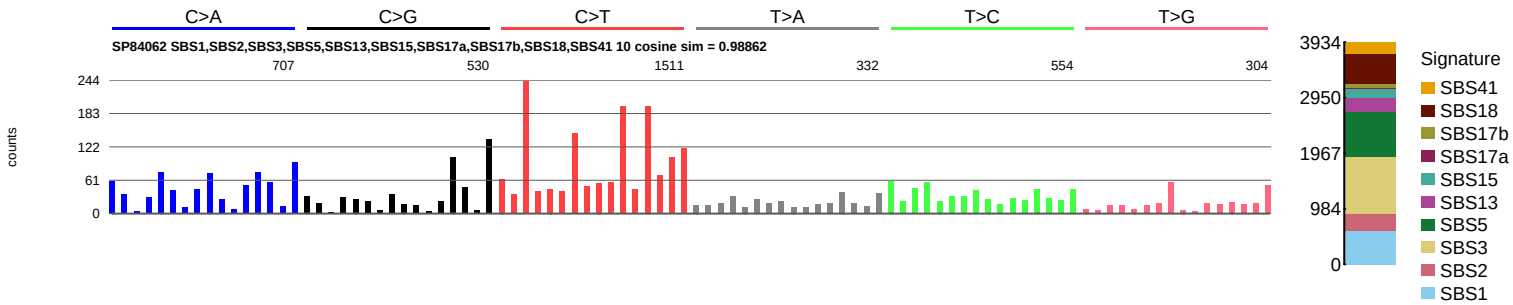

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98898
731
532
1542
295
512
327
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

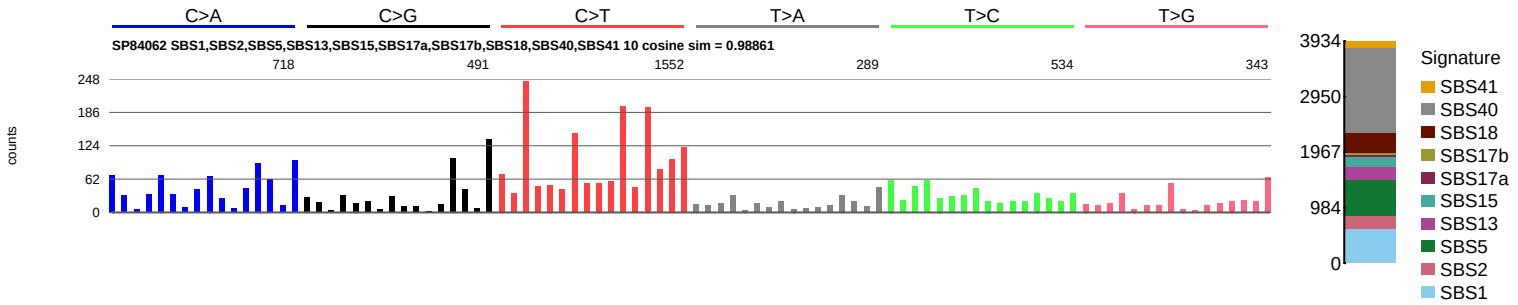

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.98892
731
534
1546
293
504
326
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

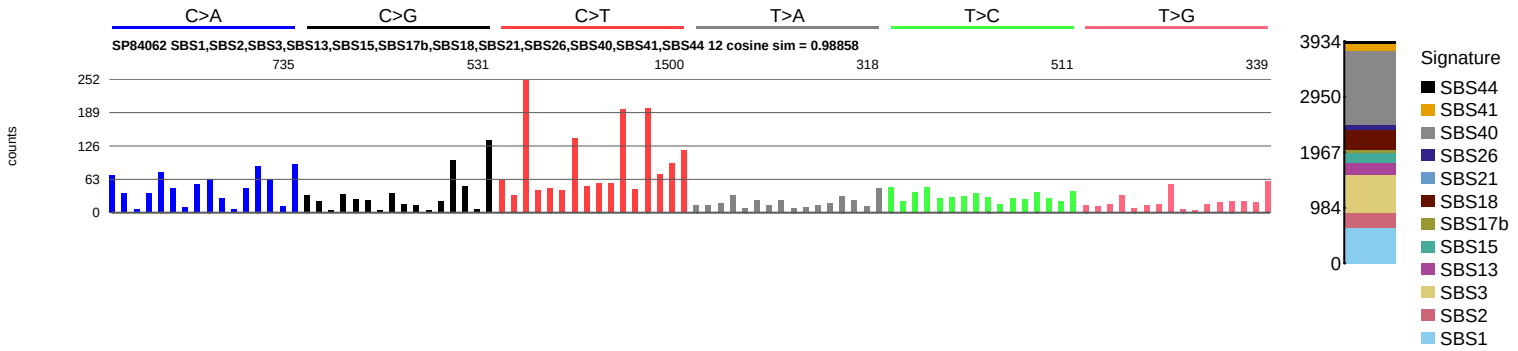

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98888
730
533
1548
294
504
326
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

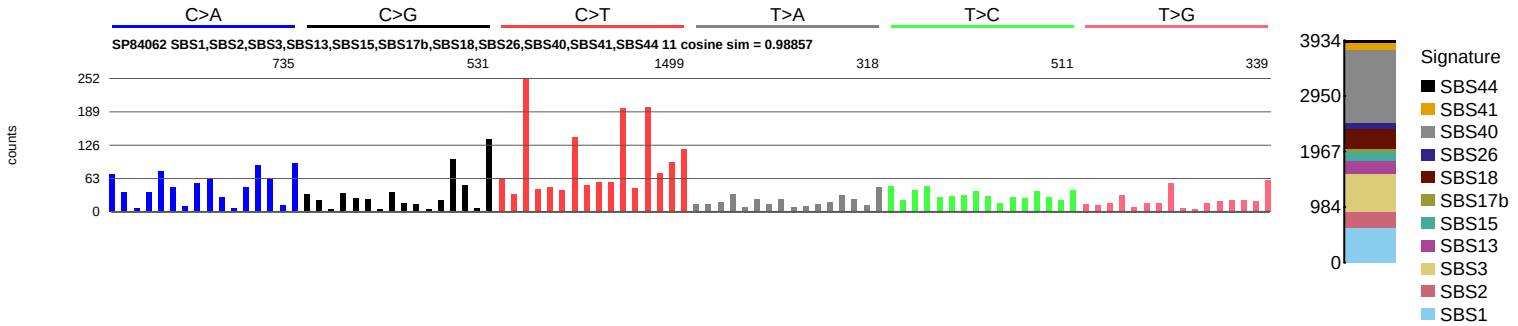

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98885
730
533
1544
292
511
326
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

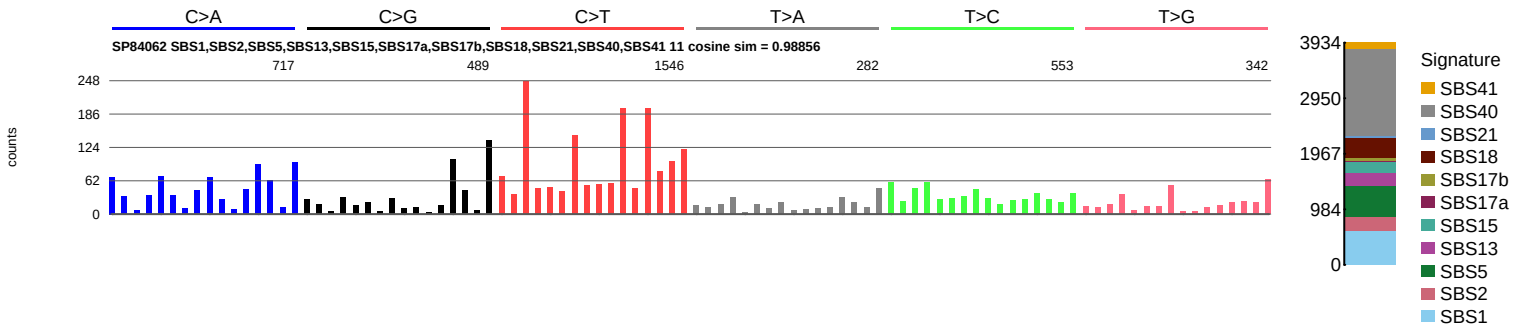

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.98869
736
525
1500
321
513
340
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

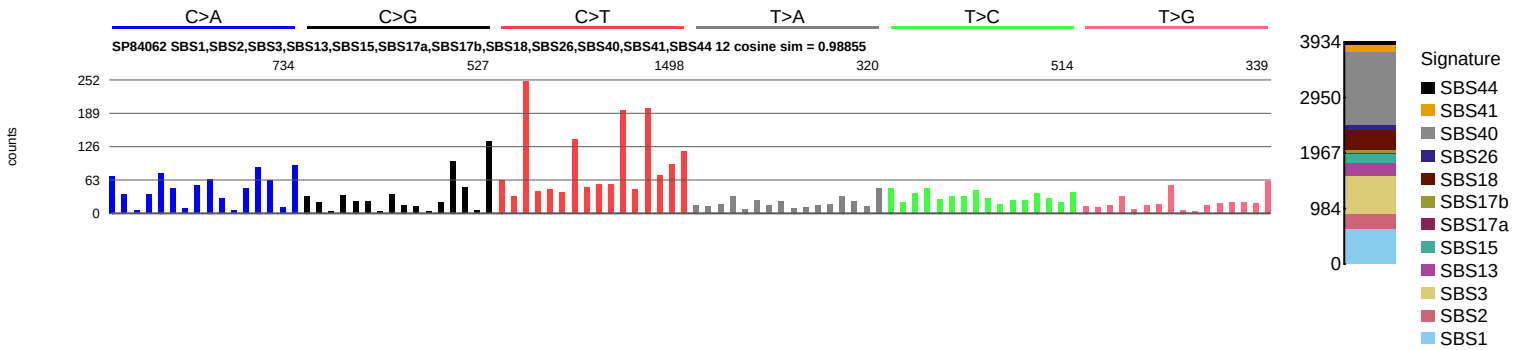

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98862
707
530
1511
332
554
304
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

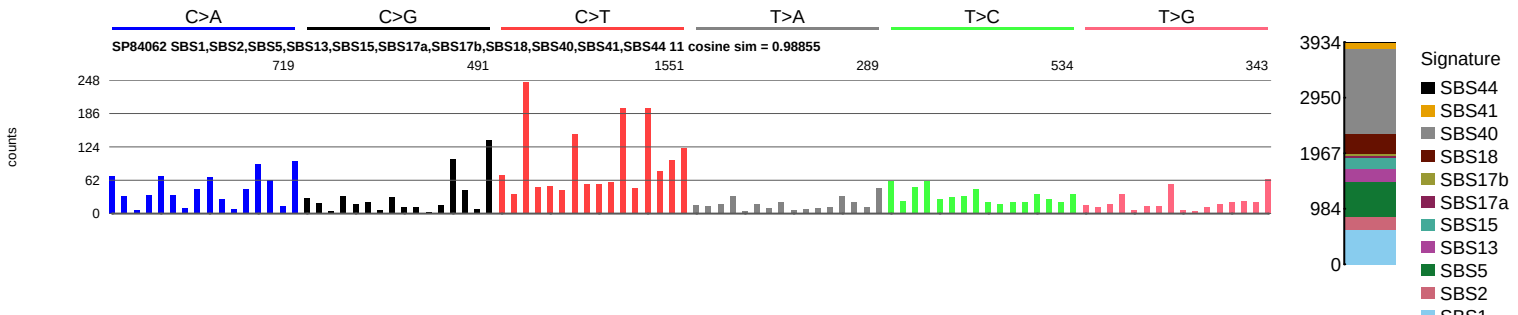

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98861
718
491
1552
289
534
343
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

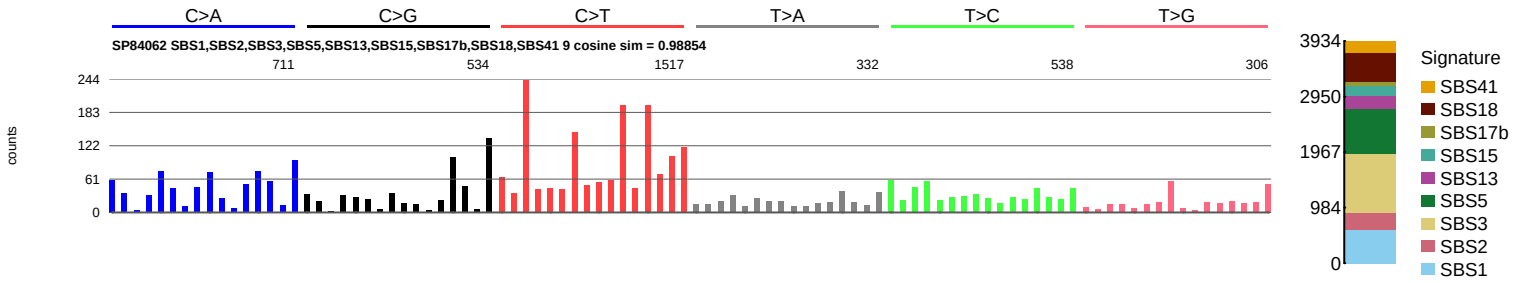

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98858
735
531
1500
318
511
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

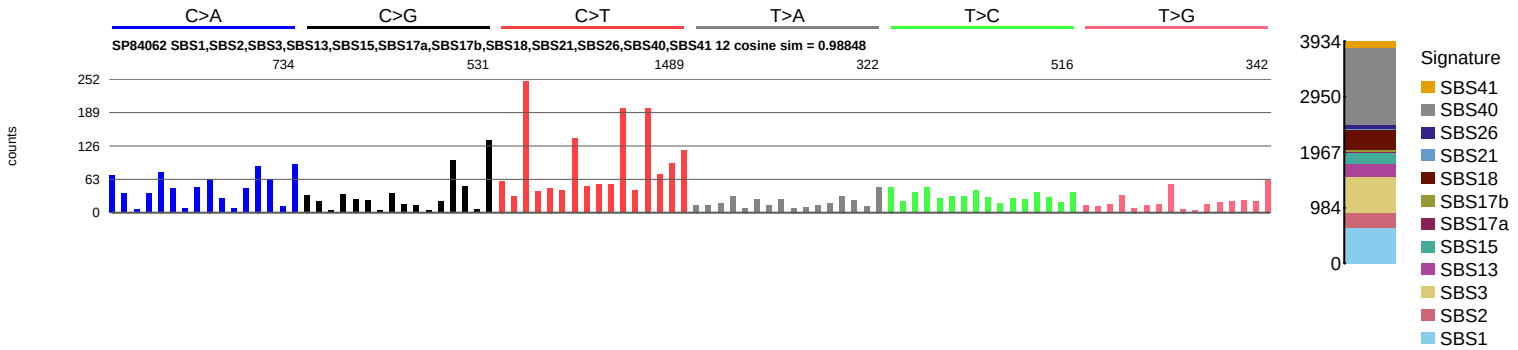

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98857
735
531
1499
318
511
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

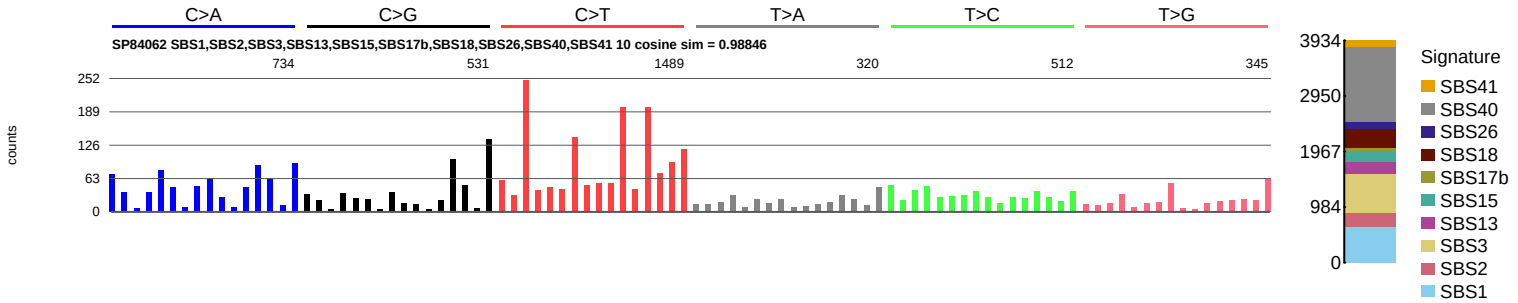

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98856
717
489
1546
282
553
342
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

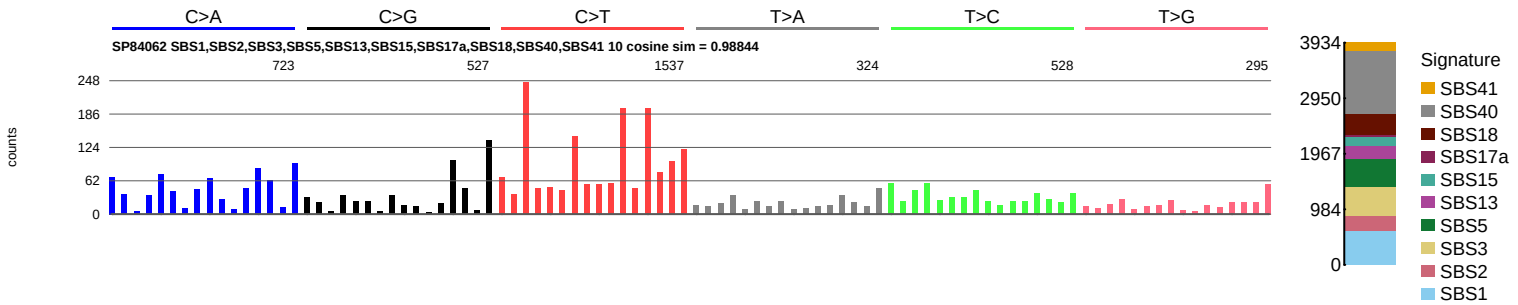

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98855
734
527
1498
320
514
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

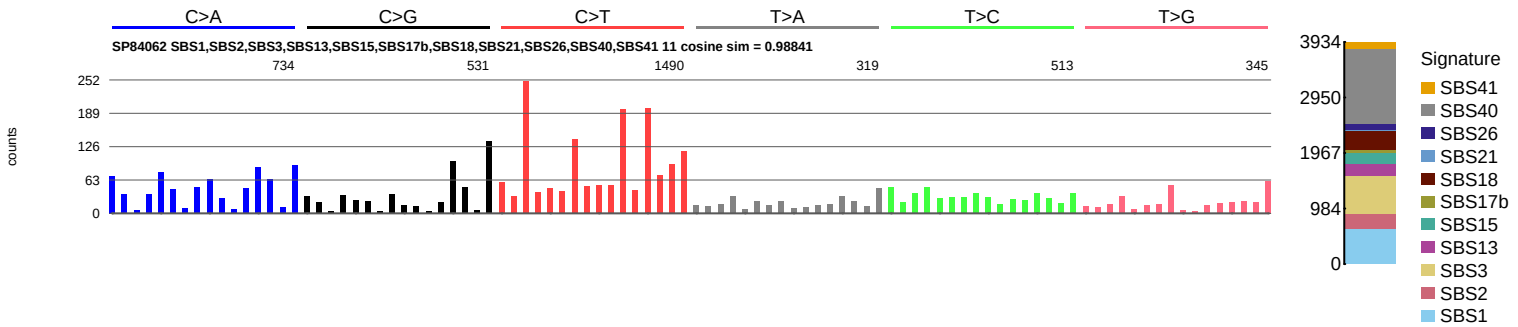

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98855
719
491
1551
289
534
343
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

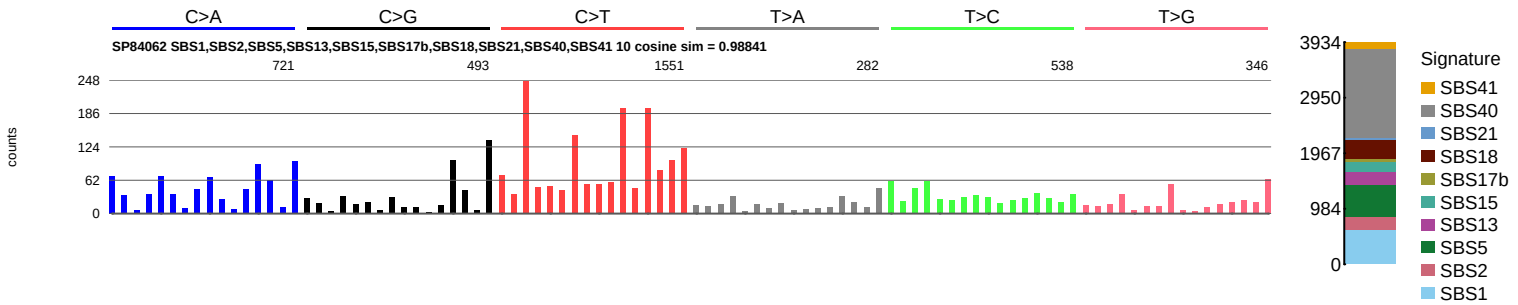

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98854
711
534
1517
332
538
306
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

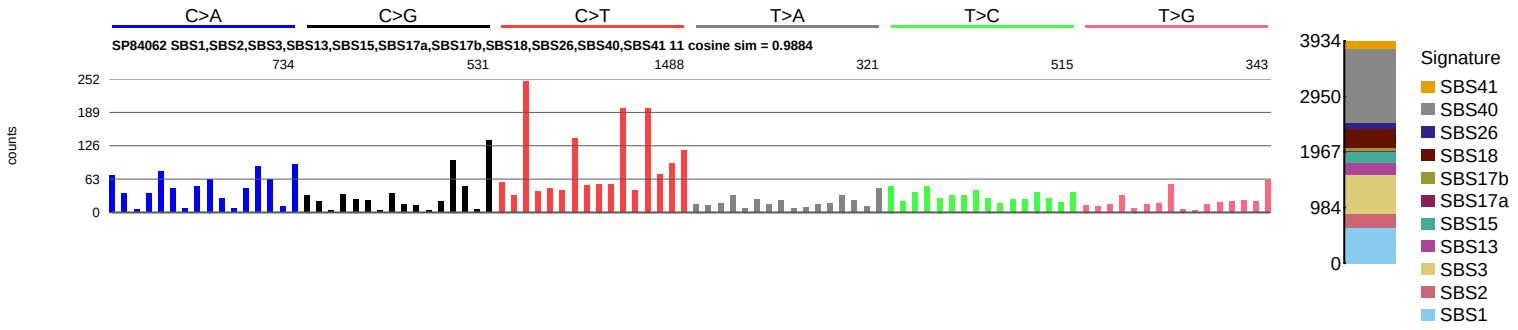

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.98848
734
531
1489
322
516
342
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

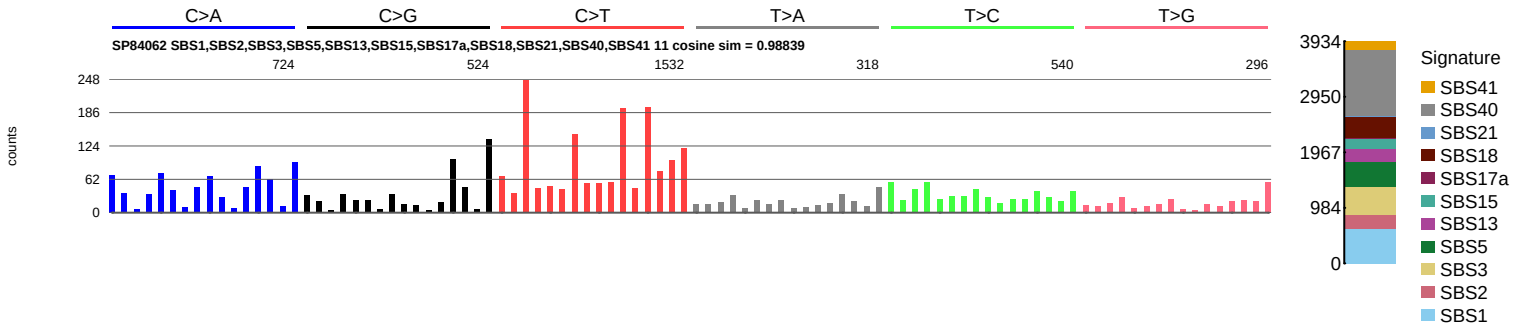

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98846
734
531
1489
320
512
345
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

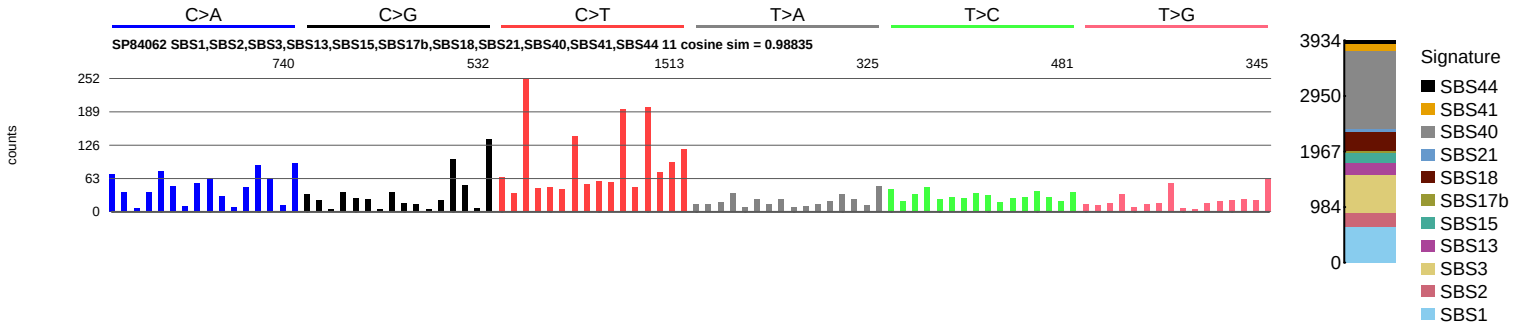

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 10 cosine sim = 0.98844
723
527
1537
324
528
295
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

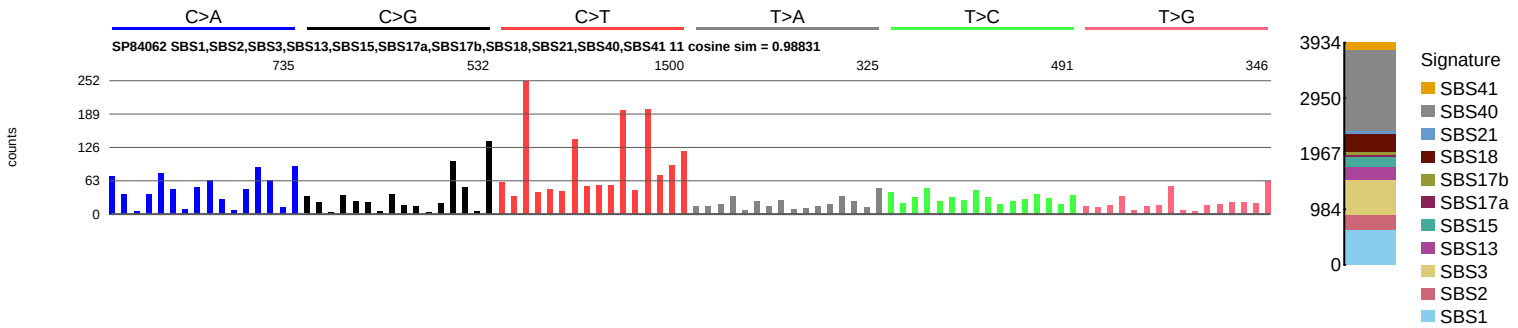

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98841
734
531
1490
319
513
345
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

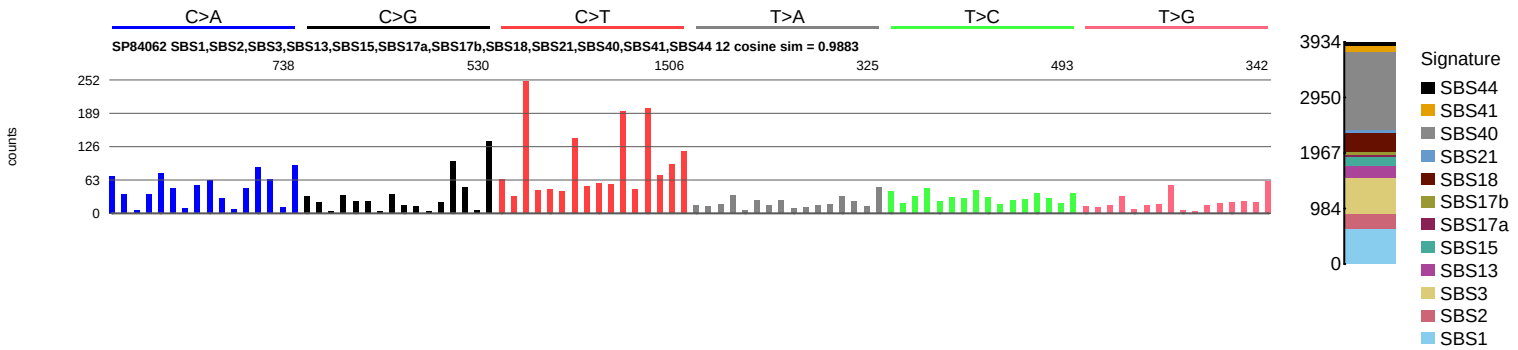

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98841
721
493
1551
282
538
346
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

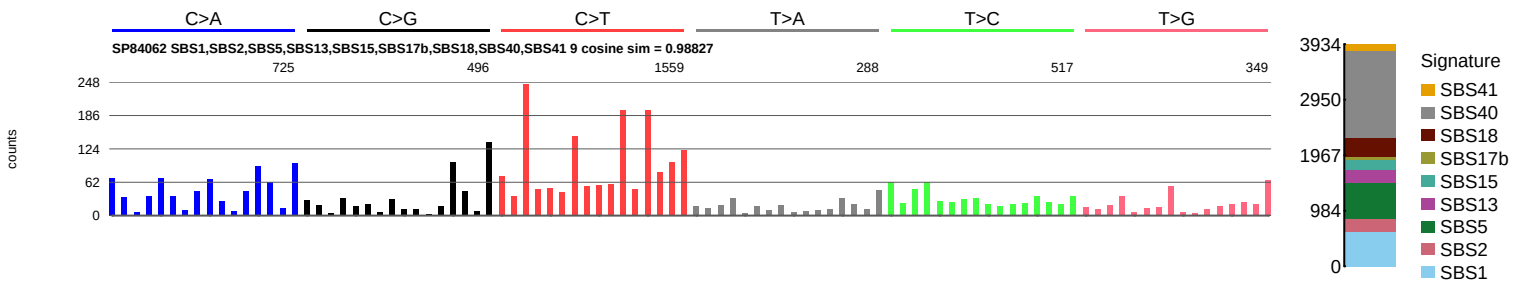

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.9884
734
531
1488
321
515
343
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

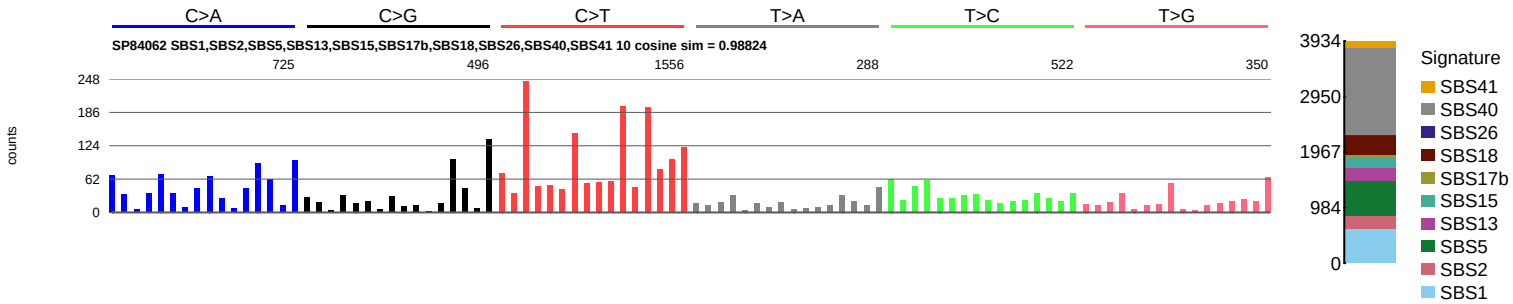

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98839
724
524
1532
318
540
296
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

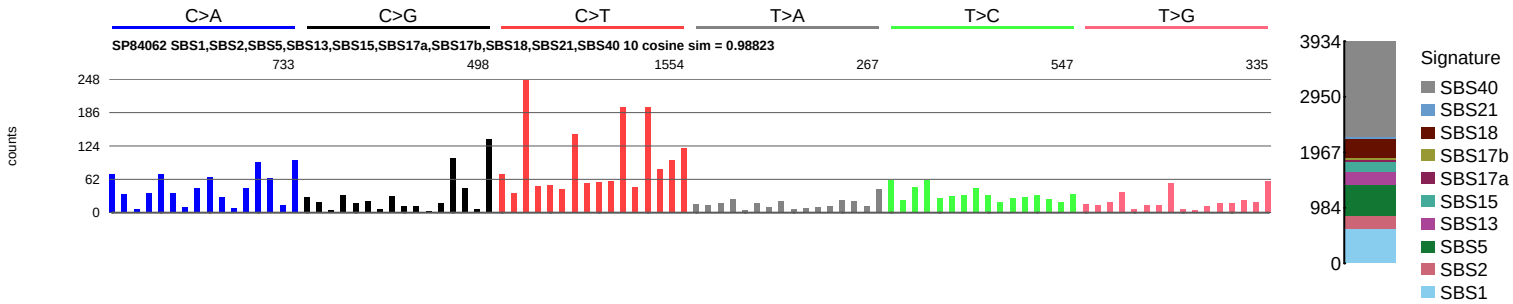

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98835
740
532
1513
325
481
345
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

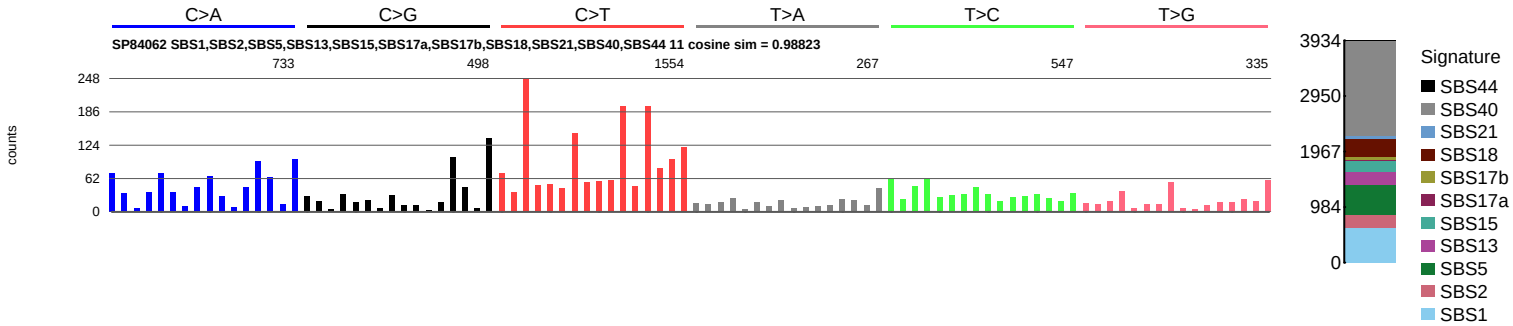

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98831
735
532
1500
325
491
346
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

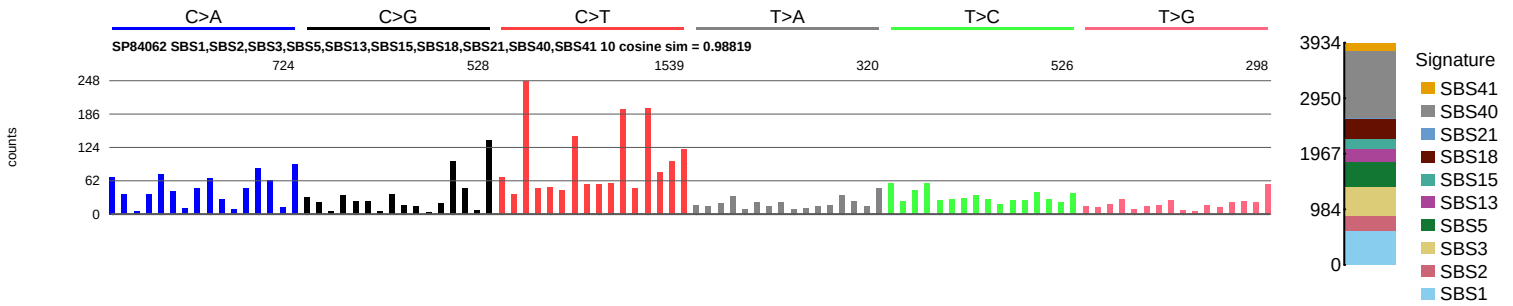

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.9883
738
530
1506
325
493
342
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

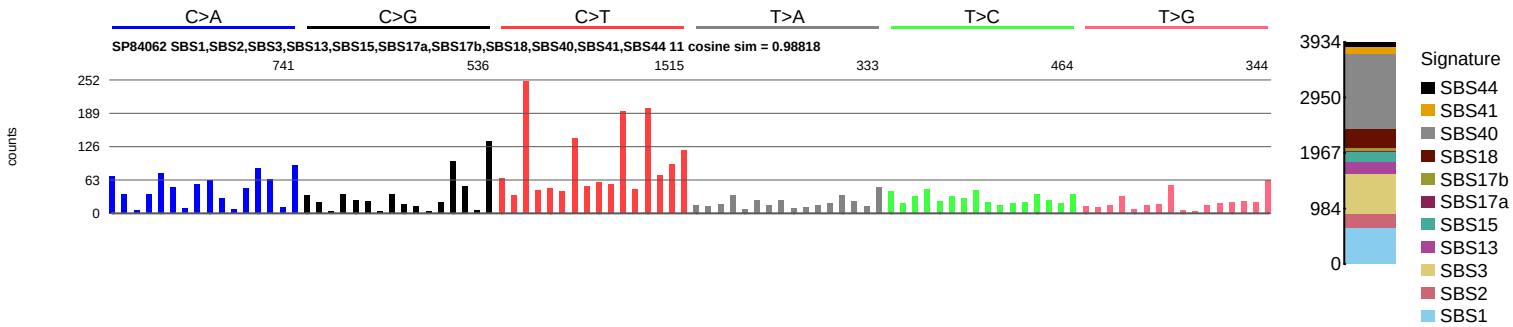

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98827
725
496
1559
288
517
349
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

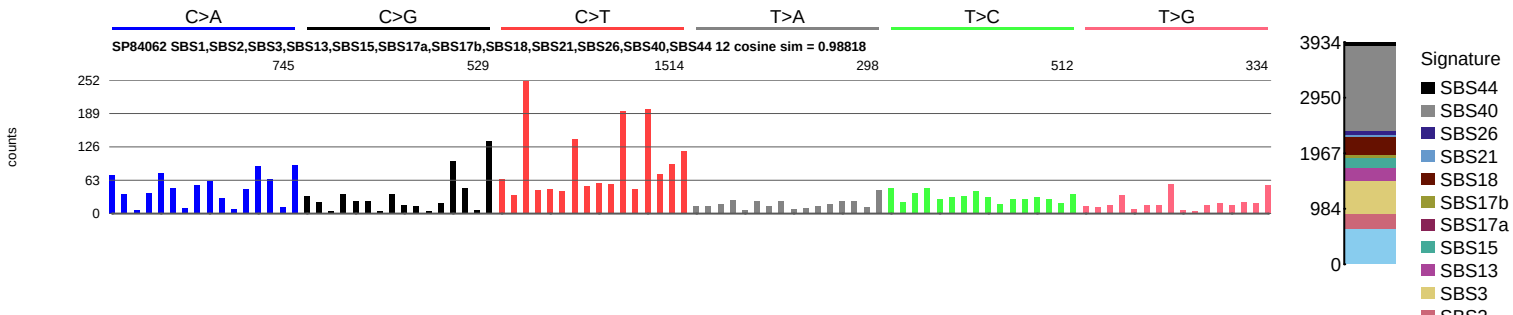

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98824
725
496
1556
288
522
350
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

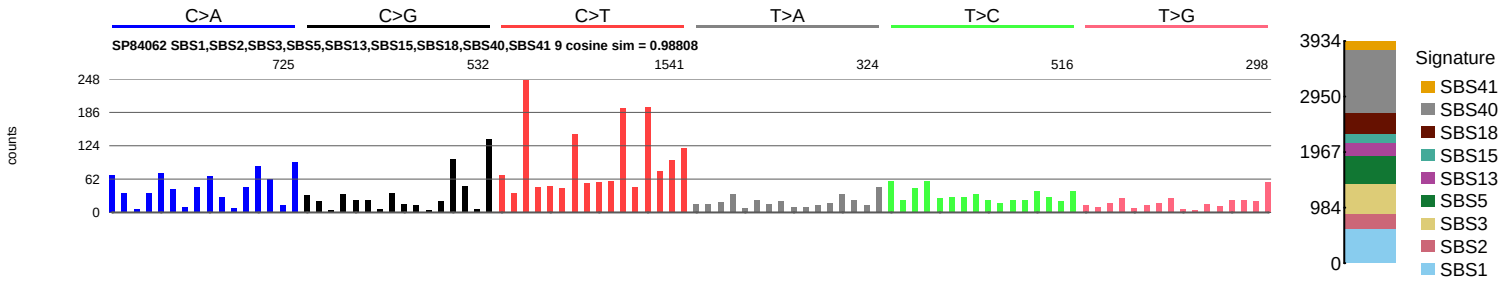

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98823
733
498
1554
267
547
335
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

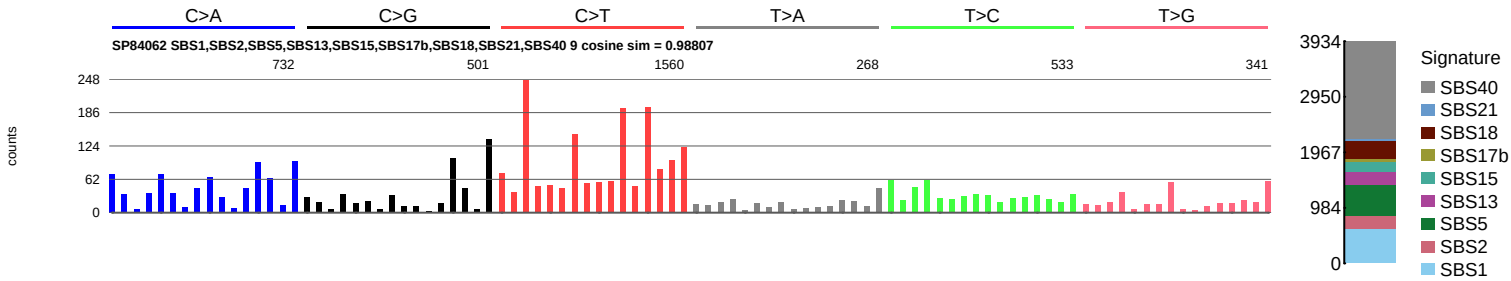

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98823
733
498
1554
267
547
335
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

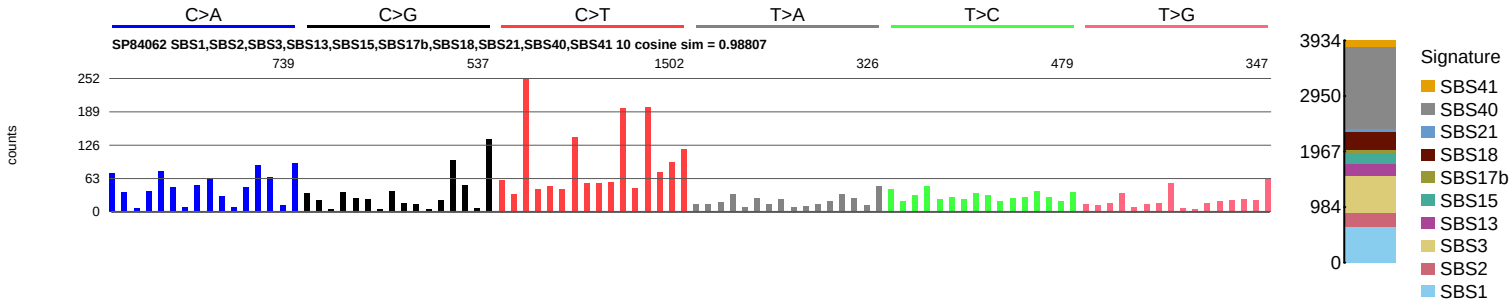

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98819
724
528
1539
320
526
298
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

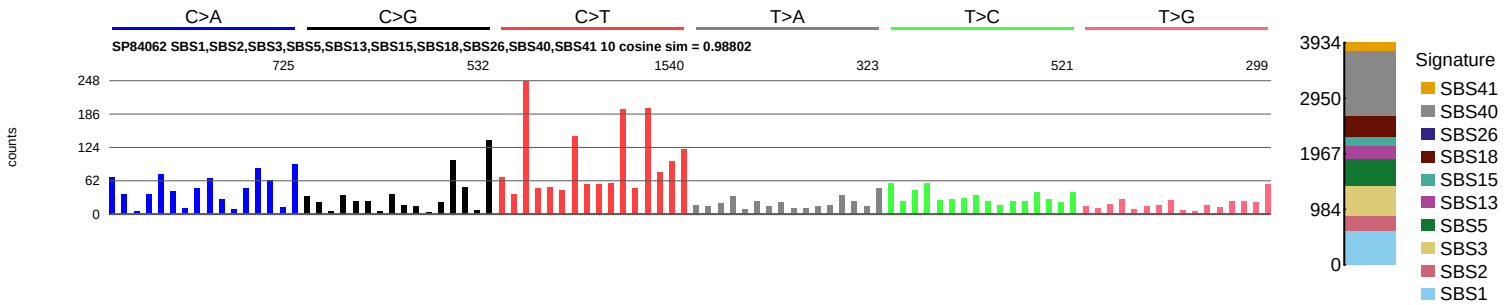

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98818
741
536
1515
333
464
344
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

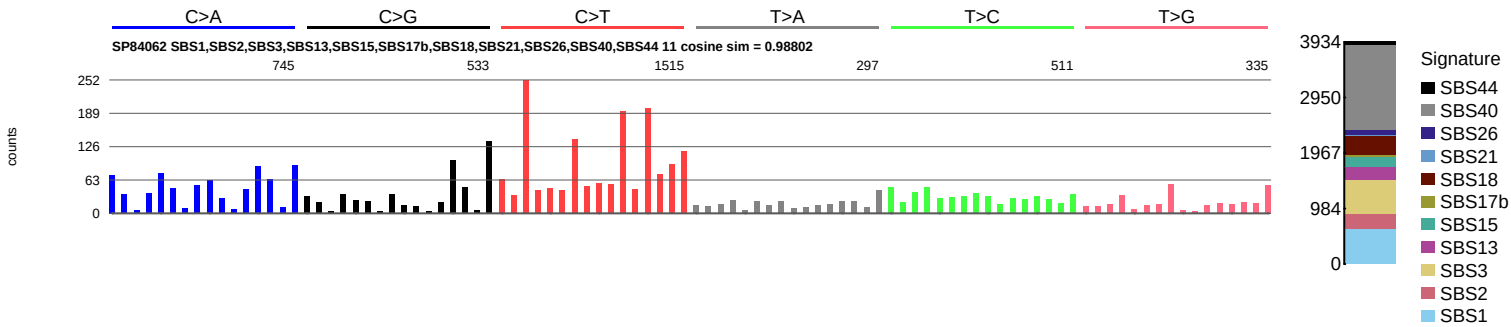

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98818
745
529
1514
298
512
334
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

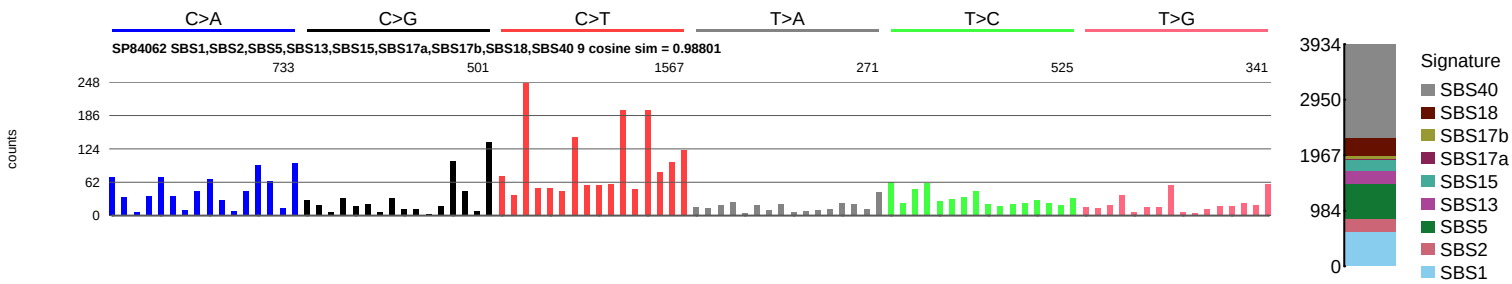

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 9 cosine sim = 0.98808
725
532
1541
324
516
298
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

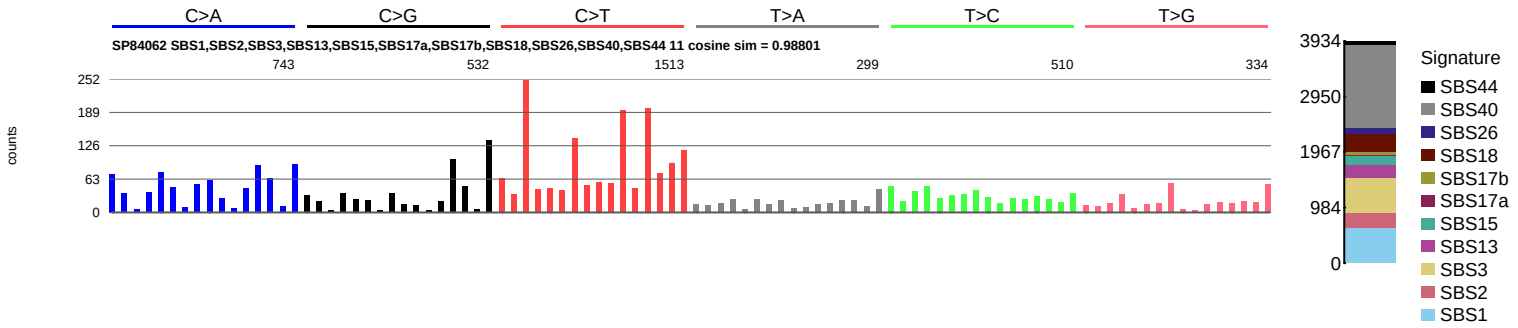

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98807
732
501
1560
268
533
341
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

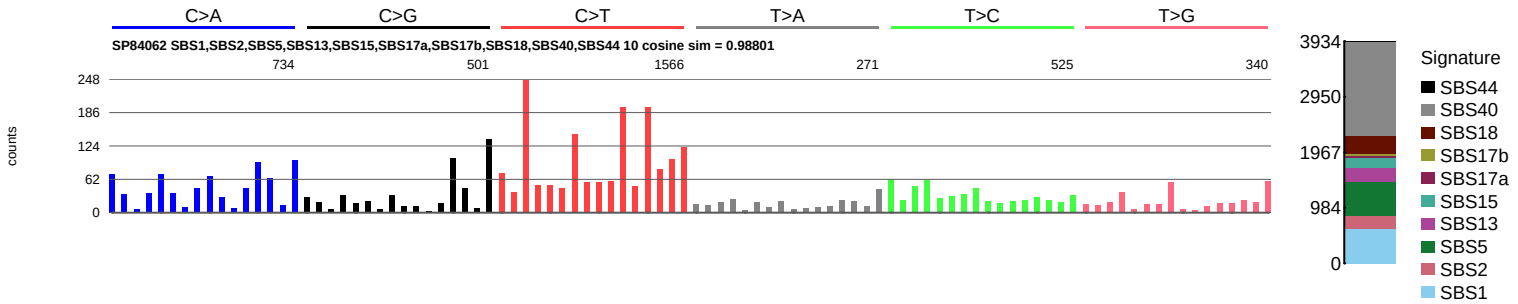

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98807
739
537
1502
326
479
347
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

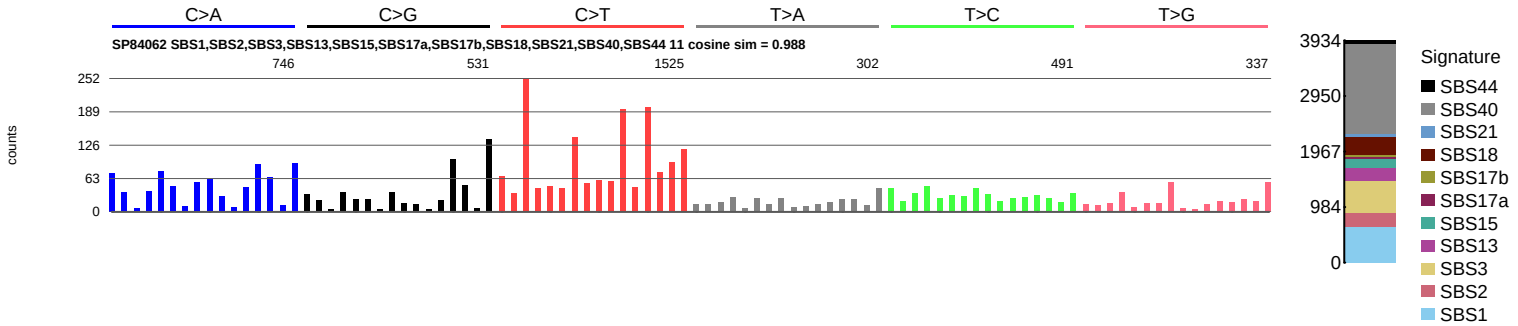

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98802
725
532
1540
323
521
299
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

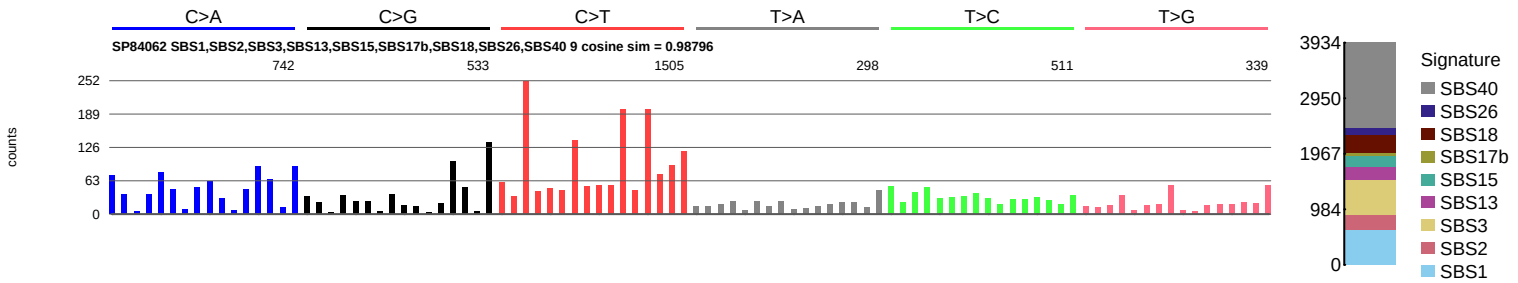

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98802
745
533
1515
297
511
335
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

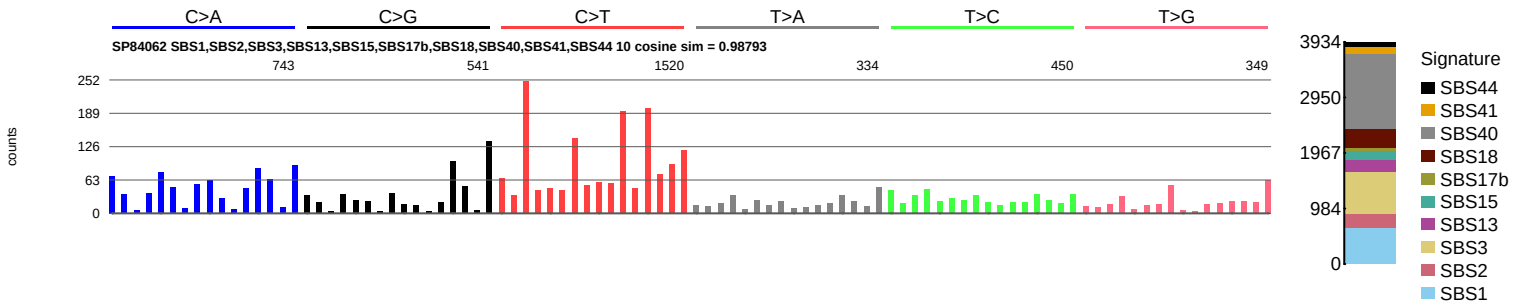

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98801
733
501
1567
271
525
341
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

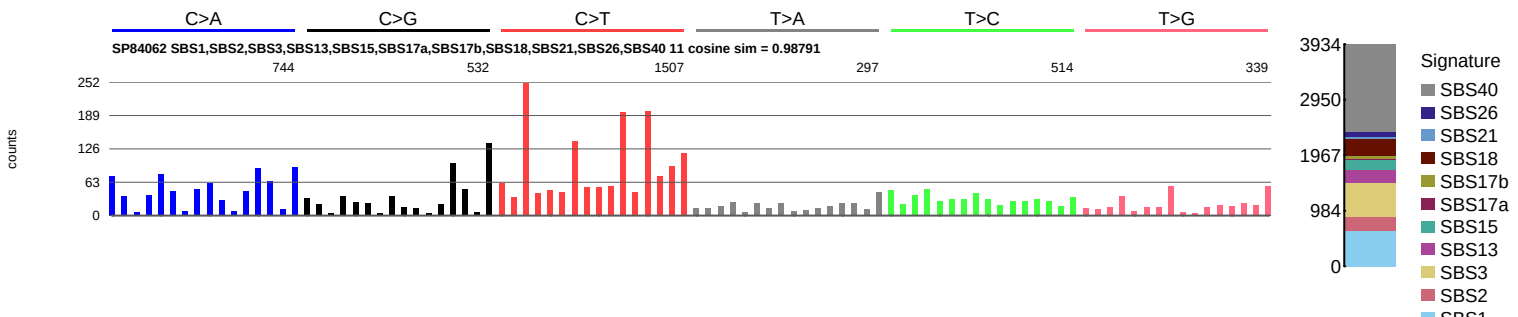

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98801
743
532
1513
299
510
334
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

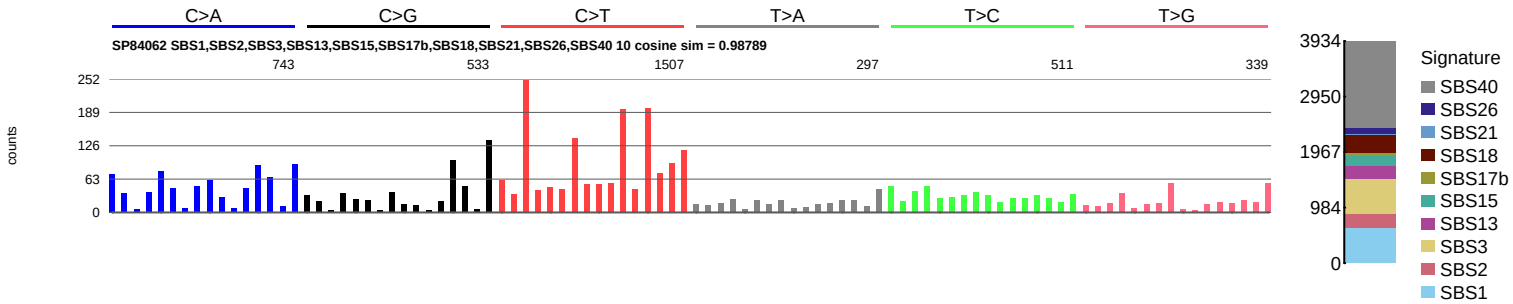

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98801
734
501
1566
271
525
340
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

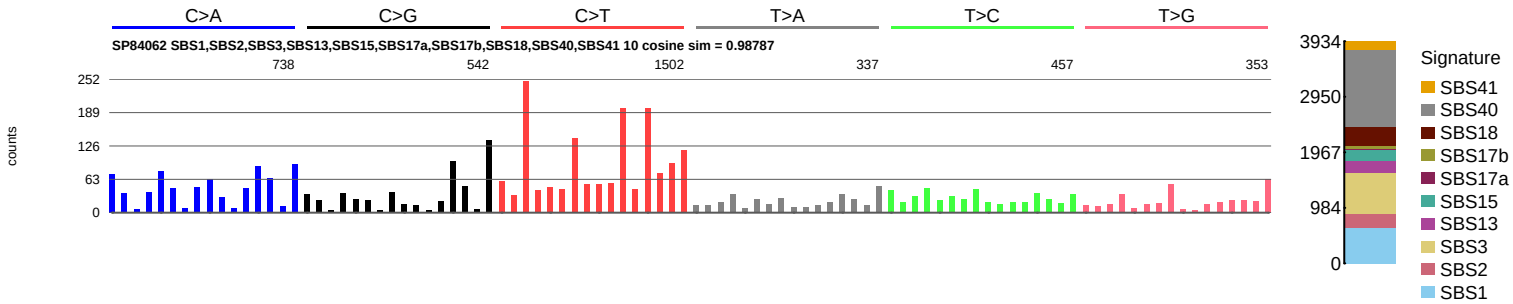

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.988
746
531
1525
302
491
337
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

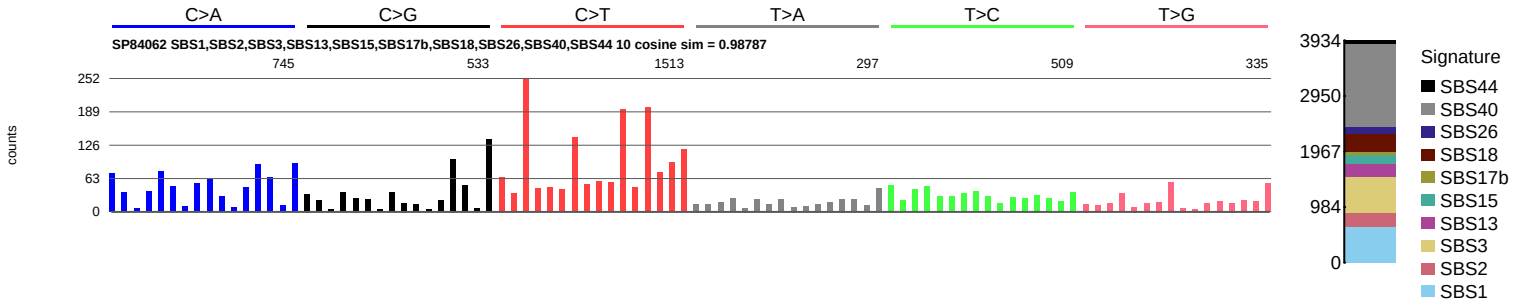

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98796
742
533
1505
298
511
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

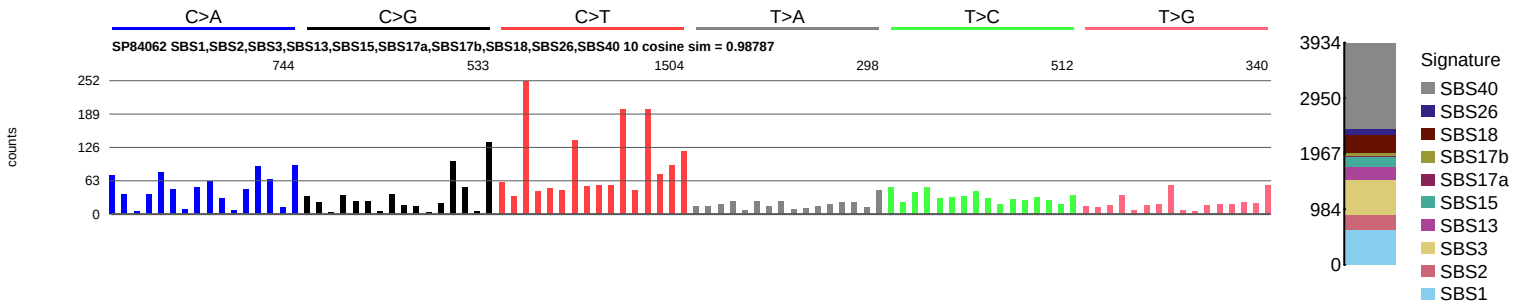

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98793
743
541
1520
334
450
349
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

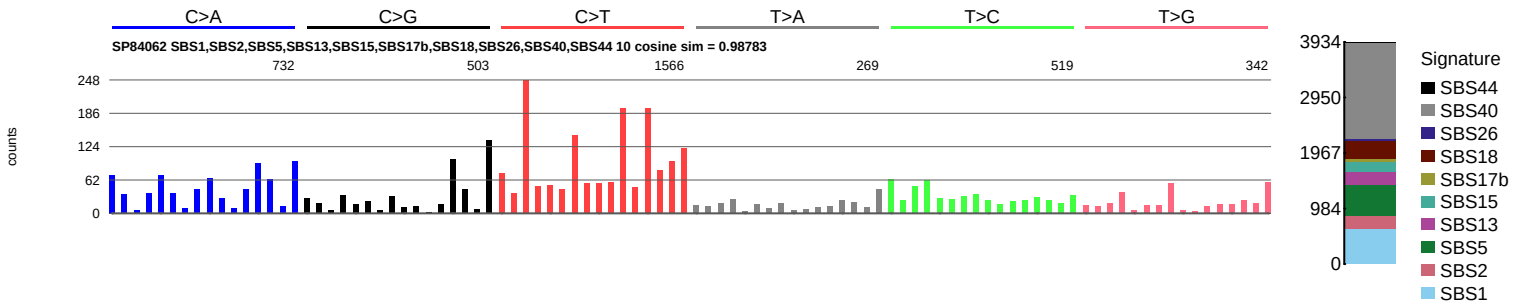

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98791
744
532
1507
297
514
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

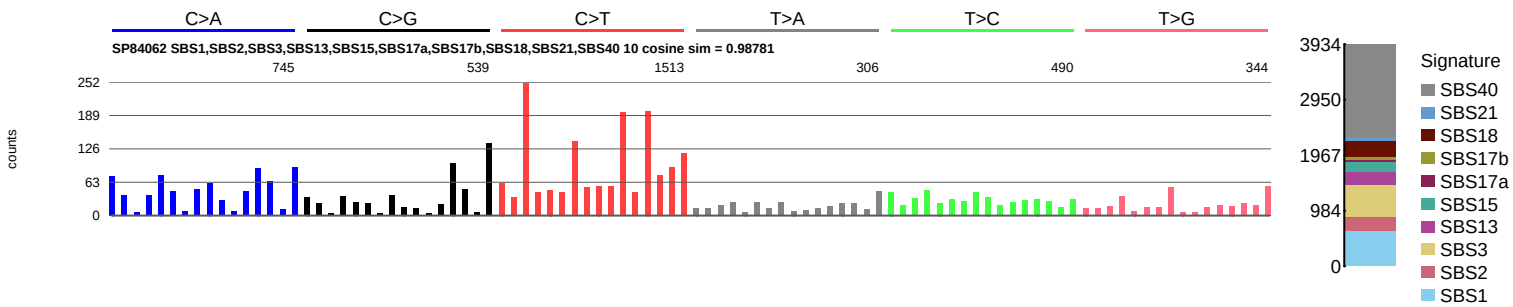

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98789
743
533
1507
297
511
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

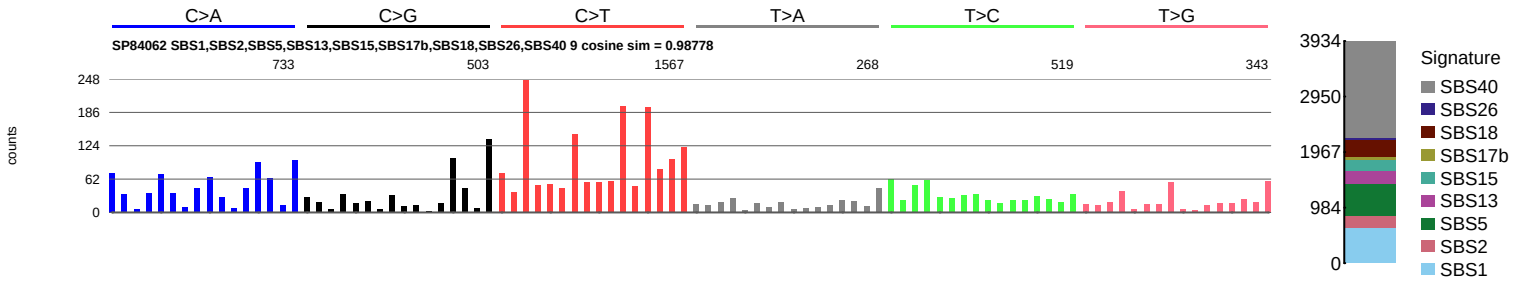

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98787
738
542
1502
337
457
353
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

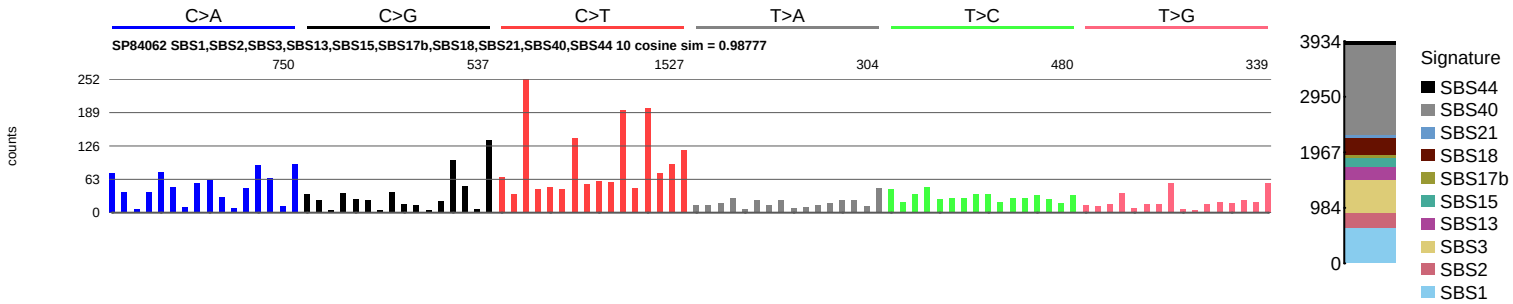

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98787
745
533
1513
297
509
335
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

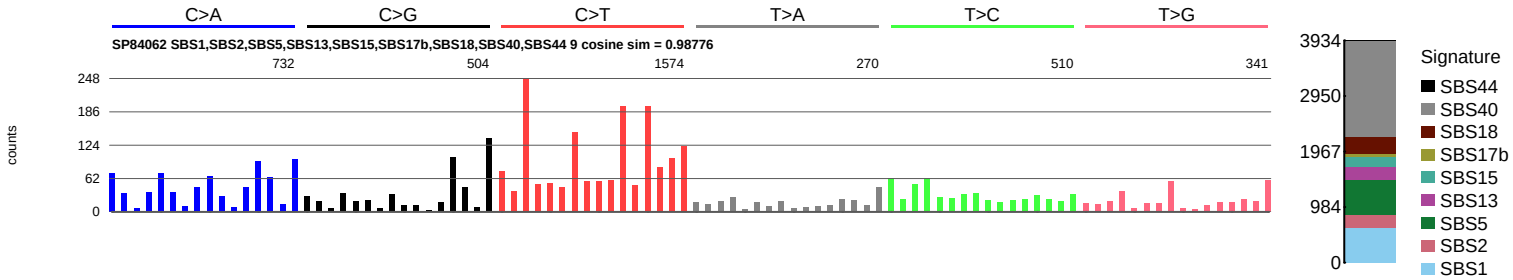

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98787
744
533
1504
298
512
340
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

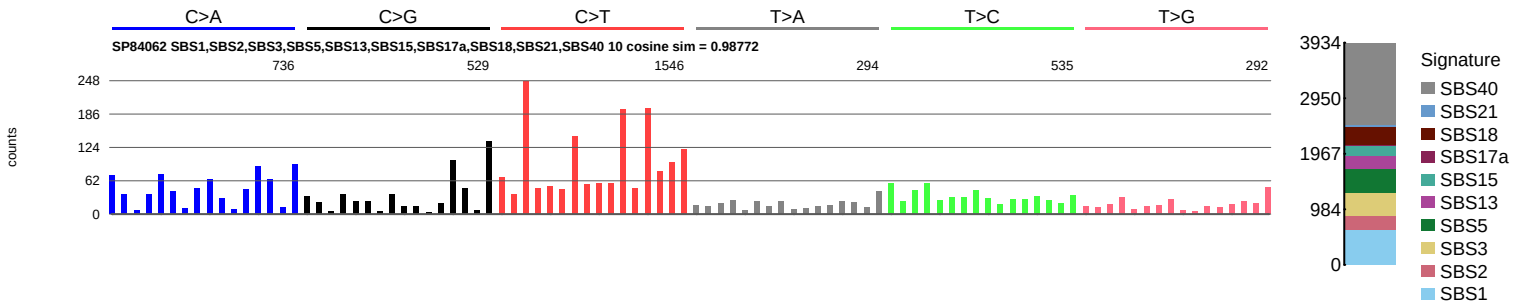

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98783
732
503
1566
269
519
342
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

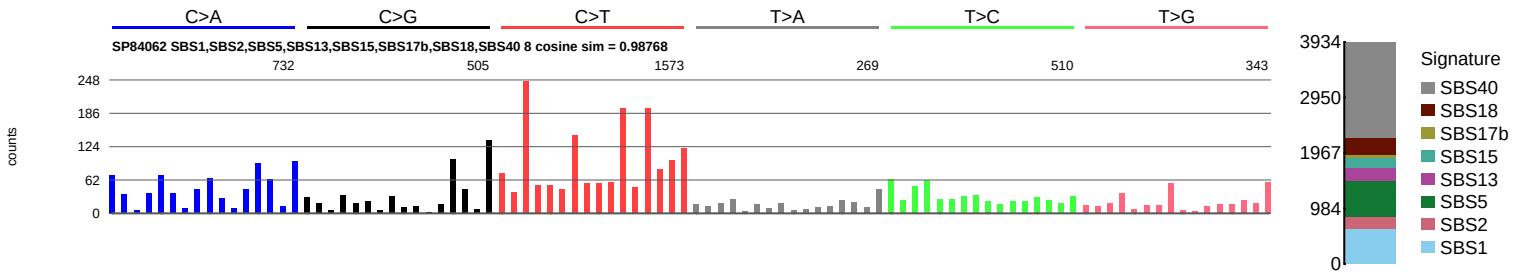

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98781
745
539
1513
306
490
344
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

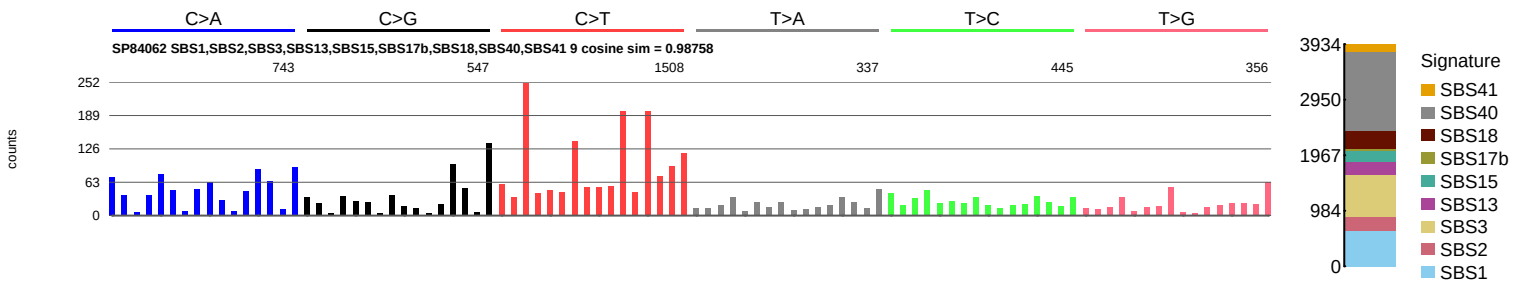

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98778
733
503
1567
268
519
343
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

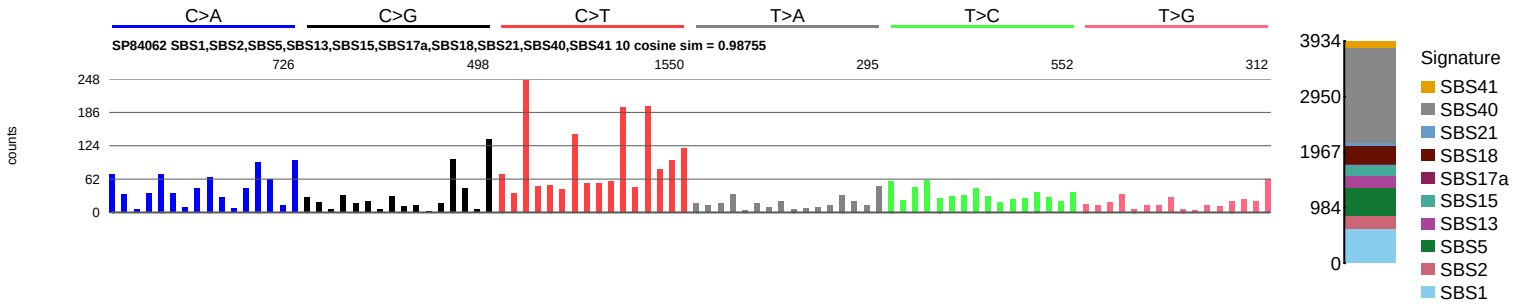

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98777
750
537
1527
304
480
339
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

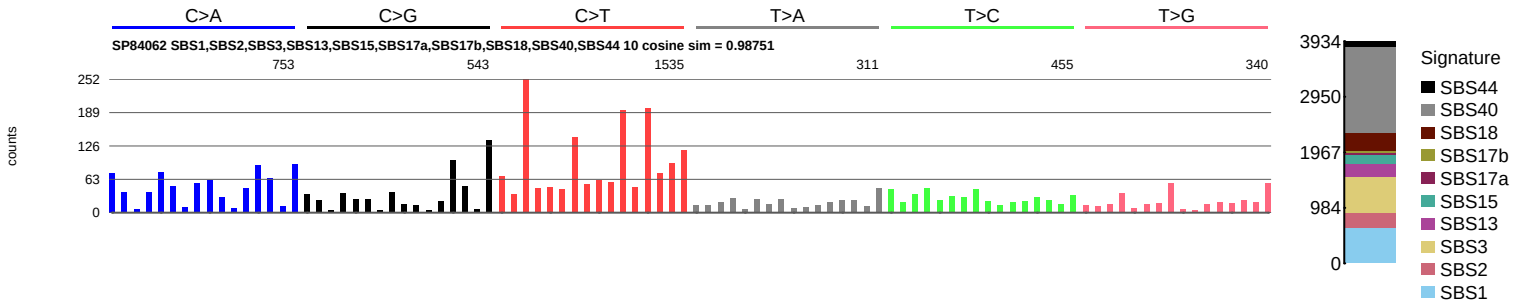

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98776
732
504
1574
270
510
341
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

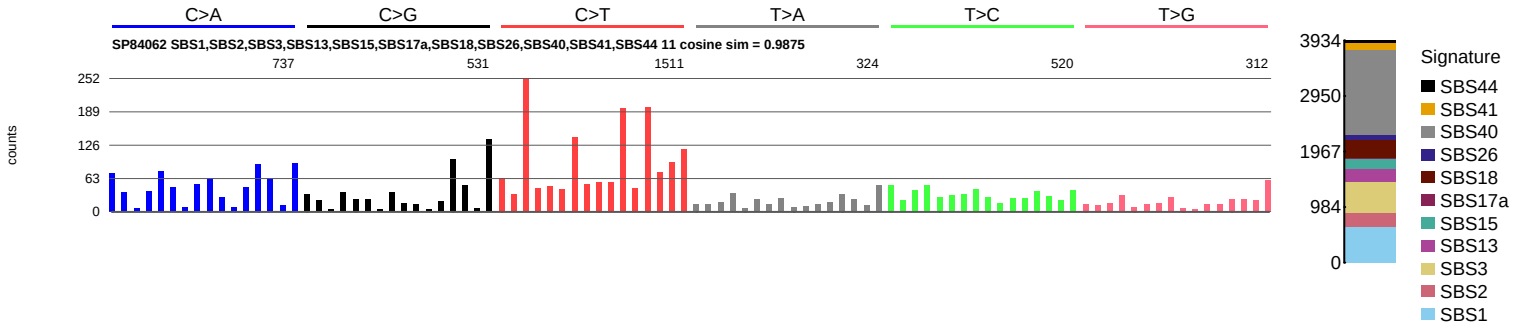

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 10 cosine sim = 0.98772
736
529
1546
294
535
292
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

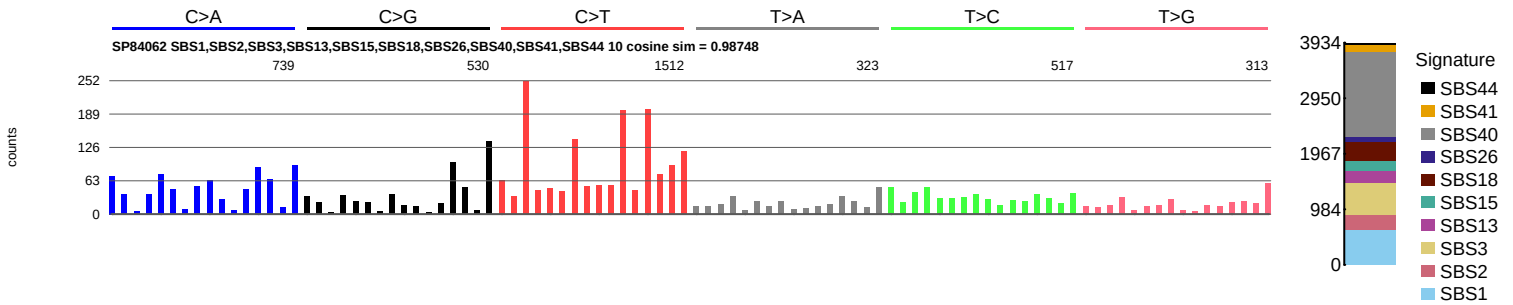

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98768
732
505
1573
269
510
343
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

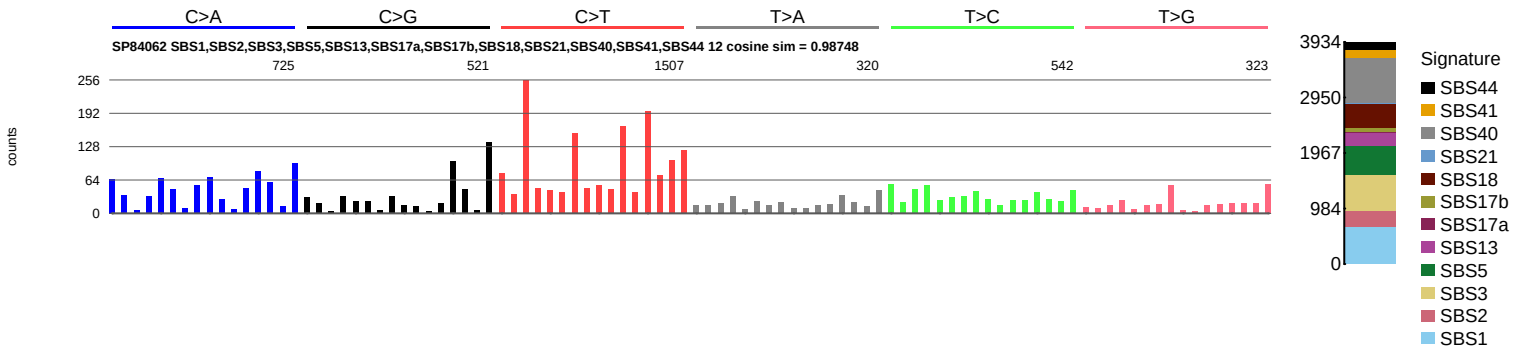

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98758
743
547
1508
337
445
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

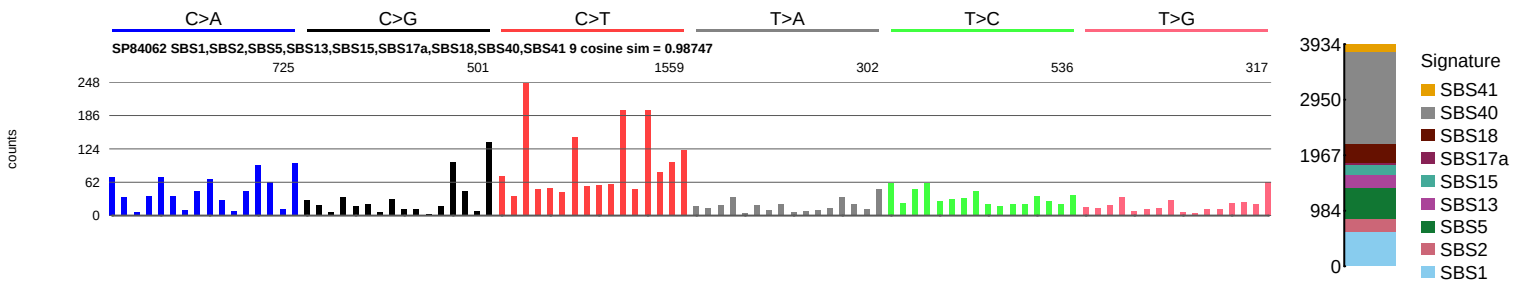

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98755
726
498
1550
295
552
312
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

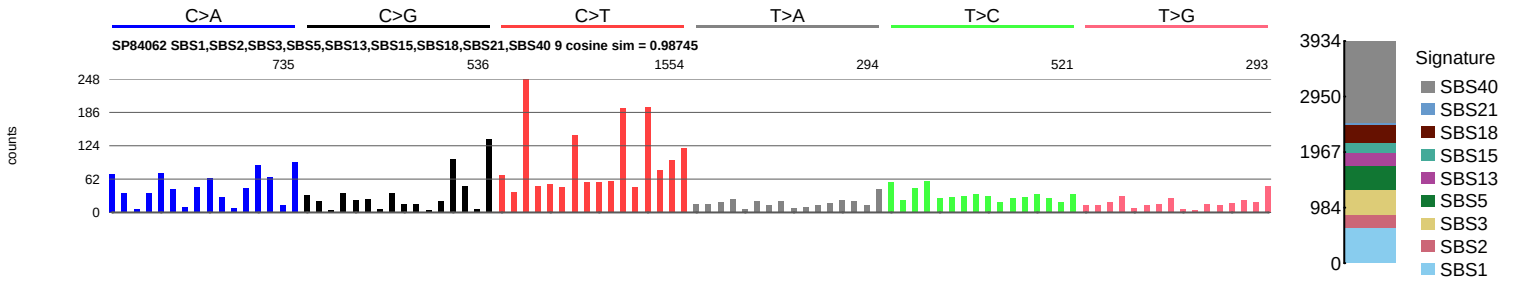

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98751
753
543
1535
311
455
340
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

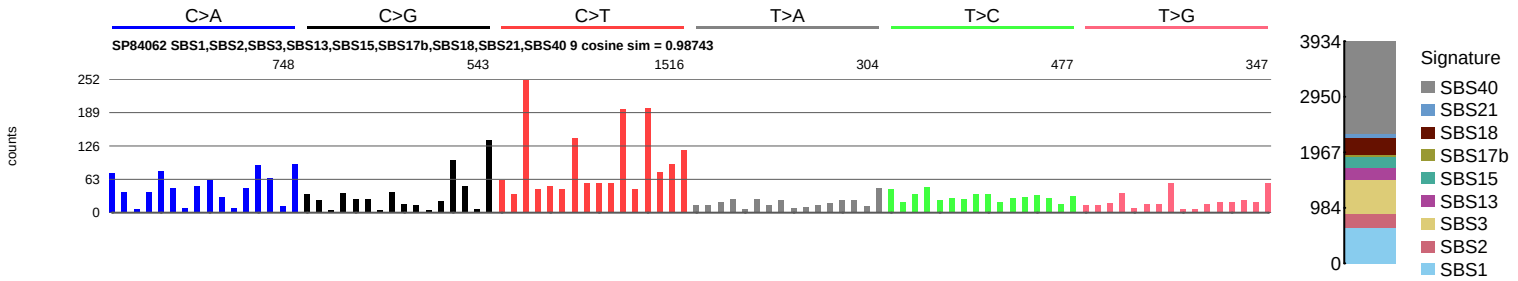

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9875
737
531
1511
324
520
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

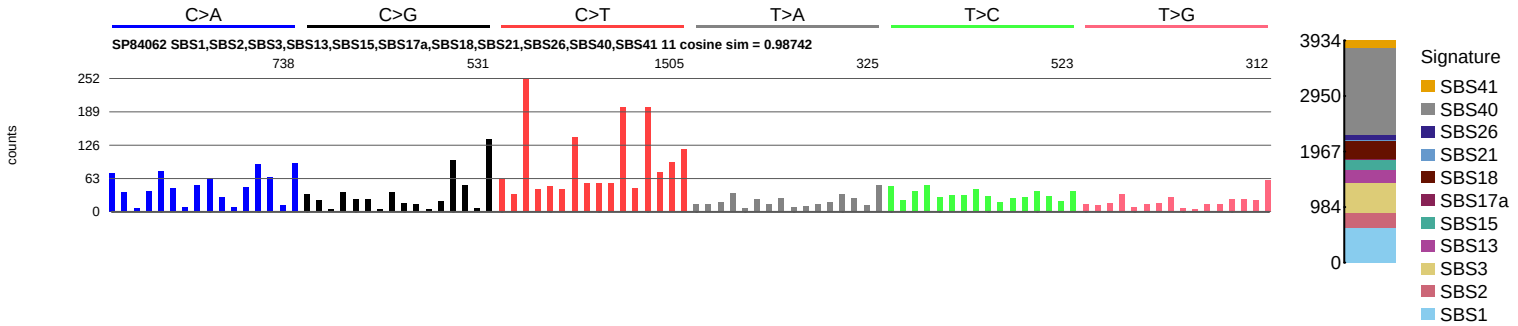

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98748
739
530
1512
323
517
313
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

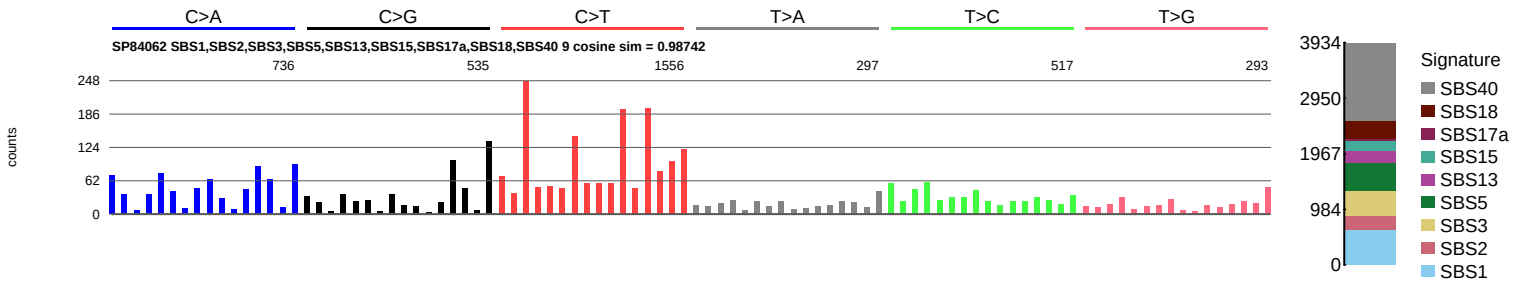

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98748
725
521
1507
320
542
323
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

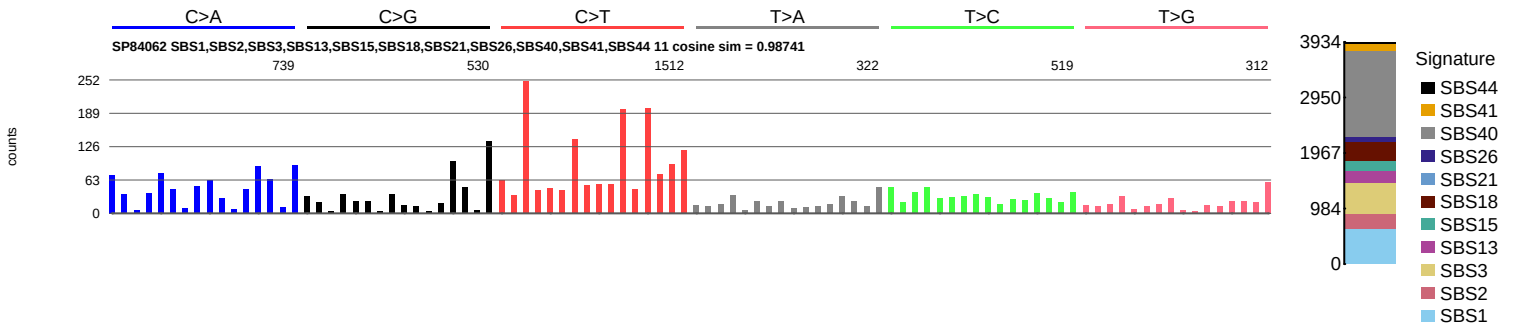

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98747
725
501
1559
302
536
317
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

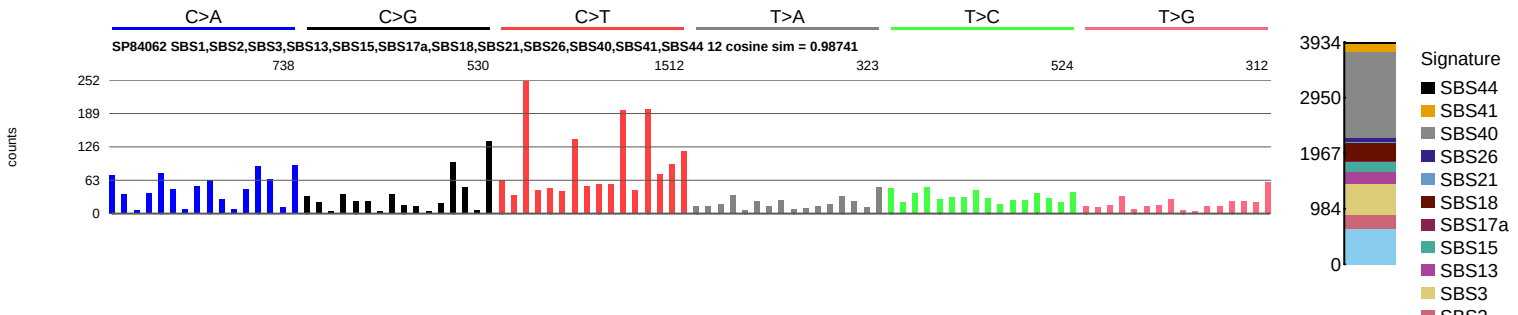

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 9 cosine sim = 0.98745
735
536
1554
294
521
293
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

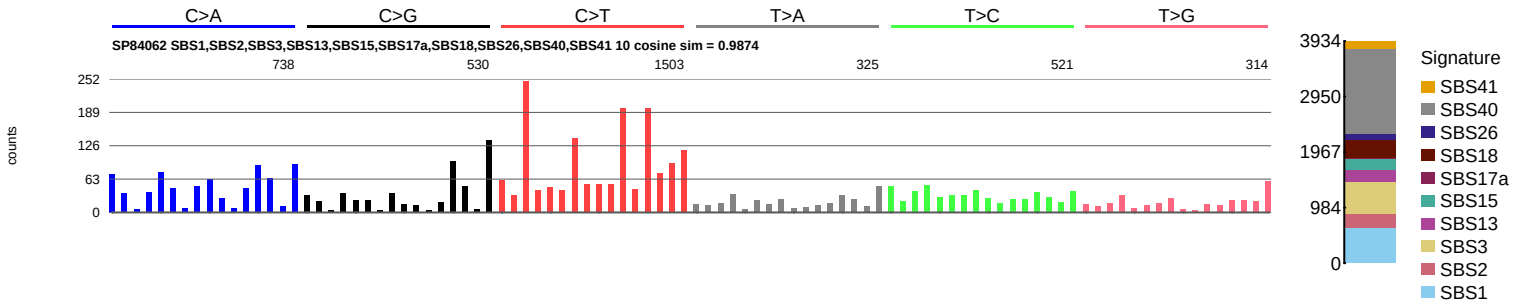

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98743
748
543
1516
304
477
347
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

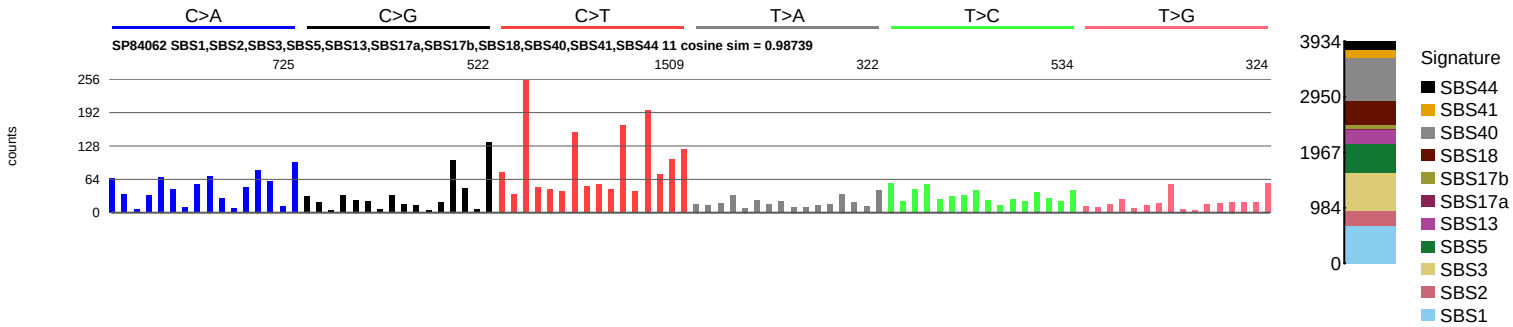

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98742
738
531
1505
325
523
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

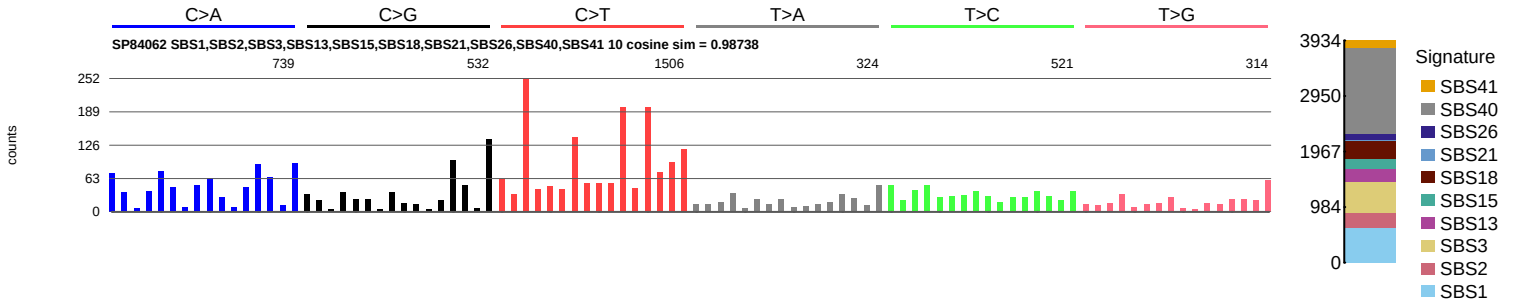

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 9 cosine sim = 0.98742
736
535
1556
297
517
293
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

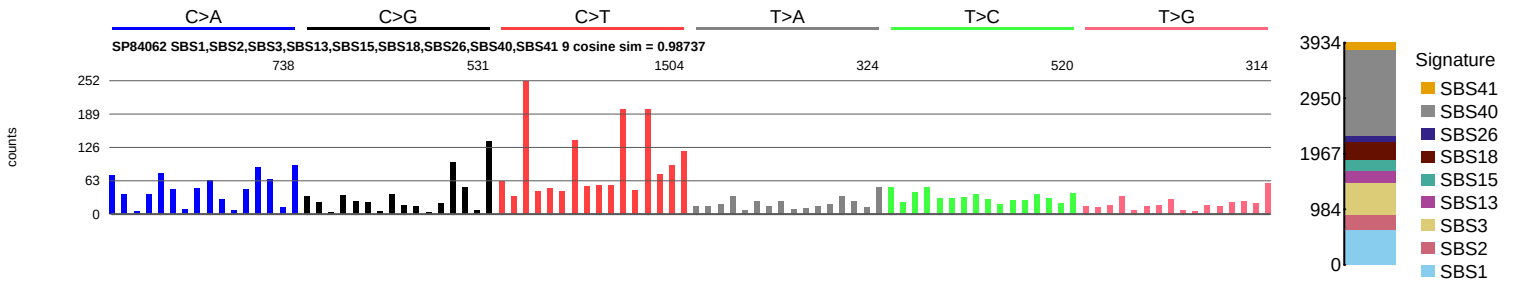

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98741
739
530
1512
322
519
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

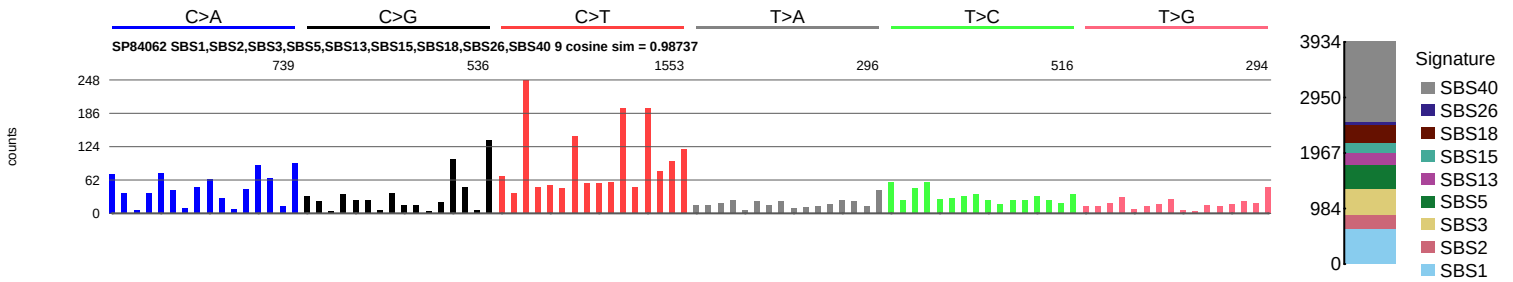

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98741
738
530
1512
323
524
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3

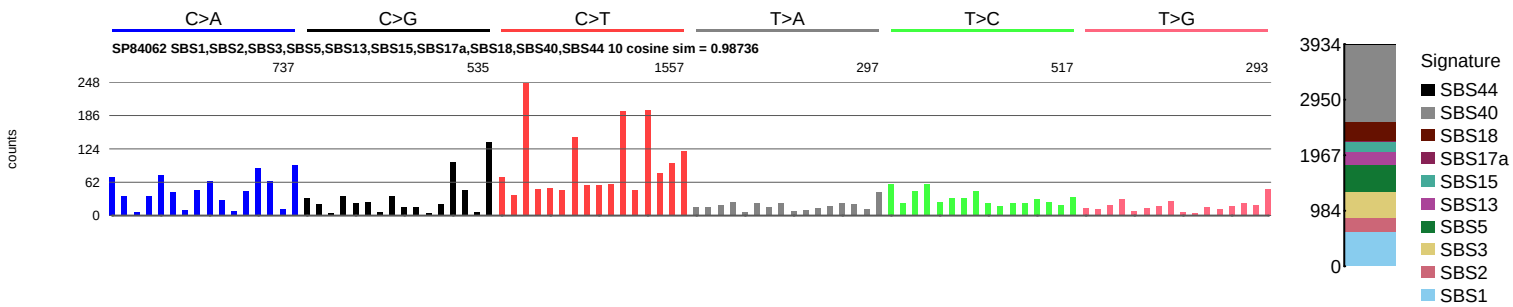

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.9874
738
530
1503
325
521
314
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

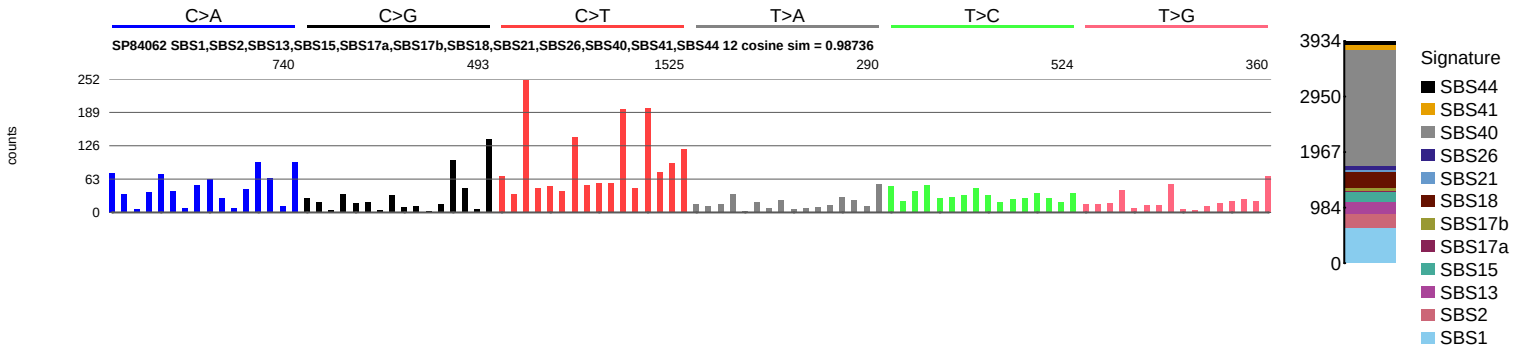

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98739
725
522
1509
322
534
324
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

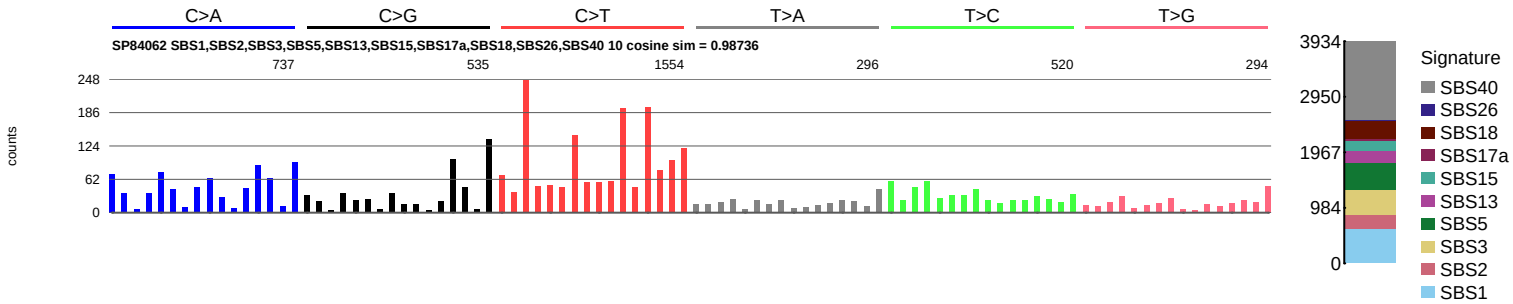

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98738
739
532
1506
324
521
314
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

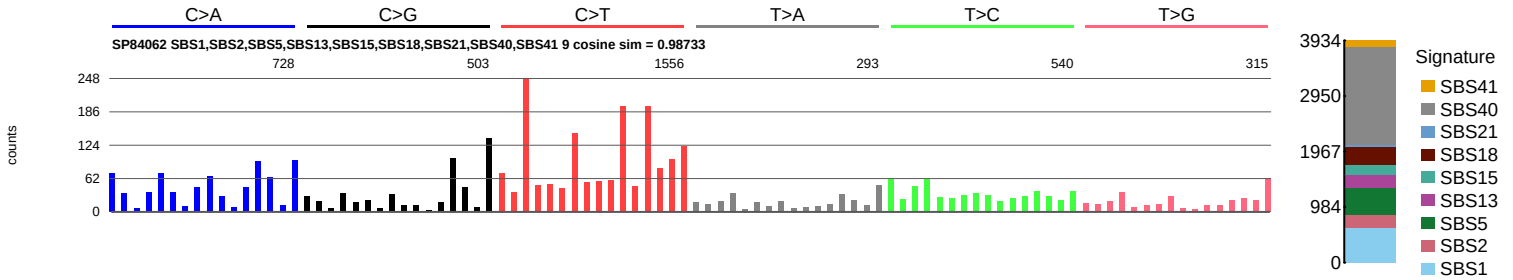

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98737
738
531
1504
324
520
314
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

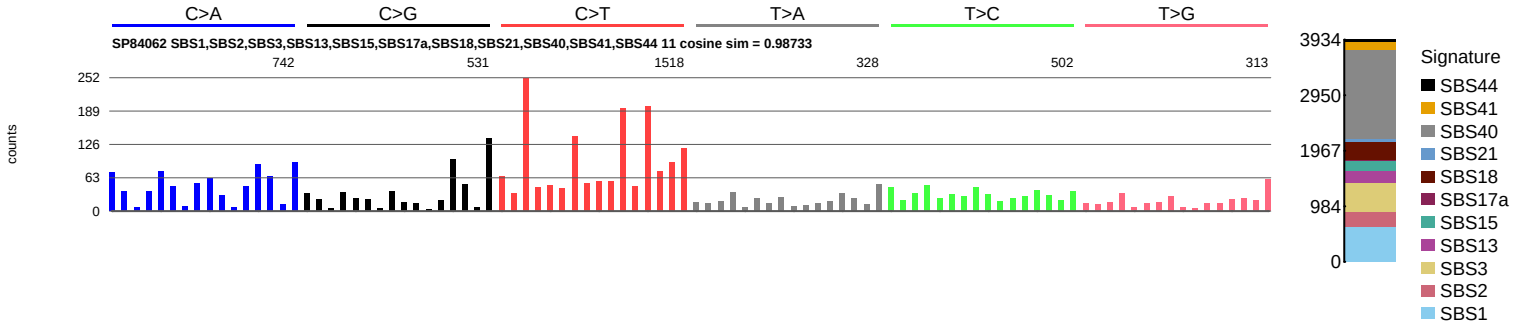

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 9 cosine sim = 0.98737
739
536
1553
296
516
294
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

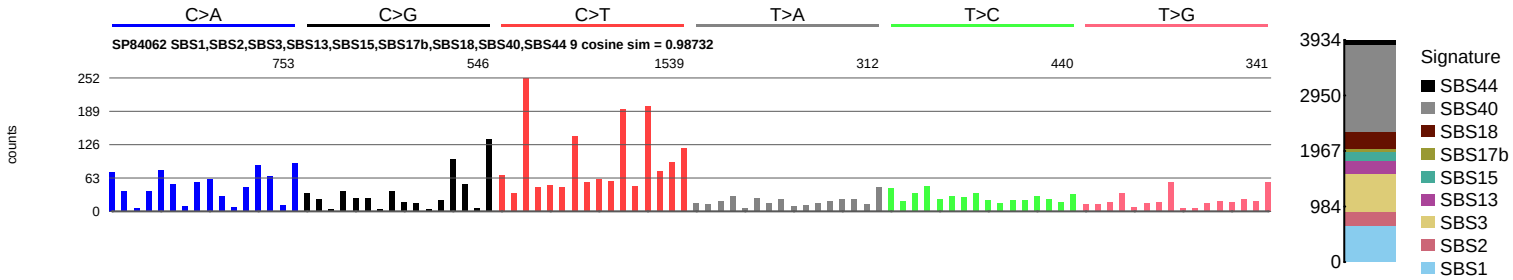

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 10 cosine sim = 0.98736
737
535
1557
297
517
293
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

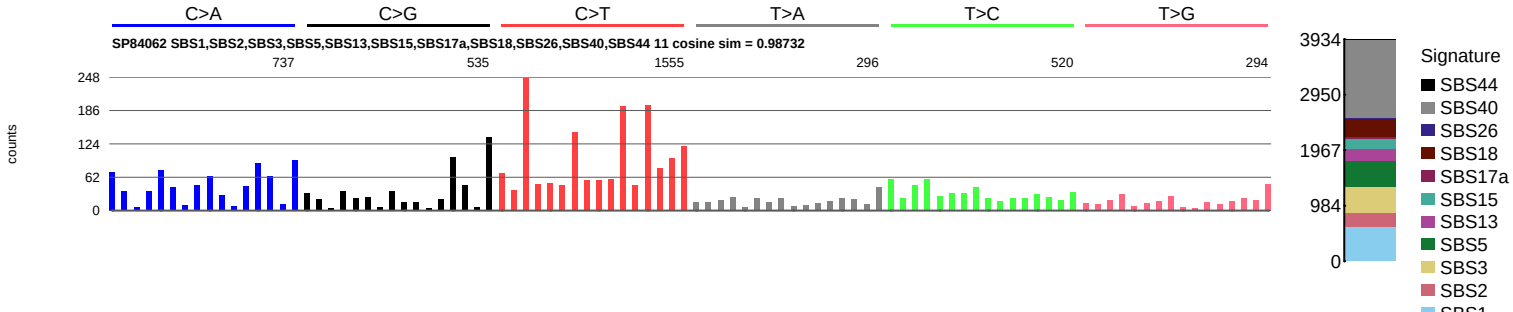

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98736
740
493
1525
290
524
360
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

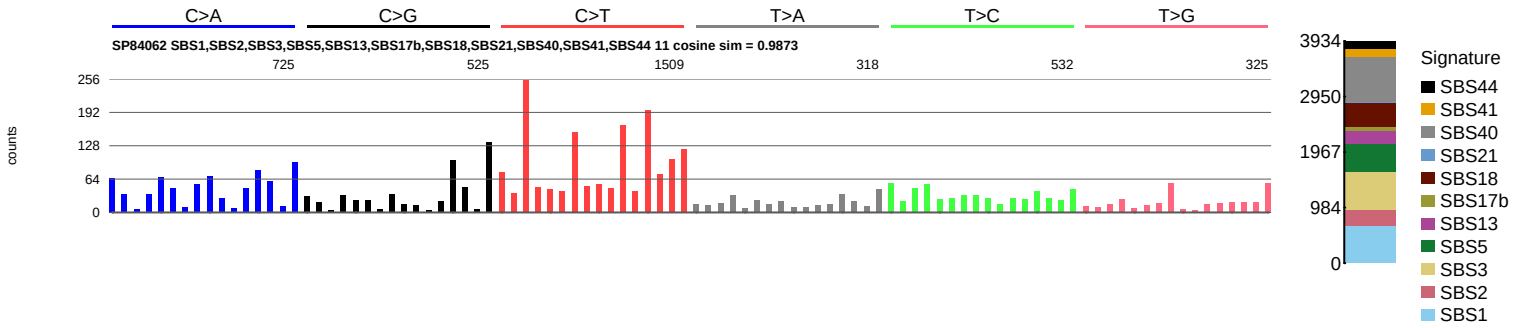

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 10 cosine sim = 0.98736
737
535
1554
296
520
294
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

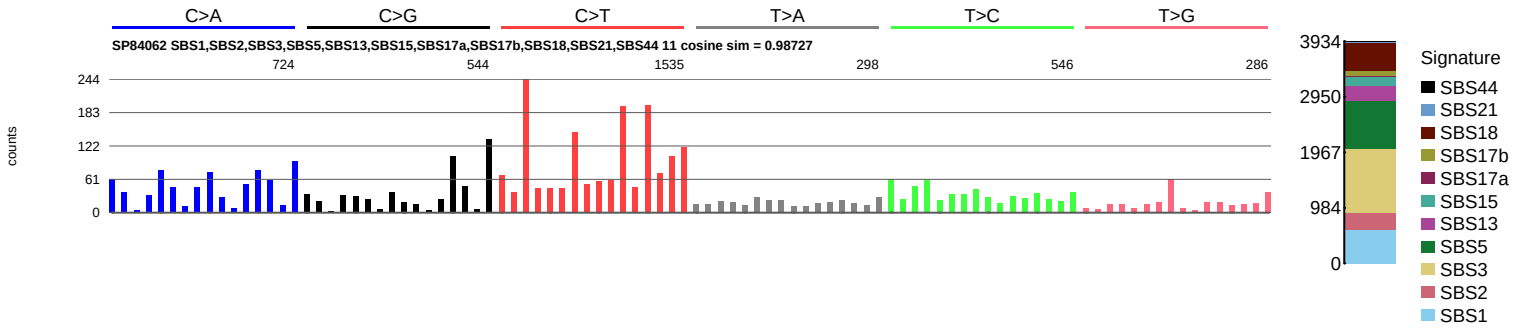

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98733
728
503
1556
293
540
315
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

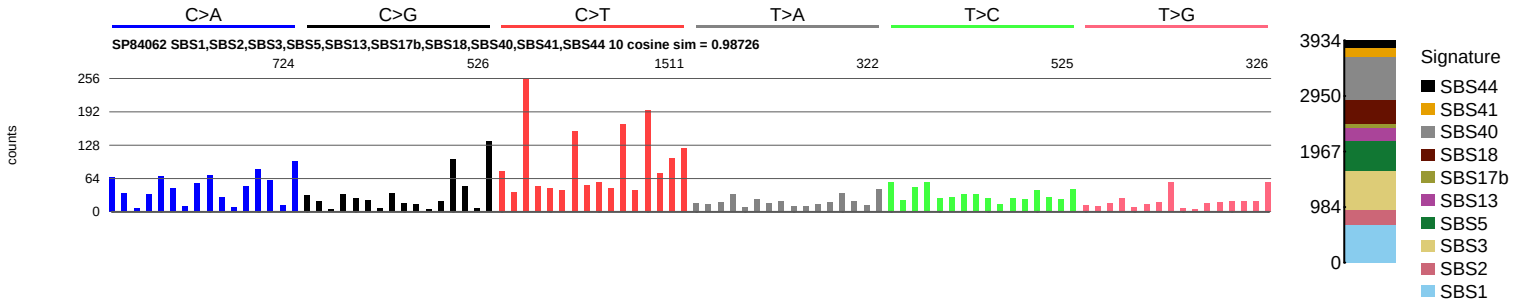

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98733
742
531
1518
328
502
313
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

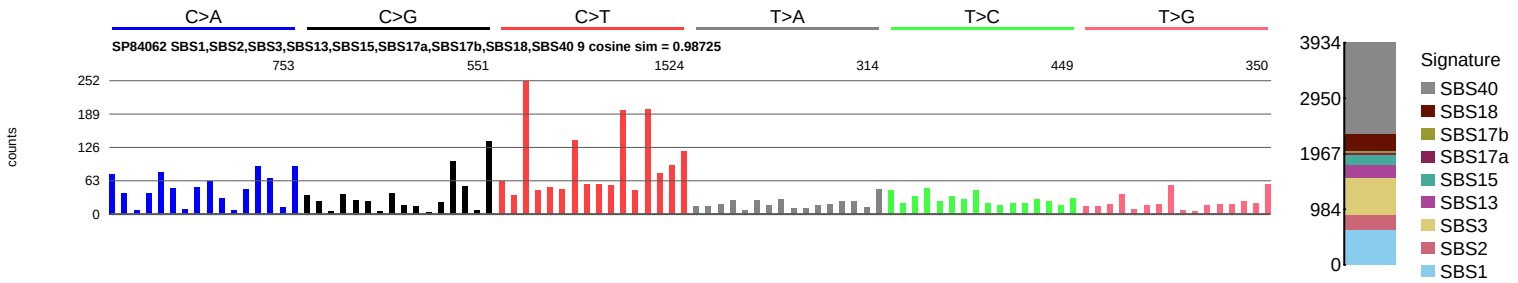

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98732
753
546
1539
312
440
341
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

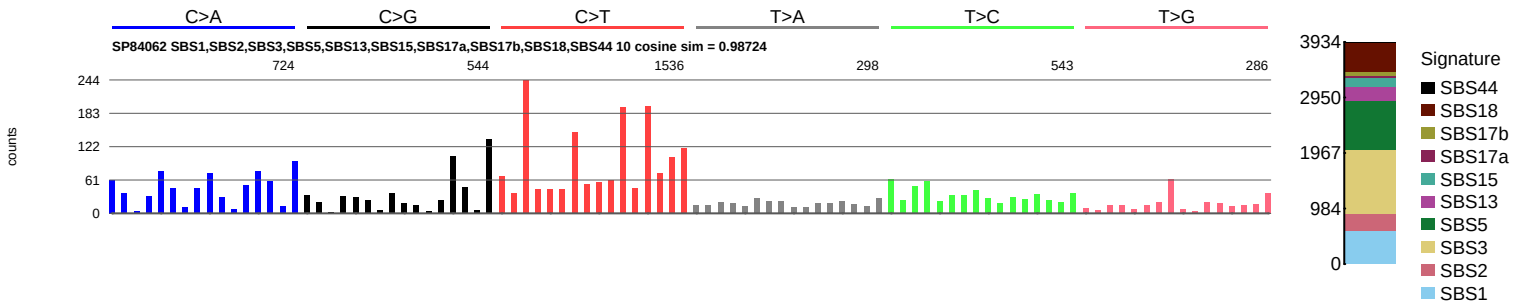

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98732
737
535
1555
296
520
294
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

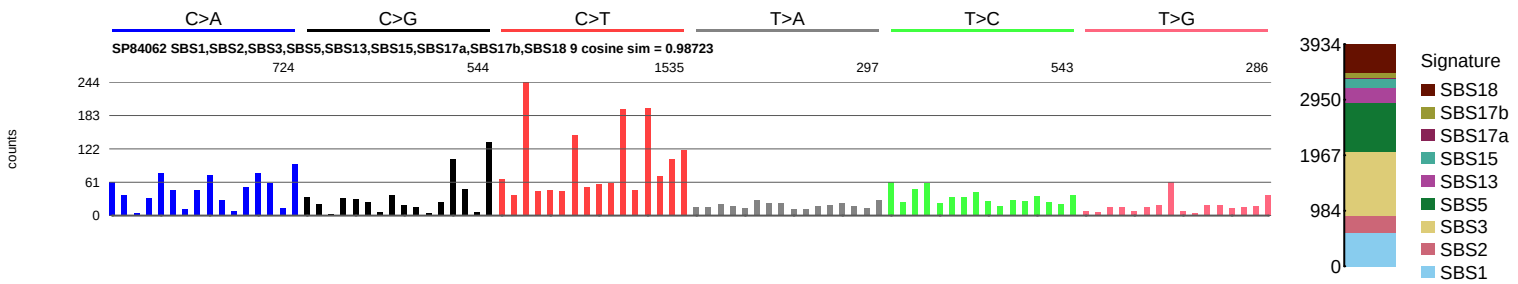

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9873
725
525
1509
318
532
325
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

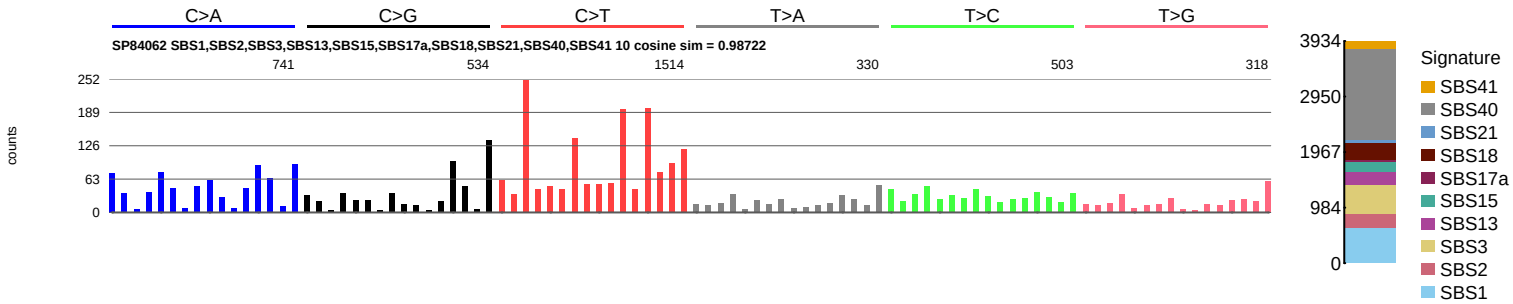

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98727
724
544
1535
298
546
286
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

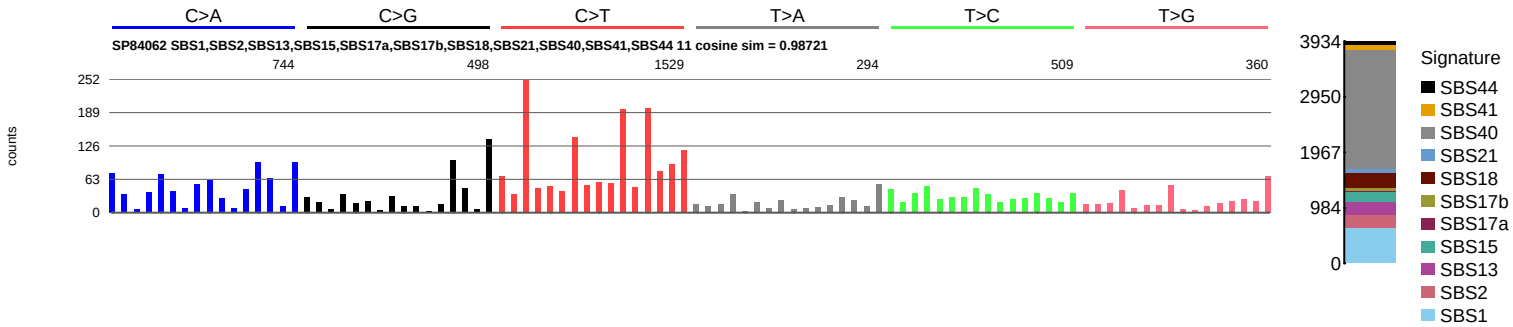

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98726
724
526
1511
322
525
326
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

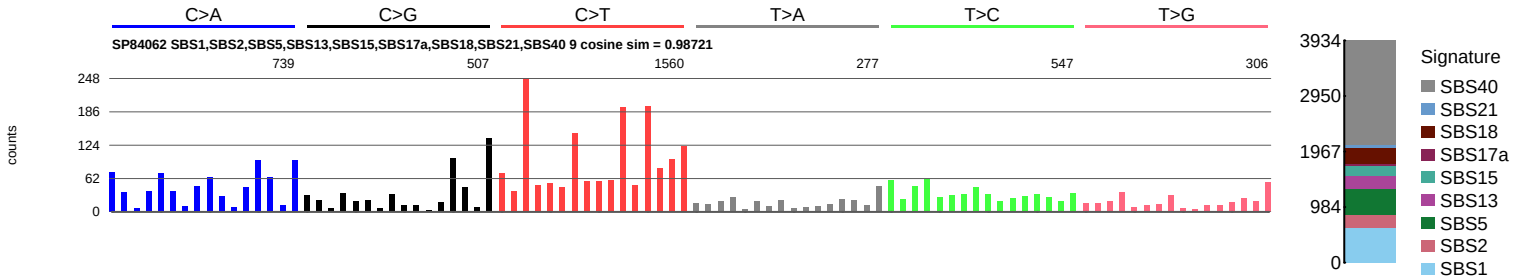

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98725
753
551
1524
314
449
350
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

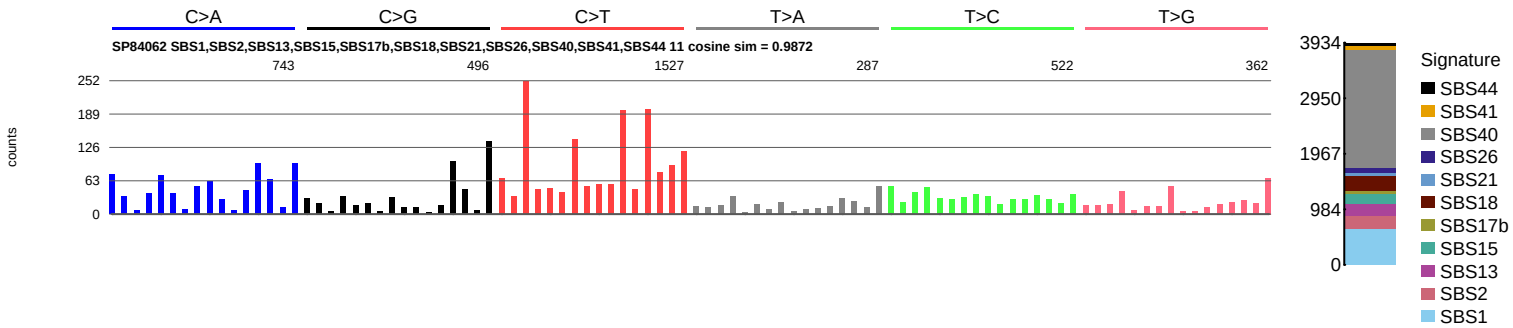

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98724
724
544
1536
298
543
286
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

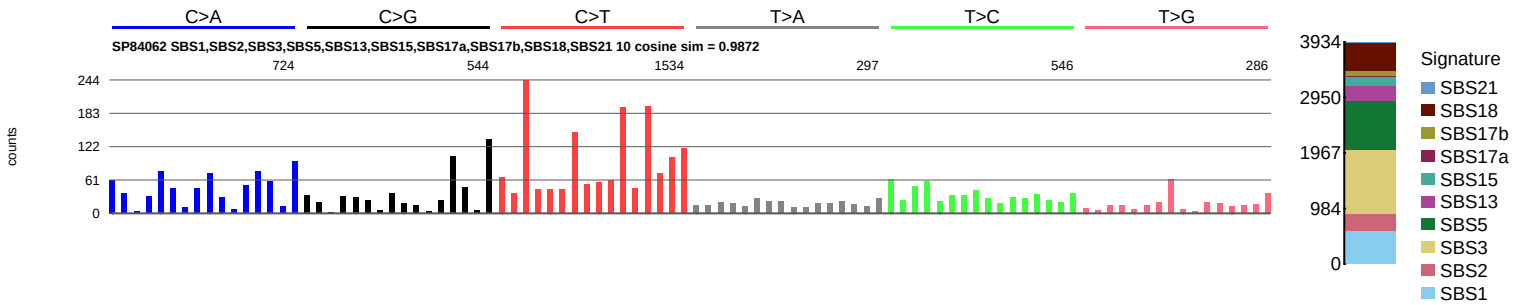

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98723
724
544
1535
297
543
286
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

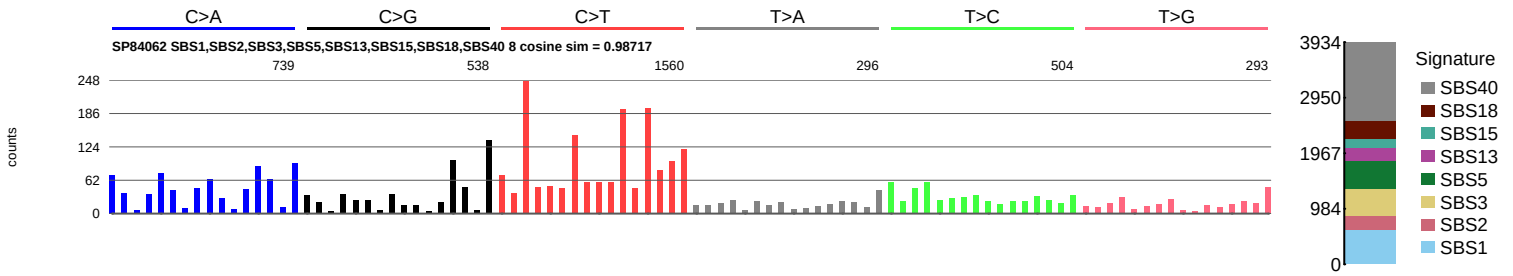

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98722
741
534
1514
330
503
318
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

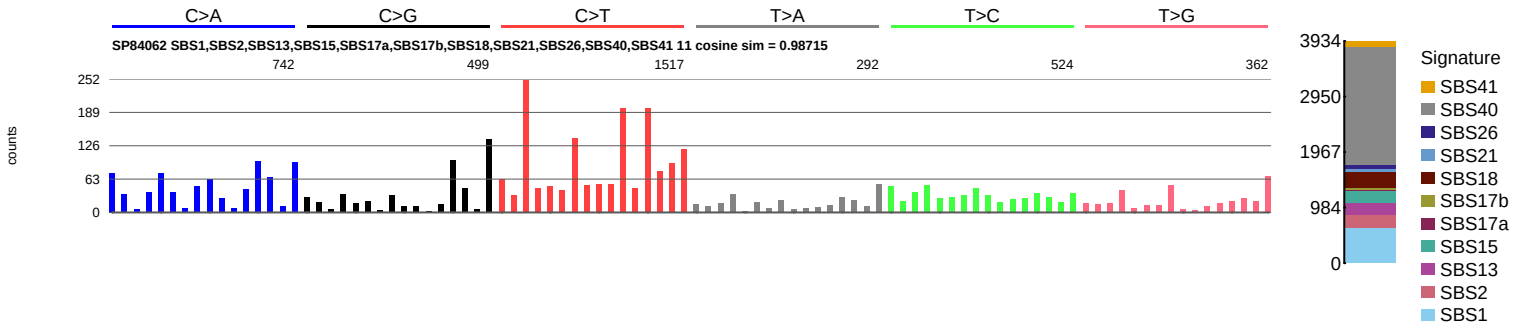

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98721
744
498
1529
294
509
360
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

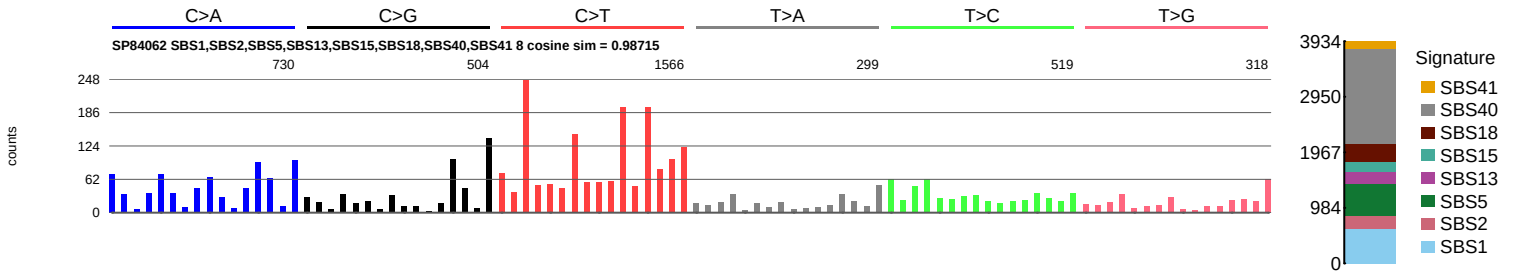

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98721
739
507
1560
277
547
306
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

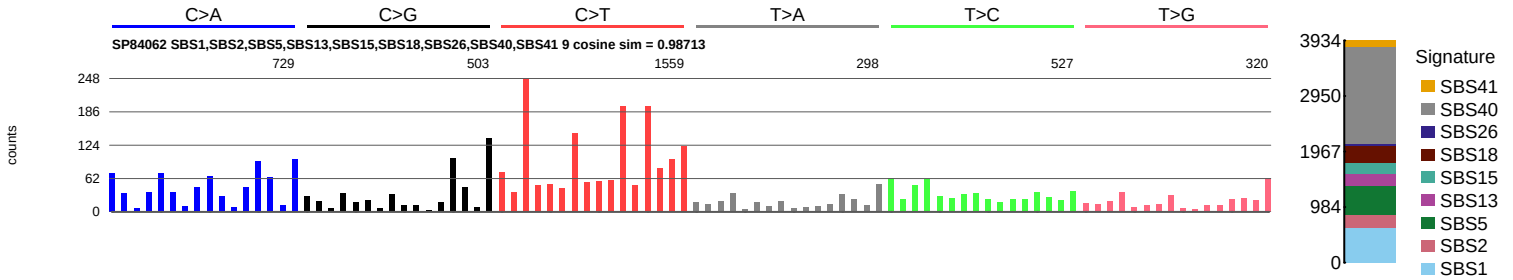

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9872
743
496
1527
287
522
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

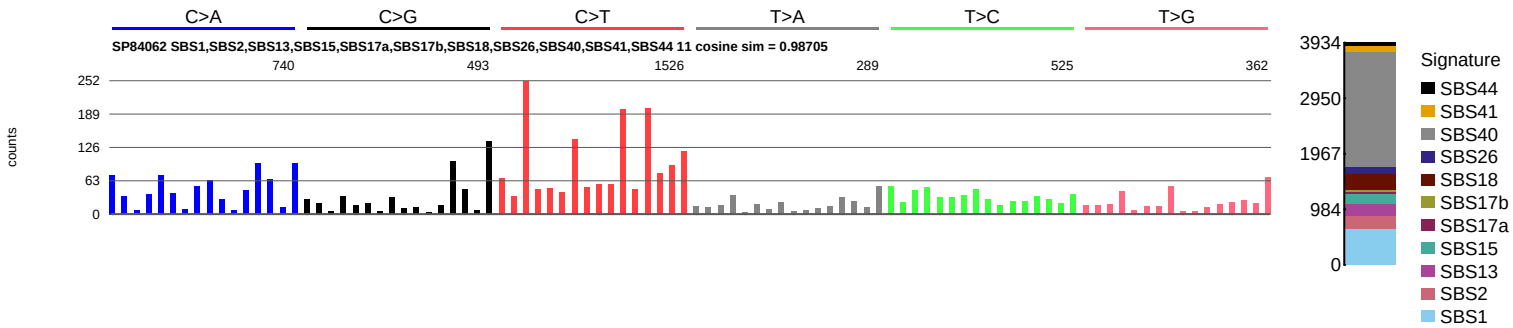

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.9872
724
544
1534
297
546
286
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

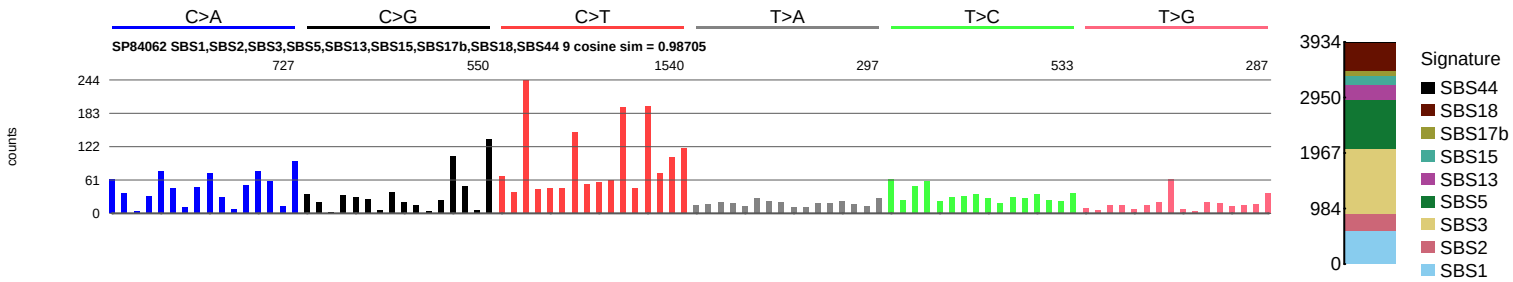

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40 8 cosine sim = 0.98717
739
538
1560
296
504
293
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

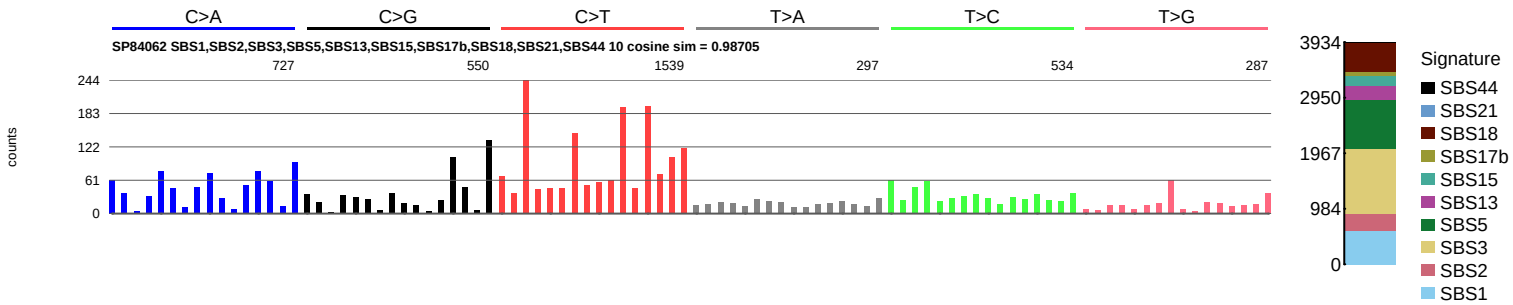

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98715
742
499
1517
292
524
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

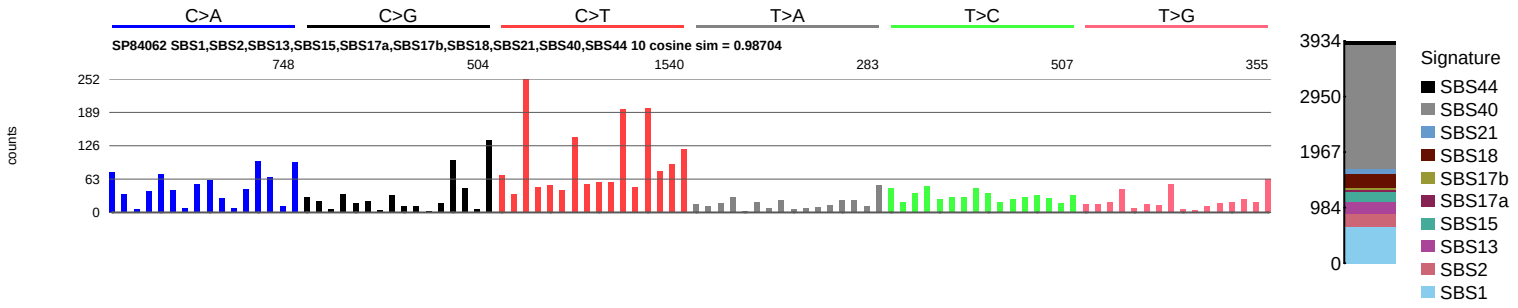

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98715
730
504
1566
299
519
318
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

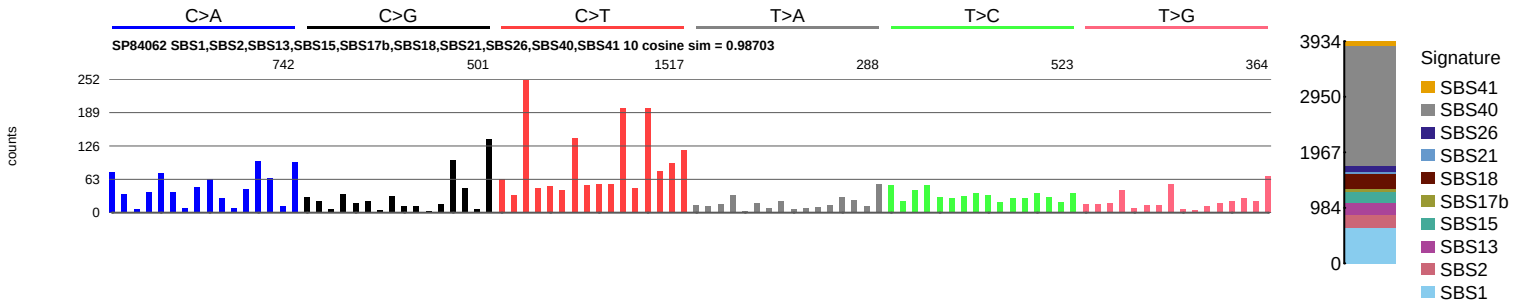

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98713
729
503
1559
298
527
320
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

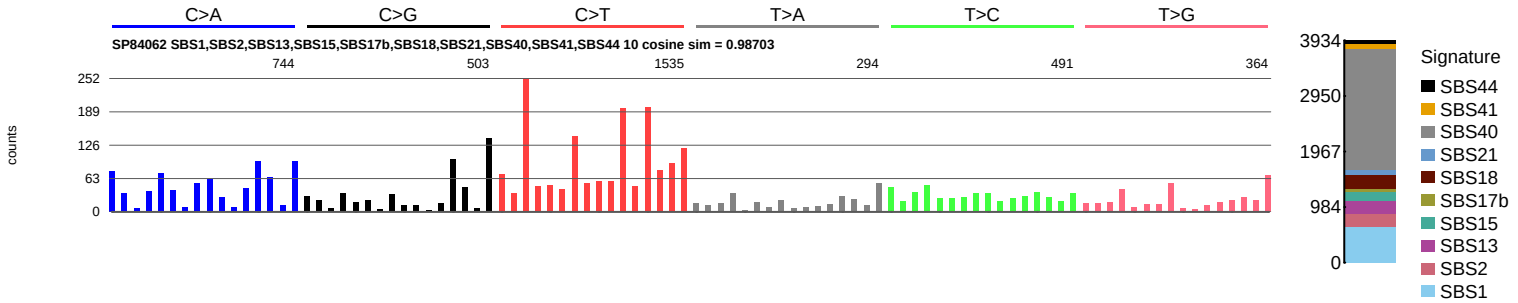

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98705
740
493
1526
289
525
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

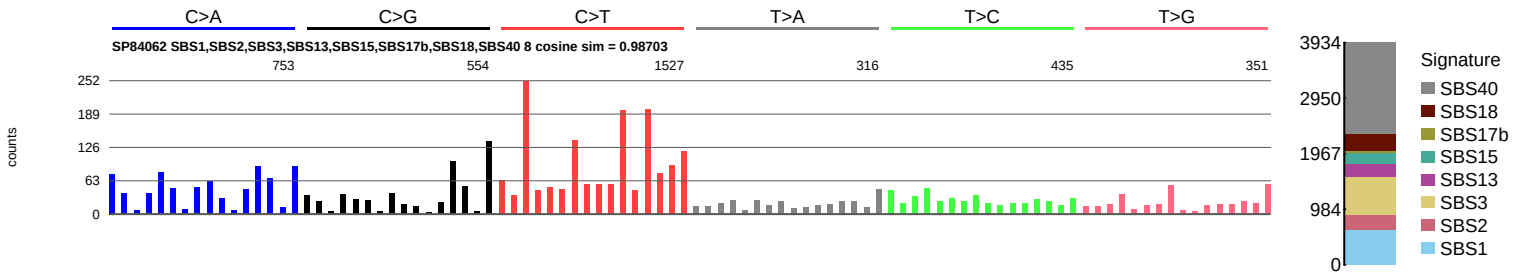

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98705
727
550
1540
297
533
287
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

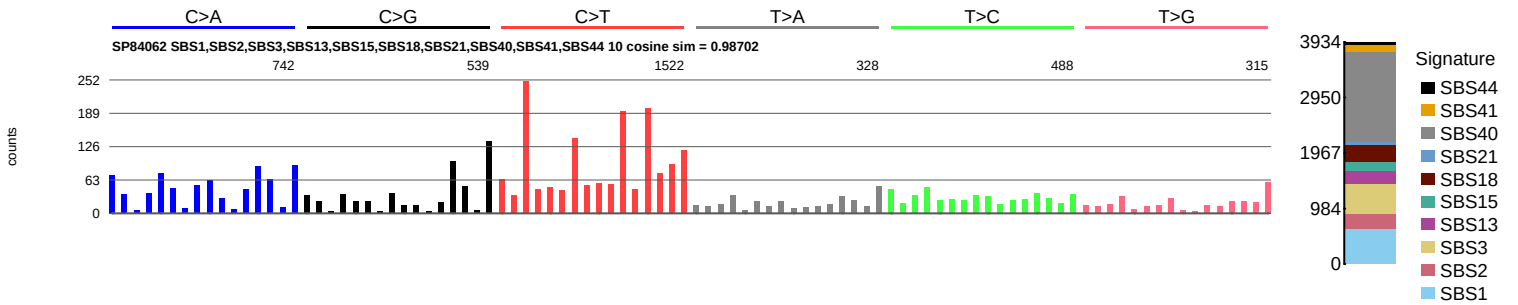

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98705
727
550
1539
297
534
287
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

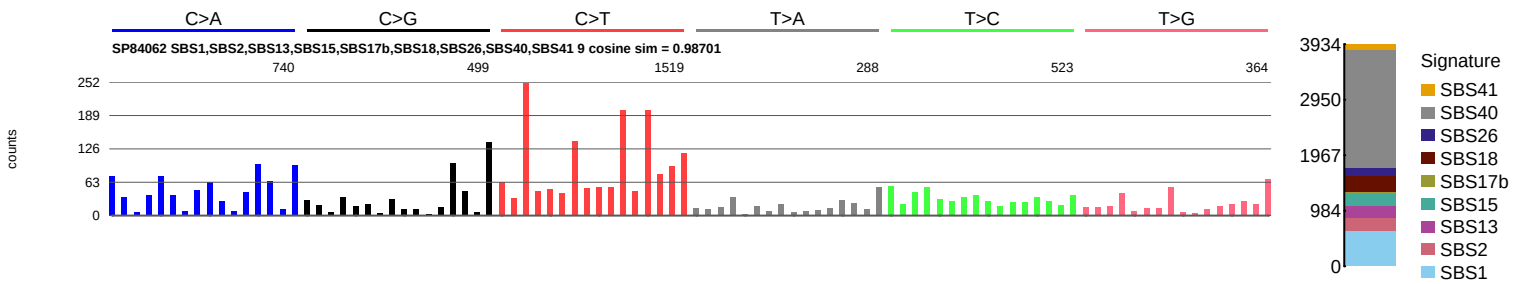

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98704
748
504
1540
283
507
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

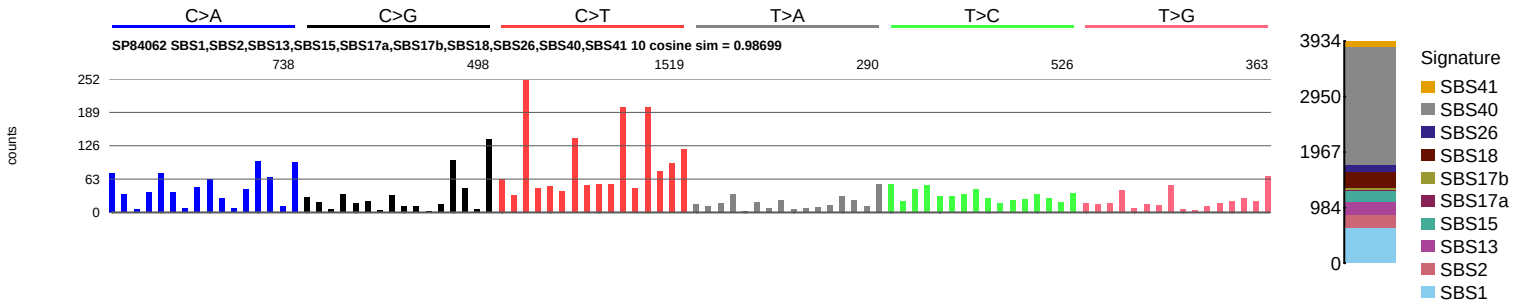

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98703
742
501
1517
288
523
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

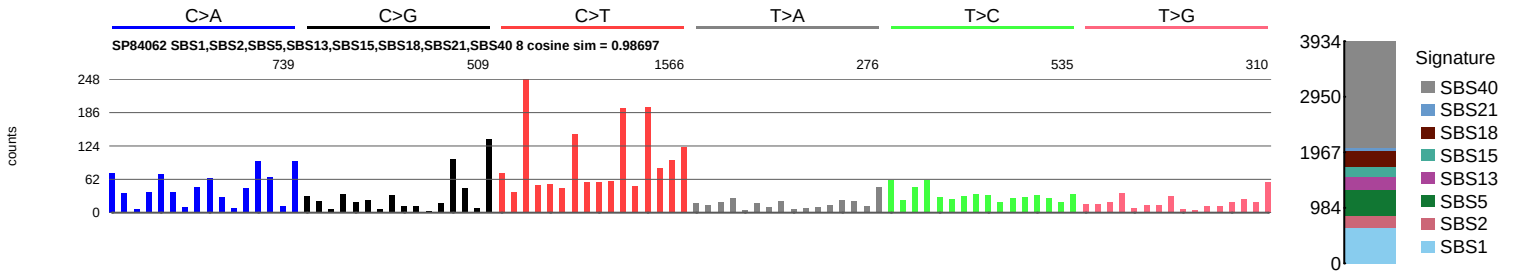

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98703
744
503
1535
294
491
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

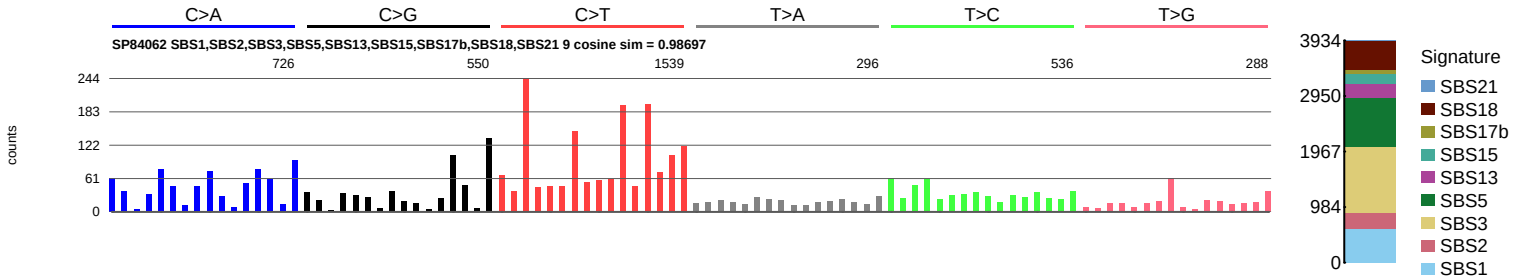

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98703
753
554
1527
316
435
351
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

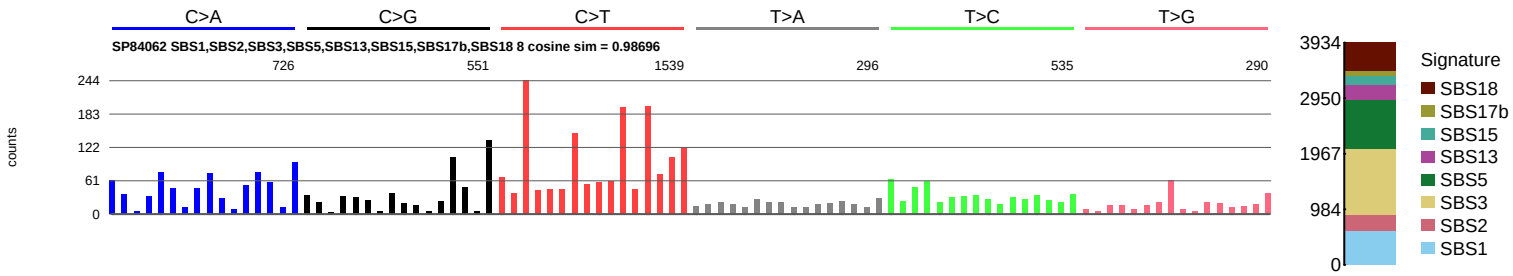

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98702
742
539
1522
328
488
315
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

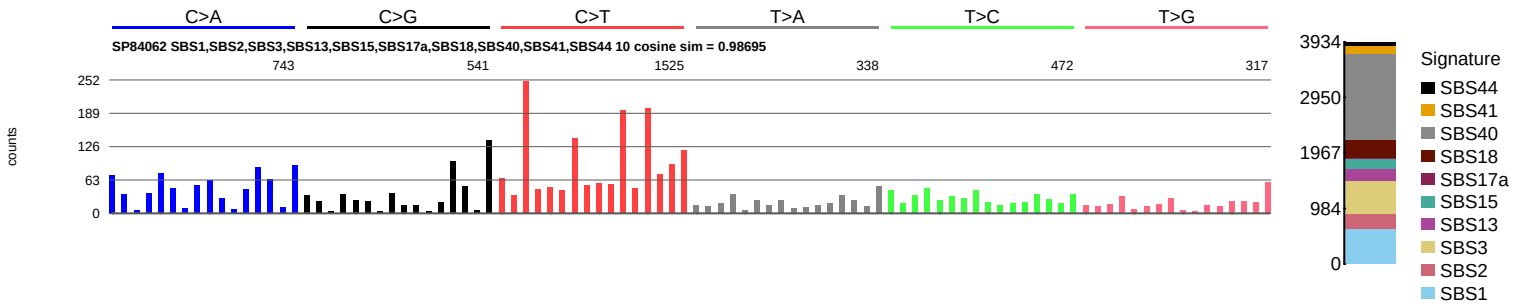

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98701
740
499
1519
288
523
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

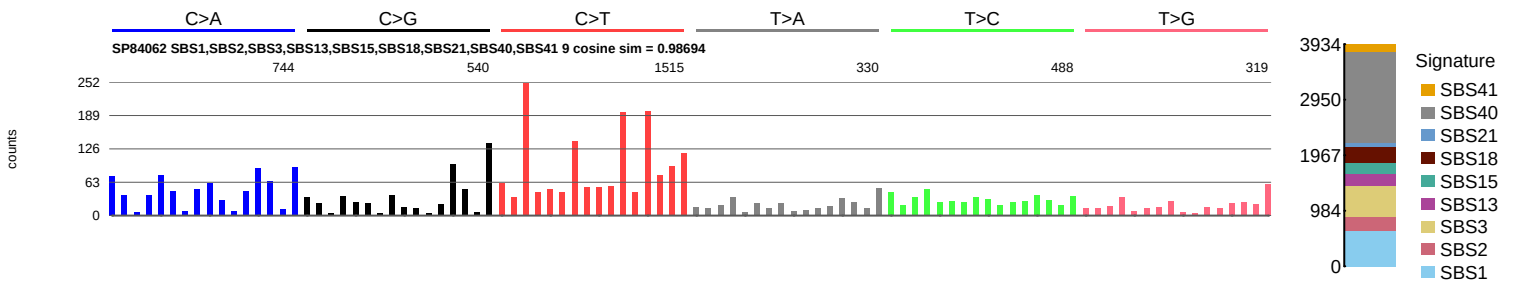

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98699
738
498
1519
290
526
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

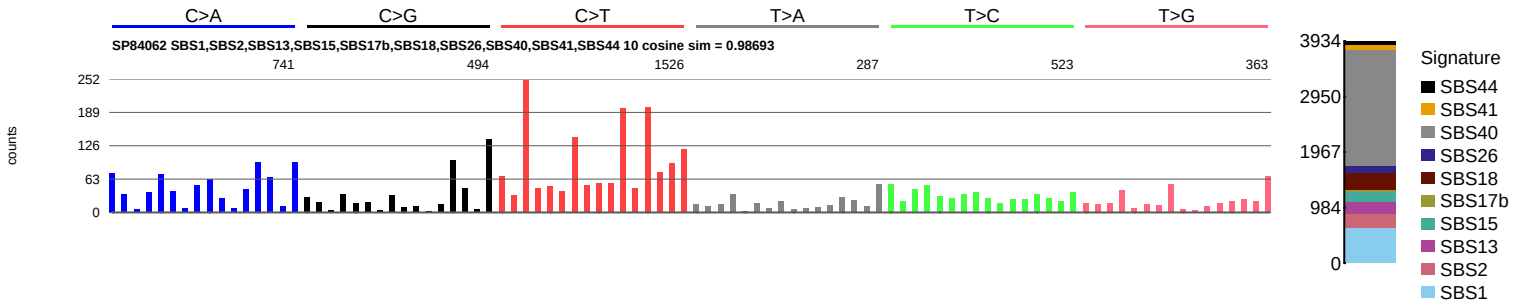

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98697
739
509
1566
276
535
310
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

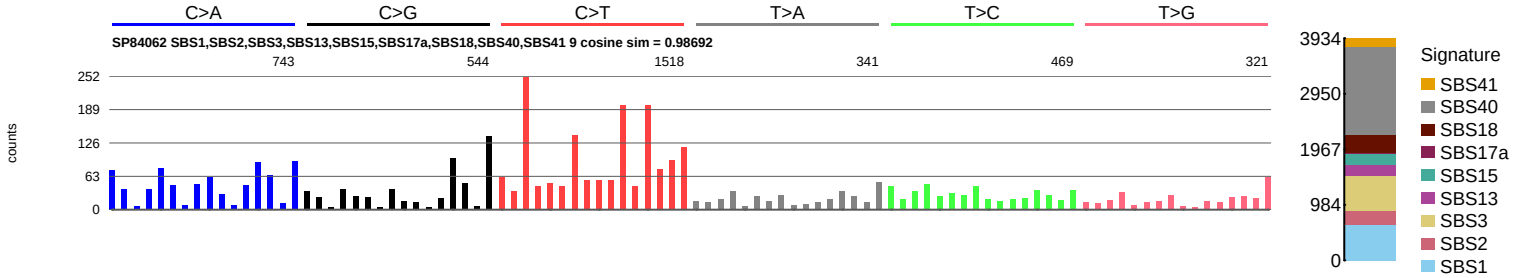

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98697
726
550
1539
296
536
288
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

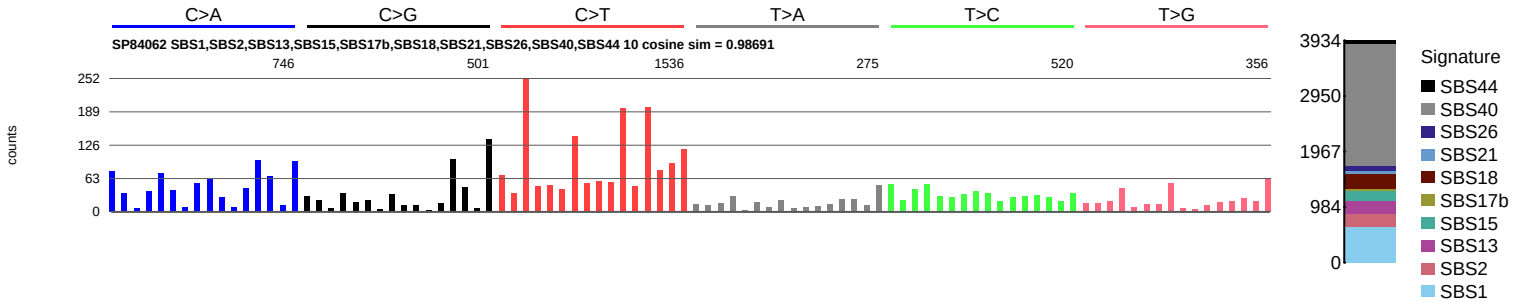

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98696
726
551
1539
296
535
290
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

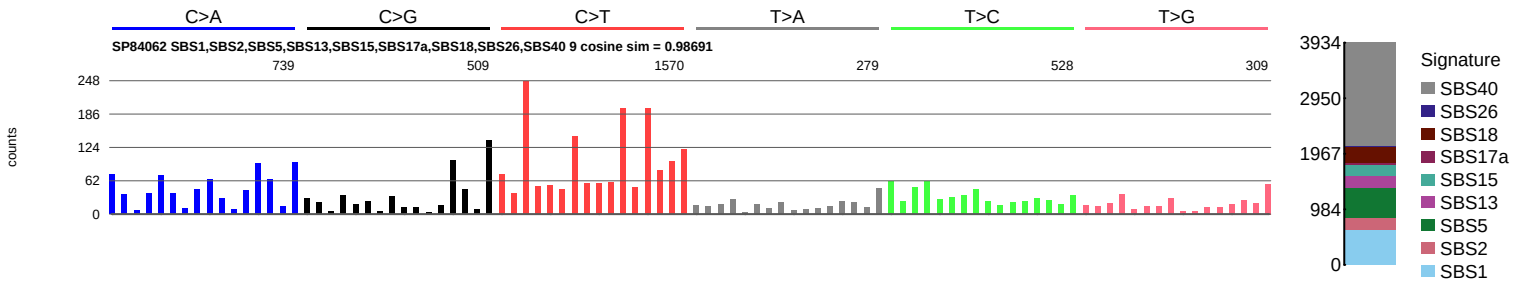

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98695
743
541
1525
338
472
317
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

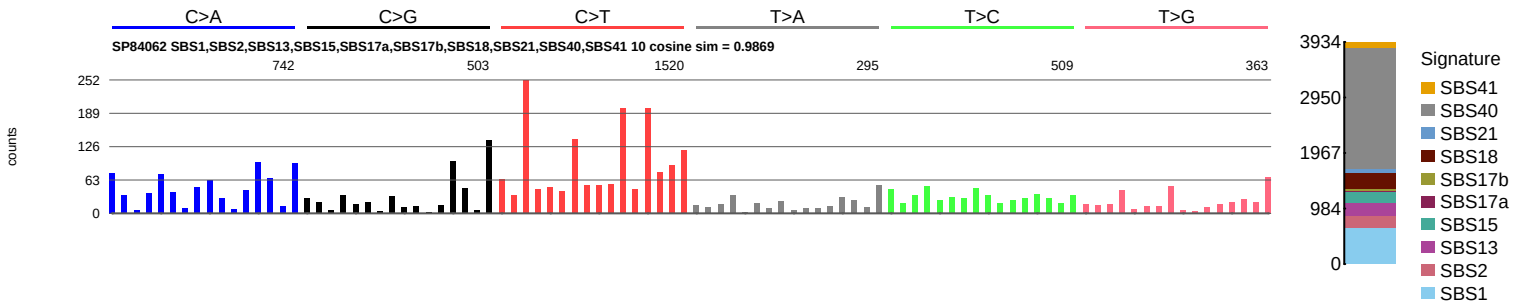

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98694
744
540
1515
330
488
319
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

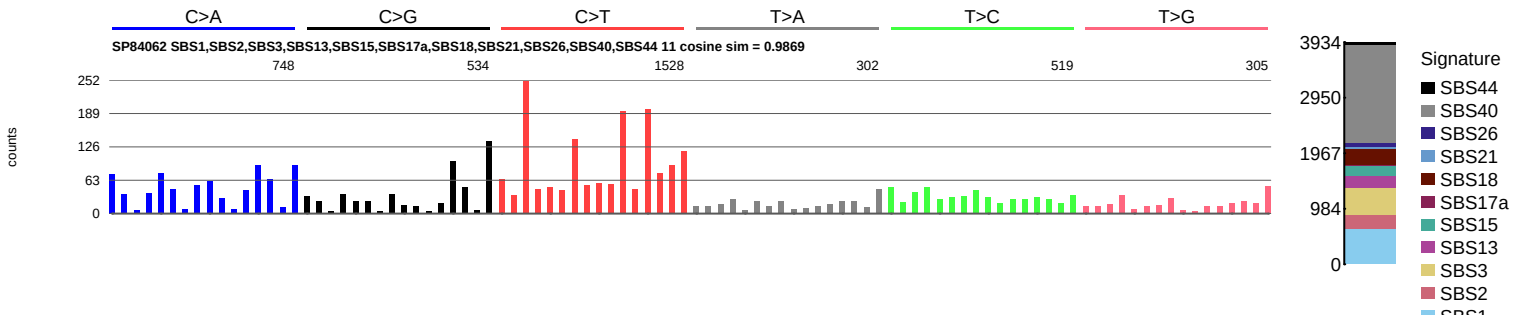

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98693
741
494
1526
287
523
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

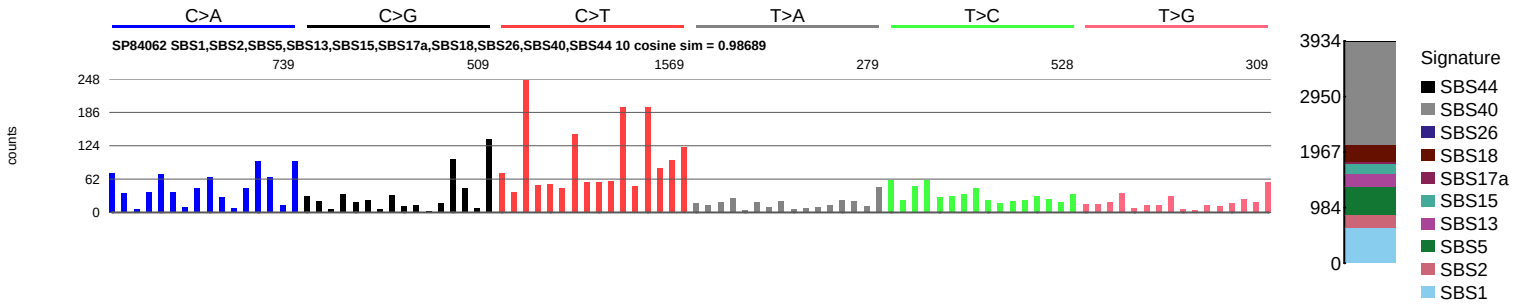

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98692
743
544
1518
341
469
321
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

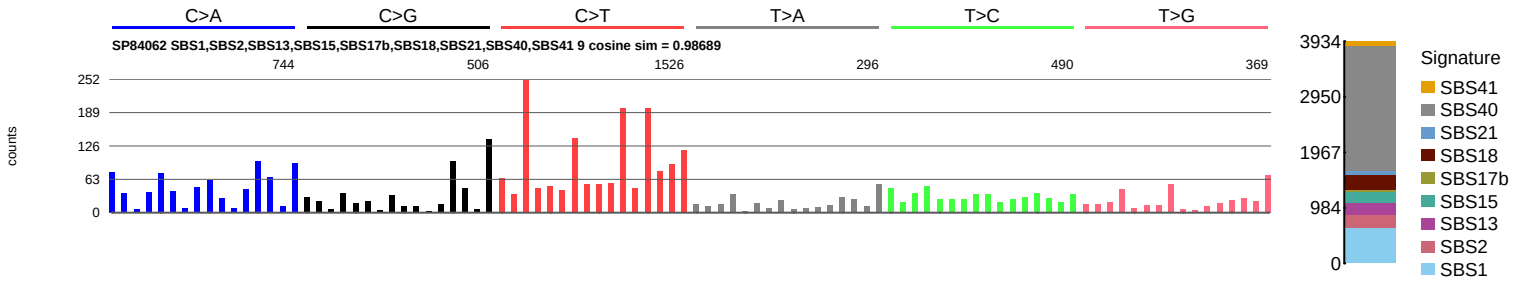

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98691
746
501
1536
275
520
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

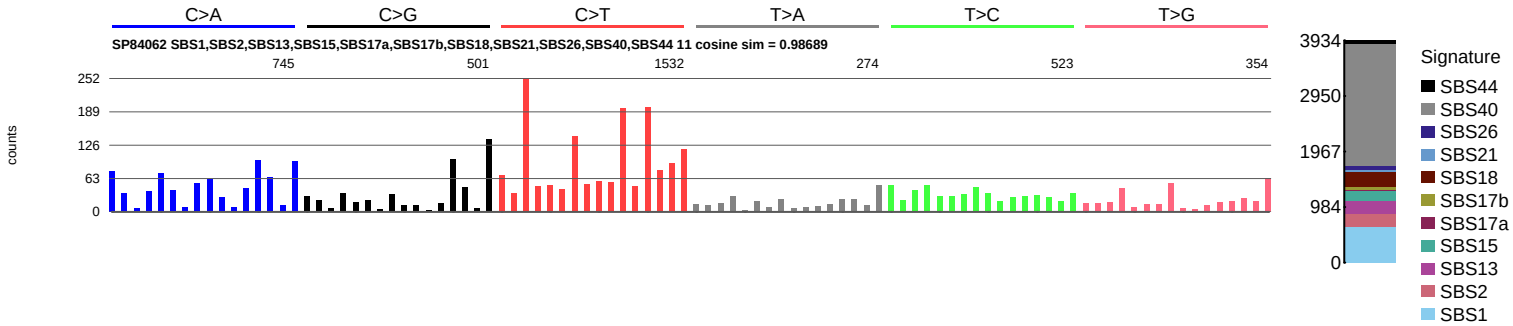

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98691
739
509
1570
279
528
309
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

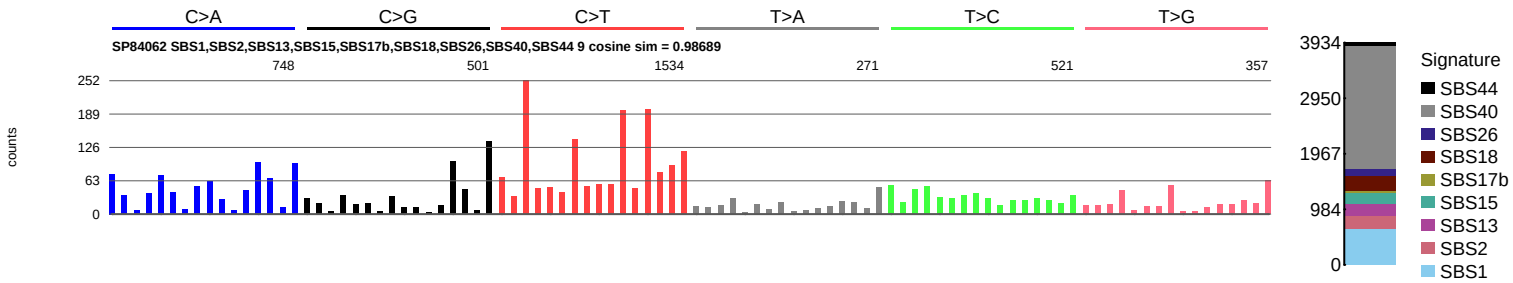

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9869
742
503
1520
295
509
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

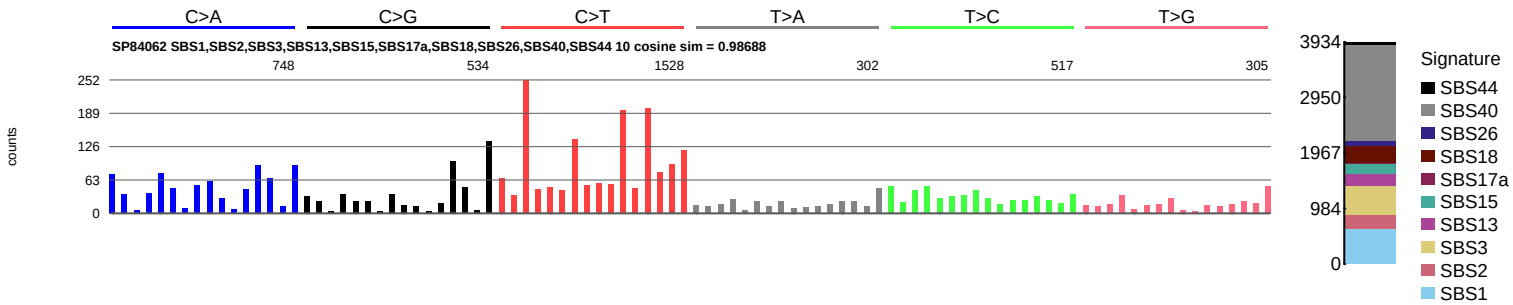

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9869
748
534
1528
302
519
305
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2

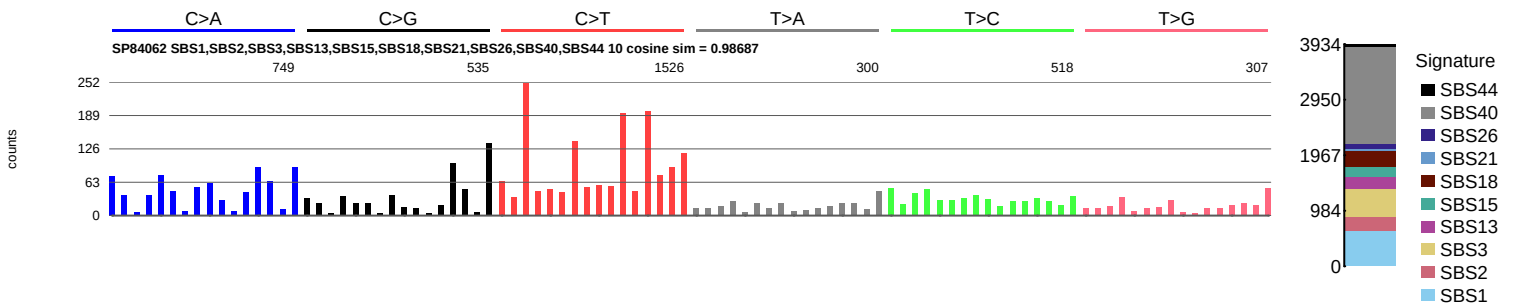

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98689
739
509
1569
279
528
309
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

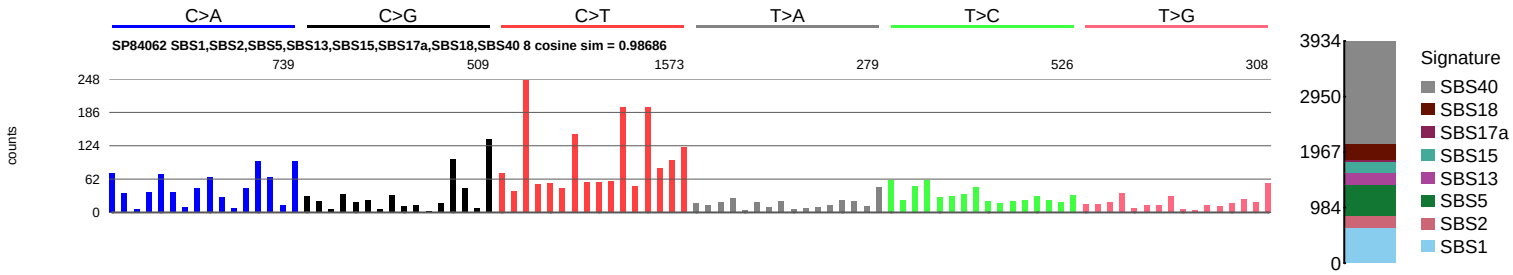

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98689
744
506
1526
296
490
369
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

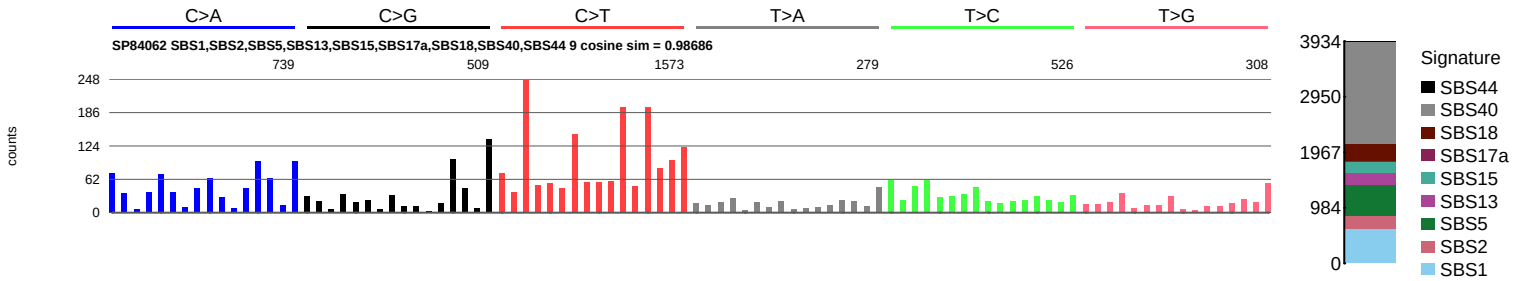

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98689
745
501
1532
274
523
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

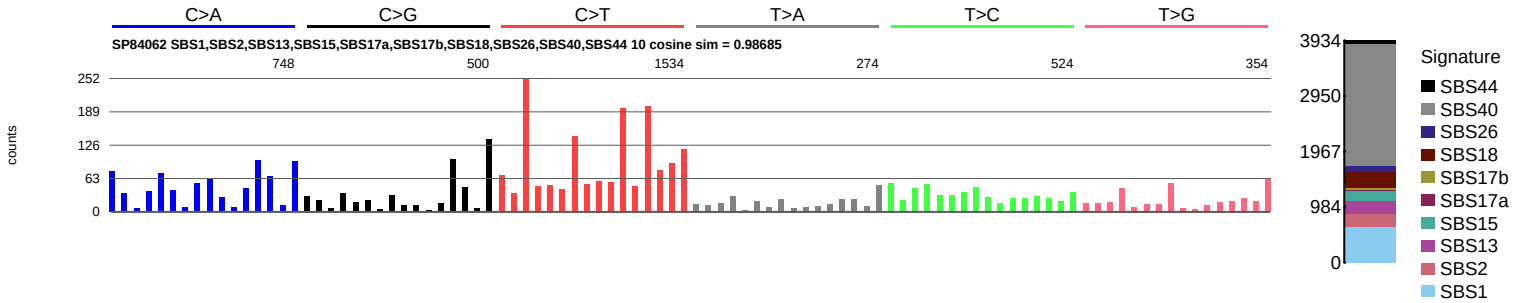

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98689
748
501
1534
271
521
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

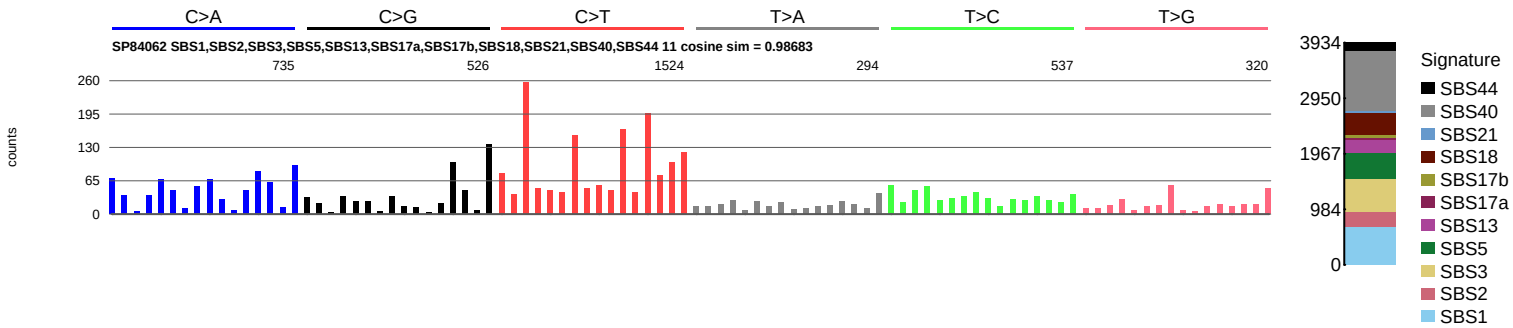

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98688
748
534
1528
302
517
305
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

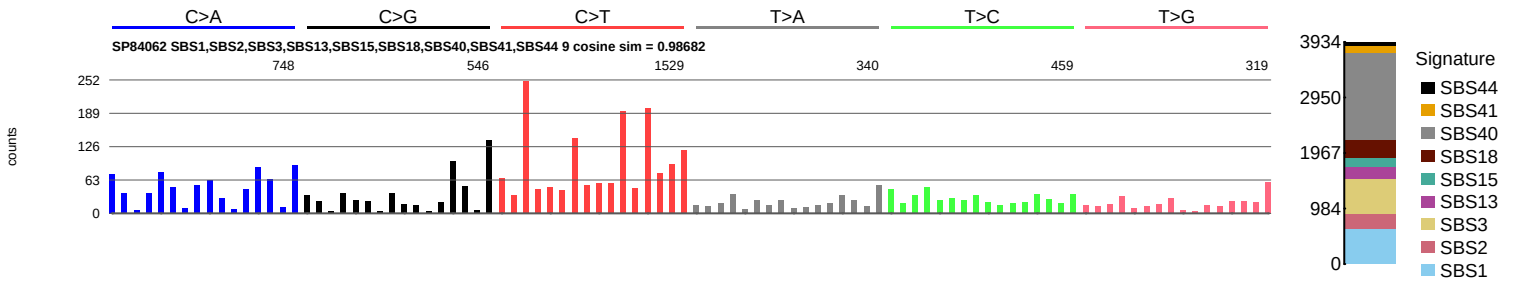

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98687
749
535
1526
300
518
307
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

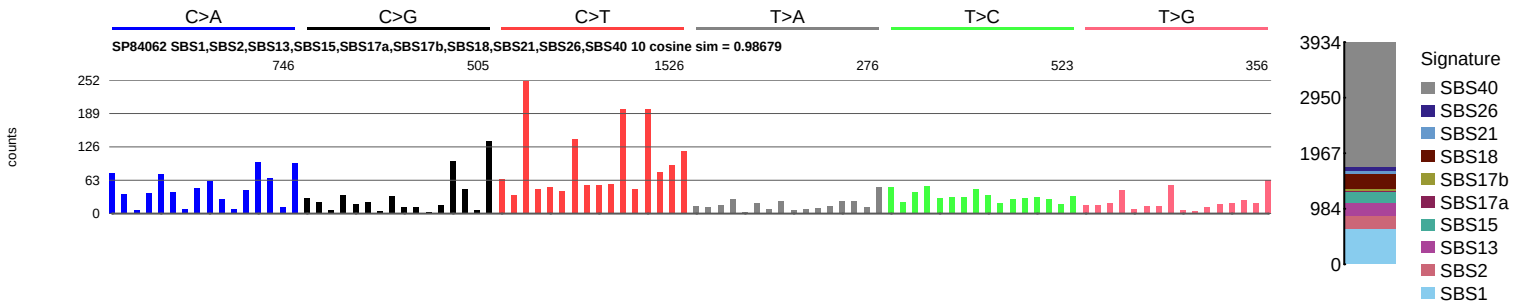

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98686
739
509
1573
279
526
308
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

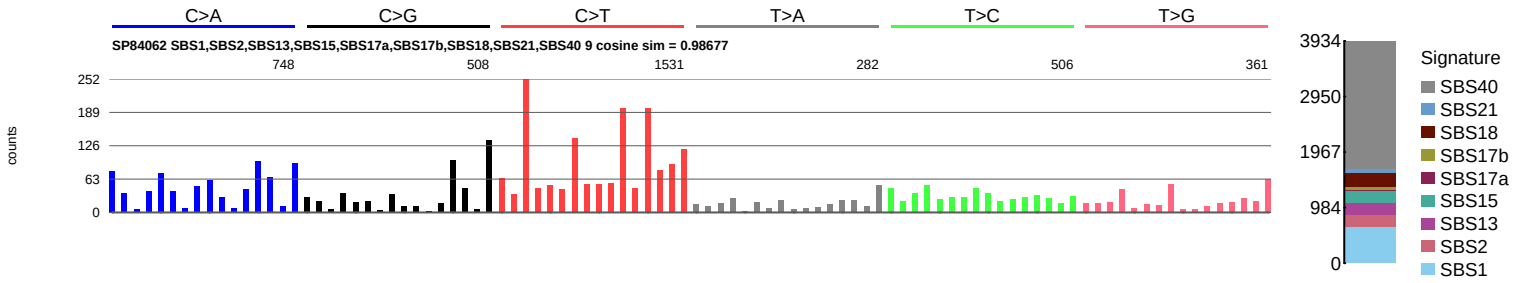

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98686
739
509
1573
279
526
308
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

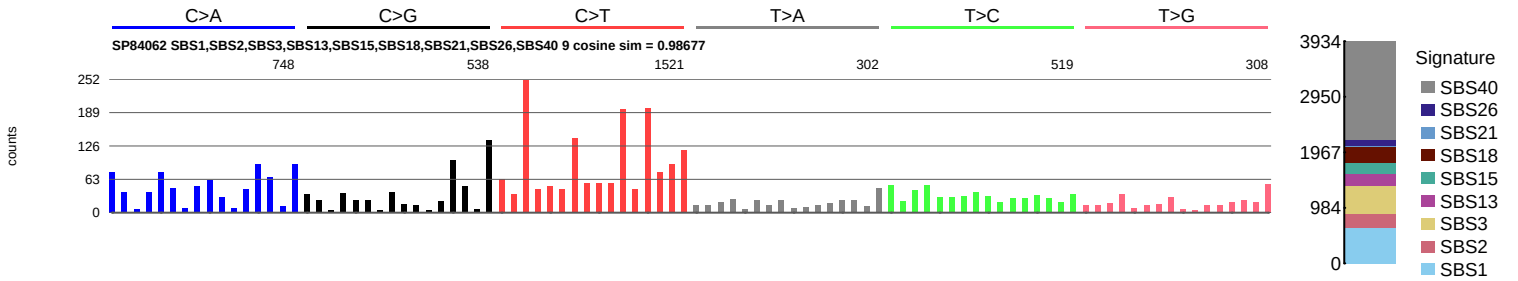

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98685
748
500
1534
274
524
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

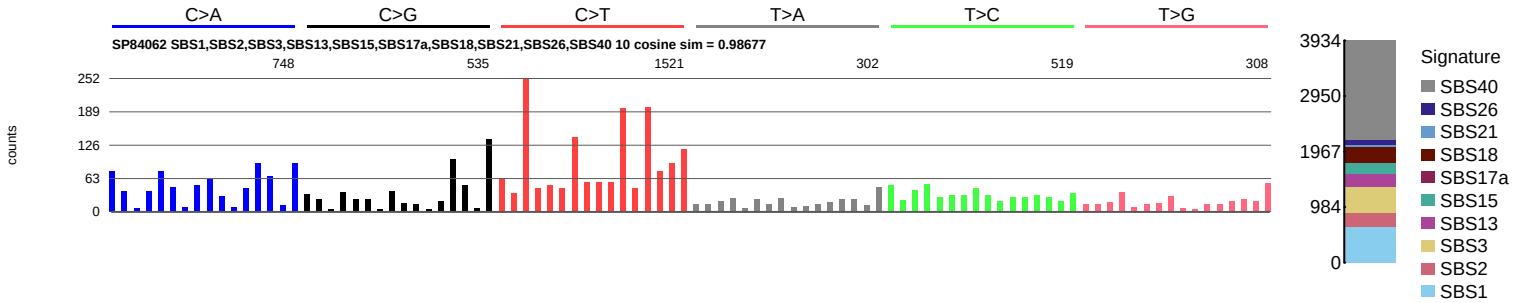

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98683
735
526
1524
294
537
320
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

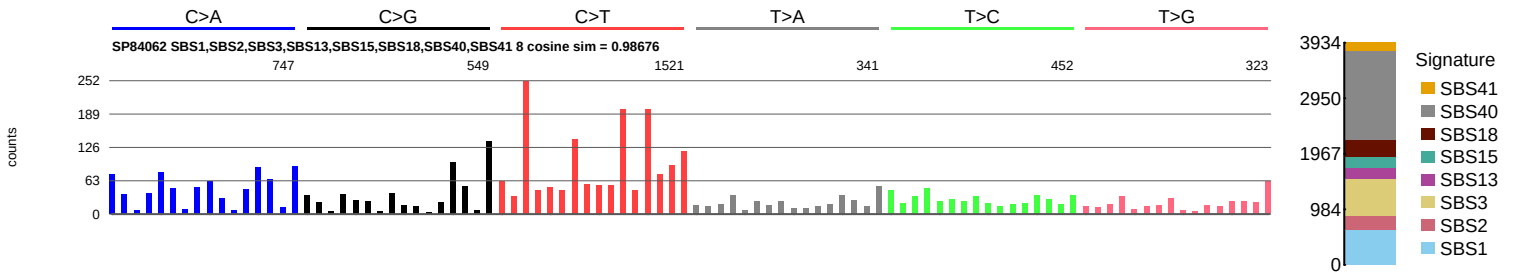

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98682
748
546
1529
340
459
319
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

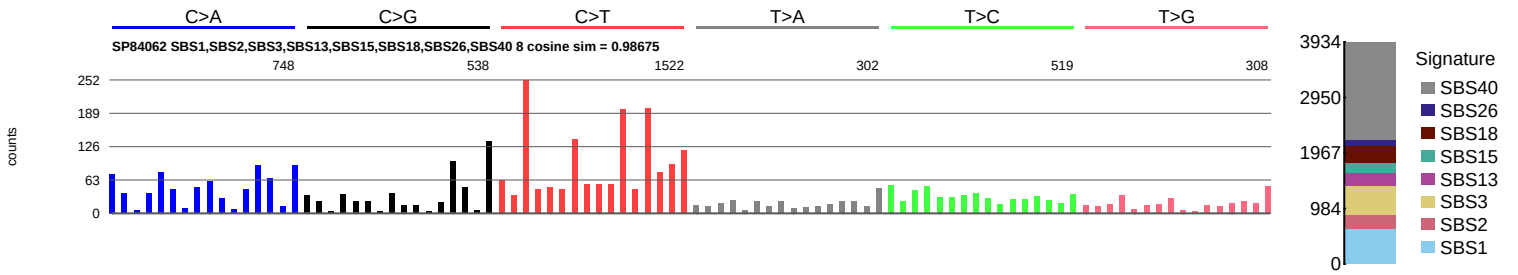

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98679
746
505
1526
276
523
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

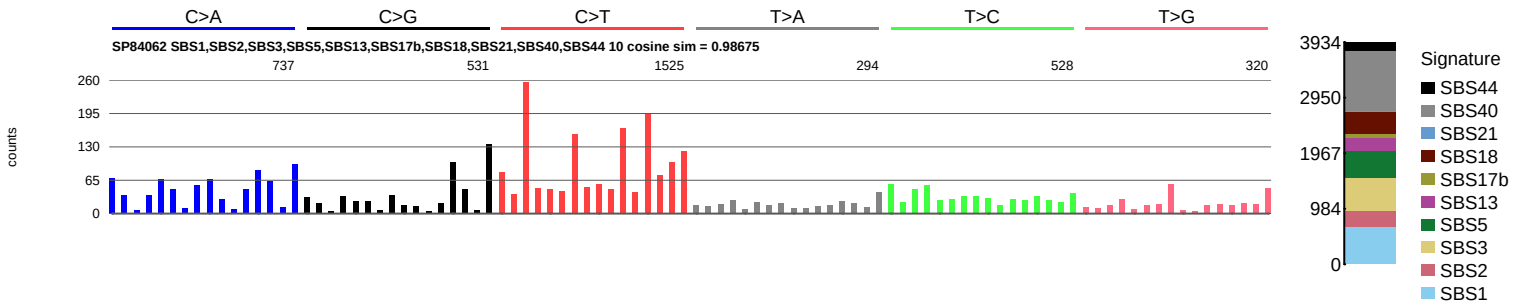

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98677
748
508
1531
282
506
361
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

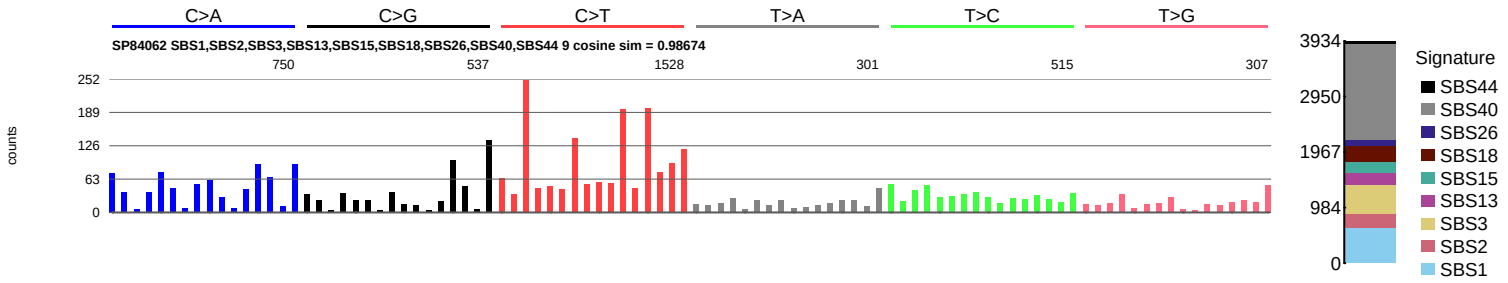

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98677
748
538
1521
302
519
308
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

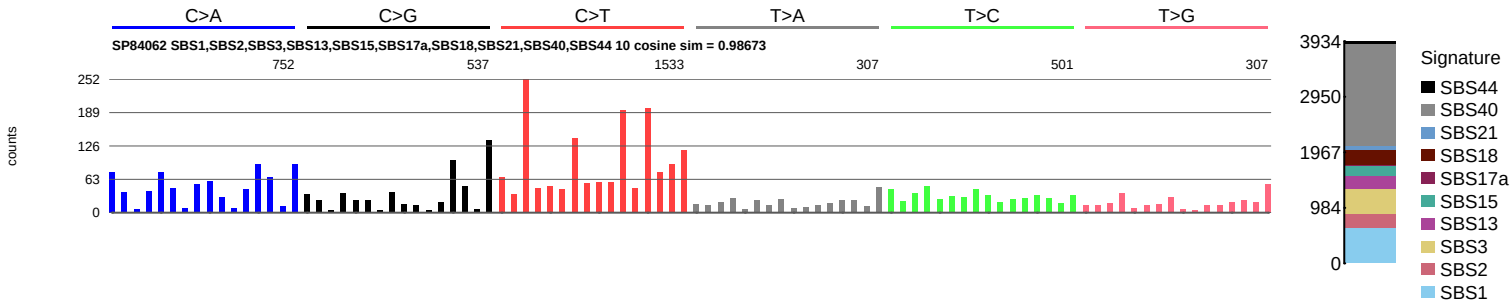

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98677
748
535
1521
302
519
308
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

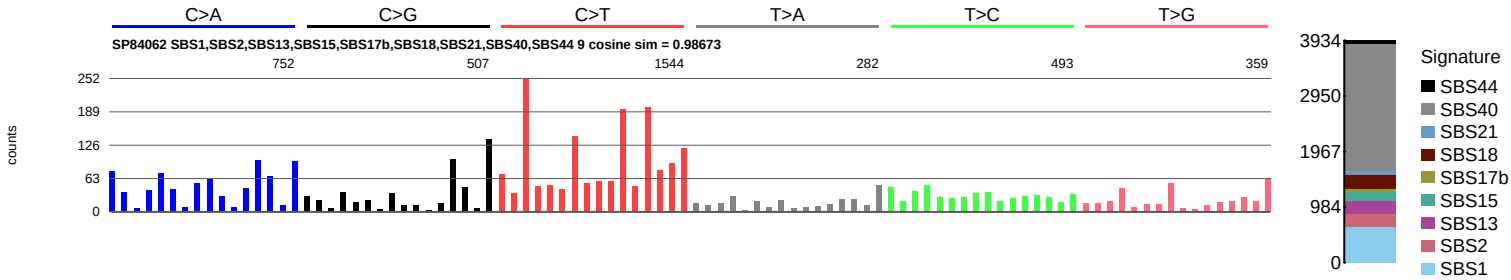

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98676
747
549
1521
341
452
323
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

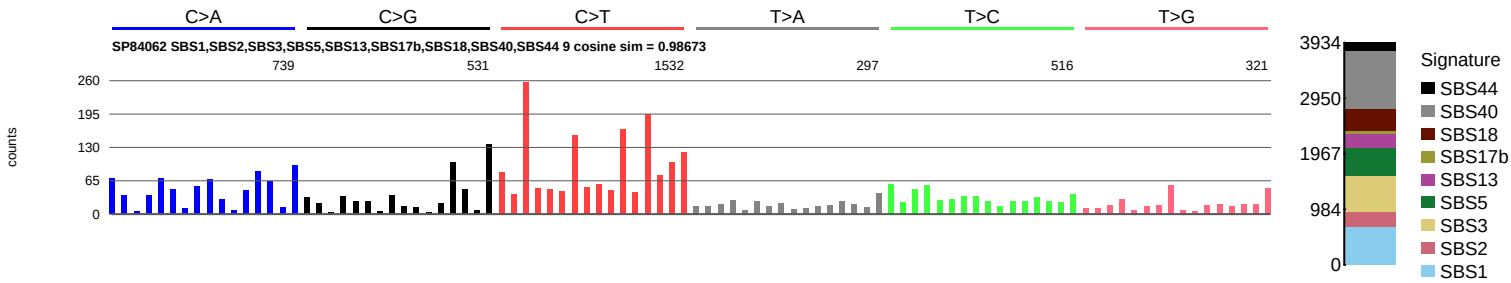

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98675
748
538
1522
302
519
308
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

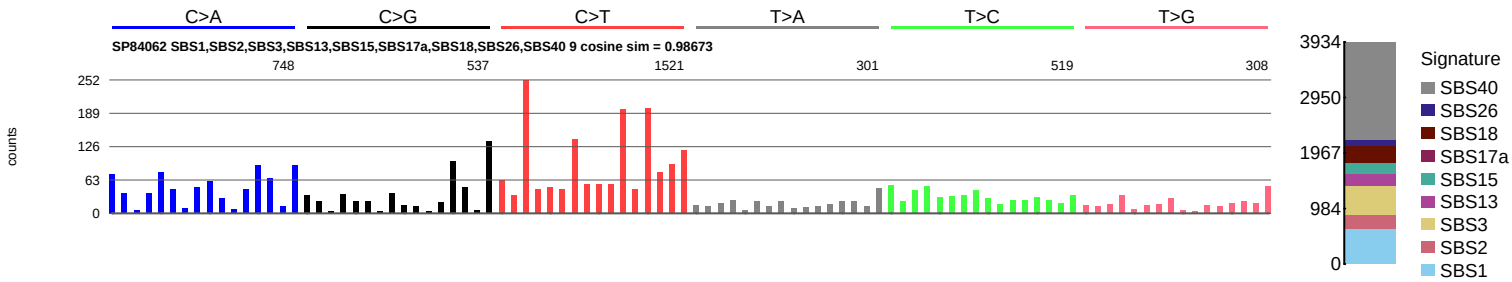

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98675
737
531
1525
294
528
320
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

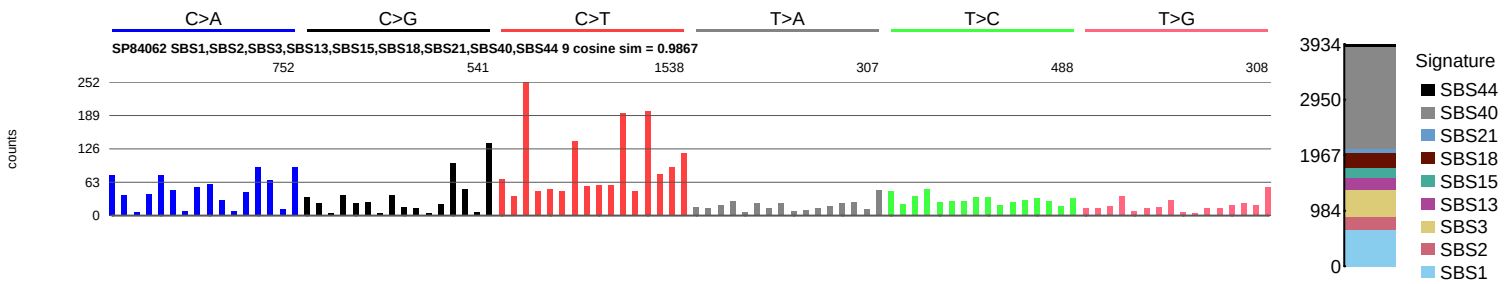

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98674
750
537
1528
301
515
307
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

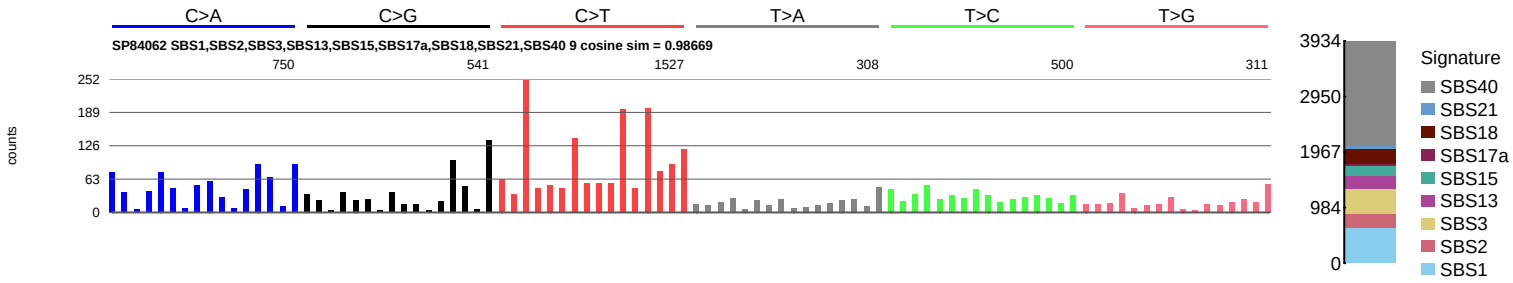

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98673
752
537
1533
307
501
307
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

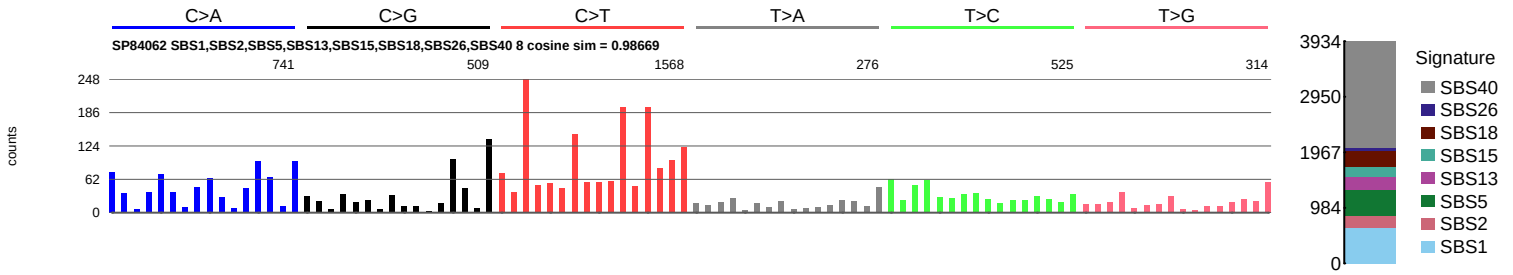

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98673
752
507
1544
282
493
359
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

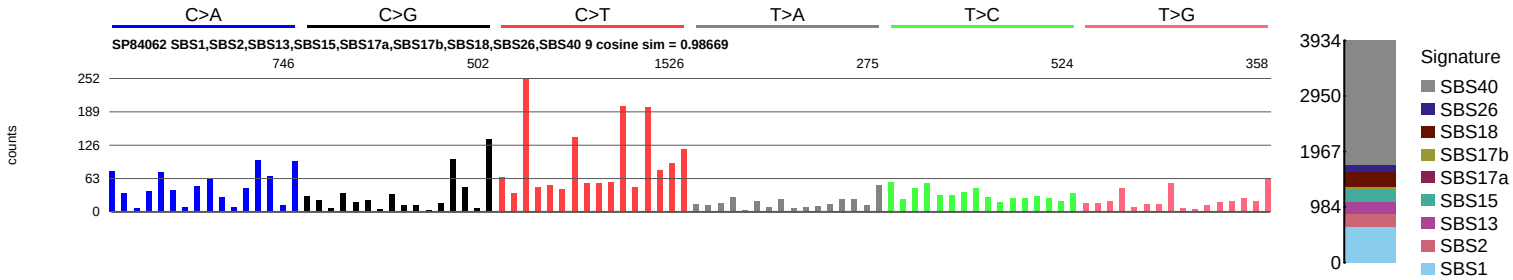

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98673
739
531
1532
297
516
321
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

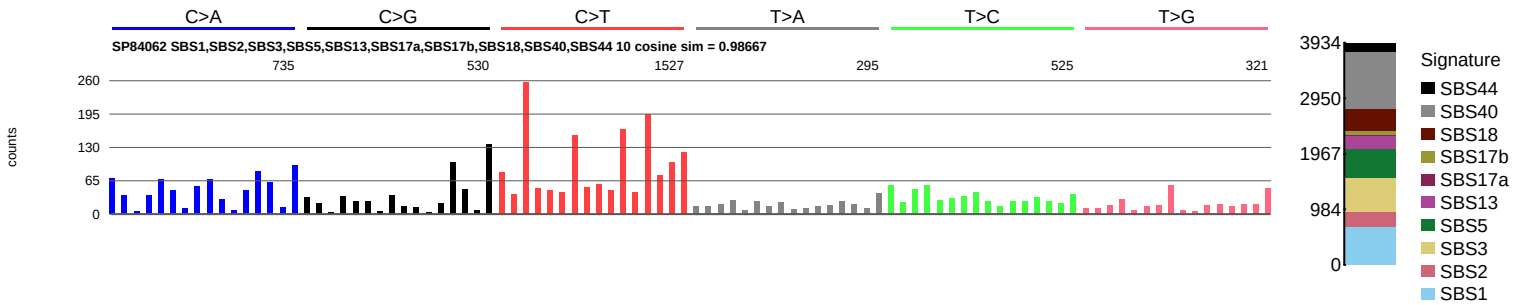

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98673
748
537
1521
301
519
308
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

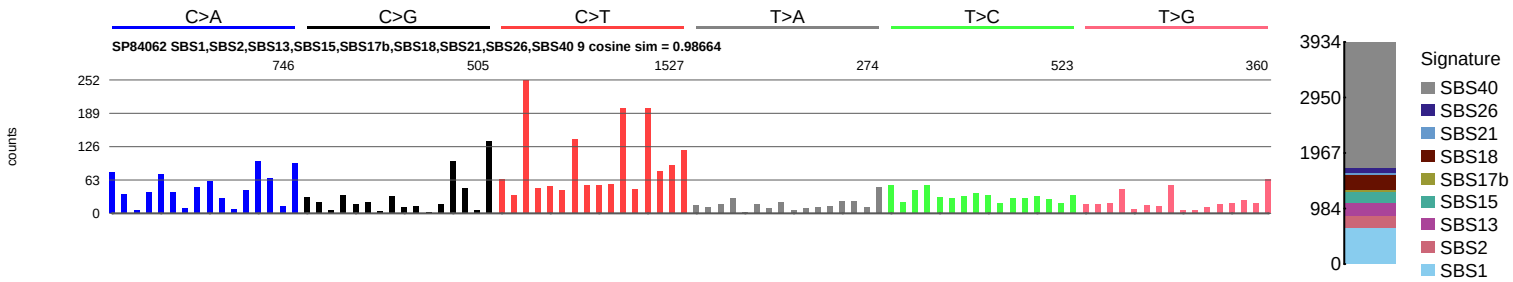

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9867
752
541
1538
307
488
308
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

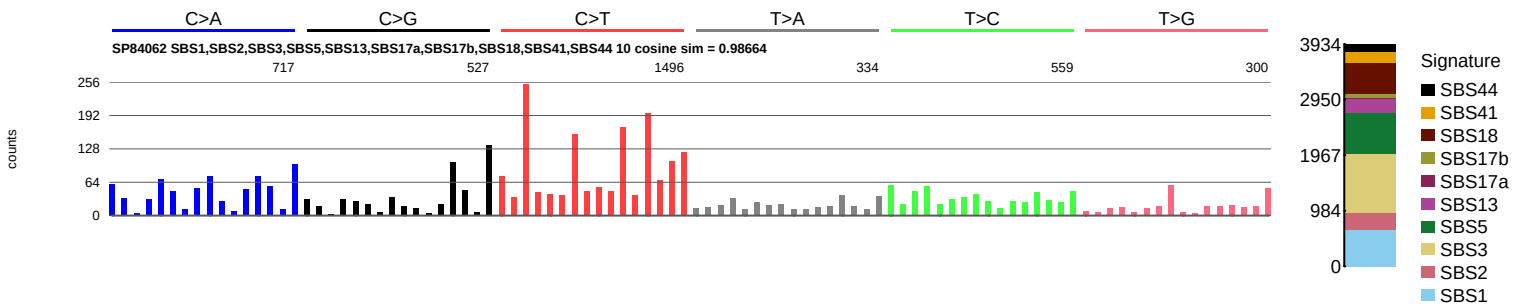

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98669
750
541
1527
308
500
311
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

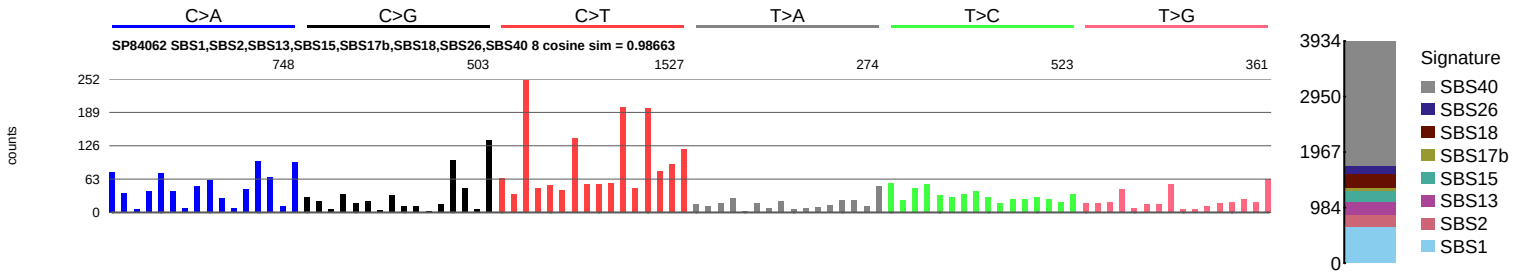

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98669
741
509
1568
276
525
314
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

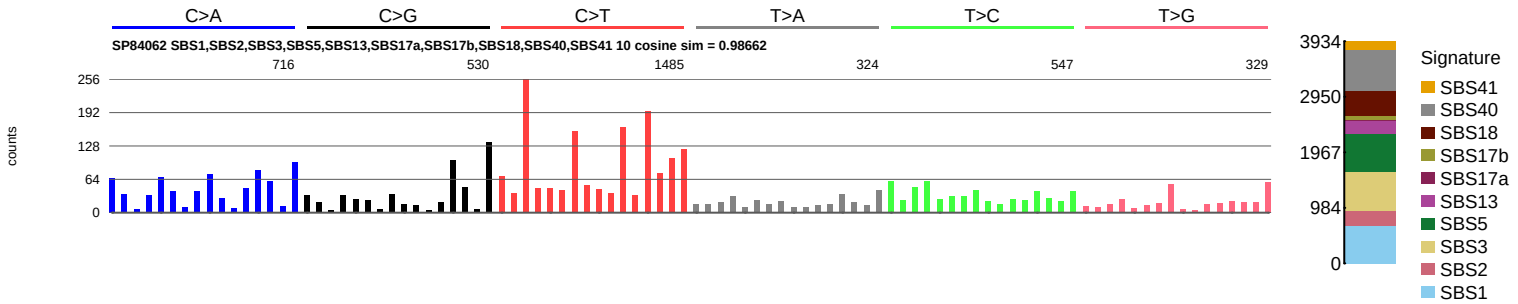

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98669
746
502
1526
275
524
358
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

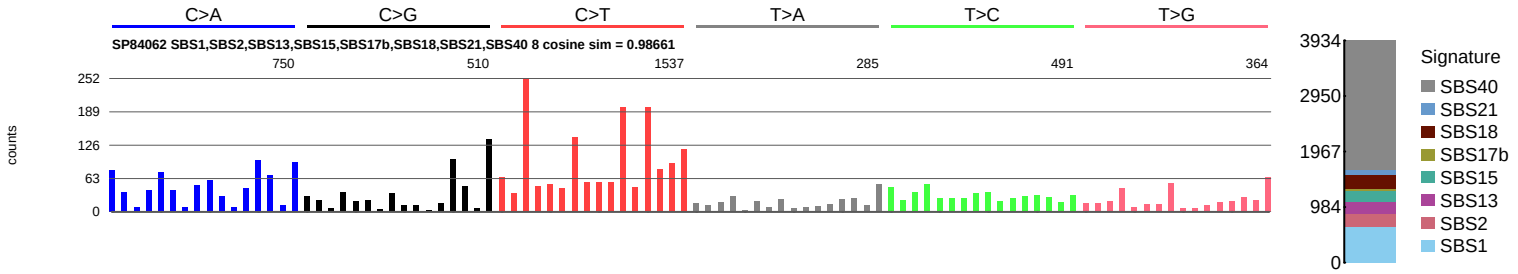

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98667
735
530
1527
295
525
321
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

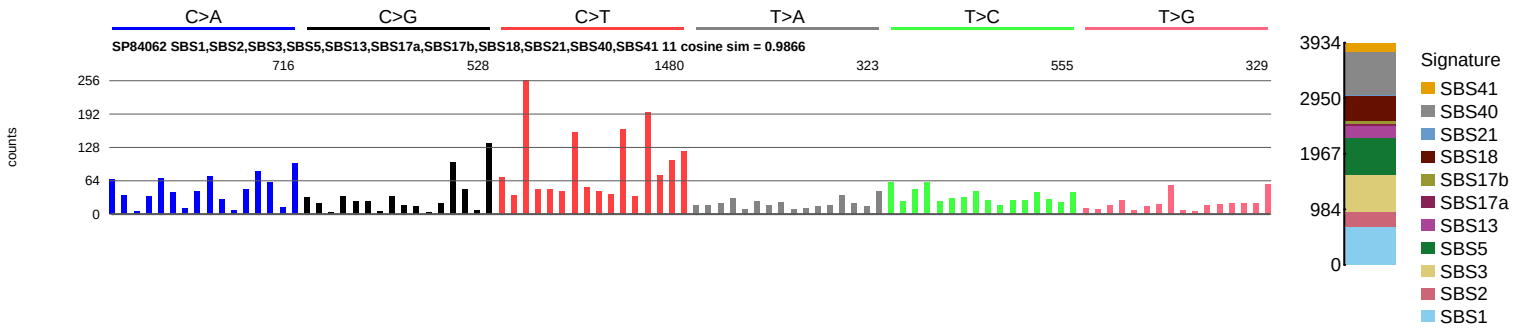

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98664
746
505
1527
274
523
360
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

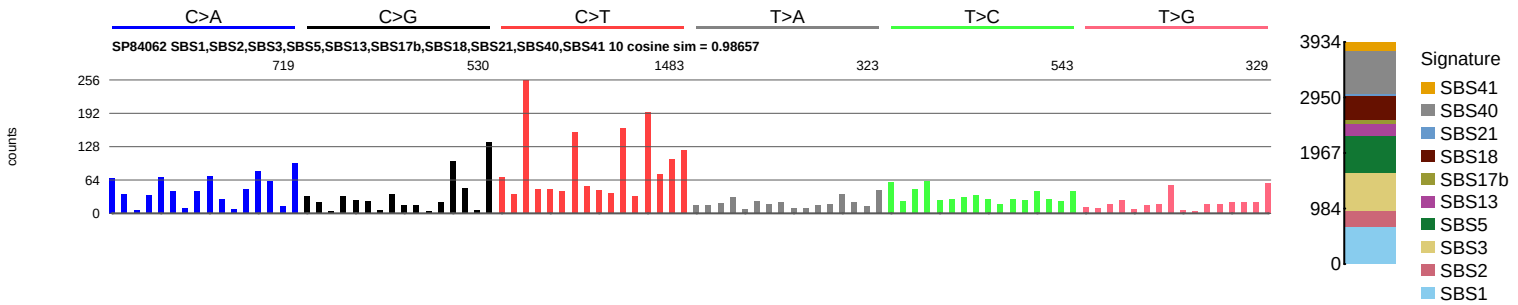

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98664
717
527
1496
334
559
300
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

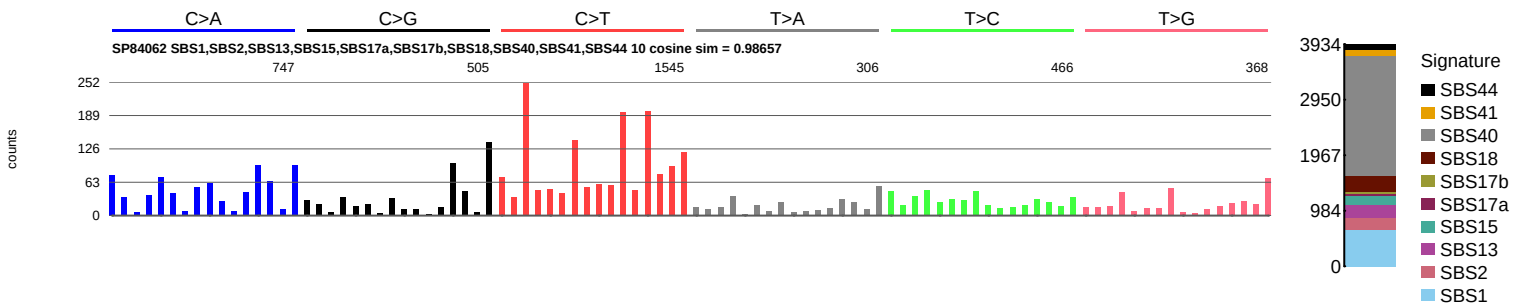

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98663
748
503
1527
274
523
361
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

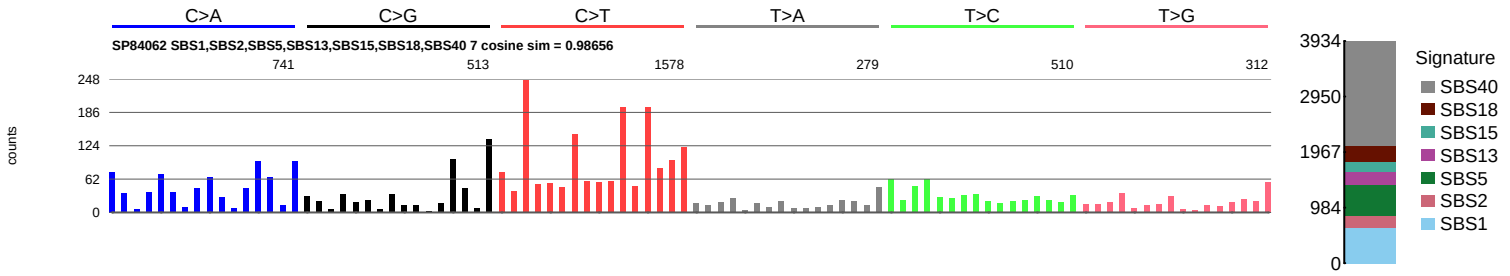

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98662
716
530
1485
324
547
329
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

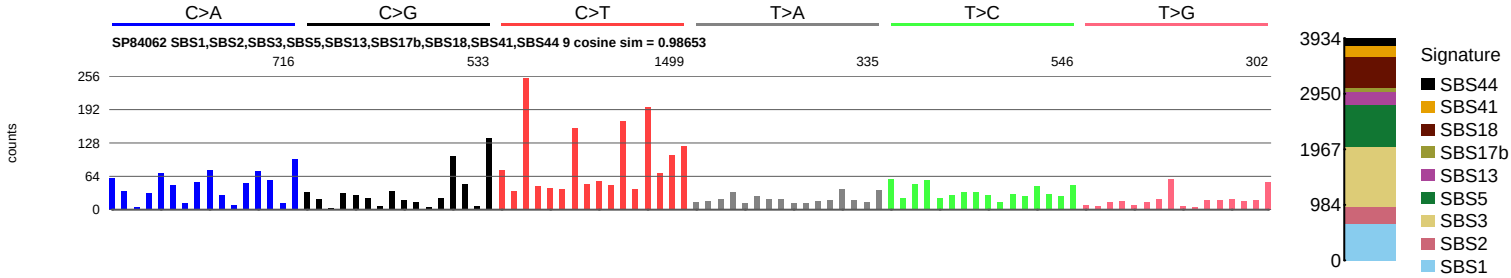

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98661
750
510
1537
285
491
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

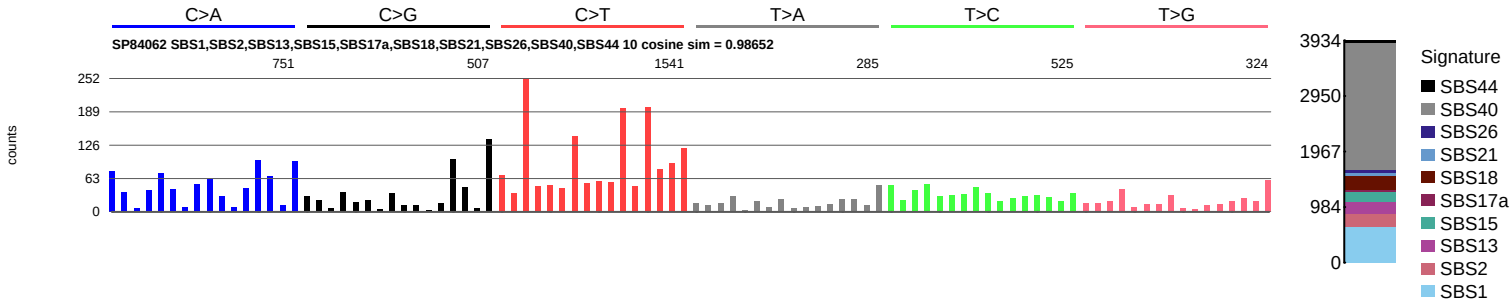

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.9866
716
528
1480
323
555
329
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

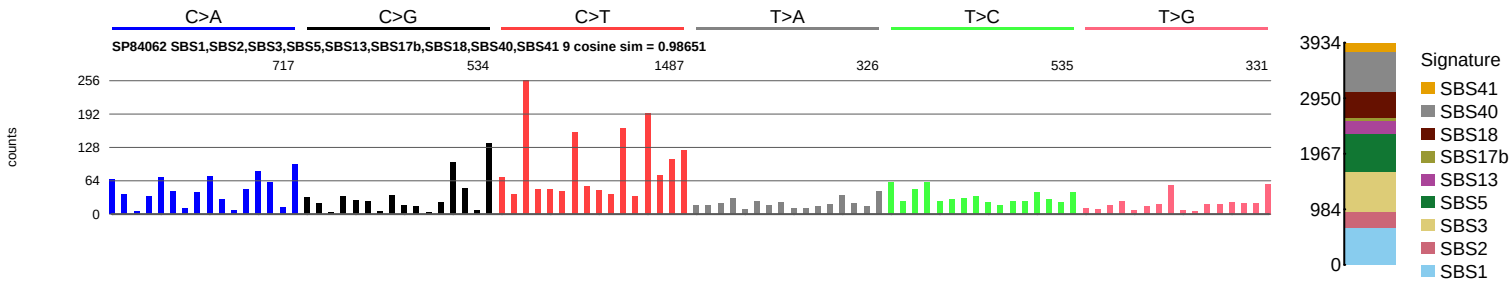

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98657
719
530
1483
323
543
329
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

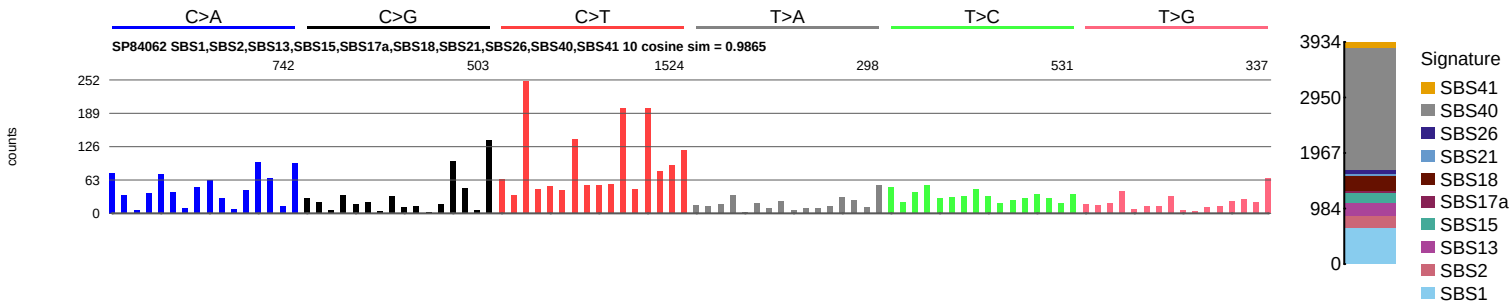

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98657
747
505
1545
306
466
368
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

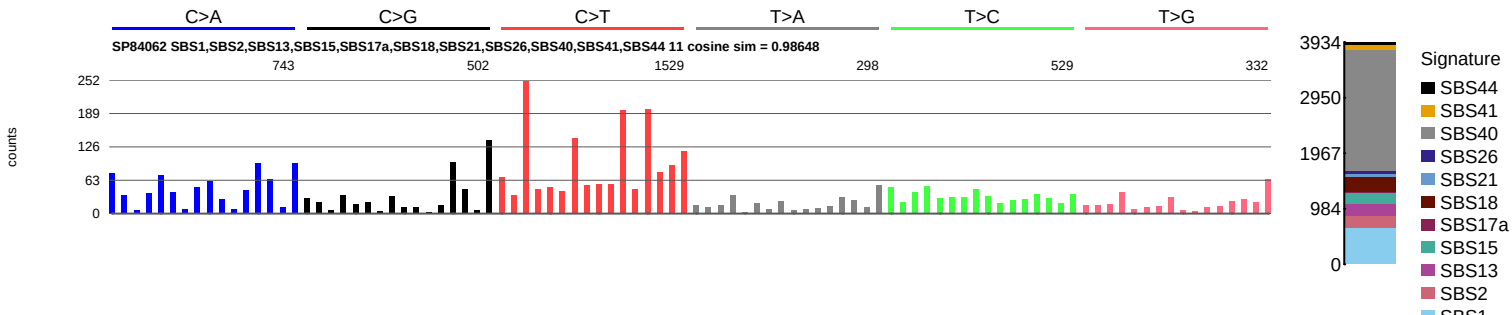

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98656
741
513
1578
279
510
312
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

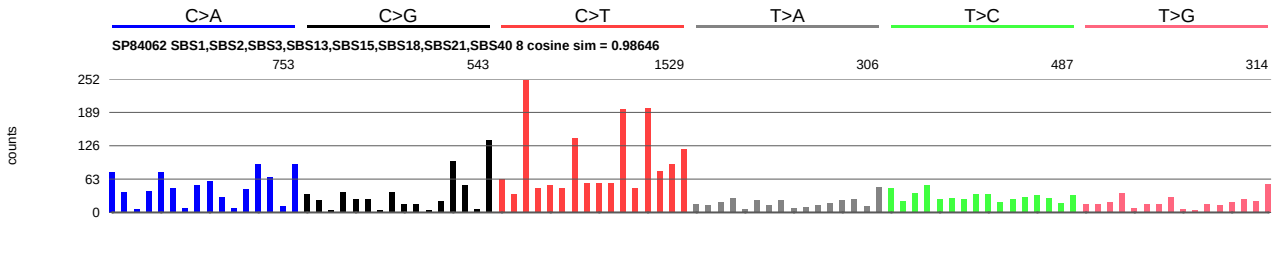

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98653
716
533
1499
335
546
302
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

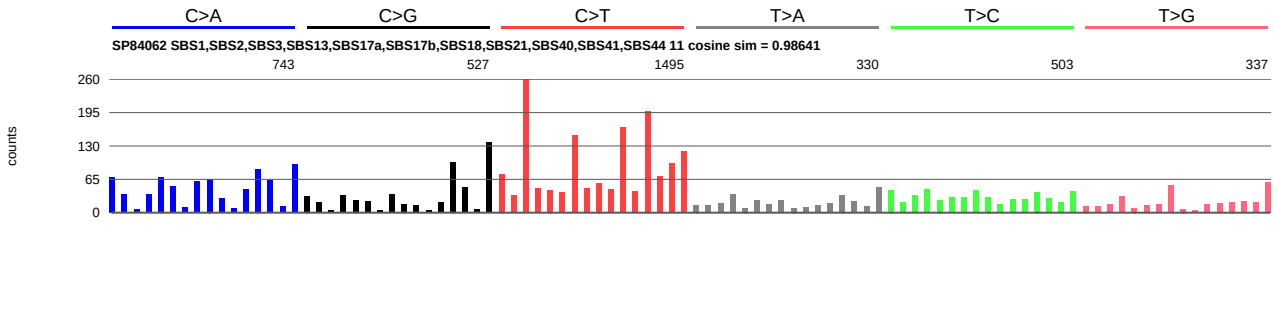

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98652
751
507
1541
285
525
324
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

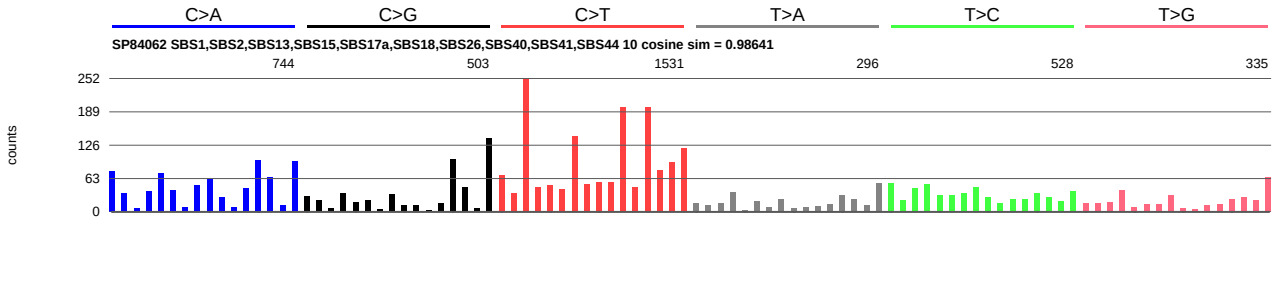

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98651
717
534
1487
326
535
331
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

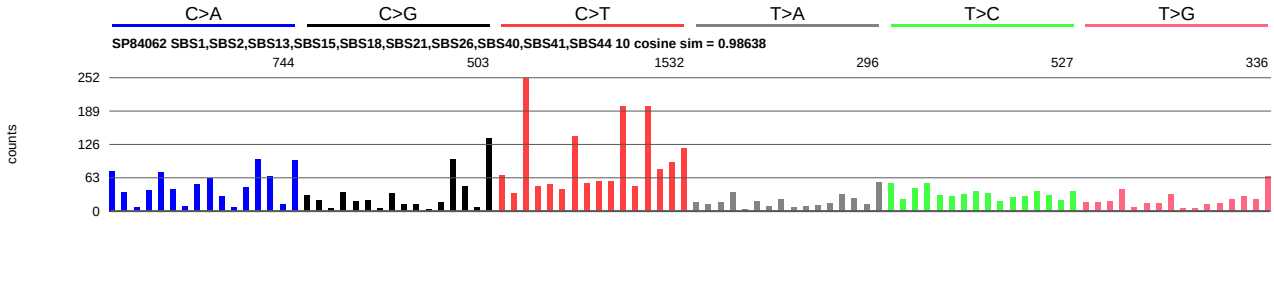

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9865
742
503
1524
298
531
337
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

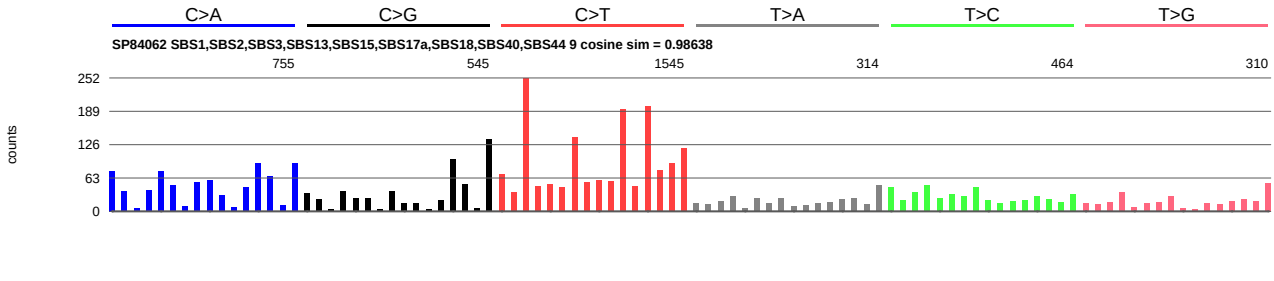

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98648
743
502
1529
298
529
332
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2

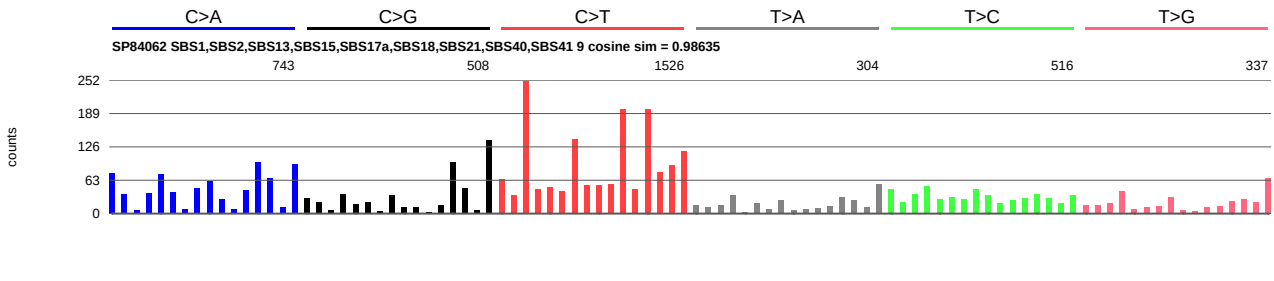

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98646
753
543
1529
306
487
314
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

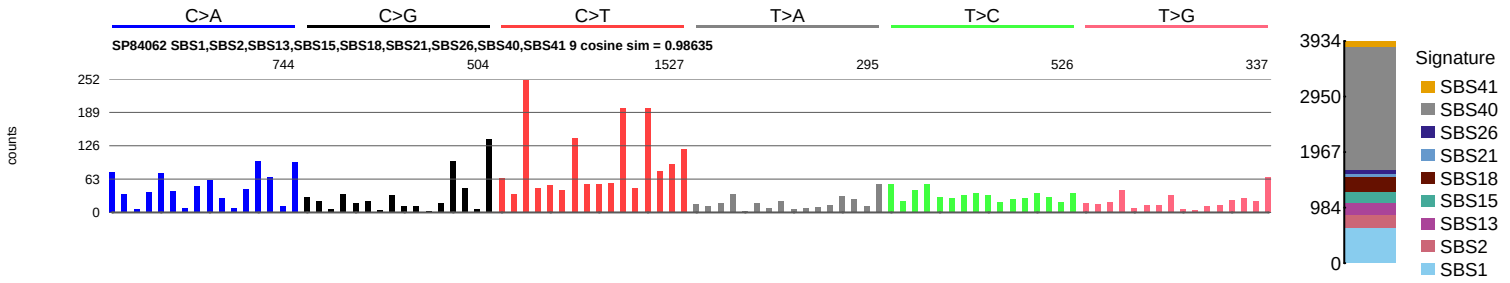

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98641
743
527
1495
330
503
337
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

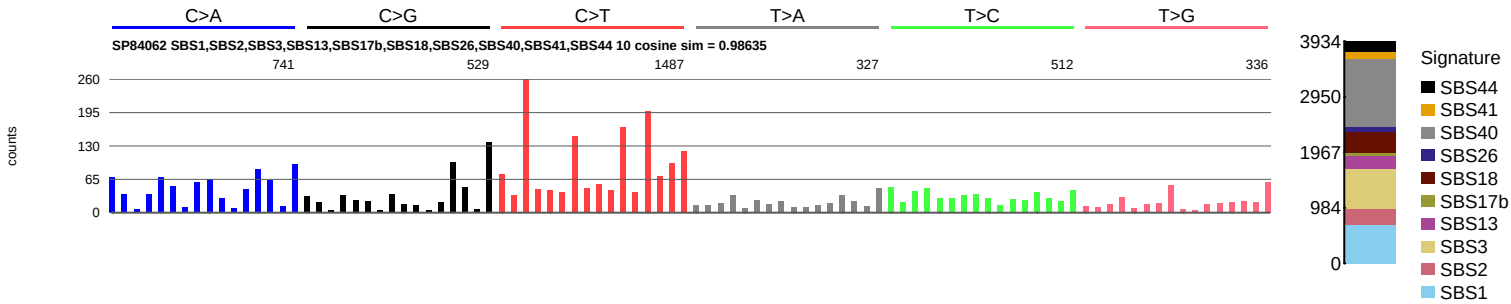

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98641
744
503
1531
296
528
335
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

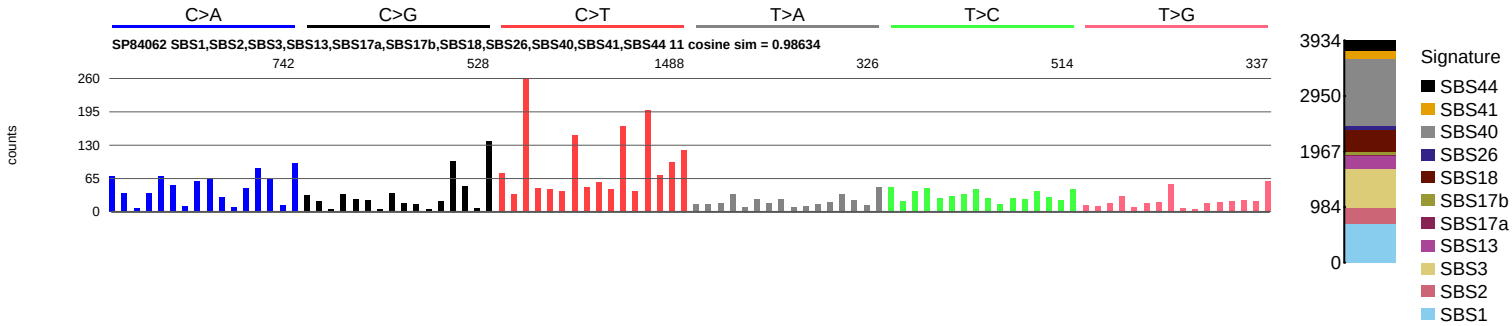

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98638
744
503
1532
296
527
336
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

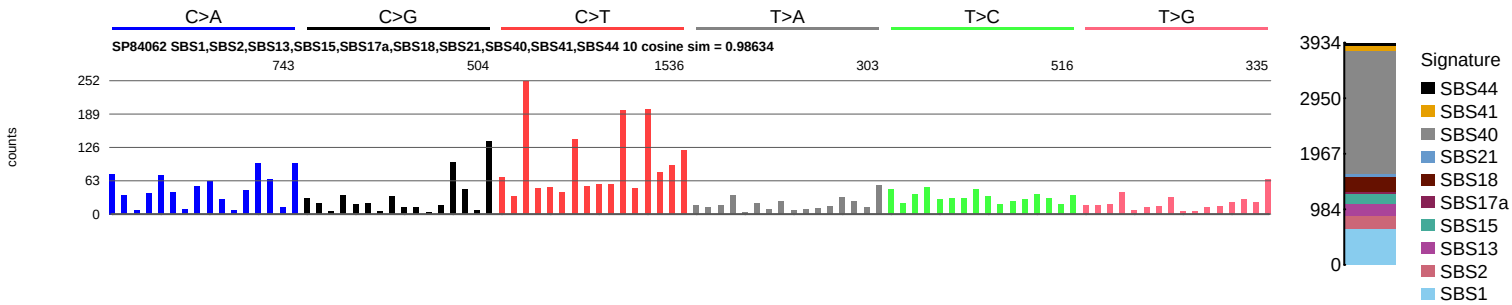

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98638
755
545
1545
314
464
310
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

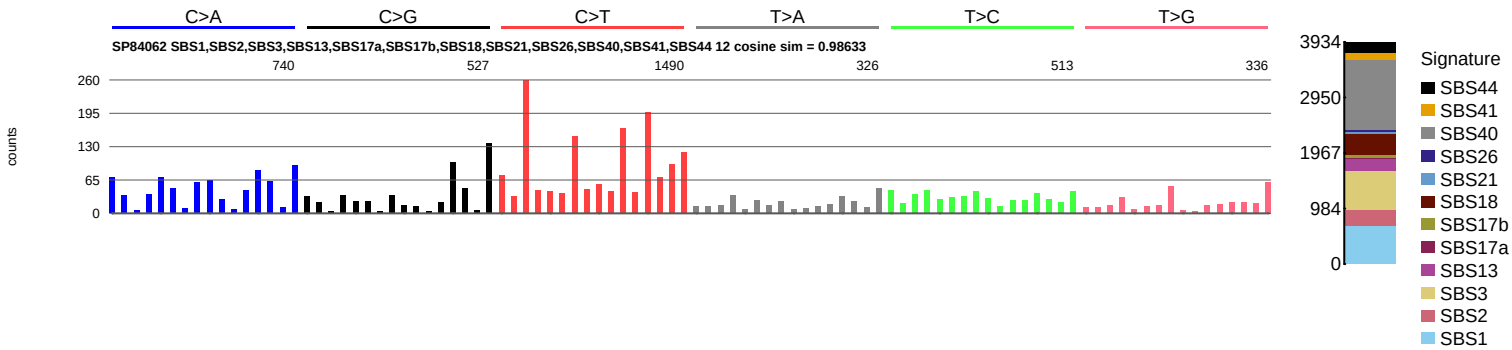

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98635
743
508
1526
304
516
337
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

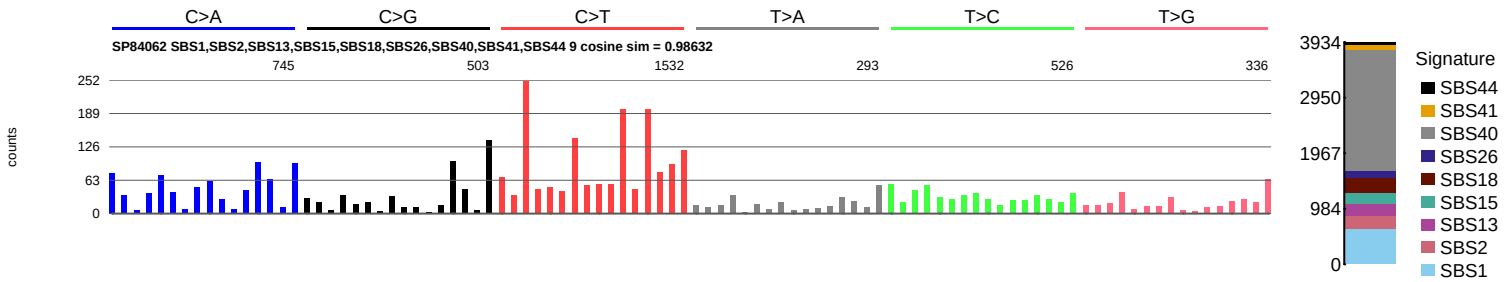

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98635
744
504
1527
295
526
337
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

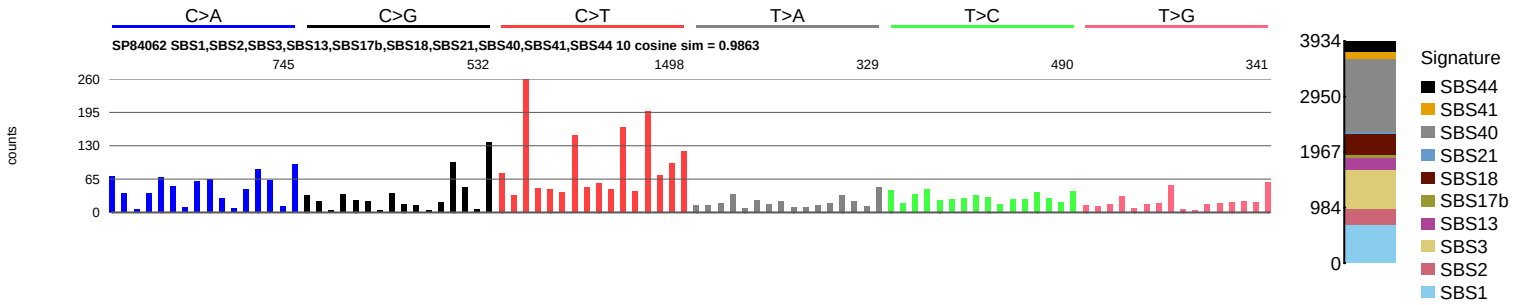

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98635
741
529
1487
327
512
336
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

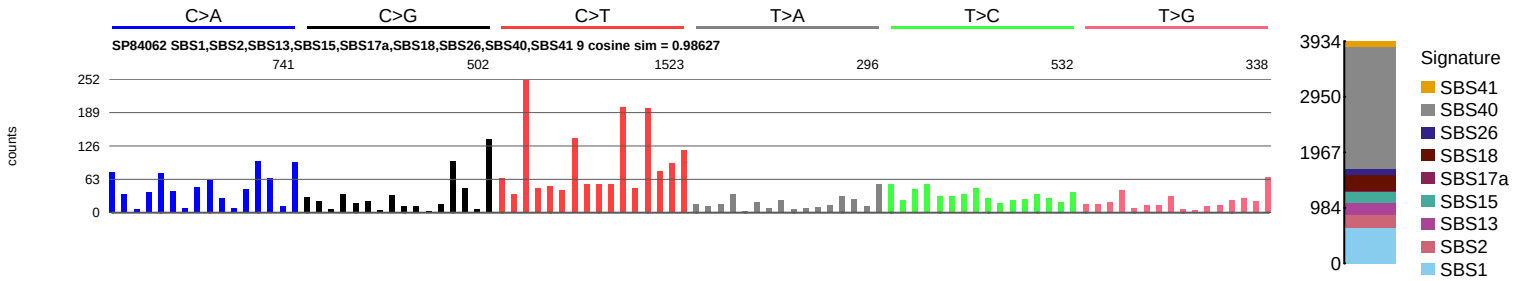

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98634
742
528
1488
326
514
337
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

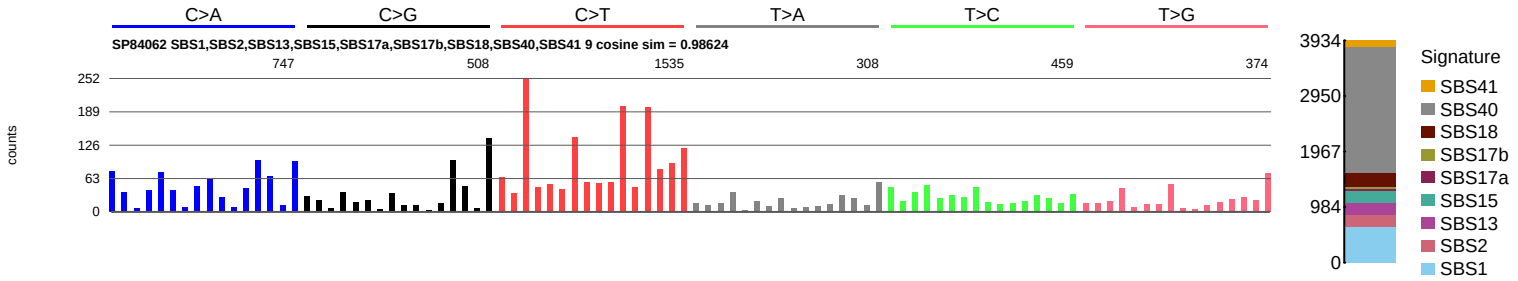

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98634
743
504
1536
303
516
335
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

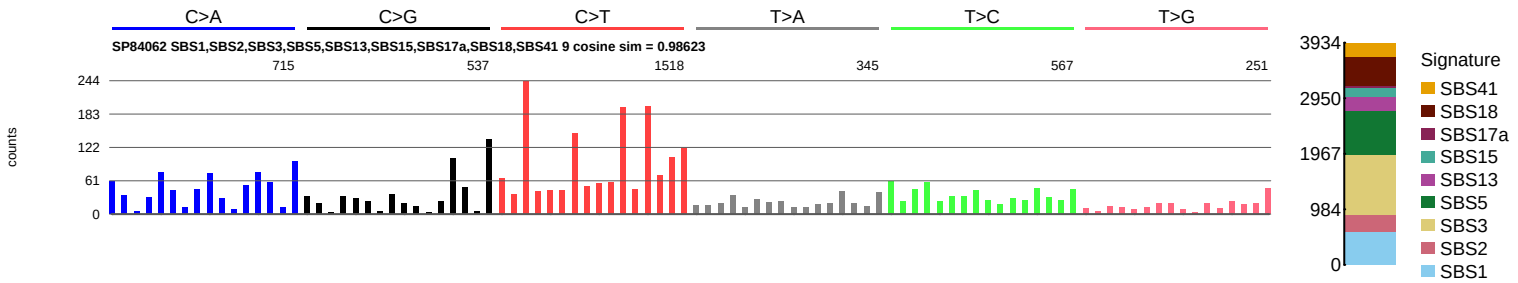

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98633
740
527
1490
326
513
336
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

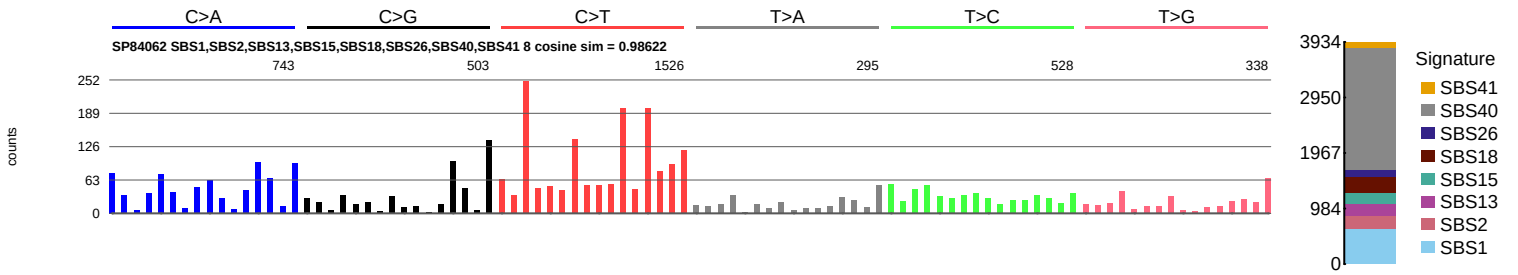

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98632
745
503
1532
293
526
336
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1

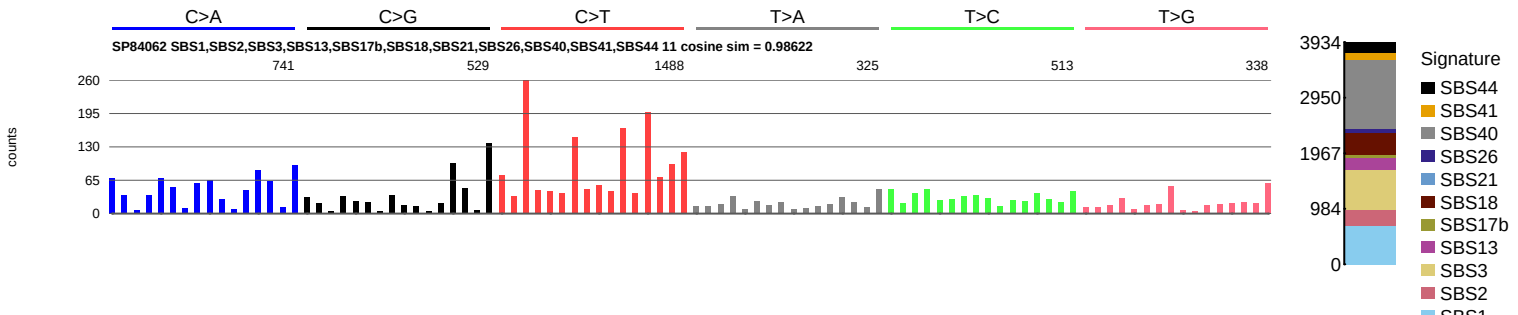

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9863
745
532
1498
329
490
341
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

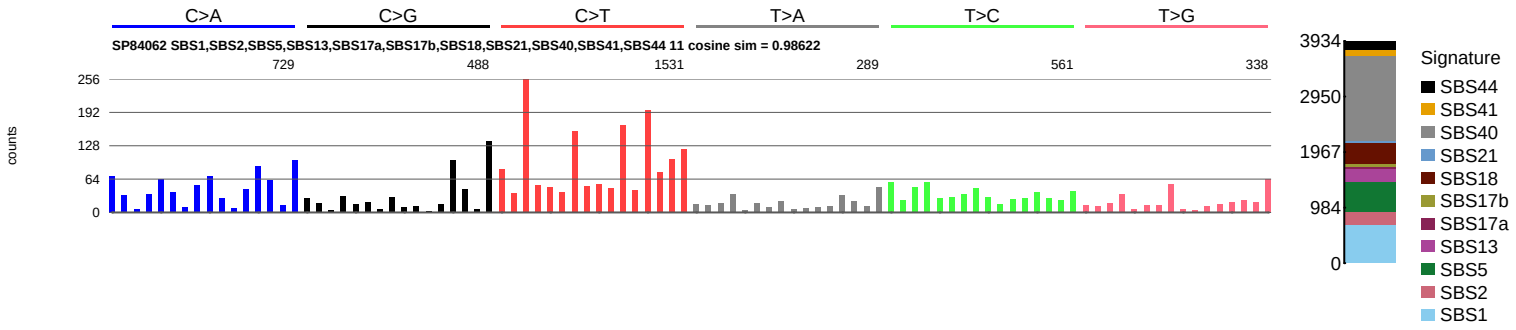

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98627
741
502
1523
296
532
338
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

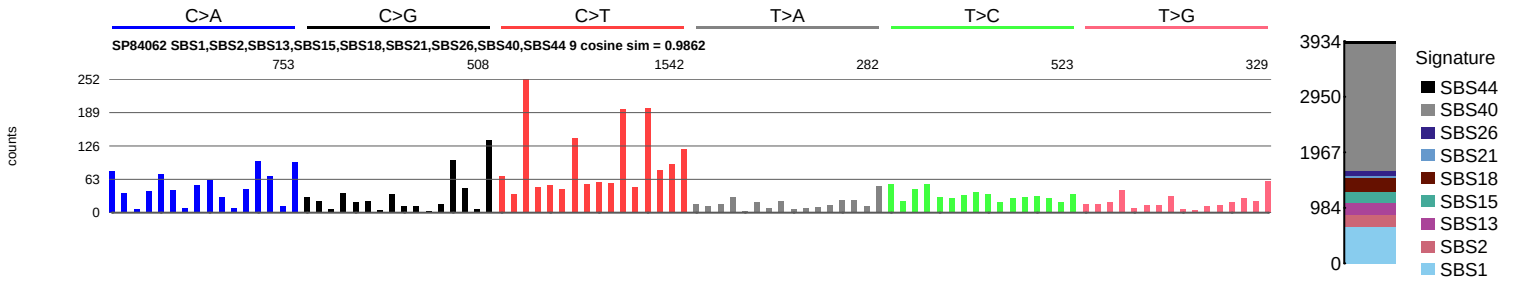

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98624
747
508
1535
308
459
374
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

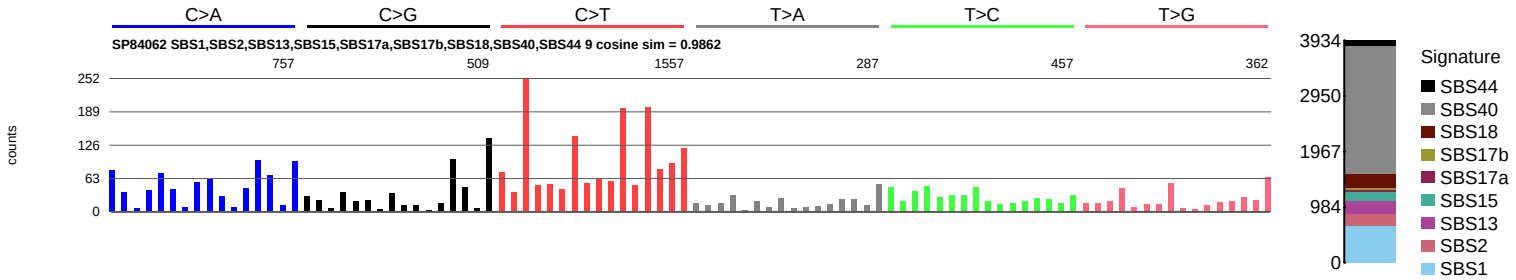

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98623
715
537
1518
345
567
251
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

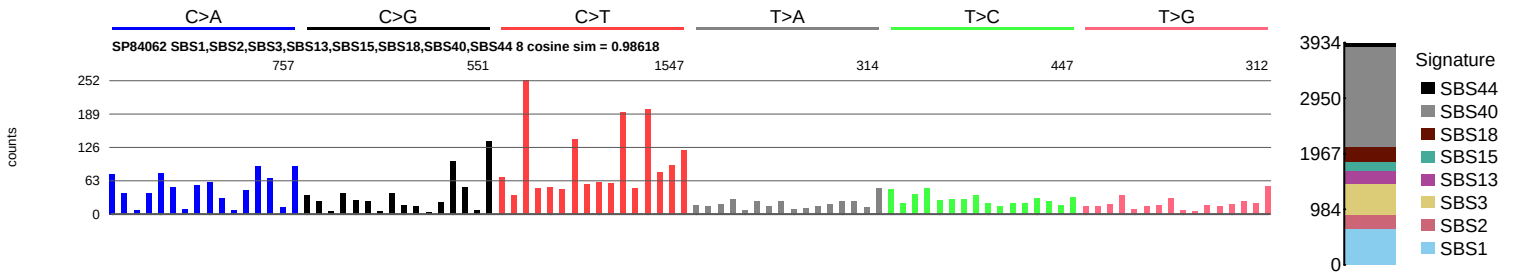

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98622
743
503
1526
295
528
338
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1

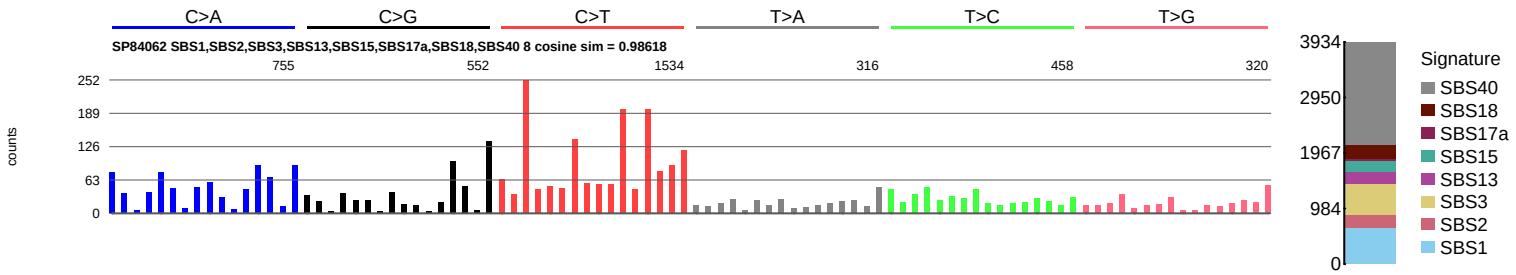

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98622
741
529
1488
325
513
338
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2

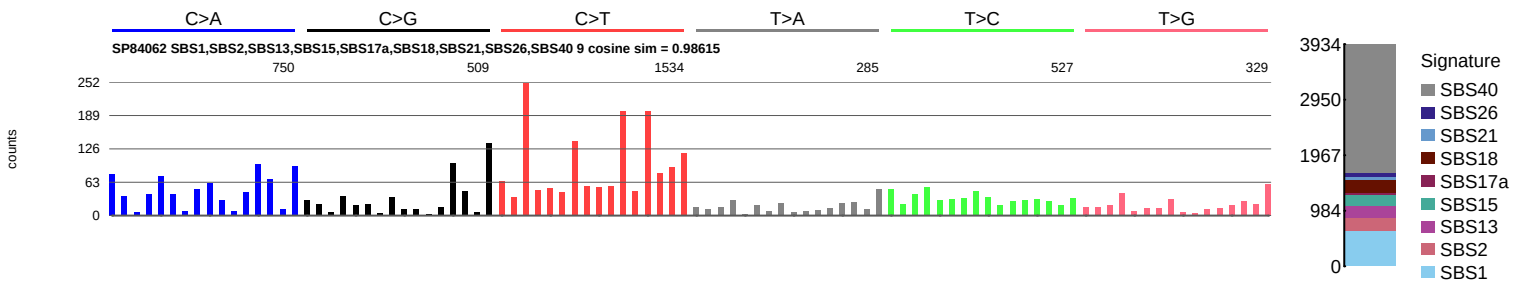

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98622
729
488
1531
289
561
338
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

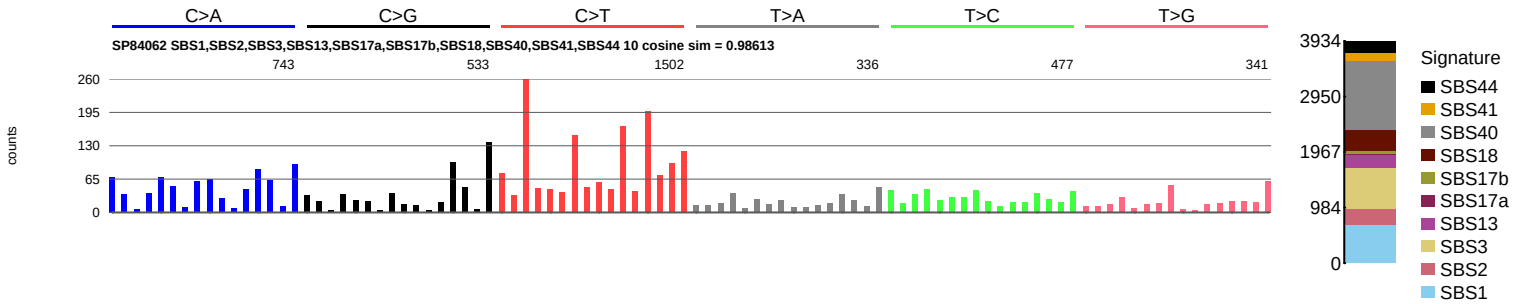

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9862
753
508
1542
282
523
329
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

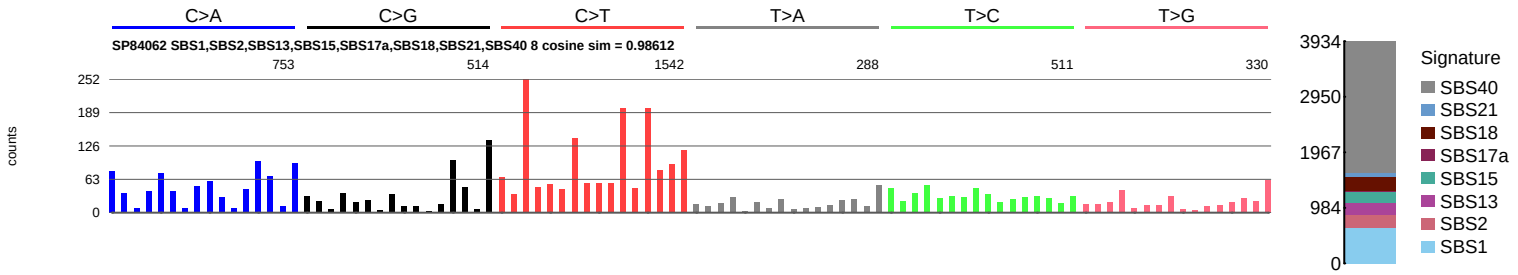

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9862
757
509
1557
287
457
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

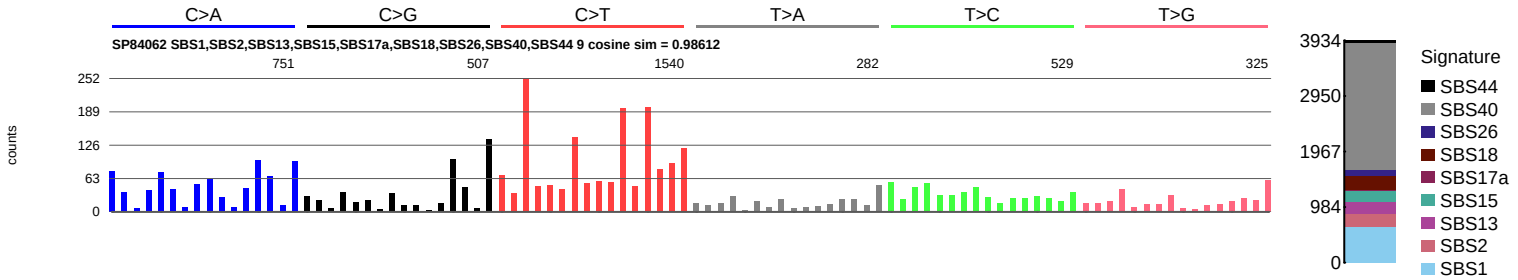

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98618
757
551
1547
314
447
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

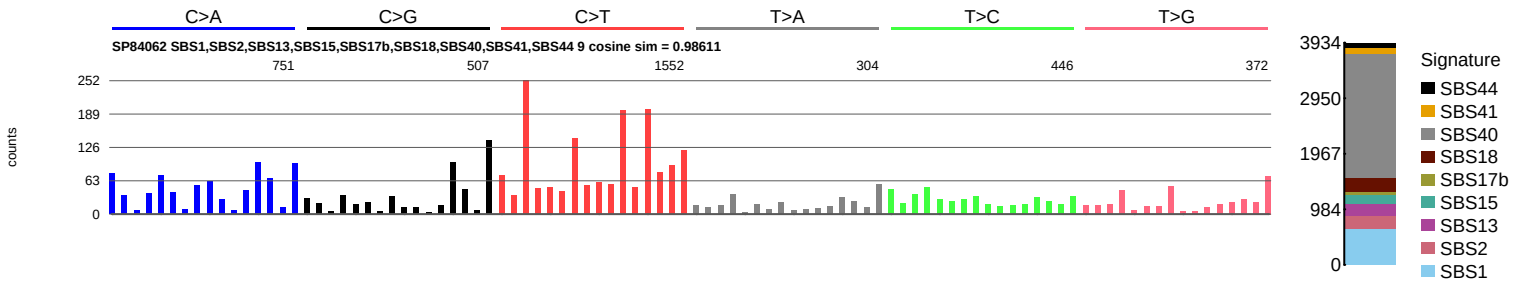

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98618
755
552
1534
316
458
320
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

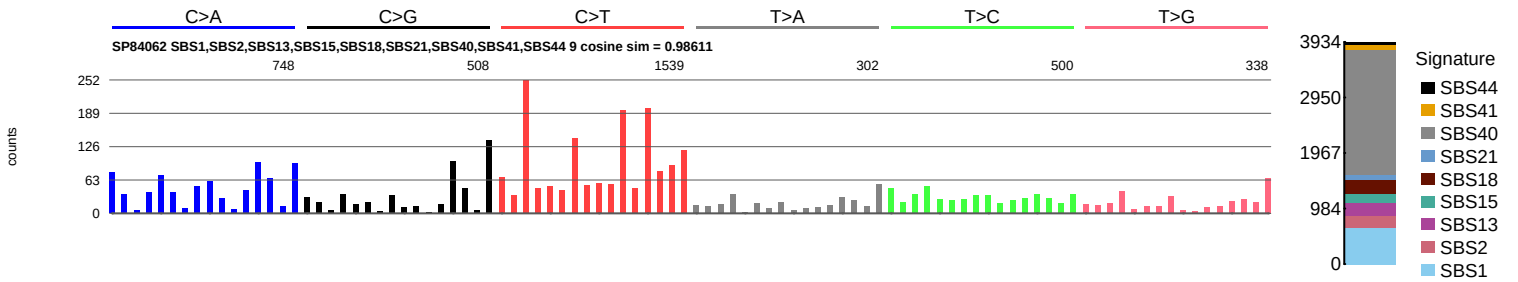

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98615
750
509
1534
285
527
329
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

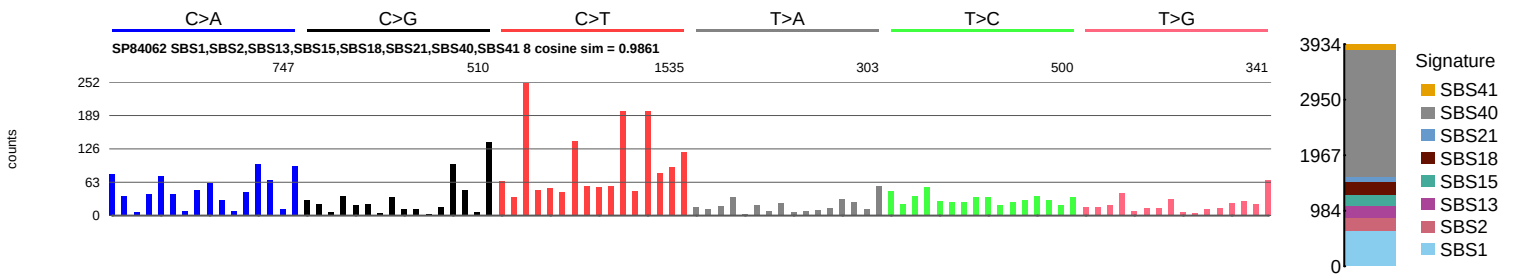

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98613
743
533
1502
336
477
341
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

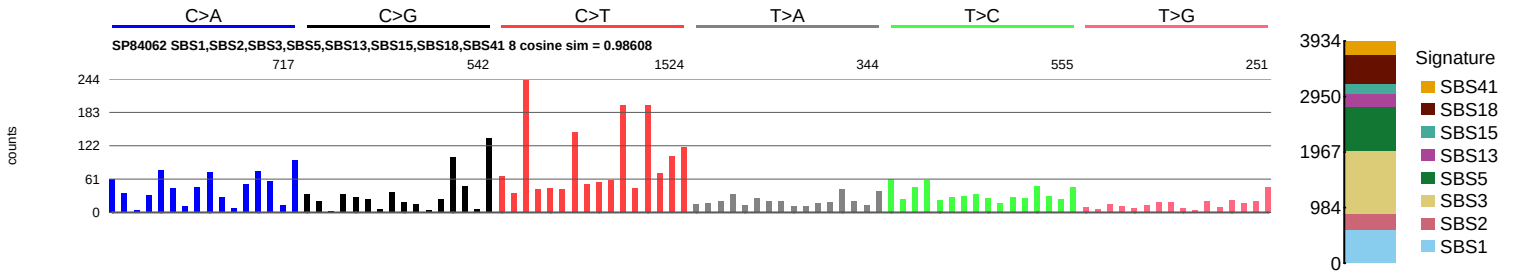

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98612
753
514
1542
288
511
330
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

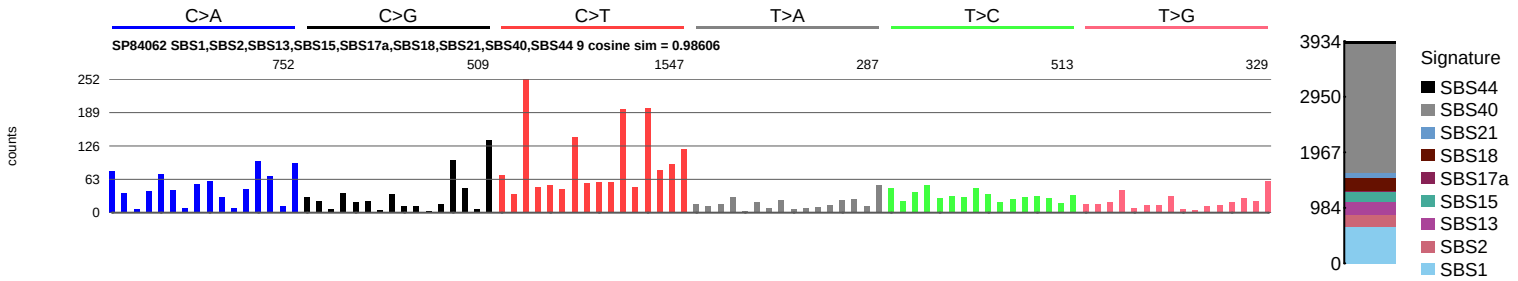

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98612
751
507
1540
282
529
325
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

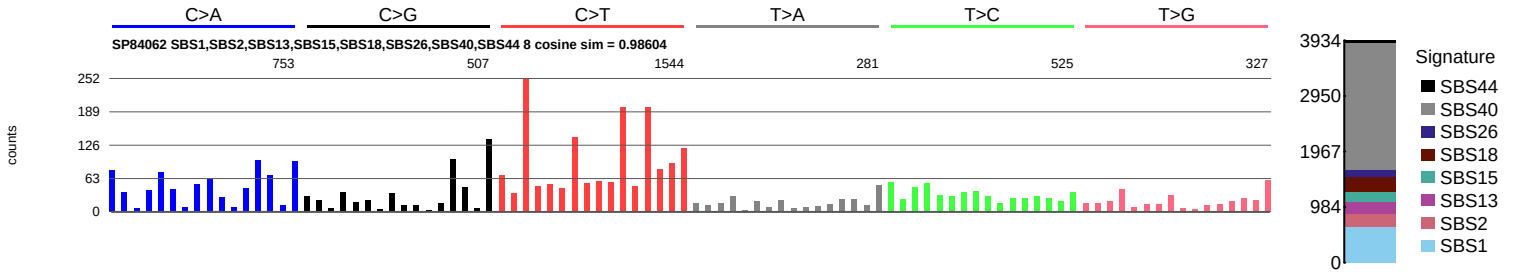

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98611
751
507
1552
304
446
372
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

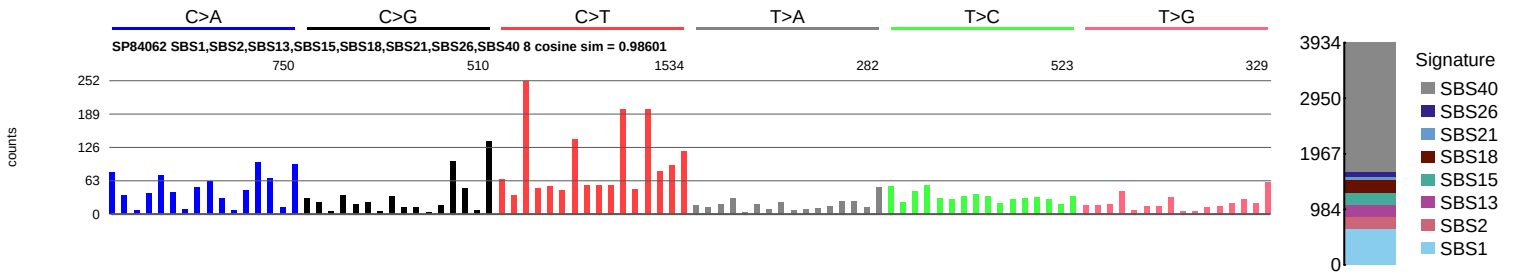

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98611
748
508
1539
302
500
338
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

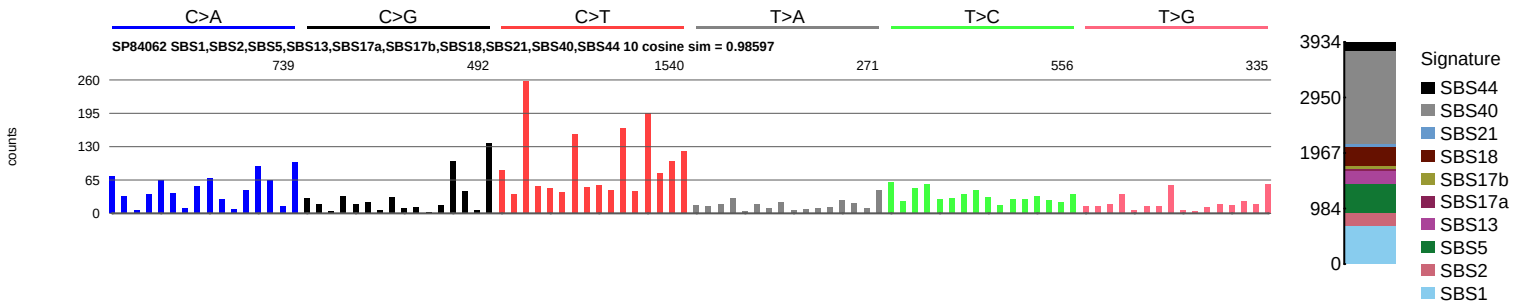

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9861
747
510
1535
303
500
341
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

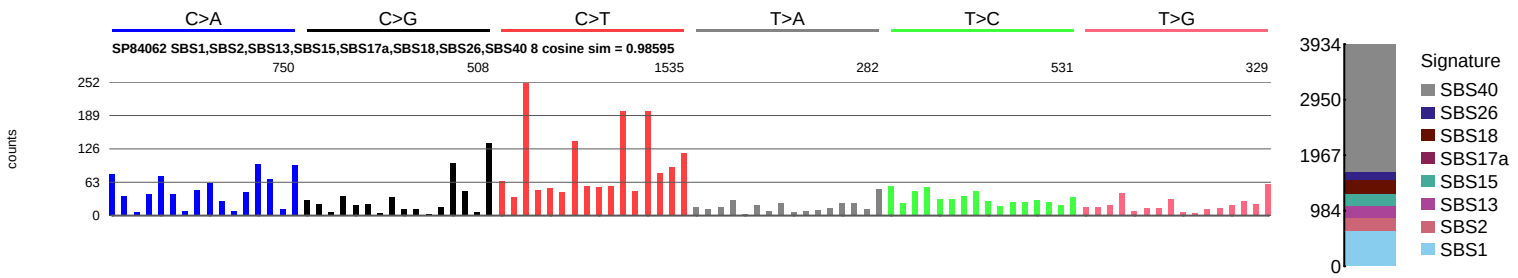

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98608
717
542
1524
344
555
251
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

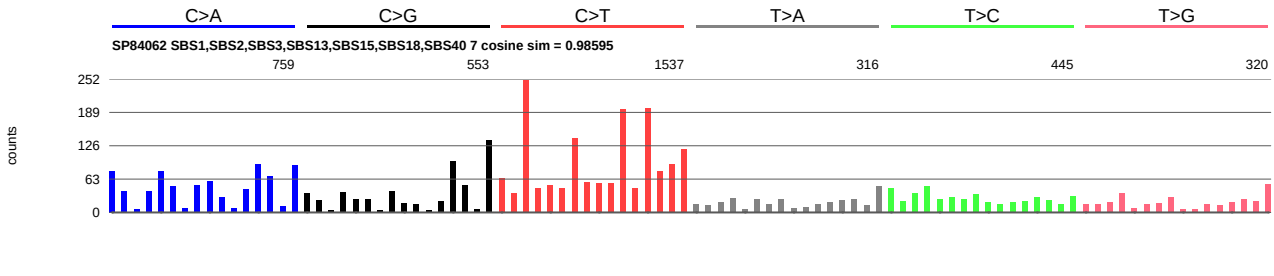

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98606
752
509
1547
287
513
329
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

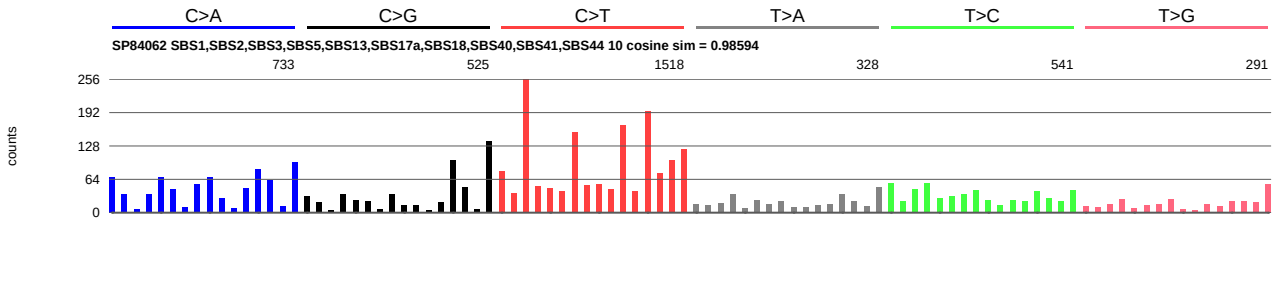

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98604
753
507
1544
281
525
327
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1

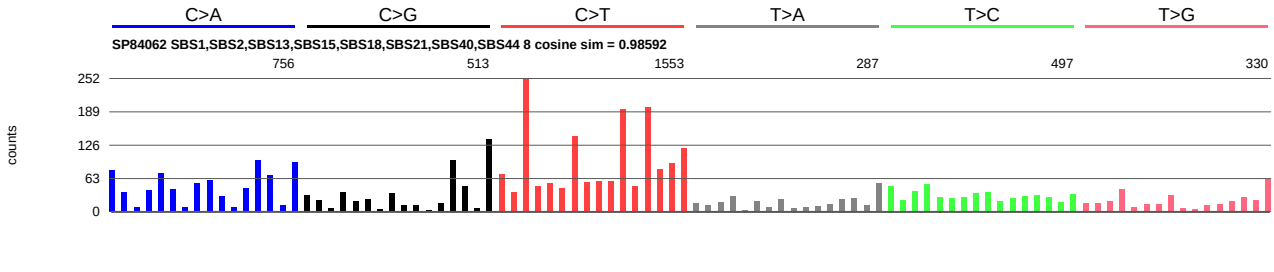

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98601
750
510
1534
282
523
329
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

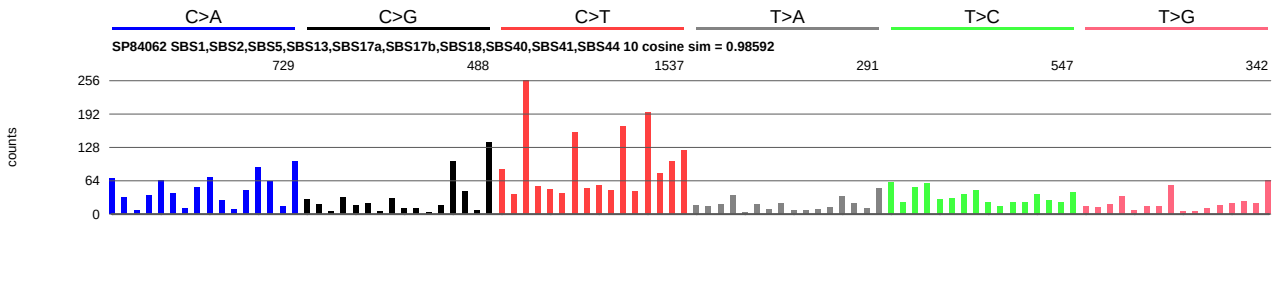

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98597
739
492
1540
271
556
335
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

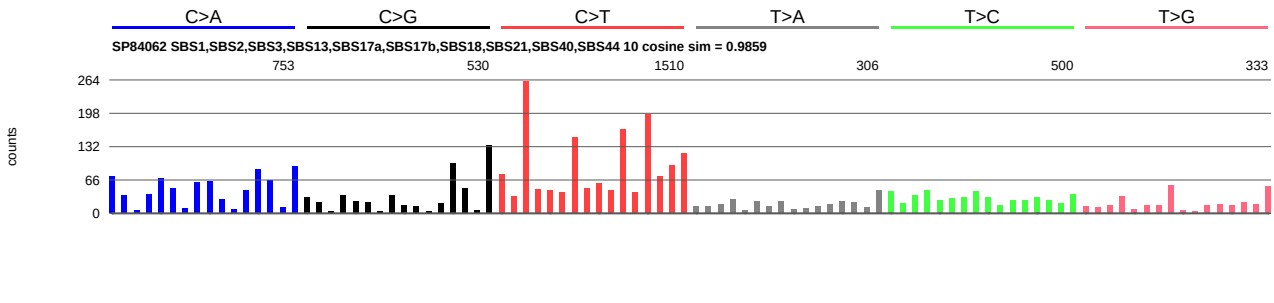

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98595
750
508
1535
282
531
329
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

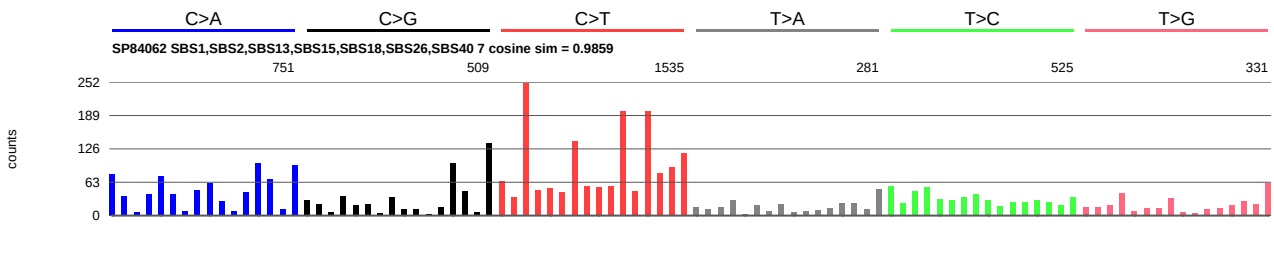

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98595
759
553
1537
316
445
320
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

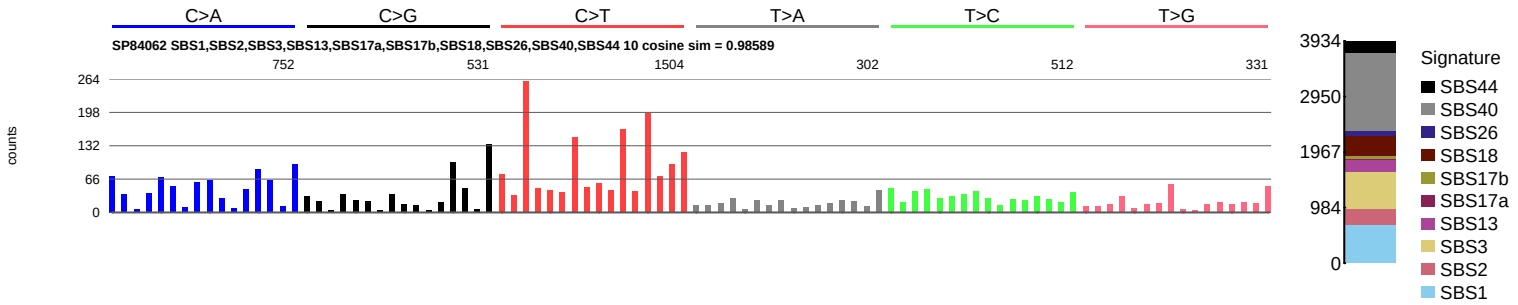

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98594
733
525
1518
328
541
291
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

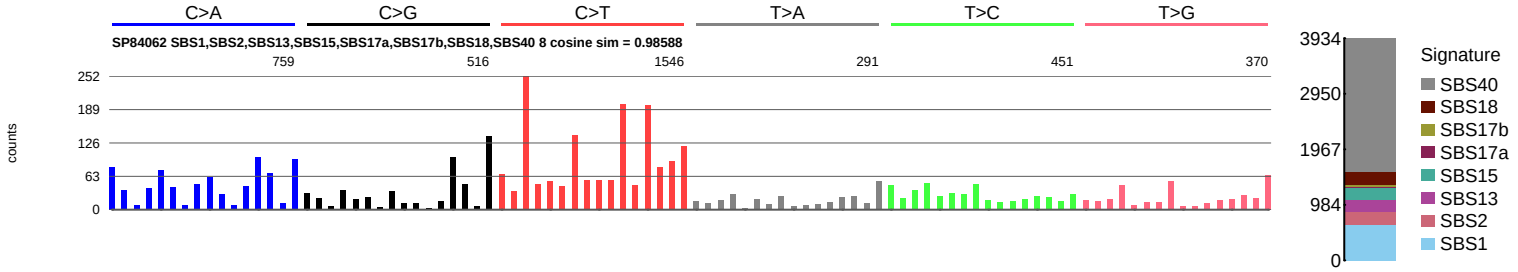

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98592
756
513
1553
287
497
330
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

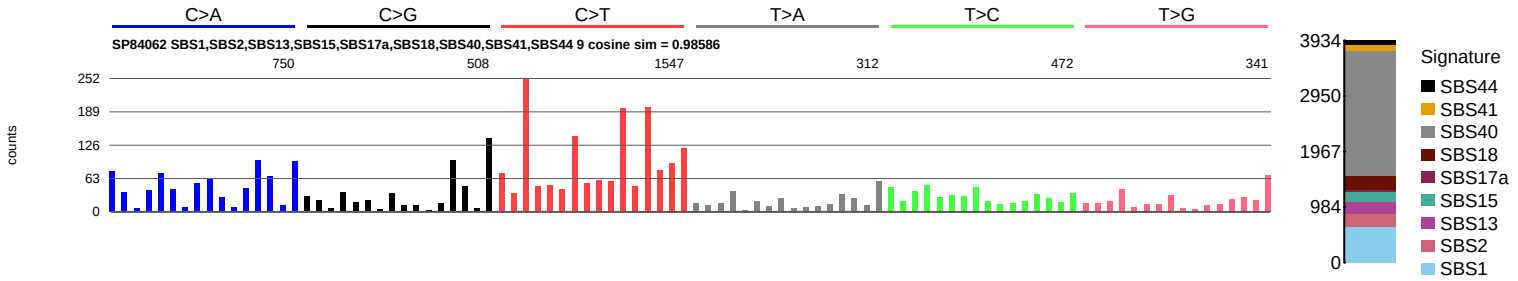

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98592
729
488
1537
291
547
342
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

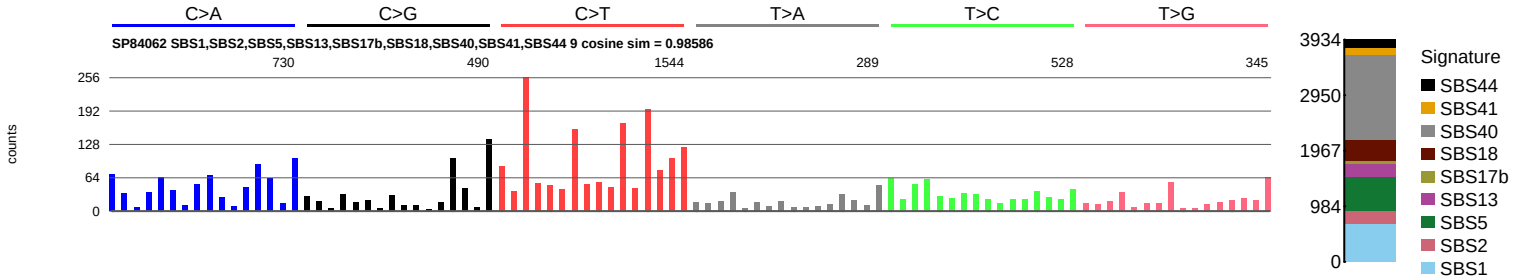

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9859
753
530
1510
306
500
333
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

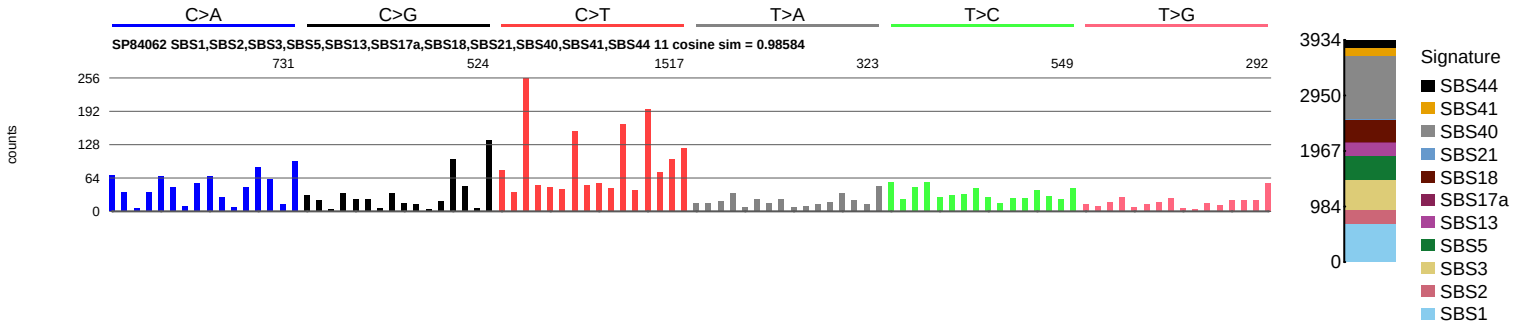

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.9859
751
509
1535
281
525
331
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1

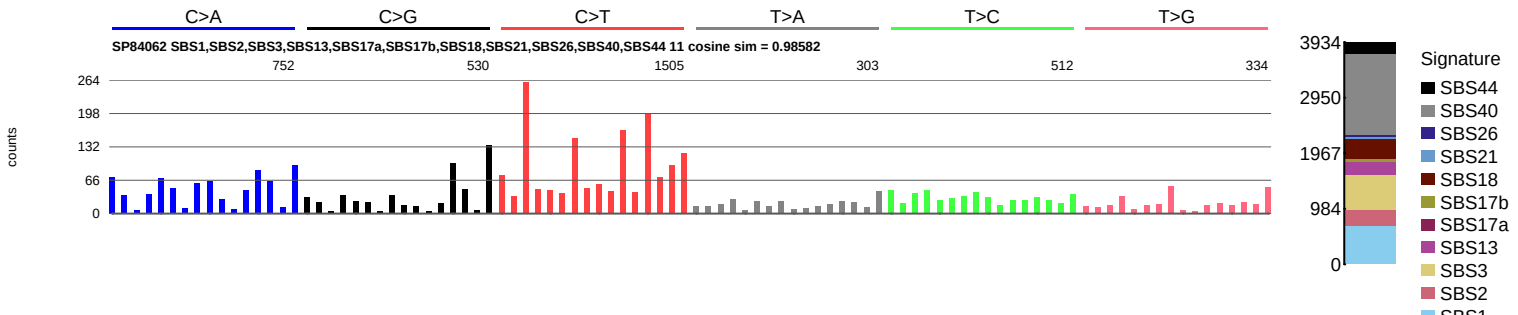

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98589
752
531
1504
302
512
331
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

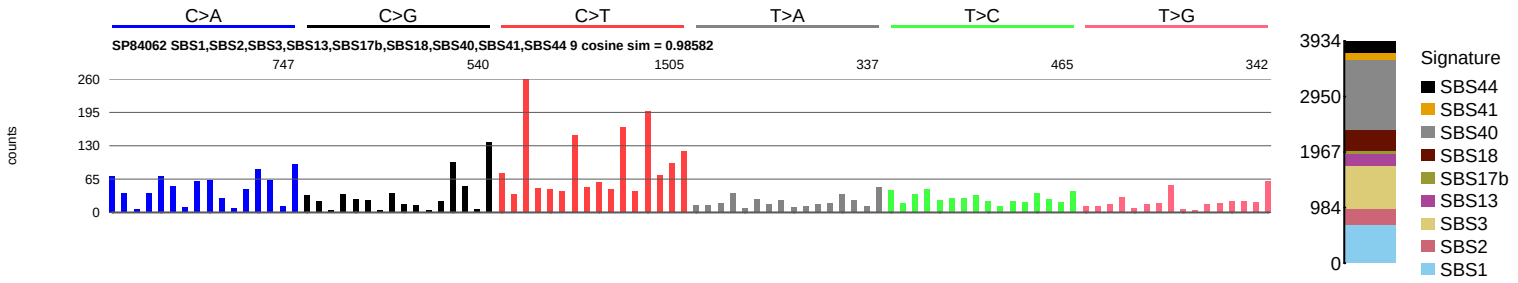

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98588
759
516
1546
291
451
370
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

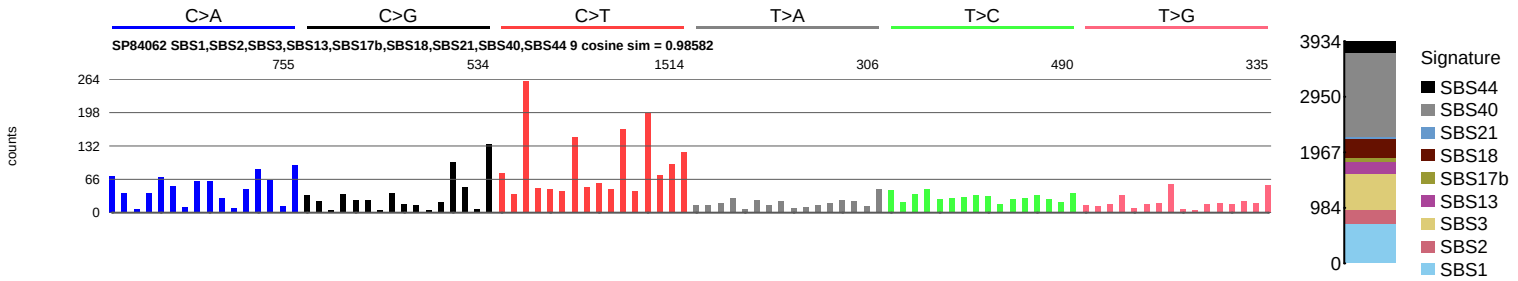

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98586
750
508
1547
312
472
341
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

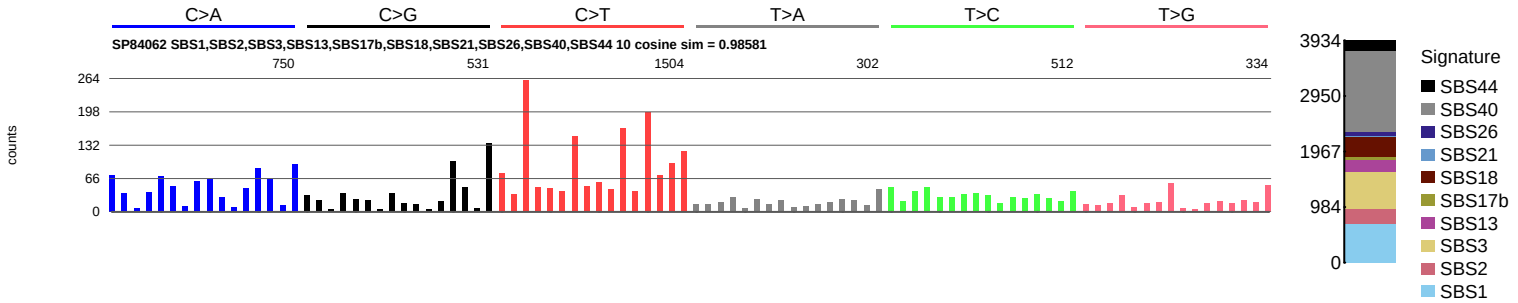

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98586
730
490
1544
289
528
345
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

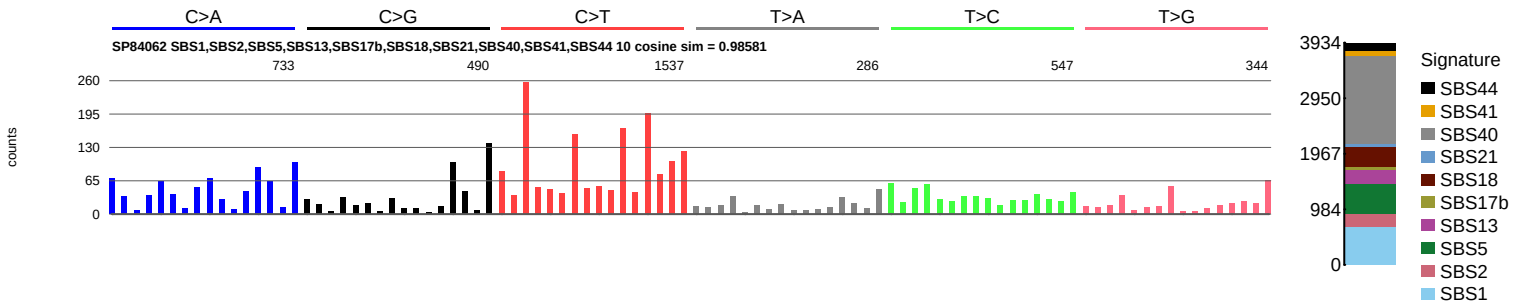

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98584
731
524
1517
323
549
292
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

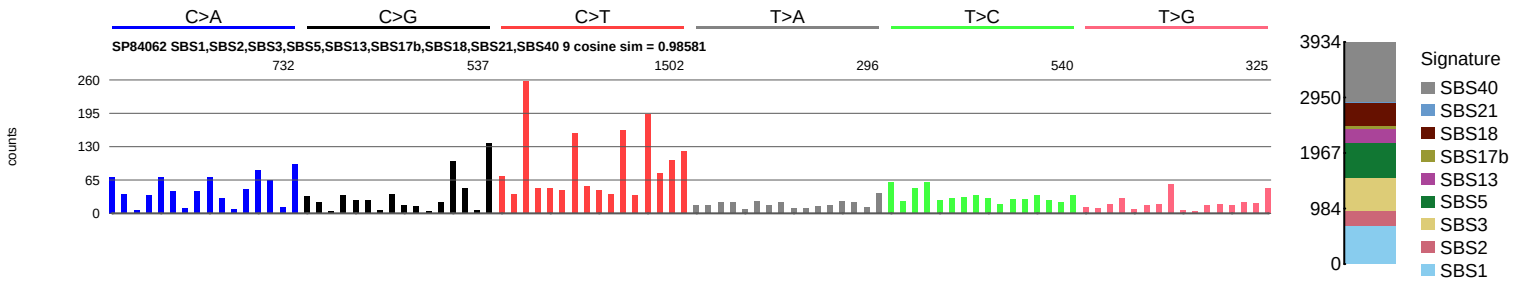

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98582
752
530
1505
303
512
334
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

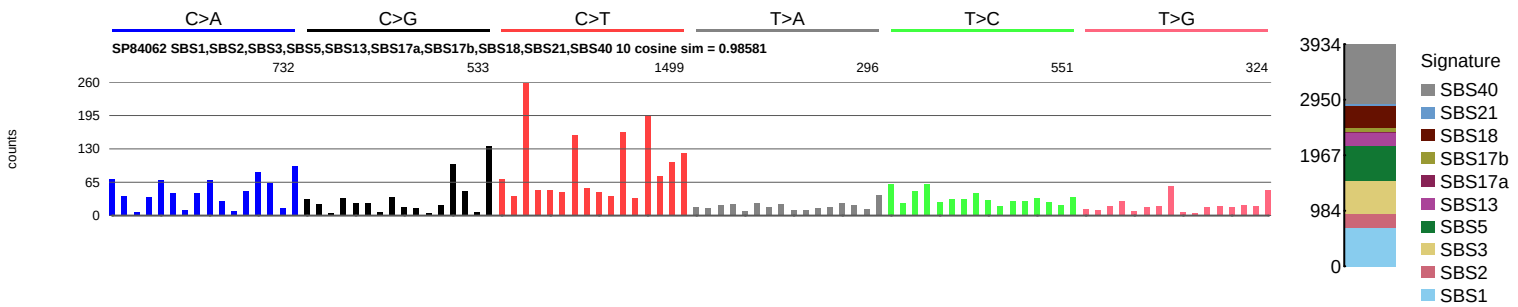

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98582
747
540
1505
337
465
342
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

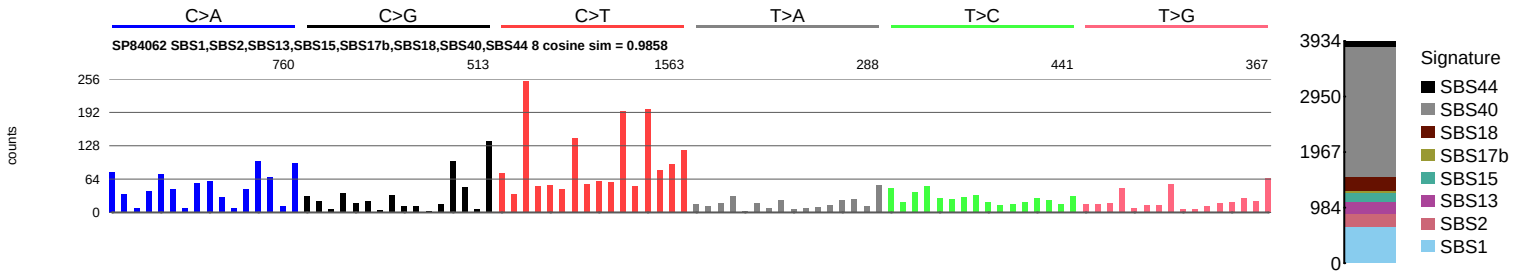

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98582
755
534
1514
306
490
335
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

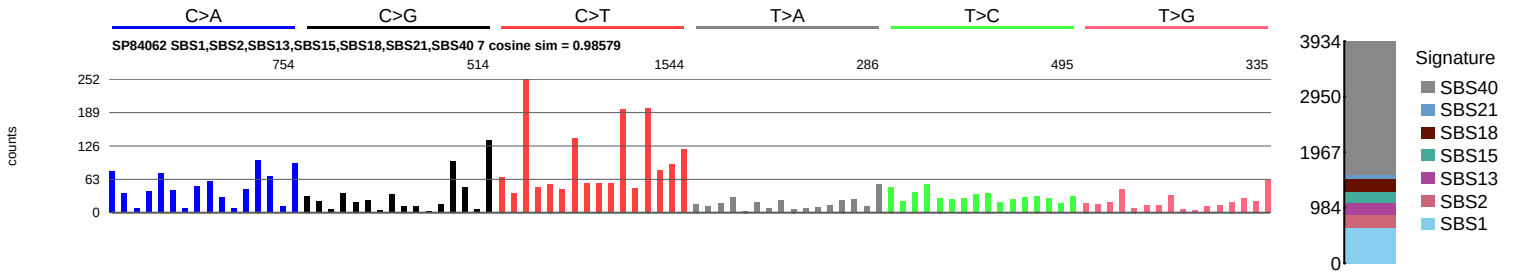

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98581
750
531
1504
302
512
334
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

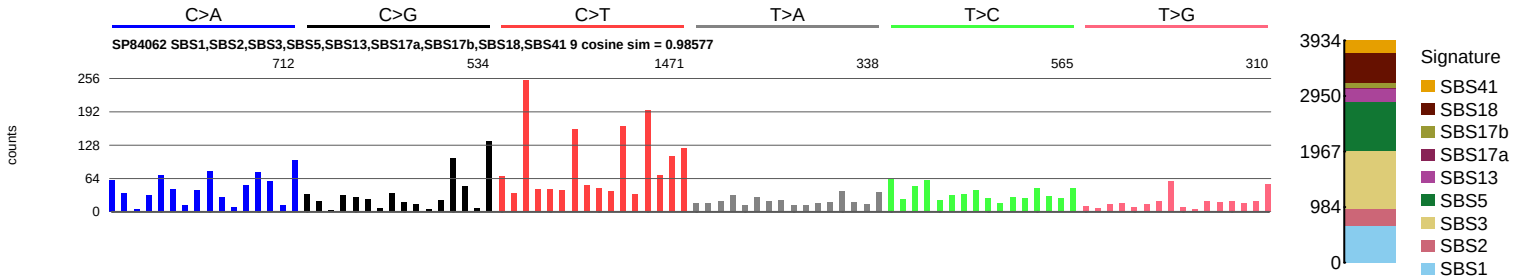

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98581
733
490
1537
286
547
344
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

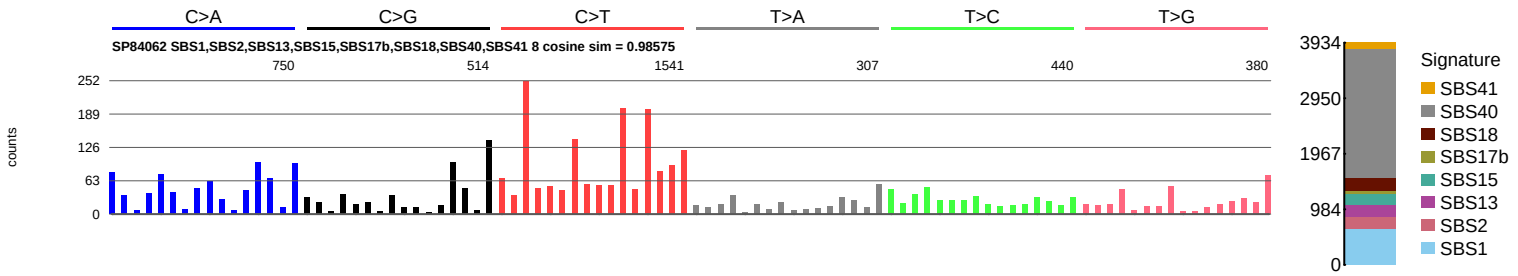

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98581
732
537
1502
296
540
325
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

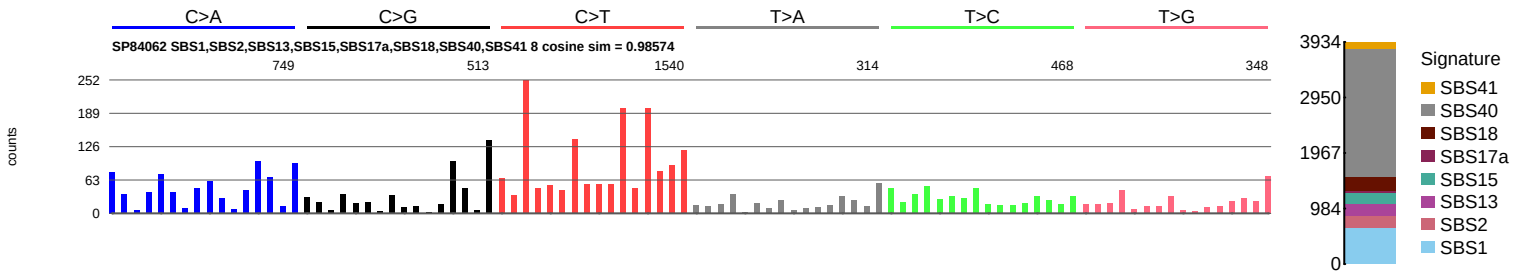

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98581
732
533
1499
296
551
324
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

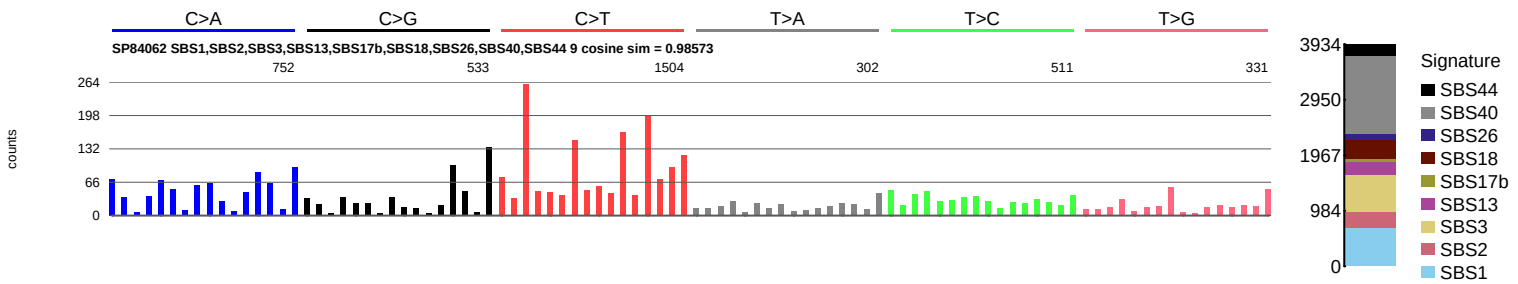

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9858
760
513
1563
288
441
367
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

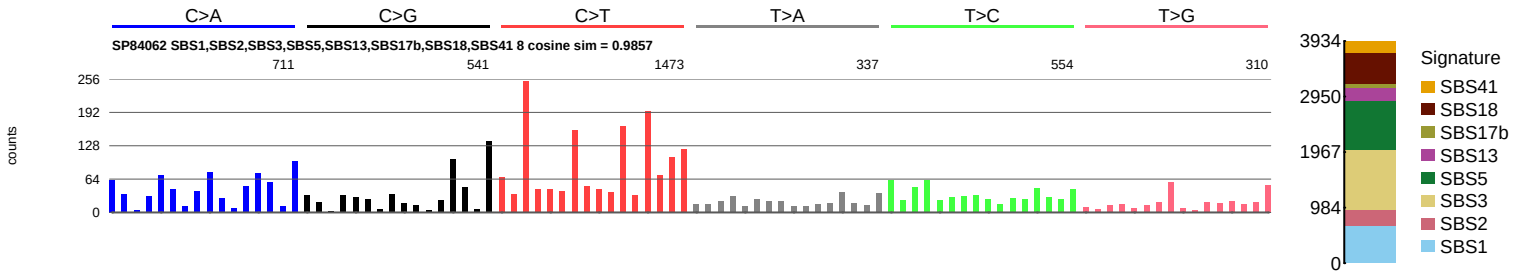

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98579
754
514
1544
286
495
335
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

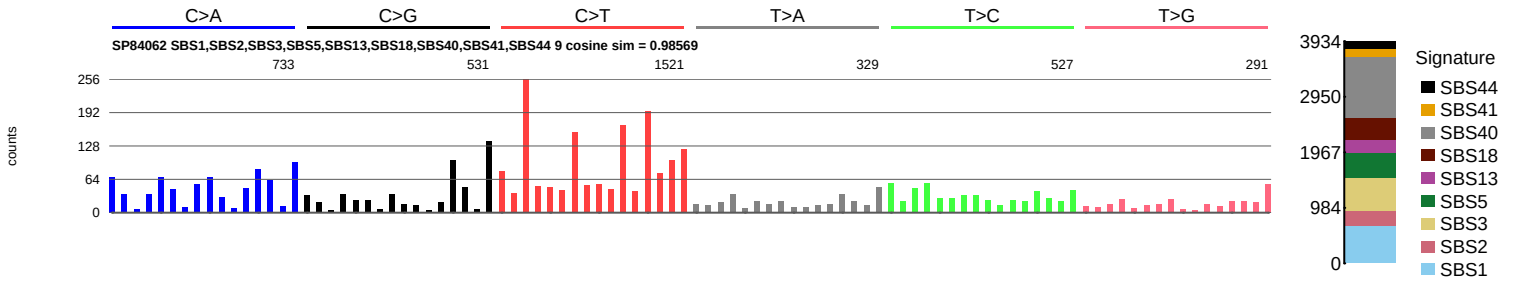

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98577
712
534
1471
338
565
310
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

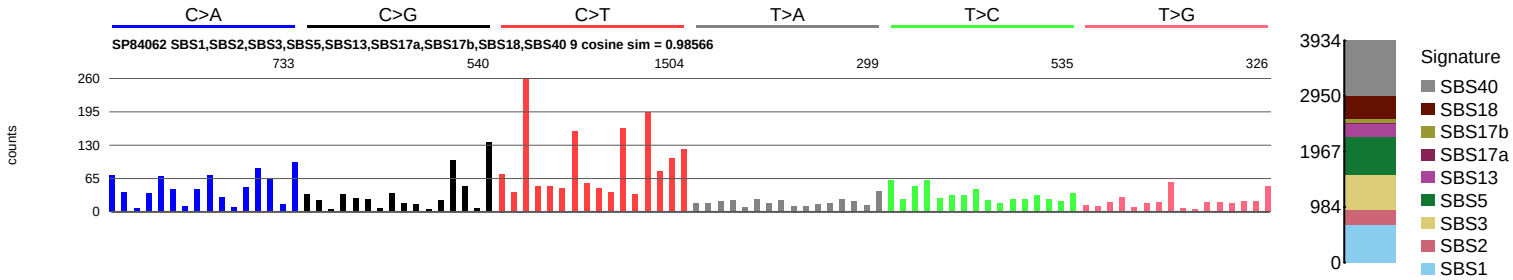

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98575
750
514
1541
307
440
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

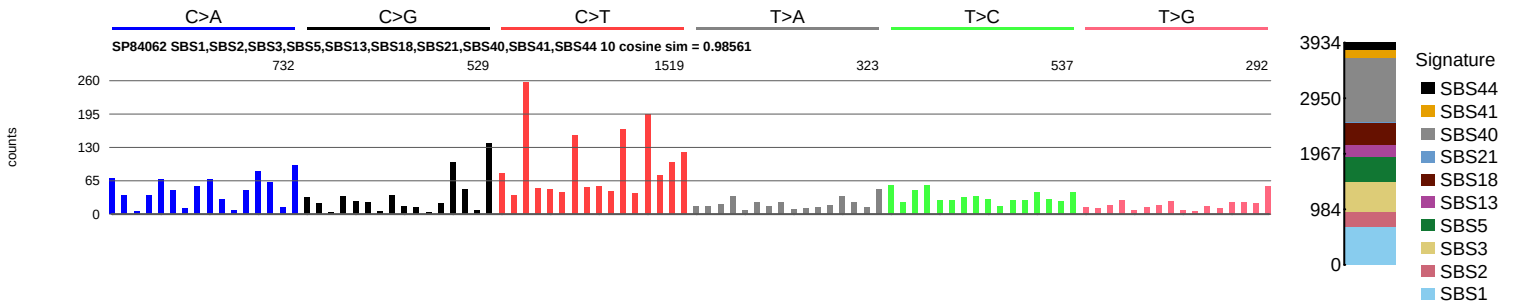

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98574
749
513
1540
314
468
348
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

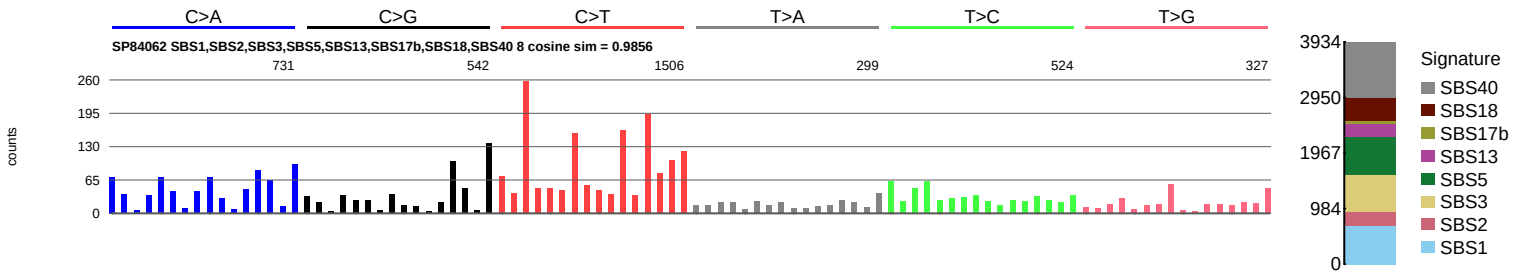

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98573
752
533
1504
302
511
331
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

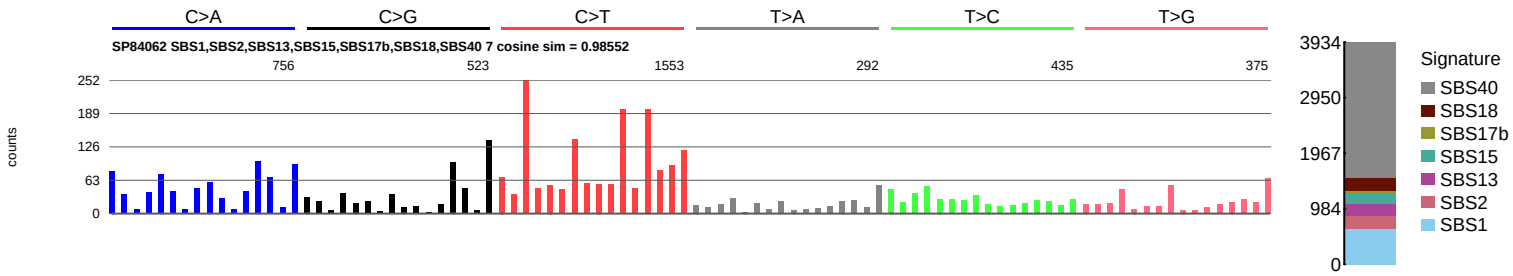

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.9857
711
541
1473
337
554
310
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

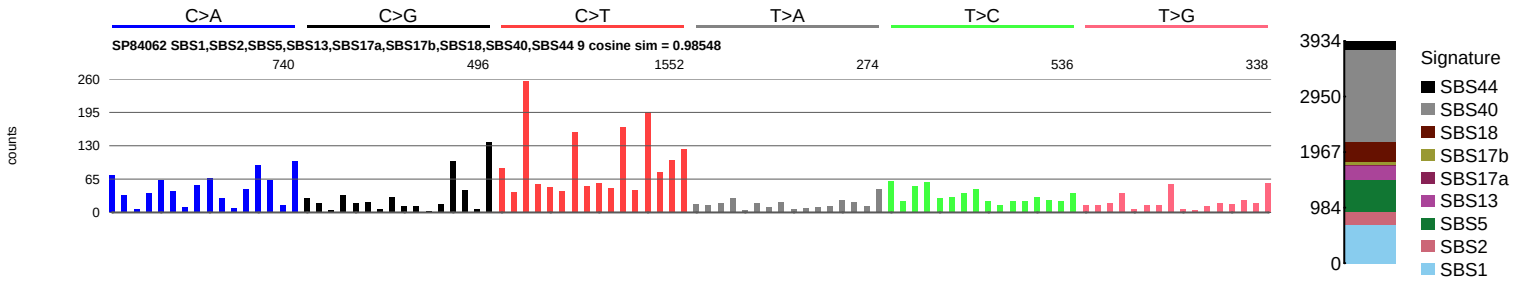

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98569
733
531
1521
329
527
291
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

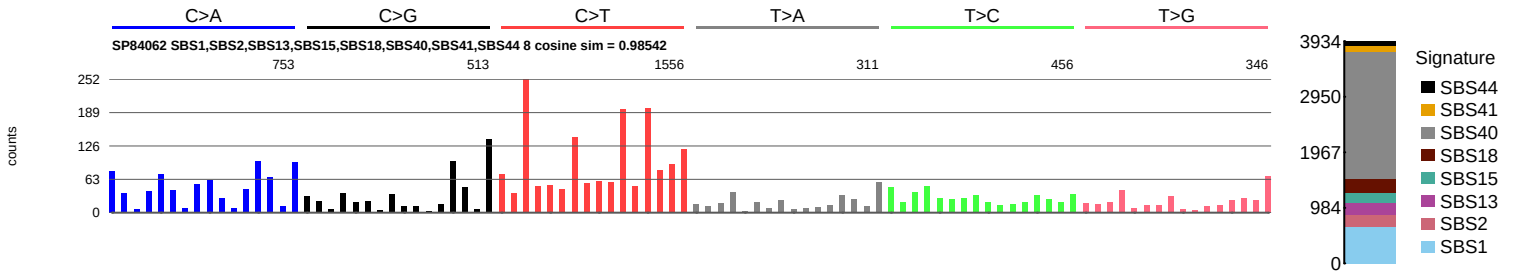

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98566
733
540
1504
299
535
326
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

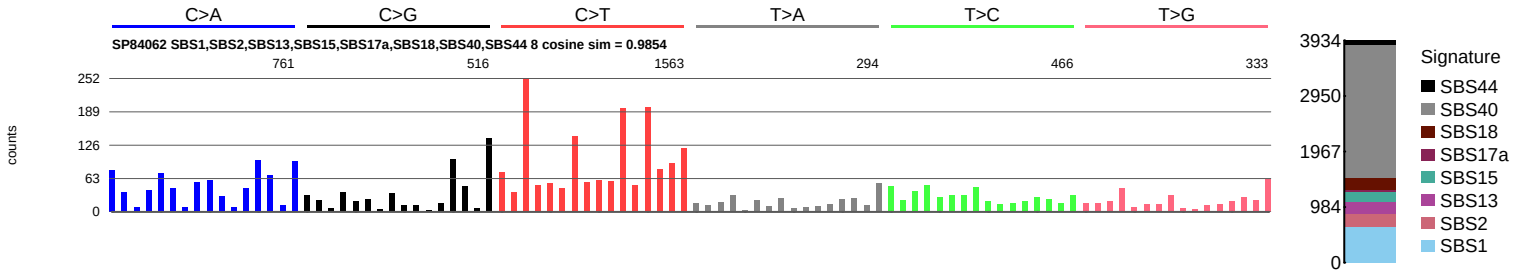

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98561
732
529
1519
323
537
292
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

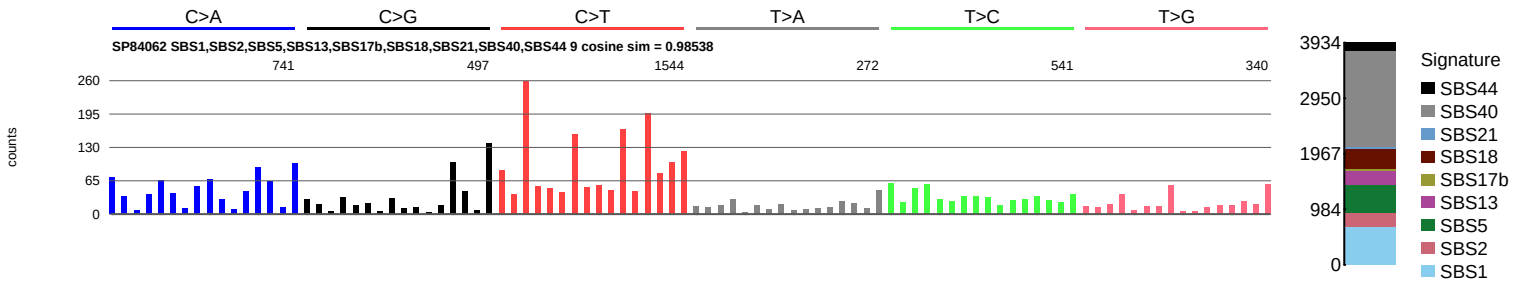

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.9856
731
542
1506
299
524
327
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

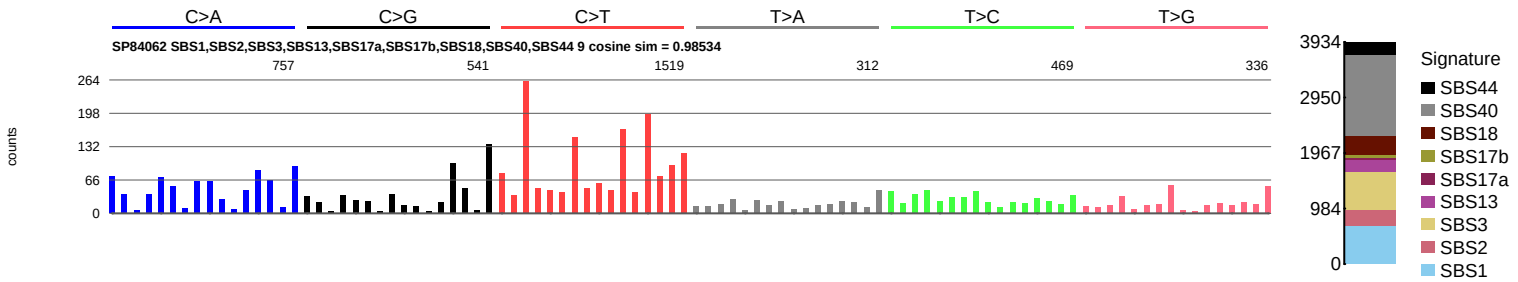

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98552
756
523
1553
292
435
375
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

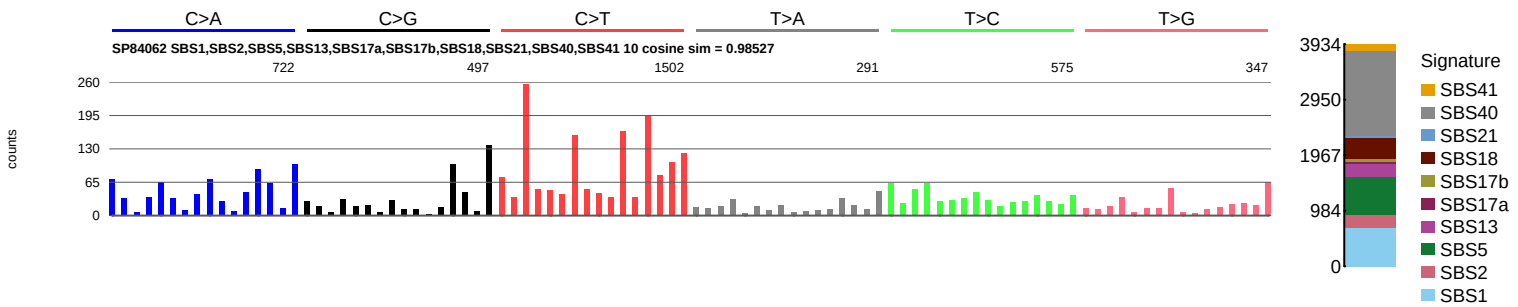

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98548
740
496
1552
274
536
338
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

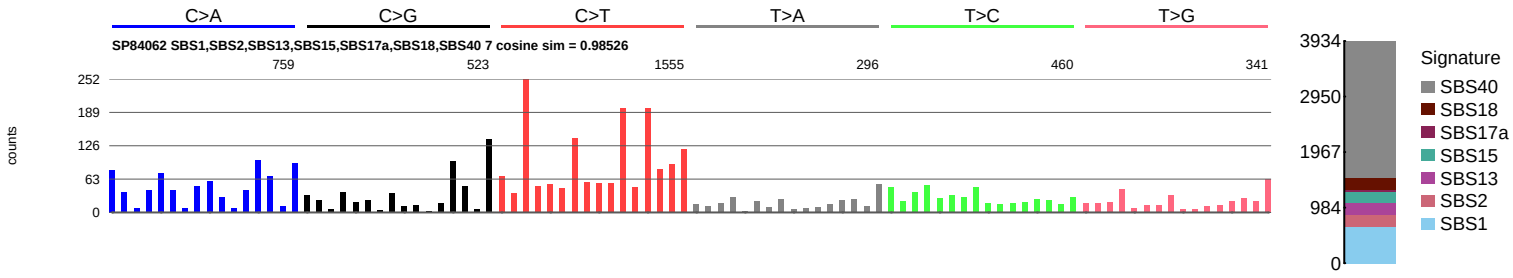

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98542
753
513
1556
311
456
346
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1

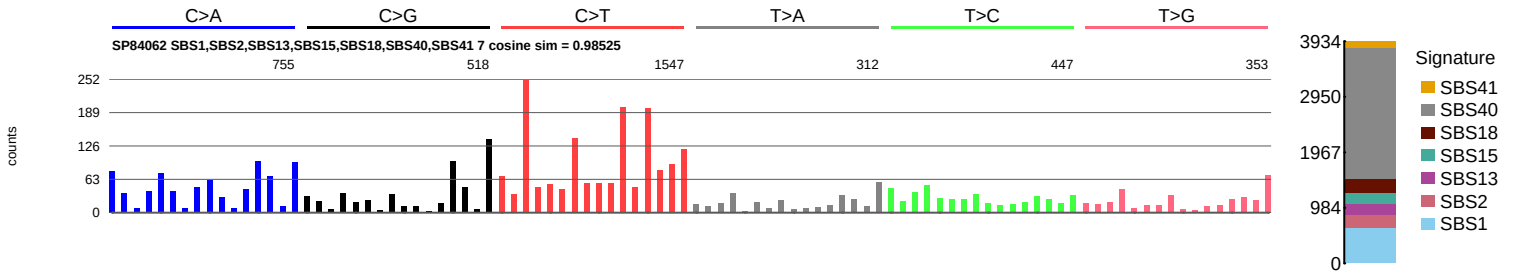

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.9854
761
516
1563
294
466
333
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

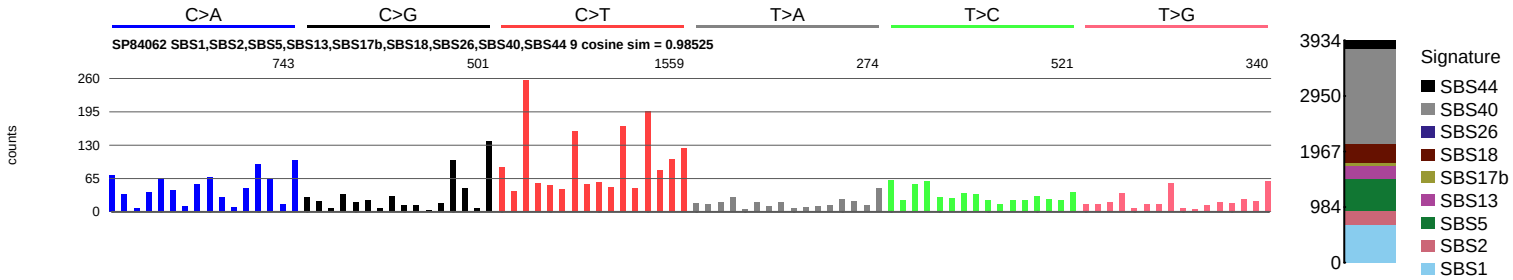

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98538
741
497
1544
272
541
340
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

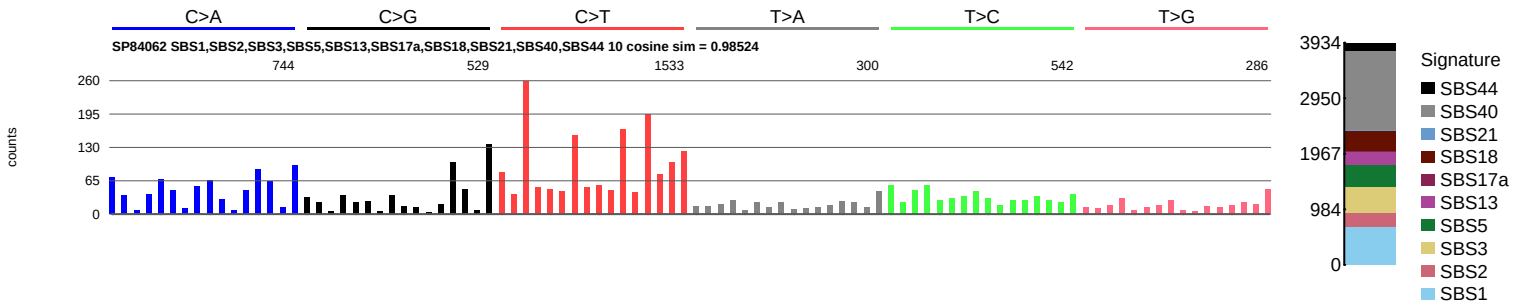

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98534
757
541
1519
312
469
336
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

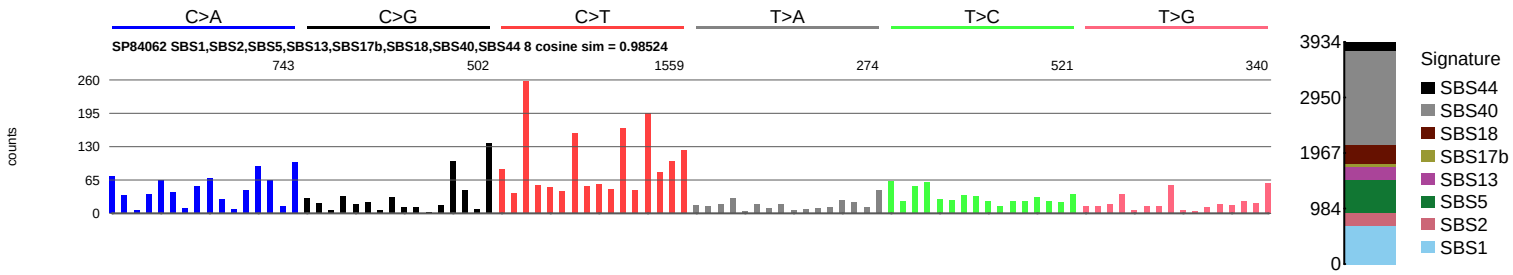

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98527
722
497
1502
291
575
347
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

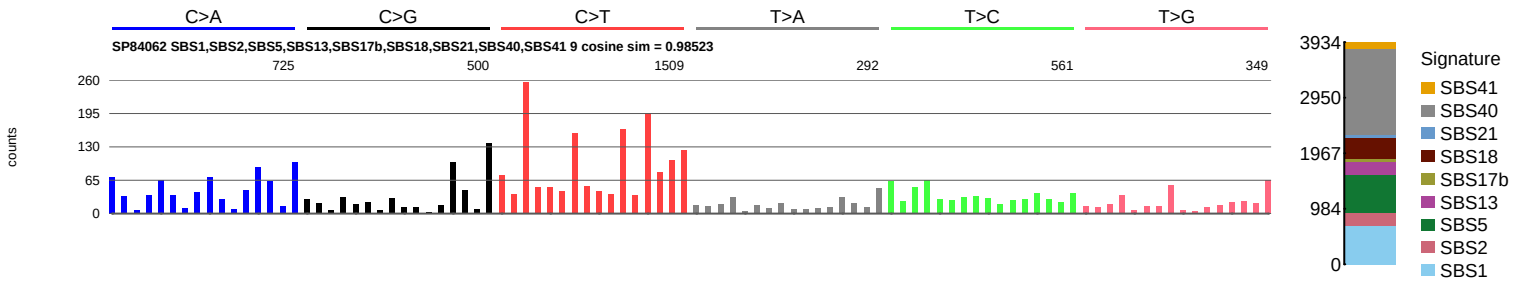

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98526
759
523
1555
296
460
341
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

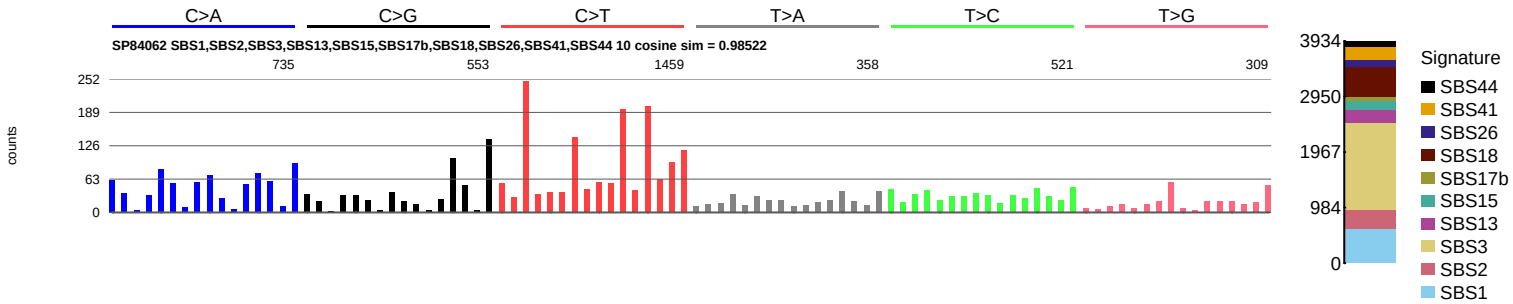

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98525
755
518
1547
312
447
353
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1

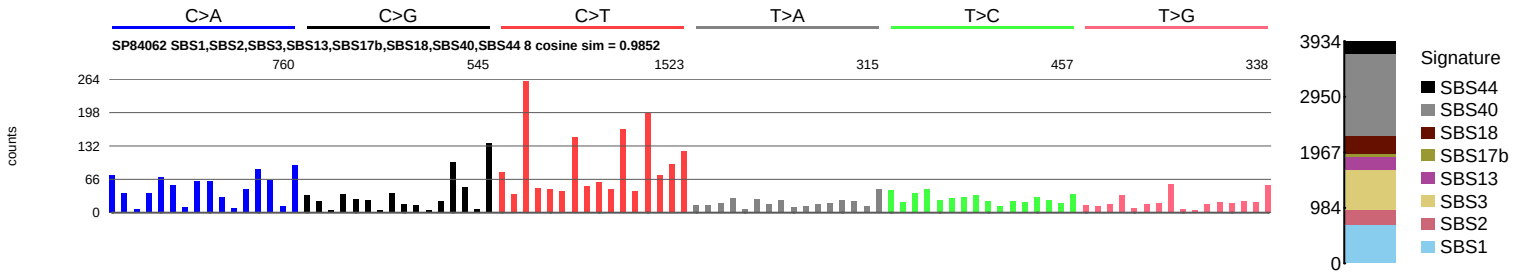

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98525
743
501
1559
274
521
340
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

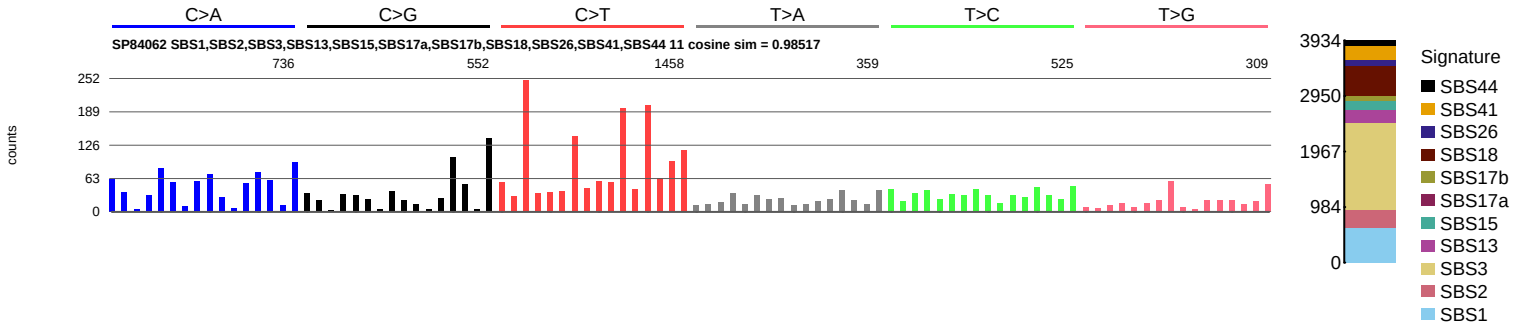

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98524
744
529
1533
300
542
286
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

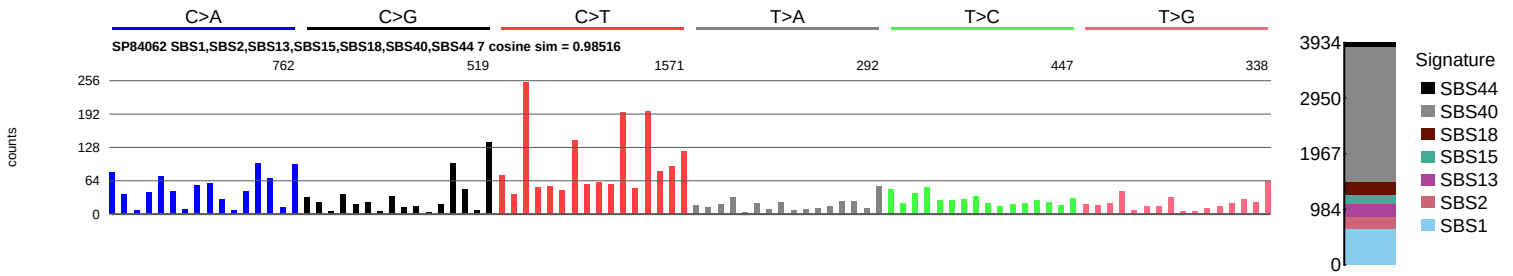

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98524
743
502
1559
274
521
340
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

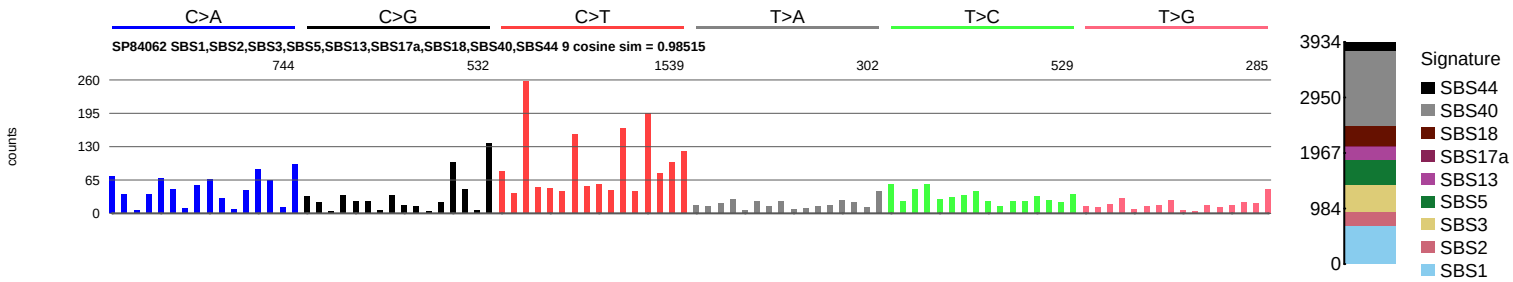

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98523
725
500
1509
292
561
349
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

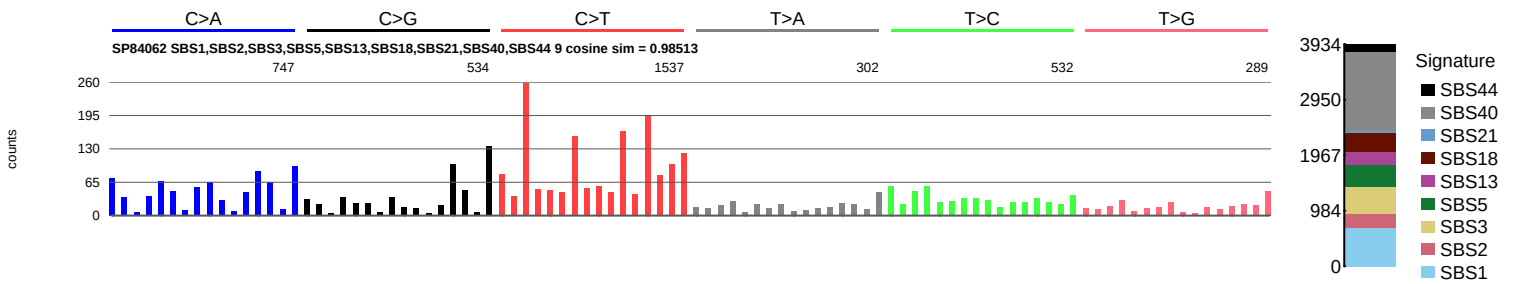

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98522
735
553
1459
358
521
309
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

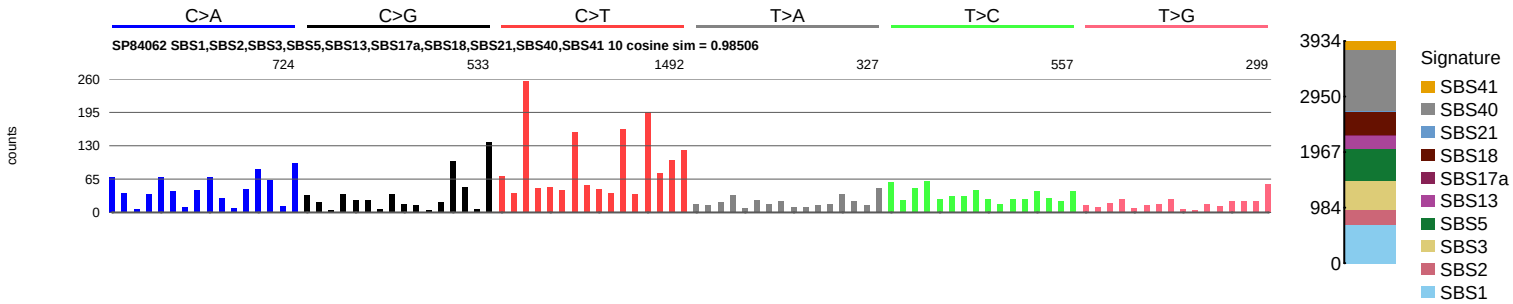

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9852
760
545
1523
315
457
338
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

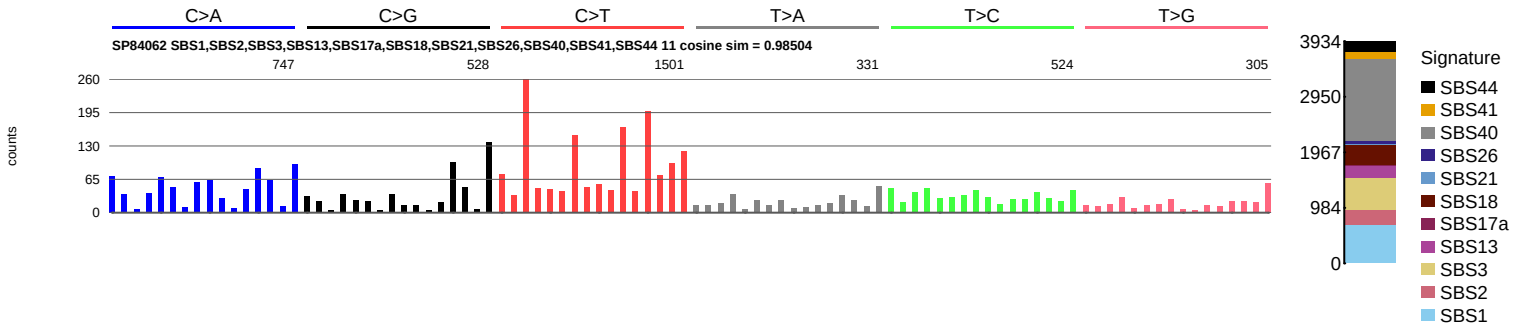

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98517
736
552
1458
359
525
309
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

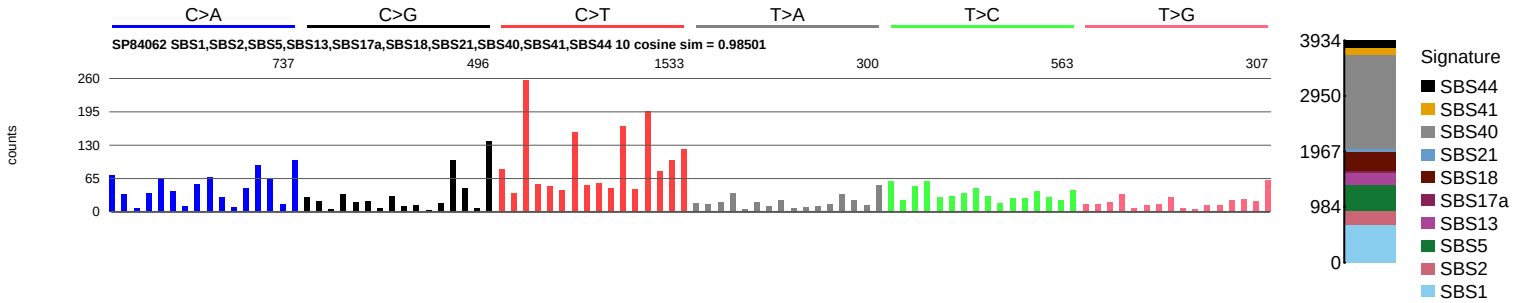

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.98516
762
519
1571
292
447
338
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1

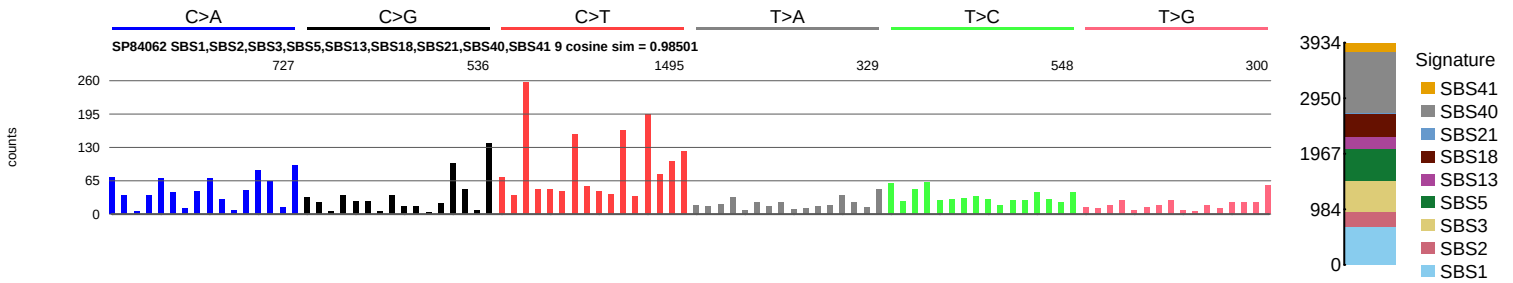

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98515
744
532
1539
302
529
285
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

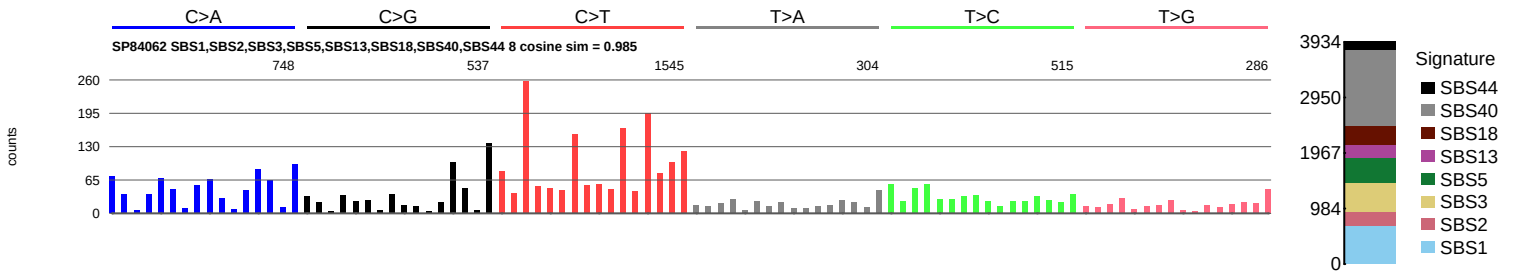

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98513
747
534
1537
302
532
289
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

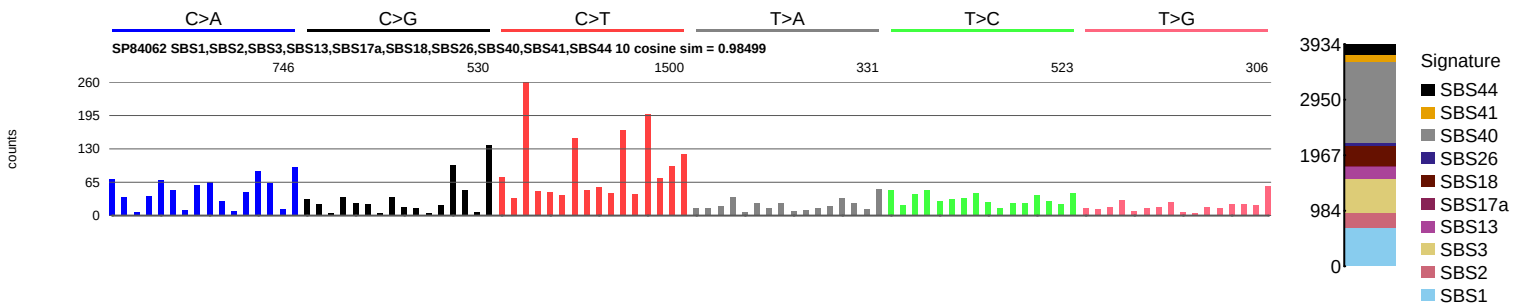

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98506
724
533
1492
327
557
299
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

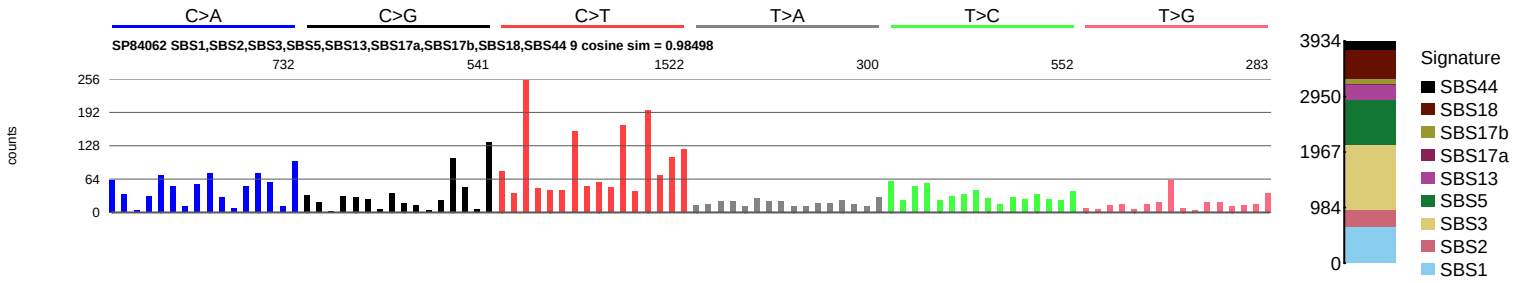

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98504
747
528
1501
331
524
305
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

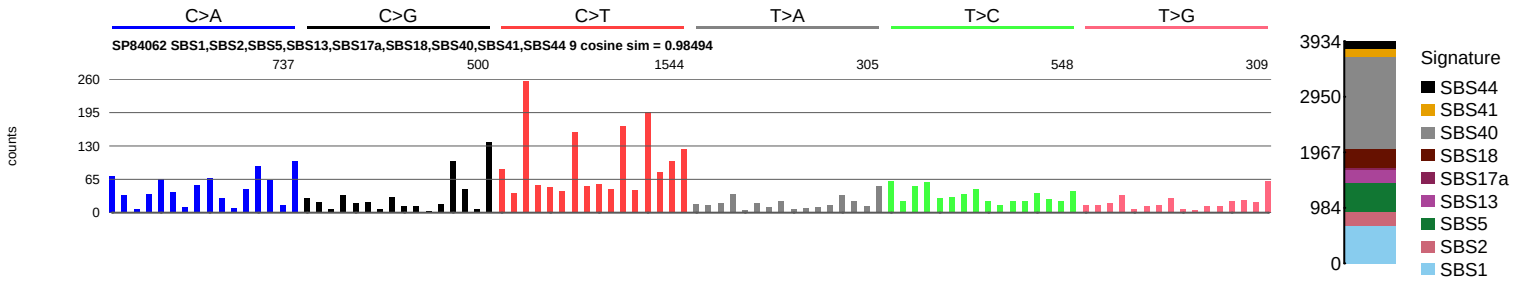

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98501
737
496
1533
300
563
307
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

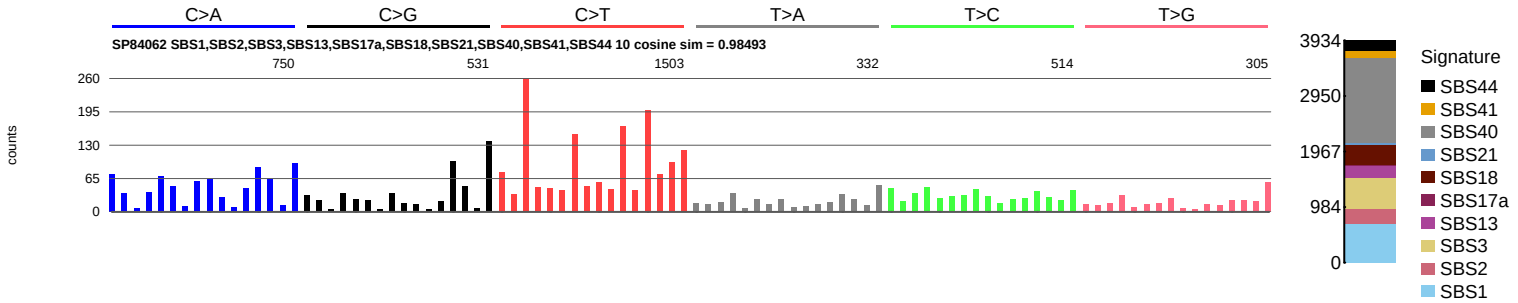

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98501
727
536
1495
329
548
300
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

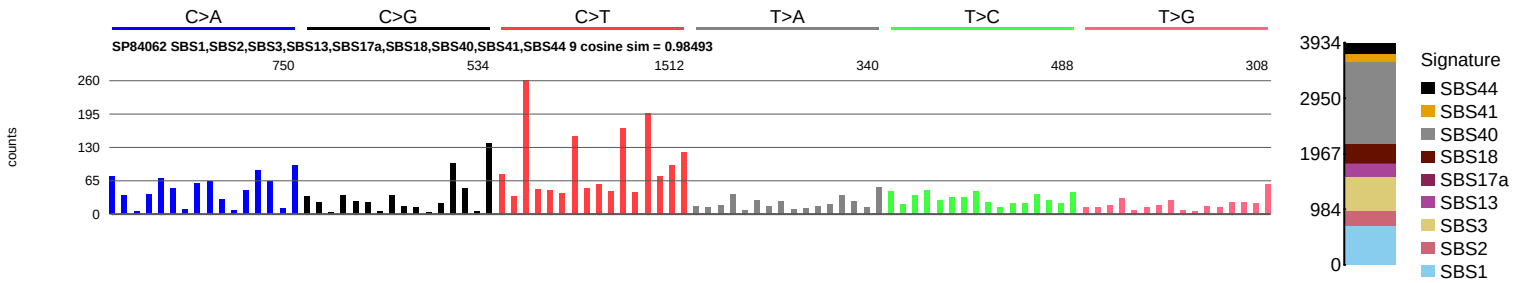

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS44 8 cosine sim = 0.985
748
537
1545
304
515
286
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

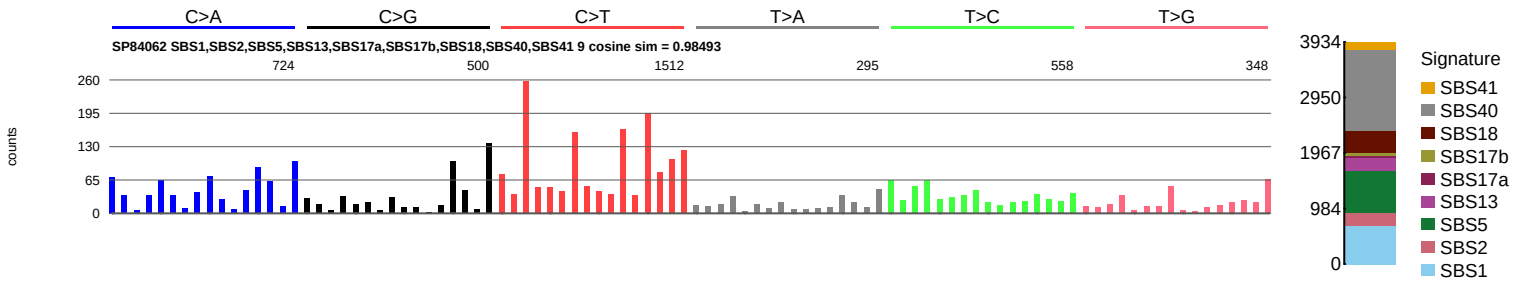

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98499
746
530
1500
331
523
306
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

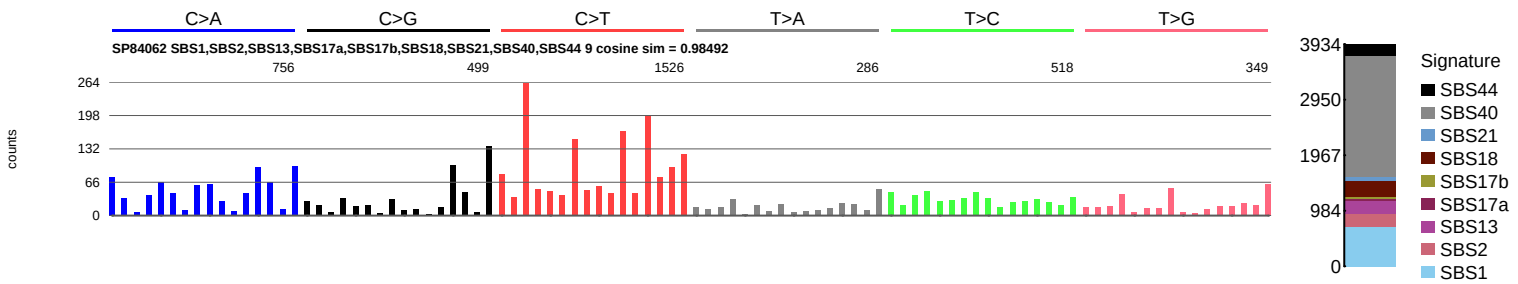

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98498
732
541
1522
300
552
283
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

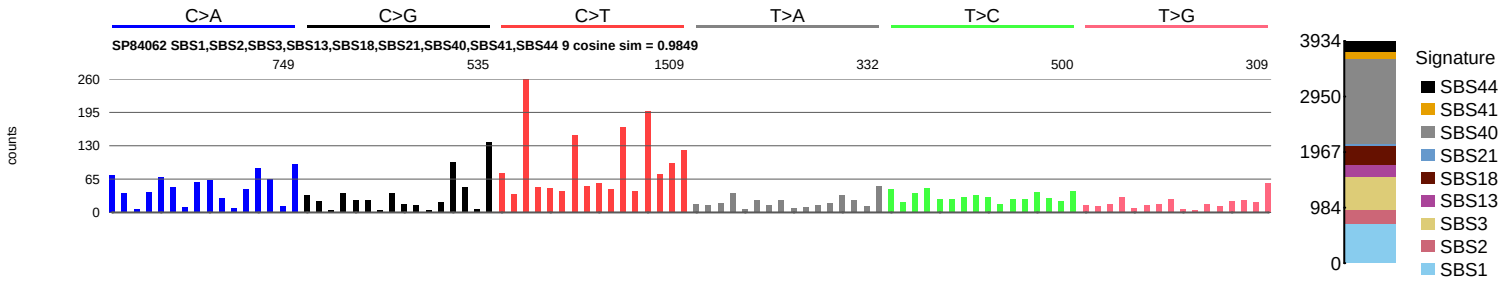

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98494
737
500
1544
305
548
309
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

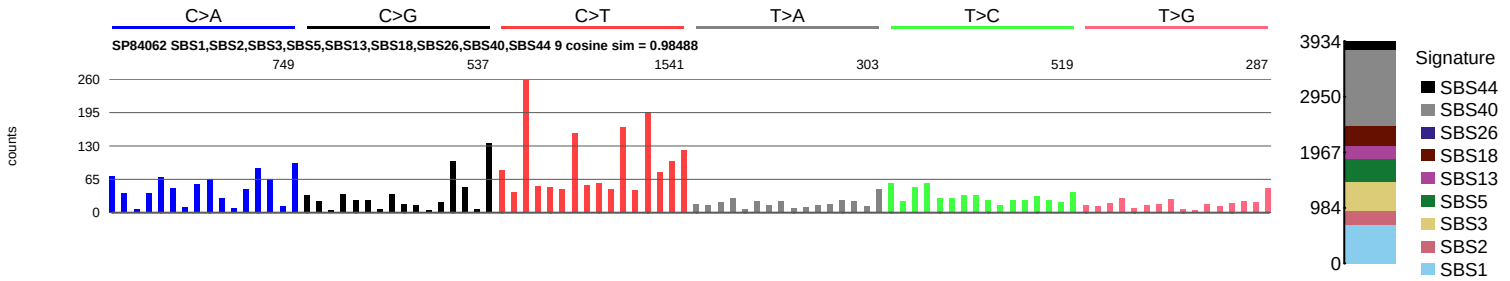

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98493
750
531
1503
332
514
305
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

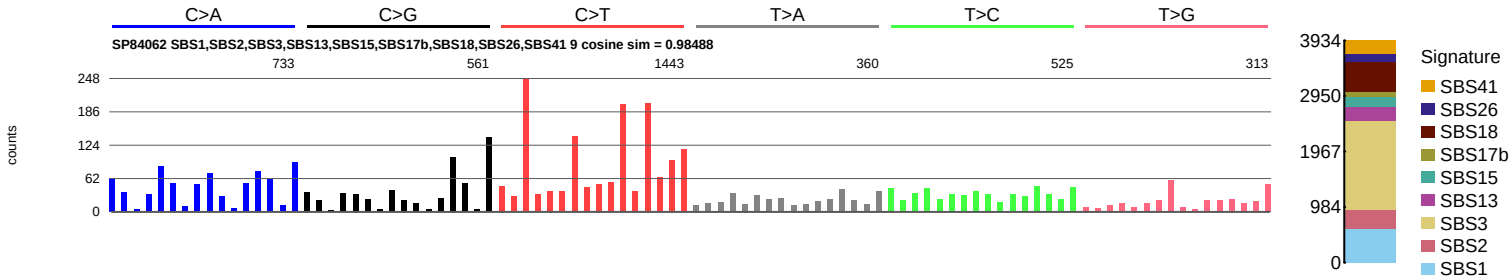

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98493
750
534
1512
340
488
308
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

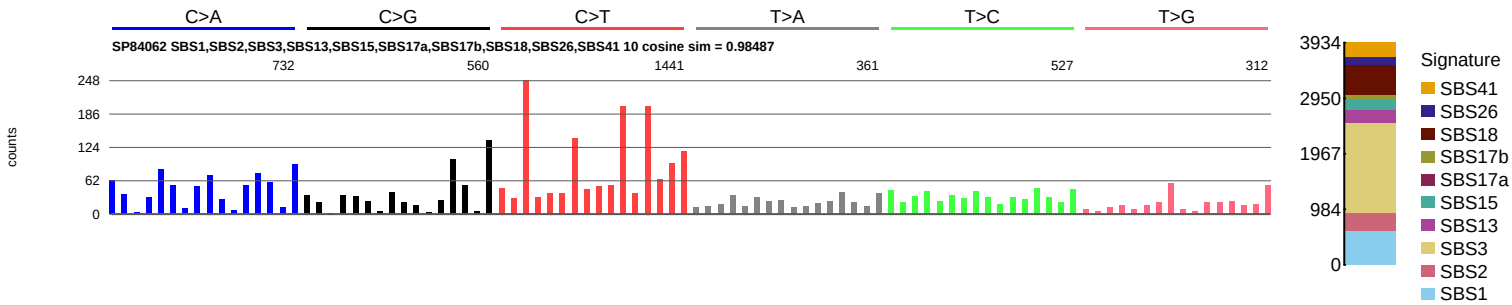

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98493
724
500
1512
295
558
348
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

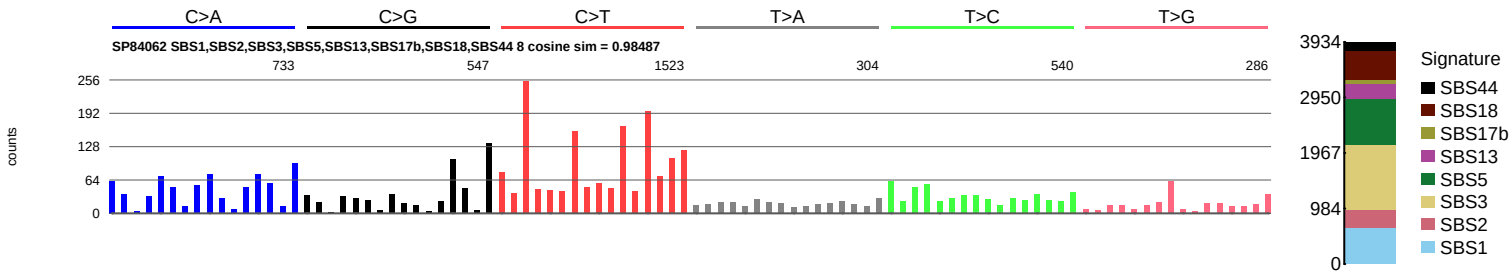

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98492
756
499
1526
286
518
349
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

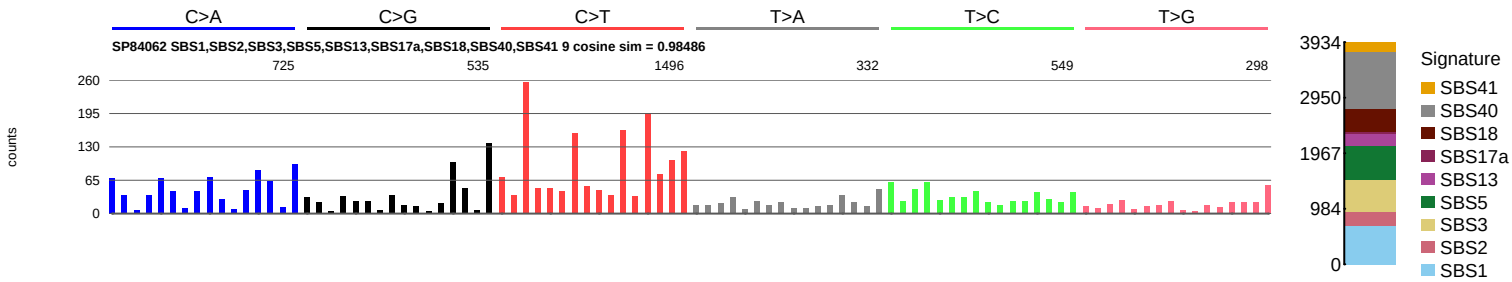

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9849
749
535
1509
332
500
309
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

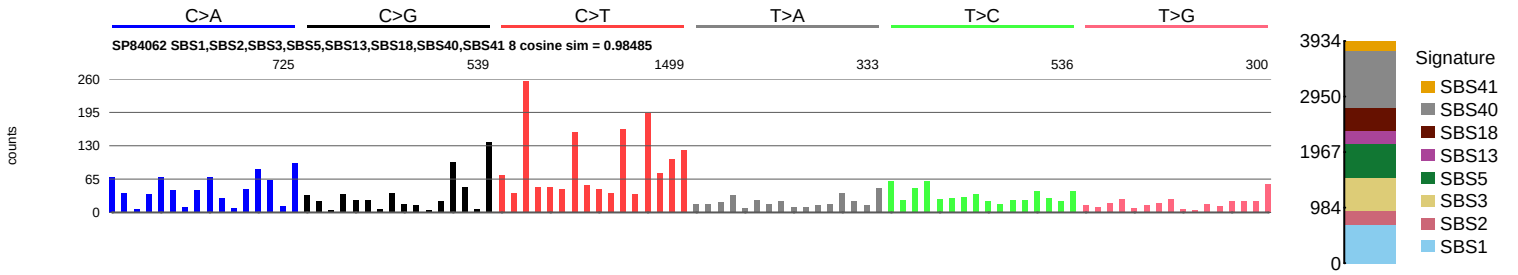

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98488
749
537
1541
303
519
287
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

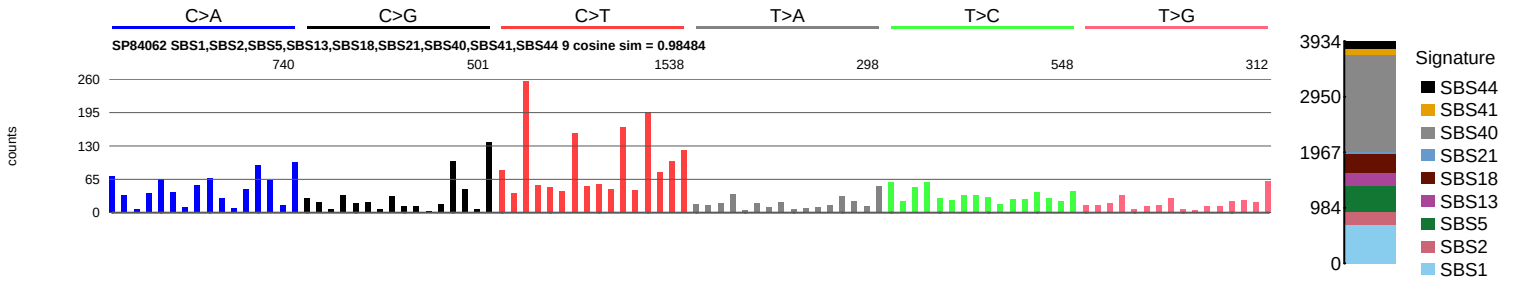

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98488
733
561
1443
360
525
313
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

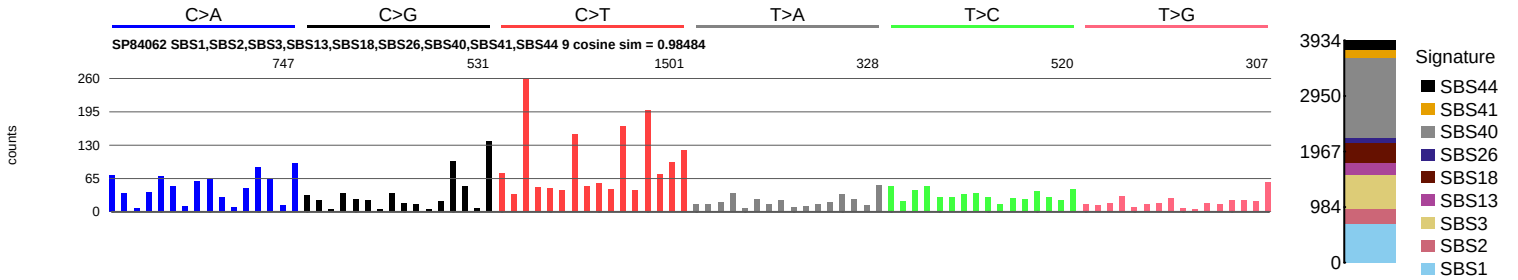

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98487
732
560
1441
361
527
312
counts
248
186
124
62
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

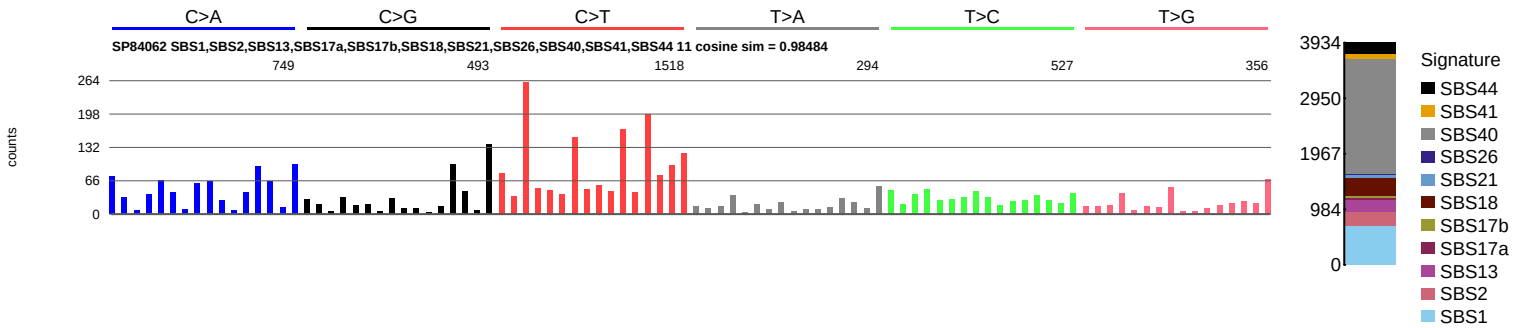

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98487
733
547
1523
304
540
286
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

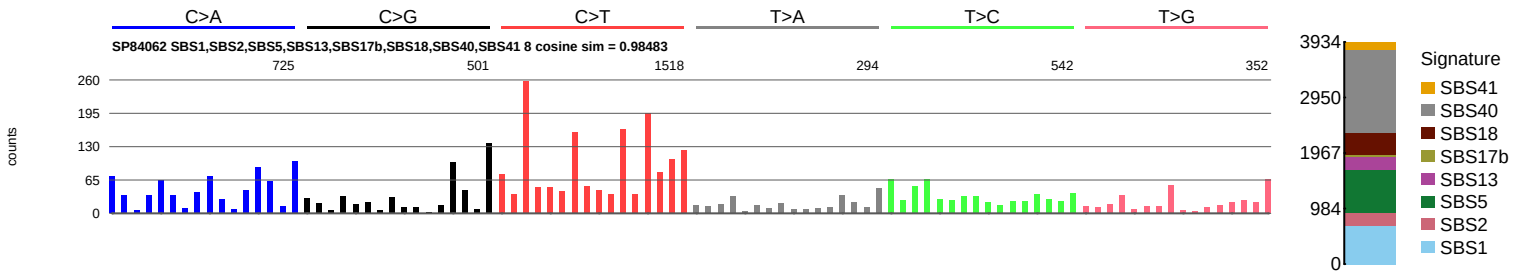

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98486
725
535
1496
332
549
298
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

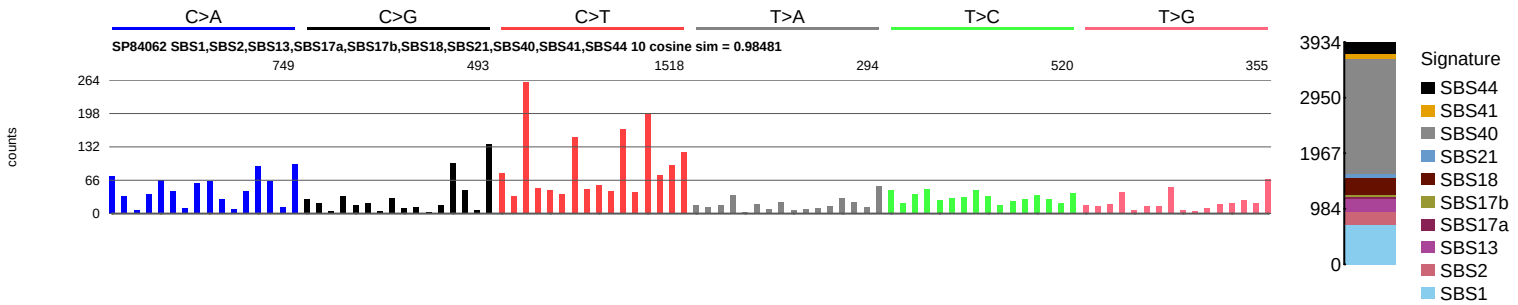

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41 8 cosine sim = 0.98485
725
539
1499
333
536
300
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

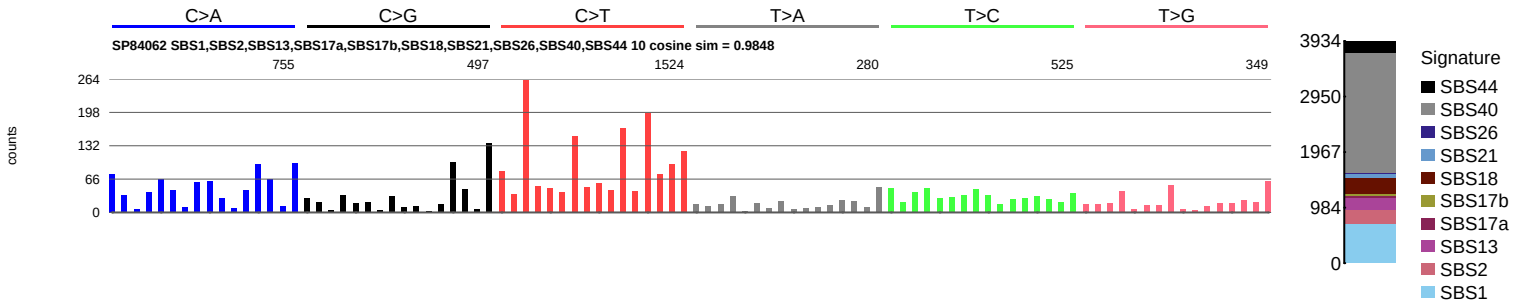

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98484
740
501
1538
298
548
312
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

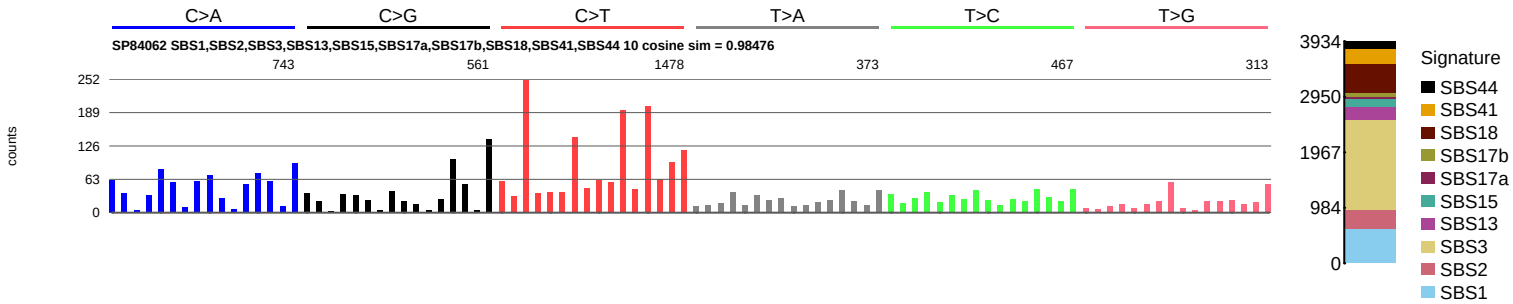

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98484
747
531
1501
328
520
307
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

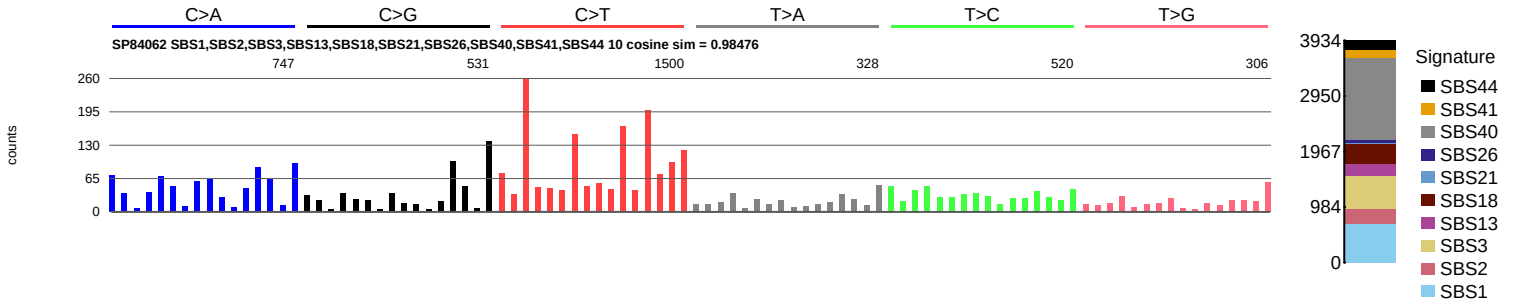

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98484
749
493
1518
294
527
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

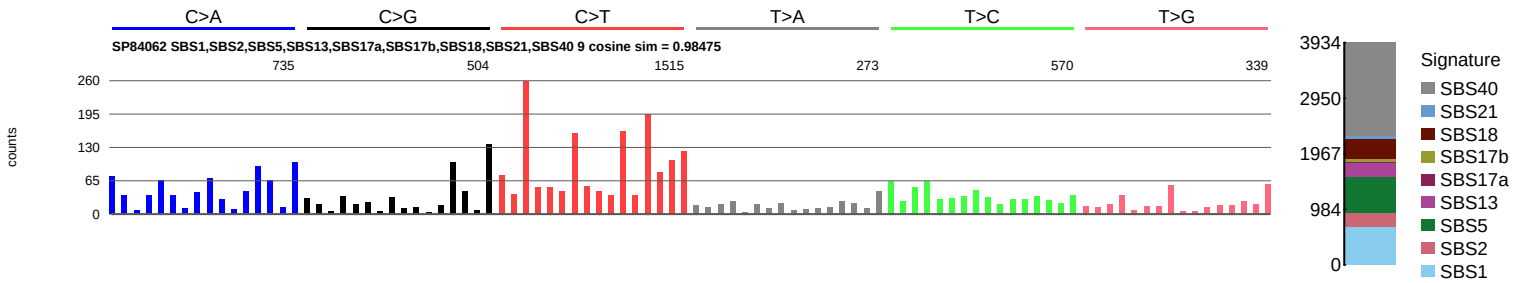

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98483
725
501
1518
294
542
352
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

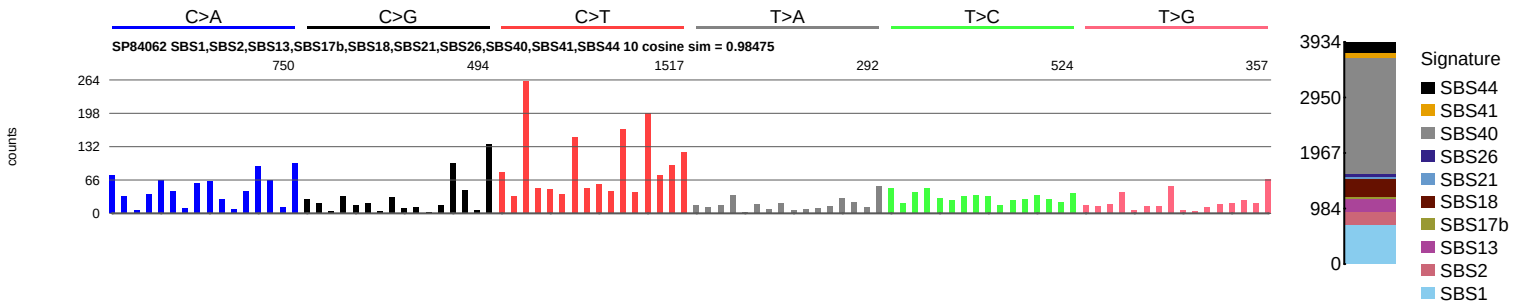

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98481
749
493
1518
294
520
355
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

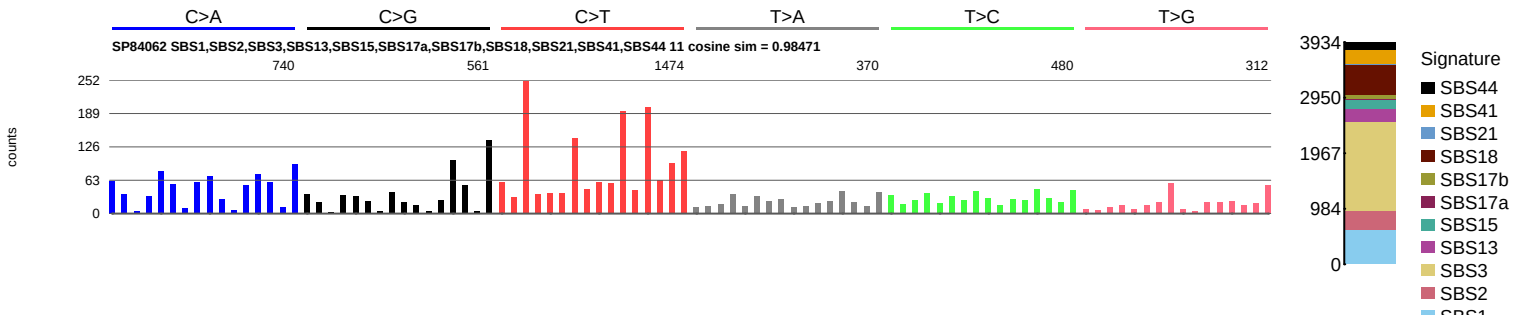

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9848
755
497
1524
280
525
349
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

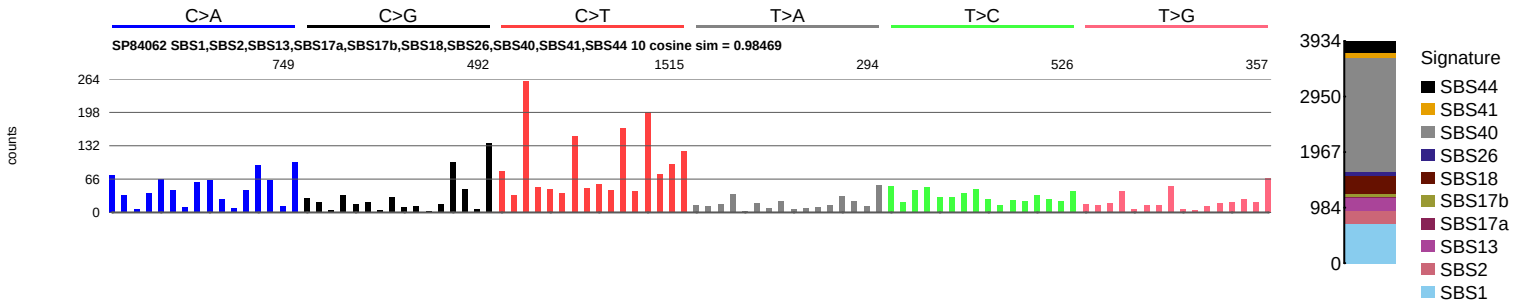

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98476
743
561
1478
373
467
313
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

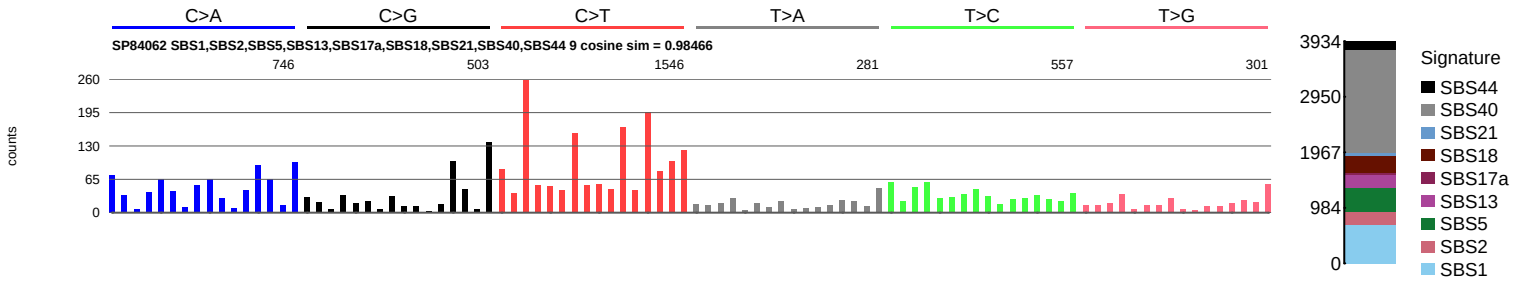

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98476
747
531
1500
328
520
306
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

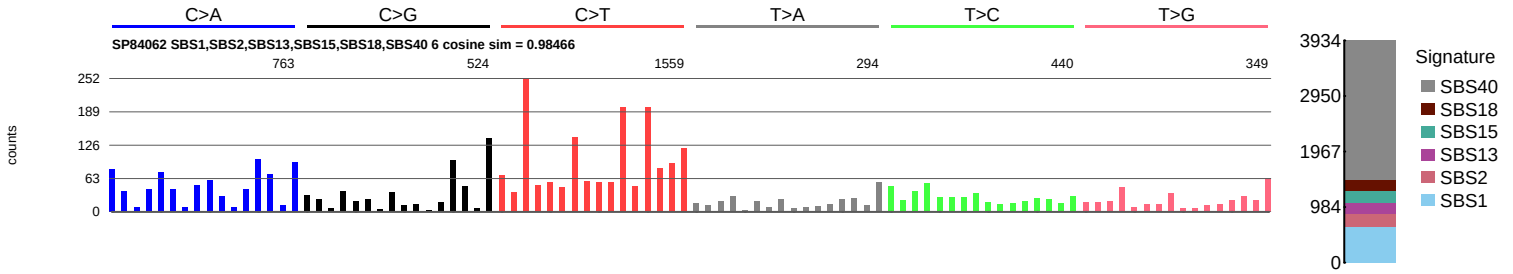

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98475
735
504
1515
273
570
339
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

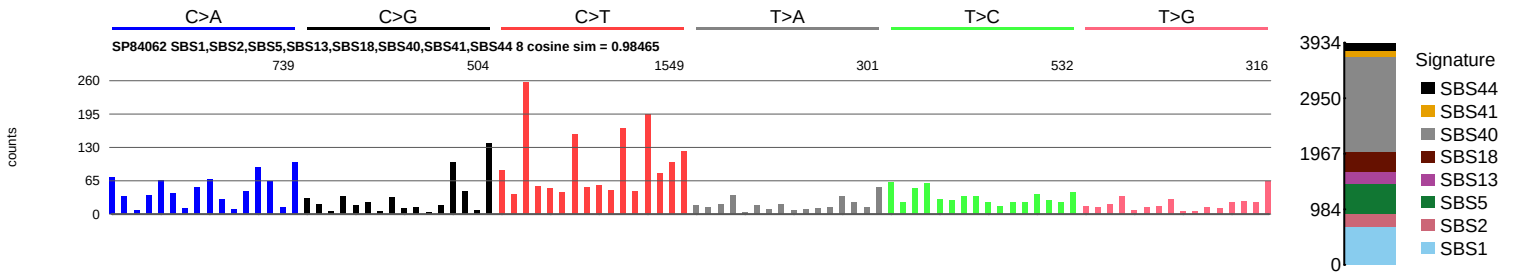

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98475
750
494
1517
292
524
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

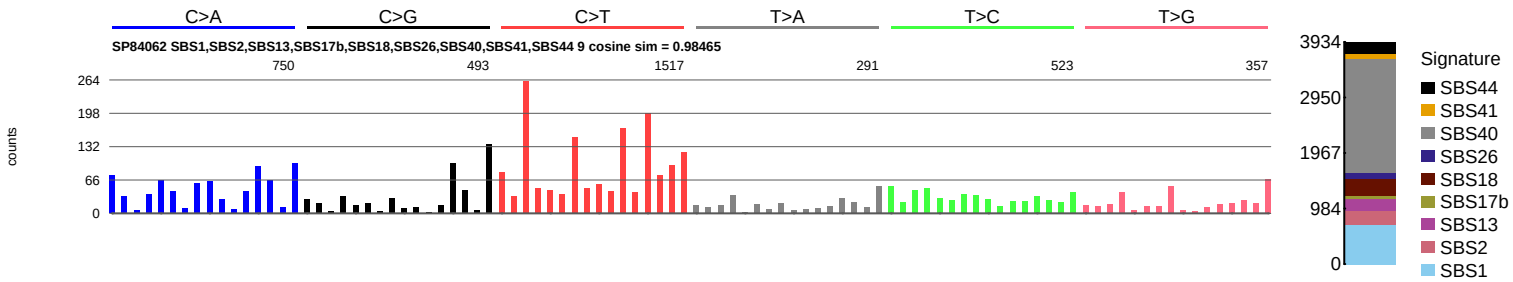

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98471
740
561
1474
370
480
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

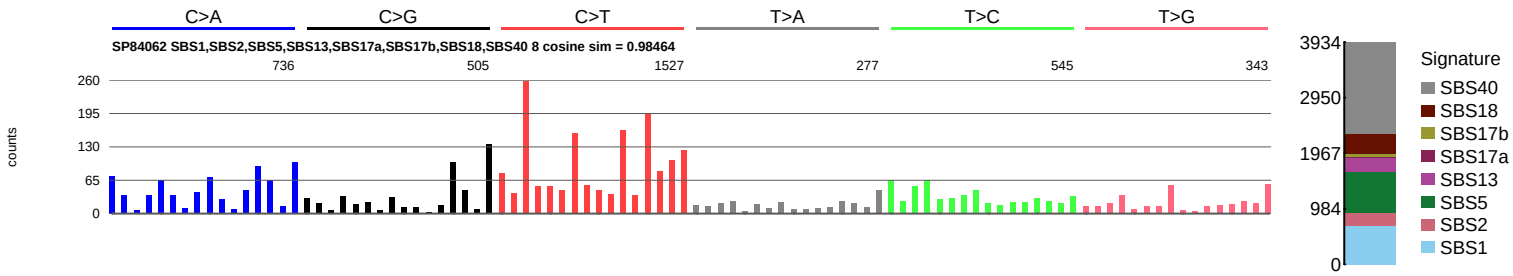

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98469
749
492
1515
294
526
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

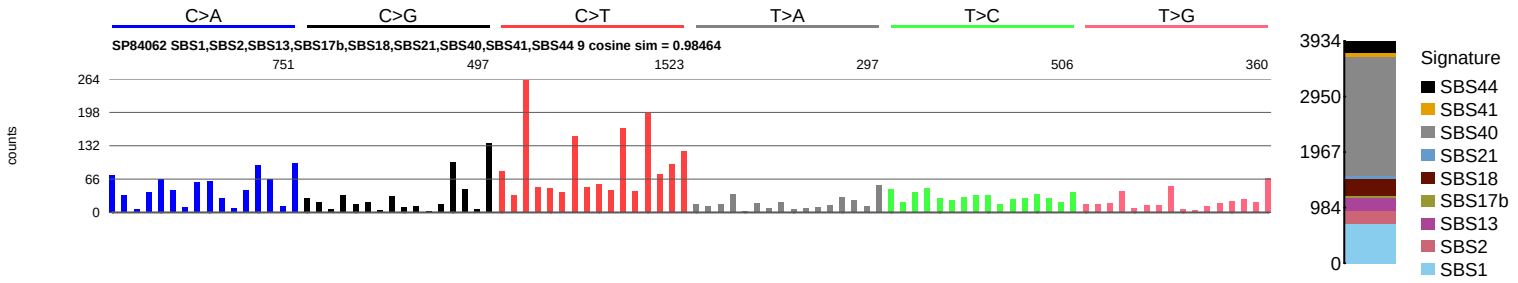

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98466
746
503
1546
281
557
301
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

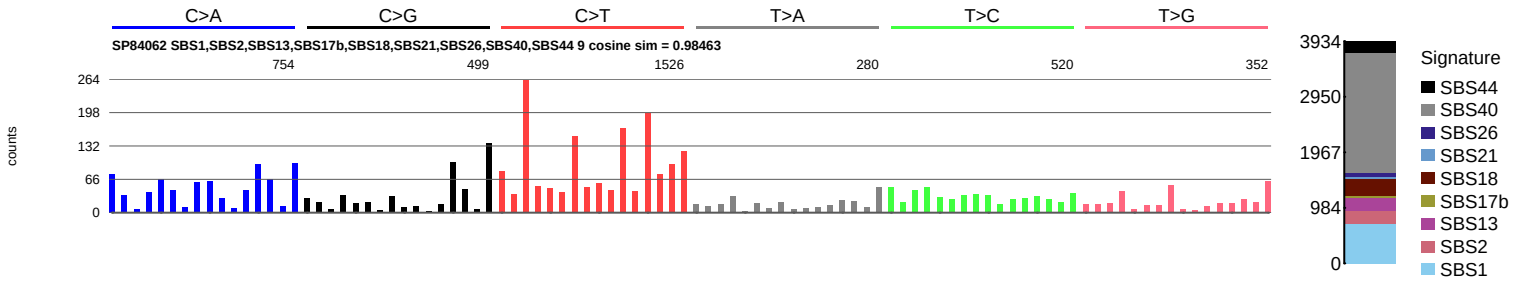

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40 6 cosine sim = 0.98466
763
524
1559
294
440
349
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1

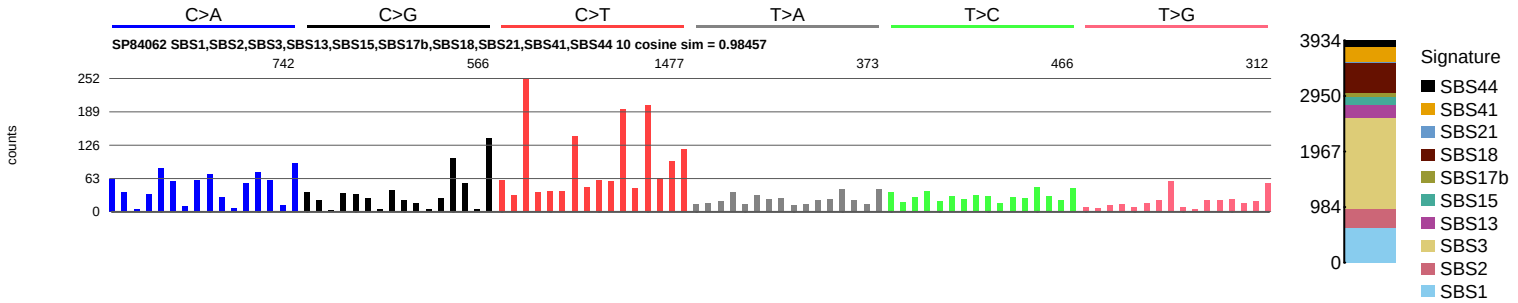

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98465
739
504
1549
301
532
316
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

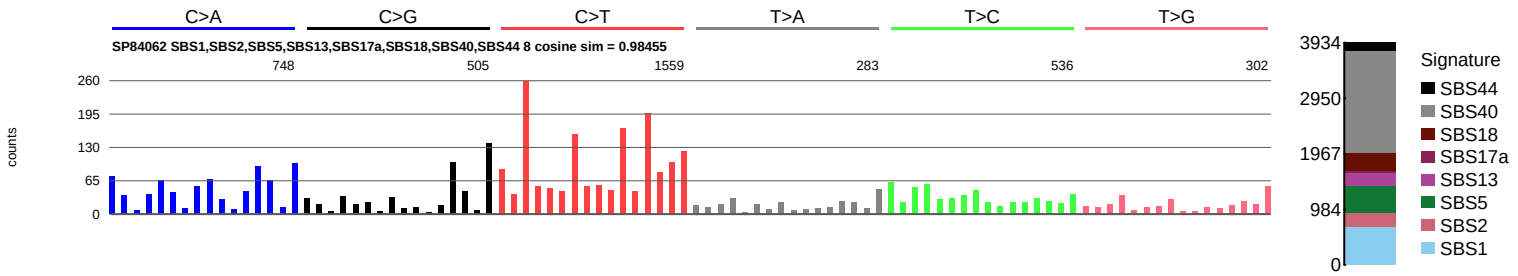

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98465
750
493
1517
291
523
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

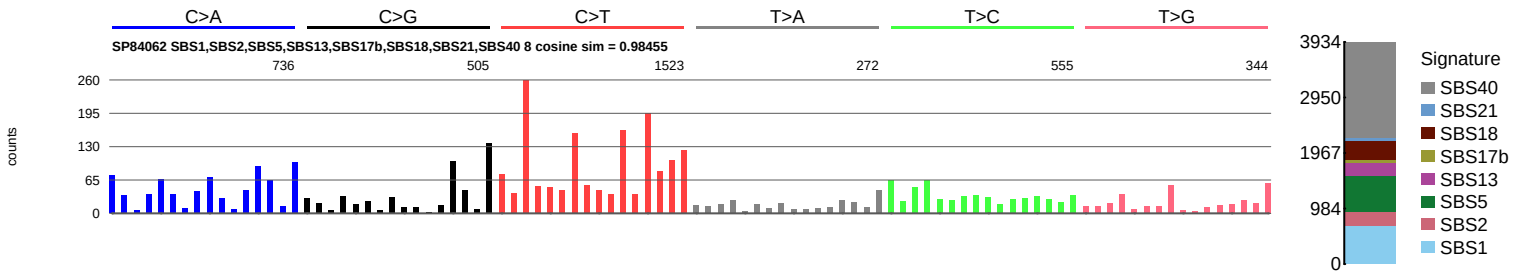

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98464
736
505
1527
277
545
343
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

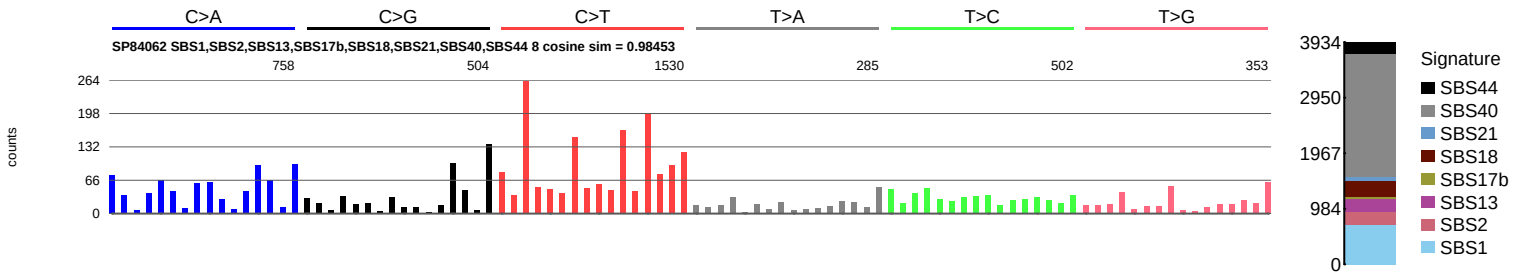

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98464
751
497
1523
297
506
360
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

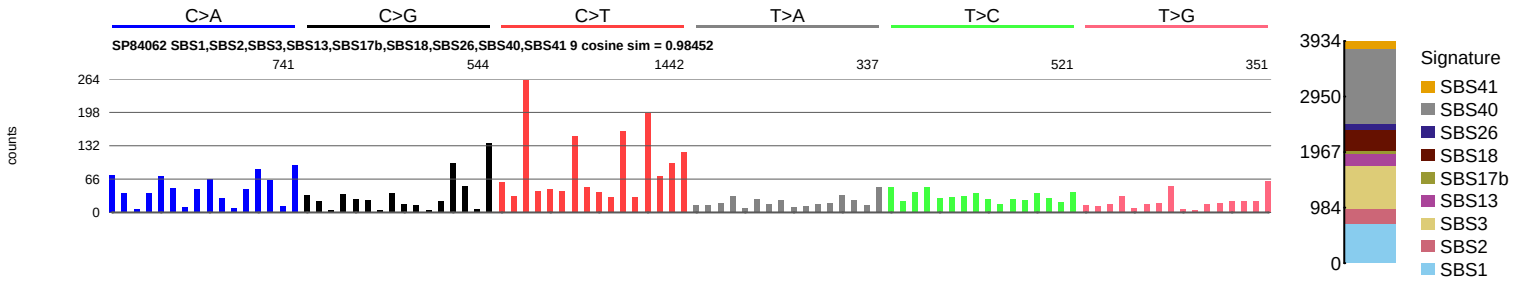

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98463
754
499
1526
280
520
352
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

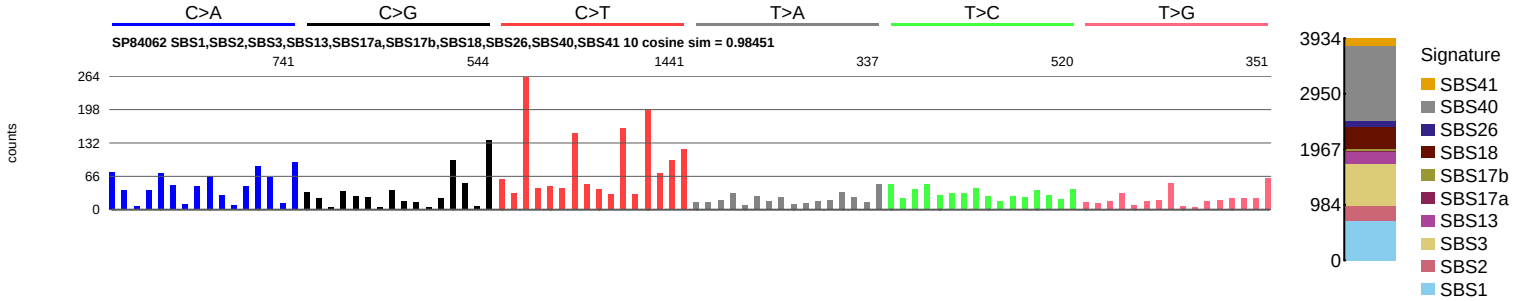

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98457
742
566
1477
373
466
312
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

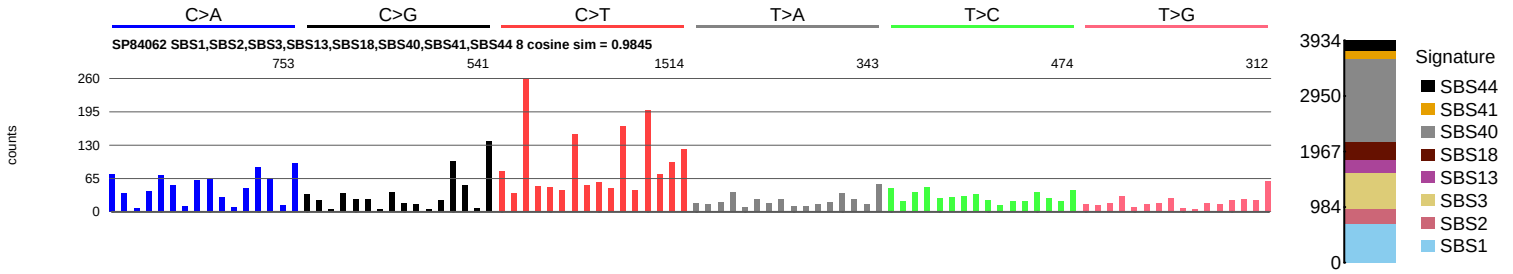

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98455
748
505
1559
283
536
302
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

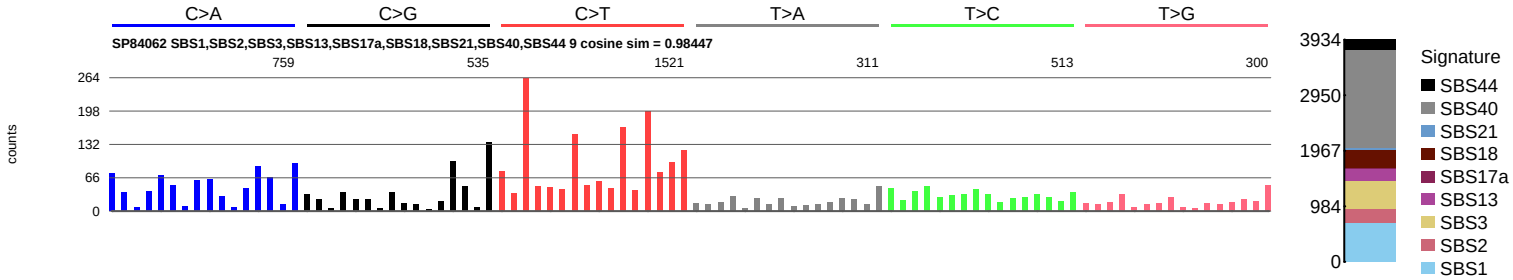

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98455
736
505
1523
272
555
344
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

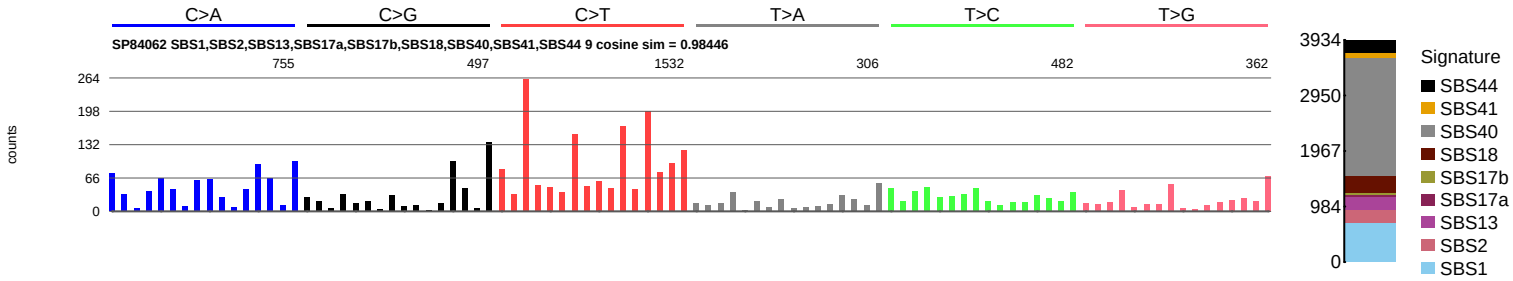

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98453
758
504
1530
285
502
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

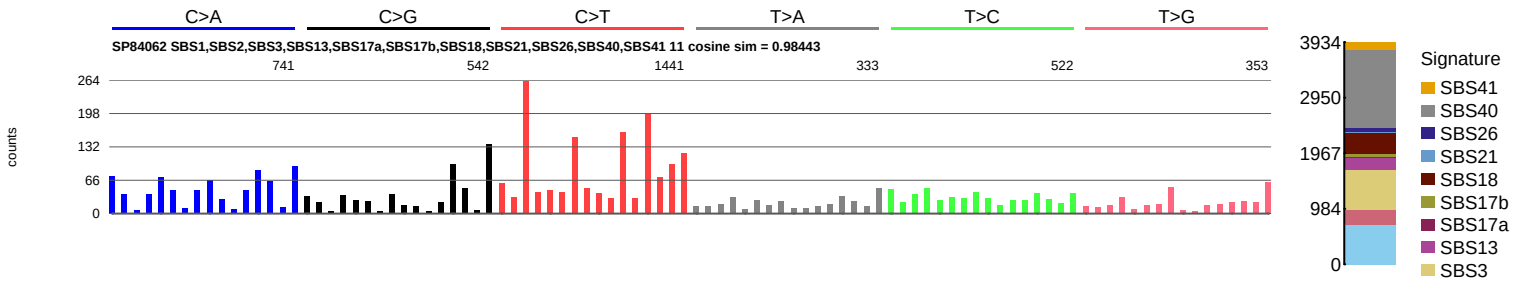

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98452
741
544
1442
337
521
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

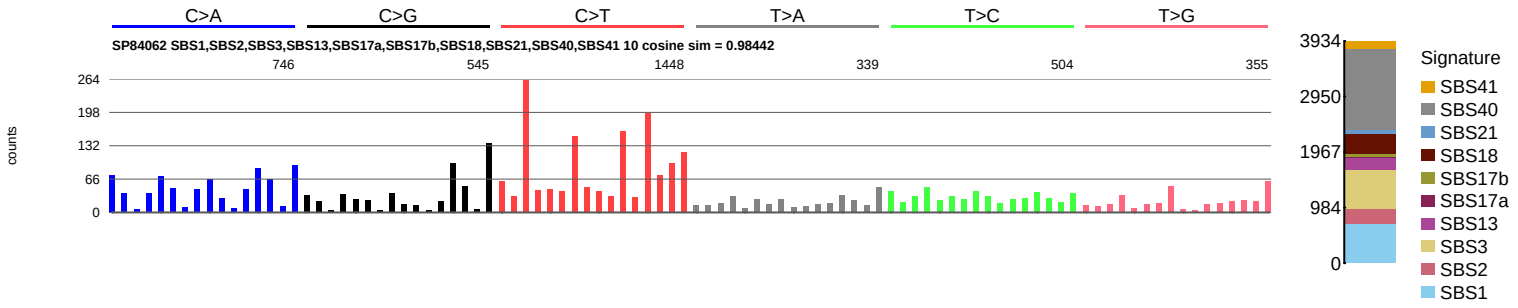

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98451
741
544
1441
337
520
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

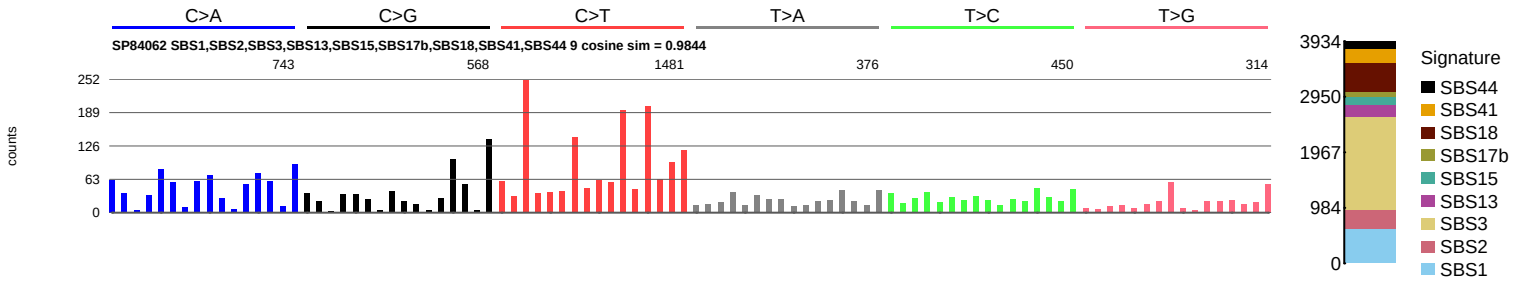

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9845
753
541
1514
343
474
312
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

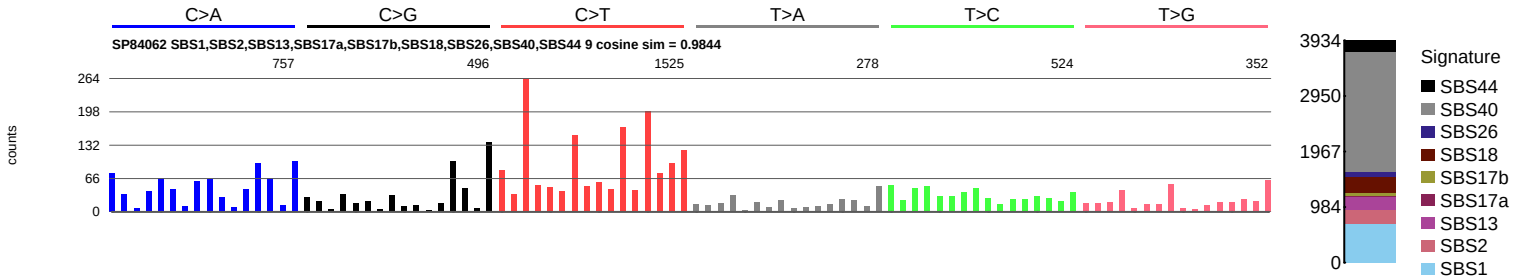

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98447
759
535
1521
311
513
300
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

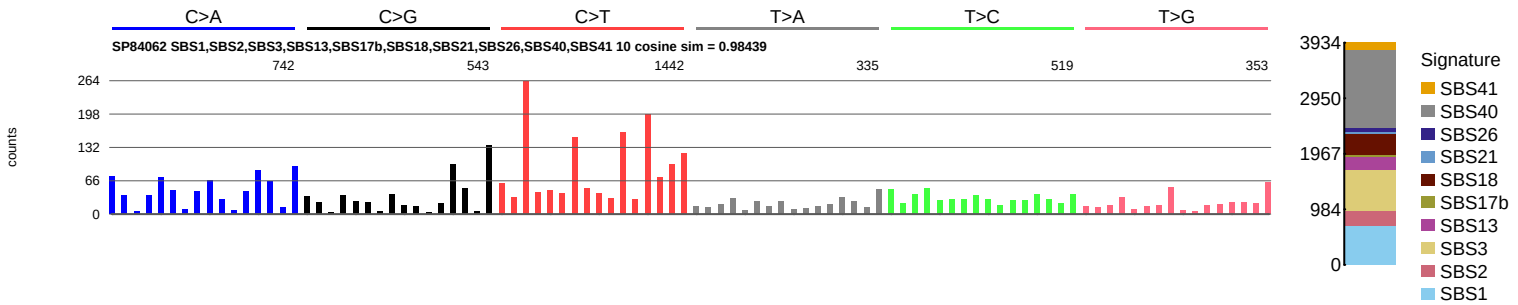

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98446
755
497
1532
306
482
362
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

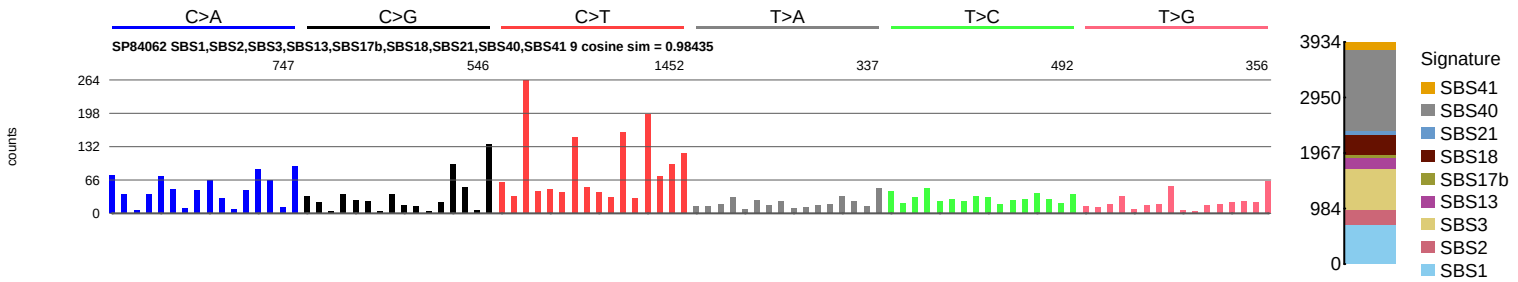

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98443
741
542
1441
333
522
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

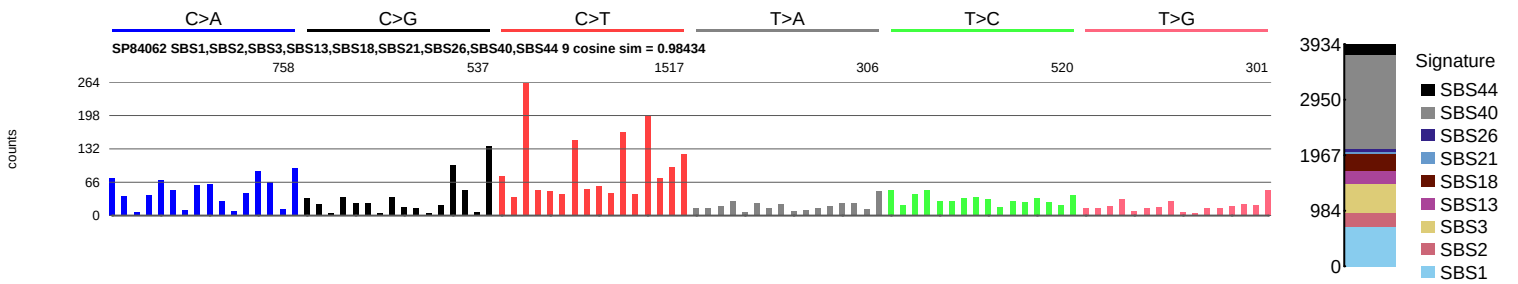

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98442
746
545
1448
339
504
355
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

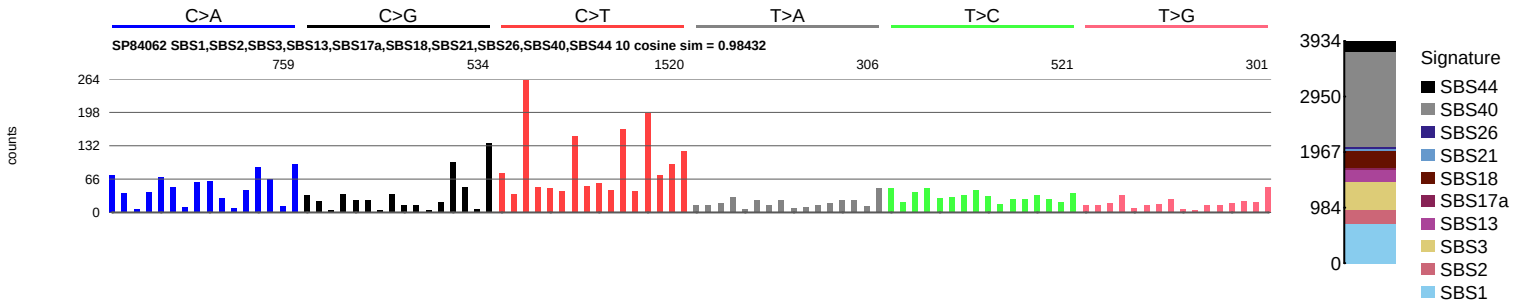

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9844
743
568
1481
376
450
314
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

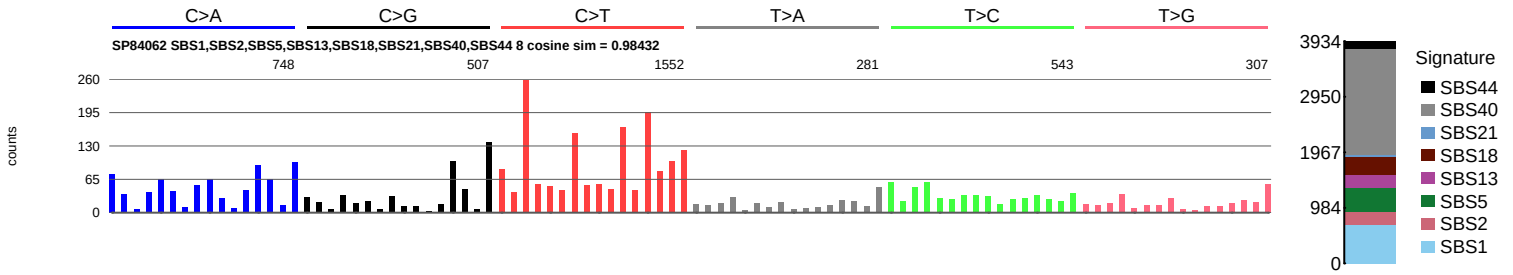

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9844
757
496
1525
278
524
352
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

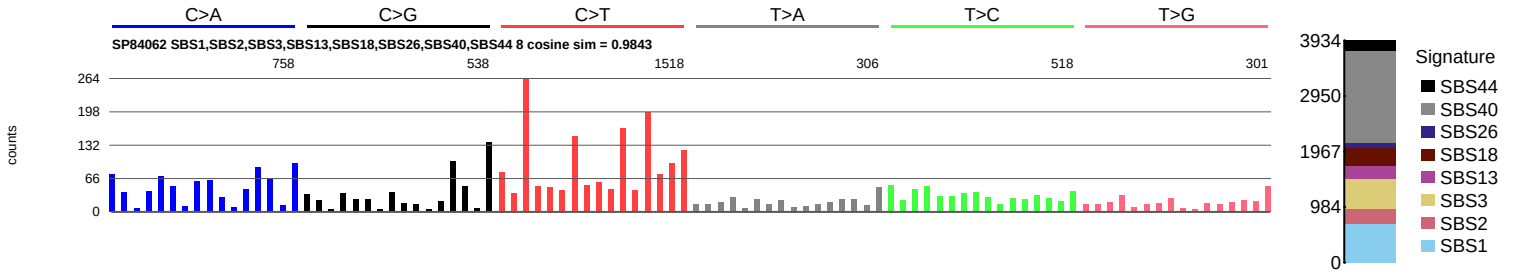

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98439
742
543
1442
335
519
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

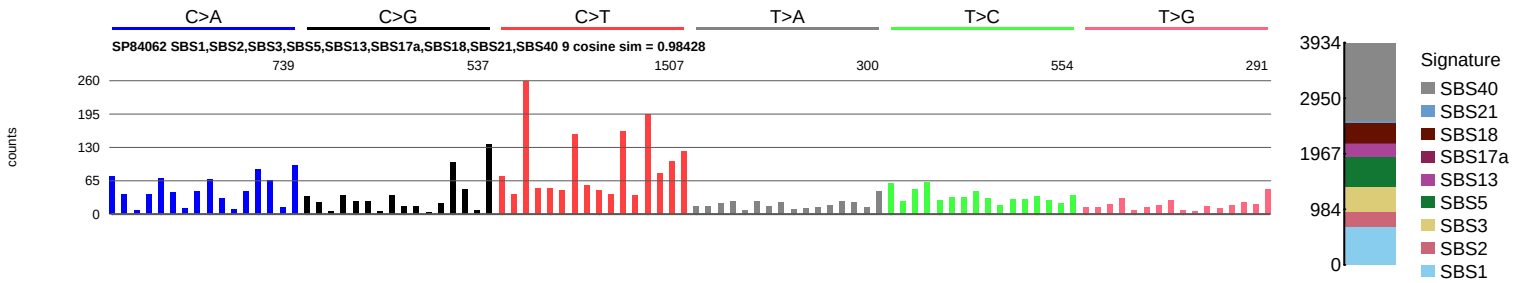

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98435
747
546
1452
337
492
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

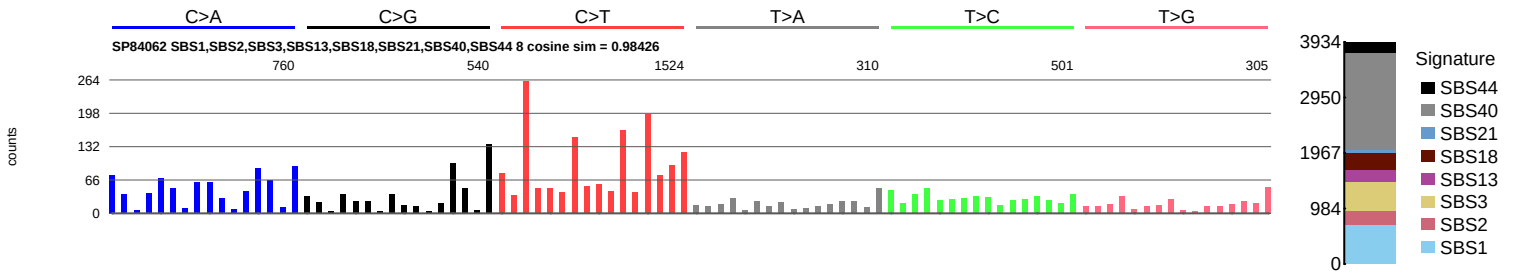

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98434
758
537
1517
306
520
301
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

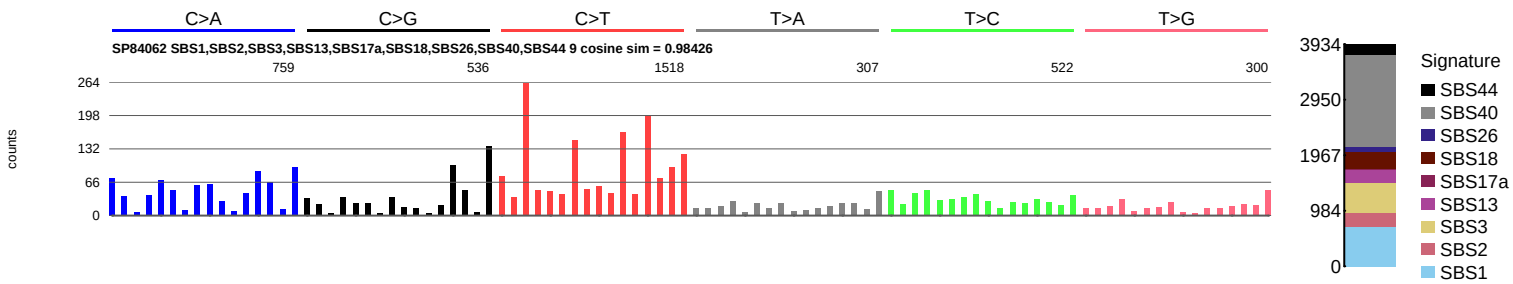

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98432
759
534
1520
306
521
301
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

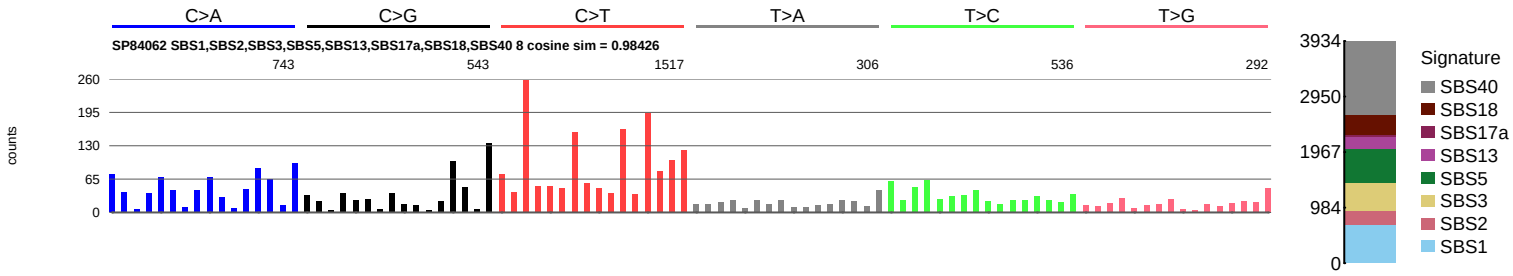

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98432
748
507
1552
281
543
307
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

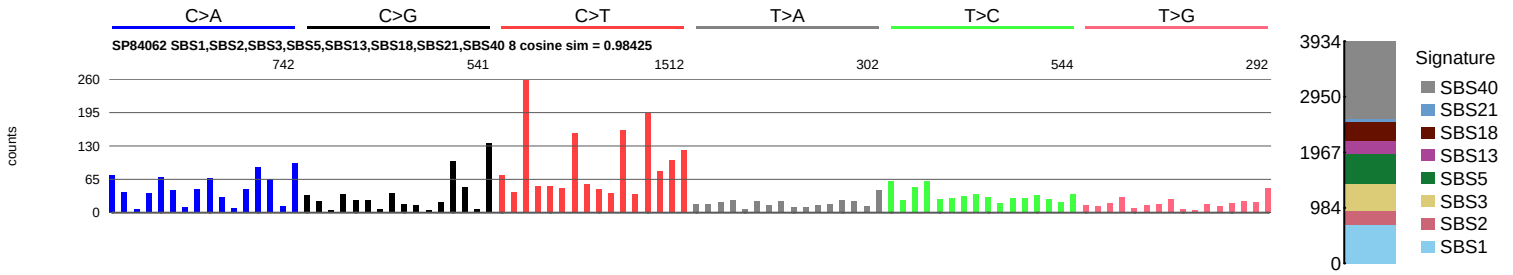

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9843
758
538
1518
306
518
301
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

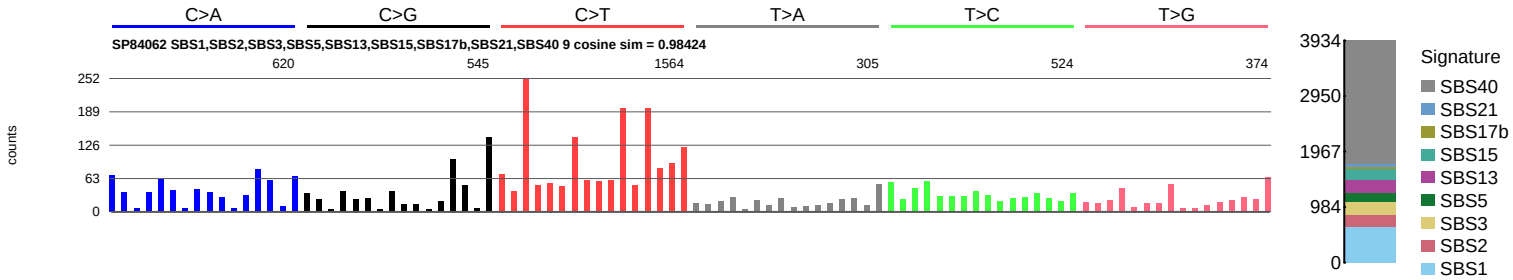

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98428
739
537
1507
300
554
291
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

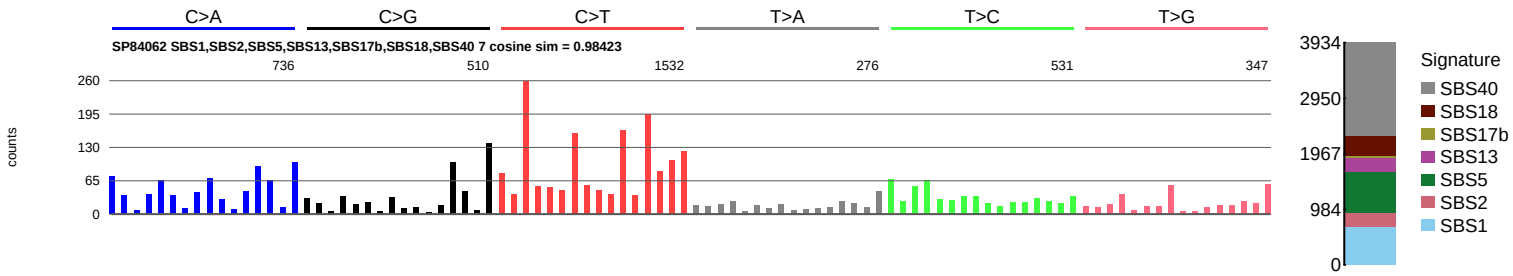

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98426
760
540
1524
310
501
305
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

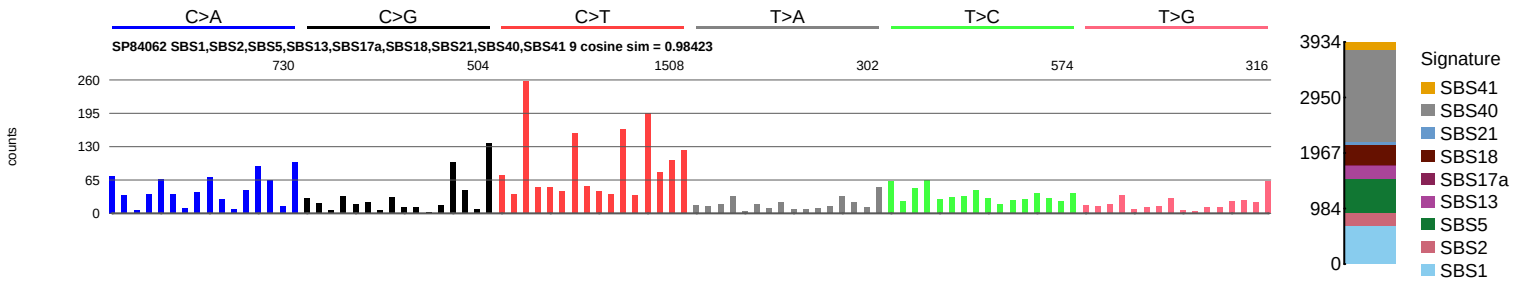

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98426
759
536
1518
307
522
300
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

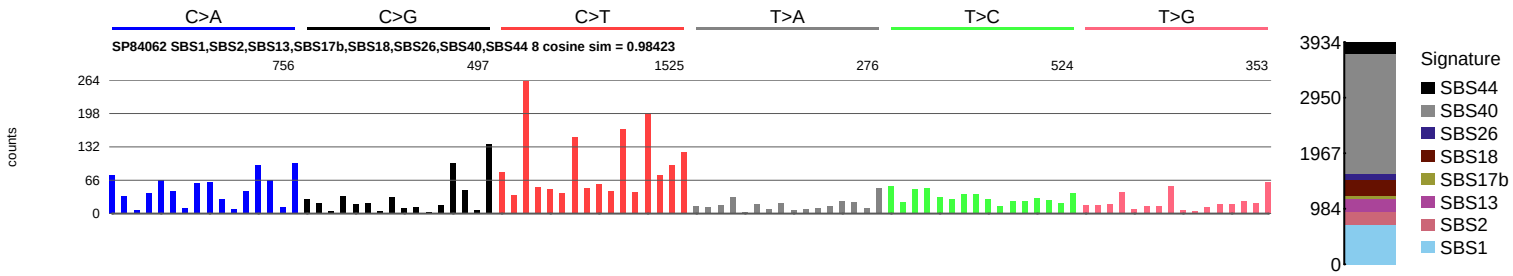

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40 8 cosine sim = 0.98426
743
543
1517
306
536
292
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

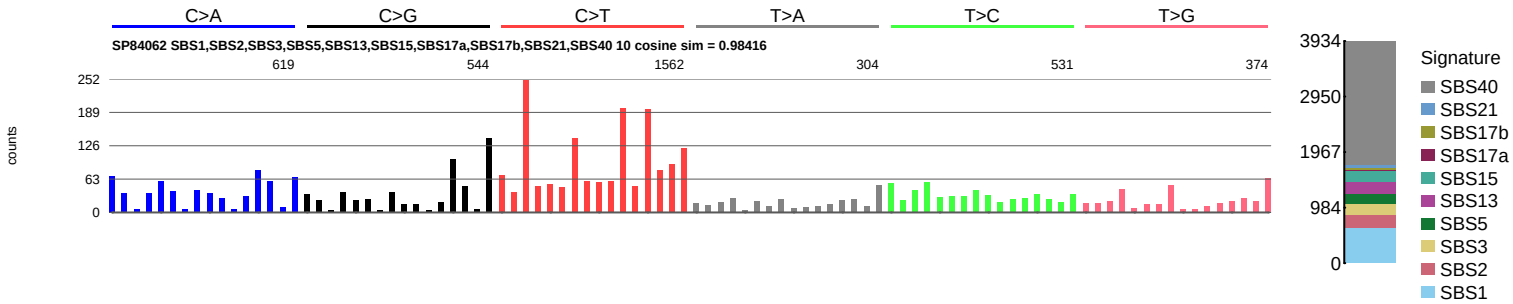

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40 8 cosine sim = 0.98425
742
541
1512
302
544
292
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

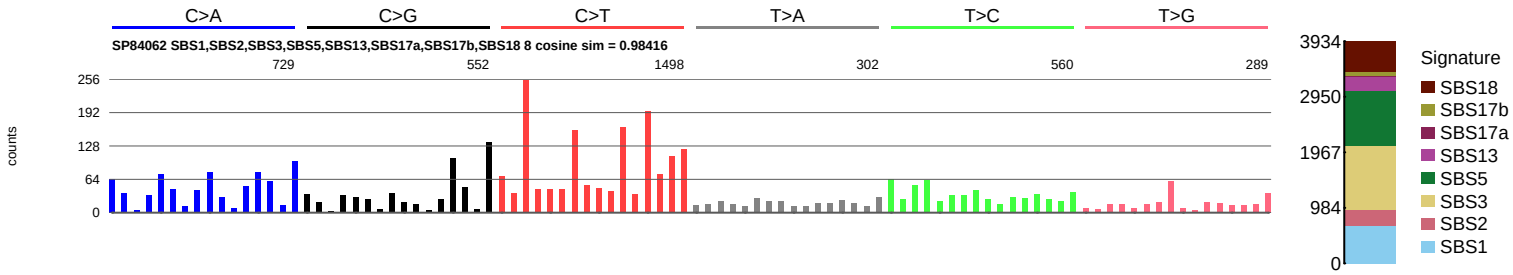

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 9 cosine sim = 0.98424
620
545
1564
305
524
374
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

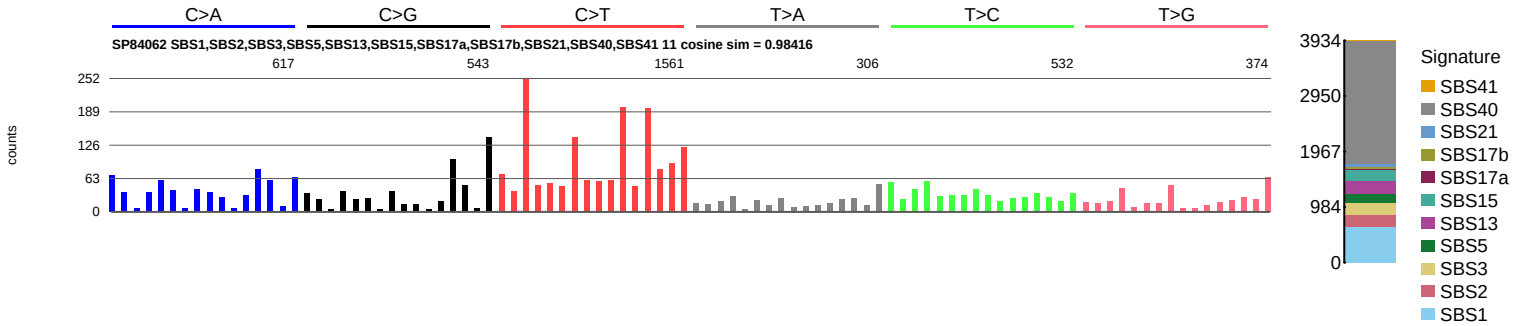

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98423
736
510
1532
276
531
347
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

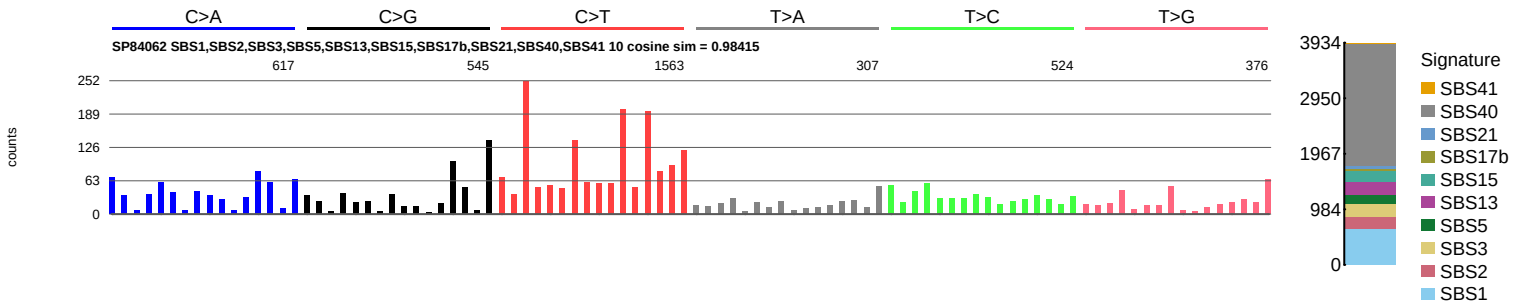

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98423
730
504
1508
302
574
316
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

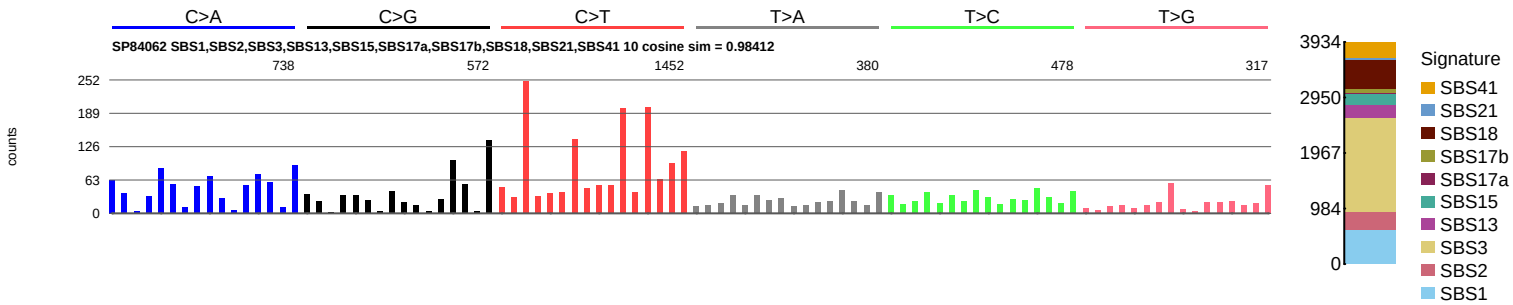

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98423
756
497
1525
276
524
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

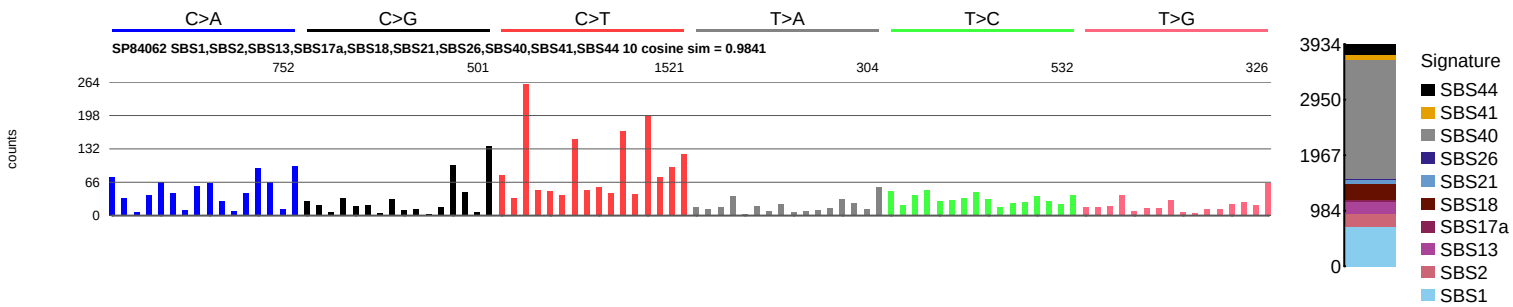

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 10 cosine sim = 0.98416
619
544
1562
304
531
374
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

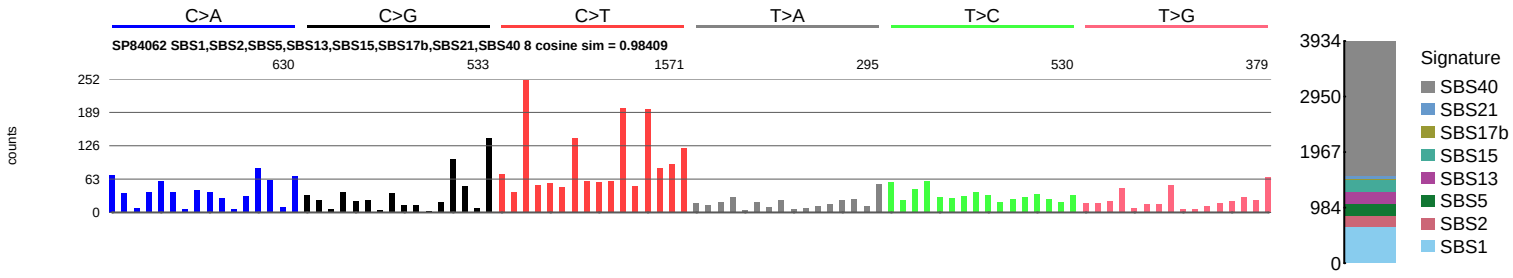

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98416
729
552
1498
302
560
289
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

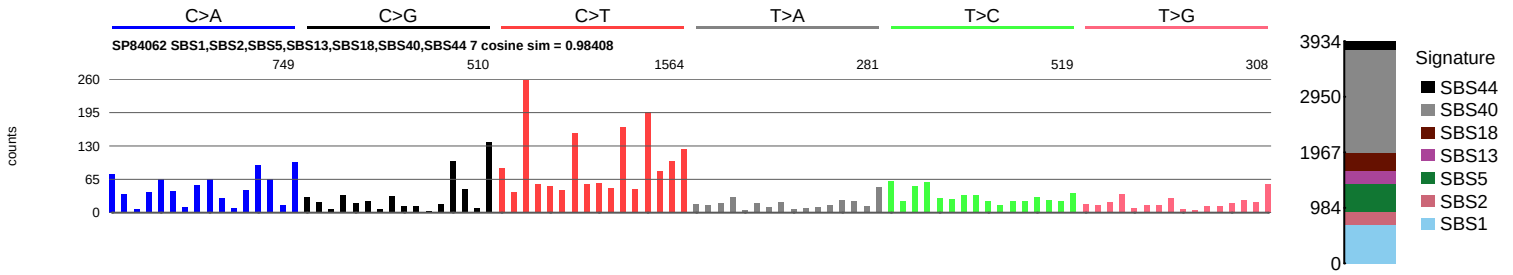

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 11 cosine sim = 0.98416
617
543
1561
306
532
374
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

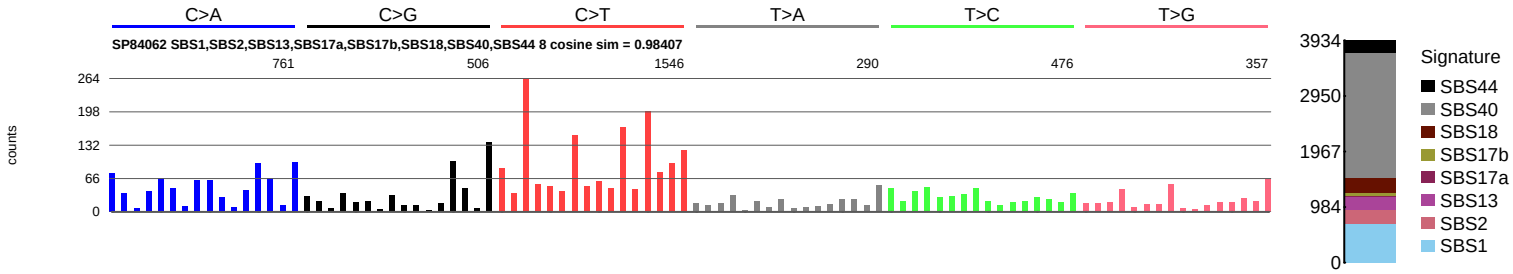

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98415
617
545
1563
307
524
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

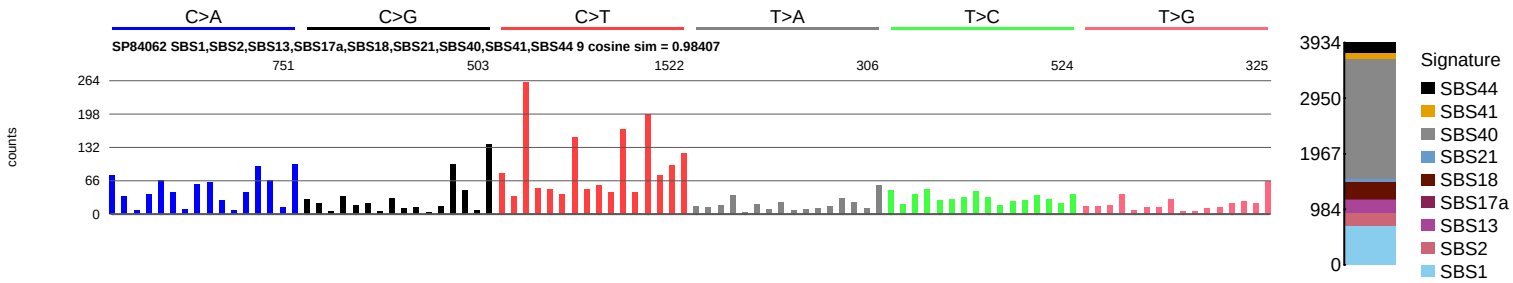

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98412
738
572
1452
380
478
317
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

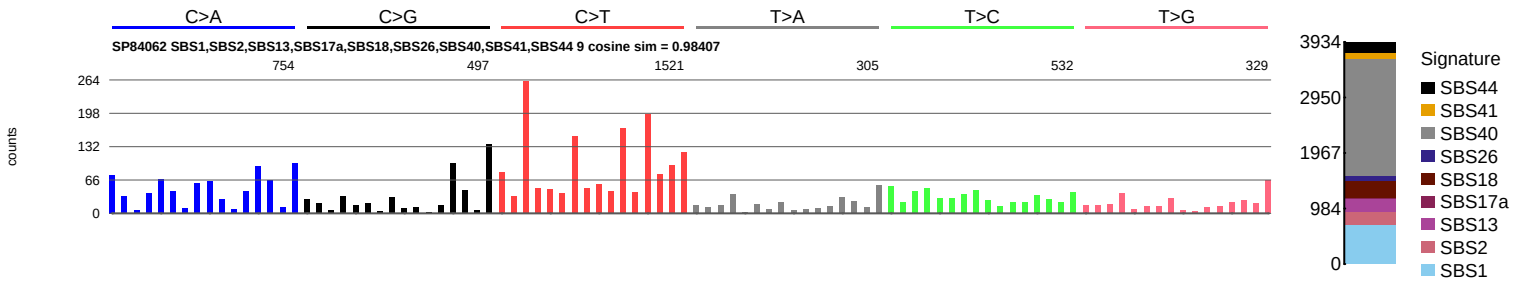

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9841
752
501
1521
304
532
326
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

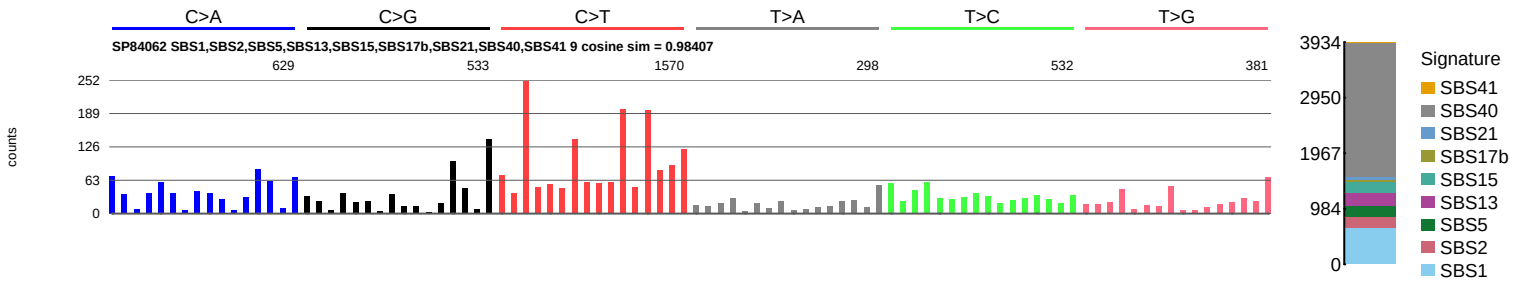

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 8 cosine sim = 0.98409
630
533
1571
295
530
379
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

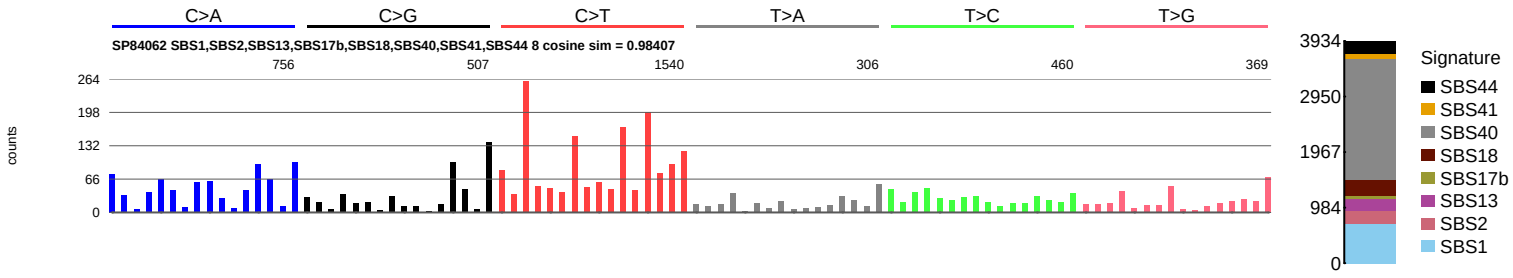

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98408
749
510
1564
281
519
308
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

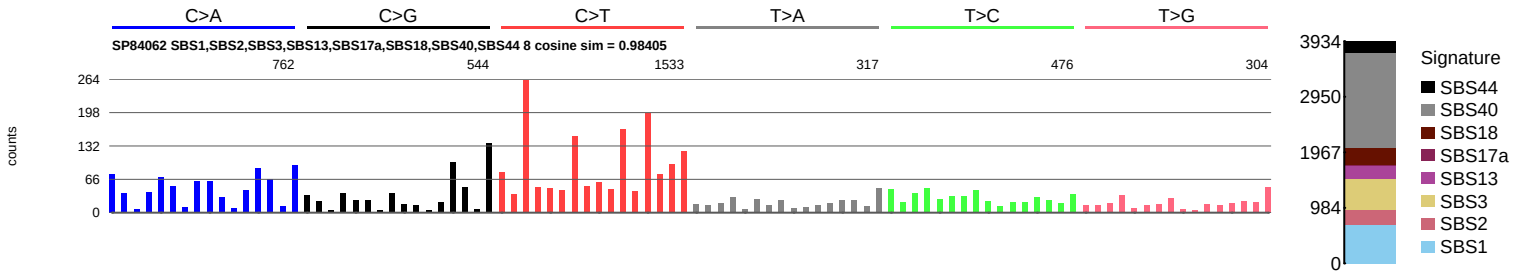

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98407
761
506
1546
290
476
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

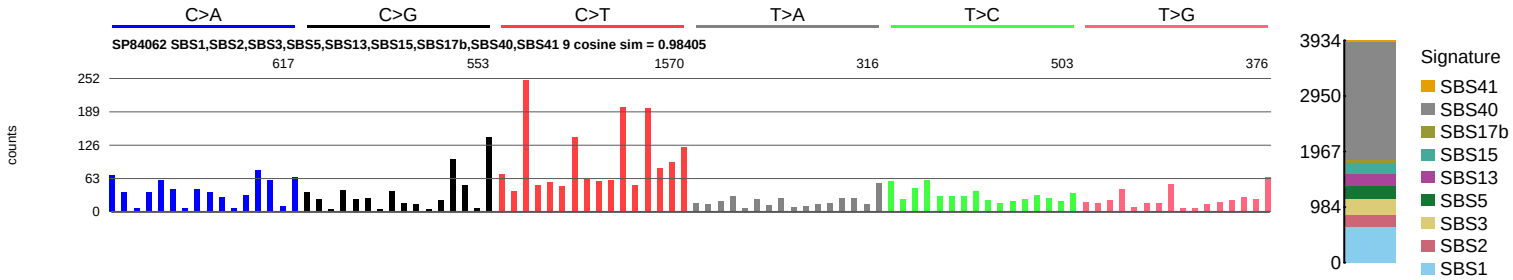

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98407
751
503
1522
306
524
325
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

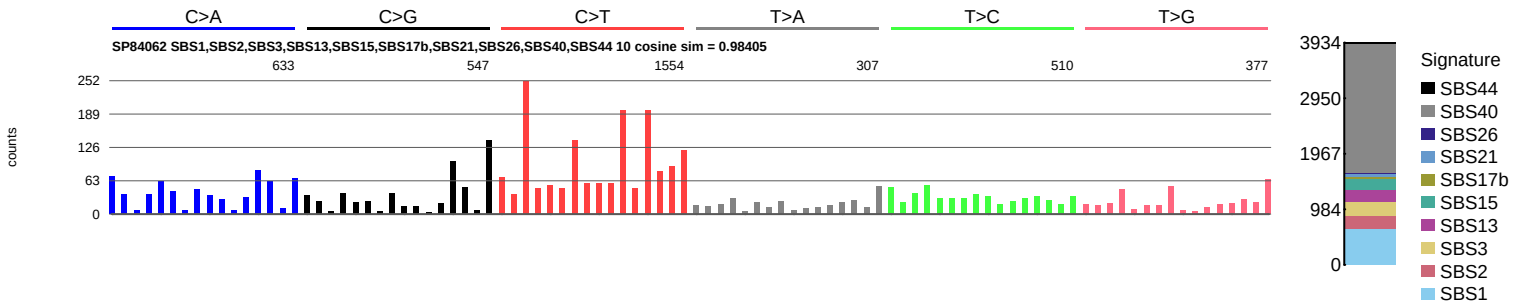

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98407
754
497
1521
305
532
329
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1

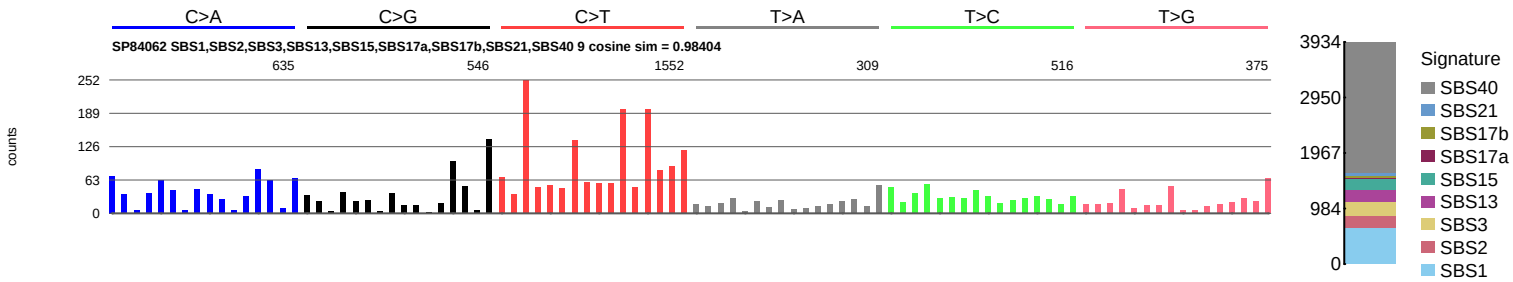

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98407
629
533
1570
298
532
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

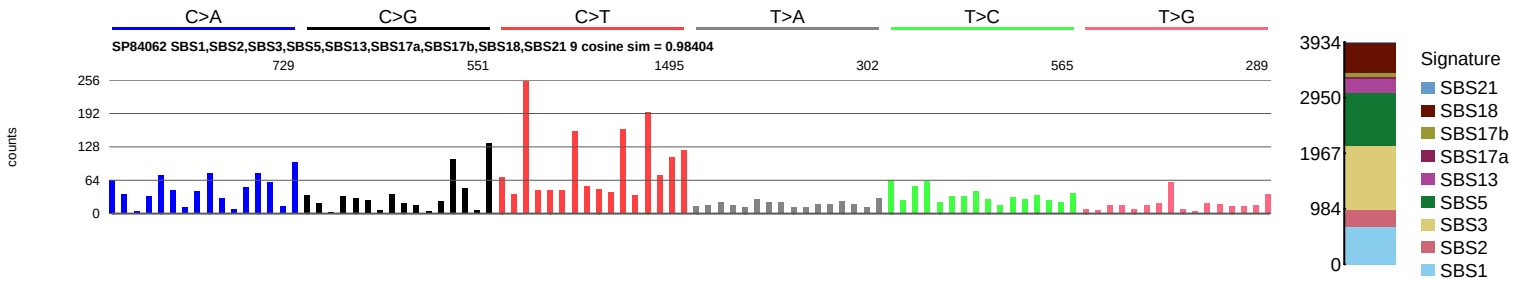

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98407
756
507
1540
306
460
369
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

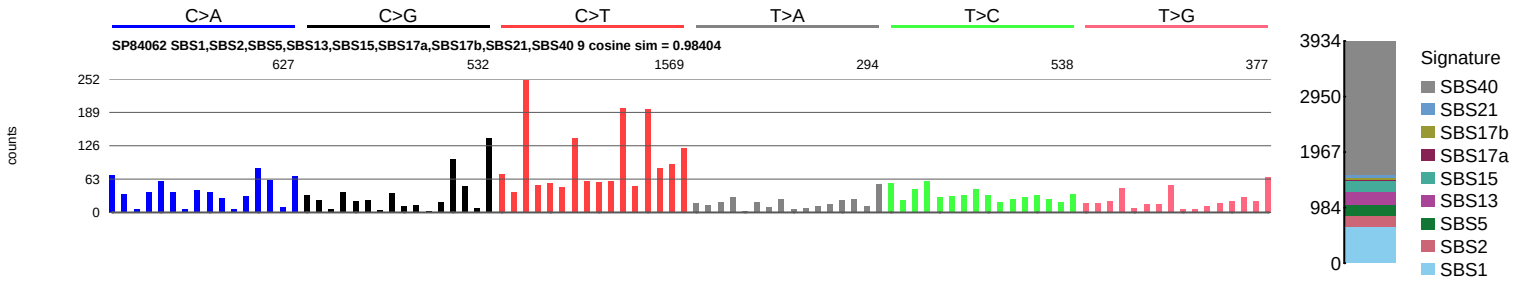

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98405
762
544
1533
317
476
304
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

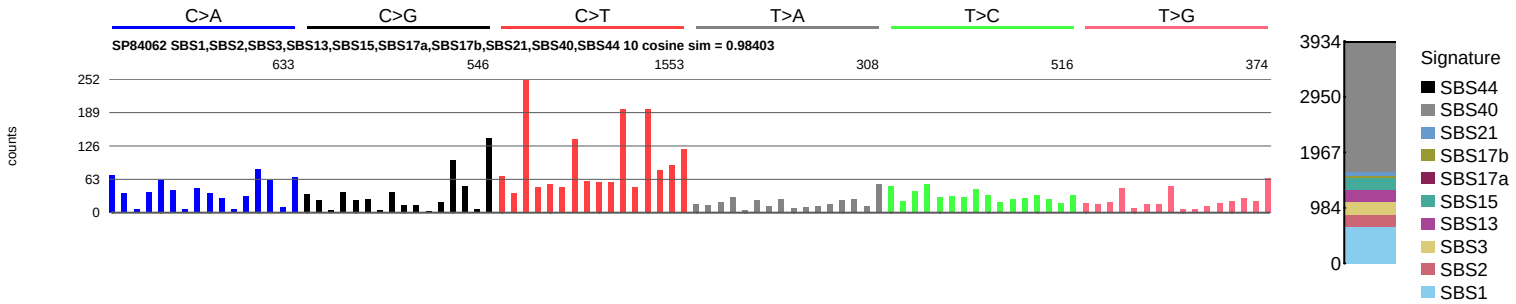

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41 9 cosine sim = 0.98405
617
553
1570
316
503
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

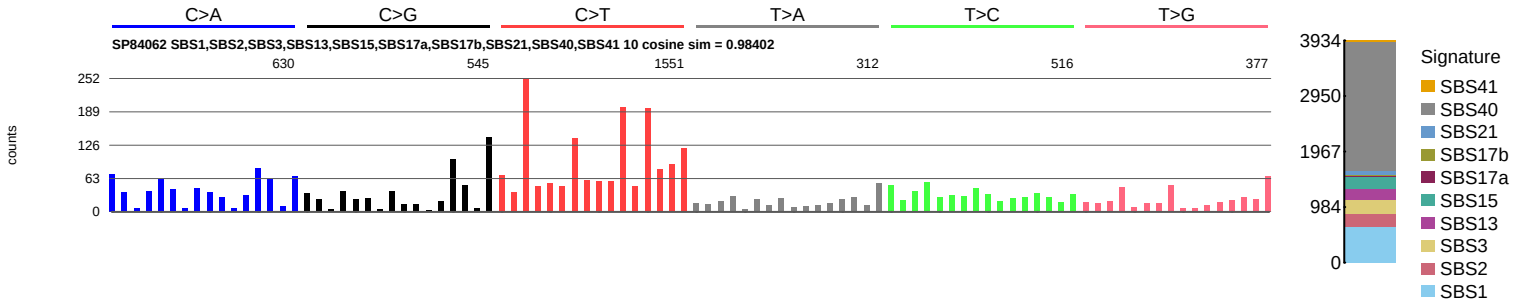

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98405
633
547
1554
307
510
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

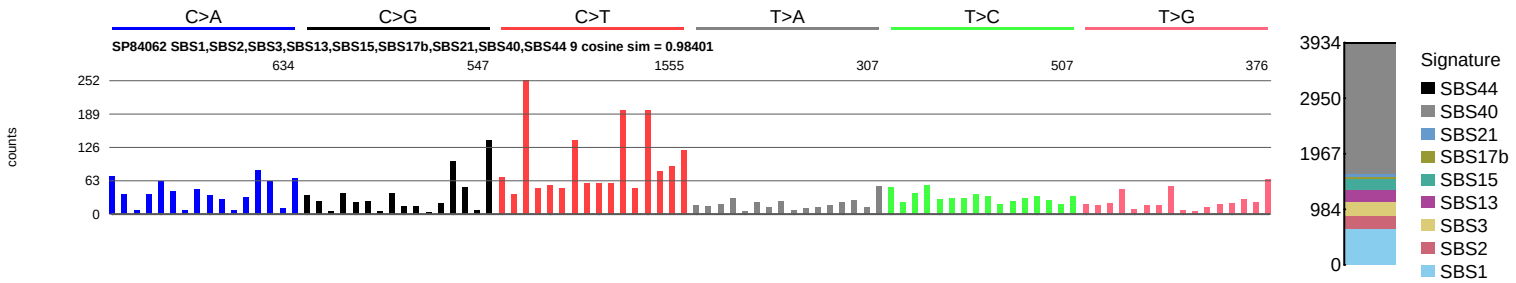

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98404
635
546
1552
309
516
375
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

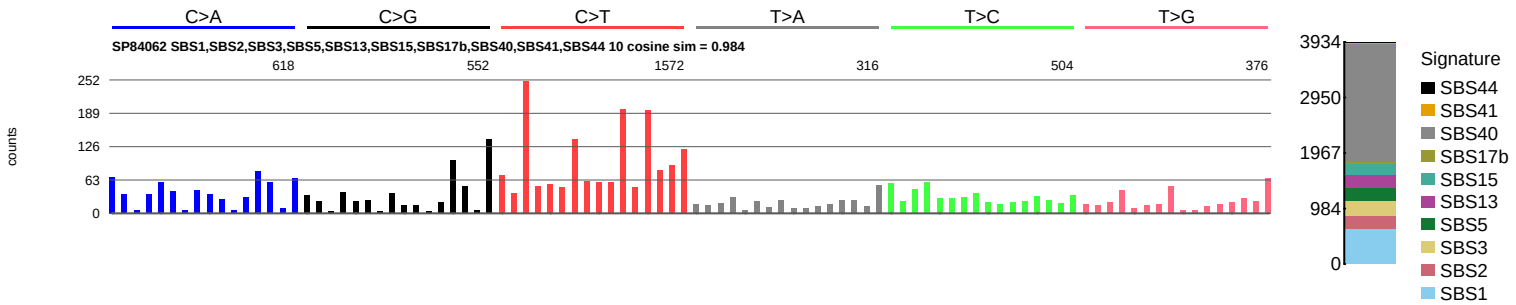

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98404
729
551
1495
302
565
289
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

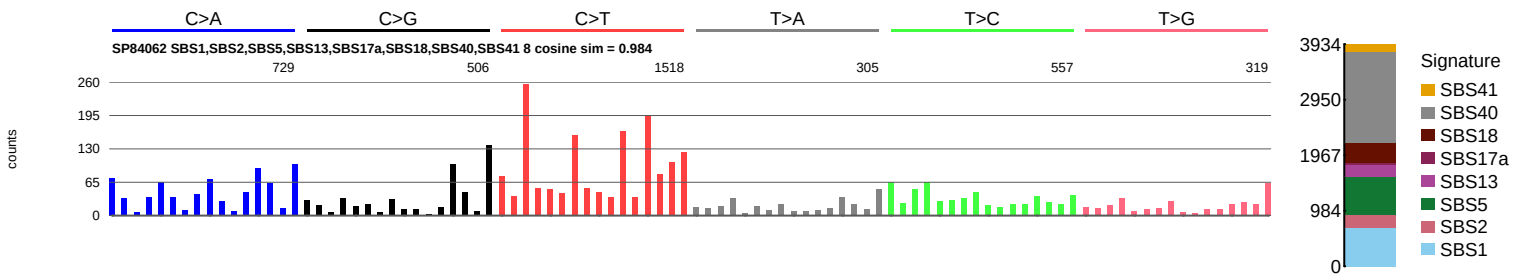

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98404
627
532
1569
294
538
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

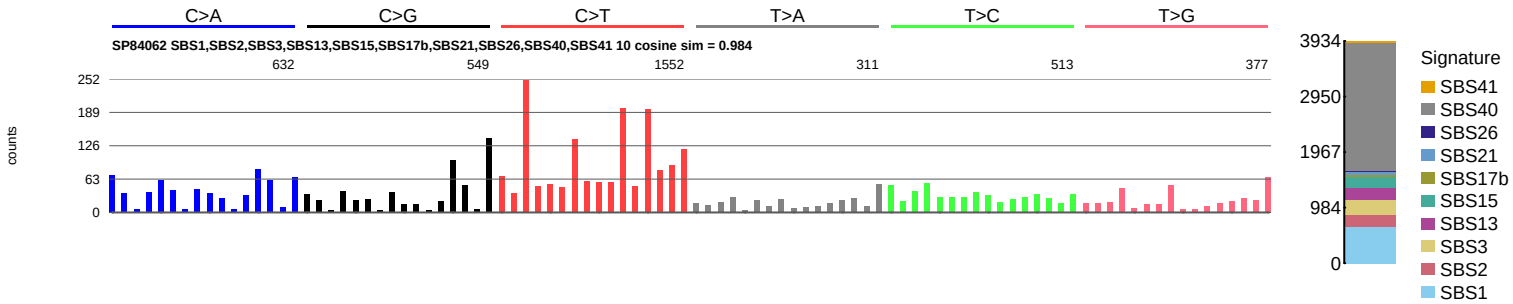

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98403
633
546
1553
308
516
374
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

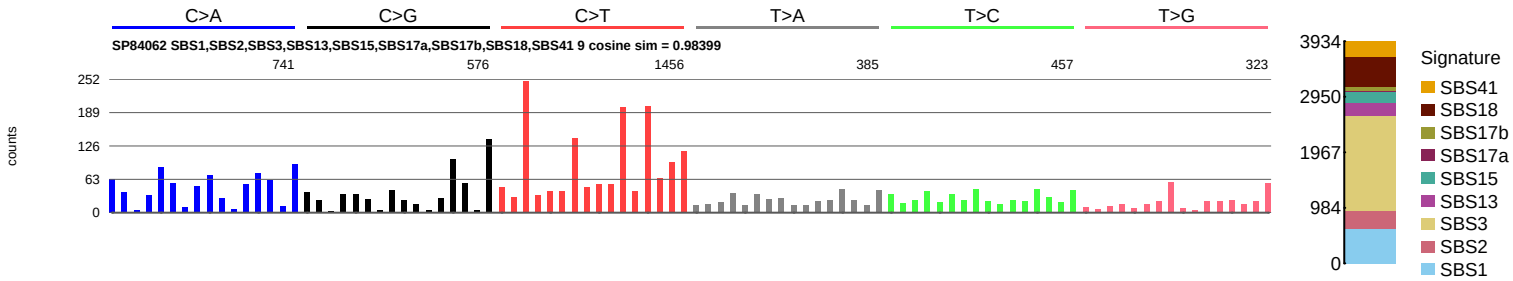

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98402
630
545
1551
312
516
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

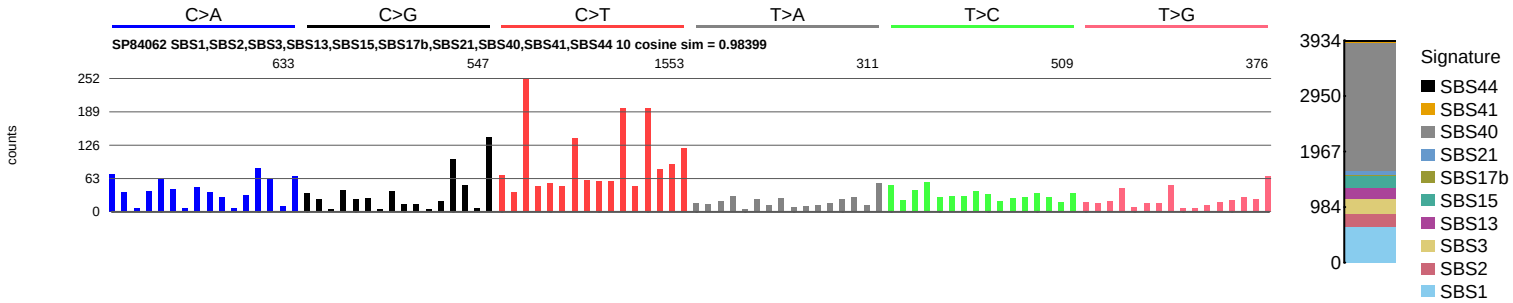

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98401
634
547
1555
307
507
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

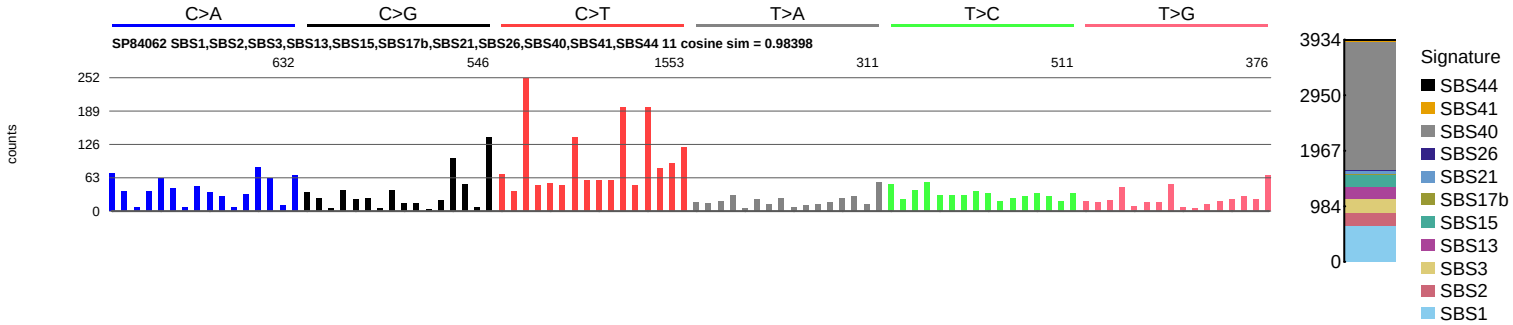

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.984
618
552
1572
316
504
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

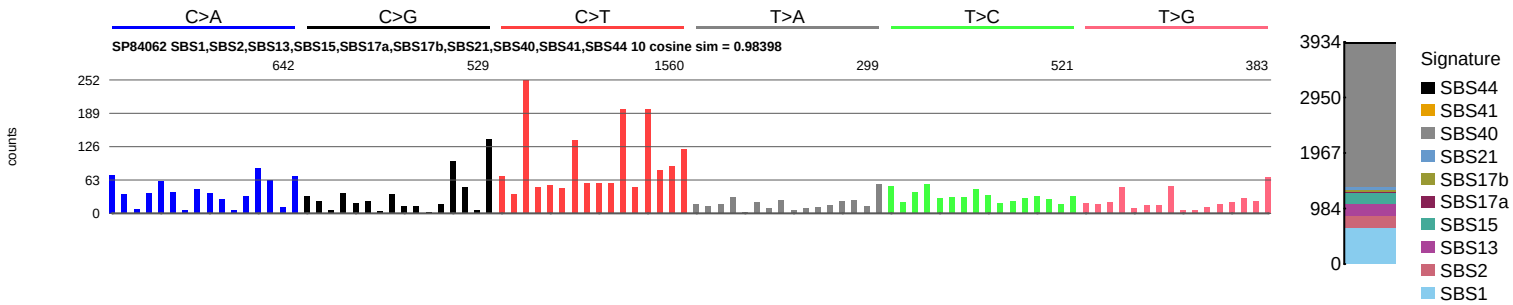

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.984
729
506
1518
305
557
319
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

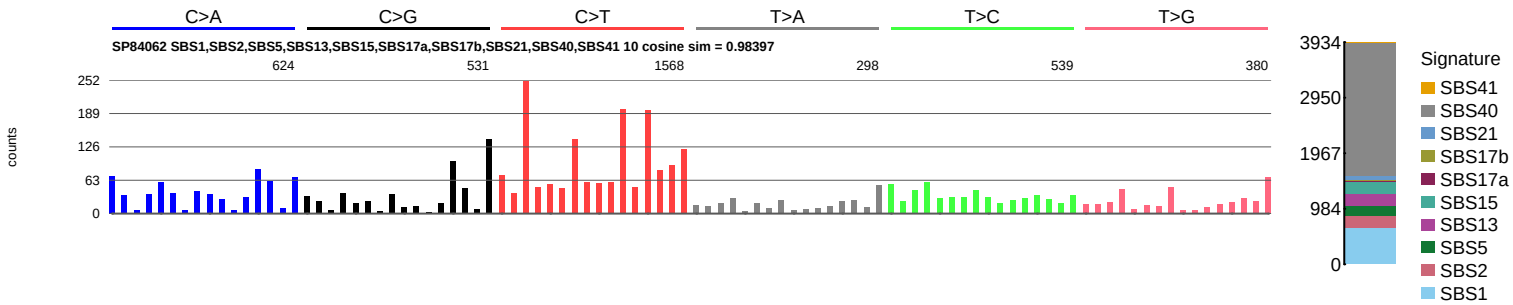

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.984
632
549
1552
311
513
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

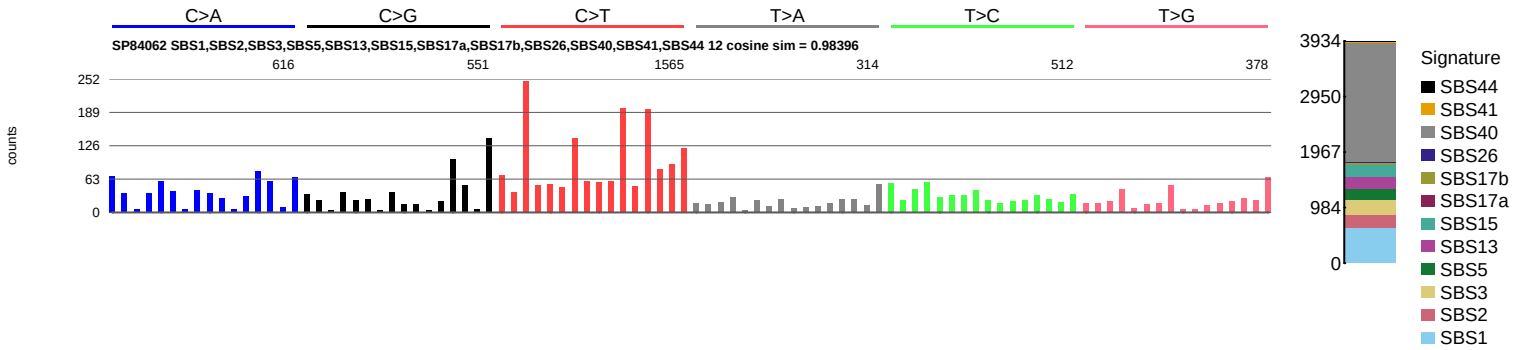

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98399
741
576
1456
385
457
323
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

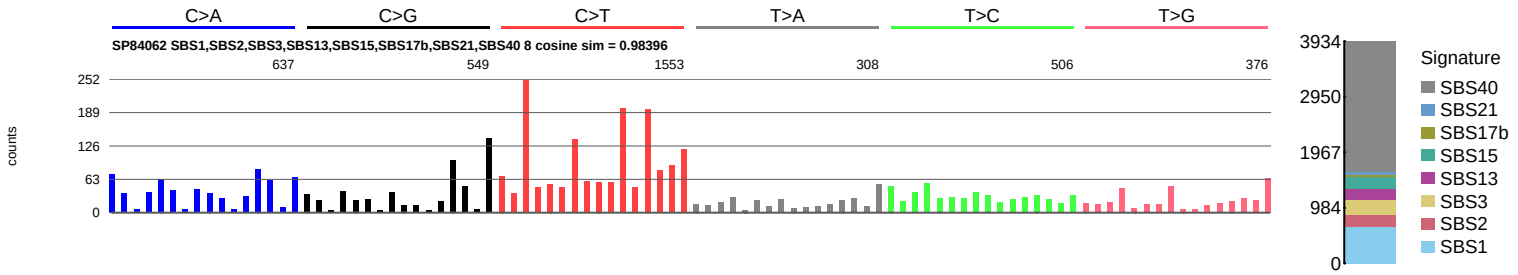

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98399
633
547
1553
311
509
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

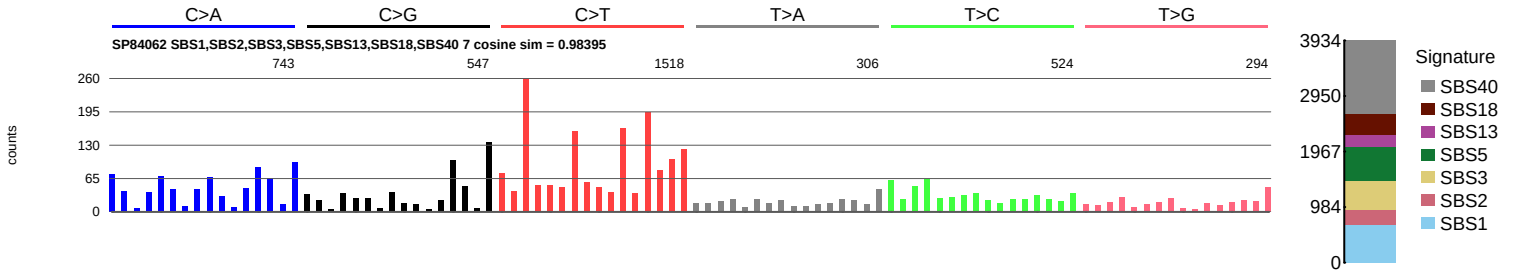

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98398
632
546
1553
311
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

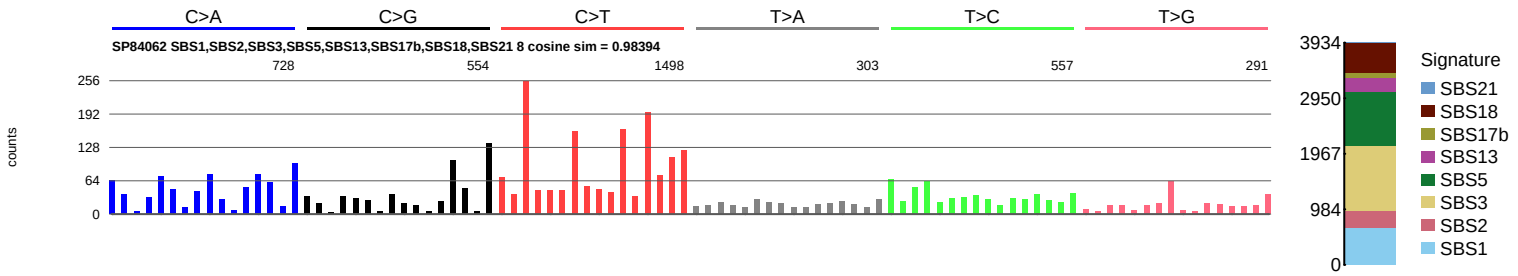

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98398
642
529
1560
299
521
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

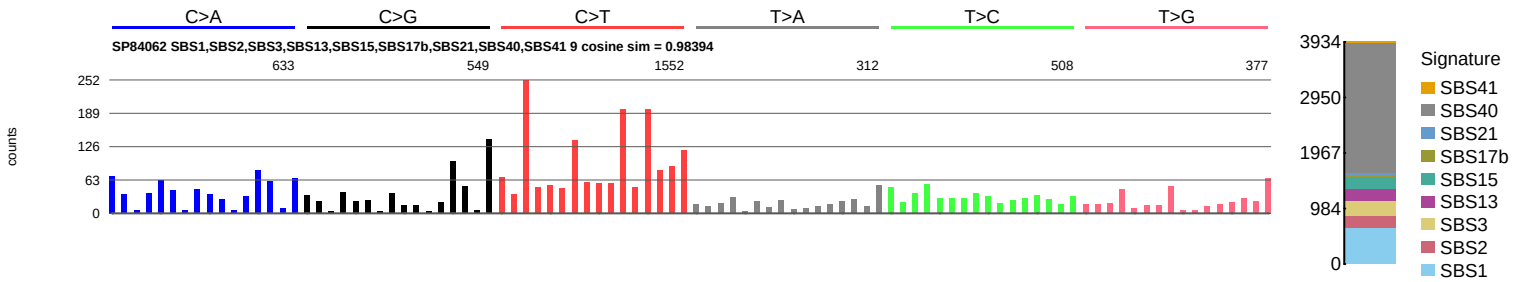

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98397
624
531
1568
298
539
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

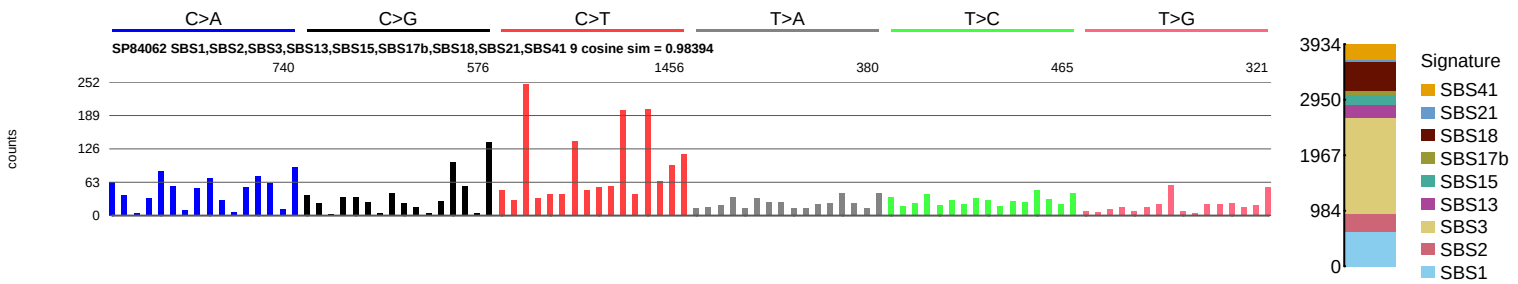

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98396
616
551
1565
314
512
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

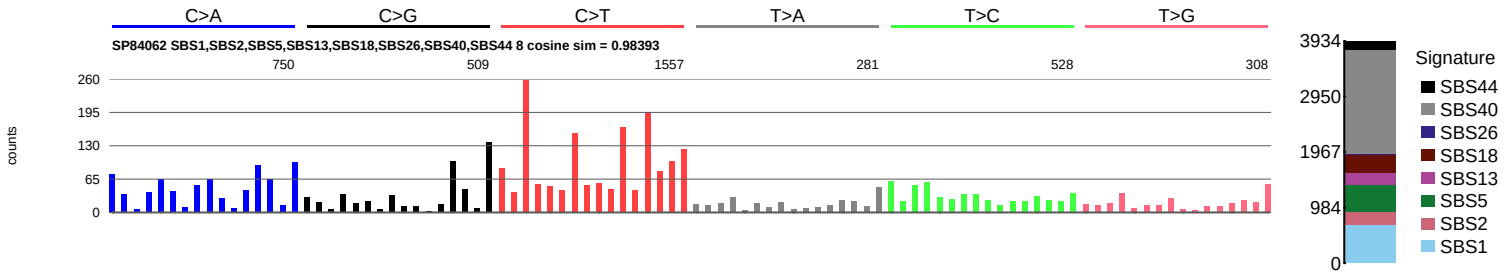

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS40 8 cosine sim = 0.98396
637
549
1553
308
506
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

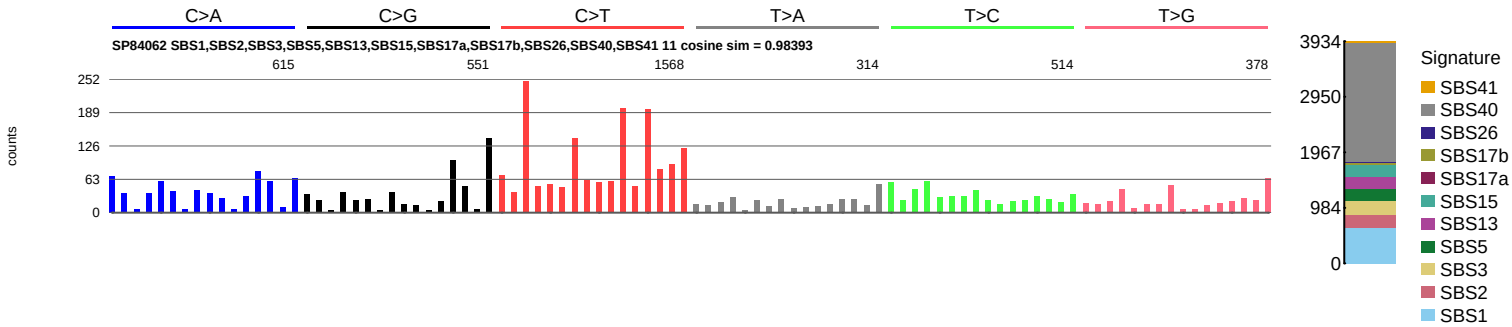

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40 7 cosine sim = 0.98395
743
547
1518
306
524
294
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

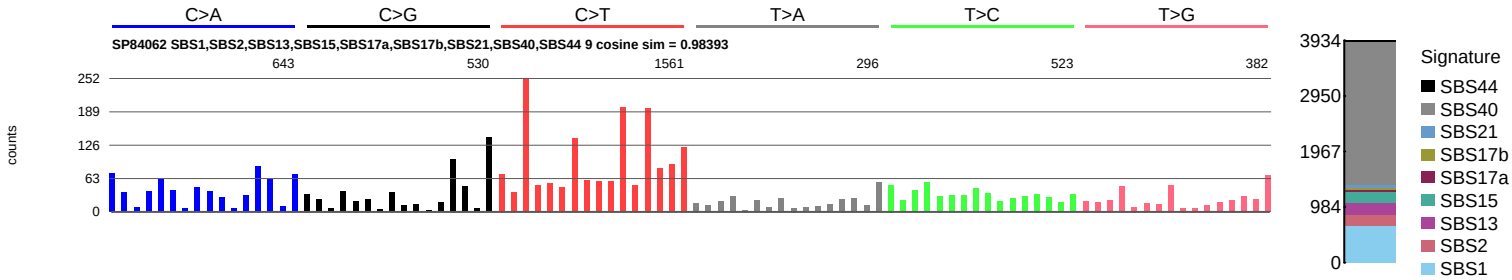

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98394
728
554
1498
303
557
291
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

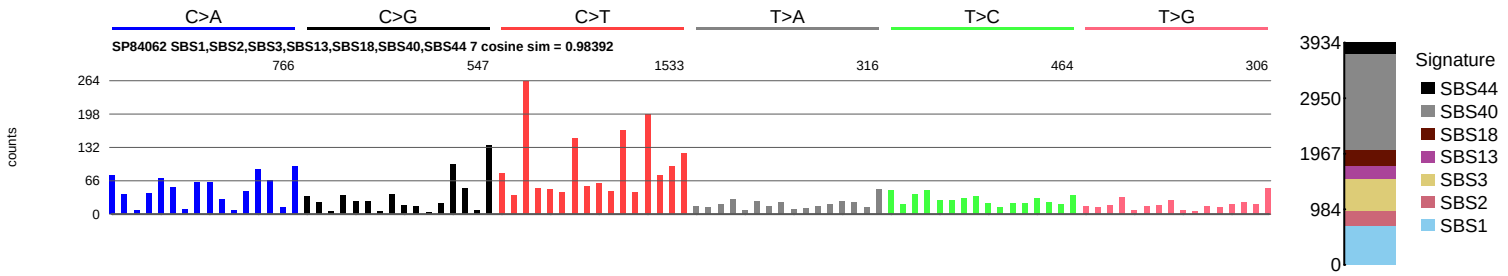

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98394
633
549
1552
312
508
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

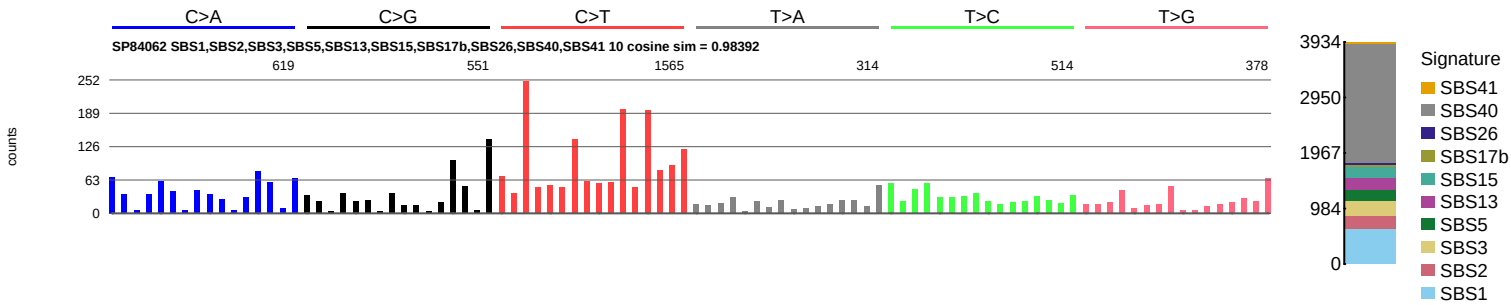

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98394
740
576
1456
380
465
321
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

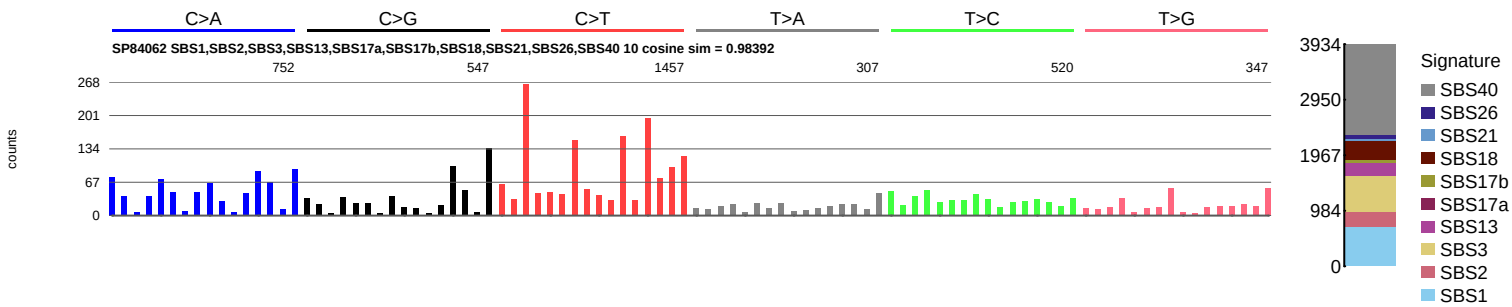

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98393
750
509
1557
281
528
308
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

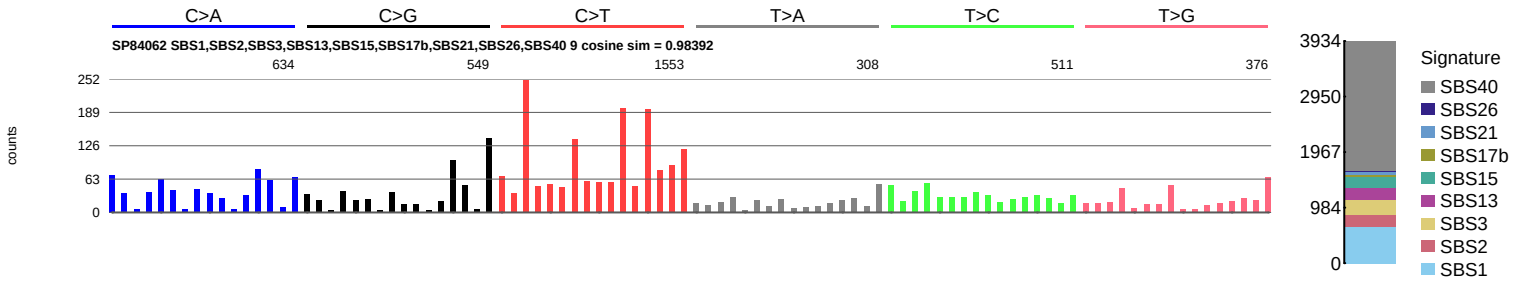

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 11 cosine sim = 0.98393
615
551
1568
314
514
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

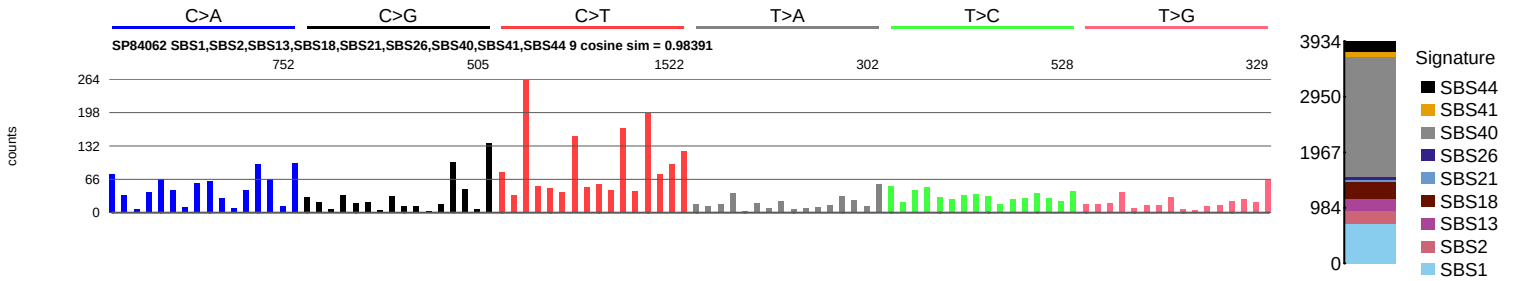

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98393
643
530
1561
296
523
382
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

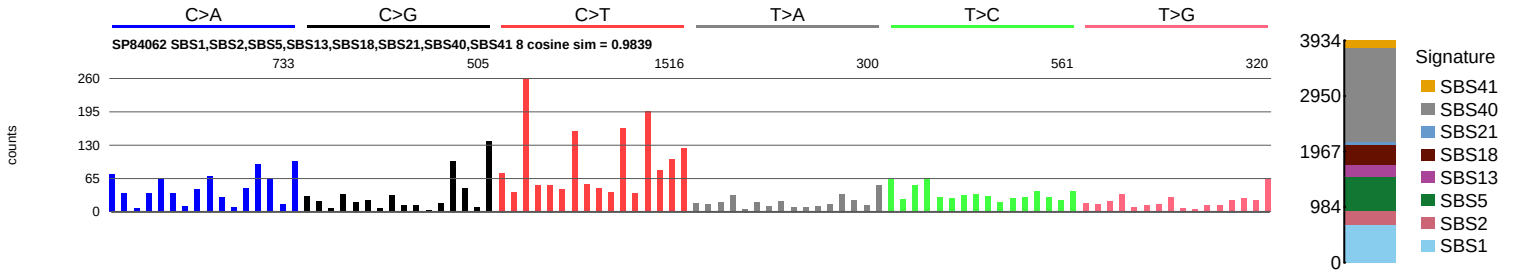

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98392
766
547
1533
316
464
306
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

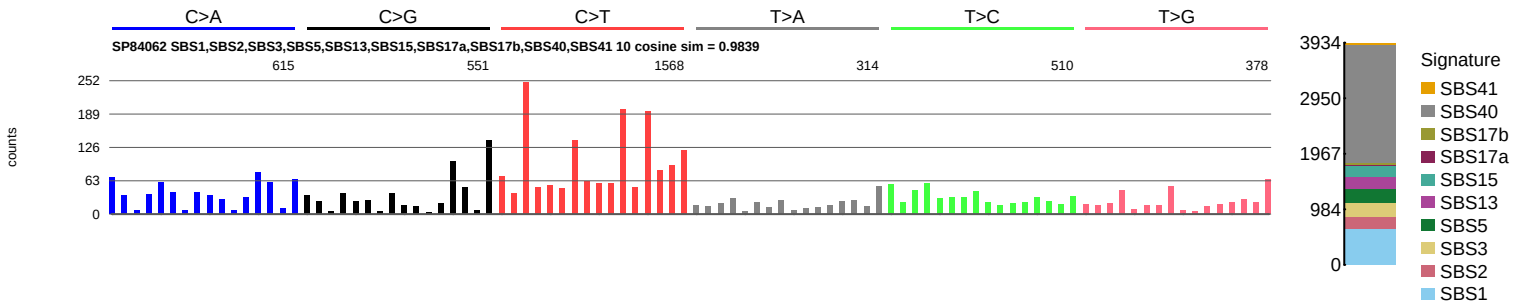

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98392
619
551
1565
314
514
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

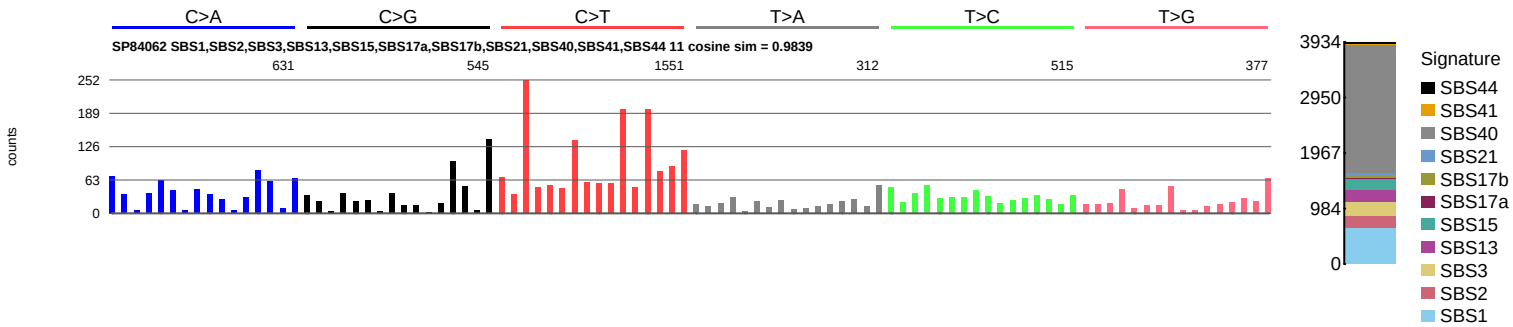

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98392
752
547
1457
307
520
347
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

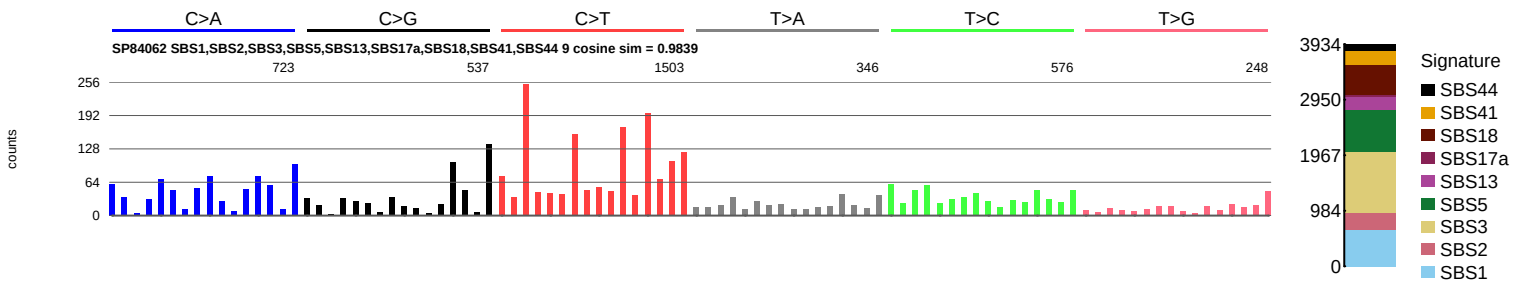

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98392
634
549
1553
308
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

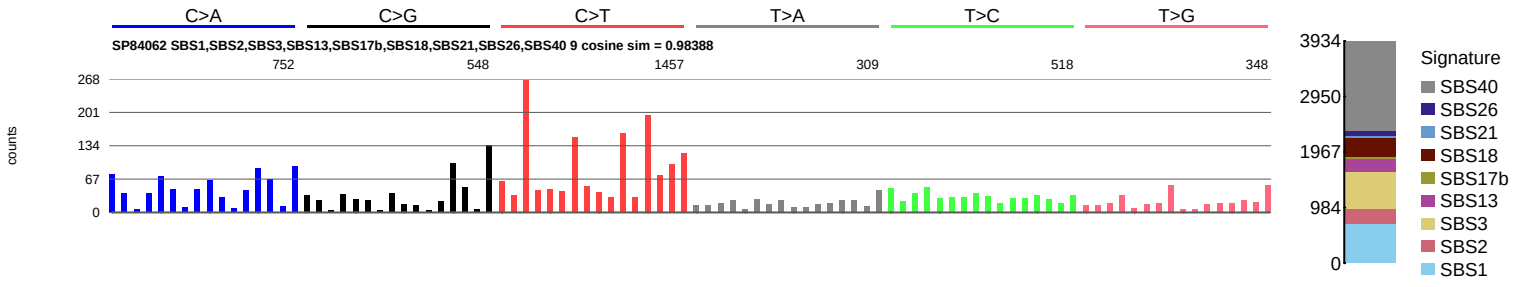

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98391
752
505
1522
302
528
329
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

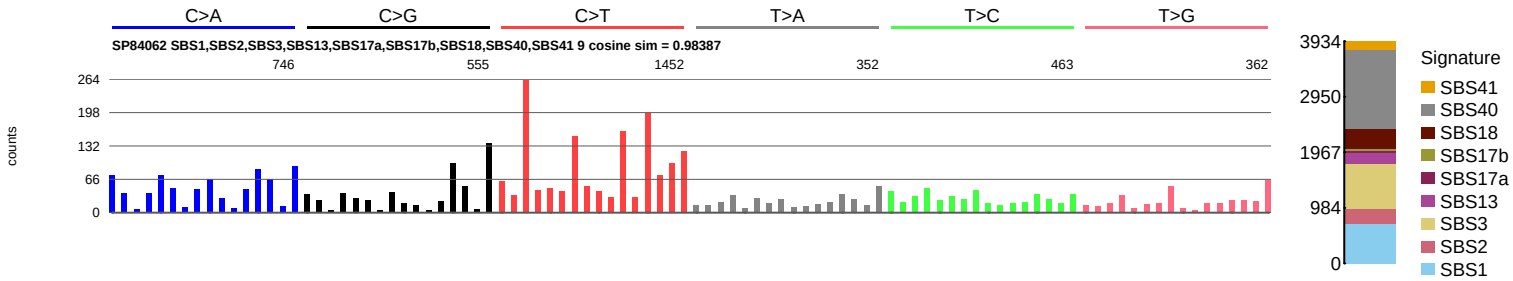

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9839
733
505
1516
300
561
320
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

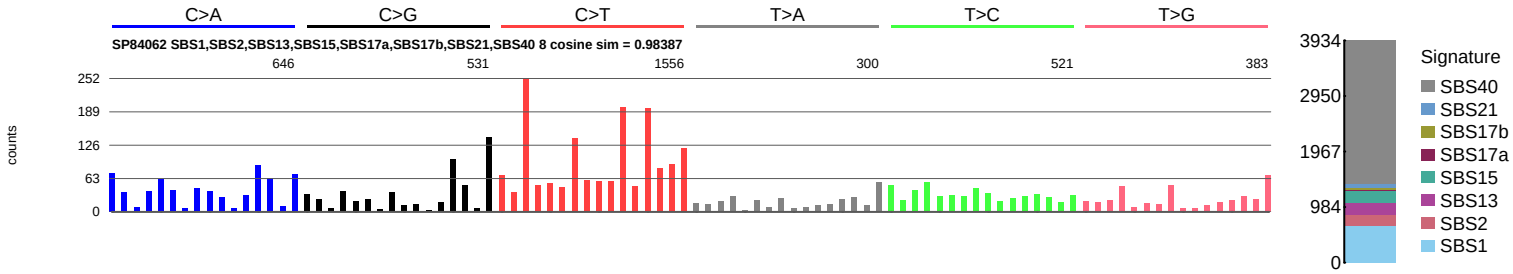

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 10 cosine sim = 0.9839
615
551
1568
314
510
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

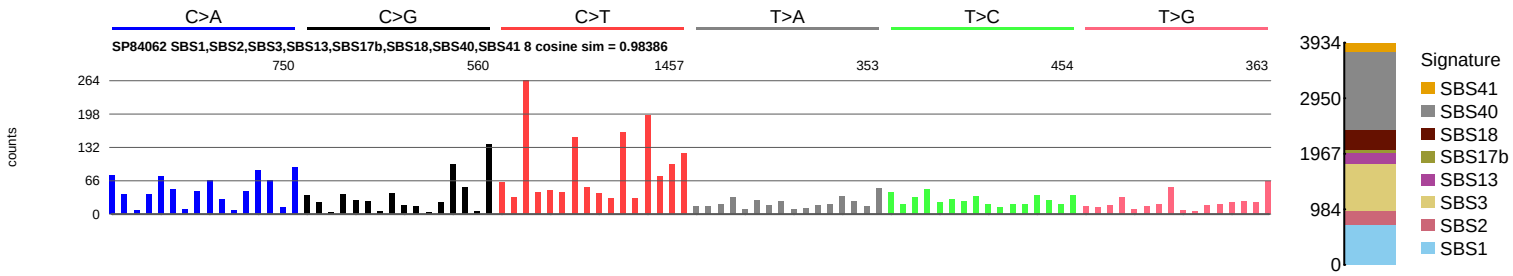

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9839
631
545
1551
312
515
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

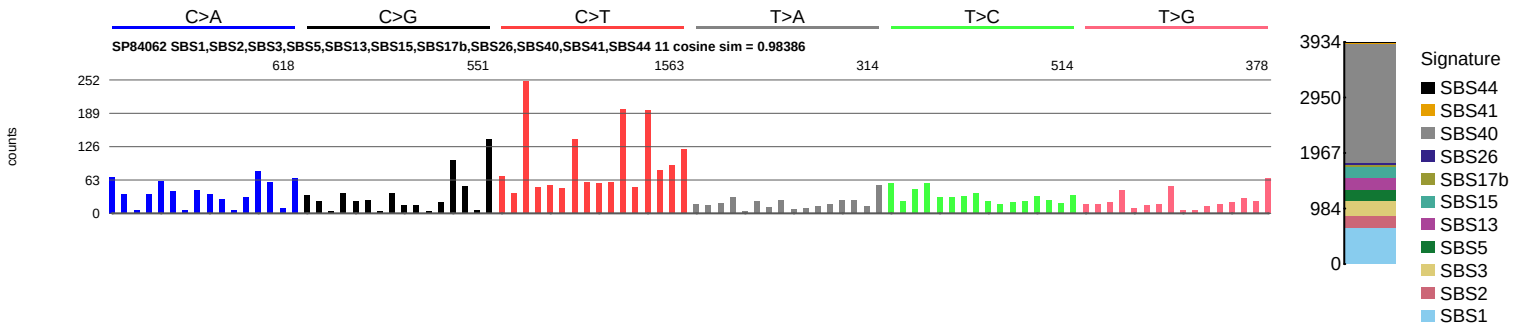

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.9839
723
537
1503
346
576
248
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

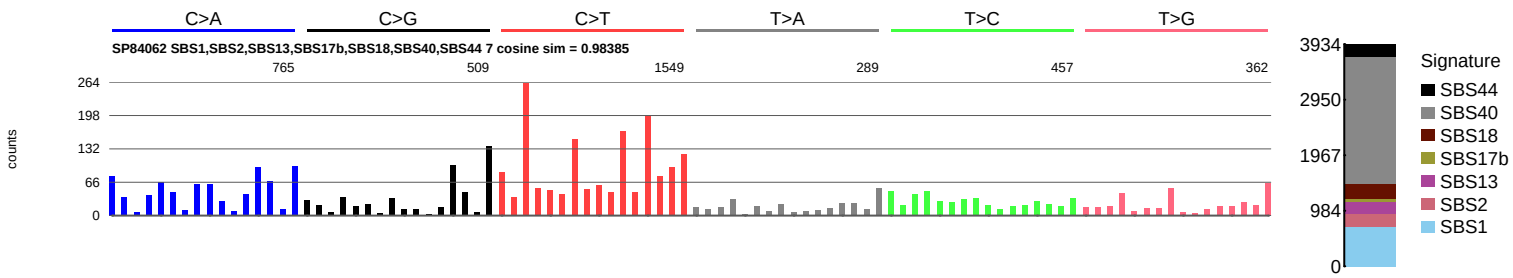

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98388
752
548
1457
309
518
348
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

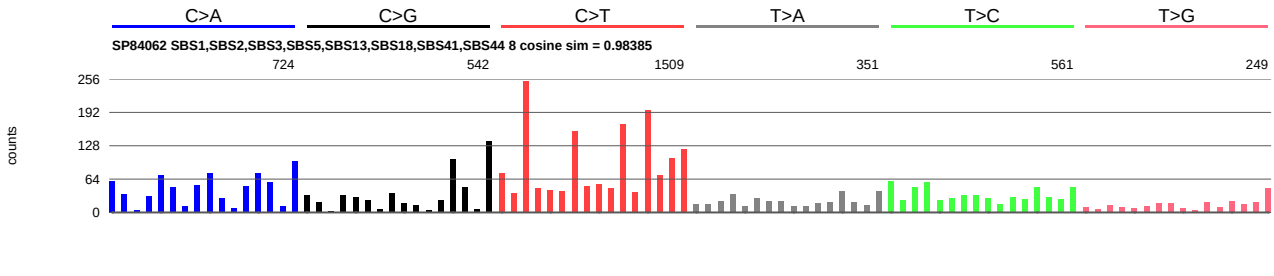

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98387
746
555
1452
352
463
362
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

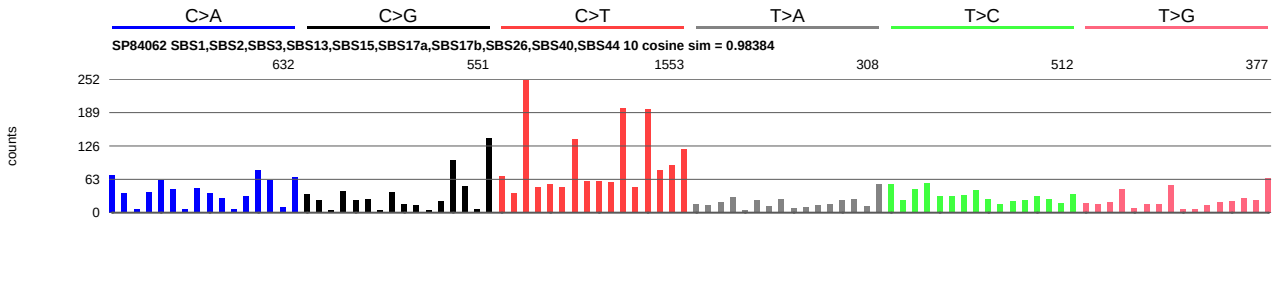

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98387
646
531
1556
300
521
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

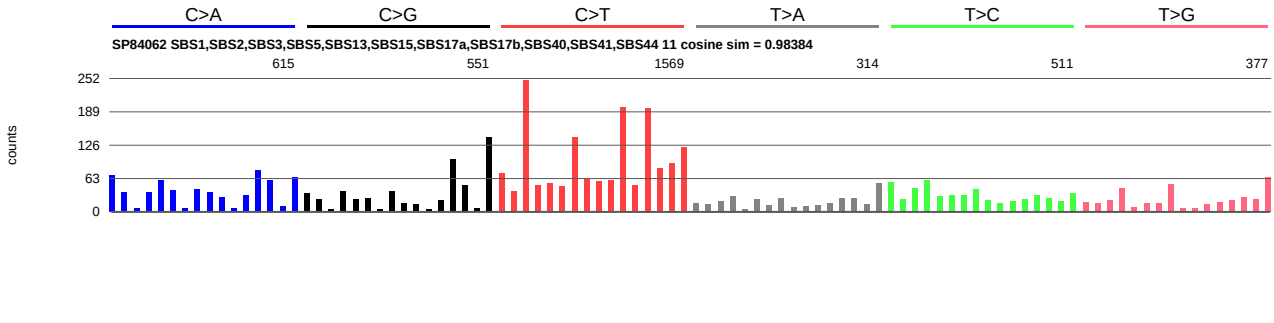

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98386
750
560
1457
353
454
363
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

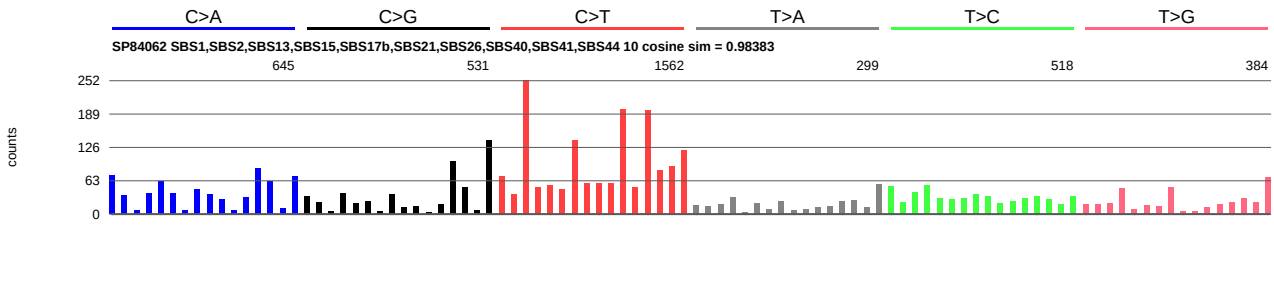

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98386
618
551
1563
314
514
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

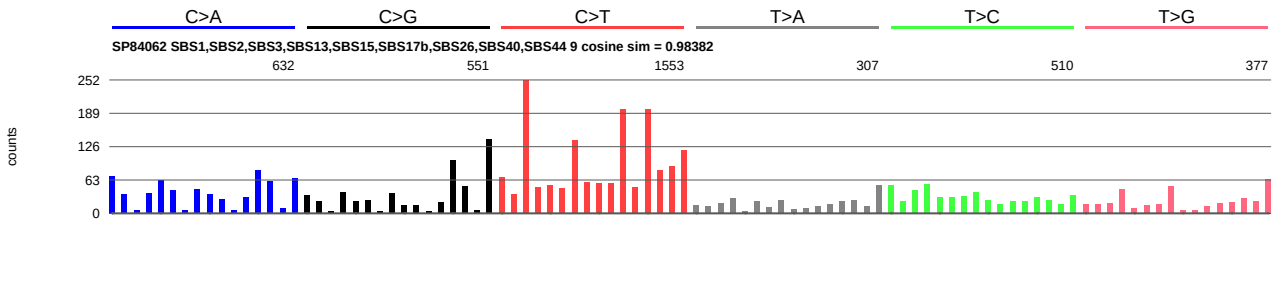

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98385
765
509
1549
289
457
362
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

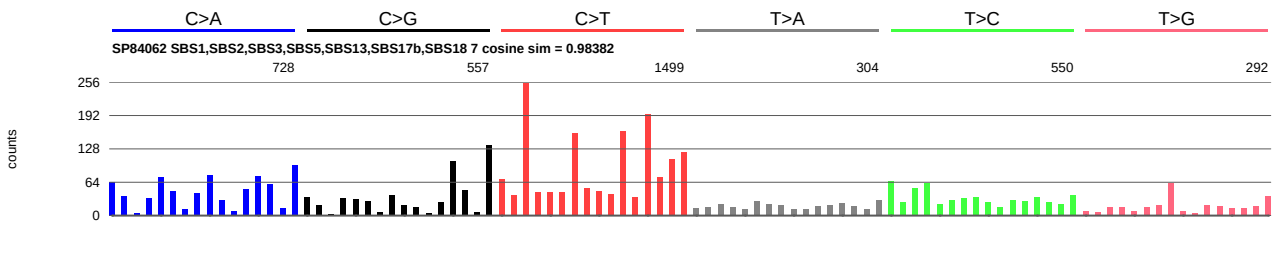

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 8 cosine sim = 0.98385
724
542
1509
351
561
249
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

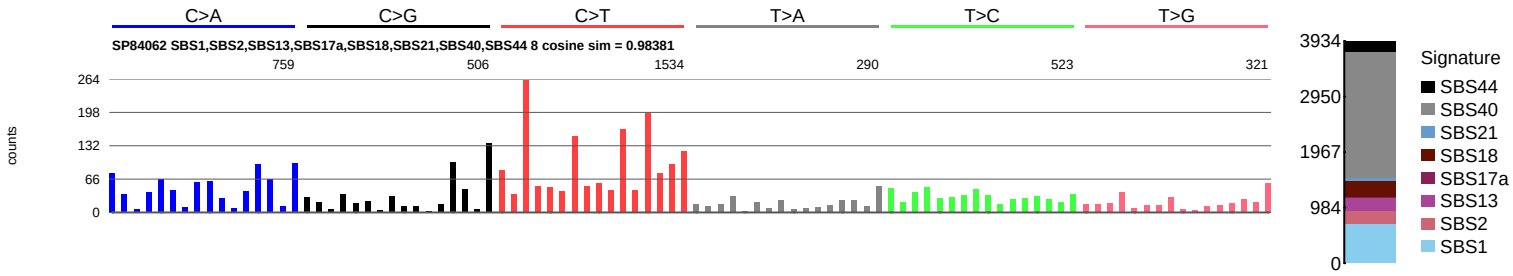

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98384
632
551
1553
308
512
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

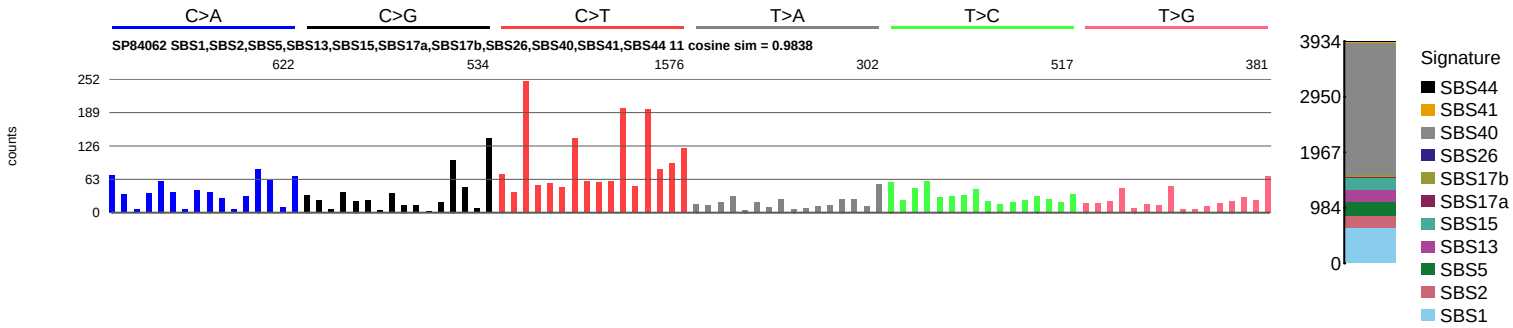

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 11 cosine sim = 0.98384
615
551
1569
314
511
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

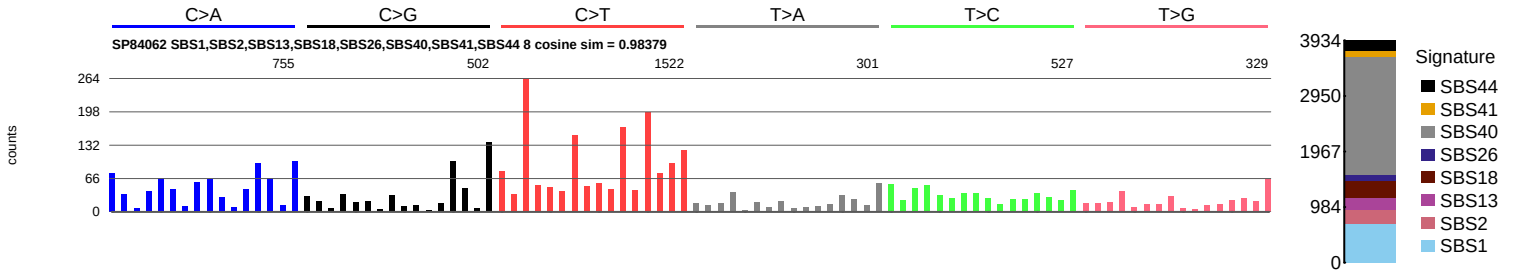

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98383
645
531
1562
299
518
384
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

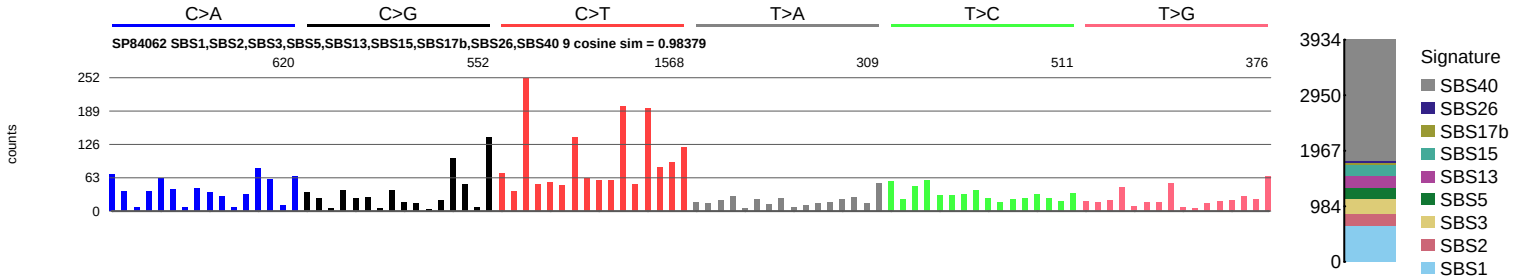

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98382
632
551
1553
307
510
377
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

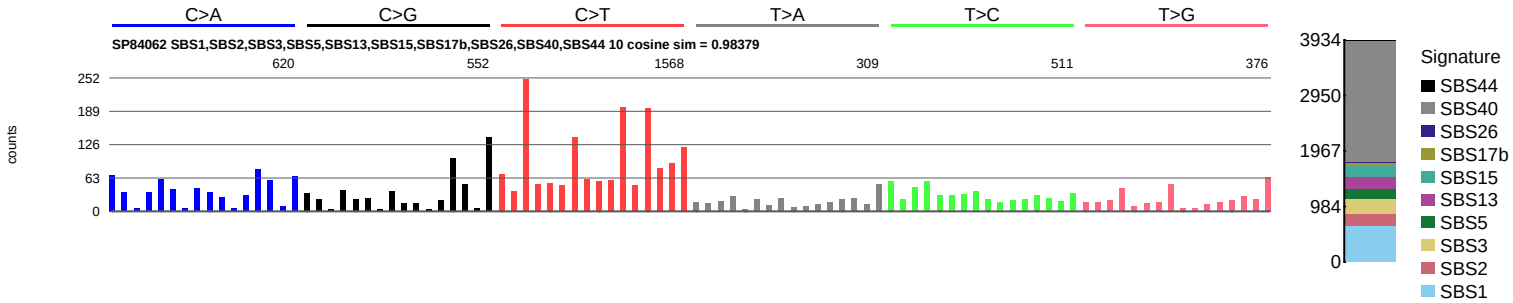

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98382
728
557
1499
304
550
292
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

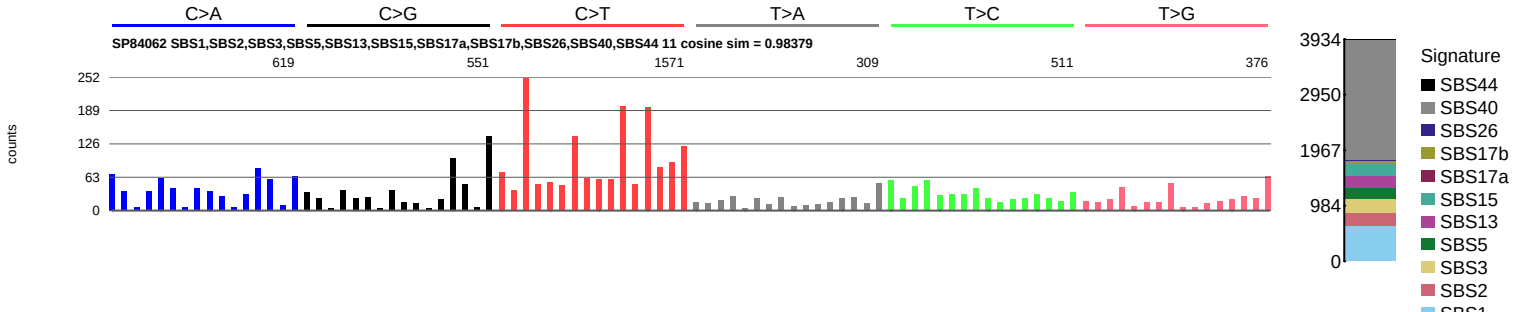

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98381
759
506
1534
290
523
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

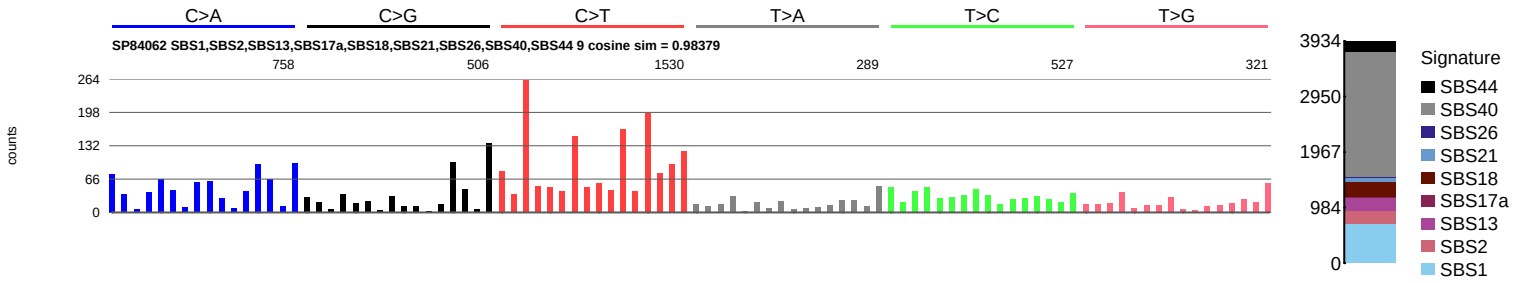

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9838
622
534
1576
302
517
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

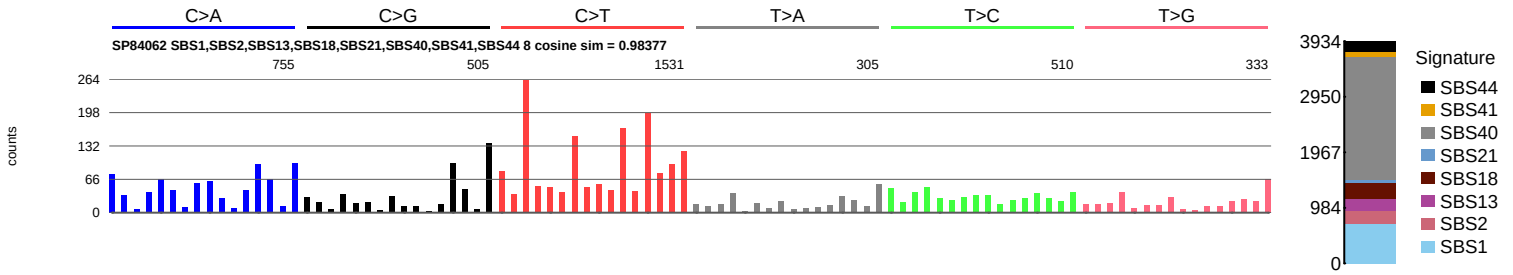

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98379
755
502
1522
301
527
329
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

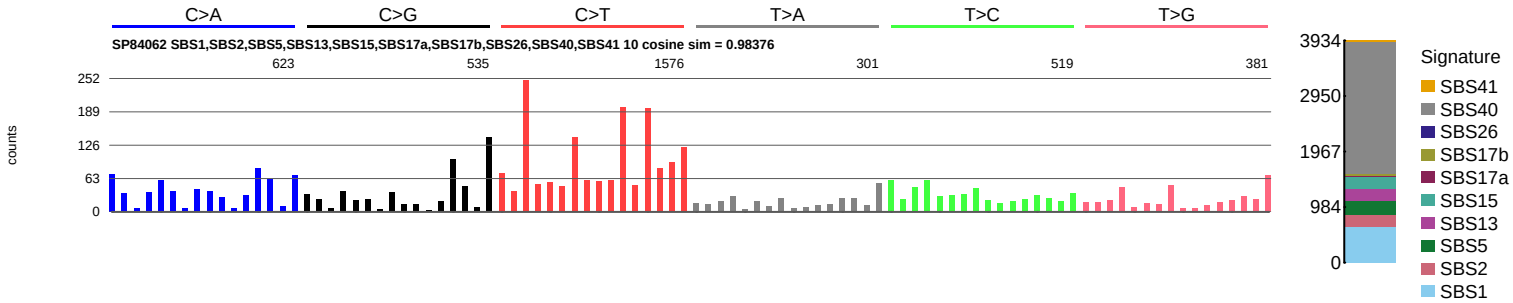

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40 9 cosine sim = 0.98379
620
552
1568
309
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

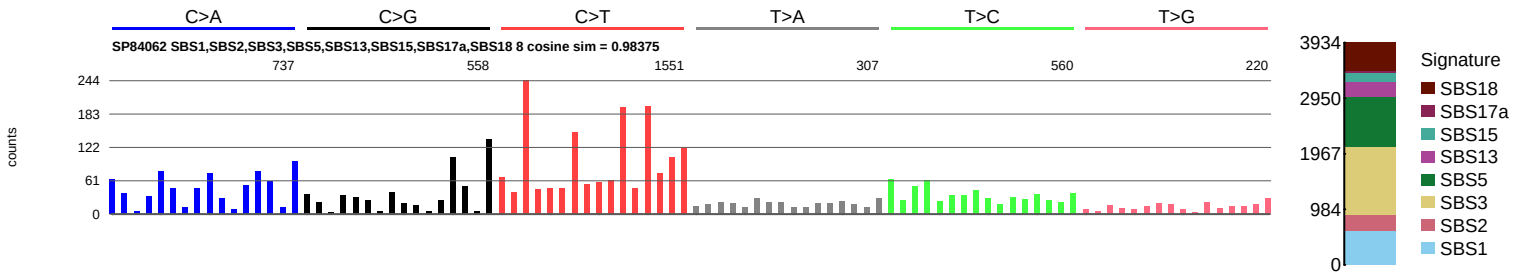

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98379
620
552
1568
309
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

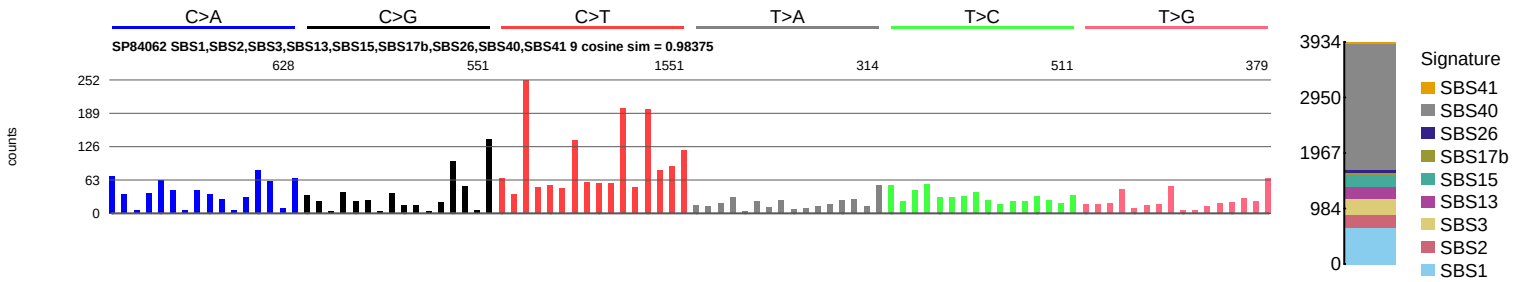

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 11 cosine sim = 0.98379
619
551
1571
309
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

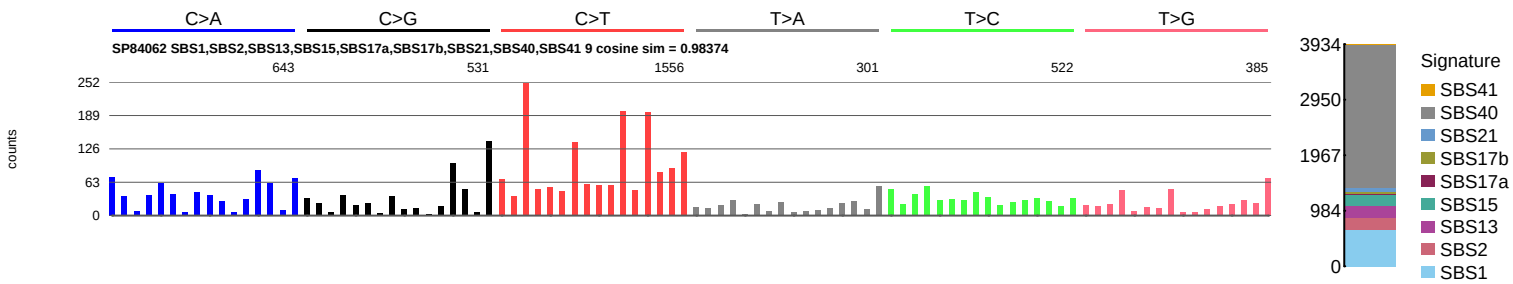

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98379
758
506
1530
289
527
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

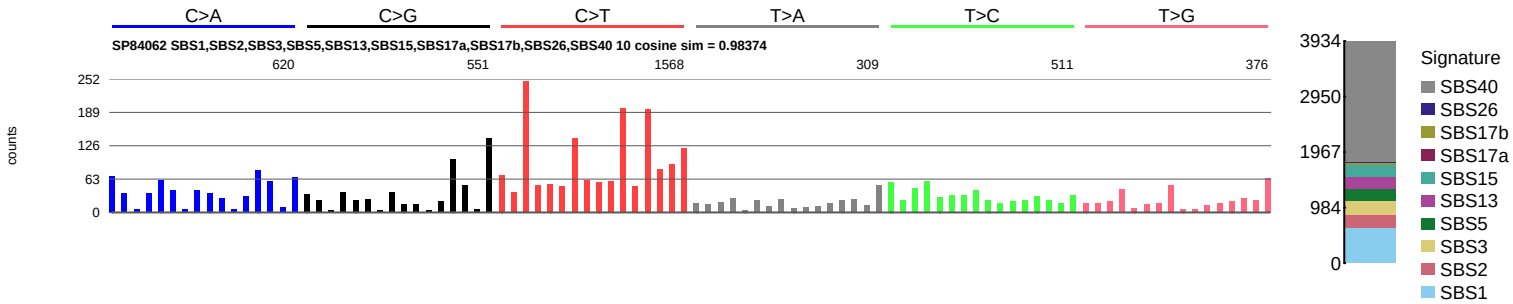

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98377
755
505
1531
305
510
333
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

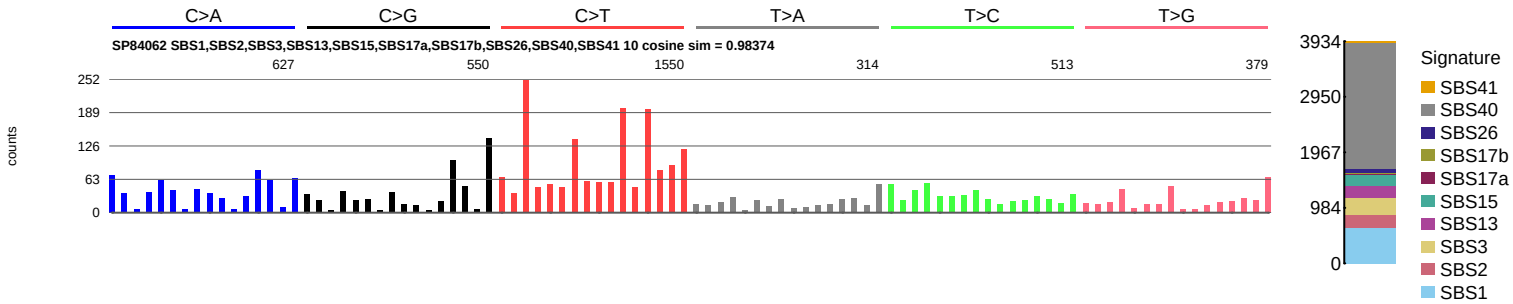

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98376
623
535
1576
301
519
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

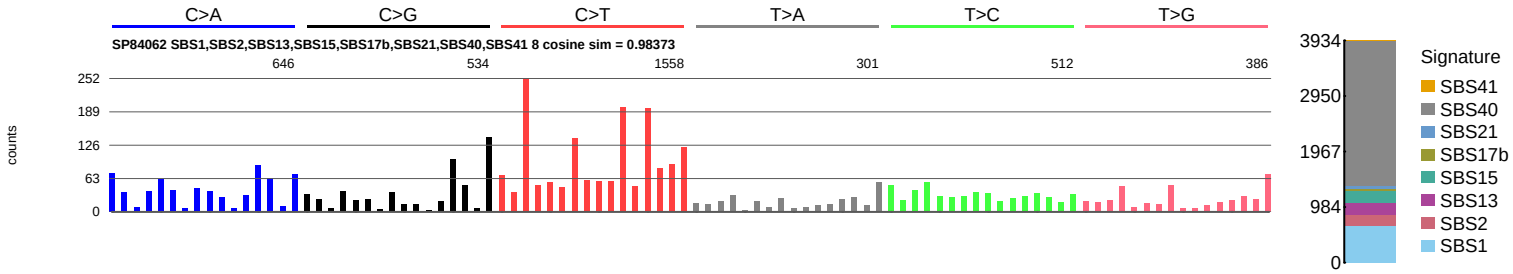

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18 8 cosine sim = 0.98375
737
558
1551
307
560
220
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

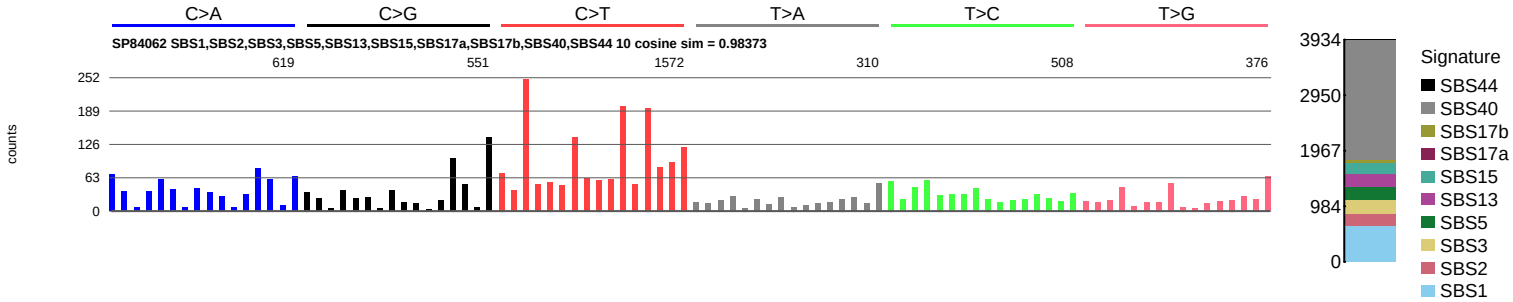

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98375
628
551
1551
314
511
379
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

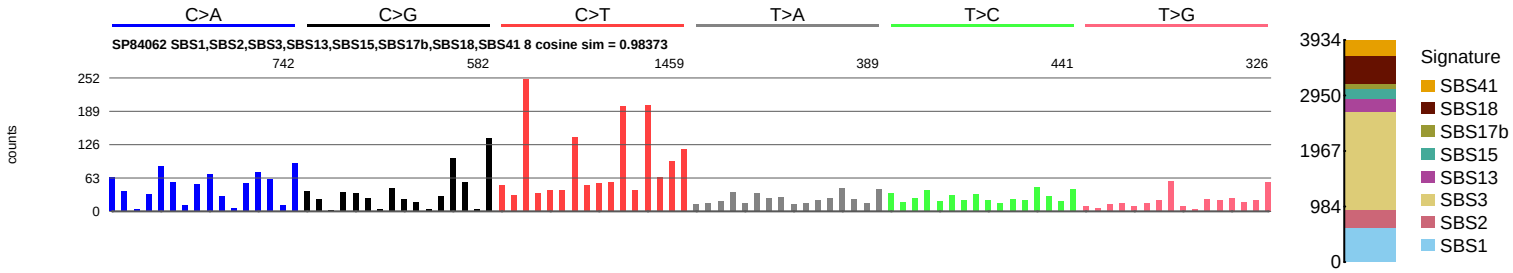

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98374
643
531
1556
301
522
385
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

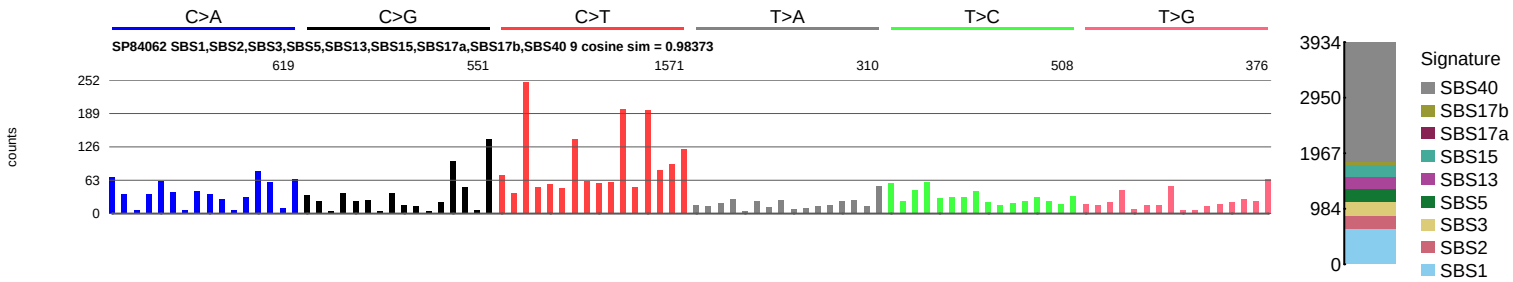

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 10 cosine sim = 0.98374
620
551
1568
309
511
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

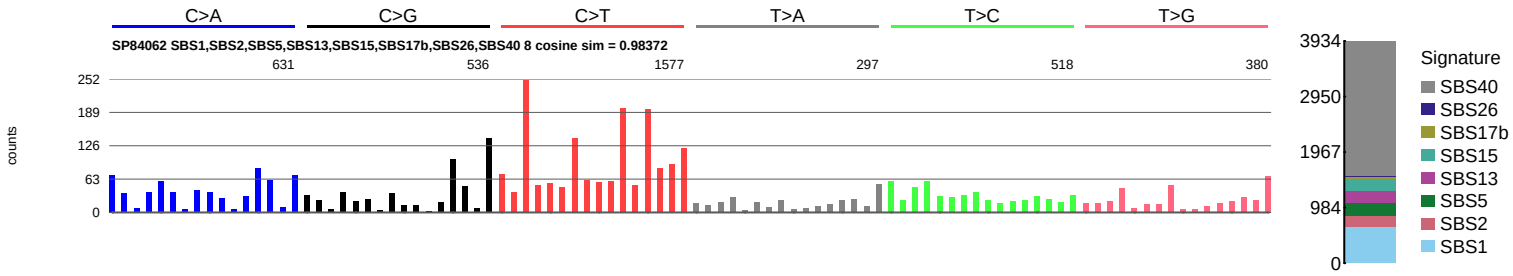

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98374
627
550
1550
314
513
379
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

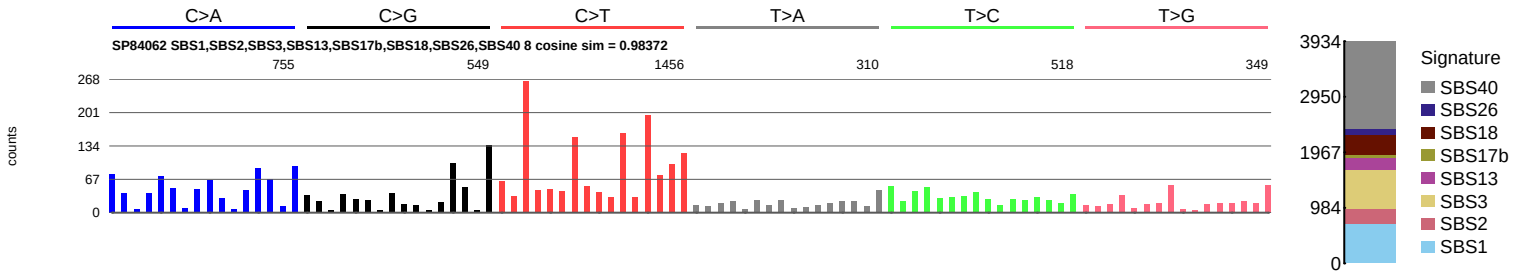

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98373
646
534
1558
301
512
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

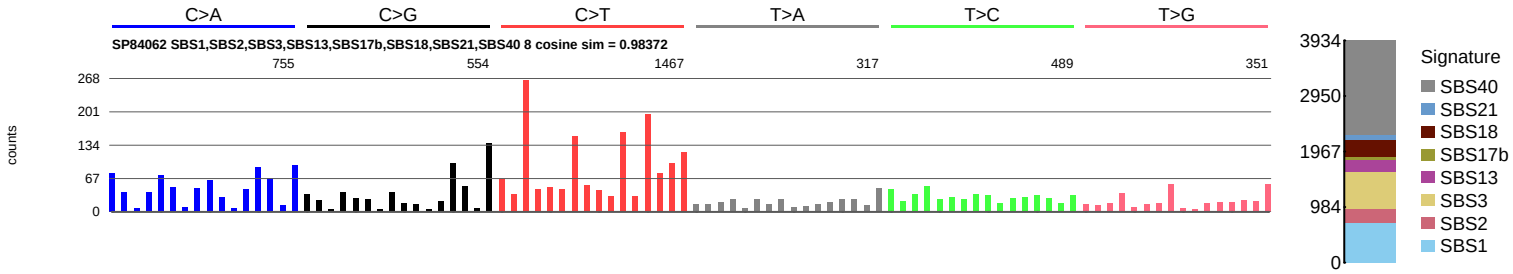

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 10 cosine sim = 0.98373
619
551
1572
310
508
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

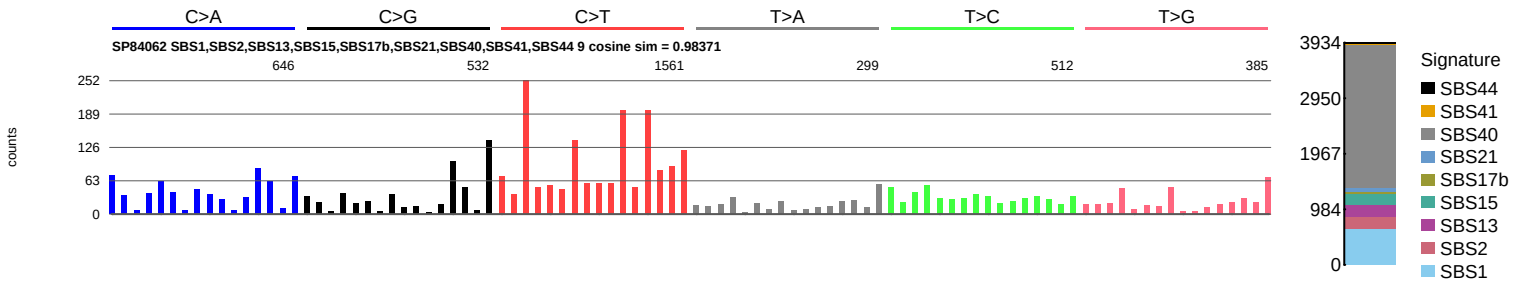

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98373
742
582
1459
389
441
326
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

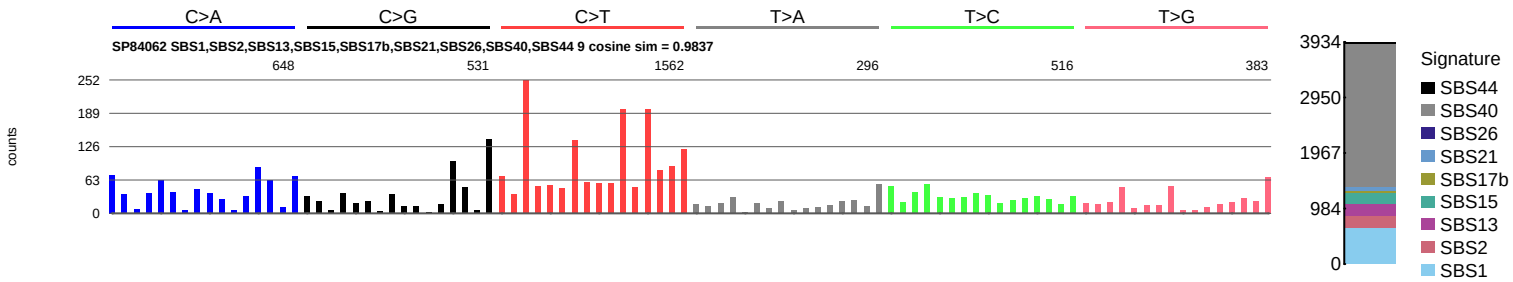

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.98373
619
551
1571
310
508
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

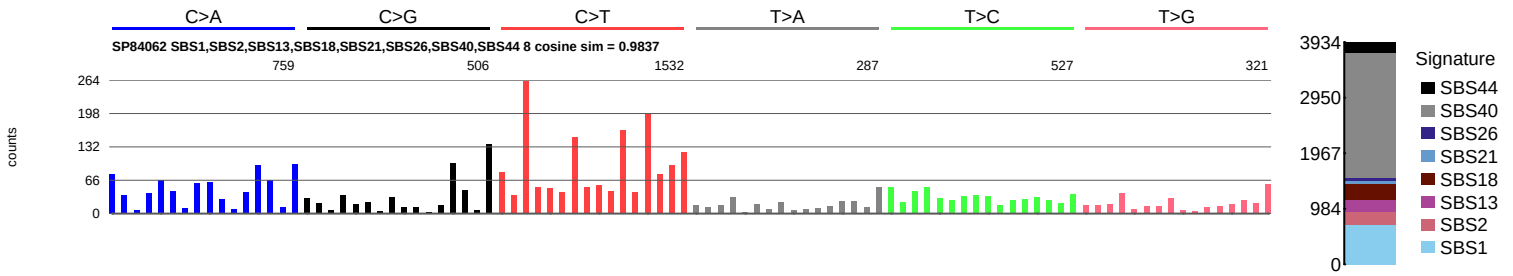

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40 8 cosine sim = 0.98372
631
536
1577
297
518
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

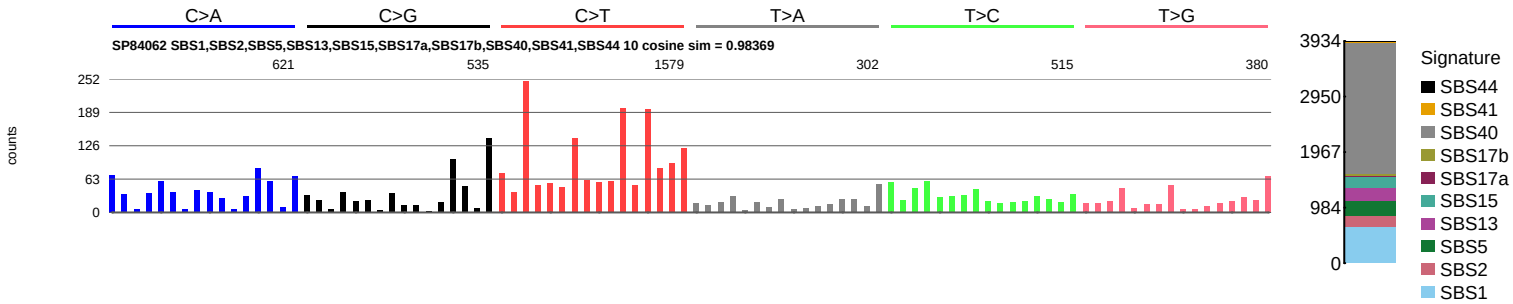

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98372
755
549
1456
310
518
349
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

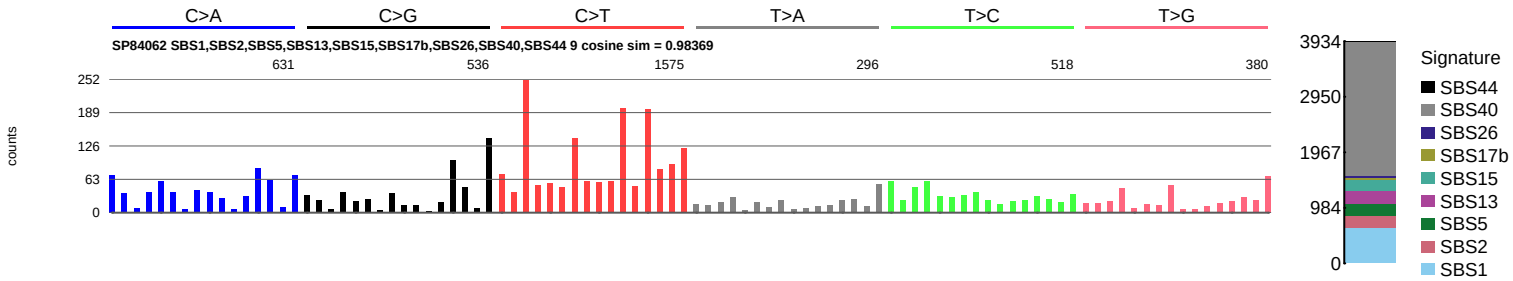

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98372
755
554
1467
317
489
351
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

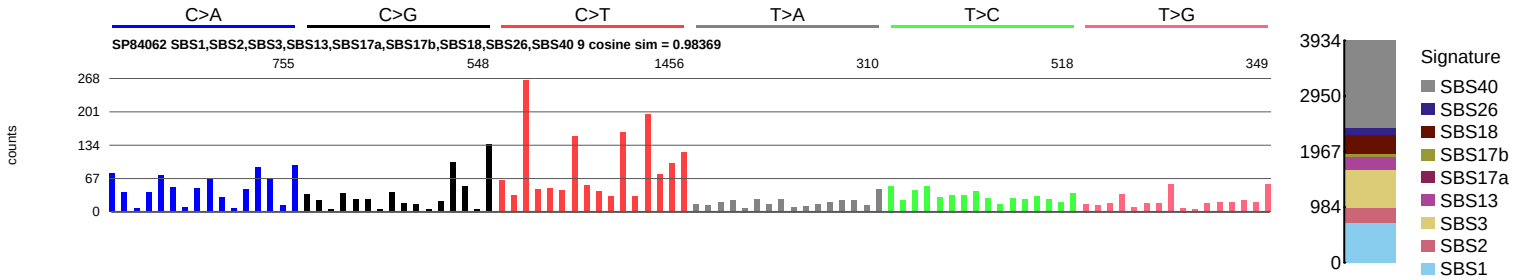

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98371
646
532
1561
299
512
385
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

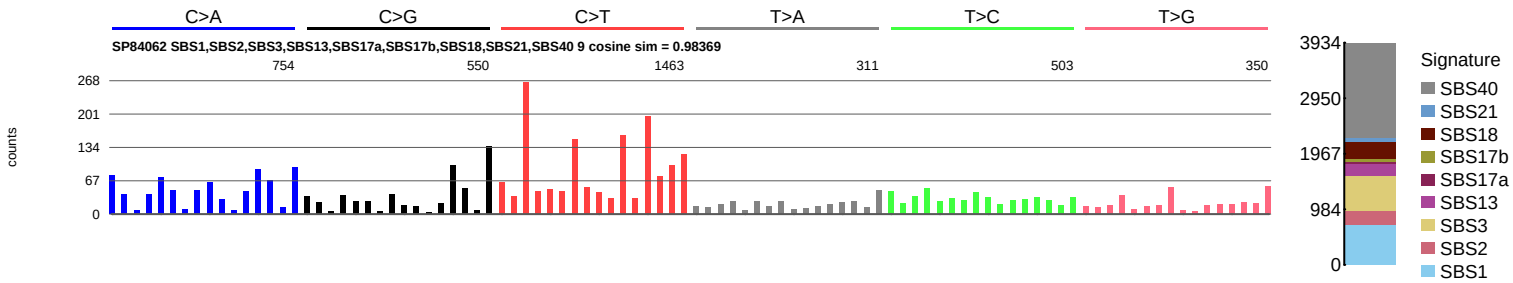

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9837
648
531
1562
296
516
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

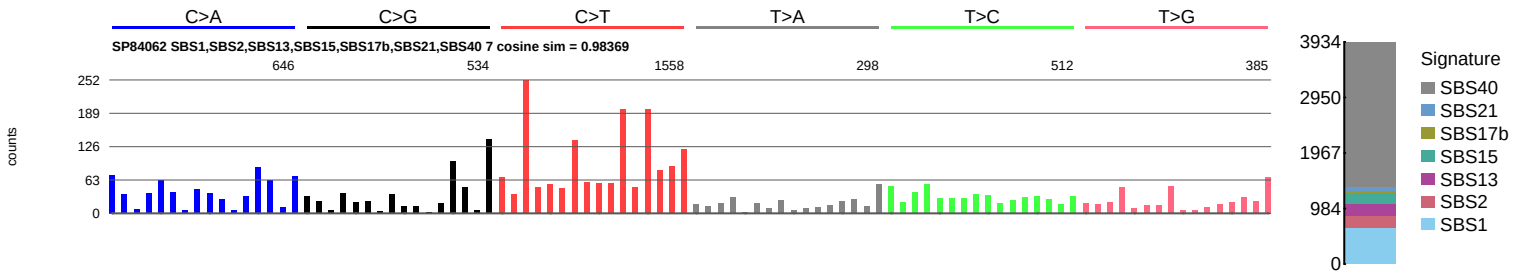

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.9837
759
506
1532
287
527
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

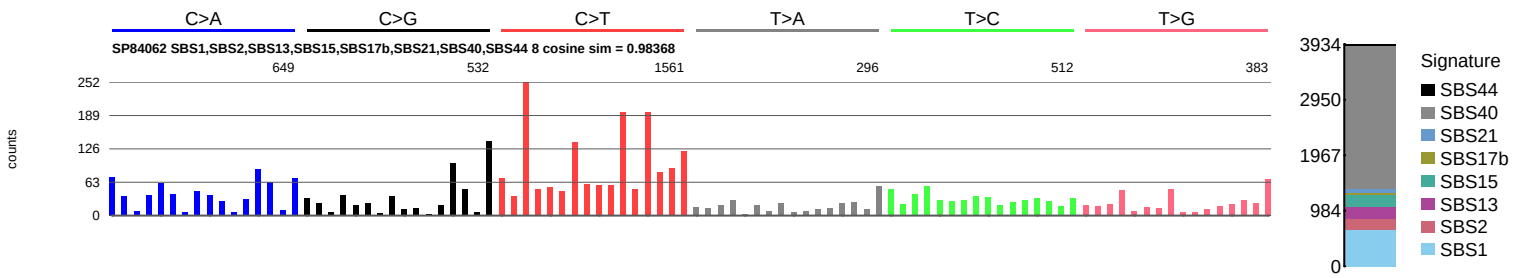

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98369
621
535
1579
302
515
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

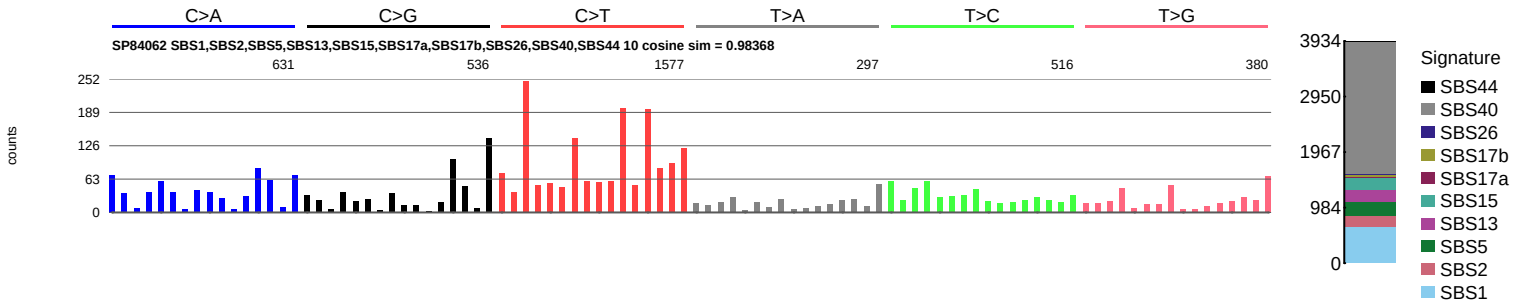

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98369
631
536
1575
296
518
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

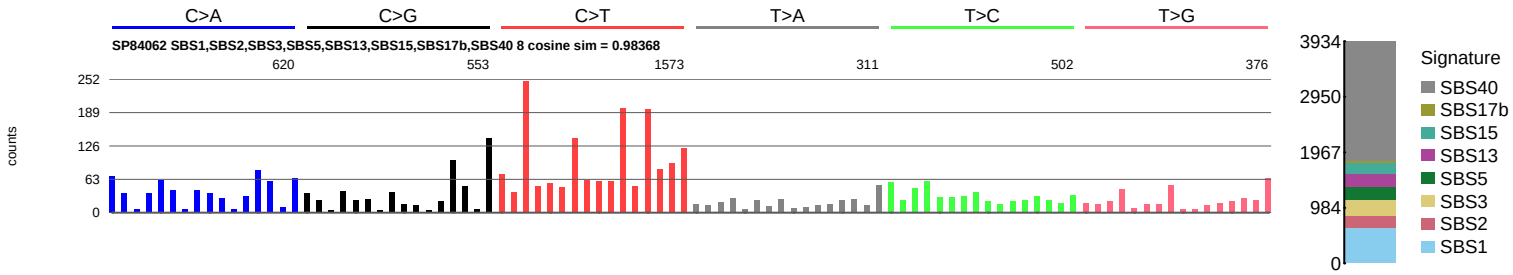

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98369
755
548
1456
310
518
349
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

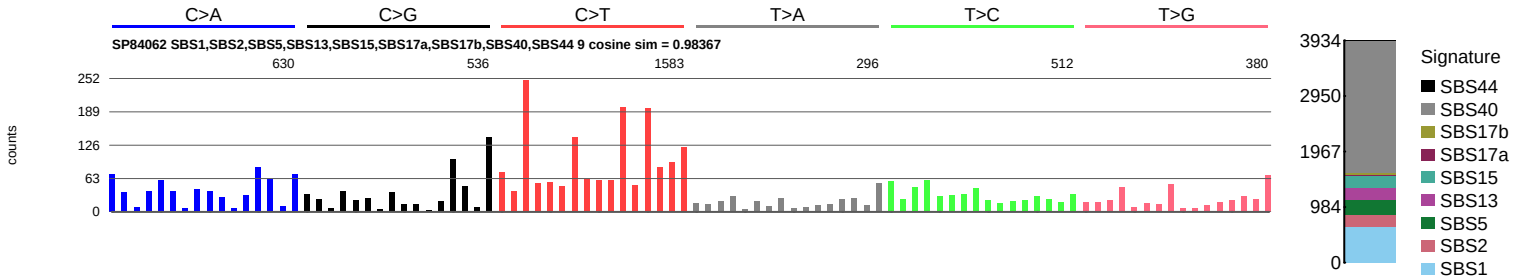

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98369
754
550
1463
311
503
350
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

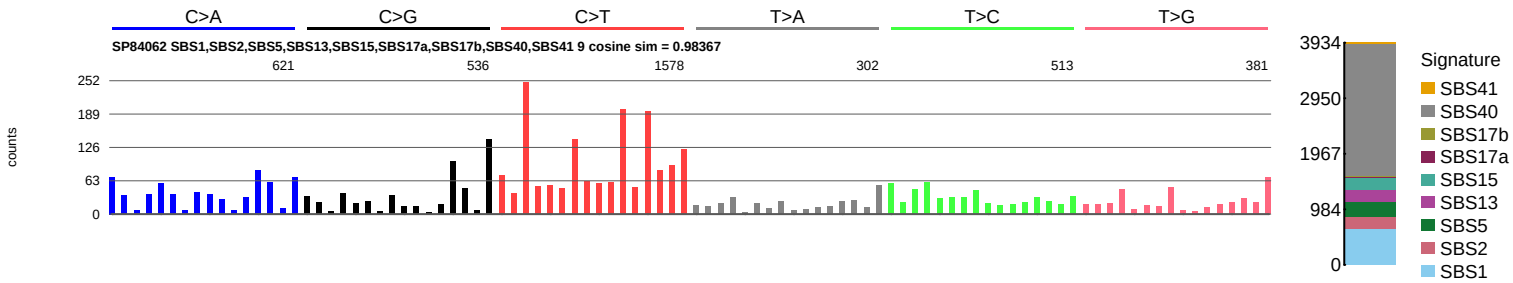

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS40 7 cosine sim = 0.98369
646
534
1558
298
512
385
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

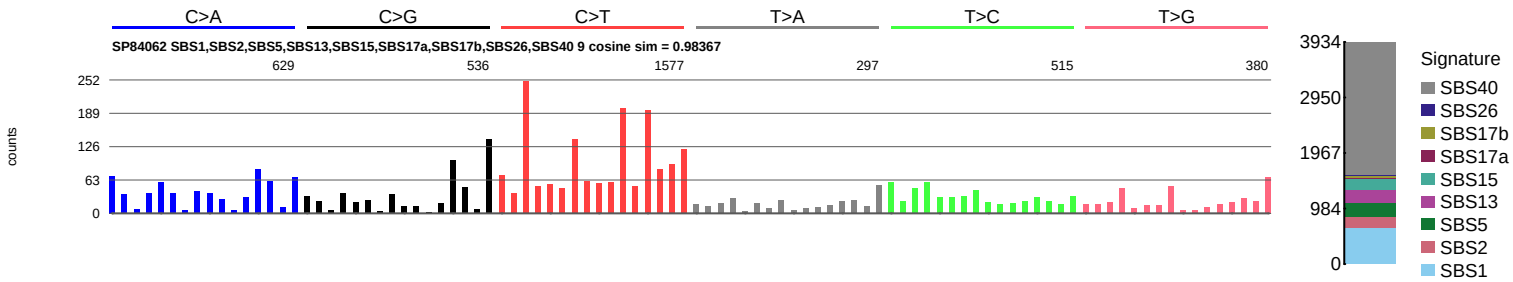

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98368
649
532
1561
296
512
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

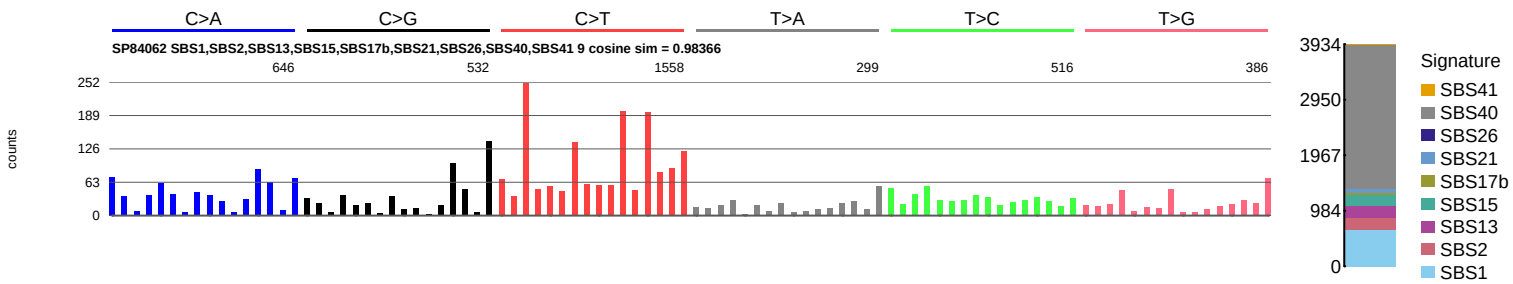

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98368
631
536
1577
297
516
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

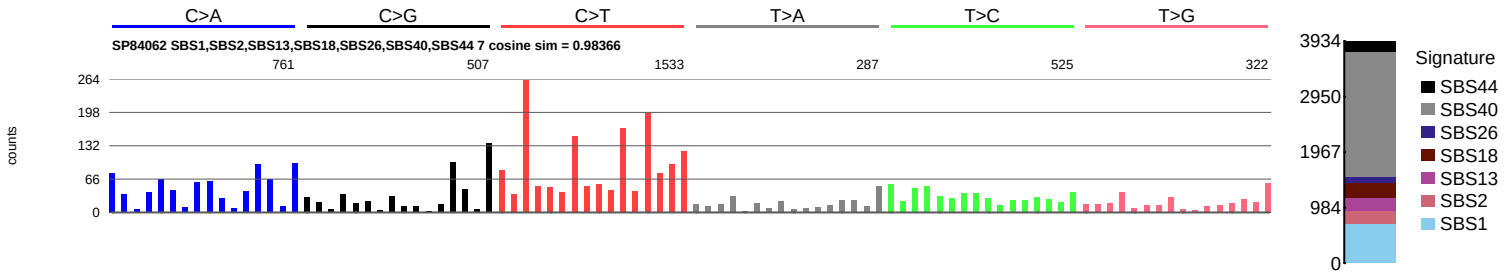

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40 8 cosine sim = 0.98368
620
553
1573
311
502
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

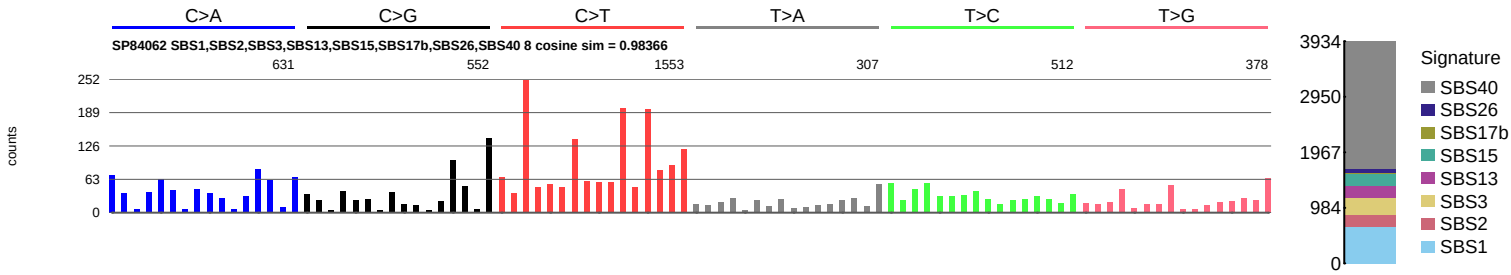

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98367
630
536
1583
296
512
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

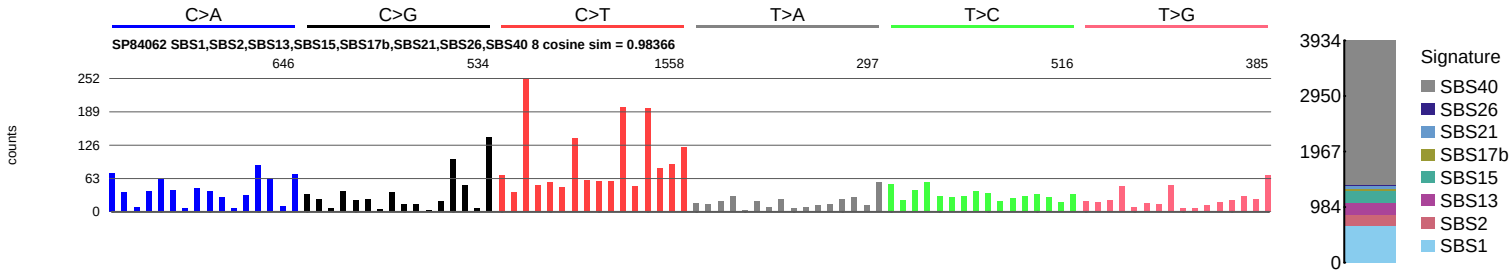

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98367
621
536
1578
302
513
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

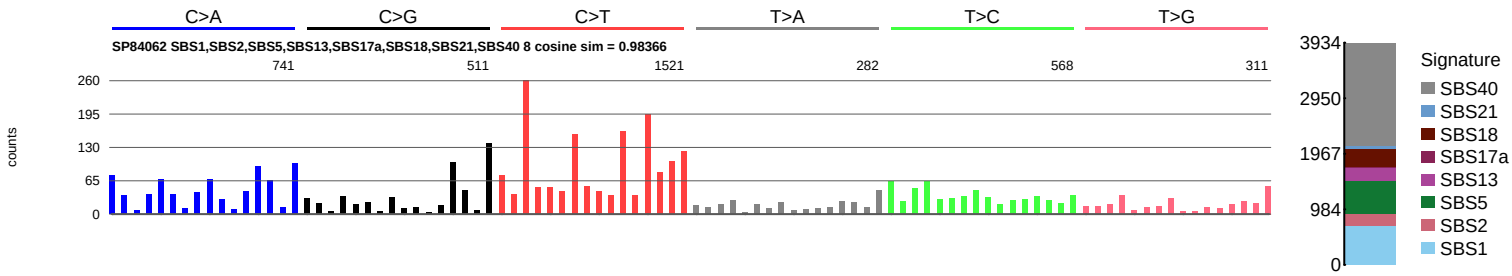

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98367
629
536
1577
297
515
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

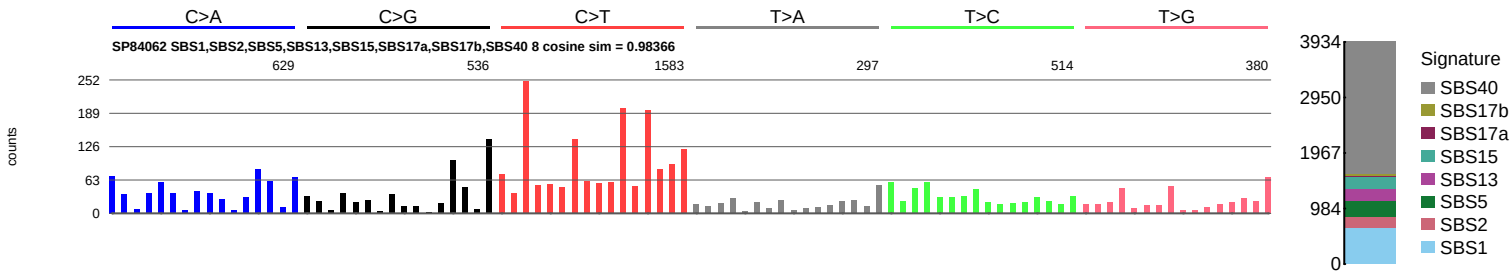

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98366
646
532
1558
299
516
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

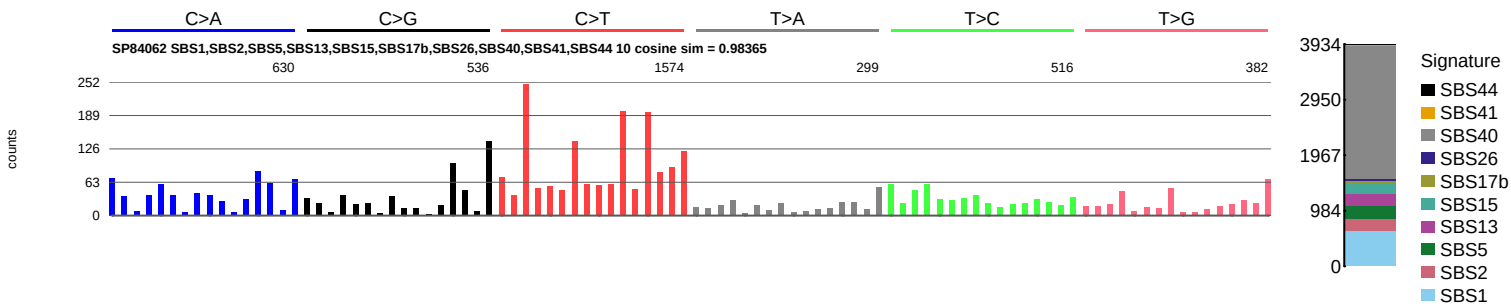

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98366
761
507
1533
287
525
322
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

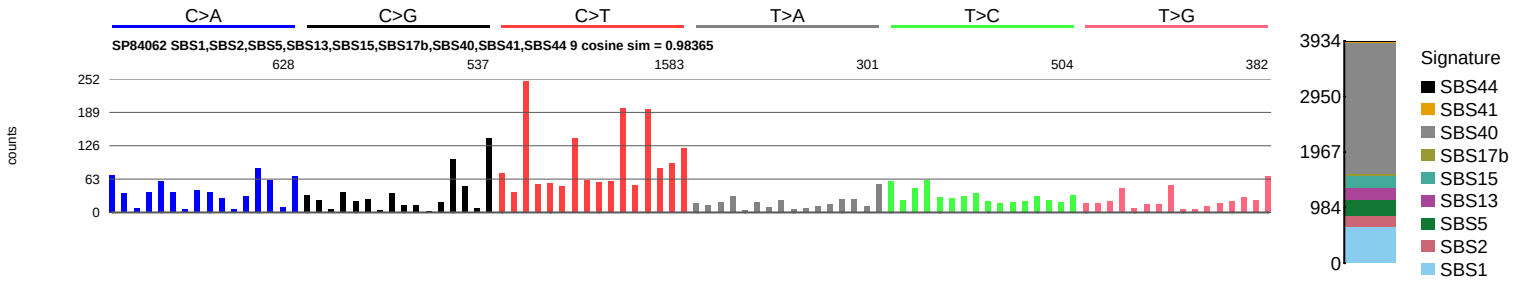

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS26,SBS40 8 cosine sim = 0.98366
631
552
1553
307
512
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

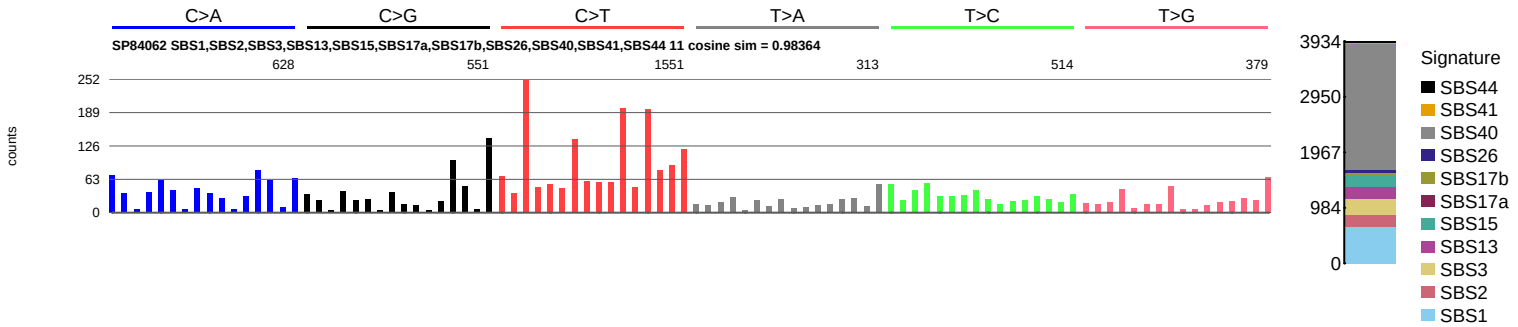

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98366
646
534
1558
297
516
385
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS13
SBS2
SBS1

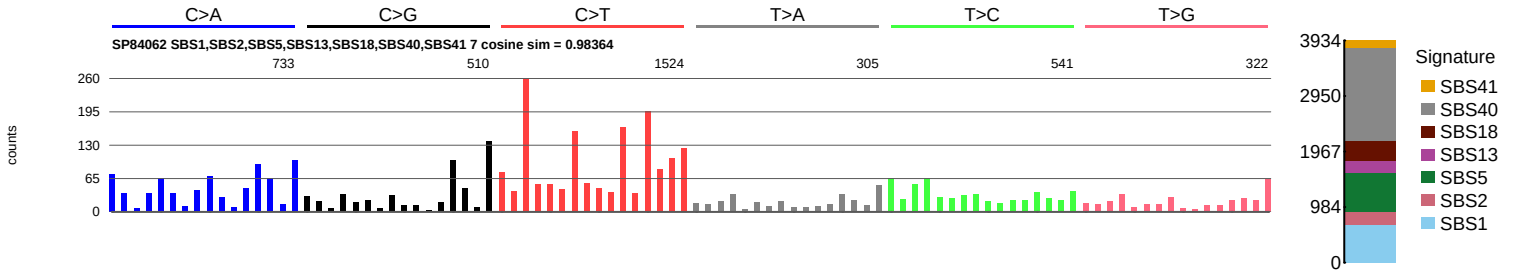

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98366
741
511
1521
282
568
311
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

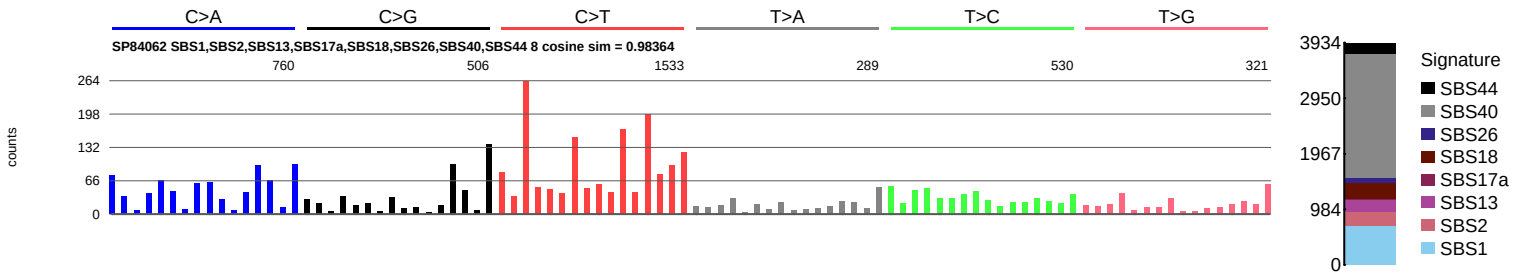

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98366
629
536
1583
297
514
380
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

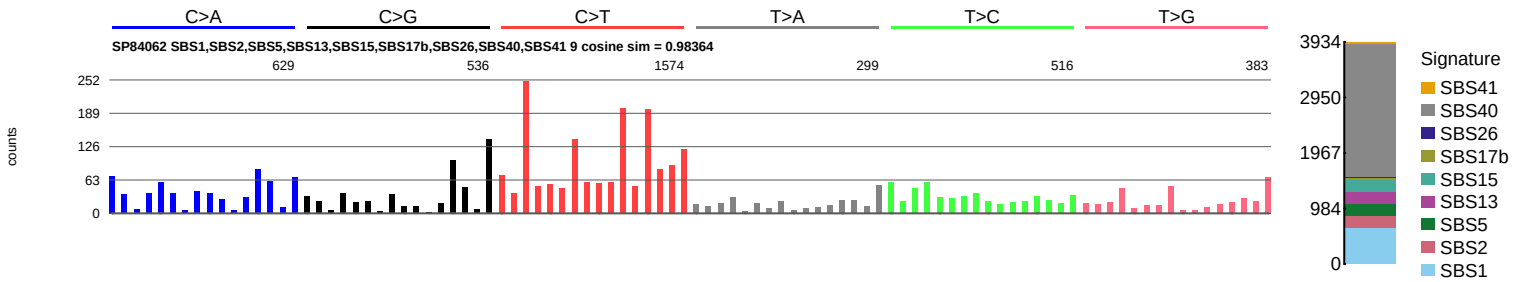

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98365
630
536
1574
299
516
382
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

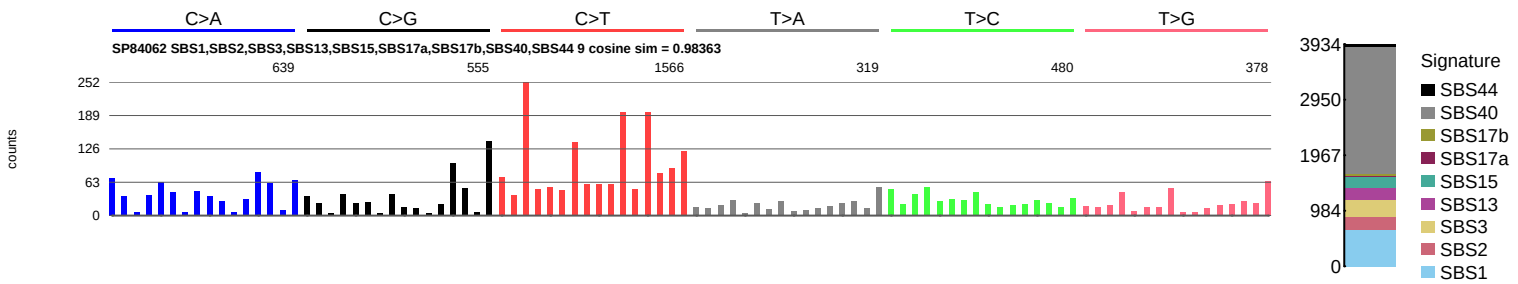

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98365
628
537
1583
301
504
382
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

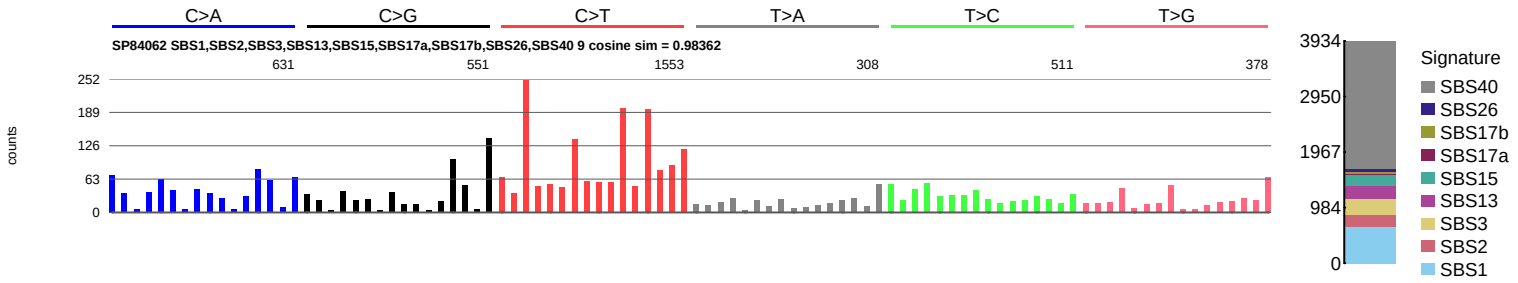

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98364
628
551
1551
313
514
379
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

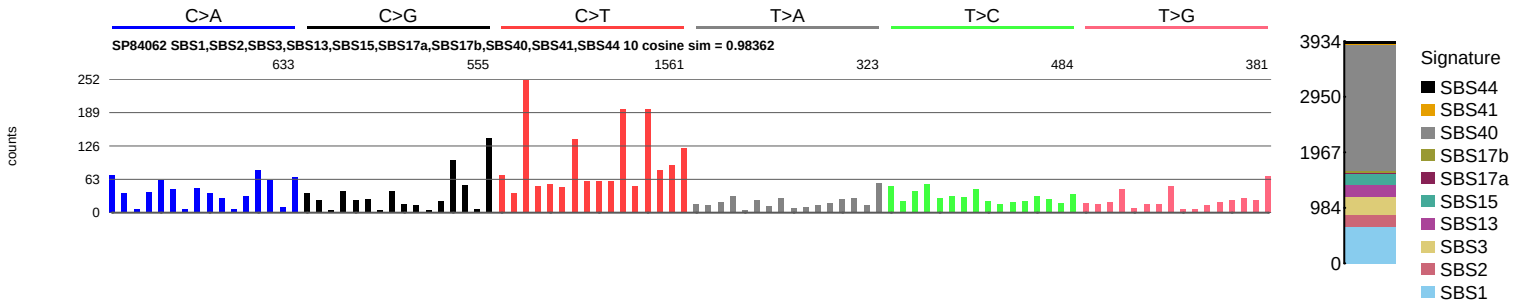

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98364
733
510
1524
305
541
322
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

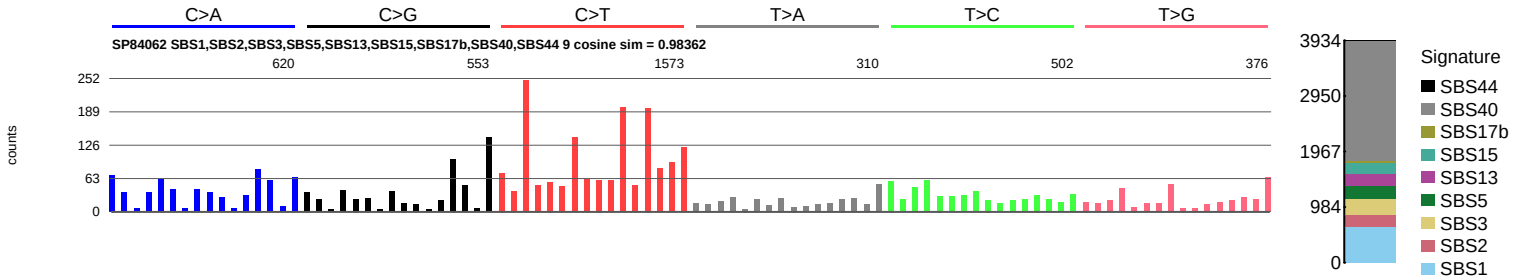

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98364
760
506
1533
289
530
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1

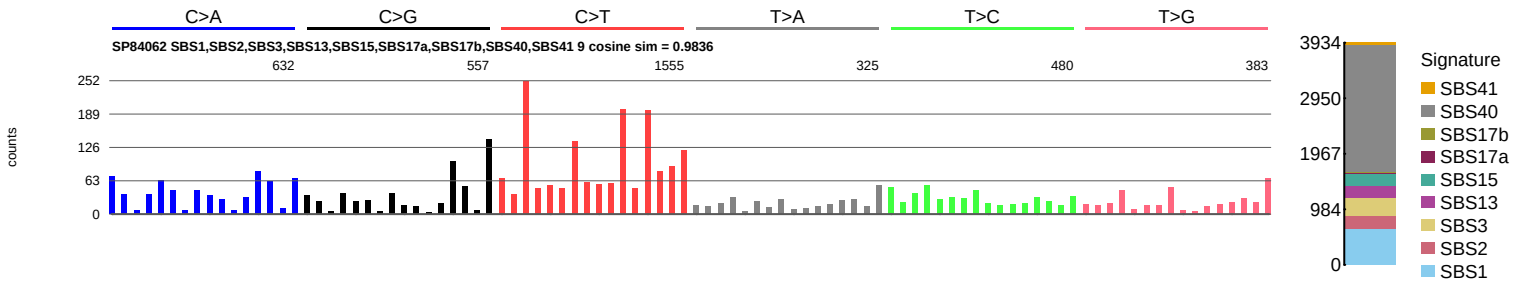

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98364
629
536
1574
299
516
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

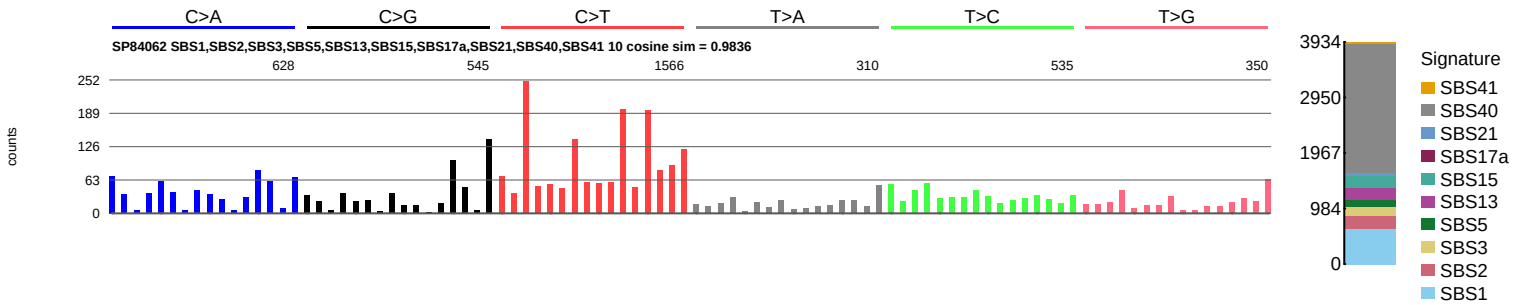

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98363
639
555
1566
319
480
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

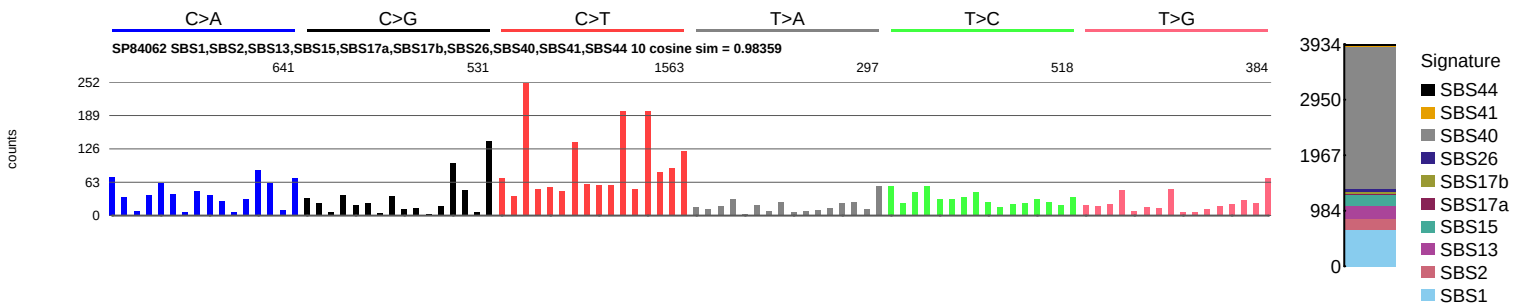

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98362
631
551
1553
308
511
378
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

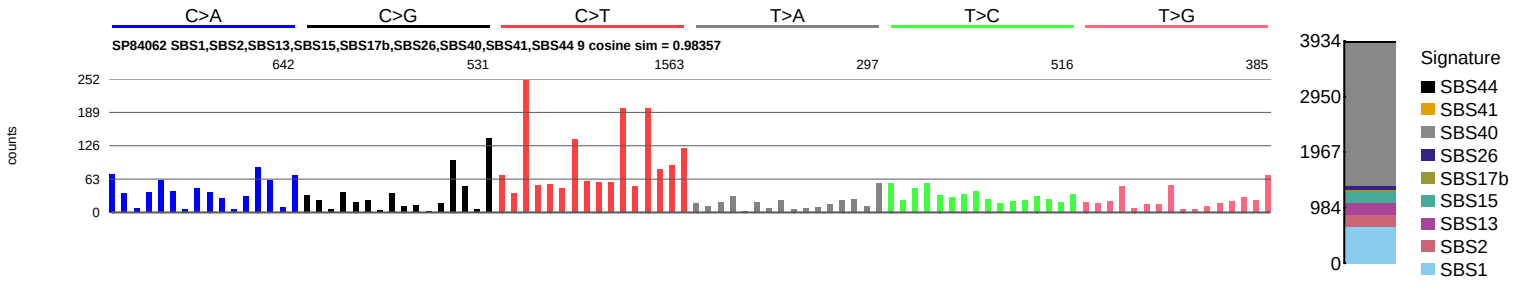

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98362
633
555
1561
323
484
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

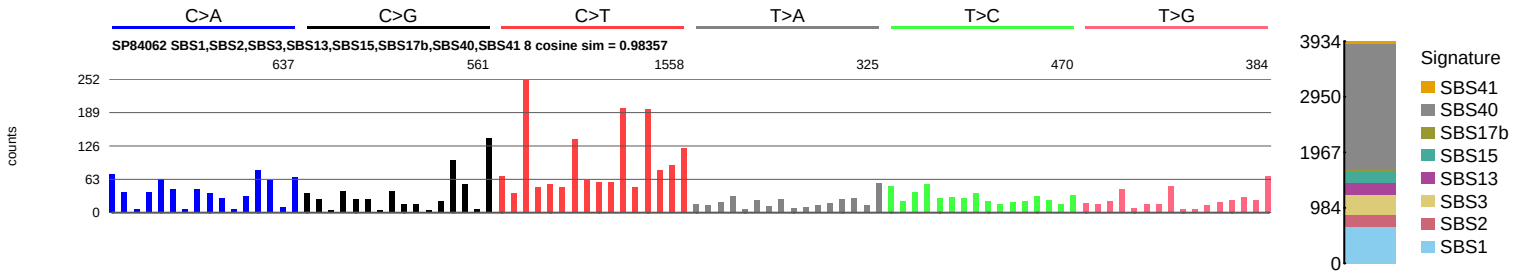

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 9 cosine sim = 0.98362
620
553
1573
310
502
376
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

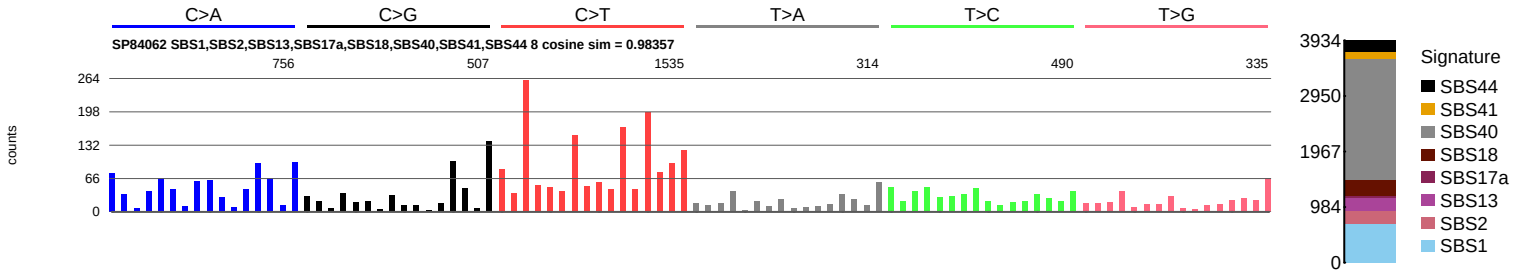

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.9836
632
557
1555
325
480
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

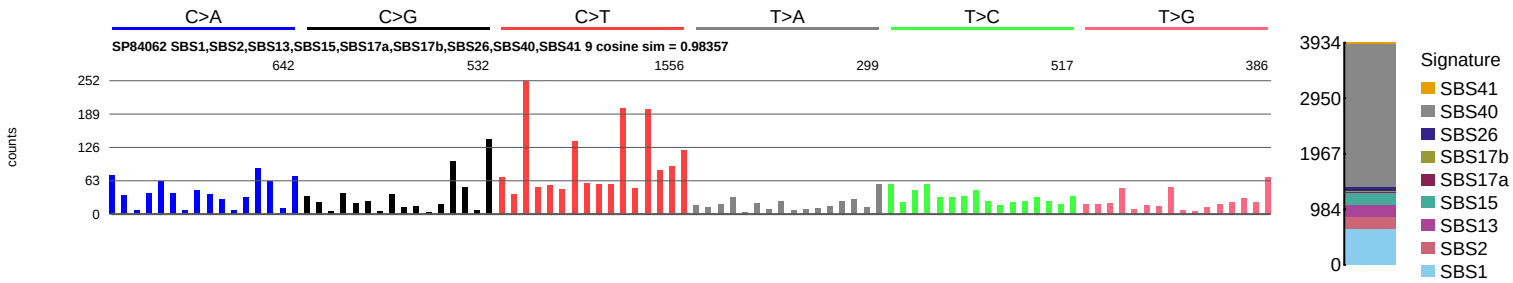

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 10 cosine sim = 0.9836
628
545
1566
310
535
350
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

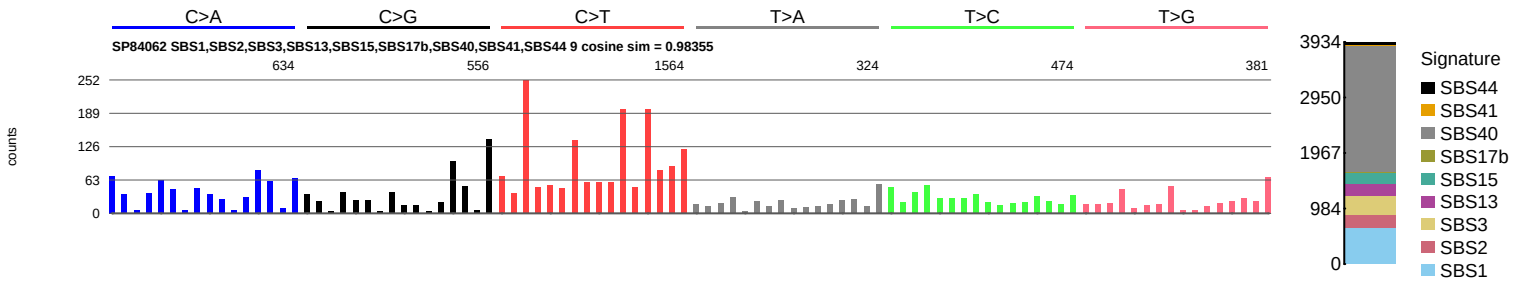

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98359
641
531
1563
297
518
384
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

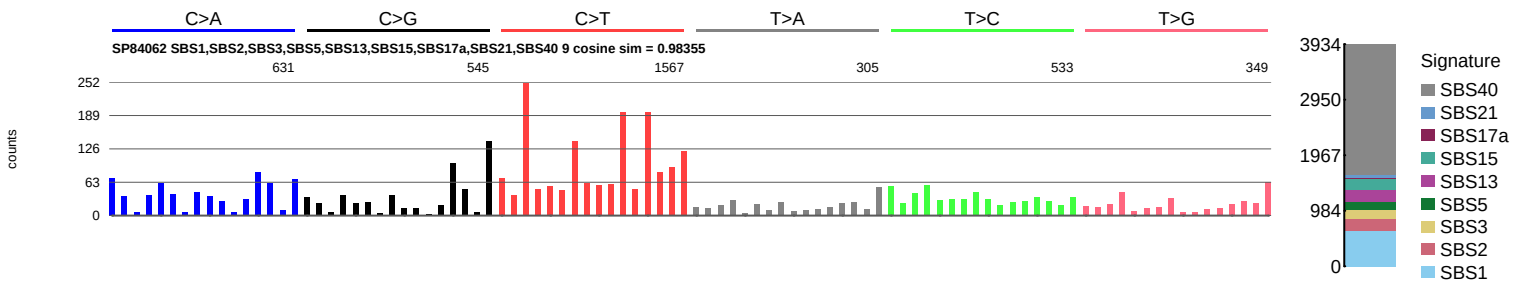

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98357
642
531
1563
297
516
385
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1

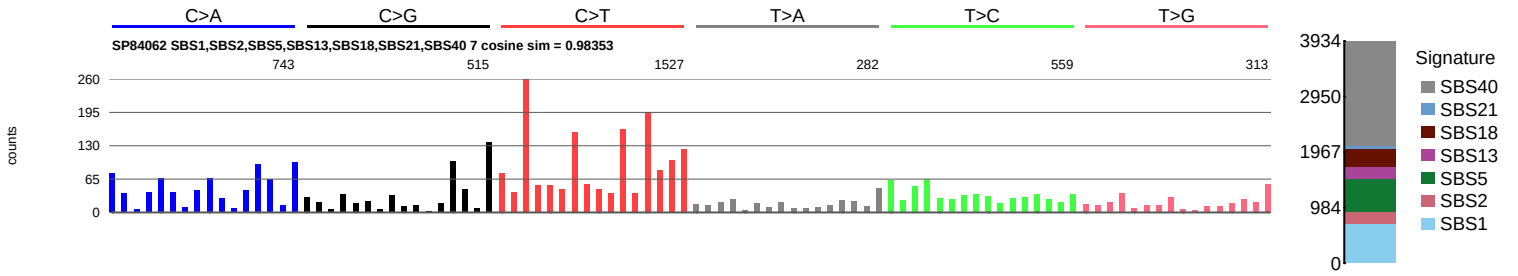

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS40,SBS41 8 cosine sim = 0.98357
637
561
1558
325
470
384
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

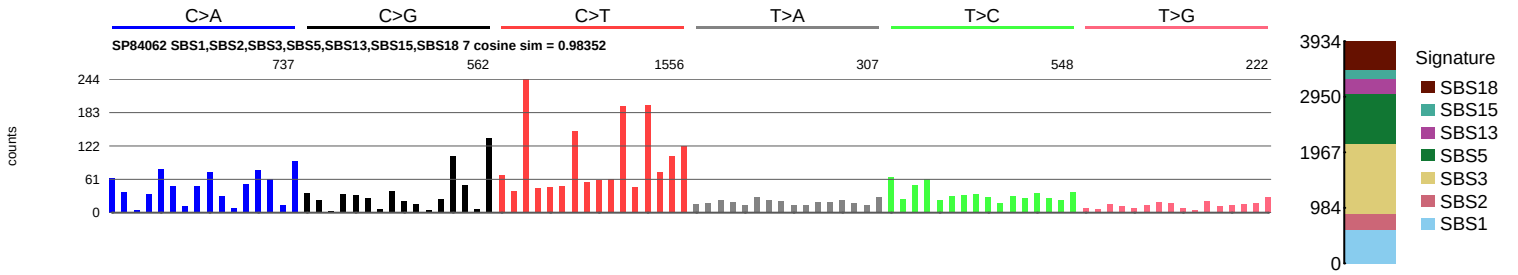

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98357
756
507
1535
314
490
335
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1

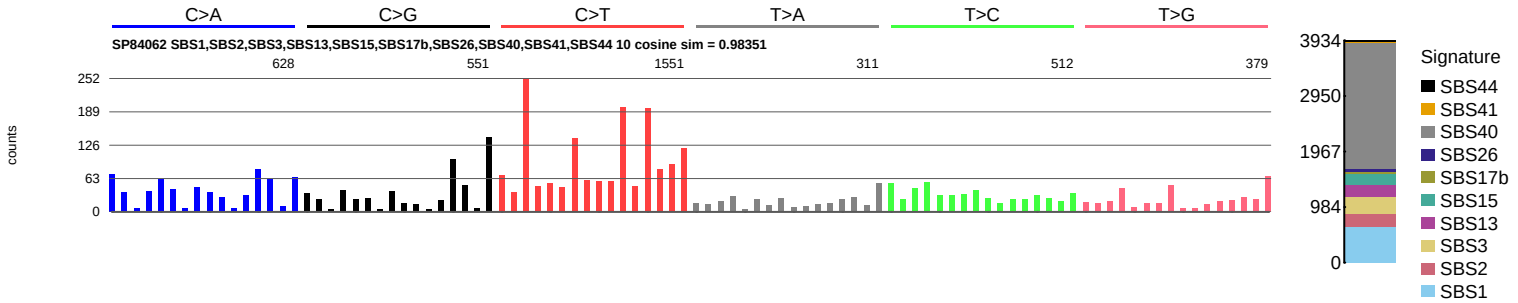

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98357
642
532
1556
299
517
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

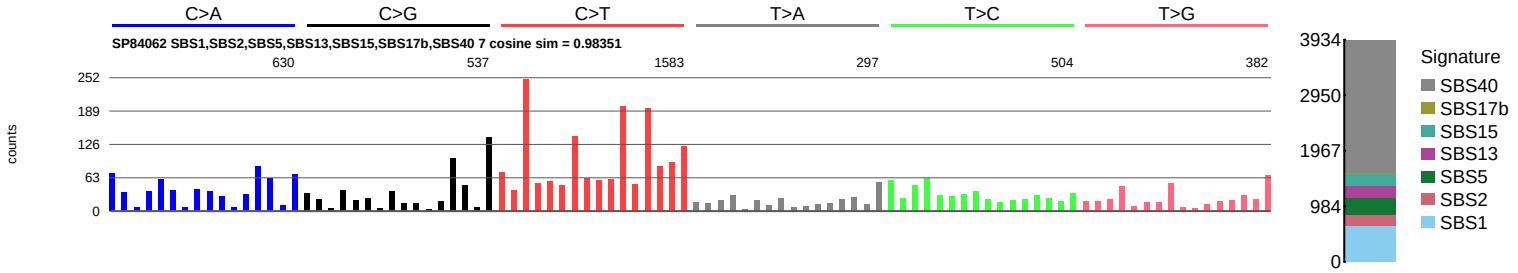

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98355
634
556
1564
324
474
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

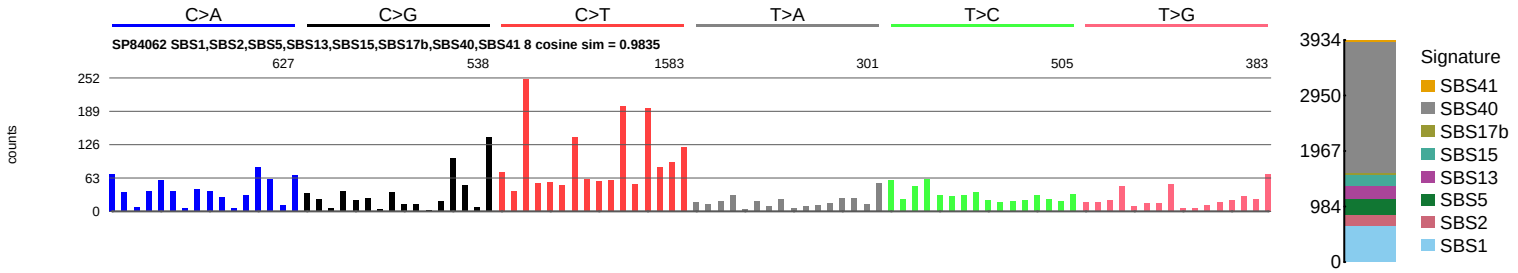

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40 9 cosine sim = 0.98355
631
545
1567
305
533
349
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

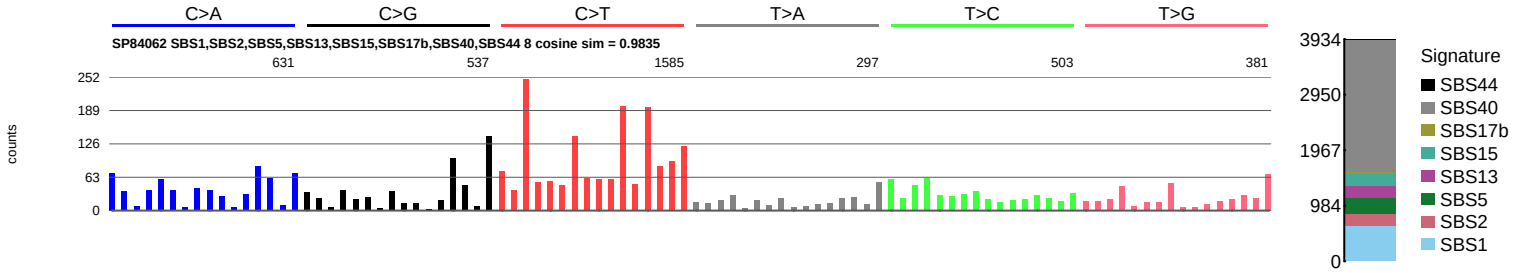

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98353
743
515
1527
282
559
313
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

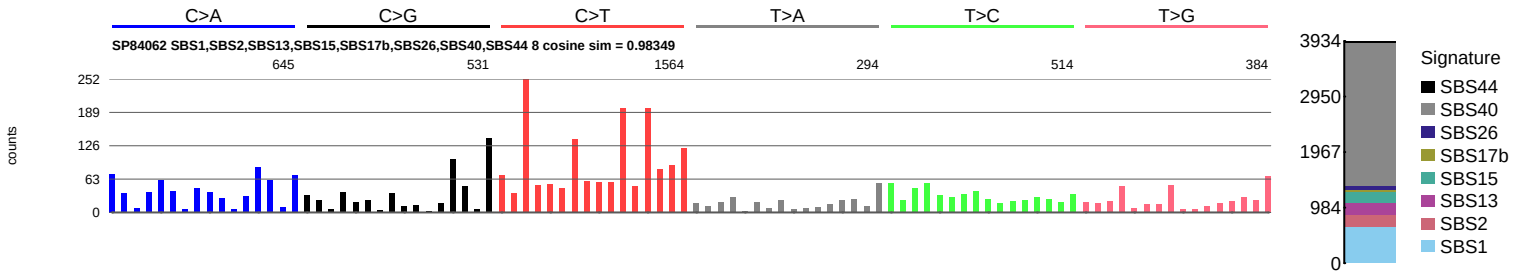

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18 7 cosine sim = 0.98352
737
562
1556
307
548
222
counts
244
183
122
61
0
3934
2950
1967
984
0
Signature
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

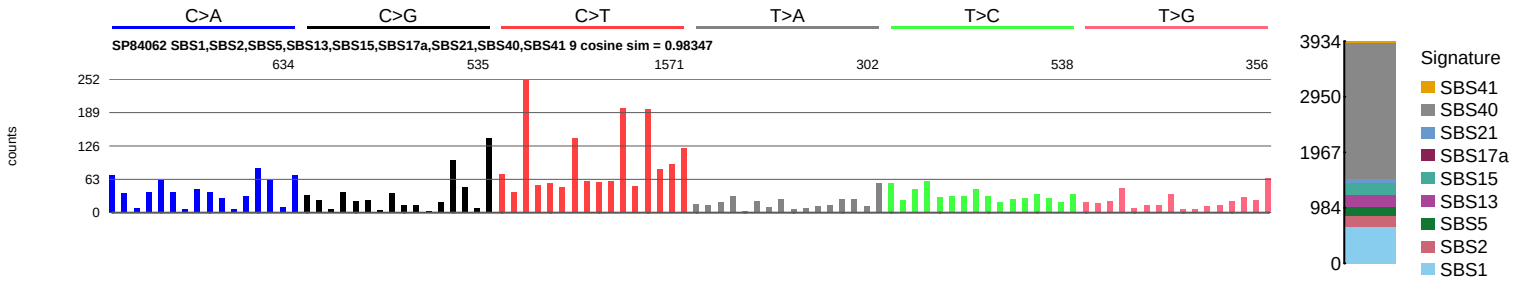

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98351
628
551
1551
311
512
379
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

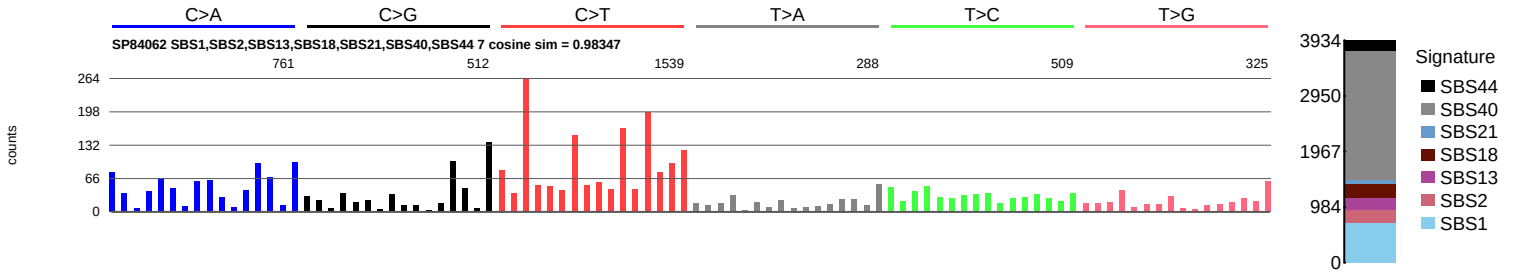

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40 7 cosine sim = 0.98351
630
537
1583
297
504
382
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

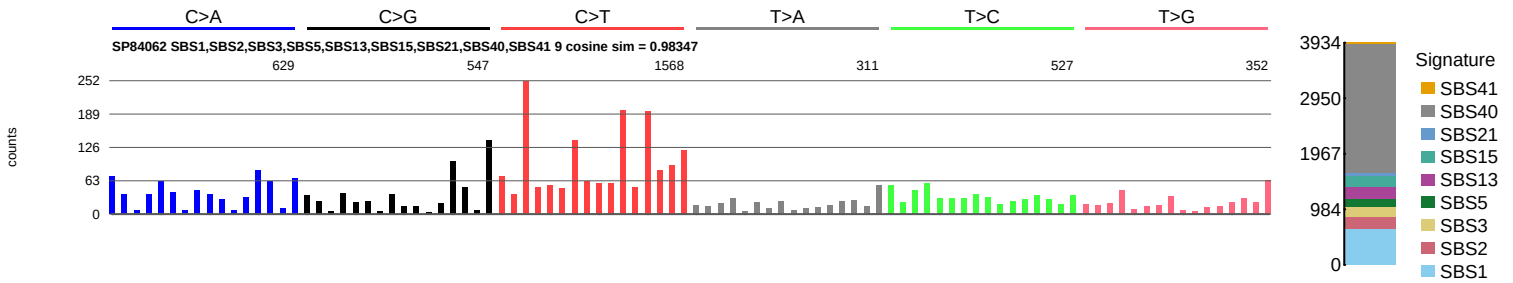

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41 8 cosine sim = 0.9835
627
538
1583
301
505
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

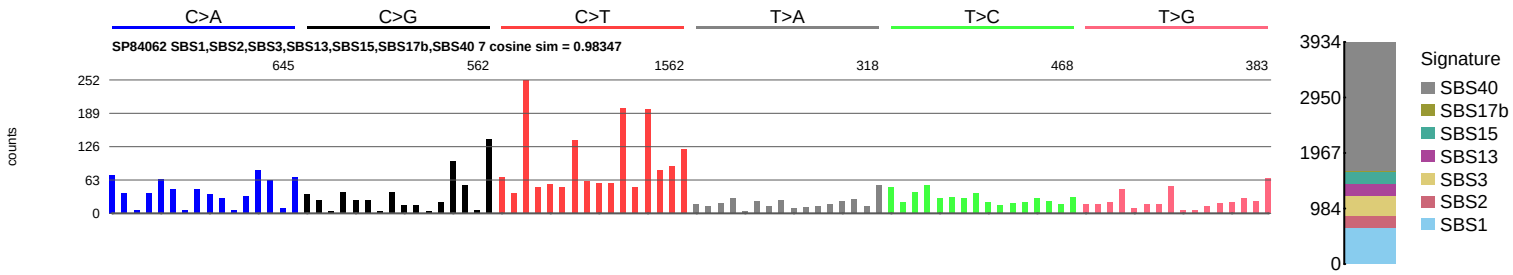

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 8 cosine sim = 0.9835
631
537
1585
297
503
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

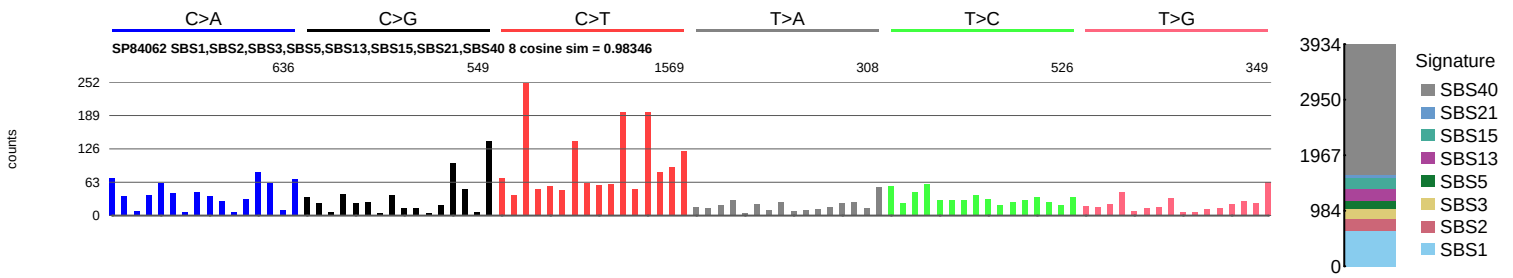

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98349
645
531
1564
294
514
384
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1

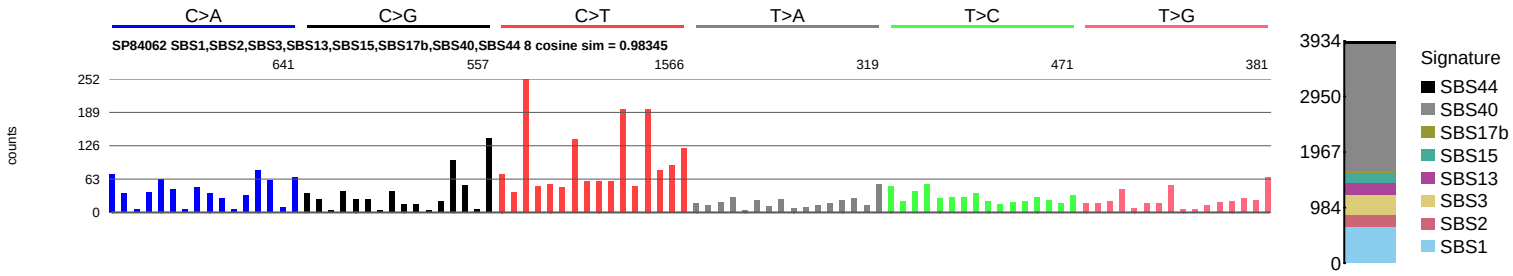

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 9 cosine sim = 0.98347
634
535
1571
302
538
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

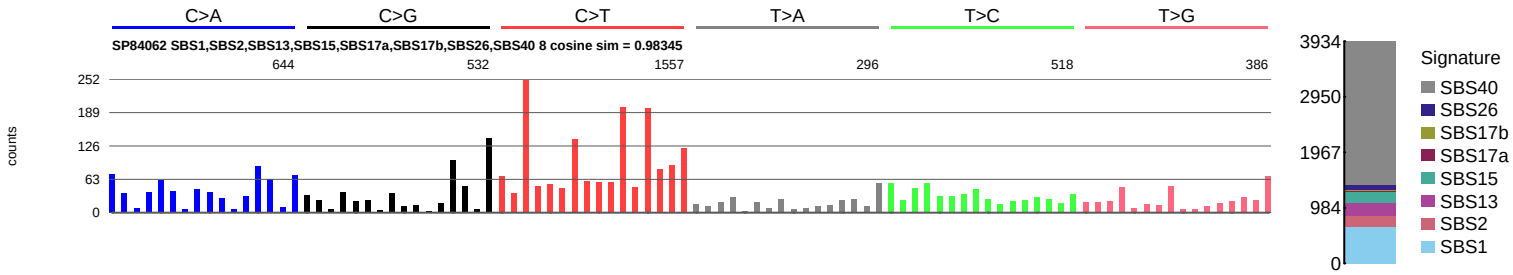

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98347
761
512
1539
288
509
325
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

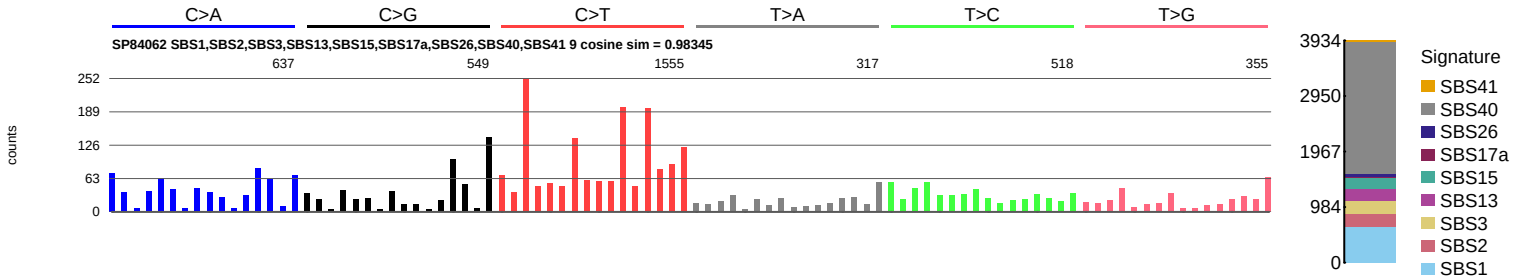

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41 9 cosine sim = 0.98347
629
547
1568
311
527
352
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

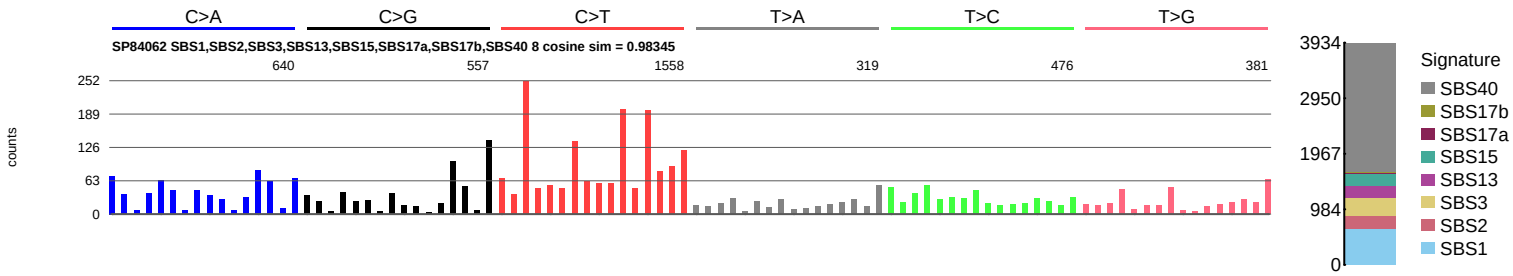

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS40 7 cosine sim = 0.98347
645
562
1562
318
468
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

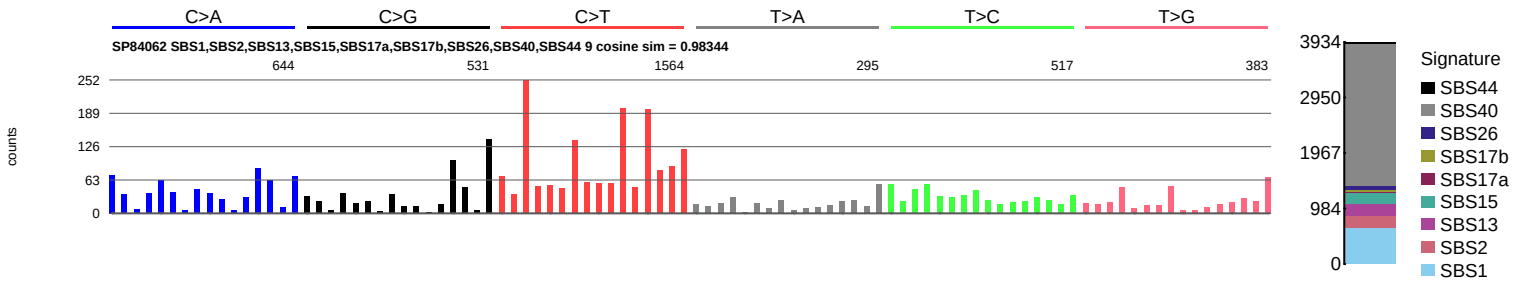

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40 8 cosine sim = 0.98346
636
549
1569
308
526
349
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

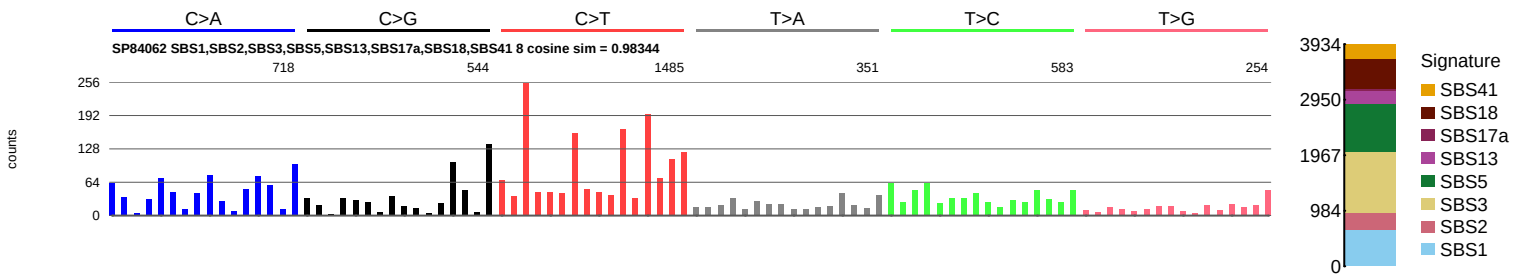

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS40,SBS44 8 cosine sim = 0.98345
641
557
1566
319
471
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

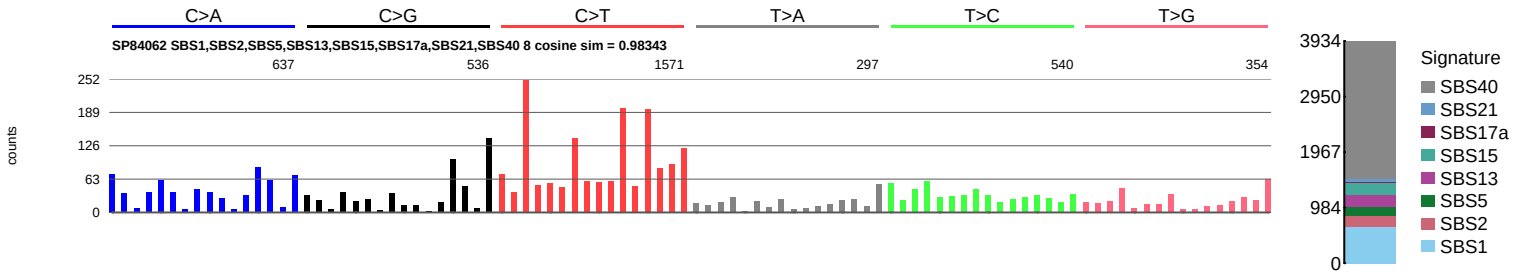

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98345
644
532
1557
296
518
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

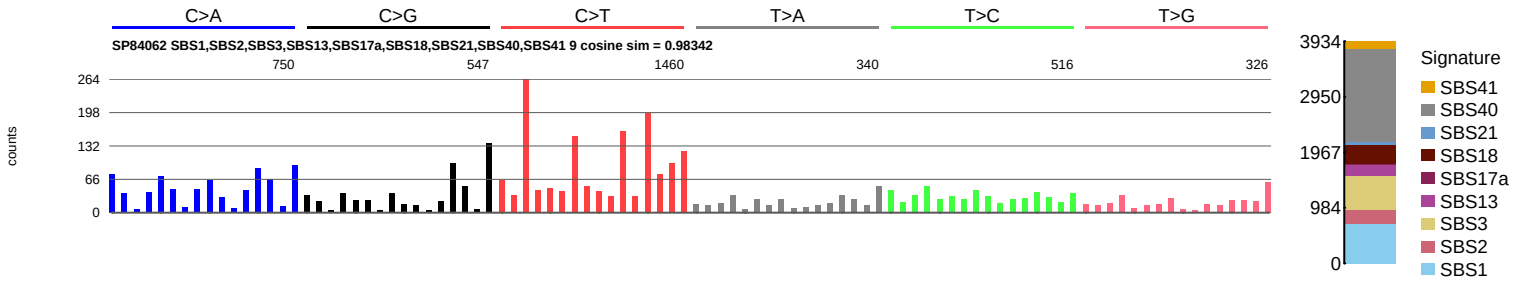

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41 9 cosine sim = 0.98345
637
549
1555
317
518
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

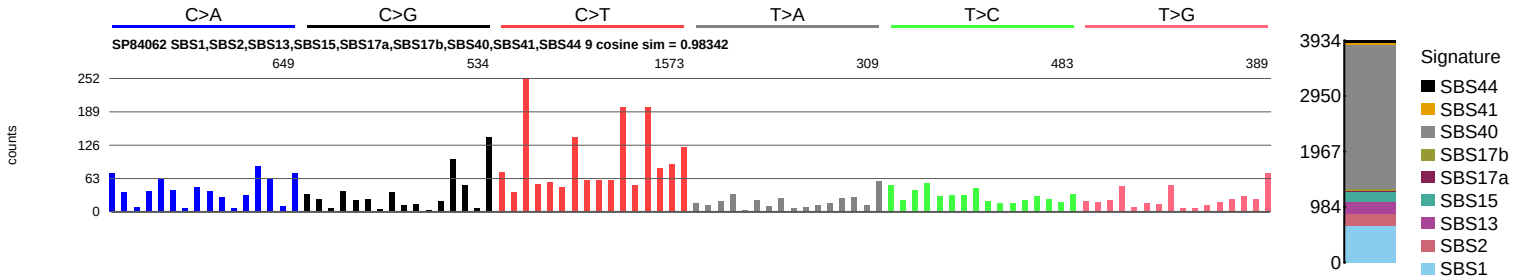

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98345
640
557
1558
319
476
381
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

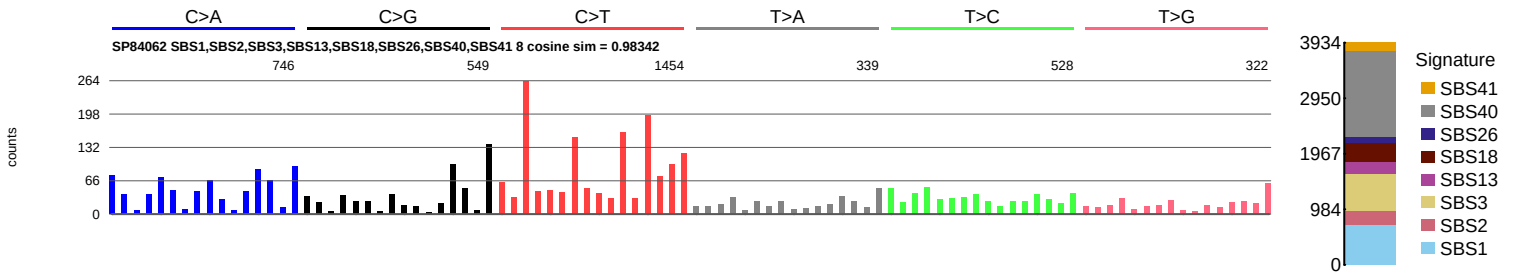

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98344
644
531
1564
295
517
383
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

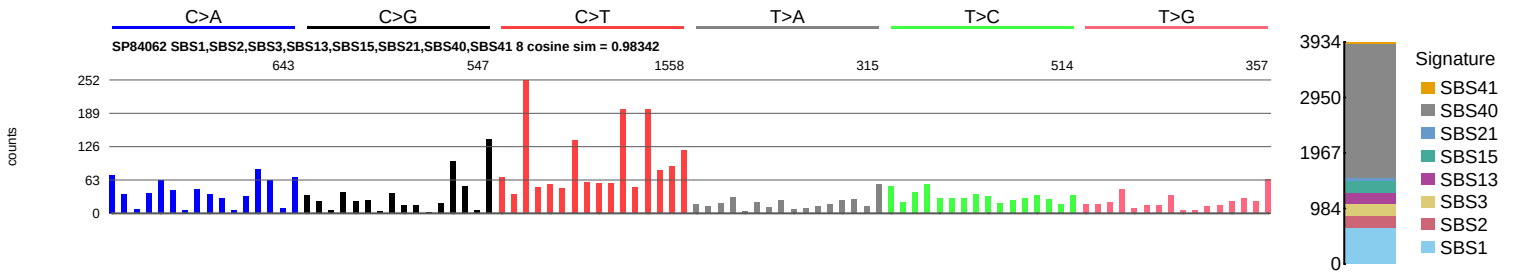

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98344
718
544
1485
351
583
254
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

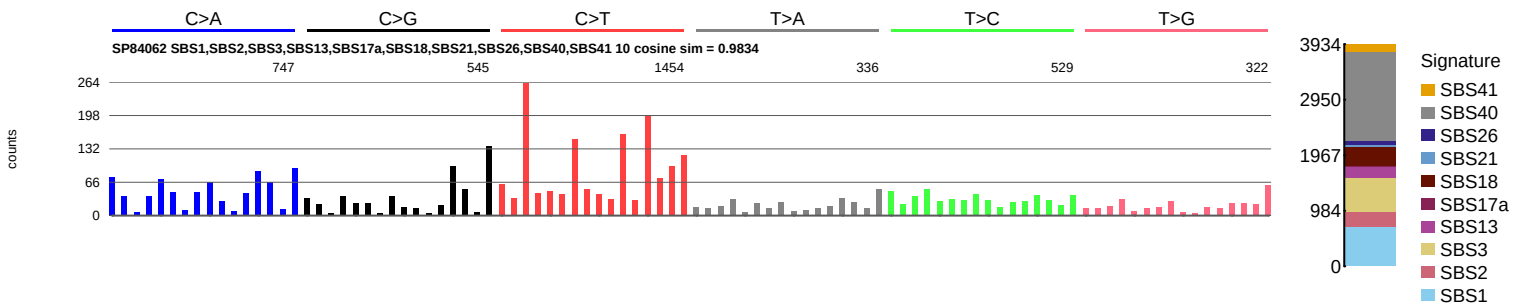

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40 8 cosine sim = 0.98343
637
536
1571
297
540
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

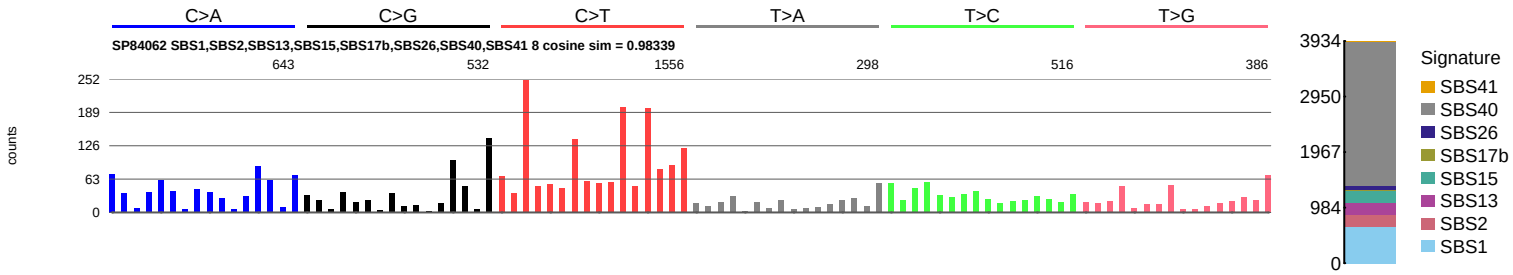

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98342
750
547
1460
340
516
326
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

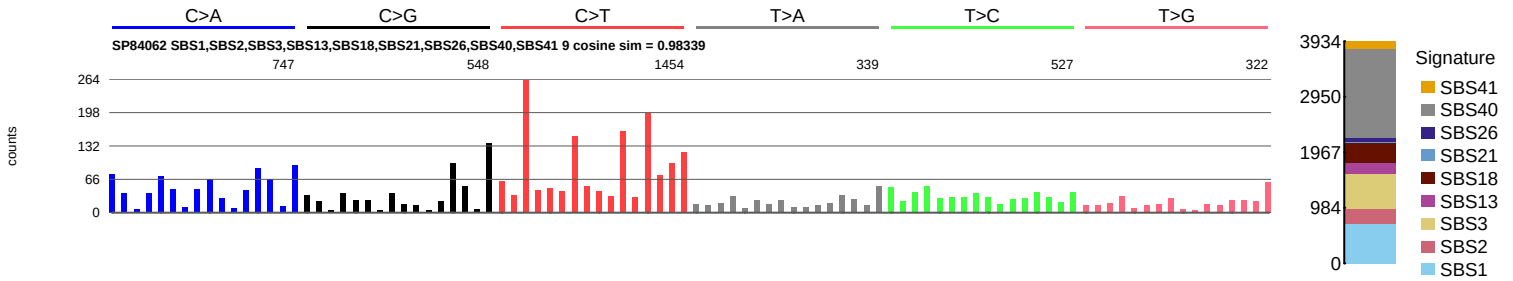

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98342
649
534
1573
309
483
389
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

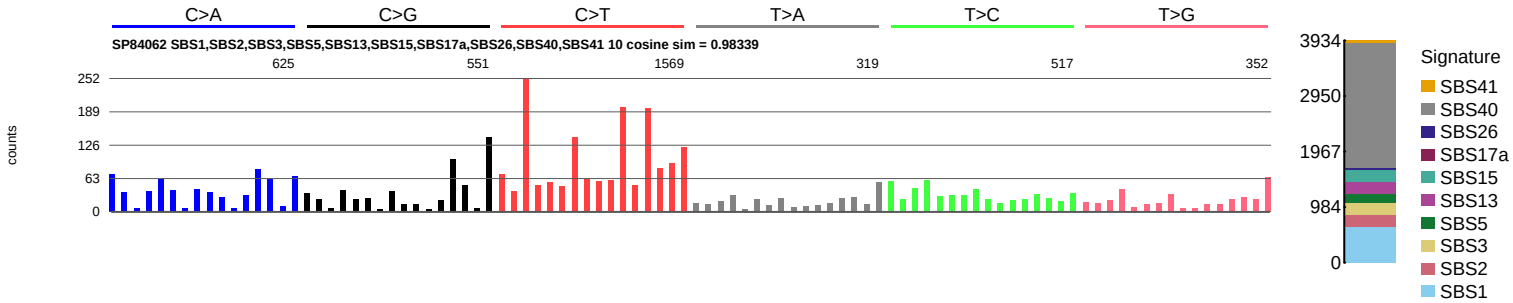

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98342
746
549
1454
339
528
322
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

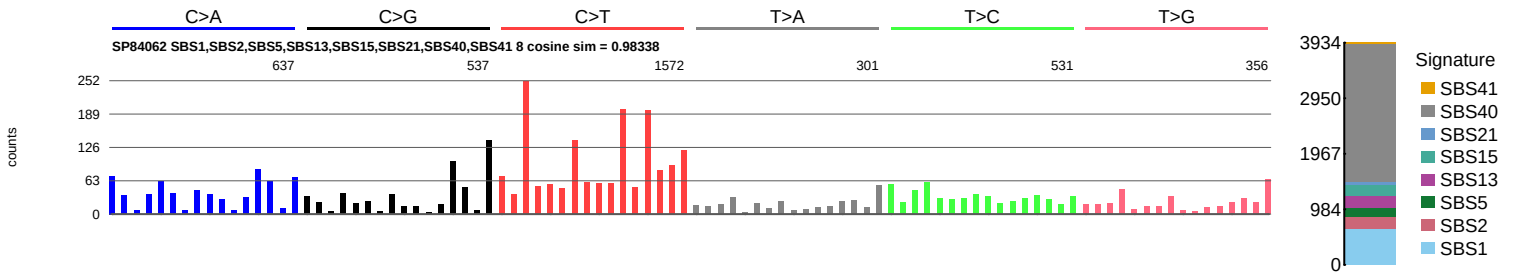

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS40,SBS41 8 cosine sim = 0.98342
643
547
1558
315
514
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

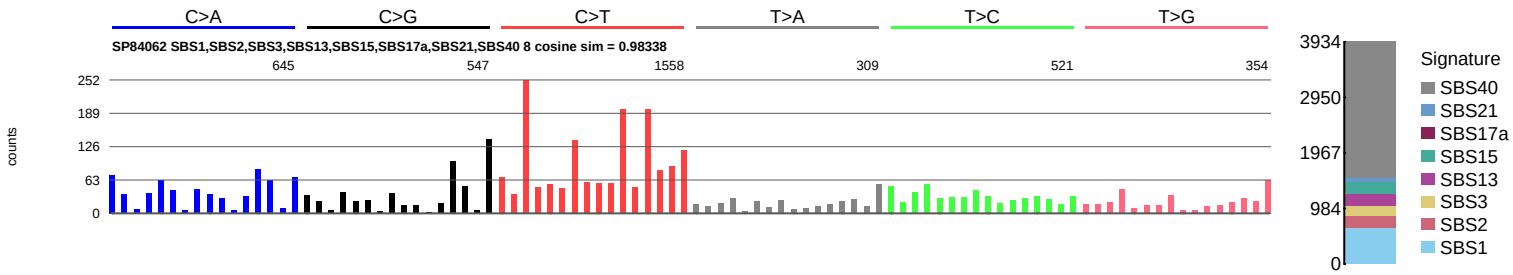

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9834
747
545
1454
336
529
322
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

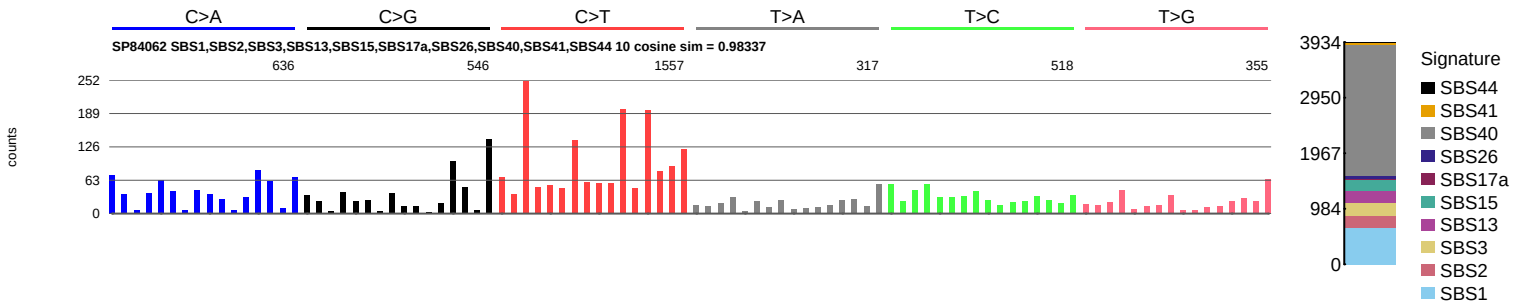

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98339
643
532
1556
298
516
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1

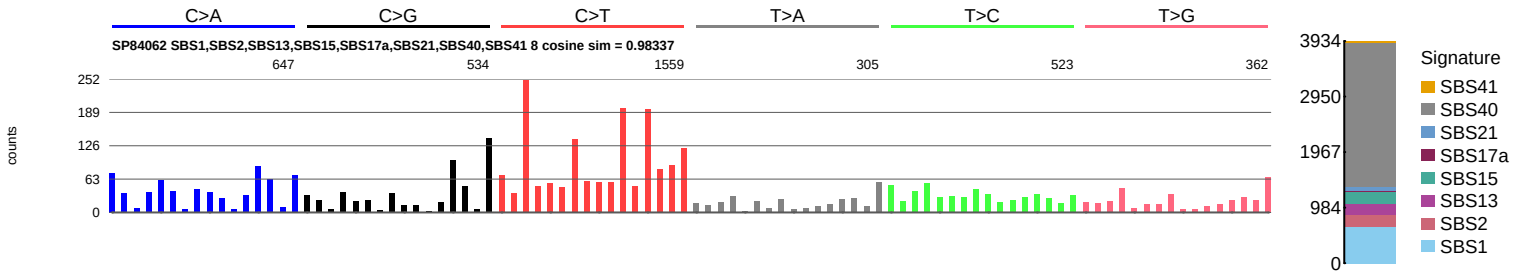

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98339
747
548
1454
339
527
322
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

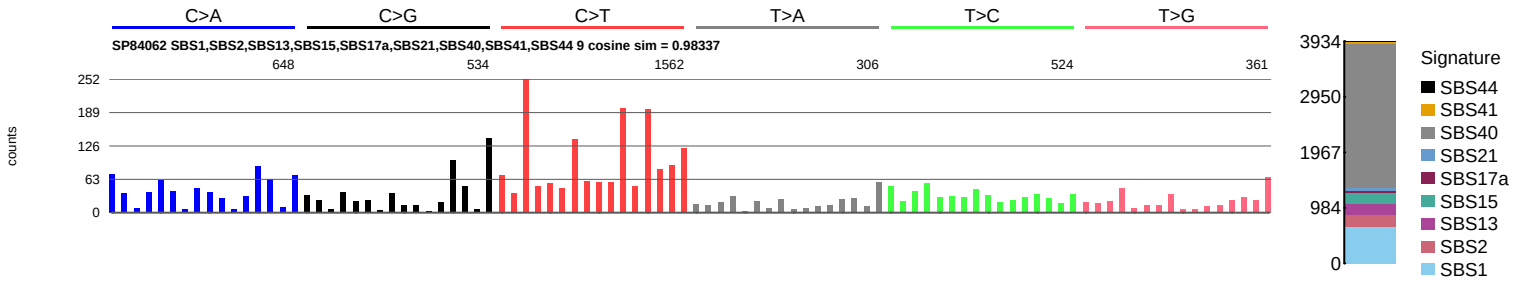

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41 10 cosine sim = 0.98339
625
551
1569
319
517
352
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

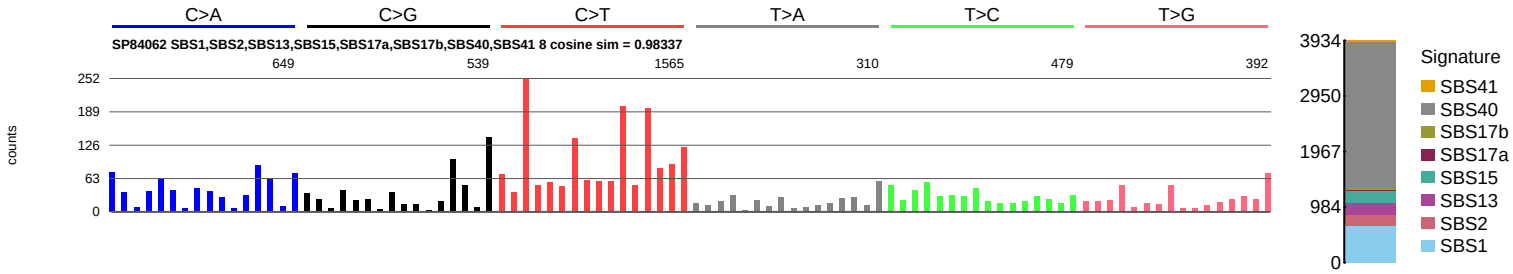

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41 8 cosine sim = 0.98338
637
537
1572
301
531
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS15
SBS13
SBS5
SBS2
SBS1

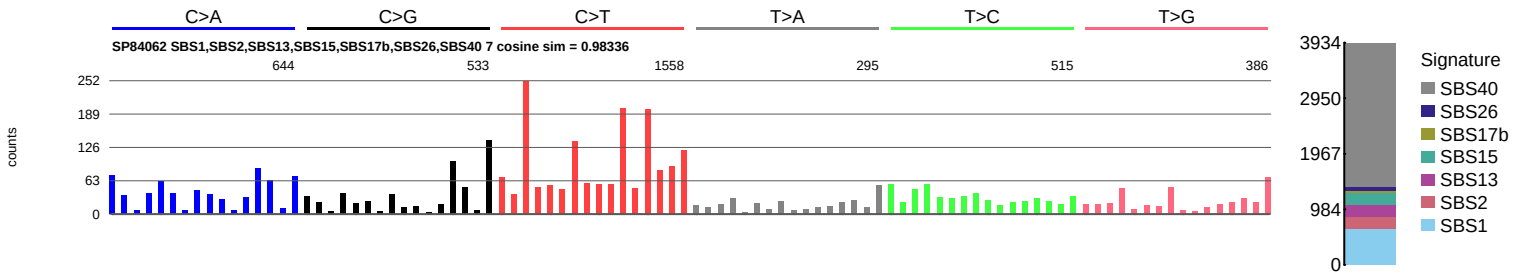

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS21,SBS40 8 cosine sim = 0.98338
645
547
1558
309
521
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

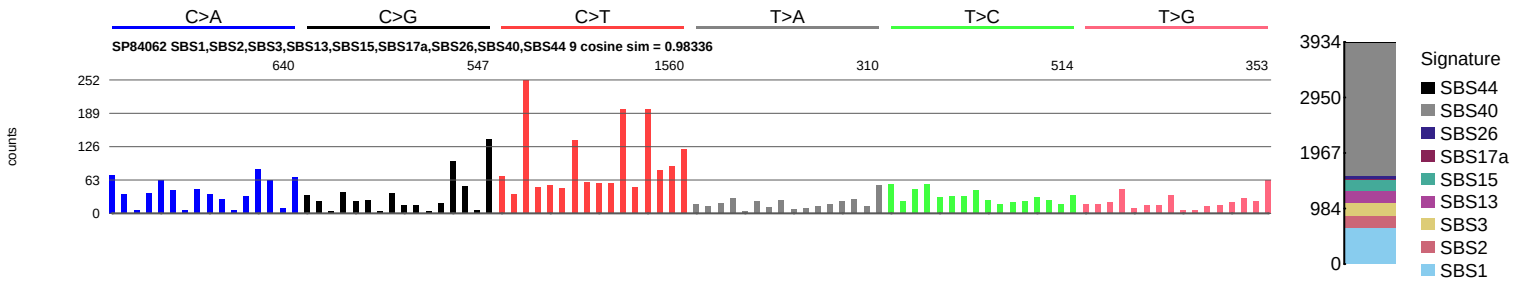

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98337
636
546
1557
317
518
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

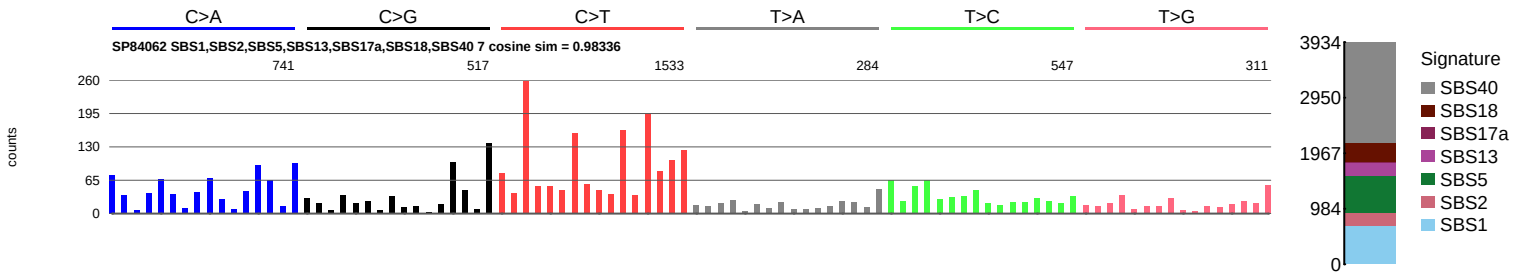

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 8 cosine sim = 0.98337
647
534
1559
305
523
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS2
SBS1

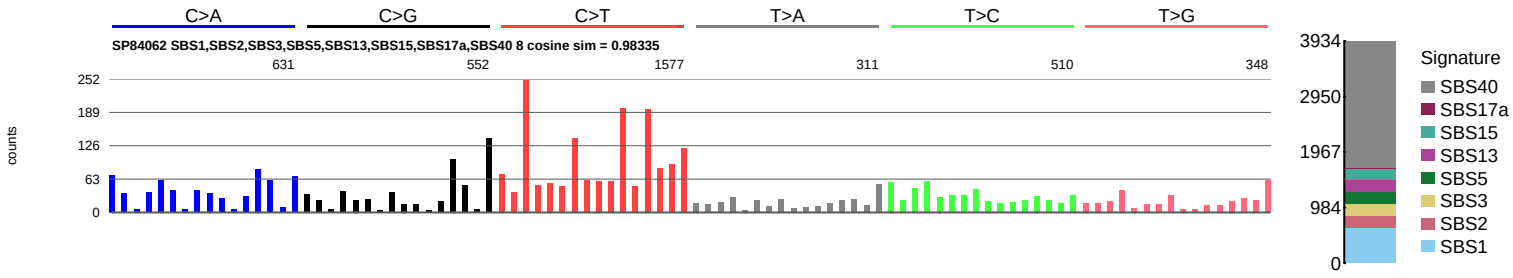

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98337
648
534
1562
306
524
361
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS2
SBS1

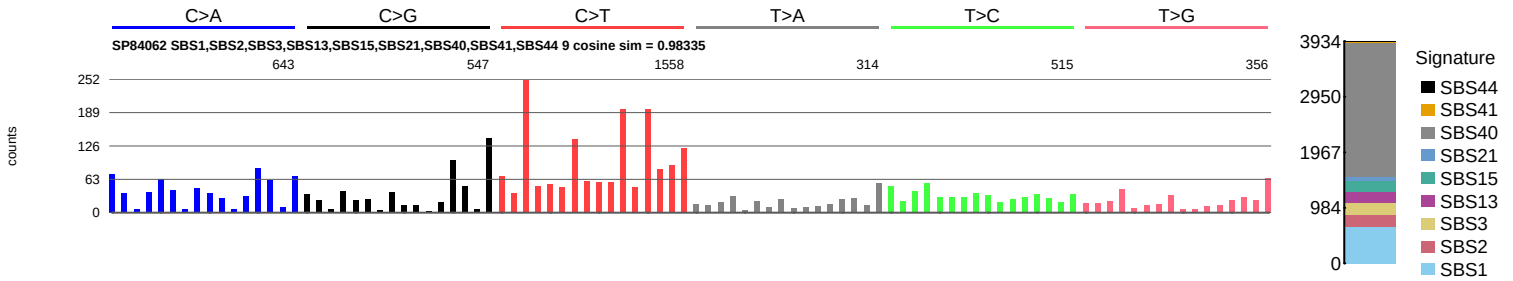

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98337
649
539
1565
310
479
392
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

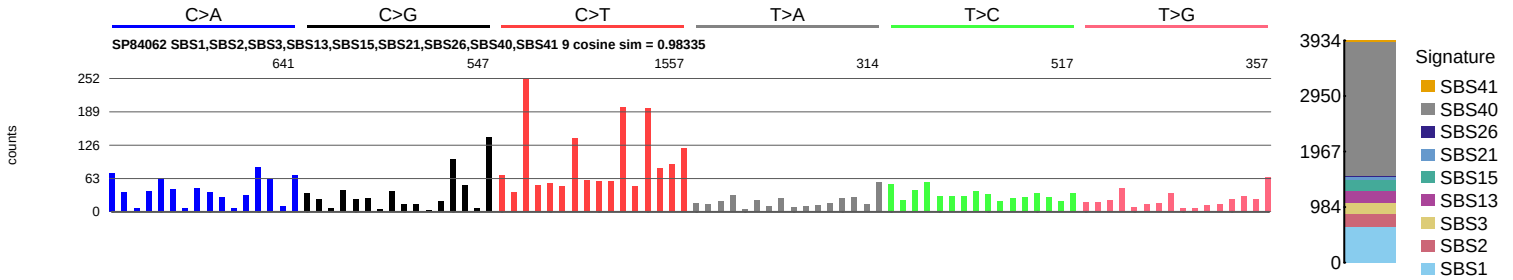

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS40 7 cosine sim = 0.98336
644
533
1558
295
515
386
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1

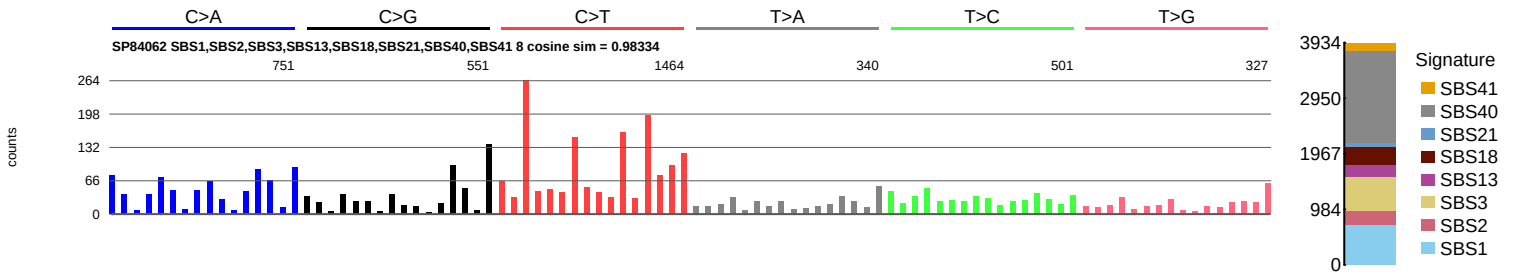

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS44 9 cosine sim = 0.98336
640
547
1560
310
514
353
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

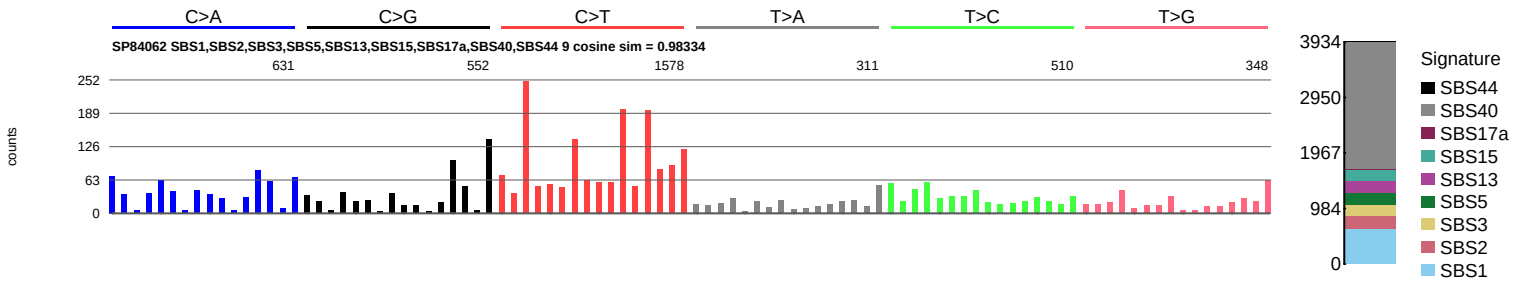

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98336
741
517
1533
284
547
311
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

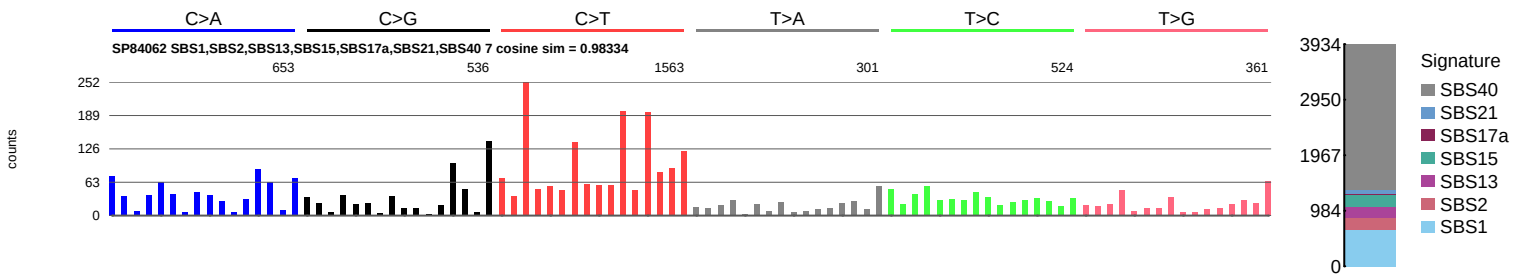

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40 8 cosine sim = 0.98335
631
552
1577
311
510
348
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

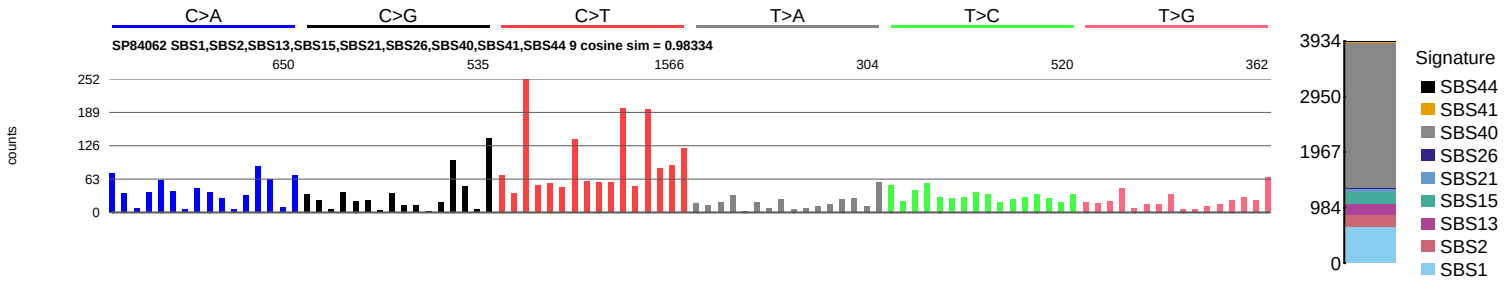

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98335
643
547
1558
314
515
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

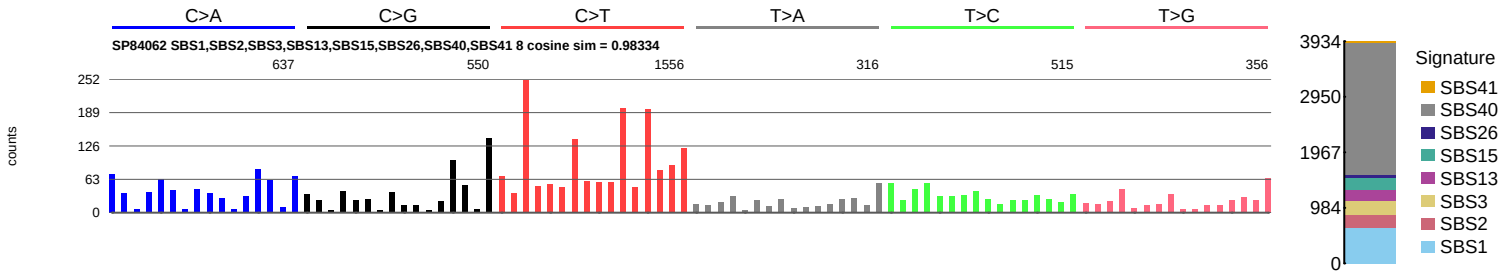

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98335
641
547
1557
314
517
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

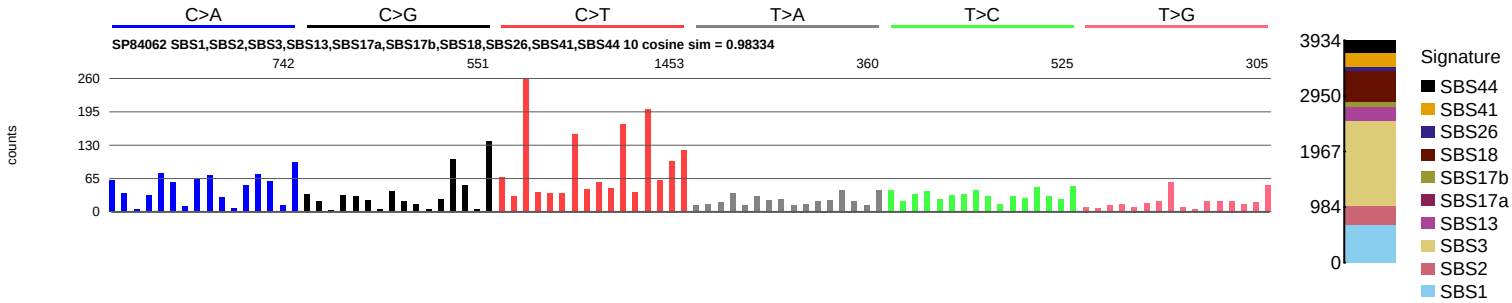

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98334
751
551
1464
340
501
327
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

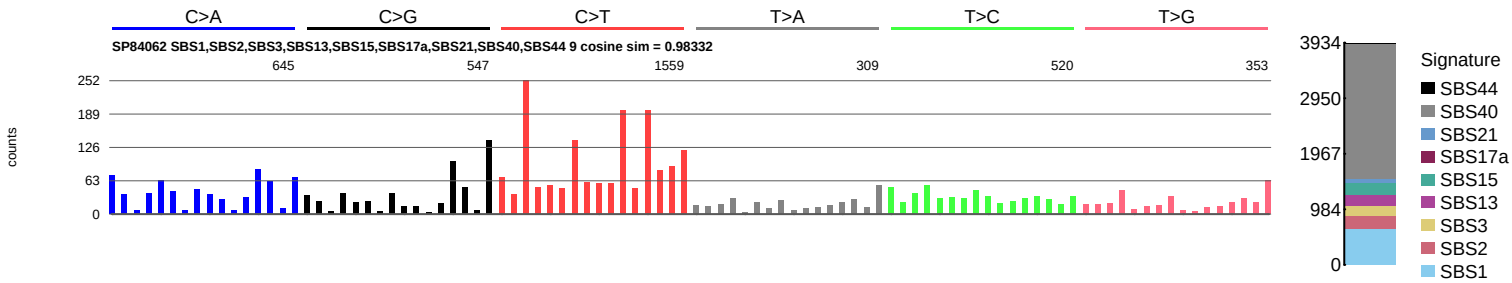

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS44 9 cosine sim = 0.98334
631
552
1578
311
510
348
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

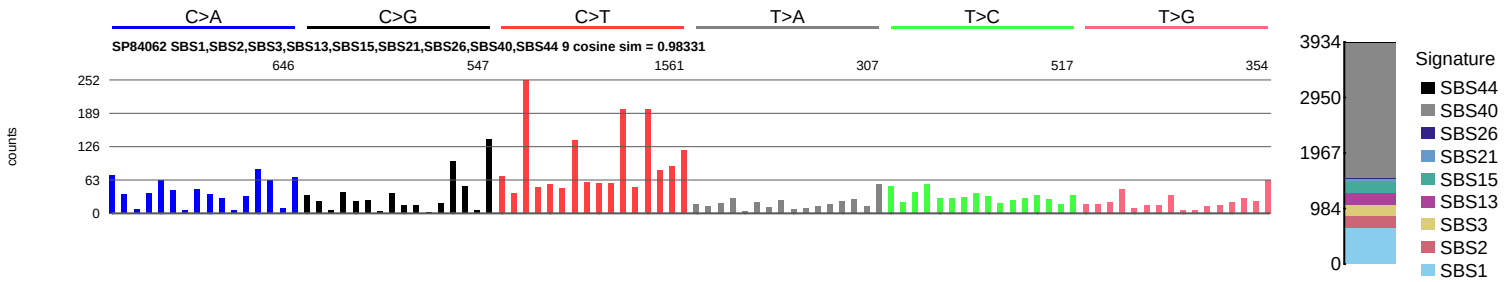

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS21,SBS40 7 cosine sim = 0.98334
653
536
1563
301
524
361
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS2
SBS1

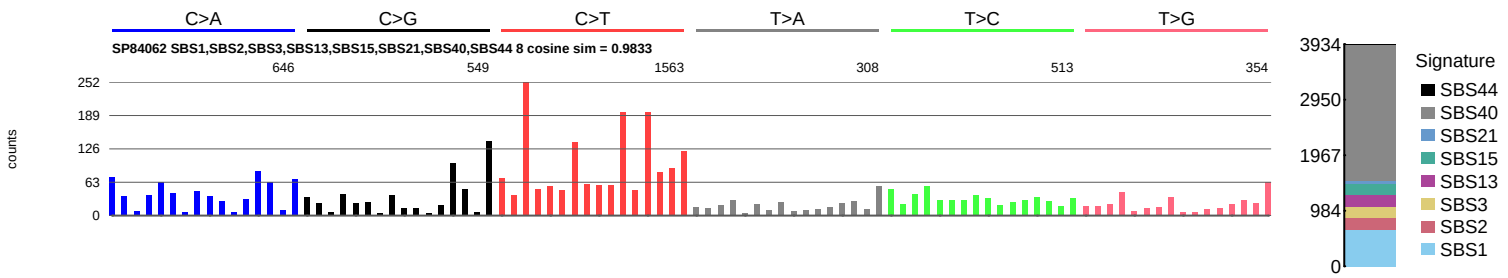

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98334
650
535
1566
304
520
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS15
SBS13
SBS2
SBS1

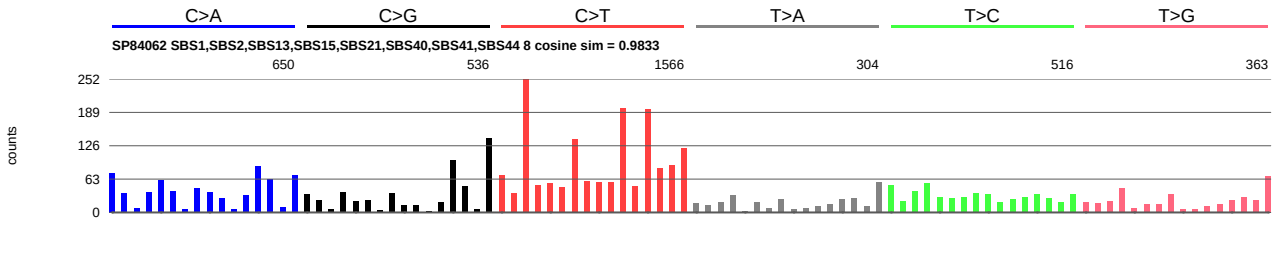

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS26,SBS40,SBS41 8 cosine sim = 0.98334
637
550
1556
316
515
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS15
SBS13
SBS3
SBS2
SBS1

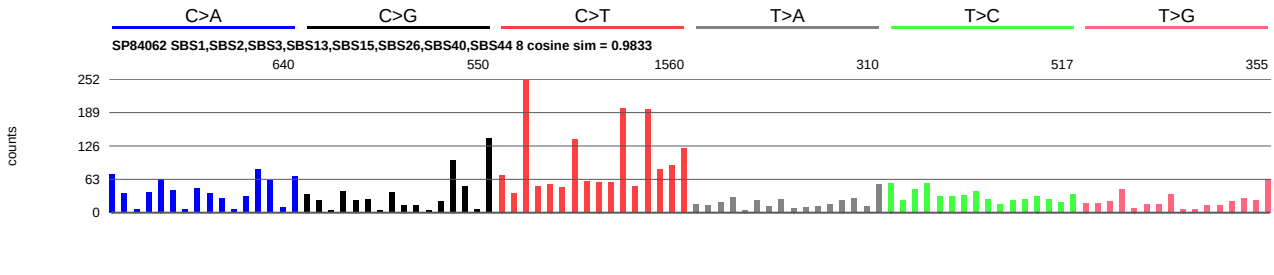

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98334
742
551
1453
360
525
305
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

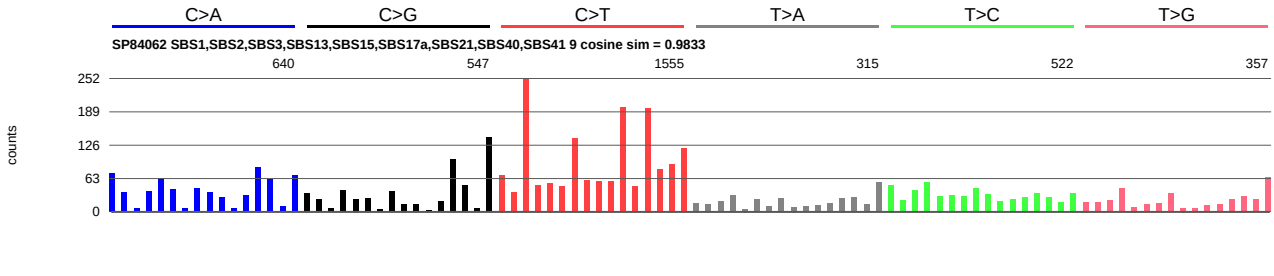

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS44 9 cosine sim = 0.98332
645
547
1559
309
520
353
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

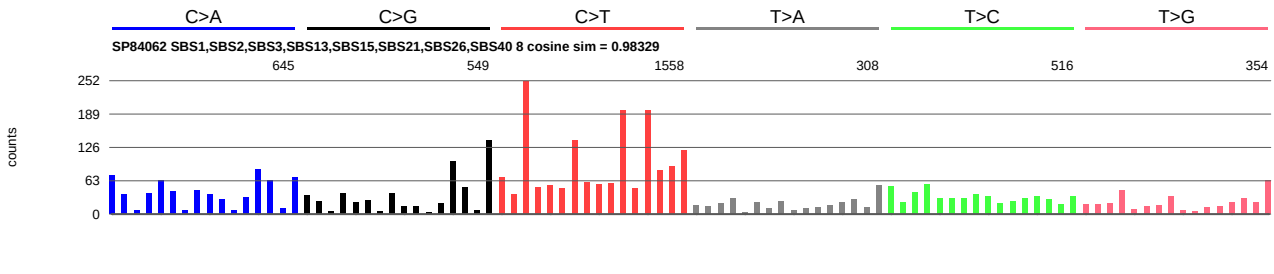

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98331
646
547
1561
307
517
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

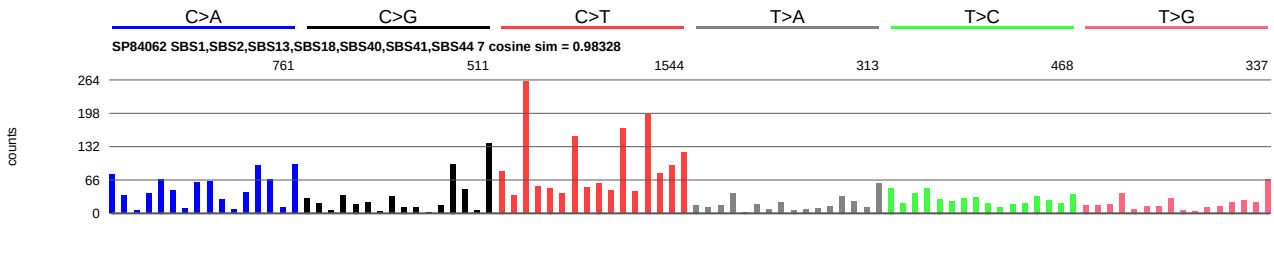

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS40,SBS44 8 cosine sim = 0.9833
646
549
1563
308
513
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

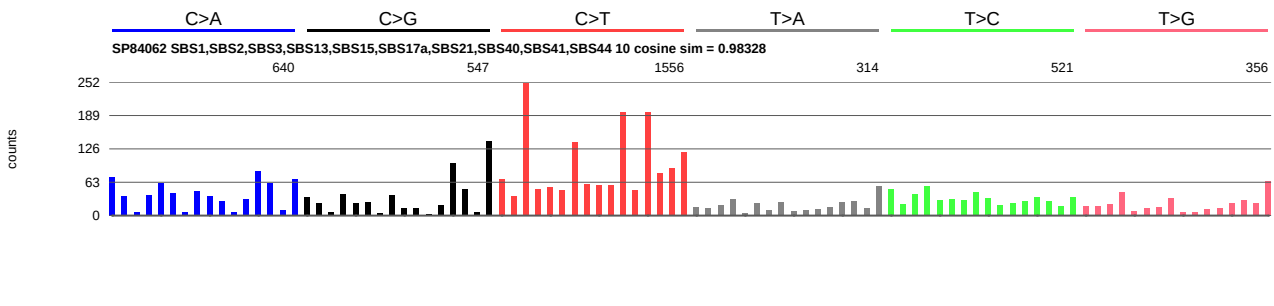

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9833
650
536
1566
304
516
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS15
SBS13
SBS2
SBS1

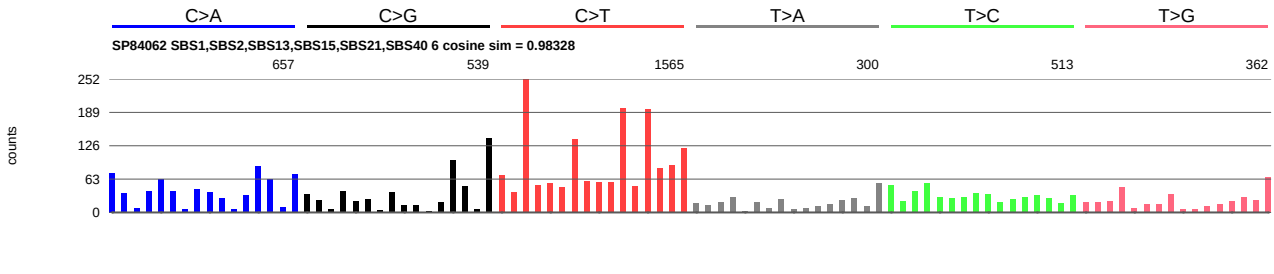

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS26,SBS40,SBS44 8 cosine sim = 0.9833
640
550
1560
310
517
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS15
SBS13
SBS3
SBS2
SBS1

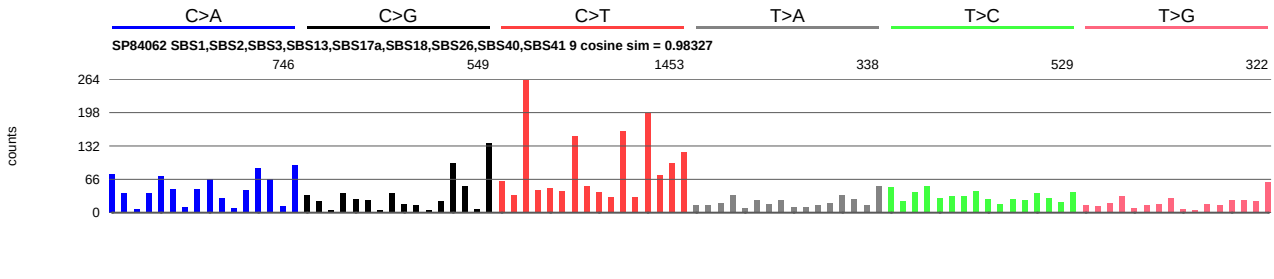

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 9 cosine sim = 0.9833
640
547
1555
315
522
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

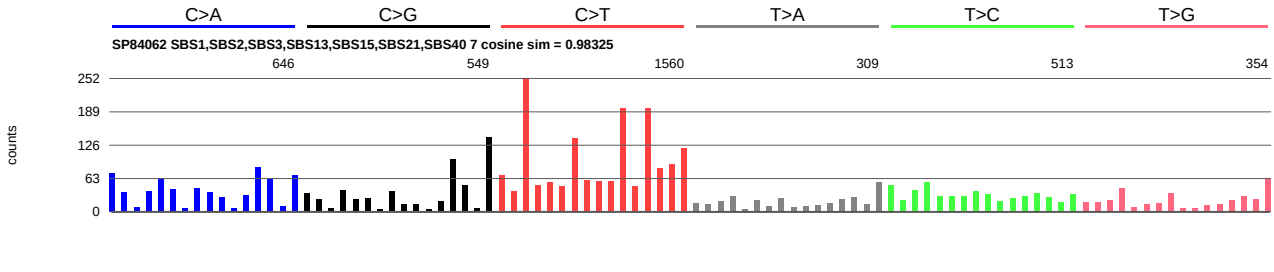

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS26,SBS40 8 cosine sim = 0.98329
645
549
1558
308
516
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

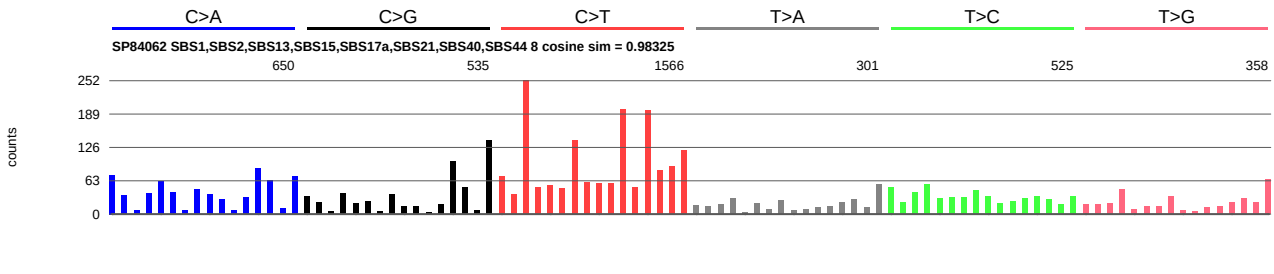

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98328
761
511
1544
313
468
337
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1

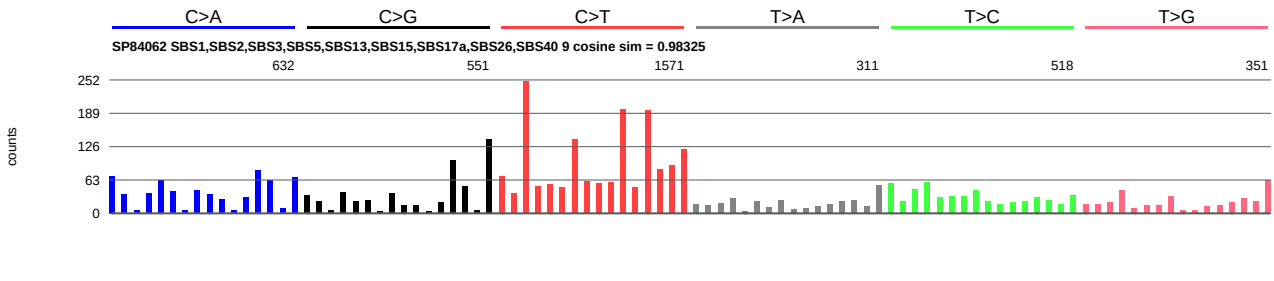

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98328
640
547
1556
314
521
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

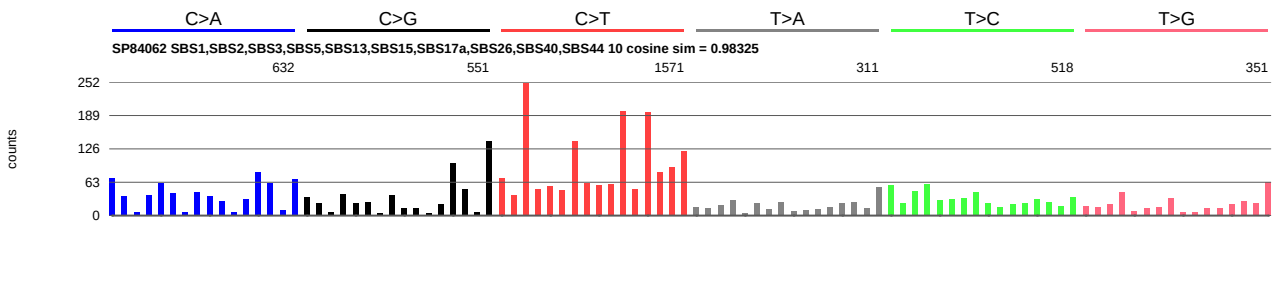

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS40 6 cosine sim = 0.98328
657
539
1565
300
513
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS15
SBS13
SBS2
SBS1

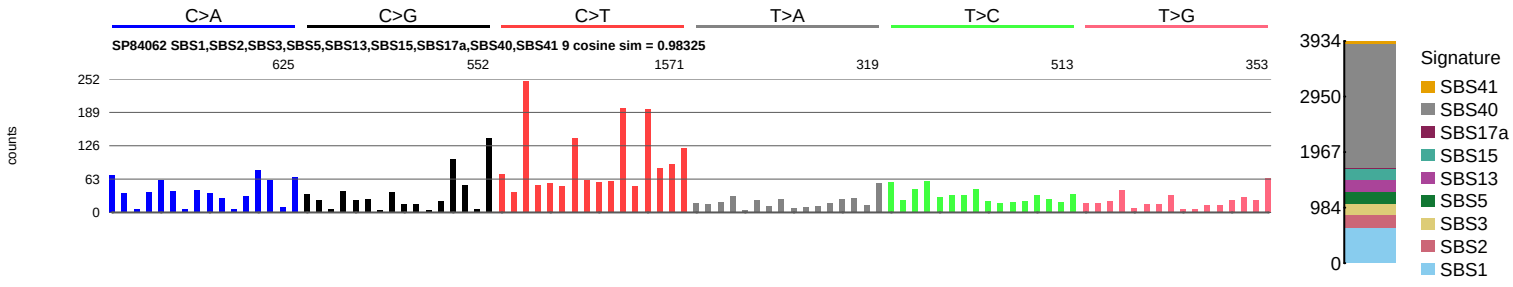

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98327
746
549
1453
338
529
322
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

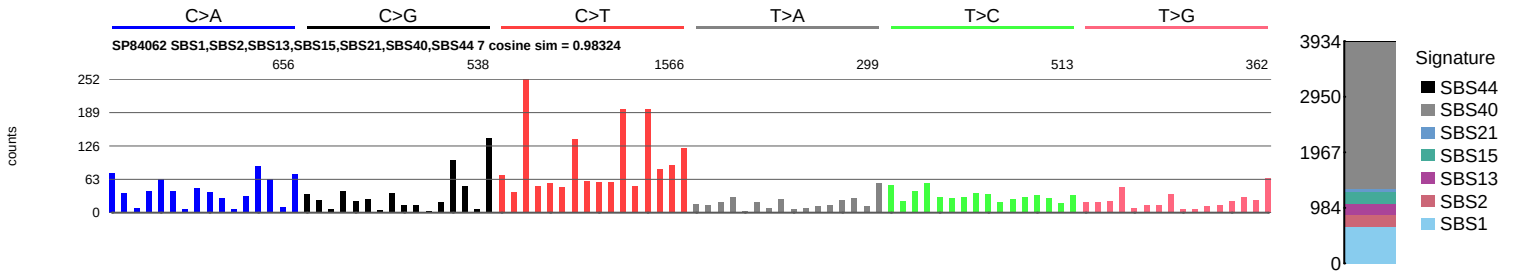

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS40 7 cosine sim = 0.98325
646
549
1560
309
513
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

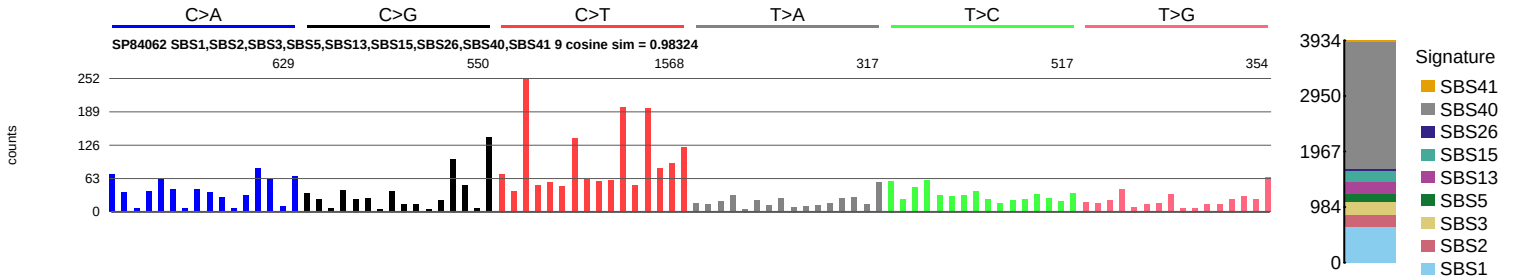

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS44 8 cosine sim = 0.98325
650
535
1566
301
525
358
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS2
SBS1

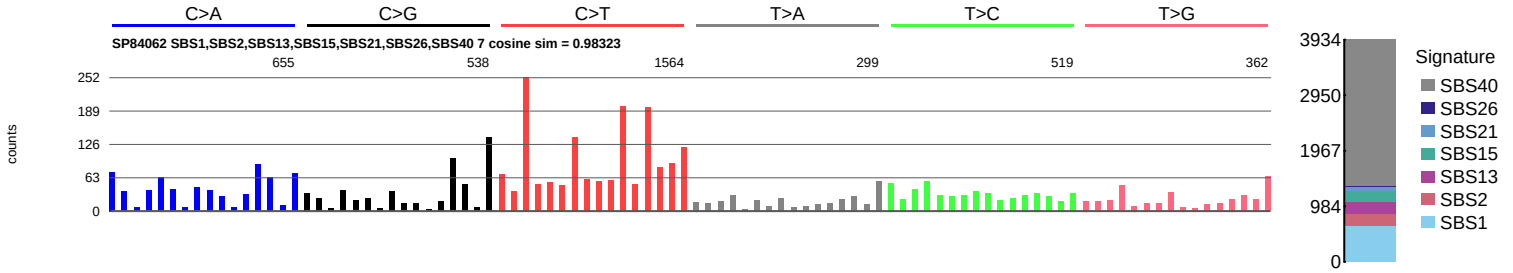

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40 9 cosine sim = 0.98325
632
551
1571
311
518
351
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

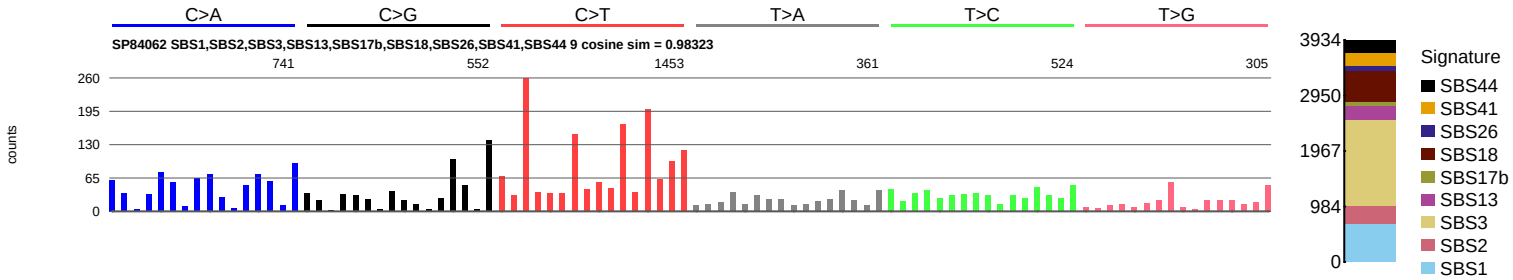

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS44 10 cosine sim = 0.98325
632
551
1571
311
518
351
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

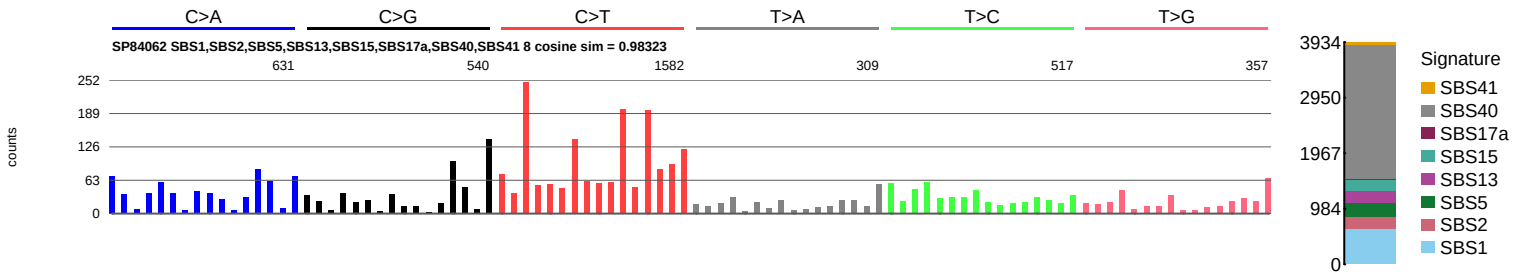

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41 9 cosine sim = 0.98325
625
552
1571
319
513
353
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

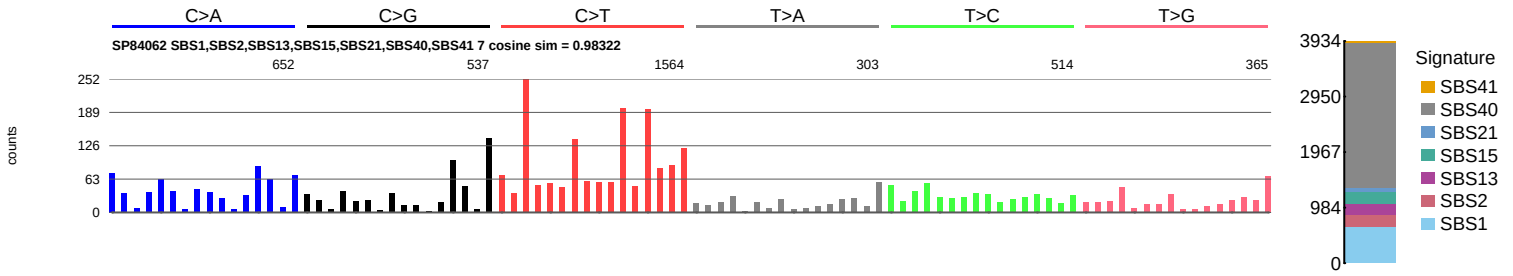

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS40,SBS44 7 cosine sim = 0.98324
656
538
1566
299
513
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS15
SBS13
SBS2
SBS1

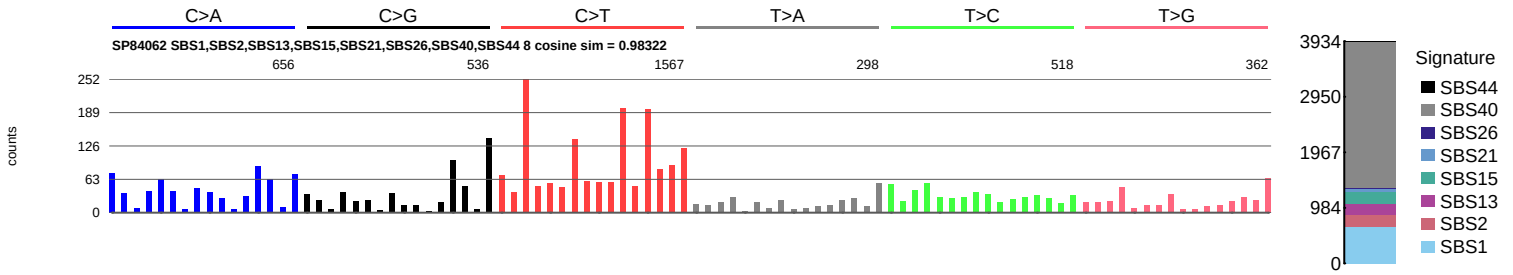

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS26,SBS40,SBS41 9 cosine sim = 0.98324
629
550
1568
317
517
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

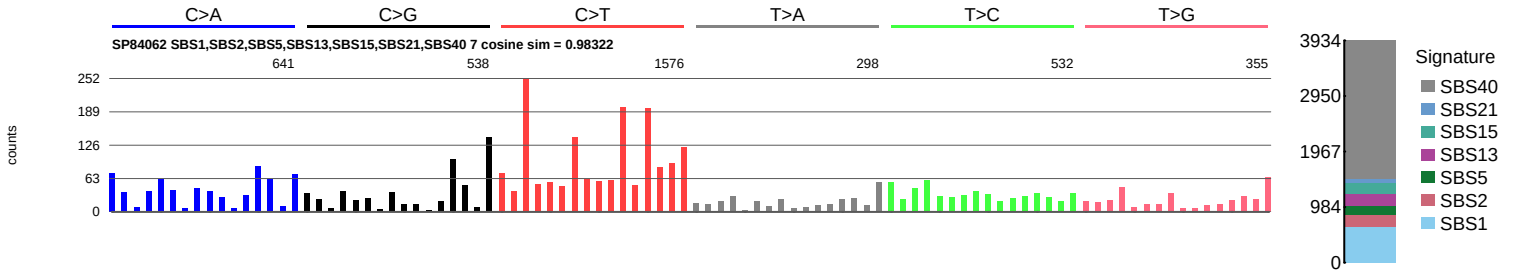

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS26,SBS40 7 cosine sim = 0.98323
655
538
1564
299
519
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS15
SBS13
SBS2
SBS1

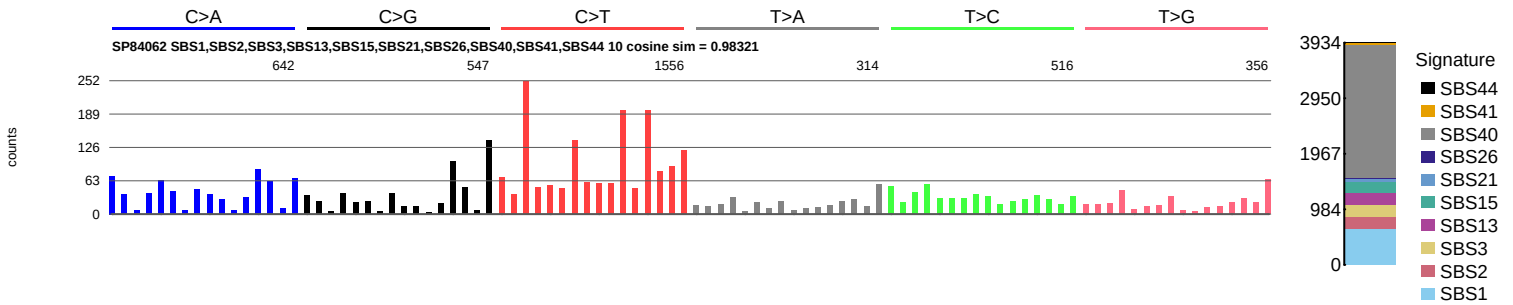

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98323
741
552
1453
361
524
305
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

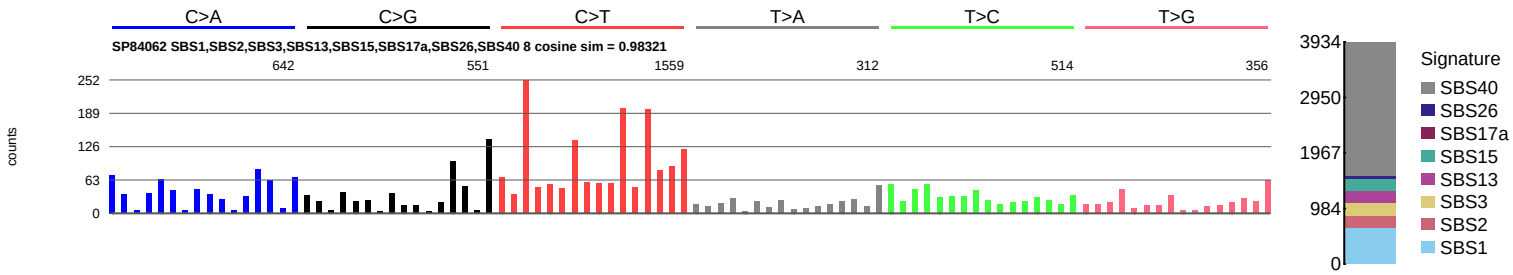

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41 8 cosine sim = 0.98323
631
540
1582
309
517
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

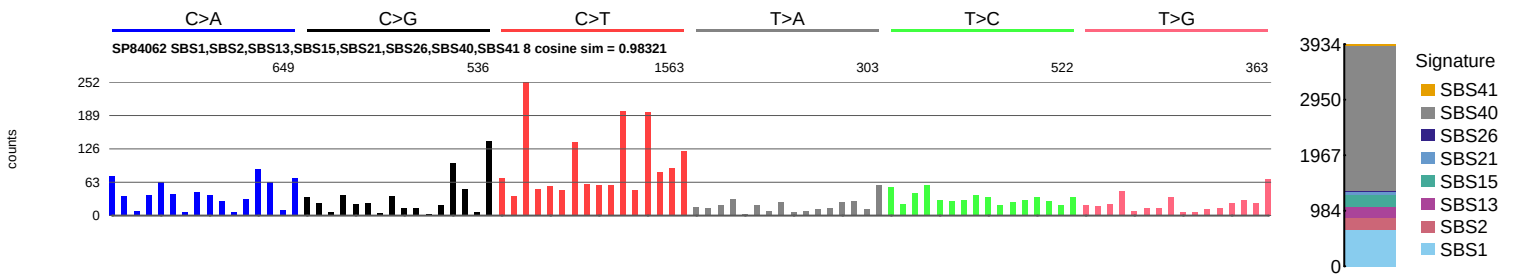

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS40,SBS41 7 cosine sim = 0.98322
652
537
1564
303
514
365
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS15
SBS13
SBS2
SBS1

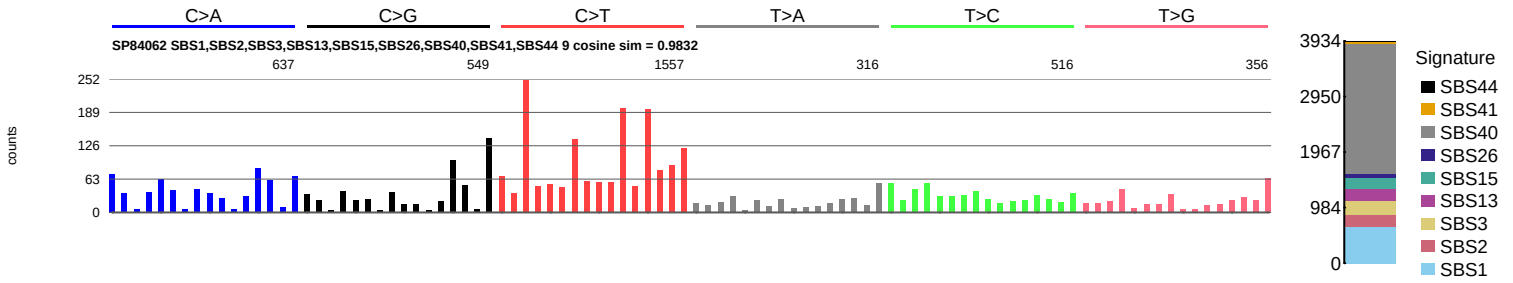

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98322
656
536
1567
298
518
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS15
SBS13
SBS2
SBS1

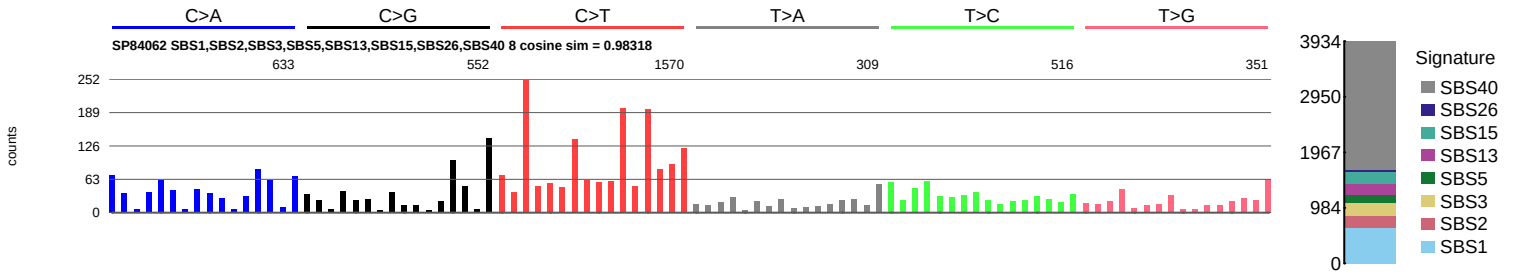

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40 7 cosine sim = 0.98322
641
538
1576
298
532
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS15
SBS13
SBS5
SBS2
SBS1

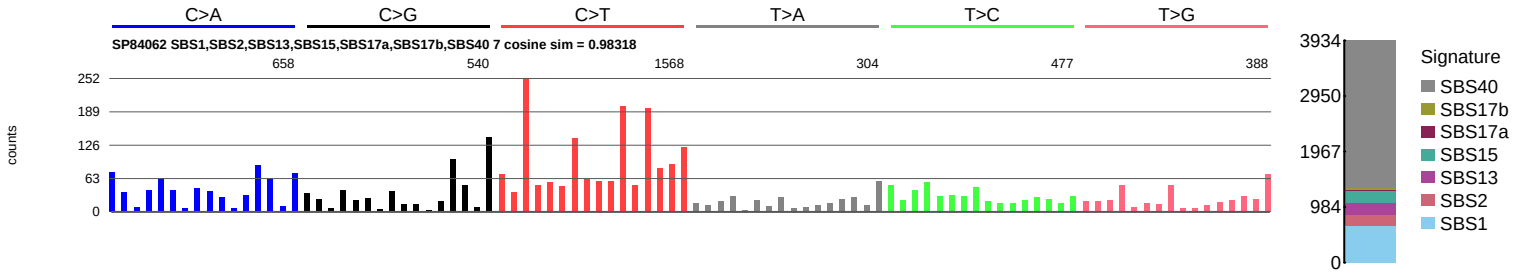

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98321
642
547
1556
314
516
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS15
SBS13
SBS3
SBS2
SBS1

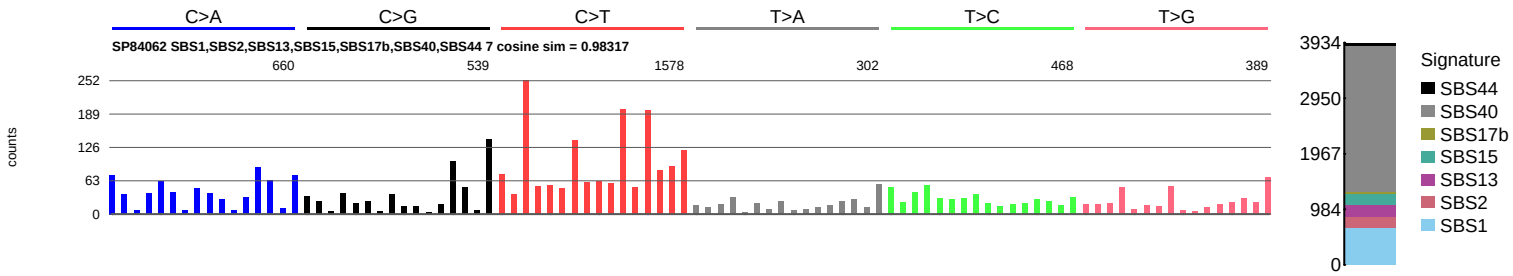

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS26,SBS40 8 cosine sim = 0.98321
642
551
1559
312
514
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

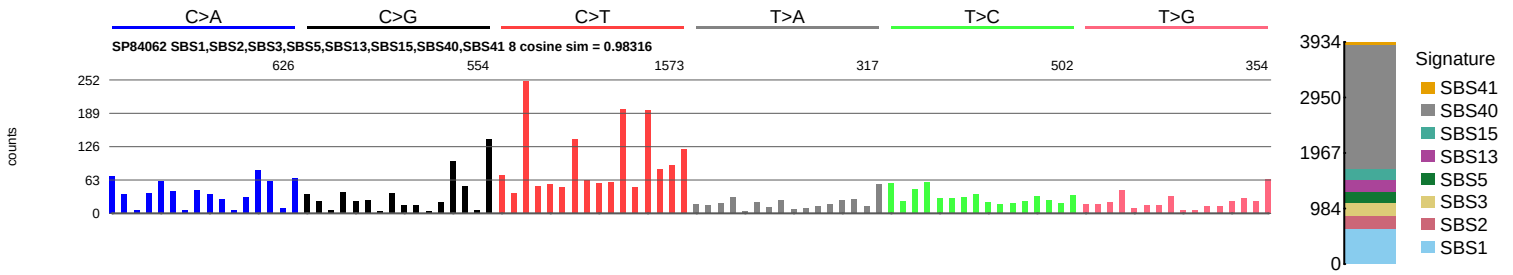

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98321
649
536
1563
303
522
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS15
SBS13
SBS2
SBS1

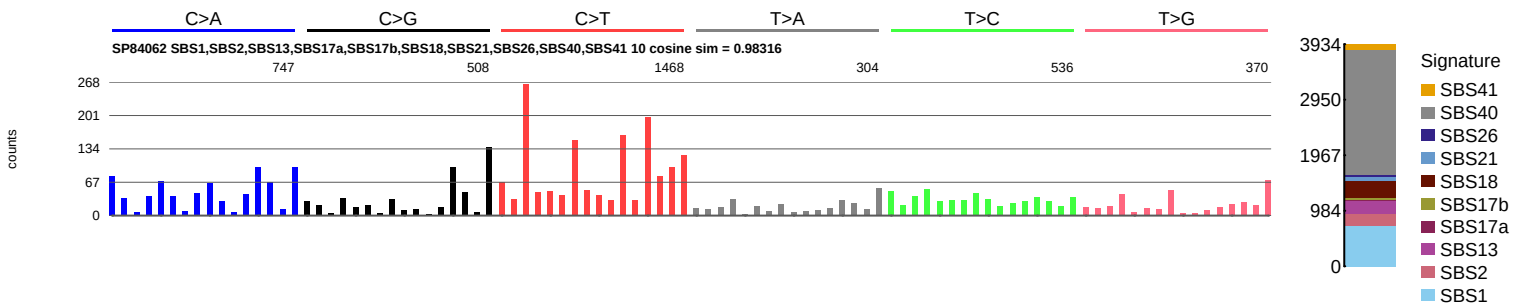

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9832
637
549
1557
316
516
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS15
SBS13
SBS3
SBS2
SBS1

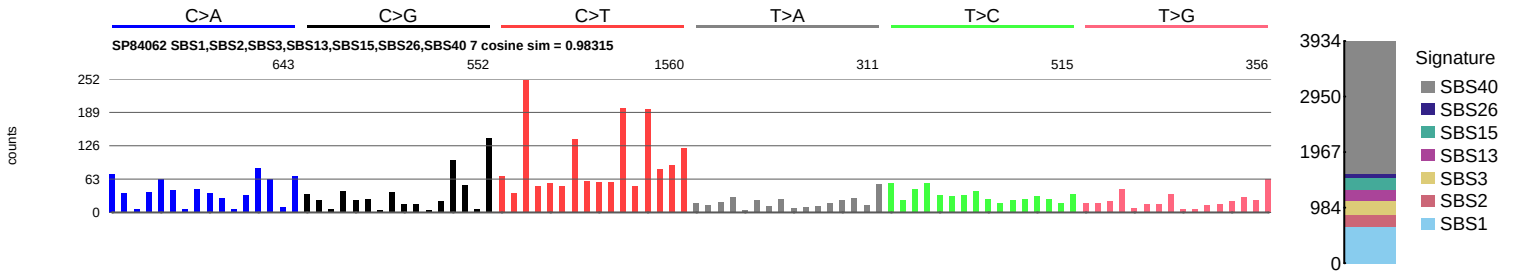

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS26,SBS40 8 cosine sim = 0.98318
633
552
1570
309
516
351
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

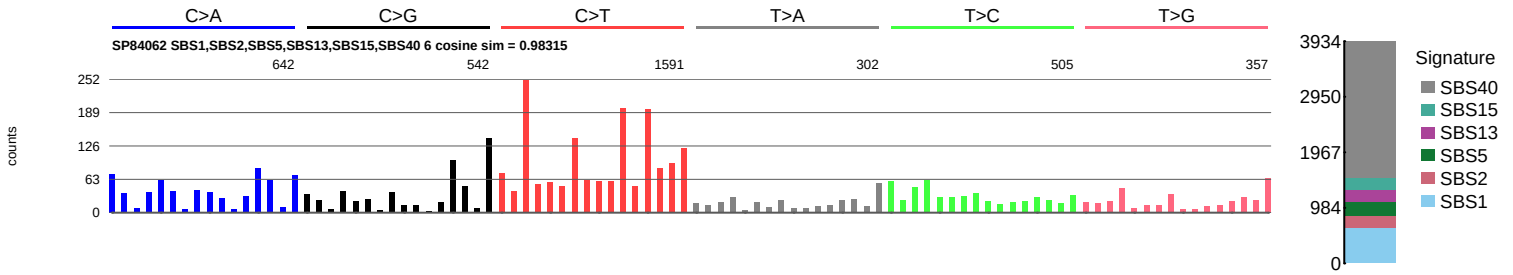

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98318
658
540
1568
304
477
388
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

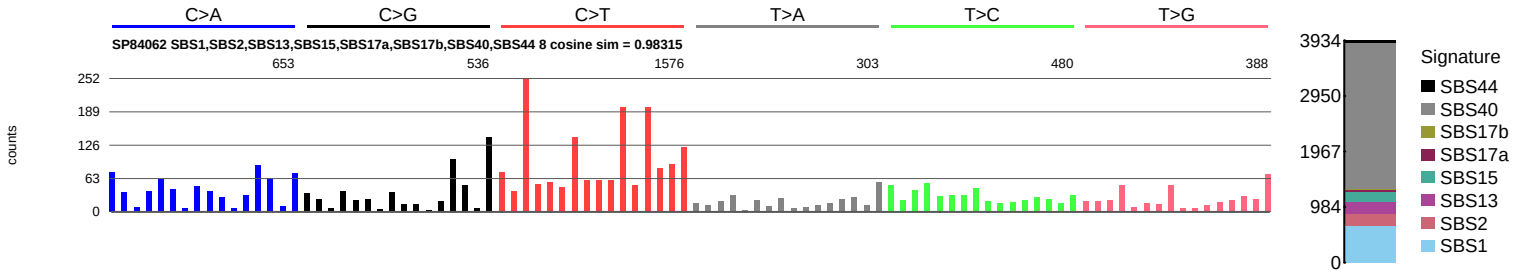

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS40,SBS44 7 cosine sim = 0.98317
660
539
1578
302
468
389
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS2
SBS1

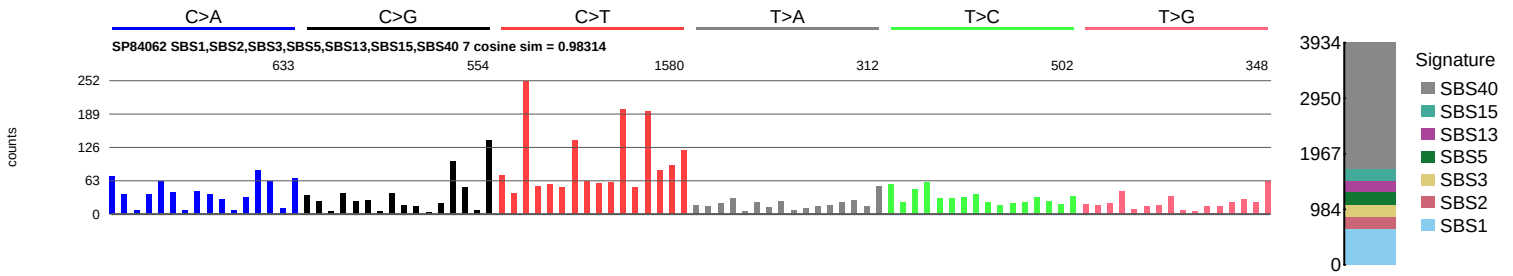

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS41 8 cosine sim = 0.98316
626
554
1573
317
502
354
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

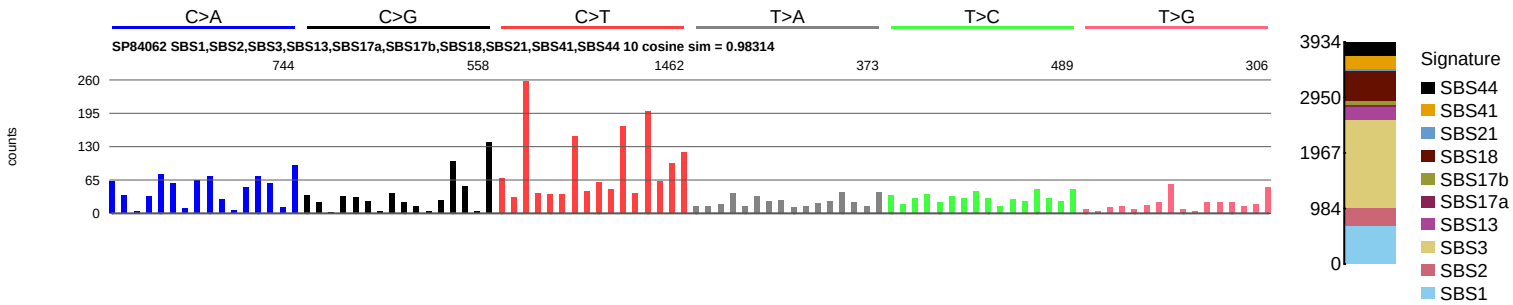

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98316
747
508
1468
304
536
370
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

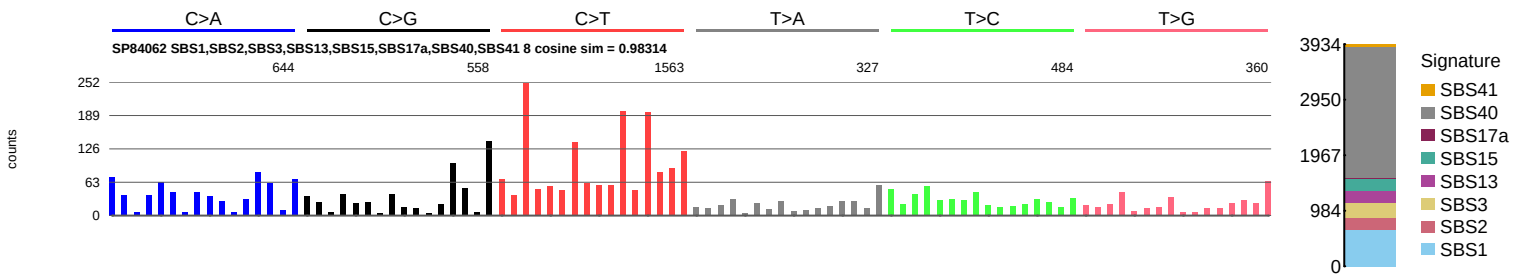

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS26,SBS40 7 cosine sim = 0.98315
643
552
1560
311
515
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS15
SBS13
SBS3
SBS2
SBS1

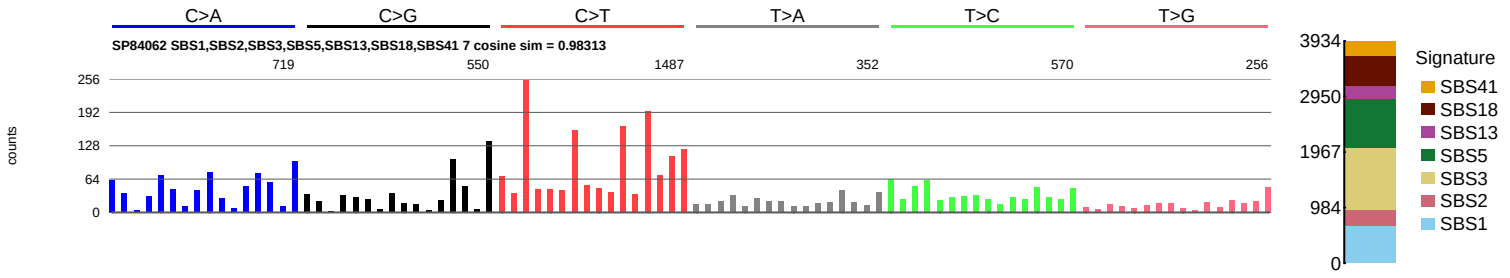

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40 6 cosine sim = 0.98315
642
542
1591
302
505
357
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1

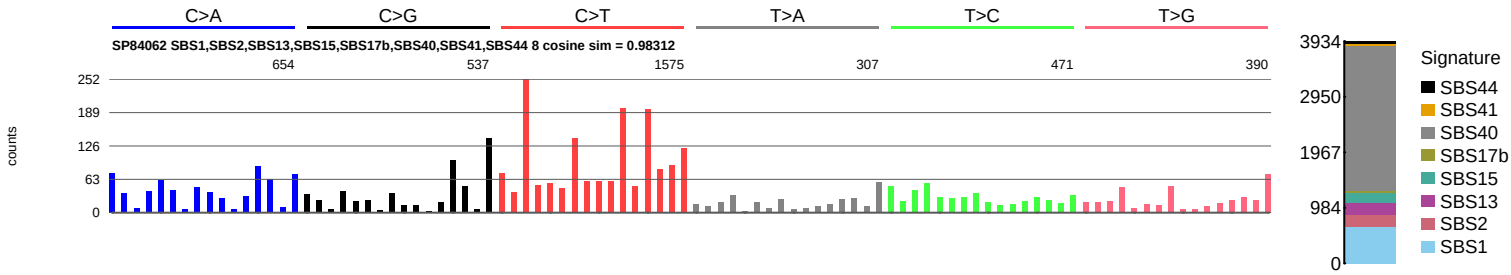

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98315
653
536
1576
303
480
388
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

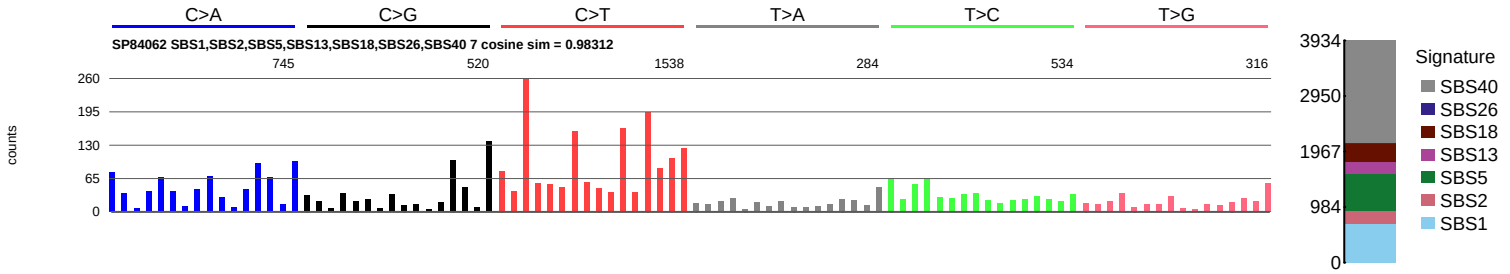

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40 7 cosine sim = 0.98314
633
554
1580
312
502
348
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

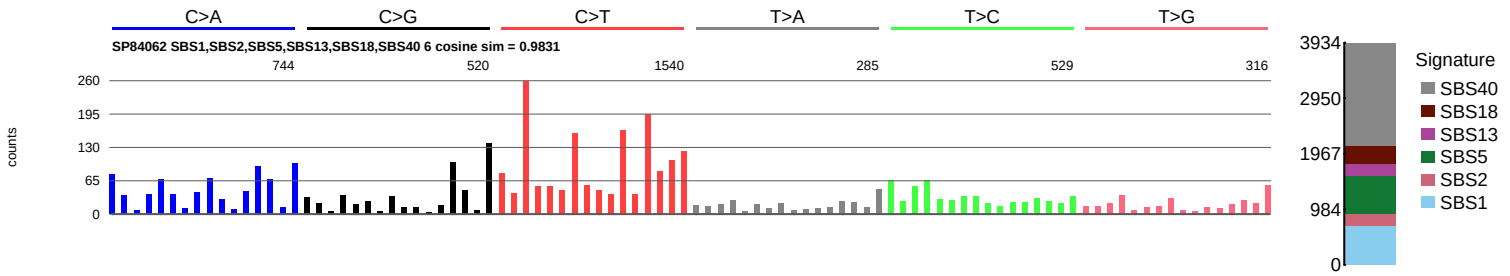

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98314
744
558
1462
373
489
306
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

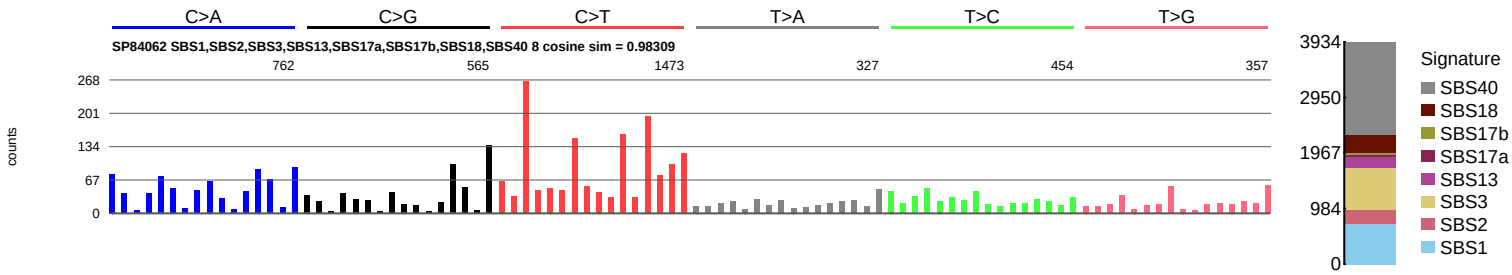

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS40,SBS41 8 cosine sim = 0.98314
644
558
1563
327
484
360
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

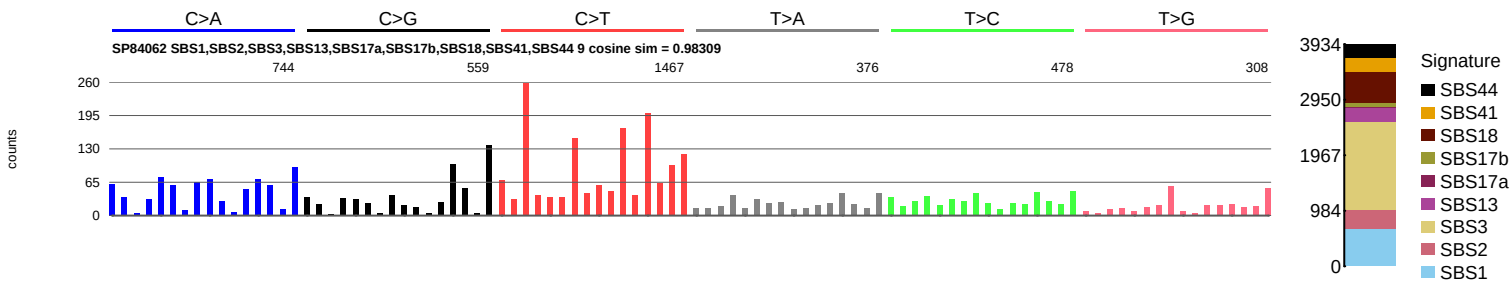

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41 7 cosine sim = 0.98313
719
550
1487
352
570
256
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

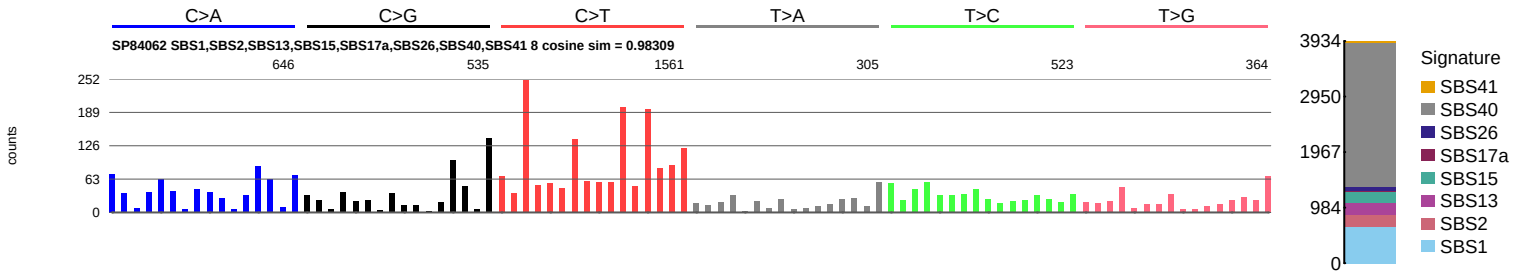

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98312
654
537
1575
307
471
390
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS2
SBS1

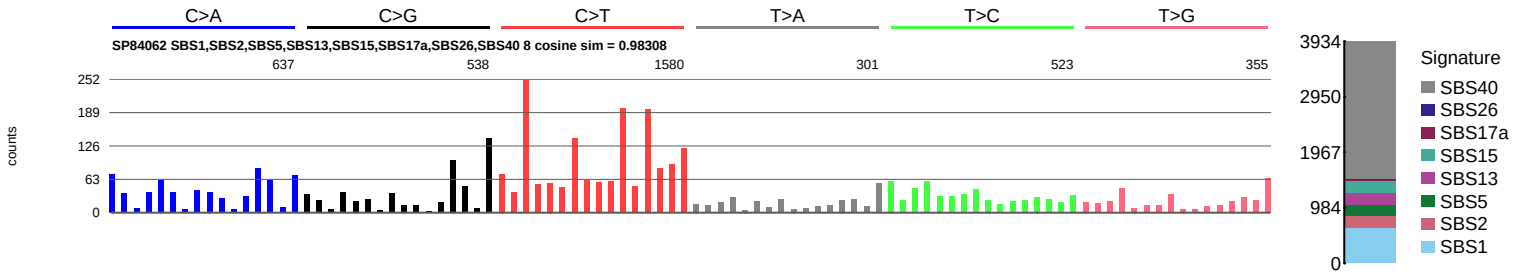

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.98312
745
520
1538
284
534
316
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

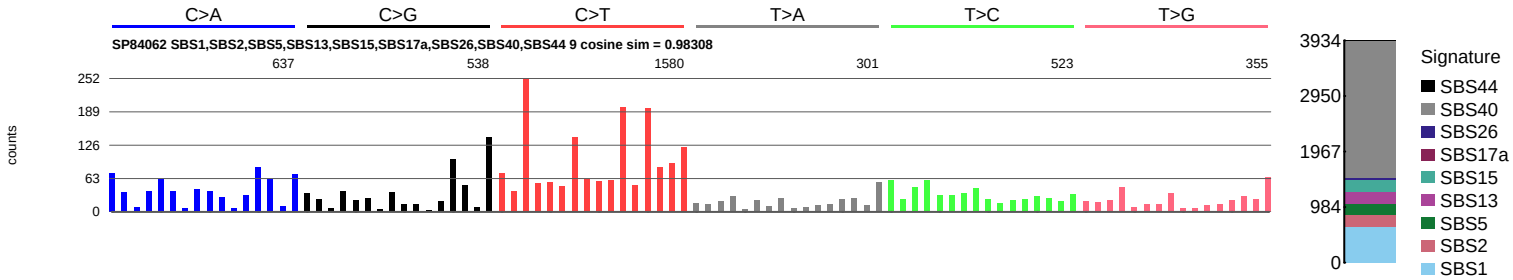

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40 6 cosine sim = 0.9831
744
520
1540
285
529
316
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

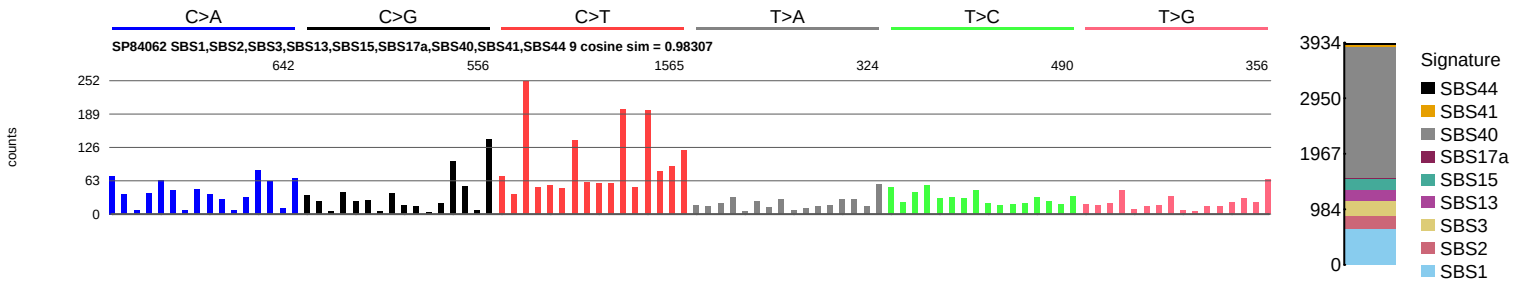

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98309
762
565
1473
327
454
357
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

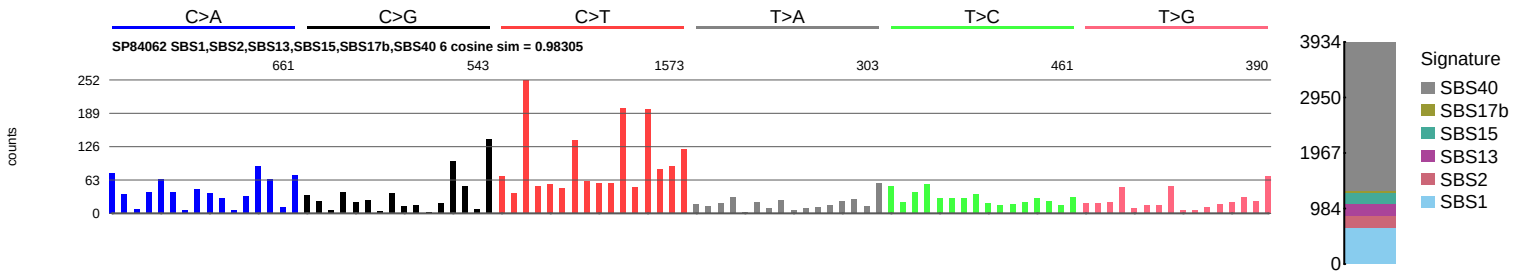

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98309
744
559
1467
376
478
308
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

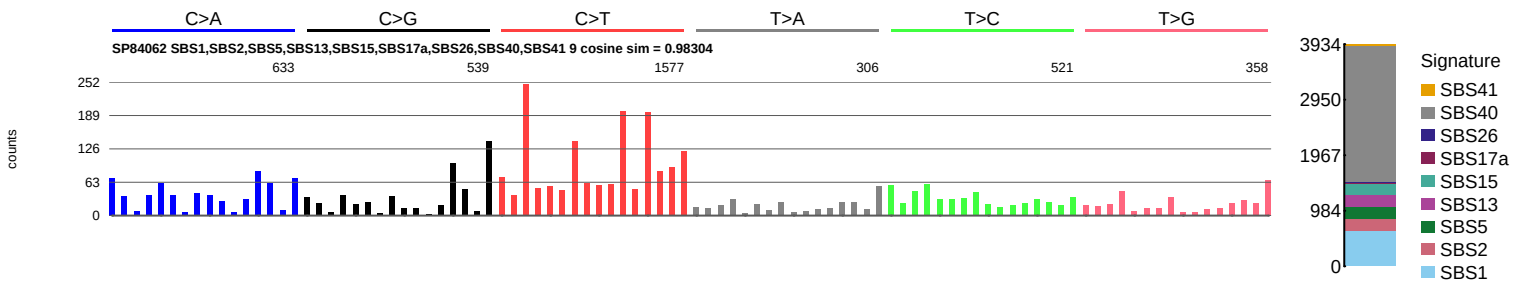

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41 8 cosine sim = 0.98309
646
535
1561
305
523
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1

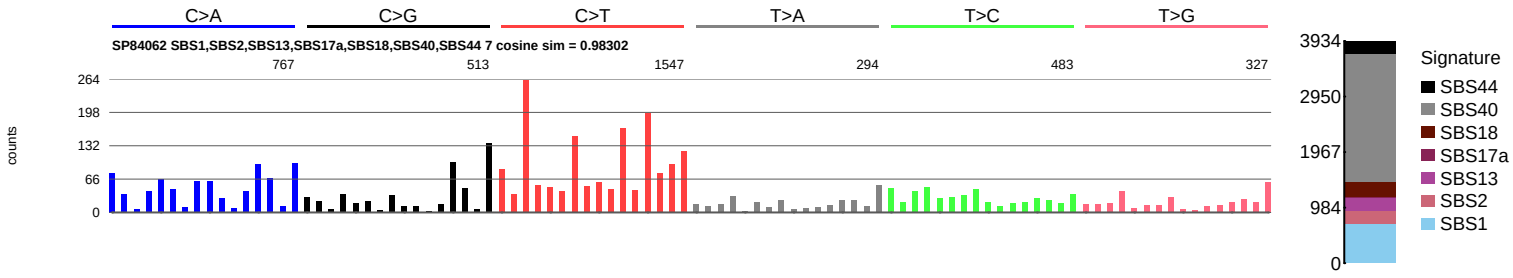

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40 8 cosine sim = 0.98308
637
538
1580
301
523
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

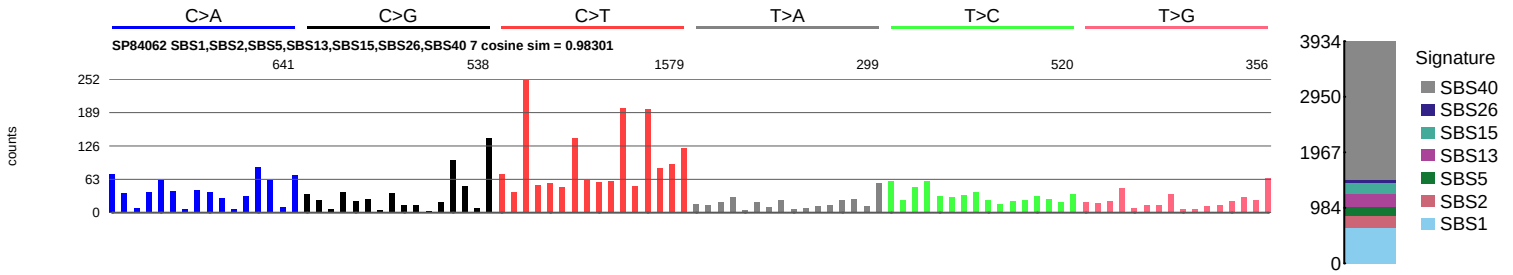

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS44 9 cosine sim = 0.98308
637
538
1580
301
523
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

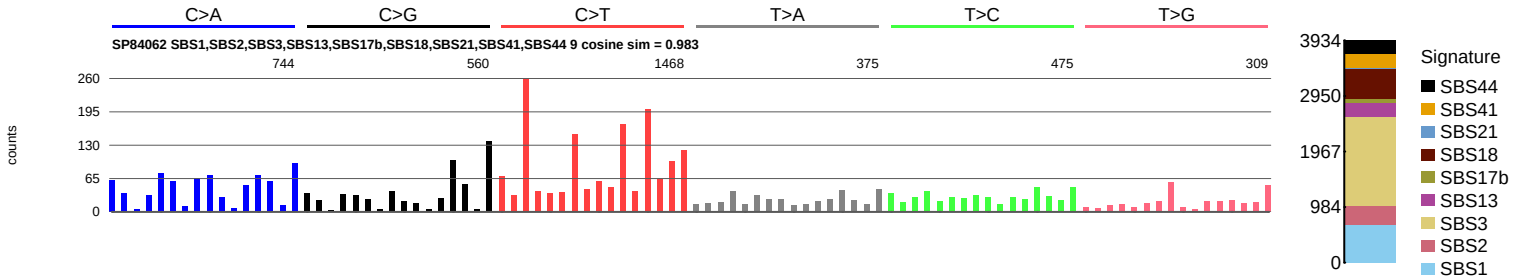

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS40,SBS41,SBS44 9 cosine sim = 0.98307
642
556
1565
324
490
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

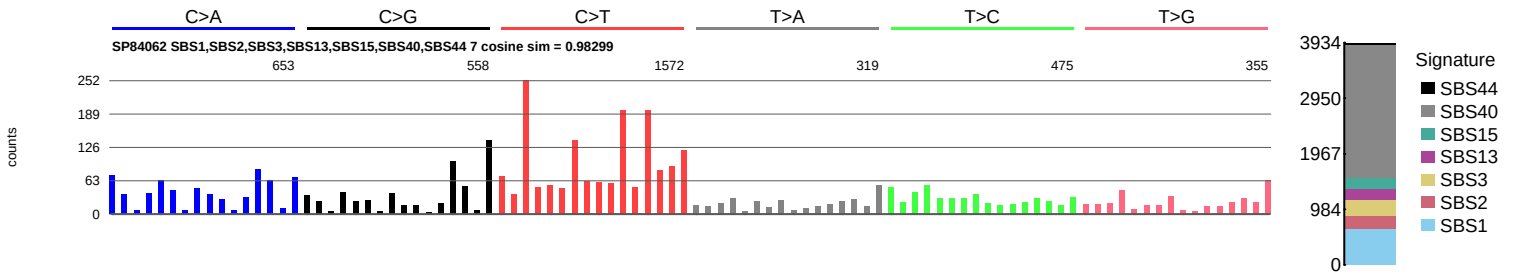

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS40 6 cosine sim = 0.98305
661
543
1573
303
461
390
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17b
SBS15
SBS13
SBS2
SBS1

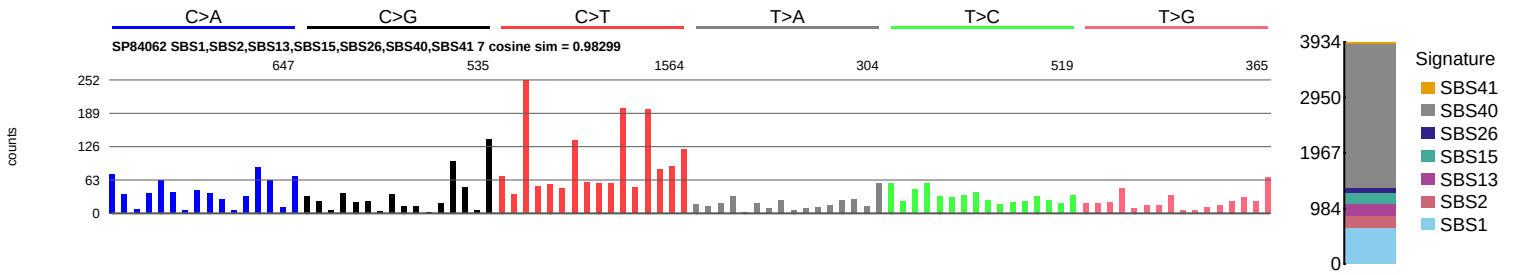

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41 9 cosine sim = 0.98304
633
539
1577
306
521
358
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

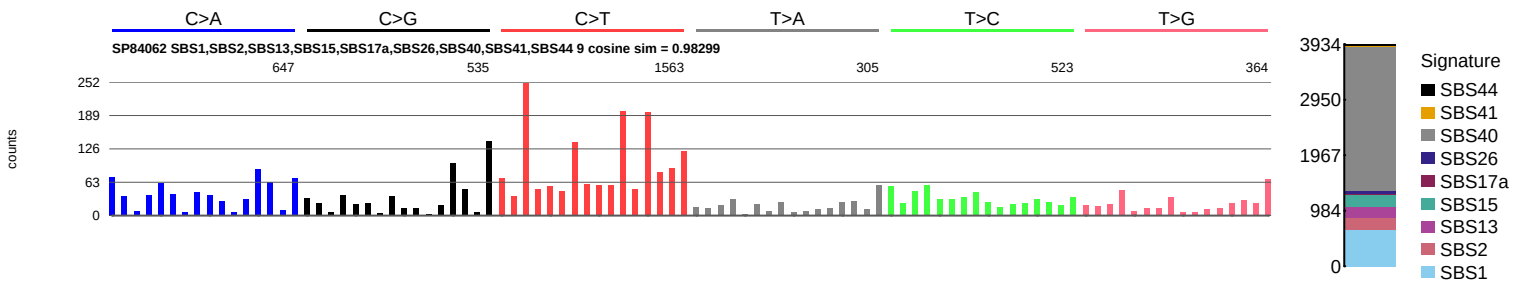

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98302
767
513
1547
294
483
327
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1

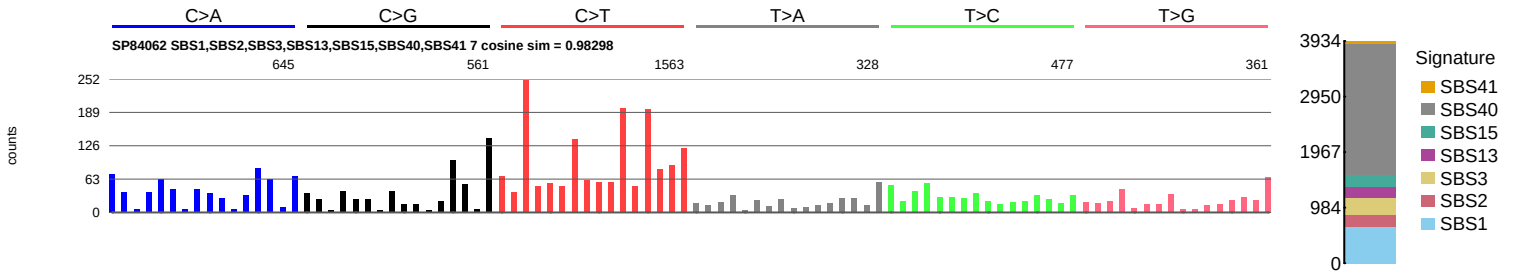

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS26,SBS40 7 cosine sim = 0.98301
641
538
1579
299
520
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS15
SBS13
SBS5
SBS2
SBS1

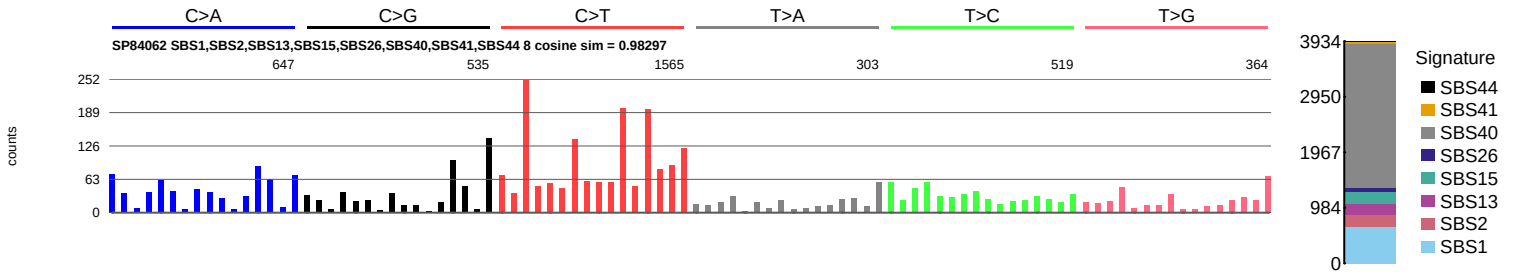

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.983
744
560
1468
375
475
309
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

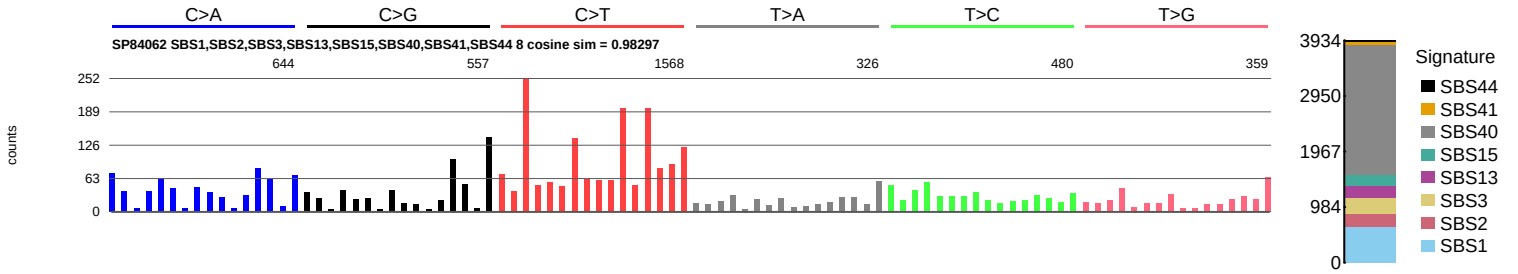

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS40,SBS44 7 cosine sim = 0.98299
653
558
1572
319
475
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS15
SBS13
SBS3
SBS2
SBS1

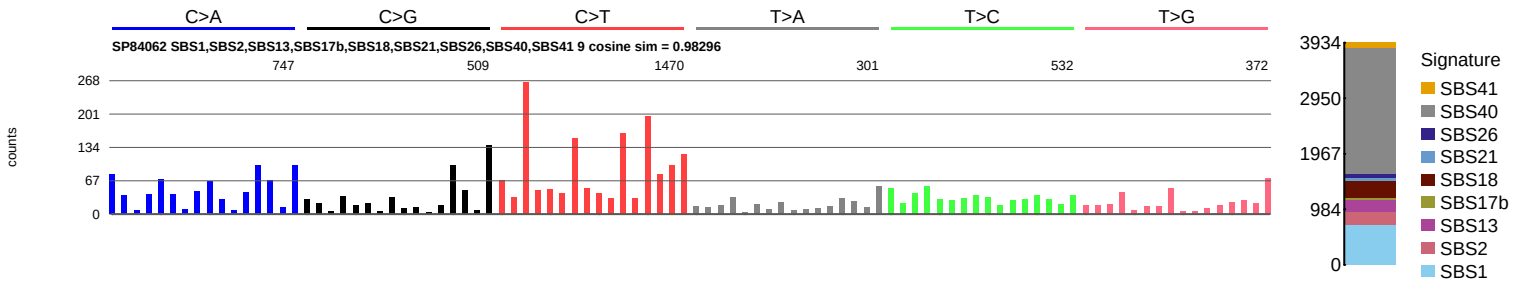

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS26,SBS40,SBS41 7 cosine sim = 0.98299
647
535
1564
304
519
365
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS15
SBS13
SBS2
SBS1

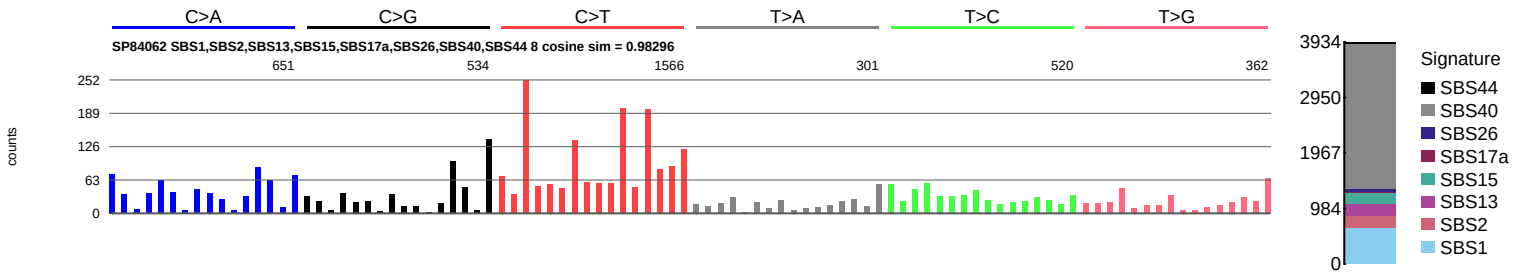

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98299
647
535
1563
305
523
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1

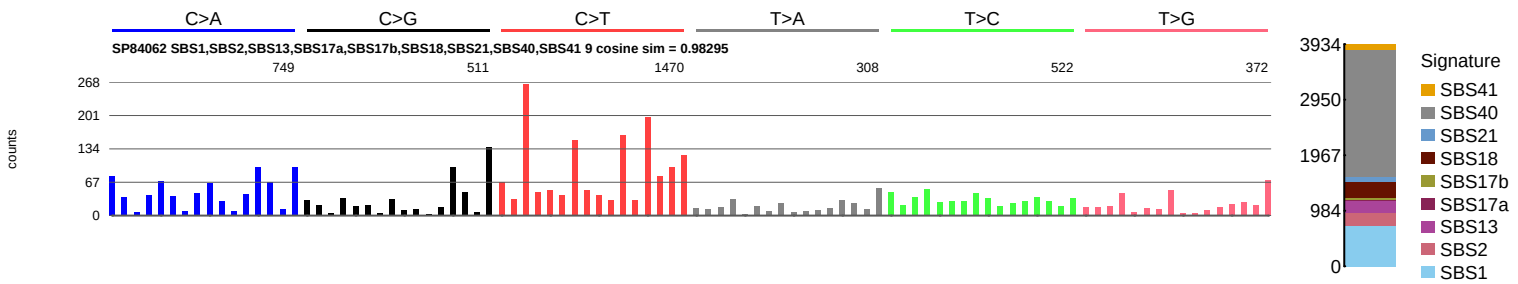

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS40,SBS41 7 cosine sim = 0.98298
645
561
1563
328
477
361
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS3
SBS2
SBS1

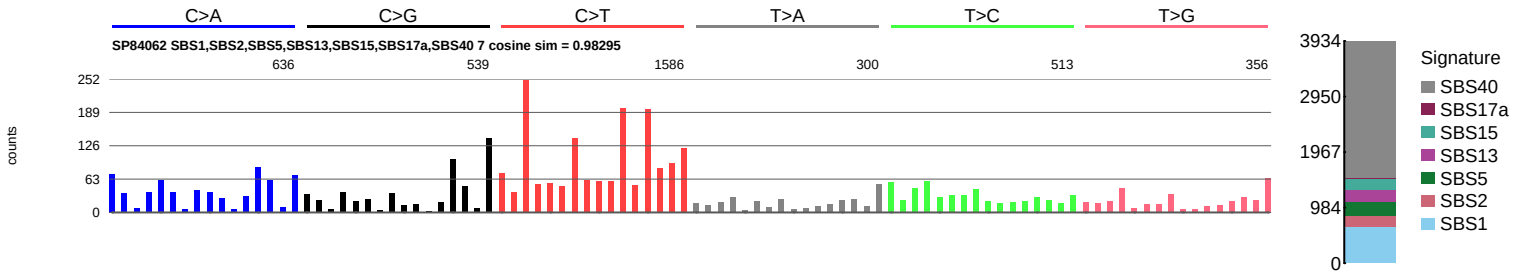

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98297
647
535
1565
303
519
364
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS15
SBS13
SBS2
SBS1

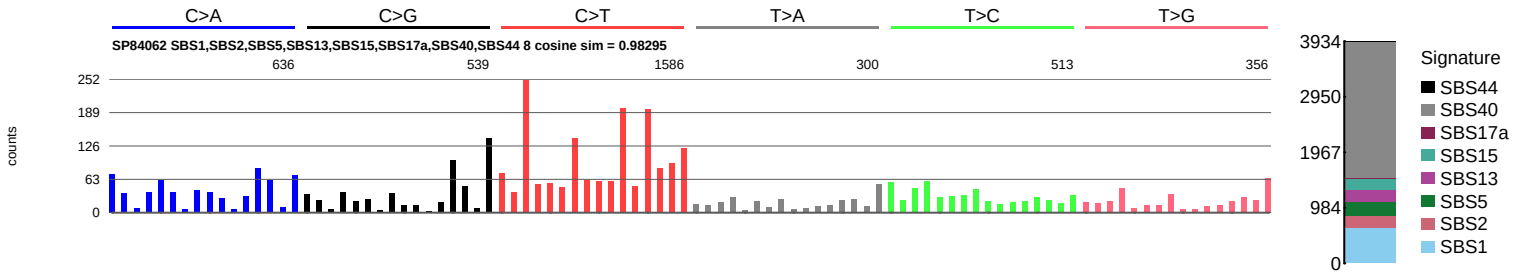

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS40,SBS41,SBS44 8 cosine sim = 0.98297
644
557
1568
326
480
359
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS15
SBS13
SBS3
SBS2
SBS1

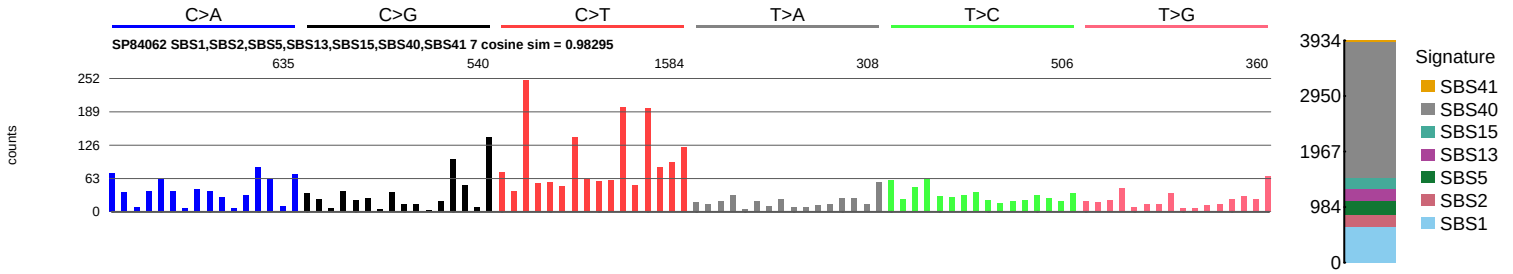

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98296
747
509
1470
301
532
372
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

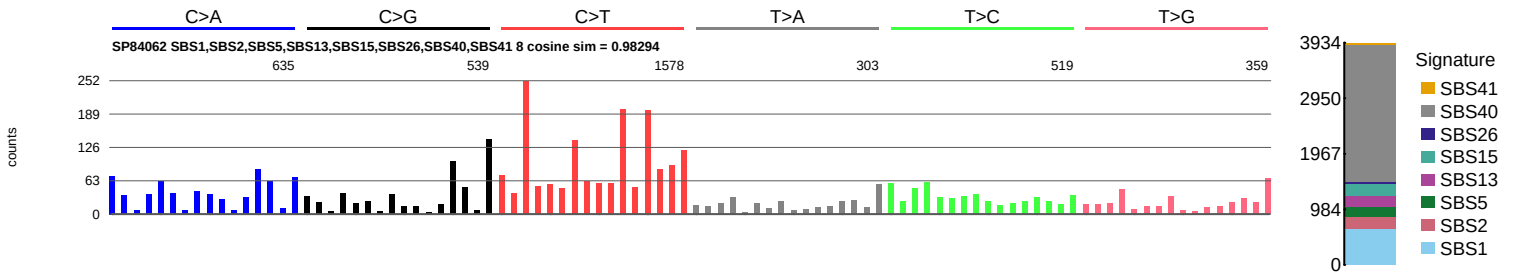

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS40,SBS44 8 cosine sim = 0.98296
651
534
1566
301
520
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1

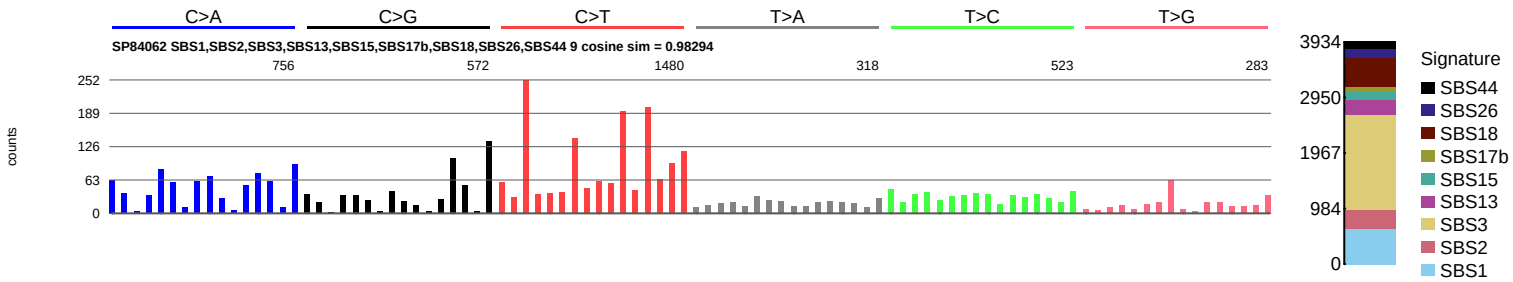

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98295
749
511
1470
308
522
372
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

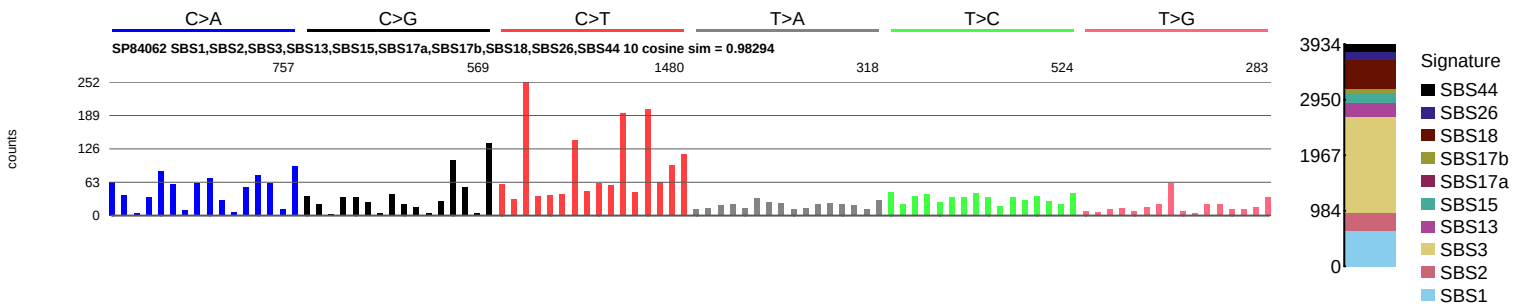

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40 7 cosine sim = 0.98295
636
539
1586
300
513
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

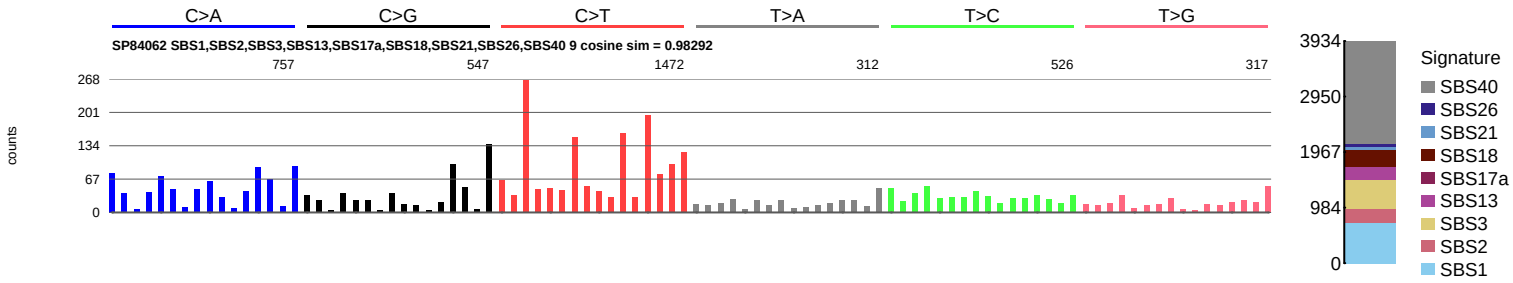

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS44 8 cosine sim = 0.98295
636
539
1586
300
513
356
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

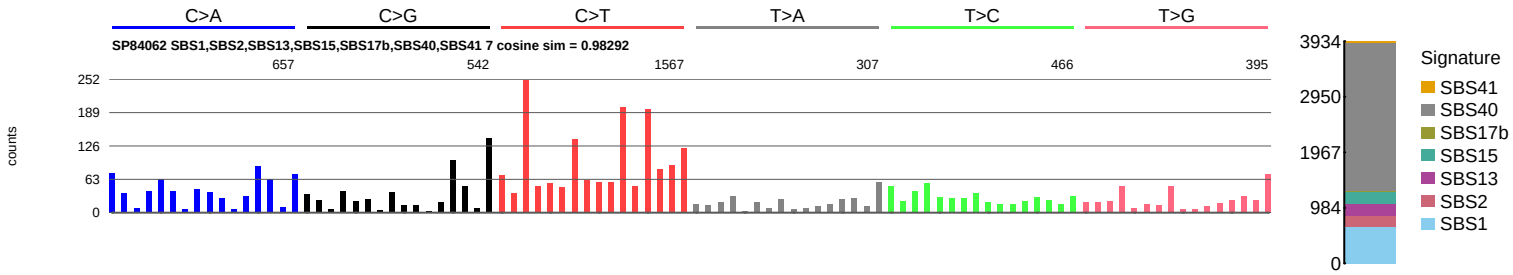

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS41 7 cosine sim = 0.98295
635
540
1584
308
506
360
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1

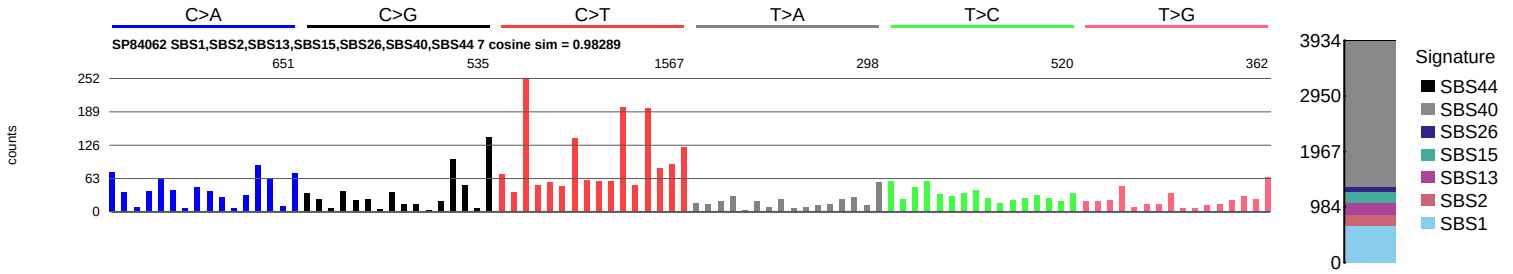

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS26,SBS40,SBS41 8 cosine sim = 0.98294
635
539
1578
303
519
359
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS15
SBS13
SBS5
SBS2
SBS1

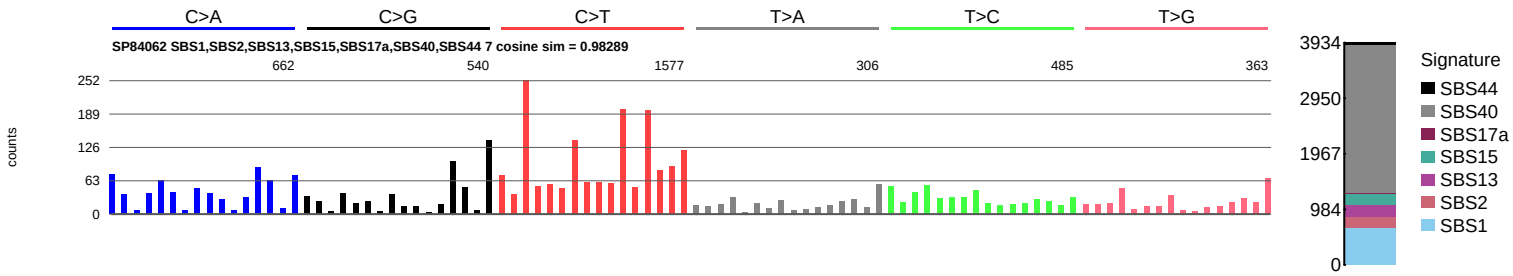

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98294
756
572
1480
318
523
283
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

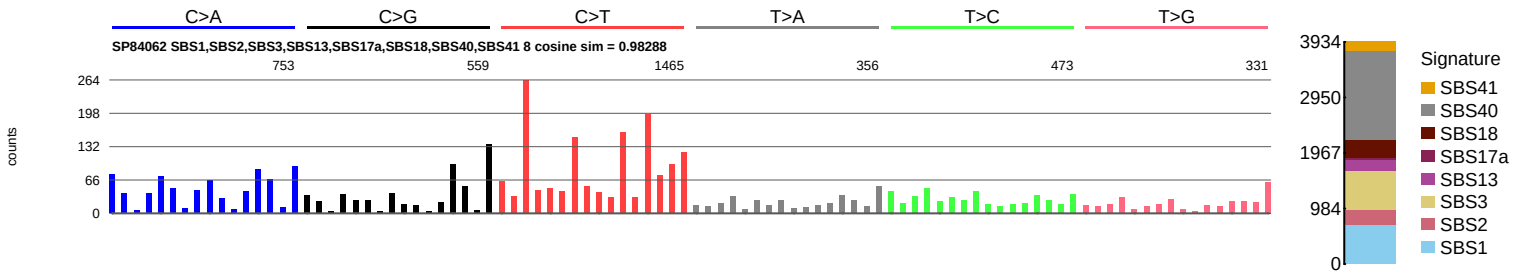

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98294
757
569
1480
318
524
283
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

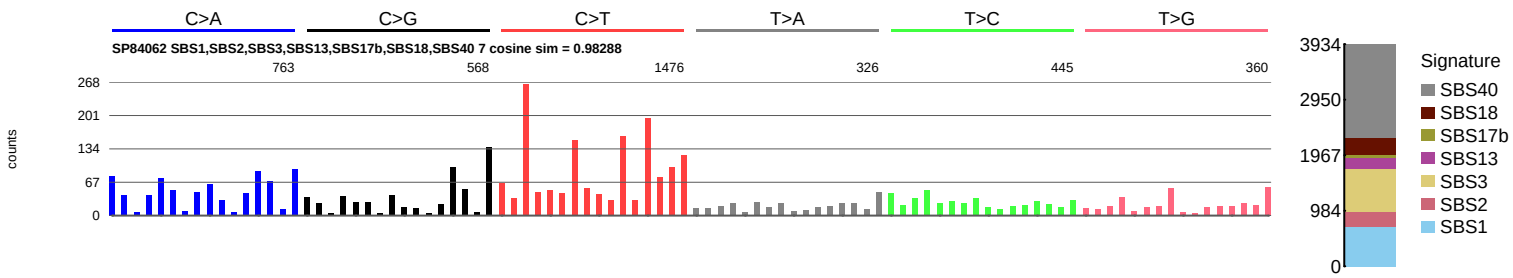

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98292
757
547
1472
312
526
317
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

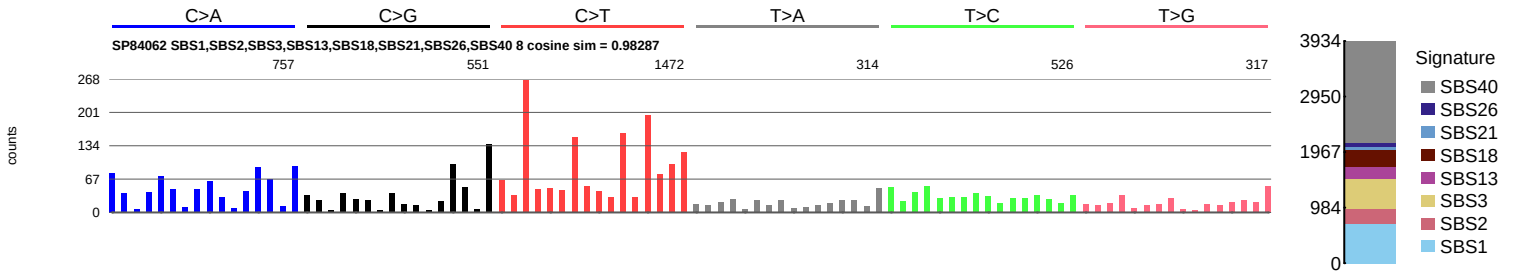

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS40,SBS41 7 cosine sim = 0.98292
657
542
1567
307
466
395
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS2
SBS1

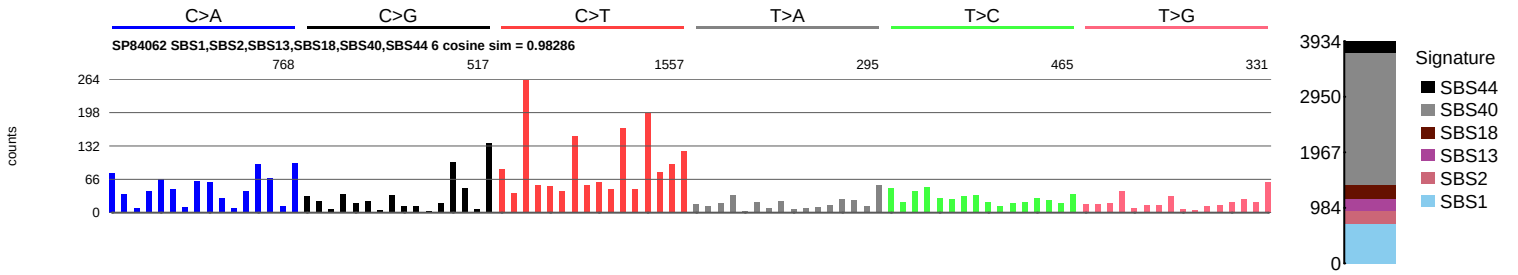

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS26,SBS40,SBS44 7 cosine sim = 0.98289
651
535
1567
298
520
362
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS15
SBS13
SBS2
SBS1

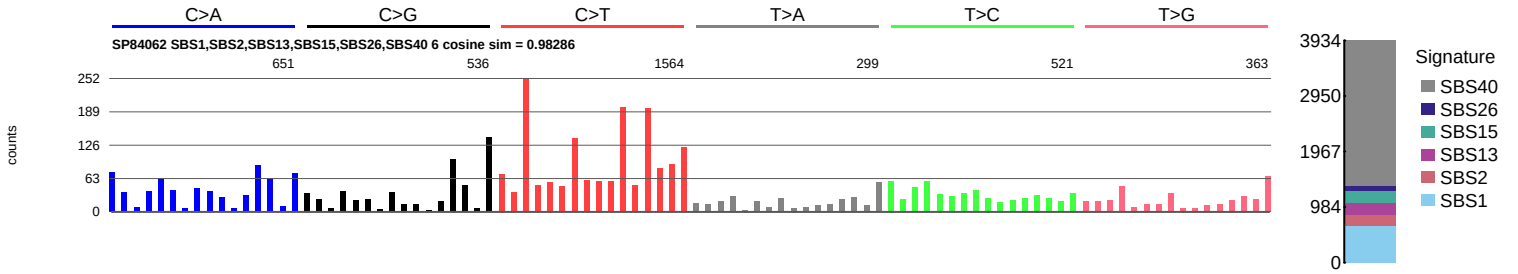

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS40,SBS44 7 cosine sim = 0.98289
662
540
1577
306
485
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS2
SBS1

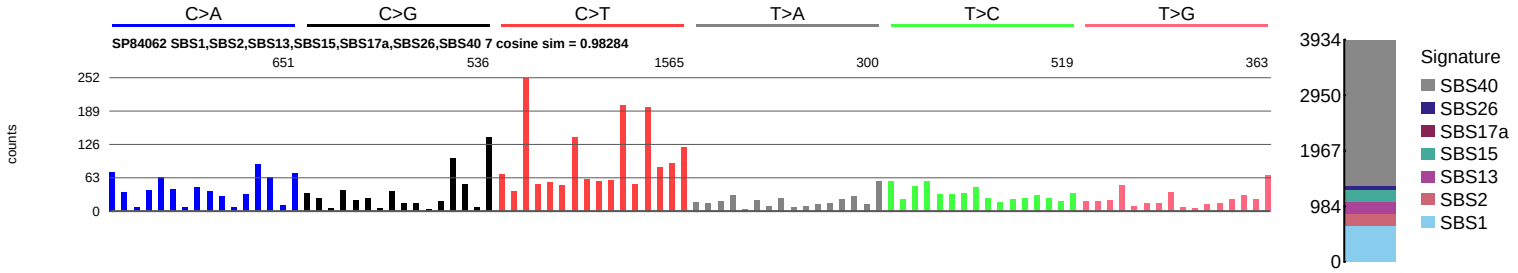

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98288
753
559
1465
356
473
331
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

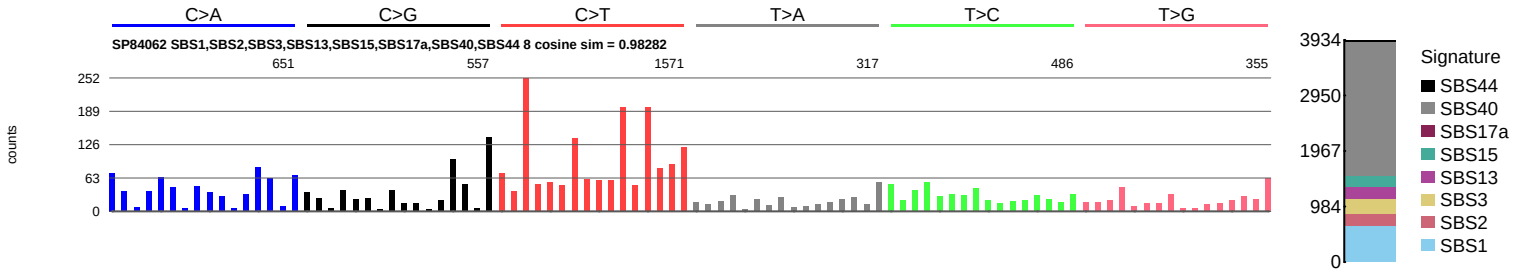

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98288
763
568
1476
326
445
360
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

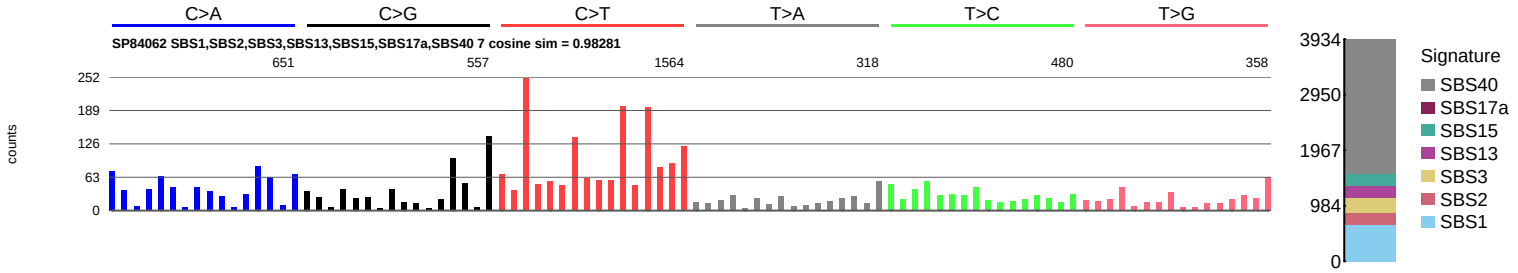

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98287
757
551
1472
314
526
317
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

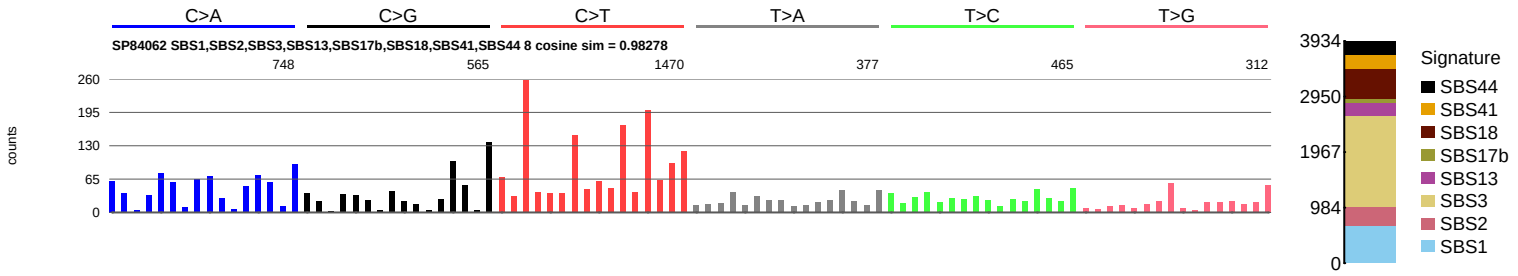

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS40,SBS44 6 cosine sim = 0.98286
768
517
1557
295
465
331
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS2
SBS1

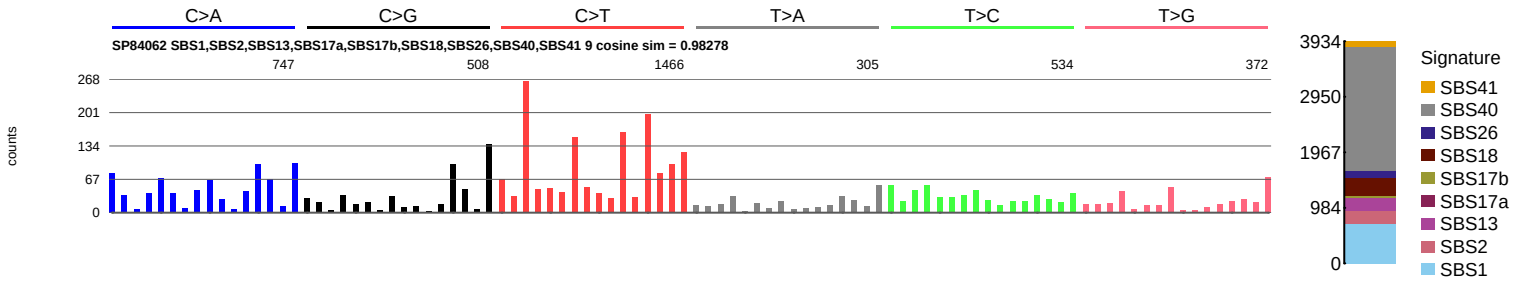

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS26,SBS40 6 cosine sim = 0.98286
651
536
1564
299
521
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS15
SBS13
SBS2
SBS1

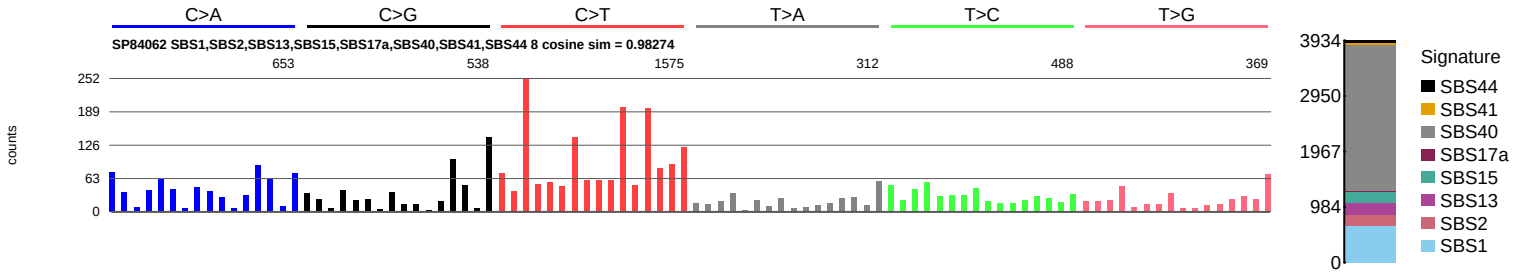

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS40 7 cosine sim = 0.98284
651
536
1565
300
519
363
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1

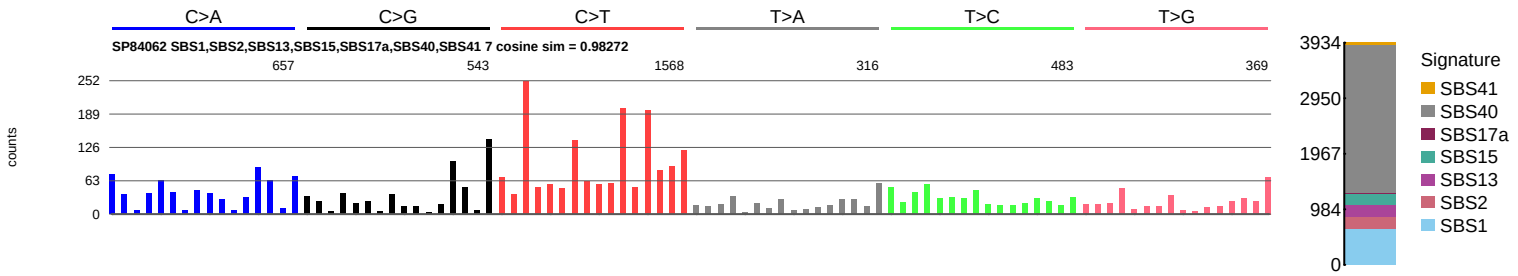

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS40,SBS44 8 cosine sim = 0.98282
651
557
1571
317
486
355
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

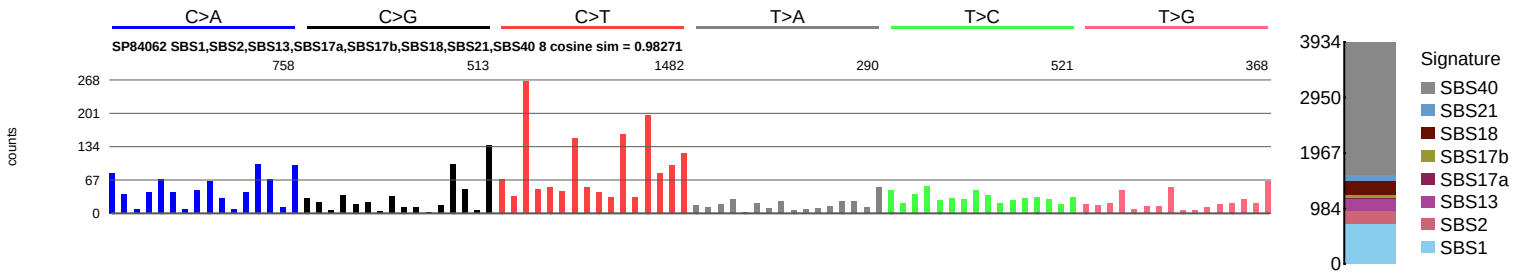

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS40 7 cosine sim = 0.98281
651
557
1564
318
480
358
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

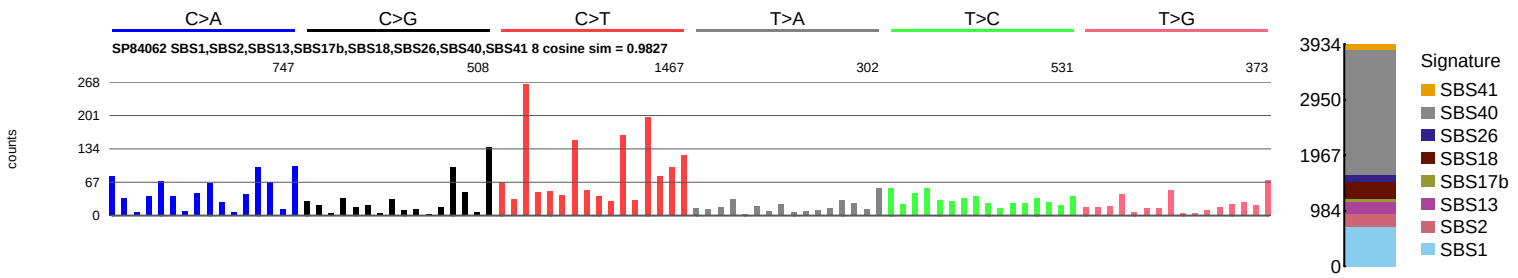

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98278
748
565
1470
377
465
312
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

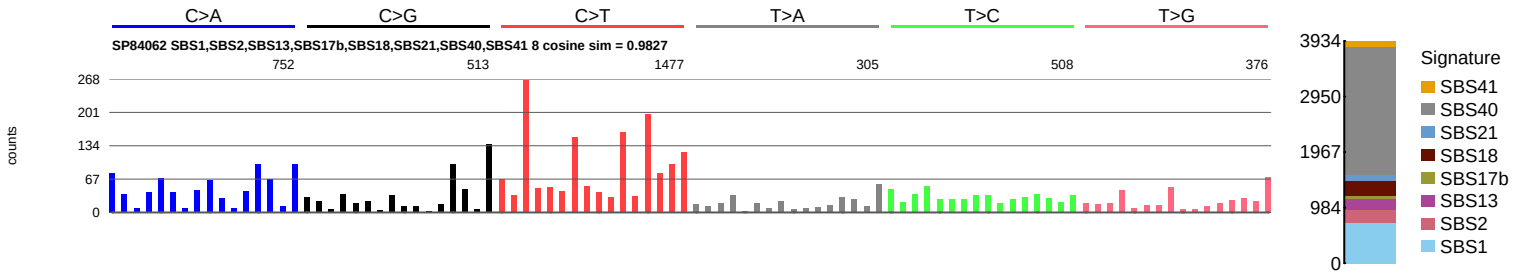

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98278
747
508
1466
305
534
372
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

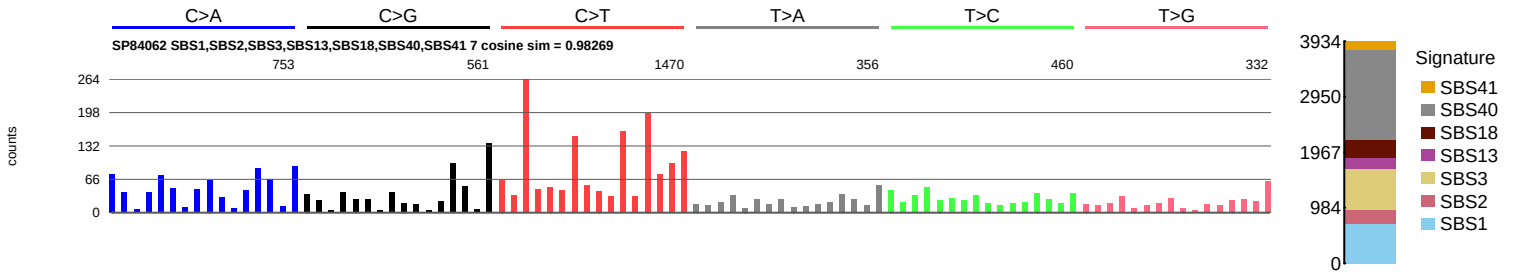

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS40,SBS41,SBS44 8 cosine sim = 0.98274
653
538
1575
312
488
369
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS2
SBS1

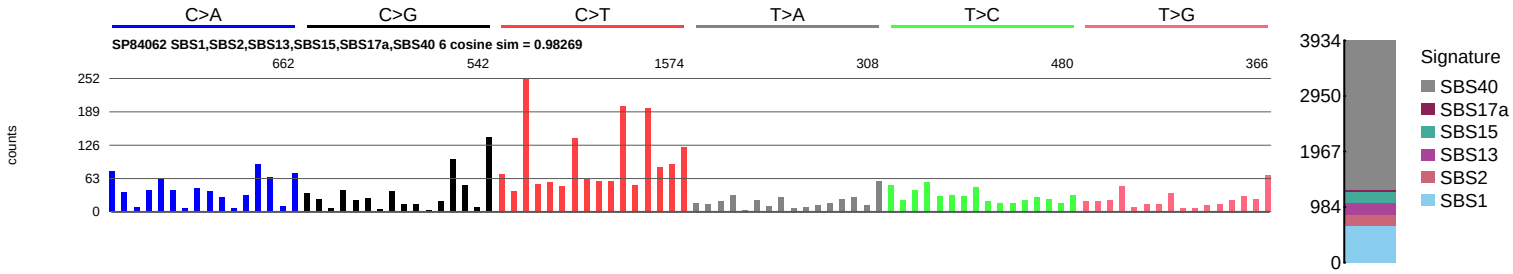

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS40,SBS41 7 cosine sim = 0.98272
657
543
1568
316
483
369
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS2
SBS1

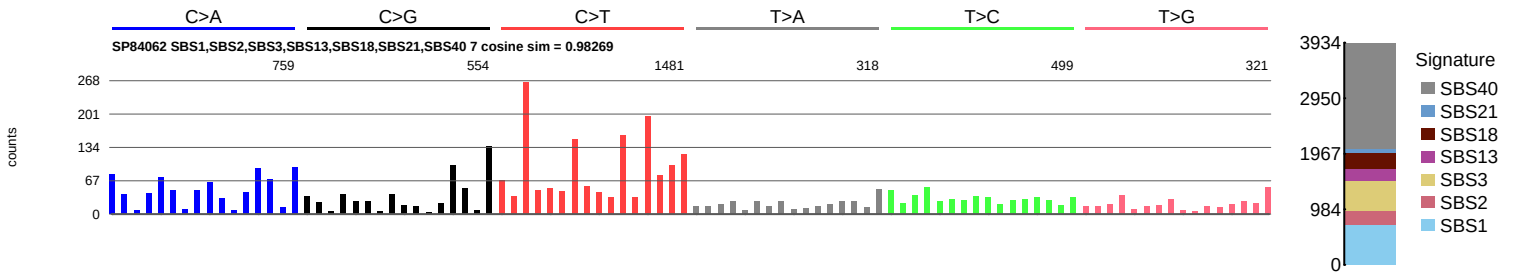

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98271
758
513
1482
290
521
368
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

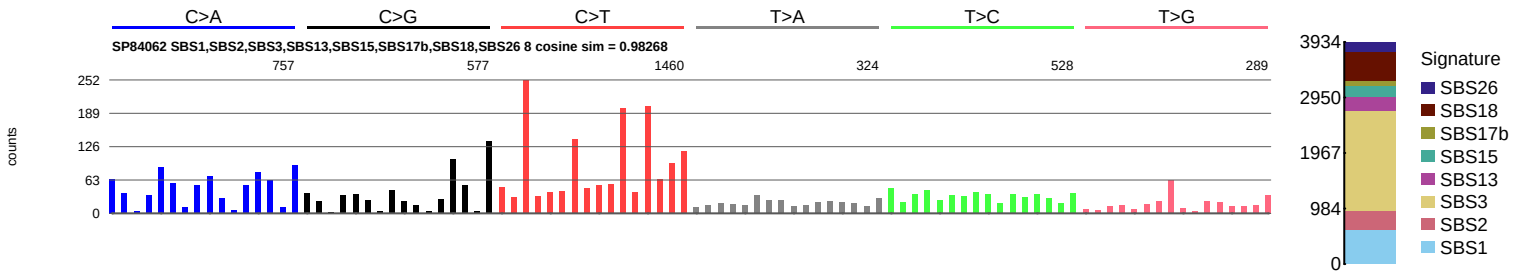

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9827
747
508
1467
302
531
373
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

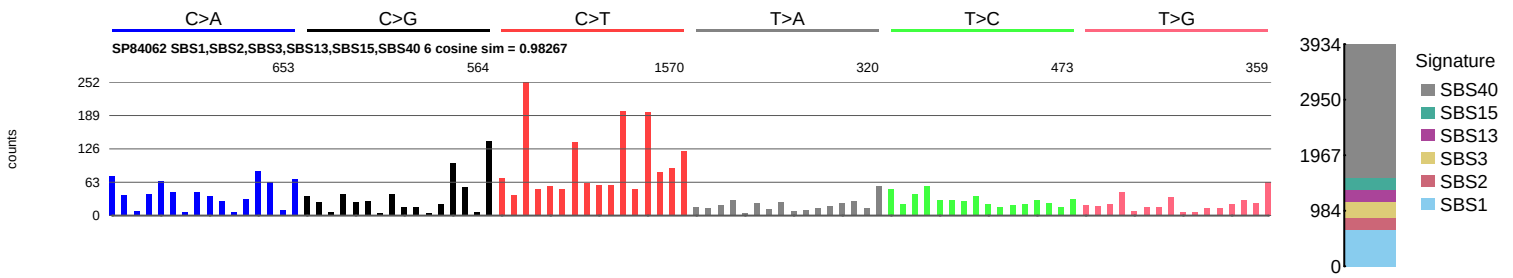

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9827
752
513
1477
305
508
376
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

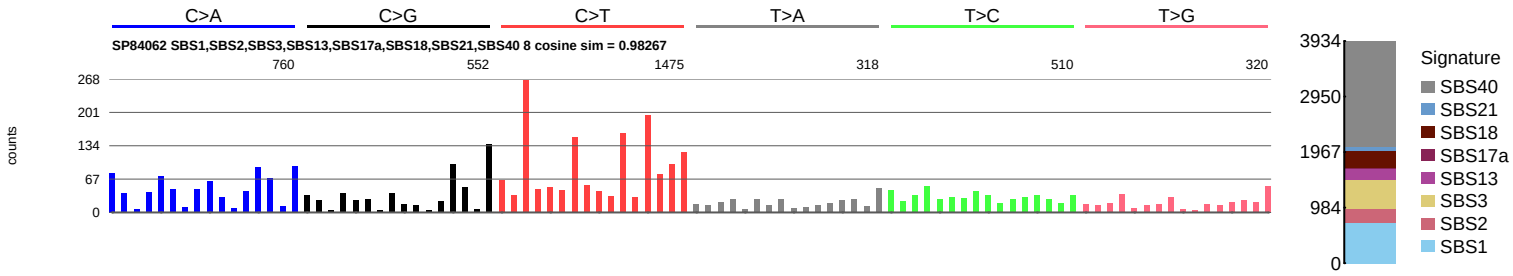

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98269
753
561
1470
356
460
332
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

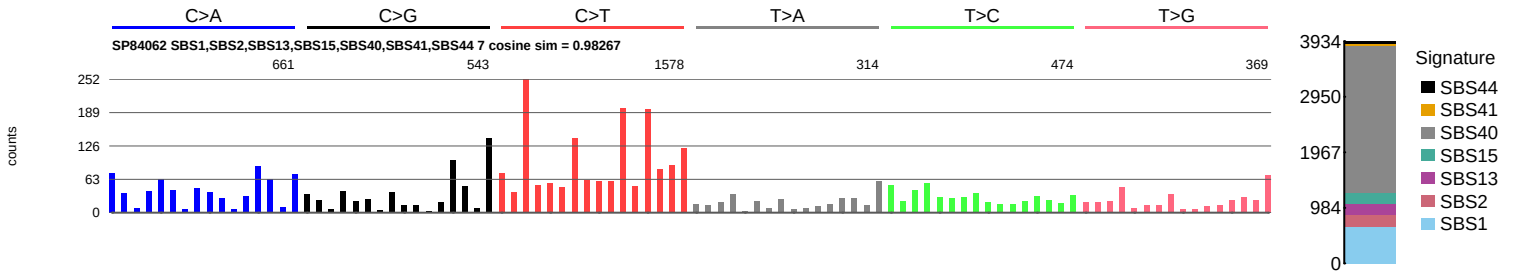

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS40 6 cosine sim = 0.98269
662
542
1574
308
480
366
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS2
SBS1

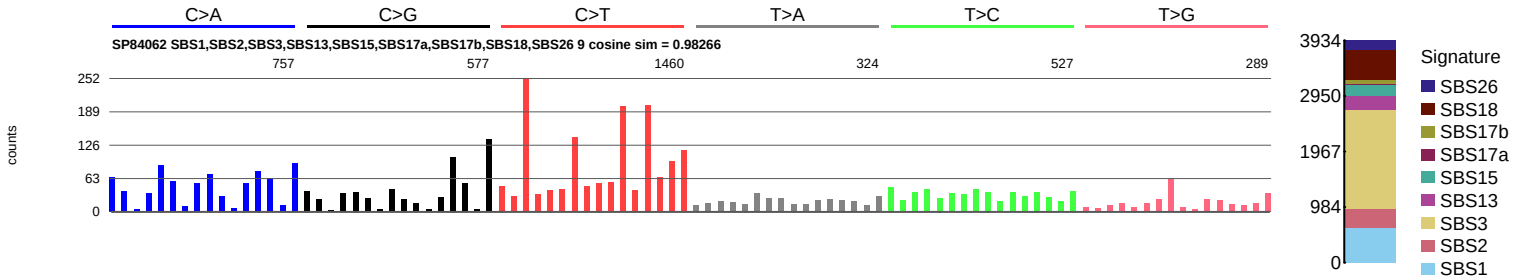

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98269
759
554
1481
318
499
321
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

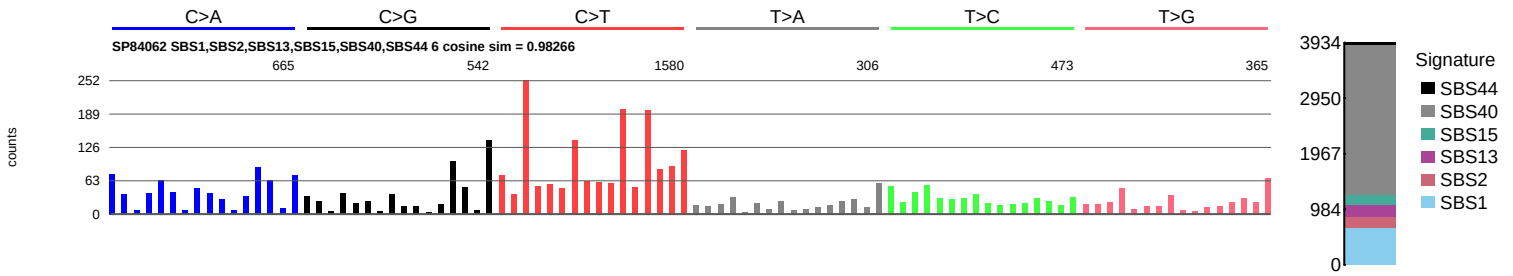

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98268
757
577
1460
324
528
289
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

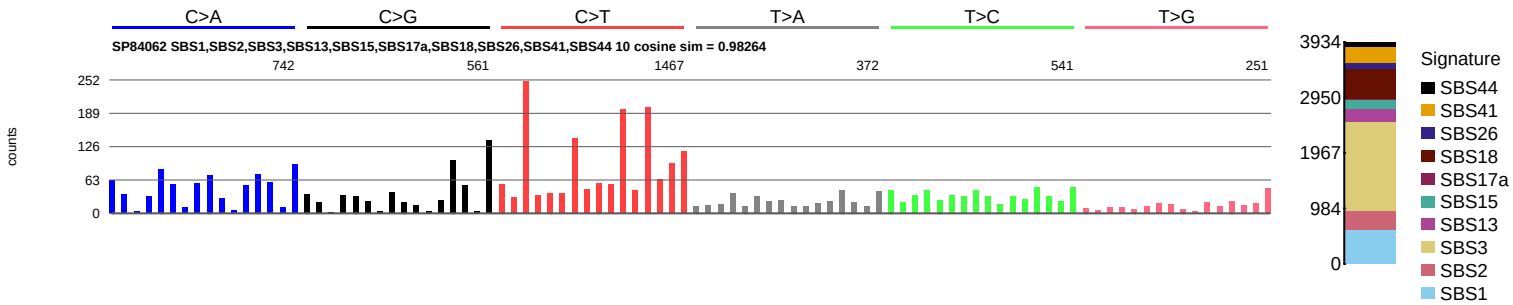

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS40 6 cosine sim = 0.98267
653
564
1570
320
473
359
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS15
SBS13
SBS3
SBS2
SBS1

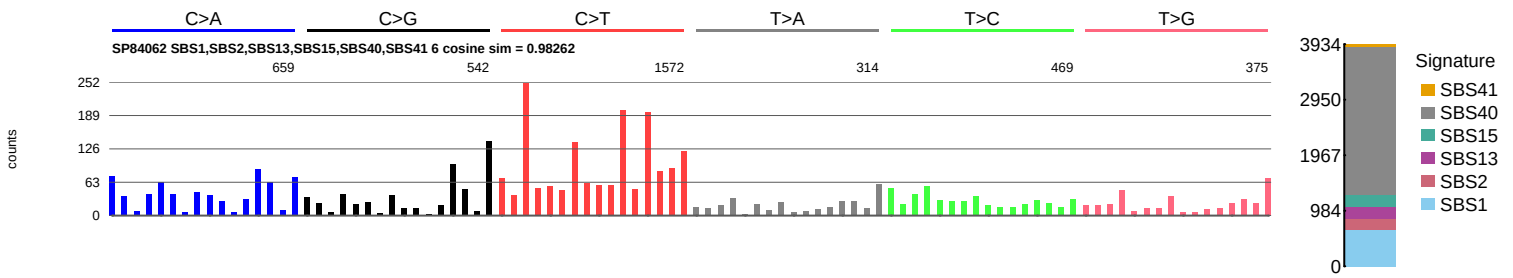

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98267
760
552
1475
318
510
320
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

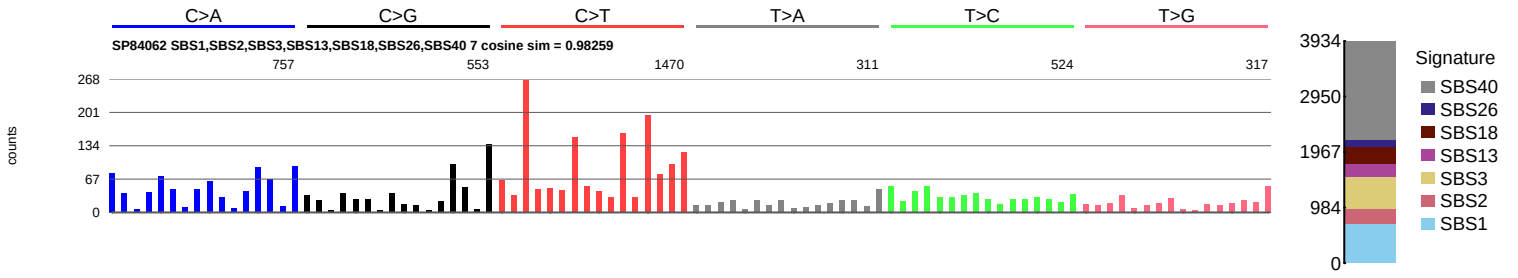

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS40,SBS41,SBS44 7 cosine sim = 0.98267
661
543
1578
314
474
369
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS15
SBS13
SBS2
SBS1

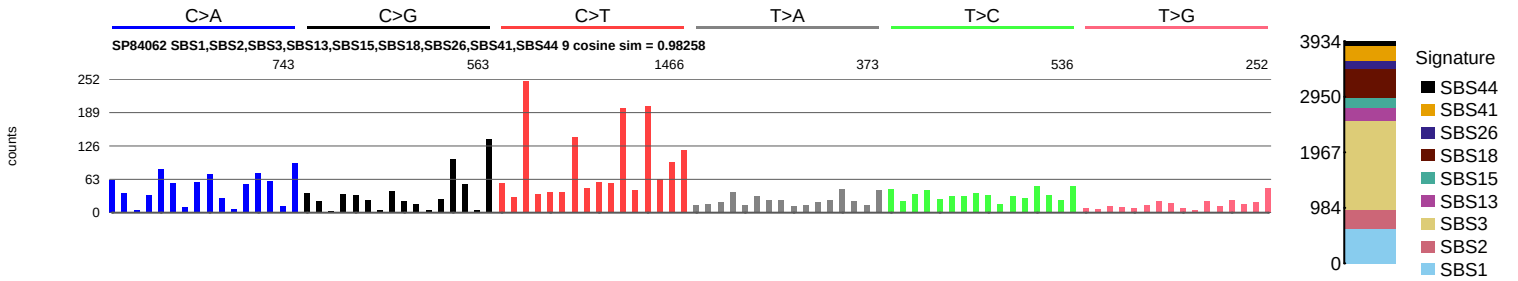

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98266
757
577
1460
324
527
289
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

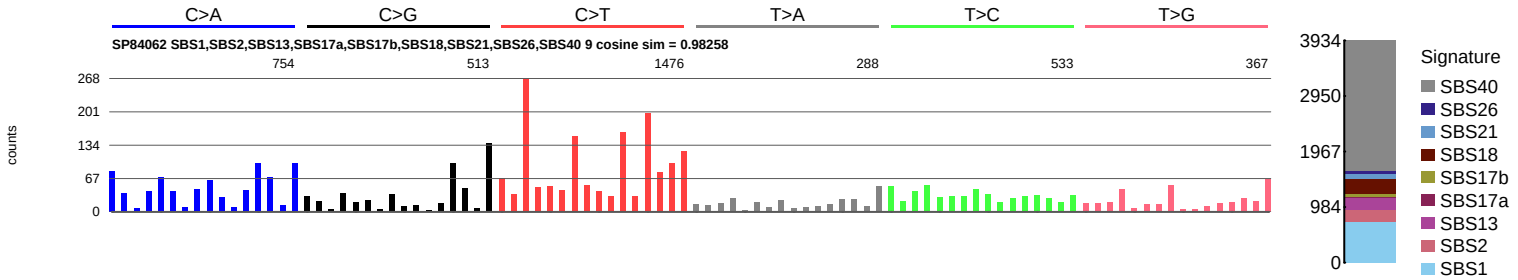

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS40,SBS44 6 cosine sim = 0.98266
665
542
1580
306
473
365
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS15
SBS13
SBS2
SBS1

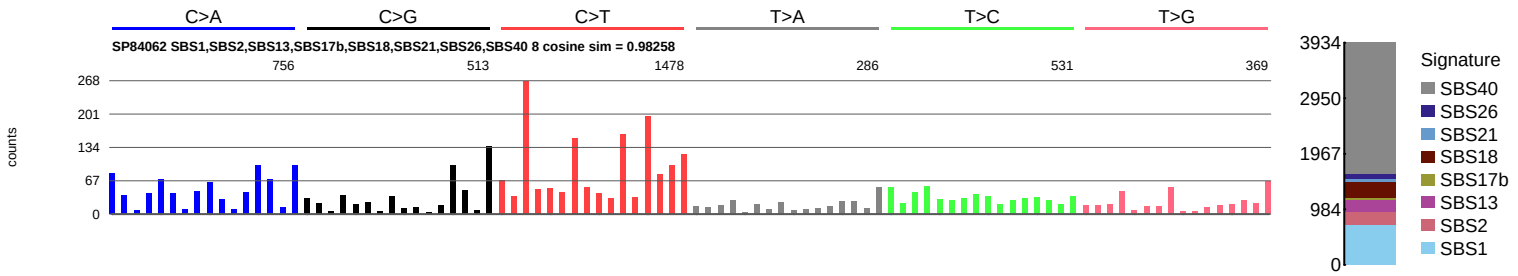

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98264
742
561
1467
372
541
251
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

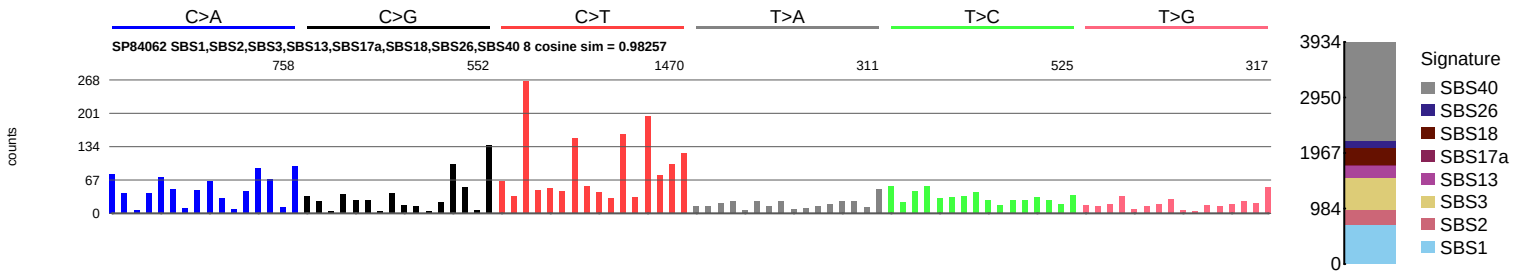

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS40,SBS41 6 cosine sim = 0.98262
659
542
1572
314
469
375
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS2
SBS1

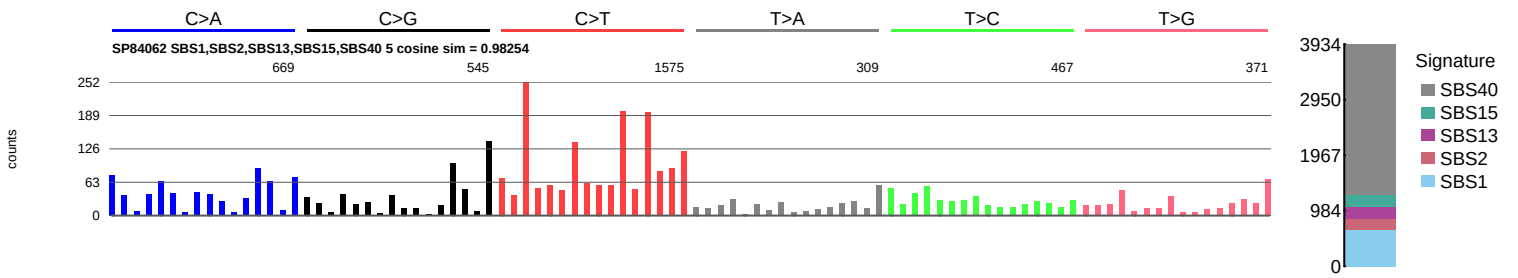

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.98259
757
553
1470
311
524
317
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

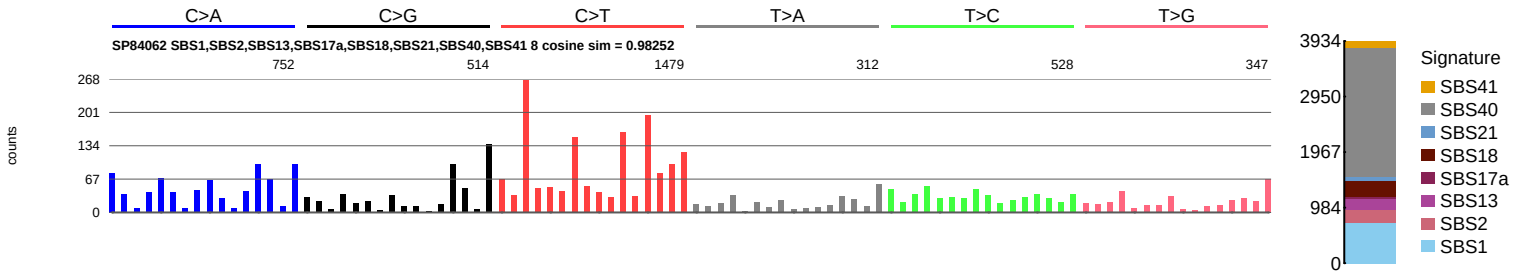

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98258
743
563
1466
373
536
252
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

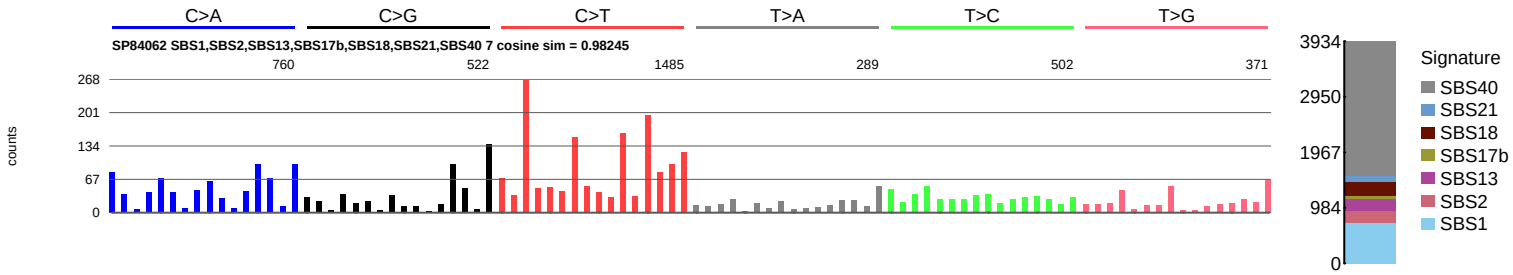

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98258
754
513
1476
288
533
367
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

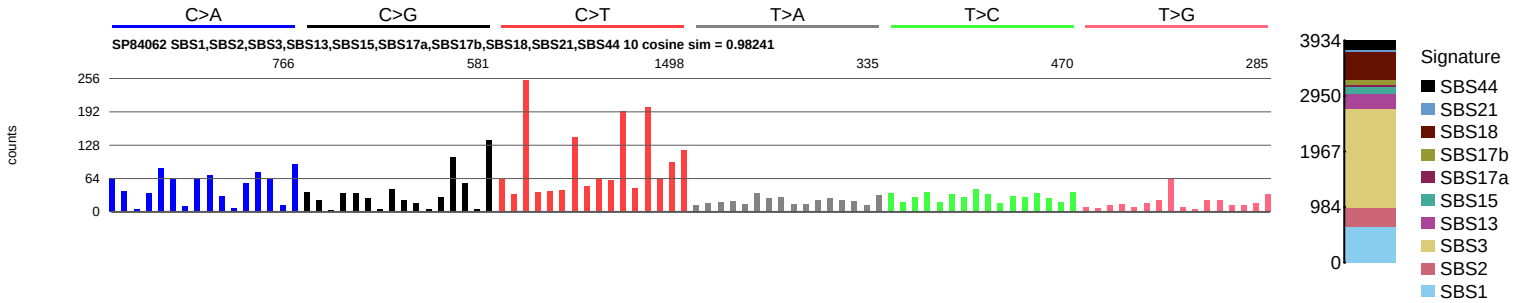

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98258
756
513
1478
286
531
369
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

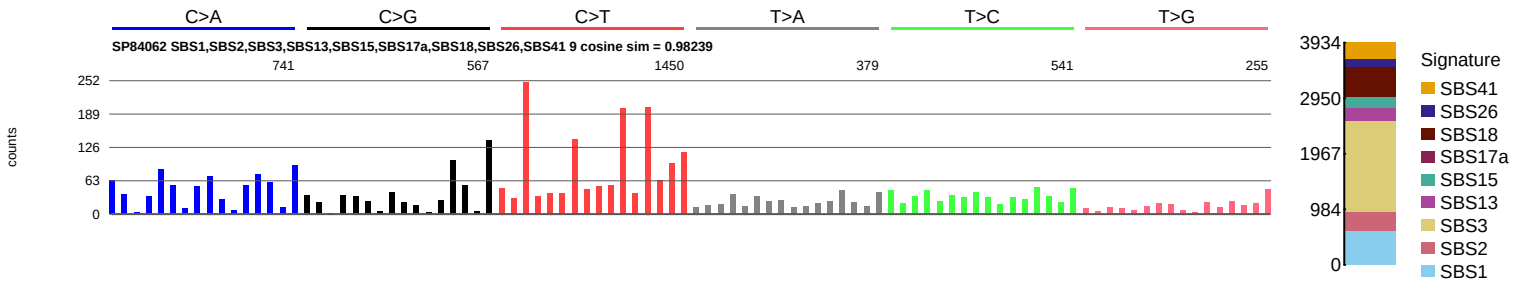

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98257
758
552
1470
311
525
317
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

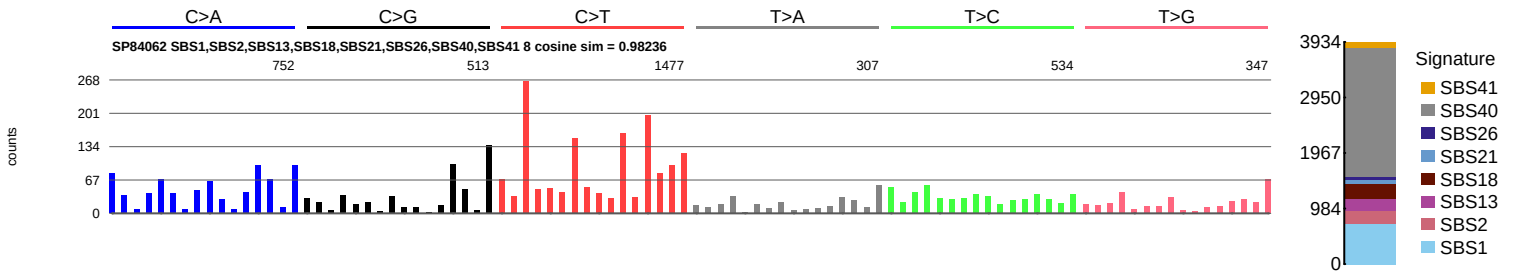

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS15,SBS40 5 cosine sim = 0.98254
669
545
1575
309
467
371
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS40
SBS15
SBS13
SBS2
SBS1

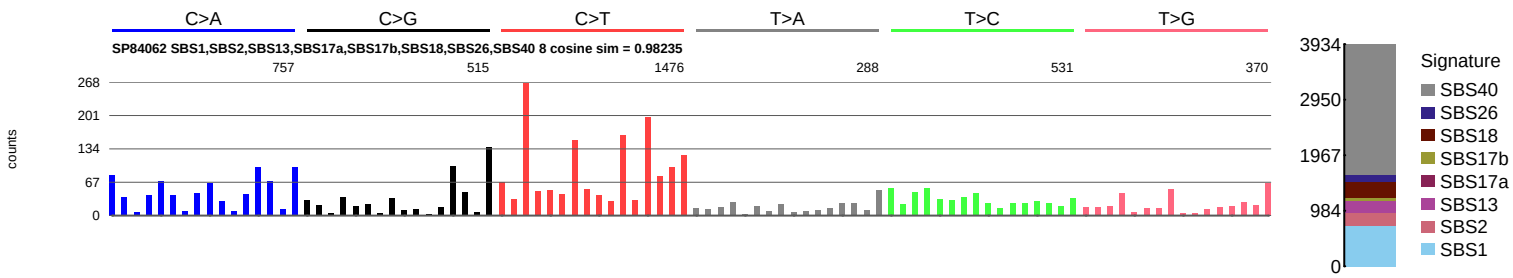

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98252
752
514
1479
312
528
347
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

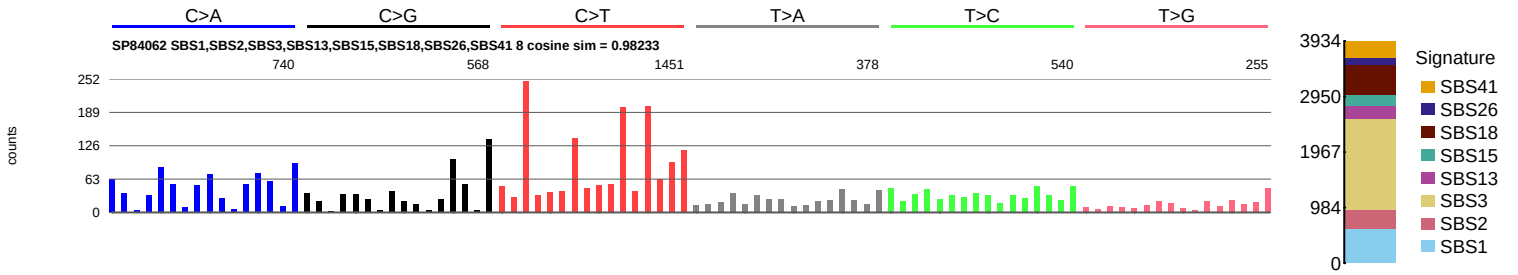

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98245
760
522
1485
289
502
371
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

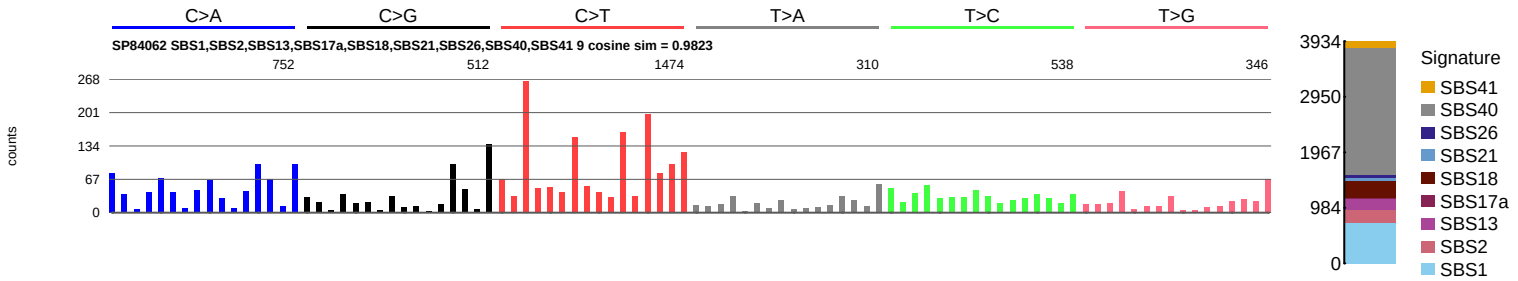

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98241
766
581
1498
335
470
285
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

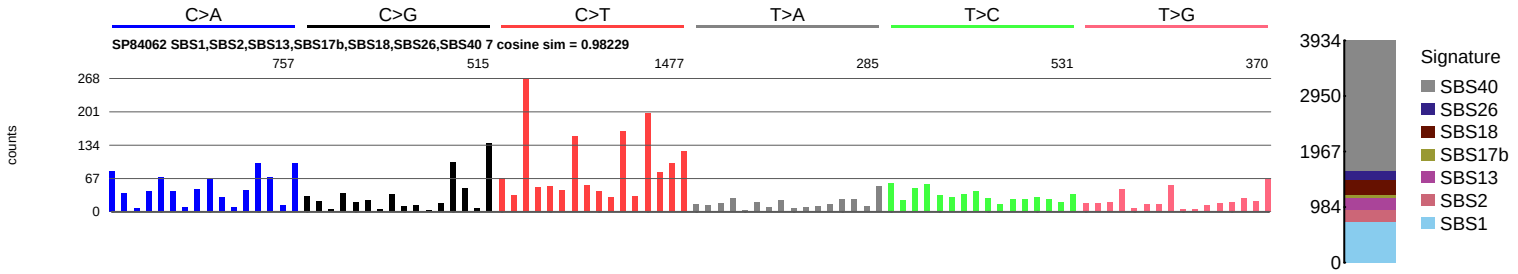

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS41 9 cosine sim = 0.98239
741
567
1450
379
541
255
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

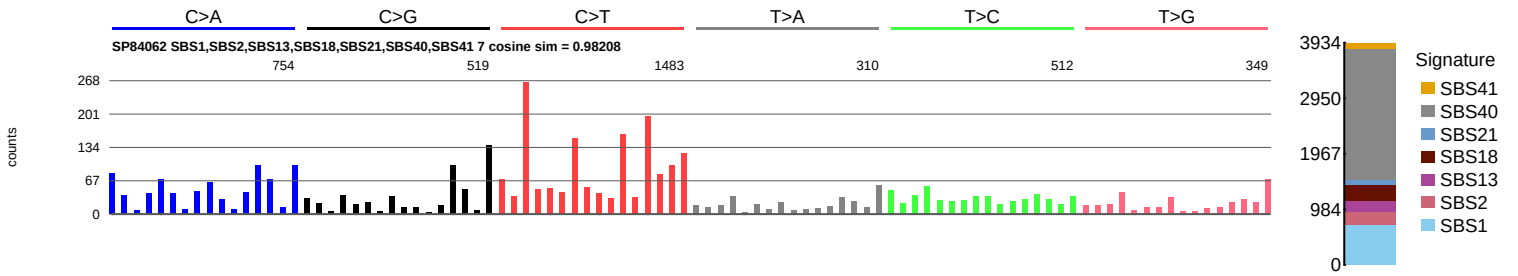

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98236
752
513
1477
307
534
347
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

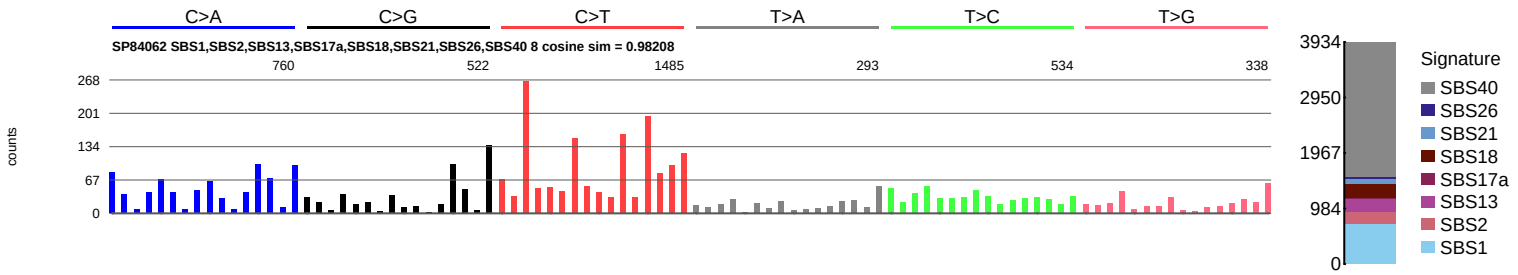

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98235
757
515
1476
288
531
370
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

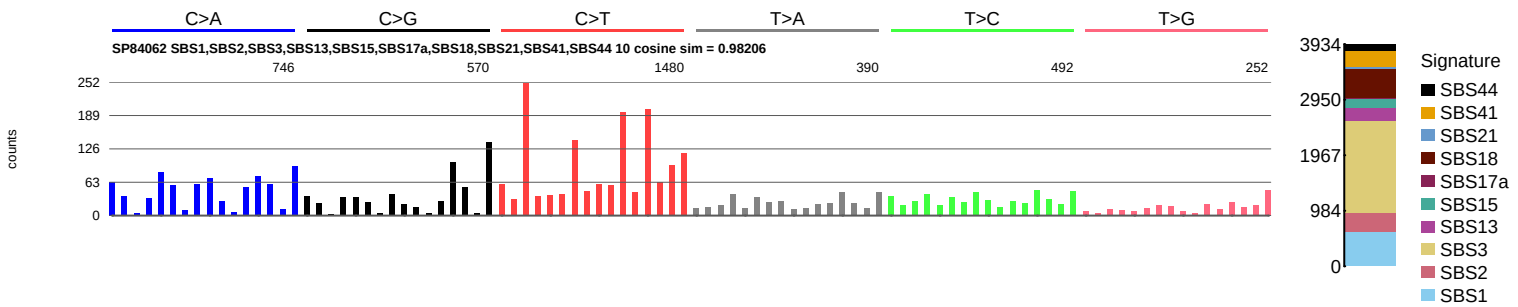

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS41 8 cosine sim = 0.98233
740
568
1451
378
540
255
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

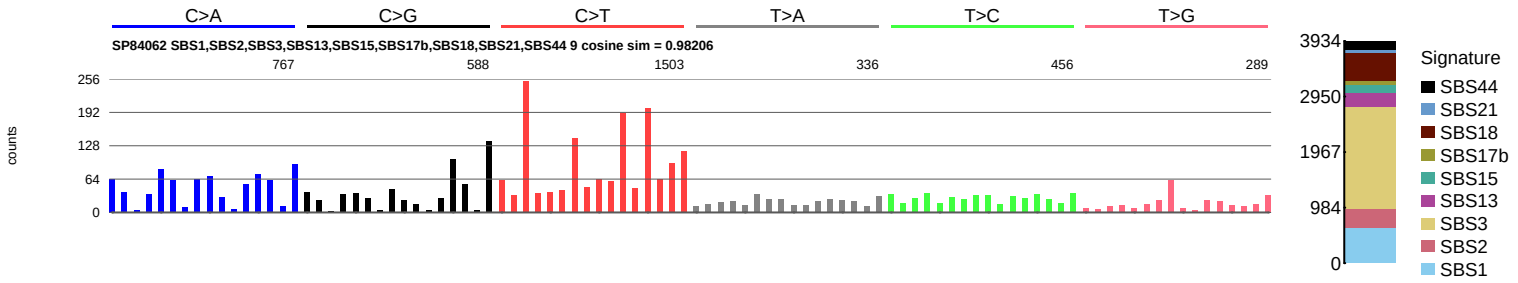

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9823
752
512
1474
310
538
346
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

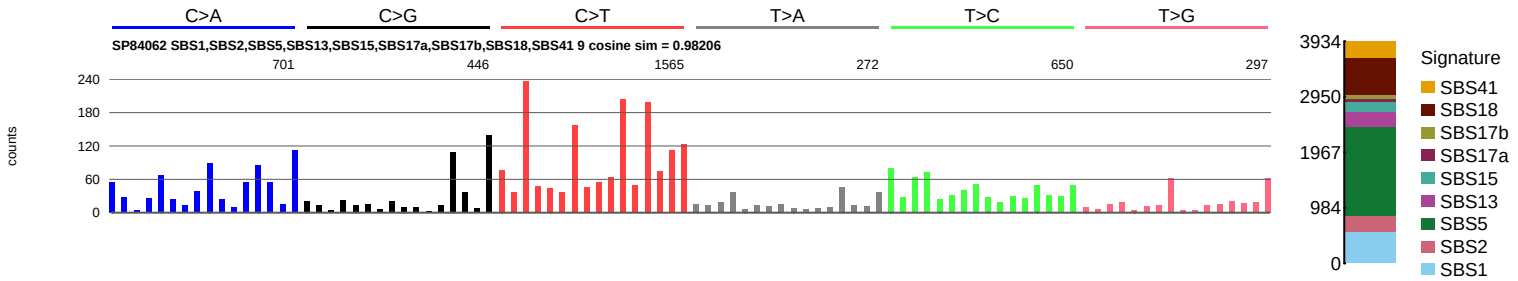

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98229
757
515
1477
285
531
370
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

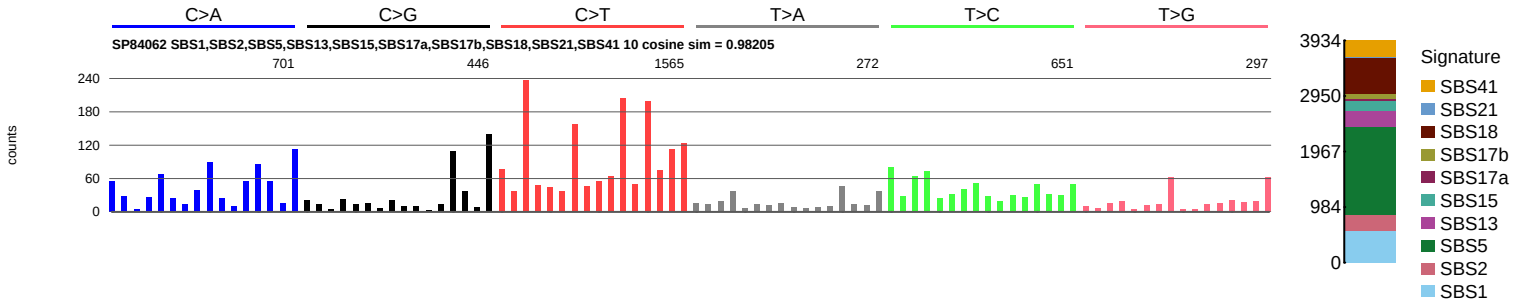

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98208
754
519
1483
310
512
349
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

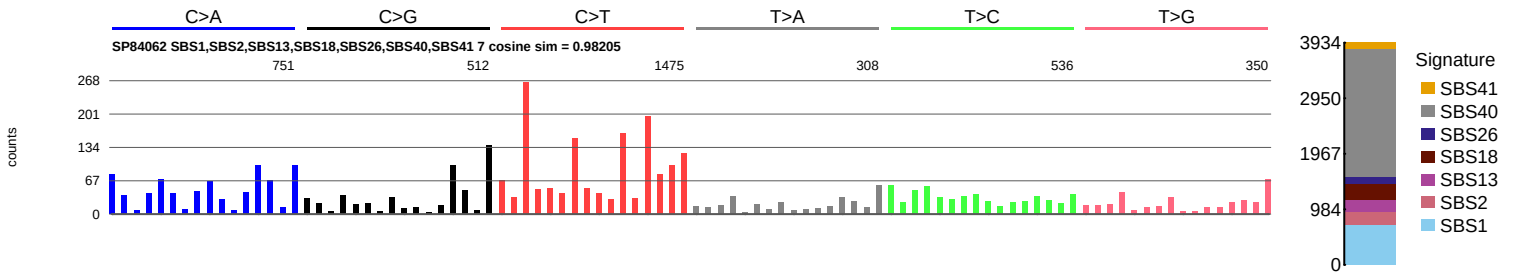

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98208
760
522
1485
293
534
338
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

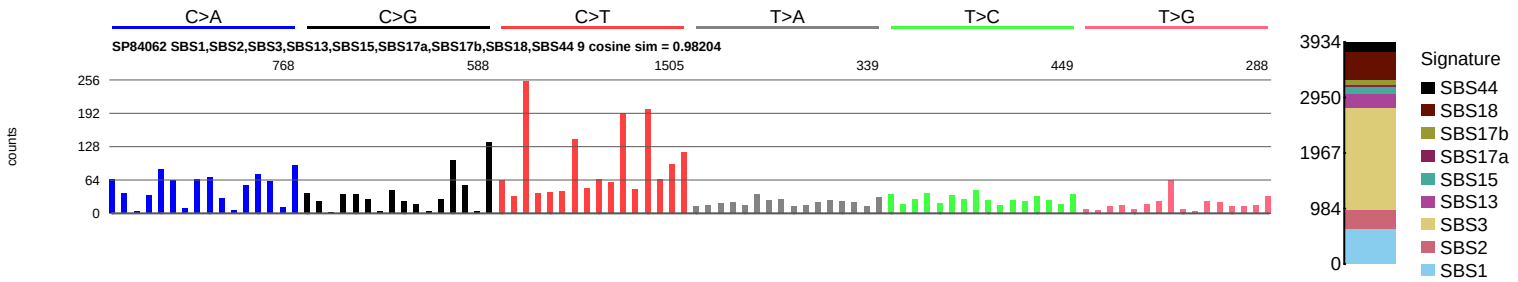

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98206
746
570
1480
390
492
252
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

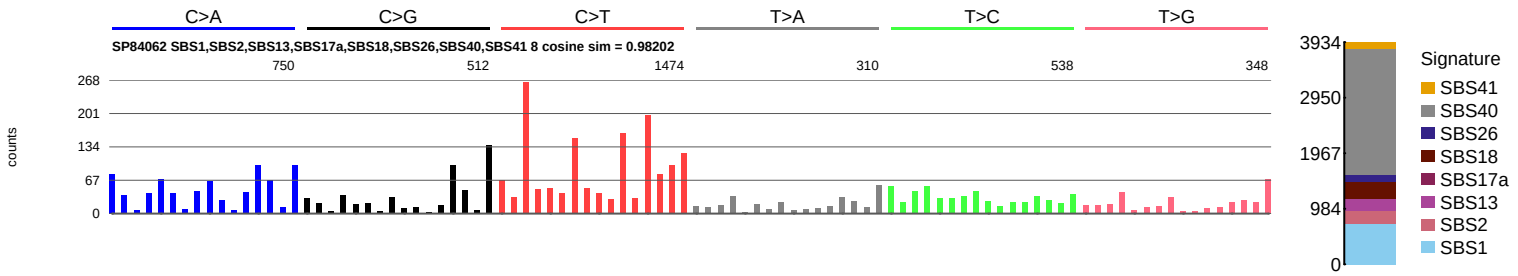

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98206
767
588
1503
336
456
289
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

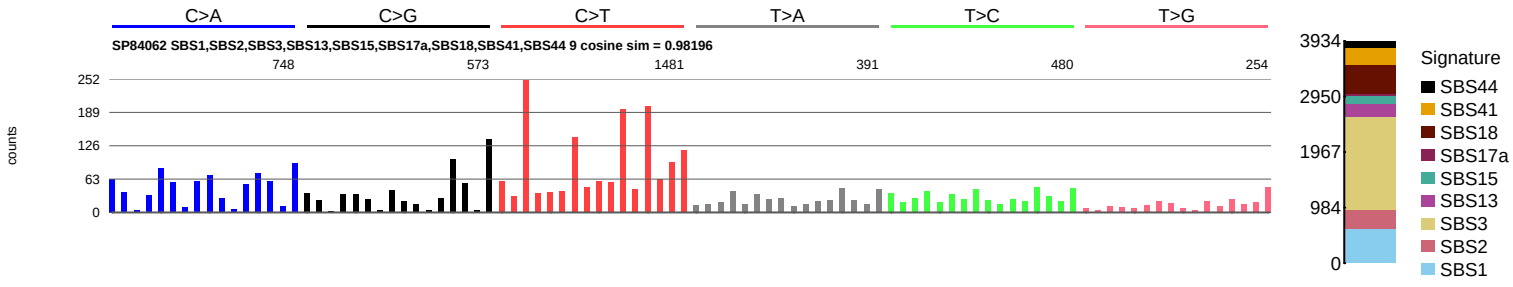

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98206
701
446
1565
272
650
297
counts
240
180
120
60
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

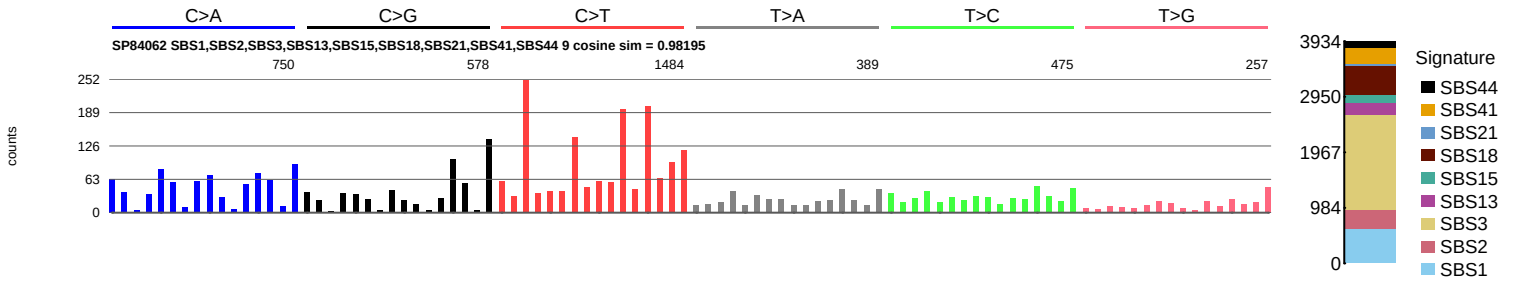

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98205
701
446
1565
272
651
297
counts
240
180
120
60
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

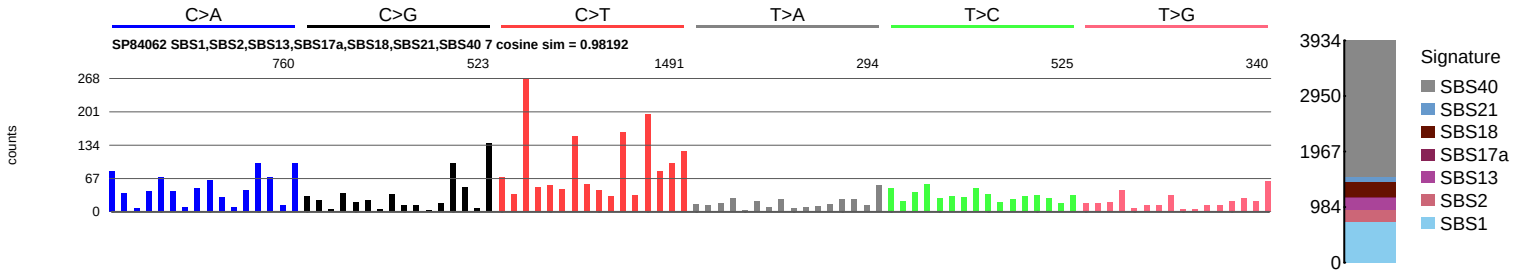

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98205
751
512
1475
308
536
350
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

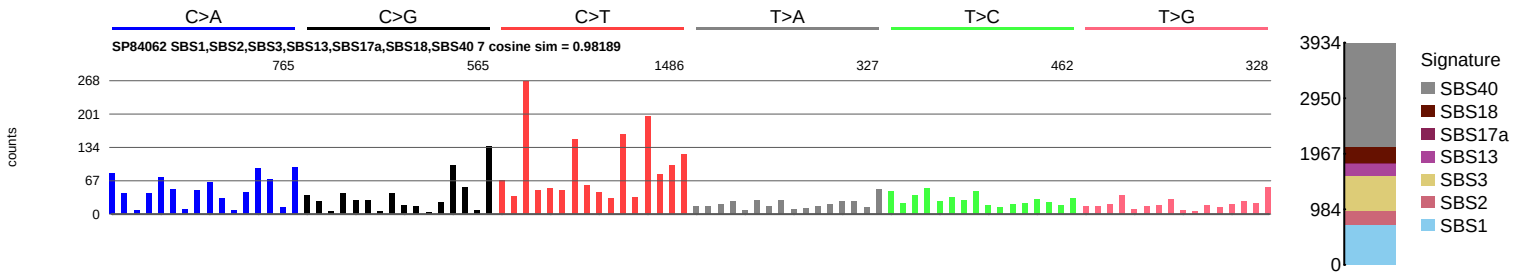

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98204
768
588
1505
339
449
288
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

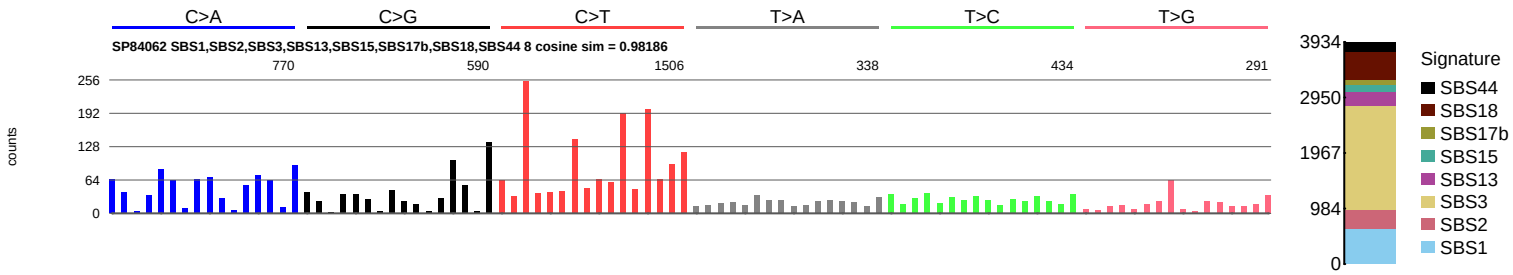

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98202
750
512
1474
310
538
348
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1

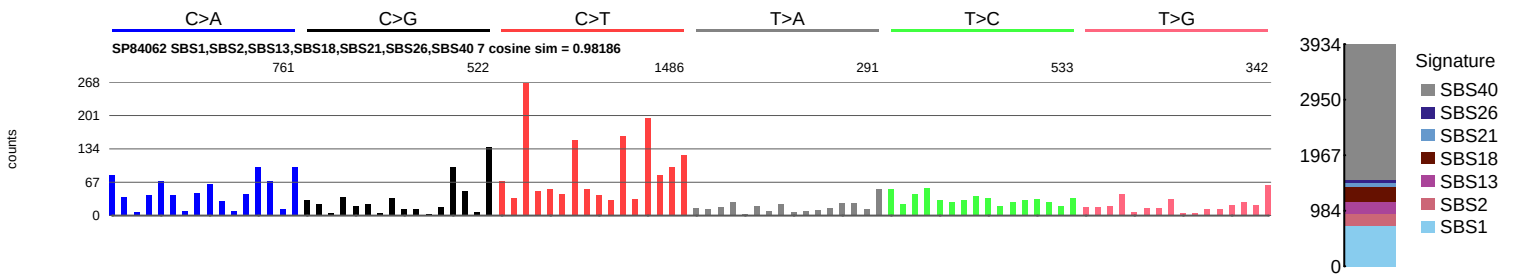

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98196
748
573
1481
391
480
254
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

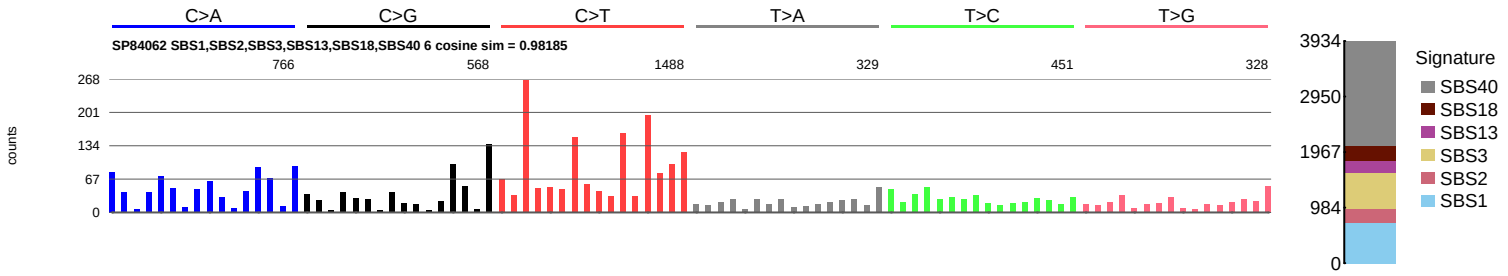

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98195
750
578
1484
389
475
257
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

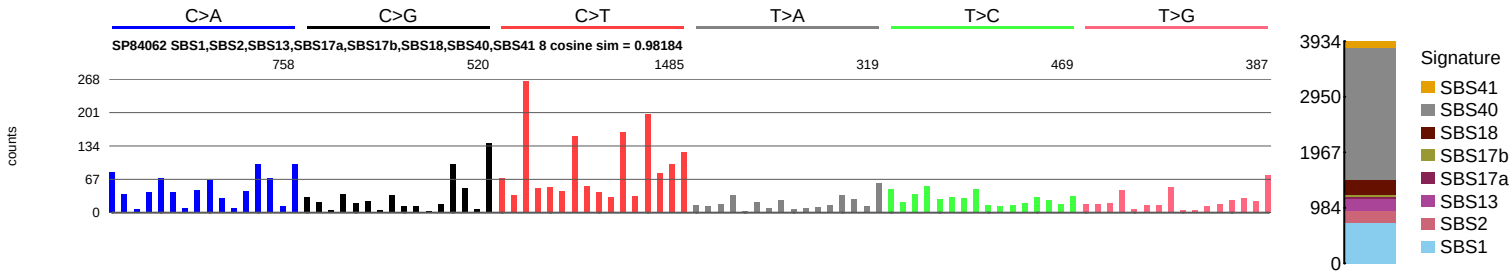

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98192
760
523
1491
294
525
340
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

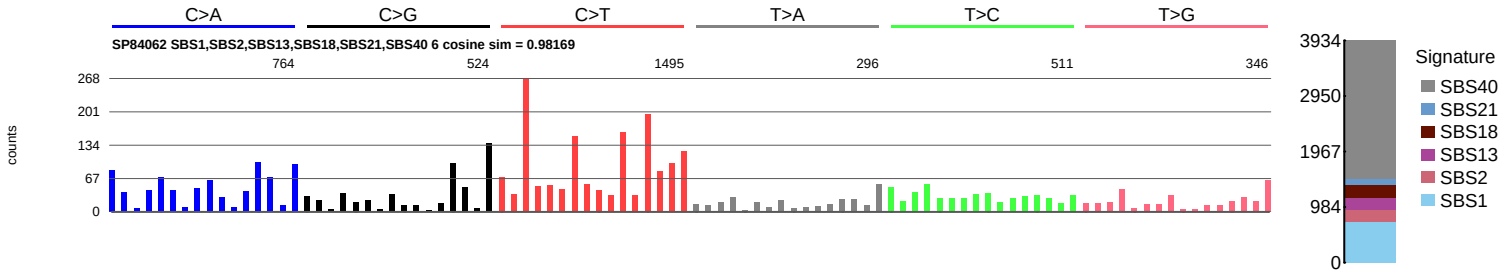

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98189
765
565
1486
327
462
328
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

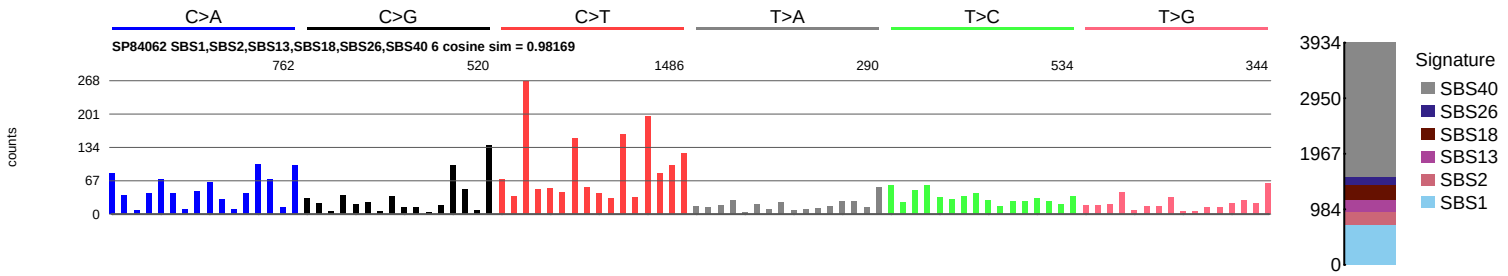

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98186
770
590
1506
338
434
291
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

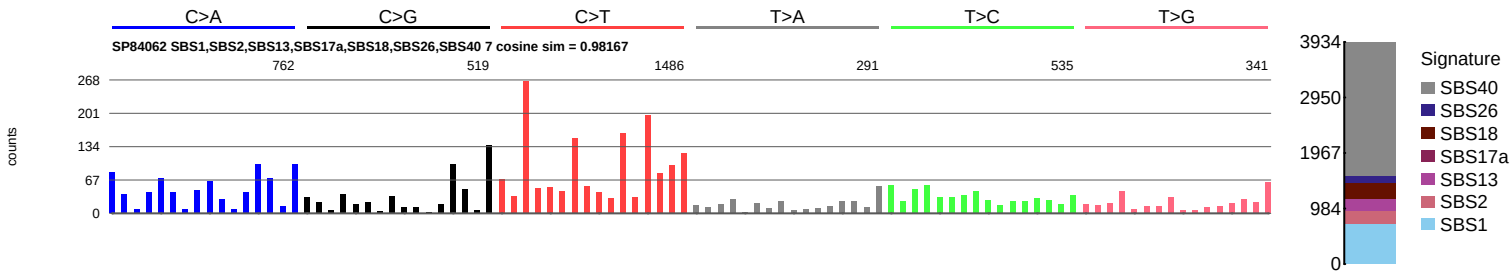

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98186
761
522
1486
291
533
342
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

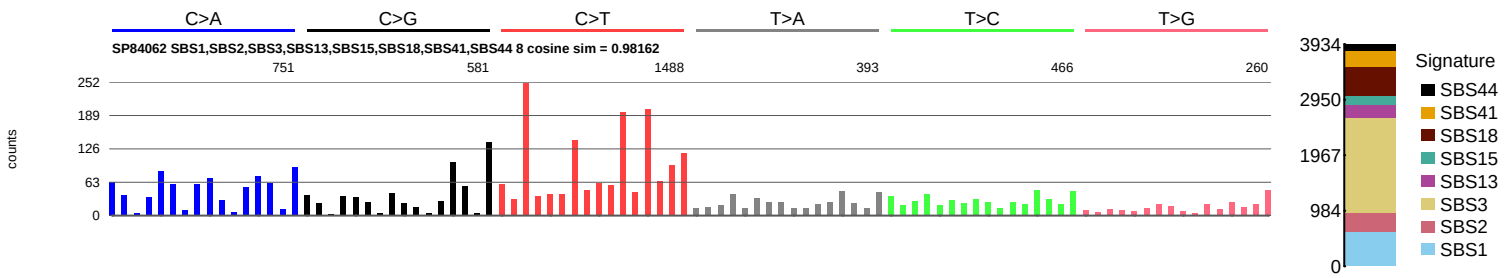

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40 6 cosine sim = 0.98185
766
568
1488
329
451
328
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

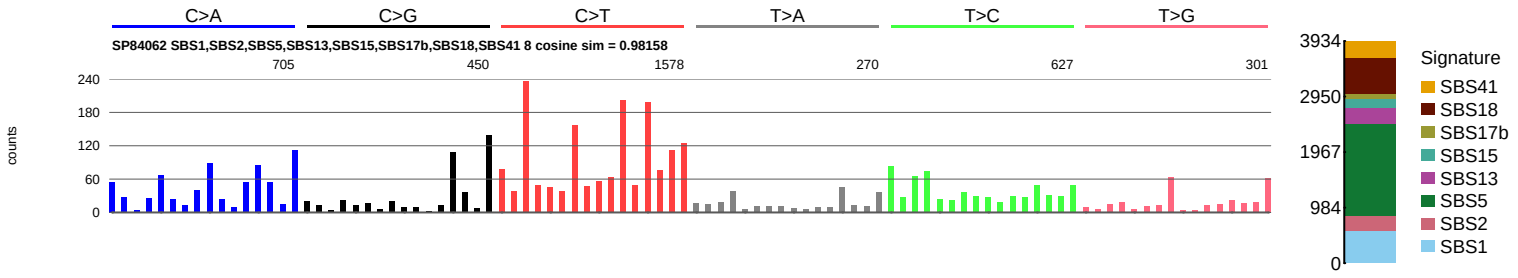

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98184
758
520
1485
319
469
387
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

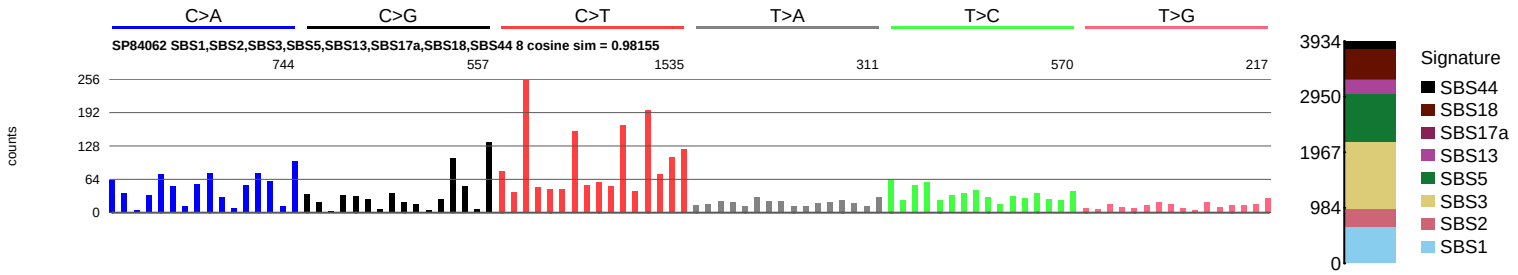

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40 6 cosine sim = 0.98169
764
524
1495
296
511
346
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

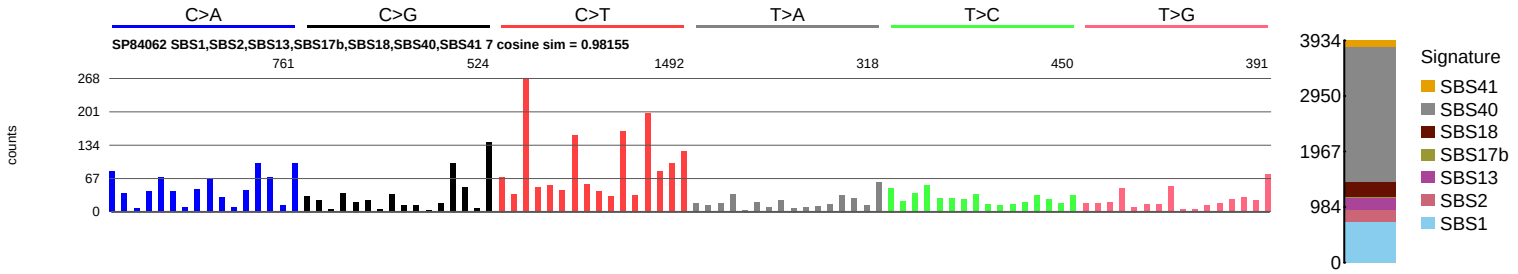

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40 6 cosine sim = 0.98169
762
520
1486
290
534
344
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

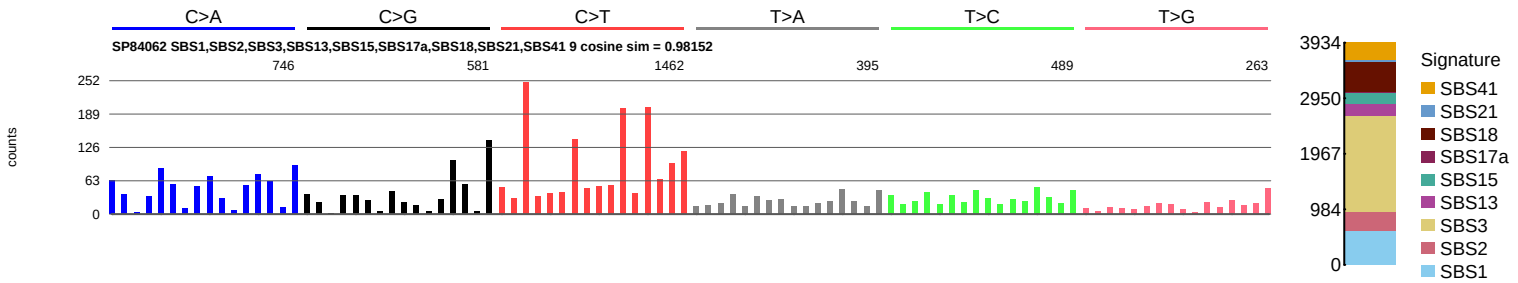

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98167
762
519
1486
291
535
341
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1

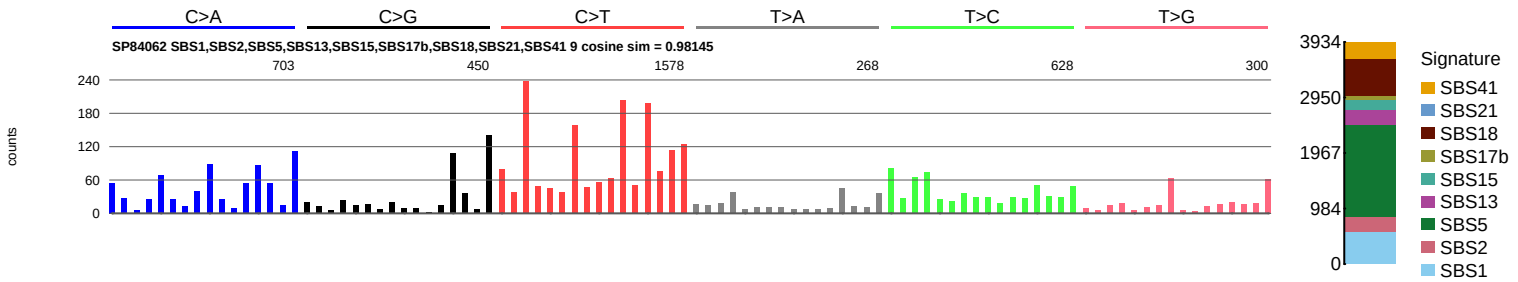

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS41,SBS44 8 cosine sim = 0.98162
751
581
1488
393
466
260
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

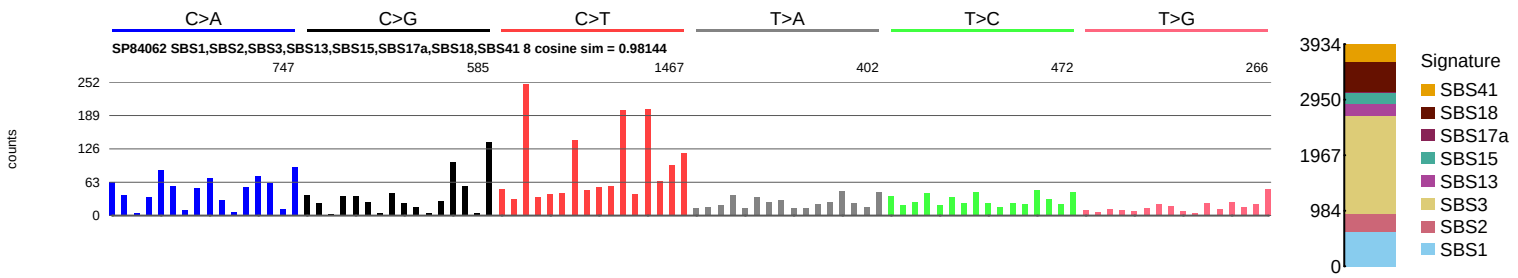

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98158
705
450
1578
270
627
301
counts
240
180
120
60
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

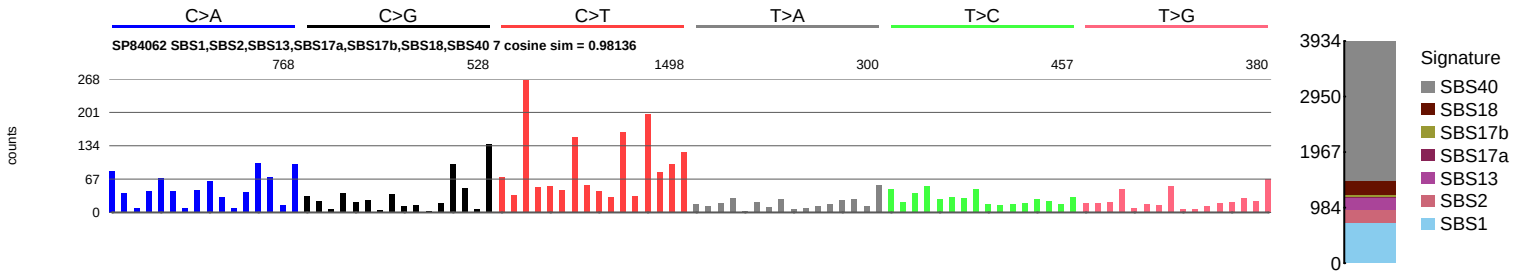

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS44 8 cosine sim = 0.98155
744
557
1535
311
570
217
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

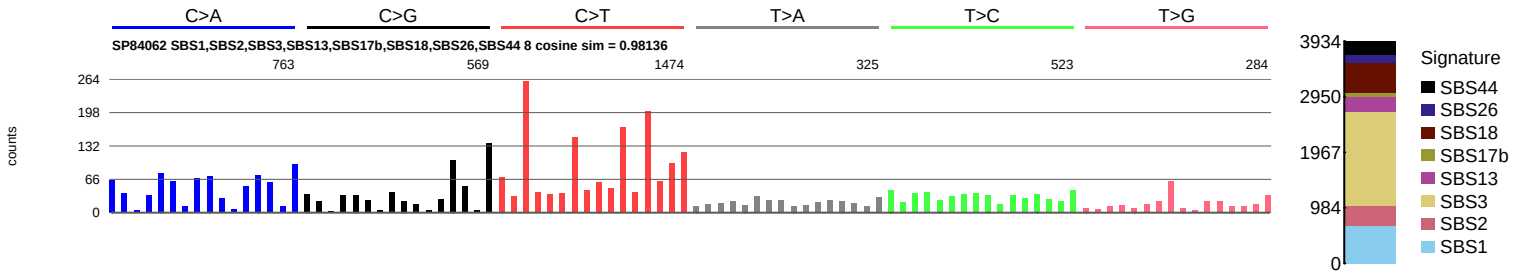

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98155
761
524
1492
318
450
391
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

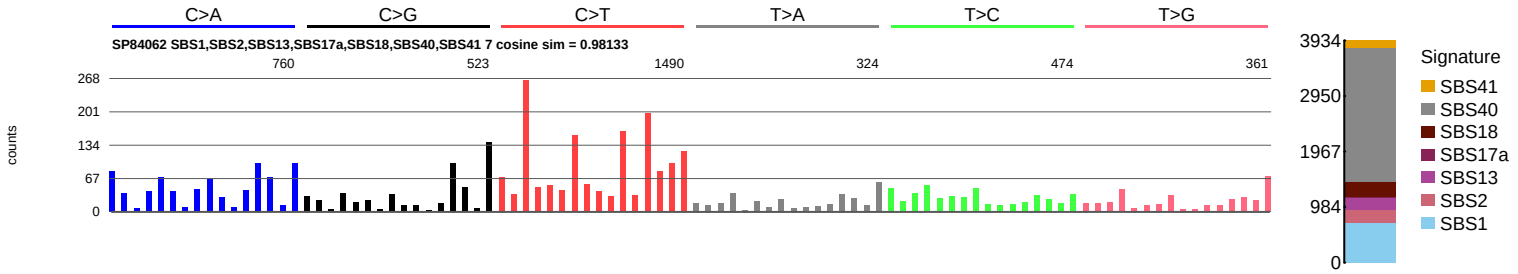

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98152
746
581
1462
395
489
263
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

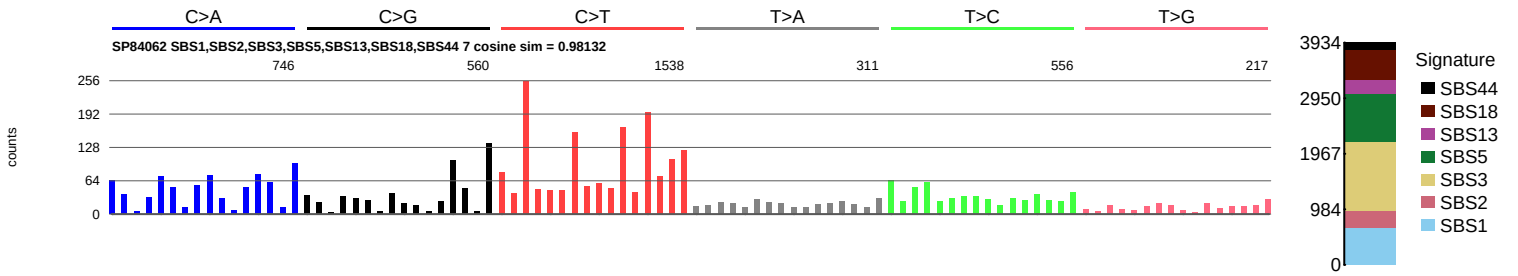

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98145
703
450
1578
268
628
300
counts
240
180
120
60
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

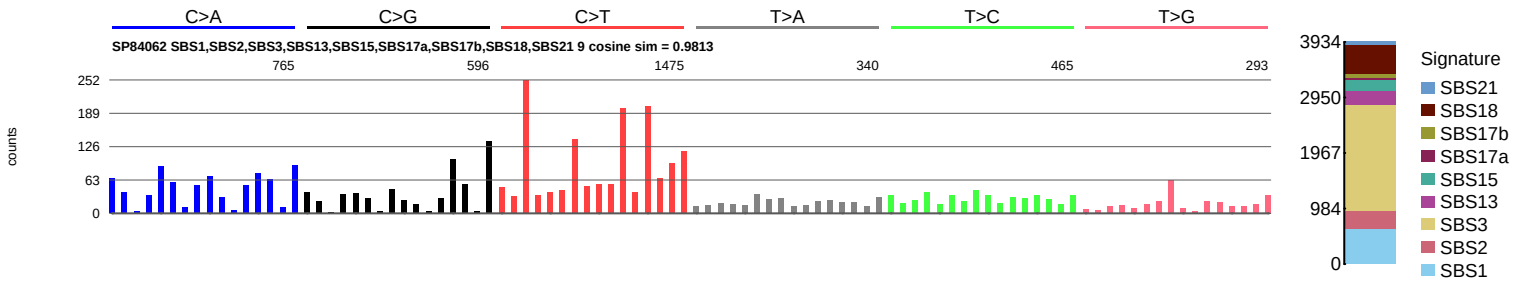

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98144
747
585
1467
402
472
266
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

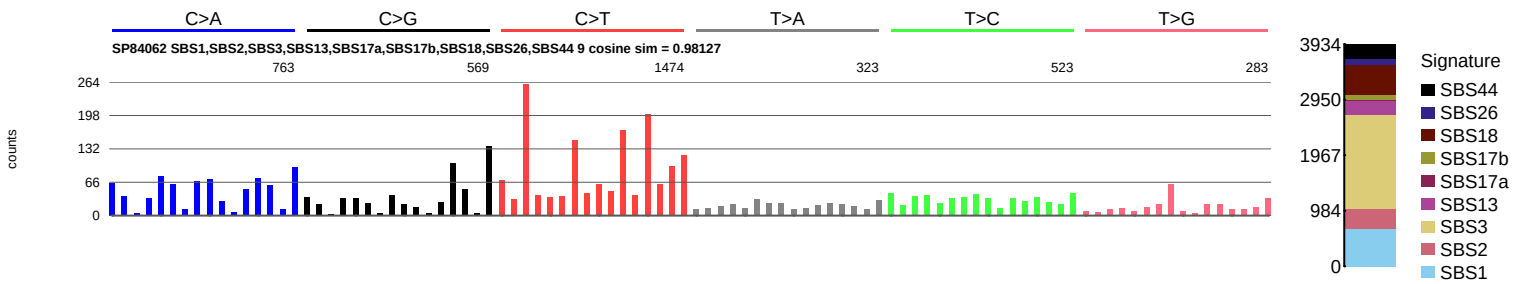

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98136
768
528
1498
300
457
380
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

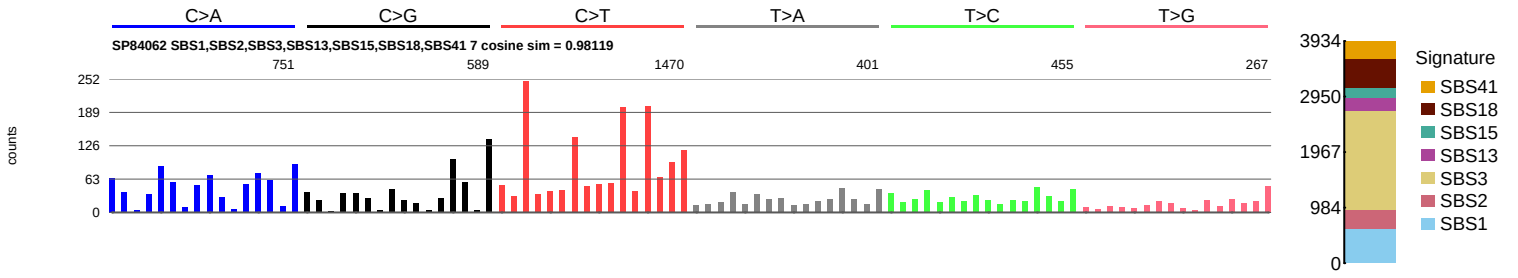

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98136
763
569
1474
325
523
284
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

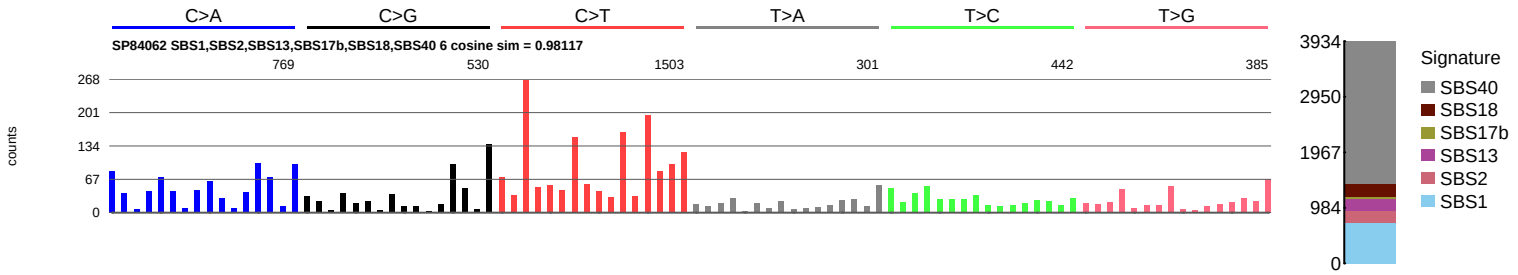

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98133
760
523
1490
324
474
361
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1

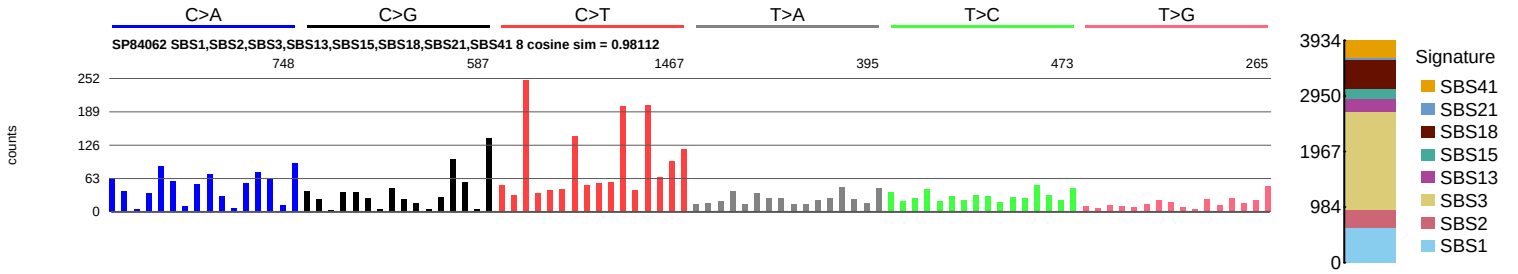

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS44 7 cosine sim = 0.98132
746
560
1538
311
556
217
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

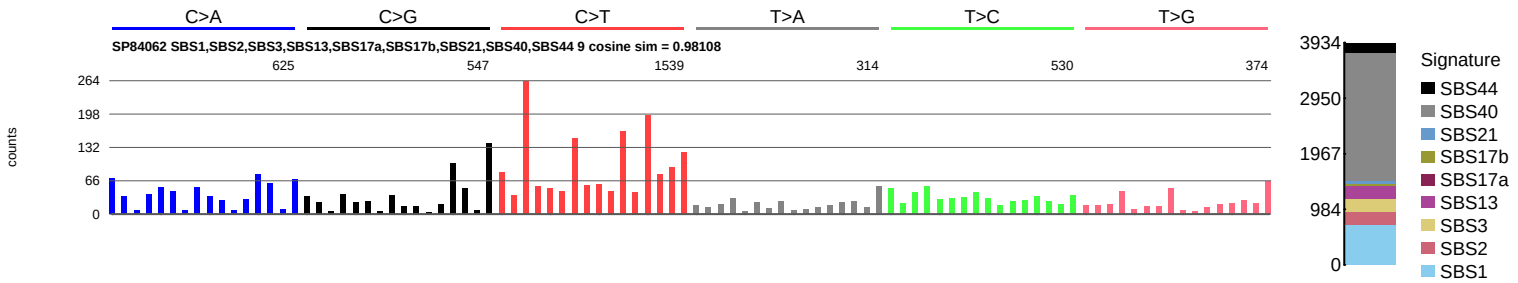

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9813
765
596
1475
340
465
293
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

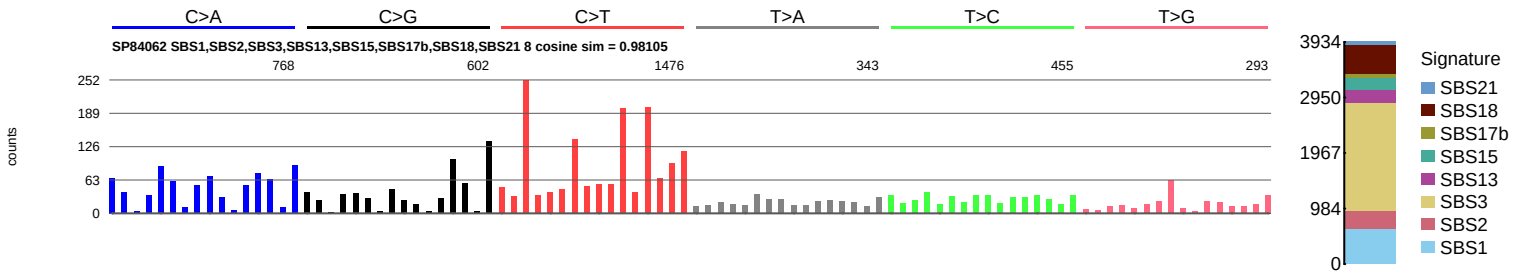

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98127
763
569
1474
323
523
283
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

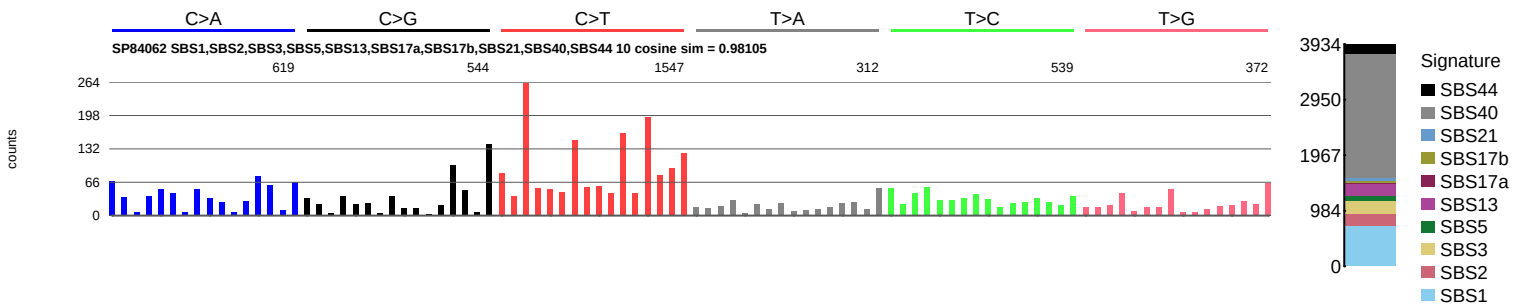

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.98119
751
589
1470
401
455
267
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

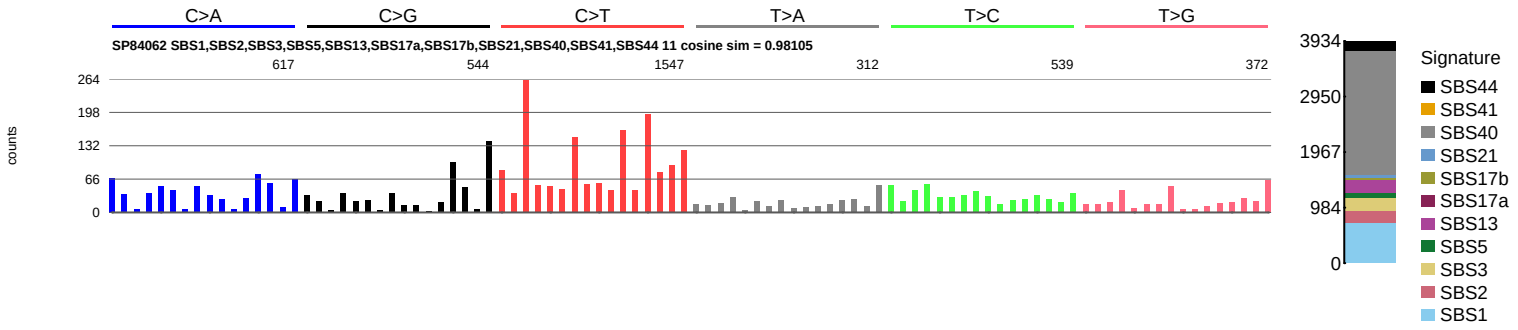

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40 6 cosine sim = 0.98117
769
530
1503
301
442
385
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

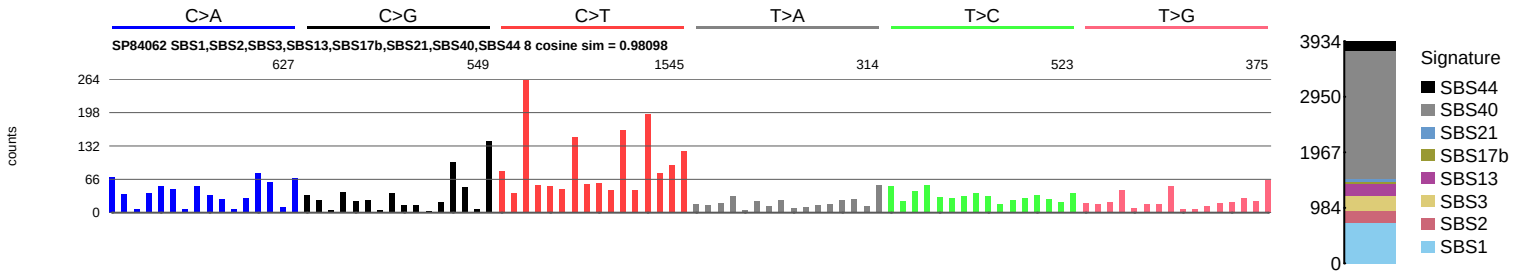

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98112
748
587
1467
395
473
265
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

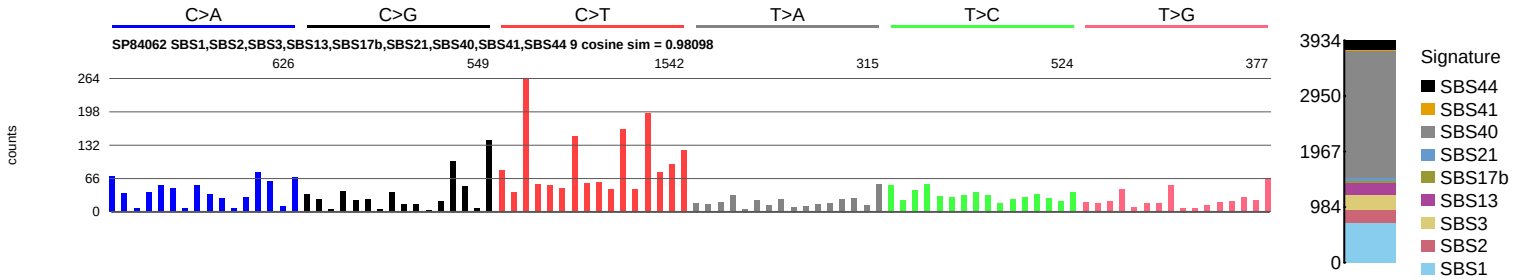

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98108
625
547
1539
314
530
374
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

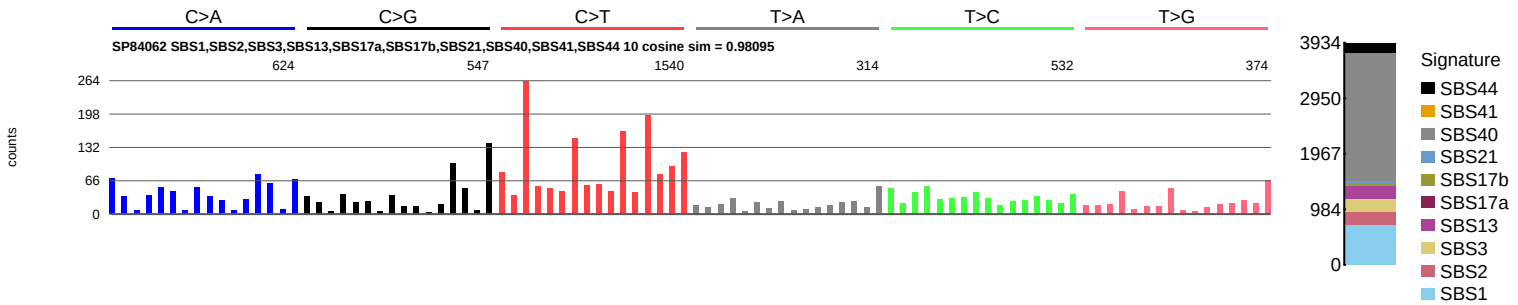

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98105
768
602
1476
343
455
293
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

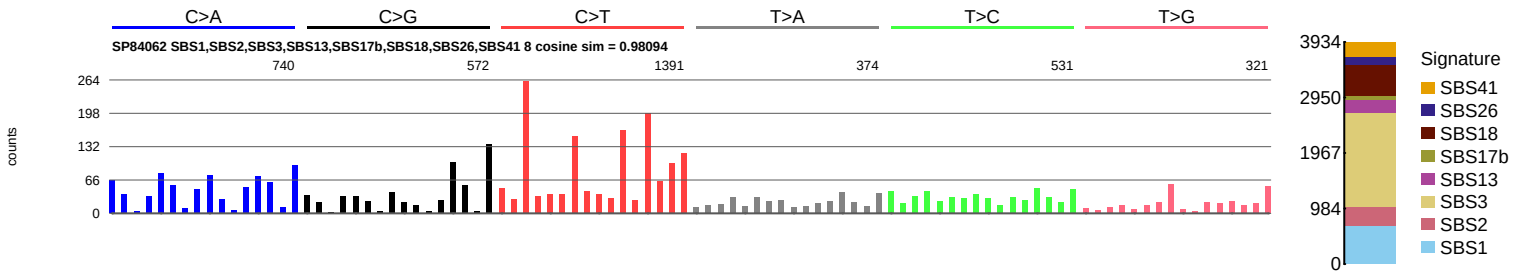

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98105
619
544
1547
312
539
372
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

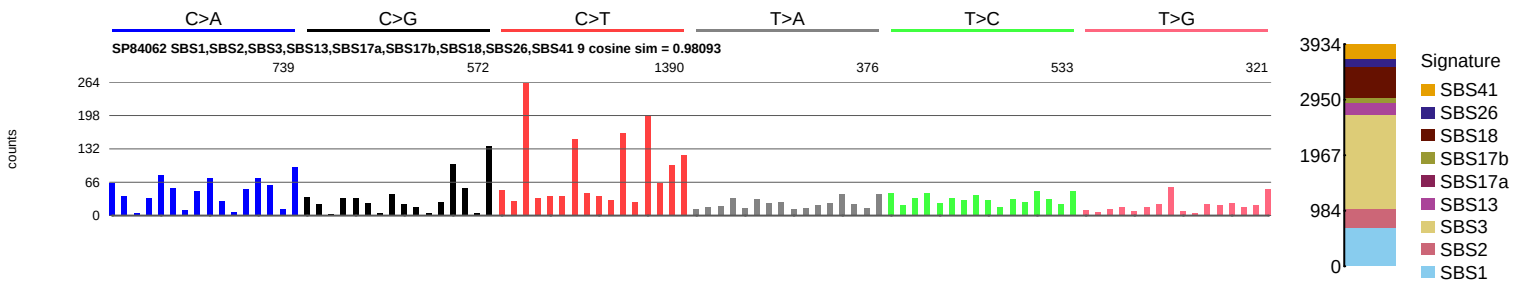

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98105
617
544
1547
312
539
372
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

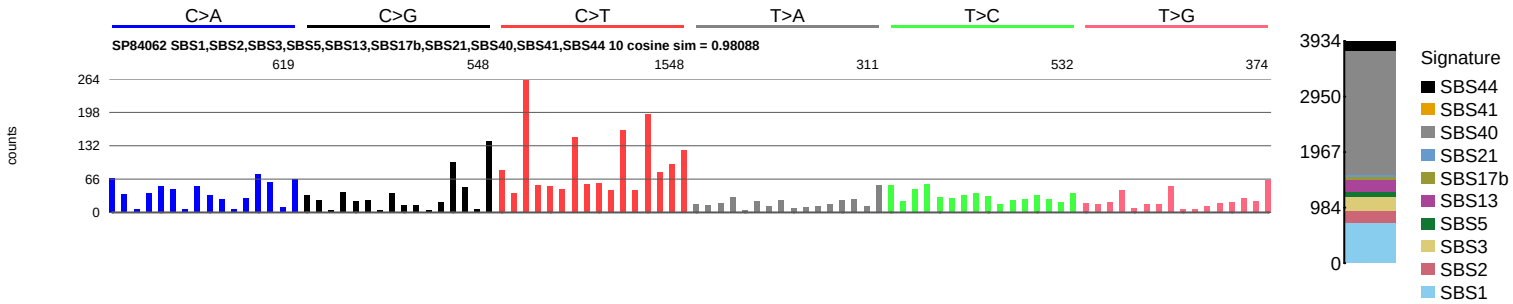

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98098
627
549
1545
314
523
375
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS3
SBS2
SBS1

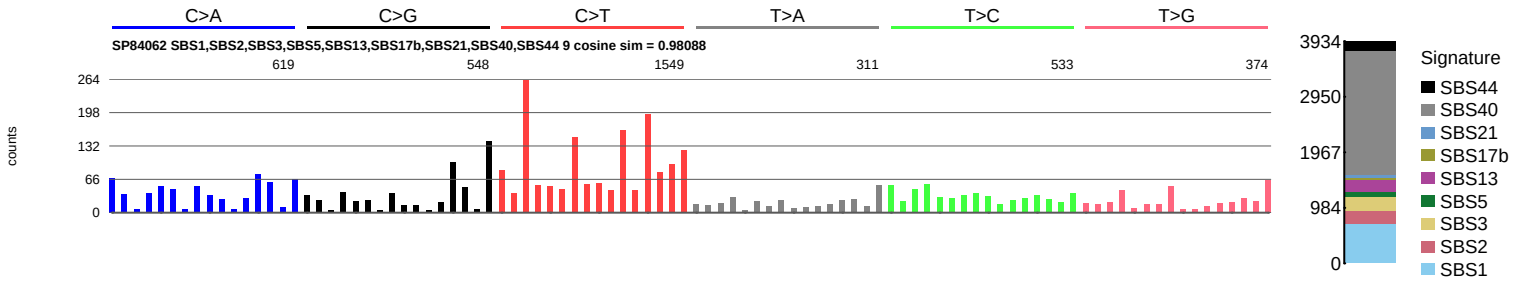

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98098
626
549
1542
315
524
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS13
SBS3
SBS2
SBS1

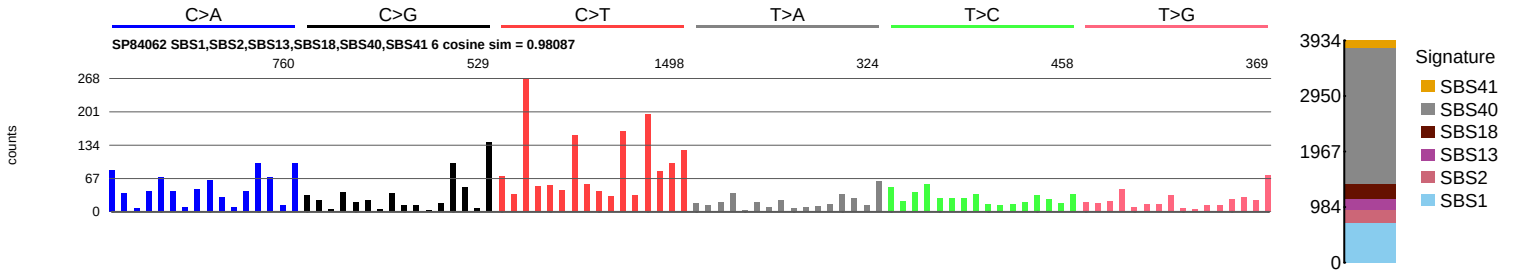

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98095
624
547
1540
314
532
374
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

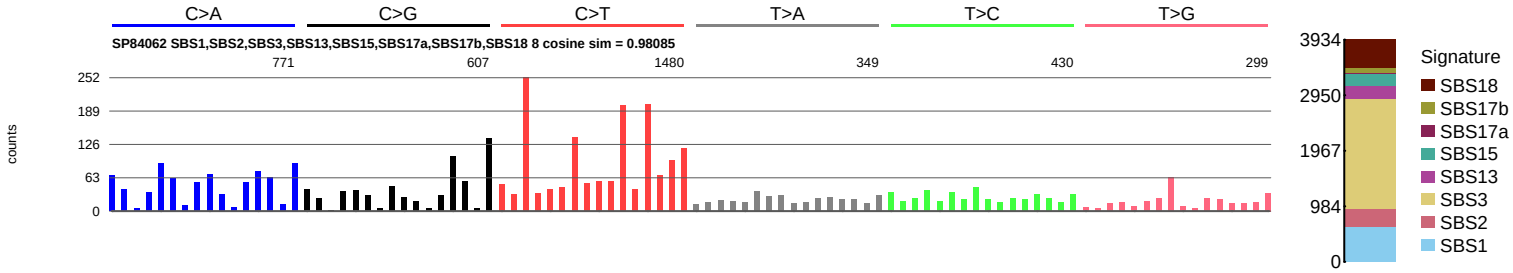

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98094
740
572
1391
374
531
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

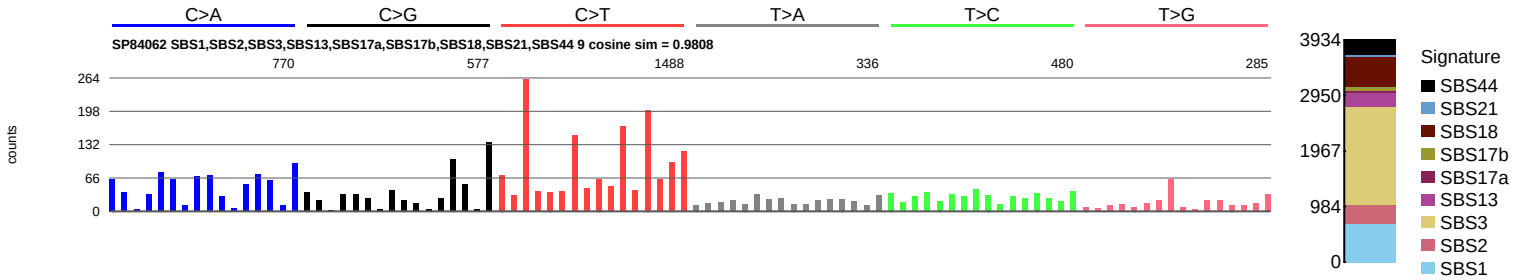

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98093
739
572
1390
376
533
321
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

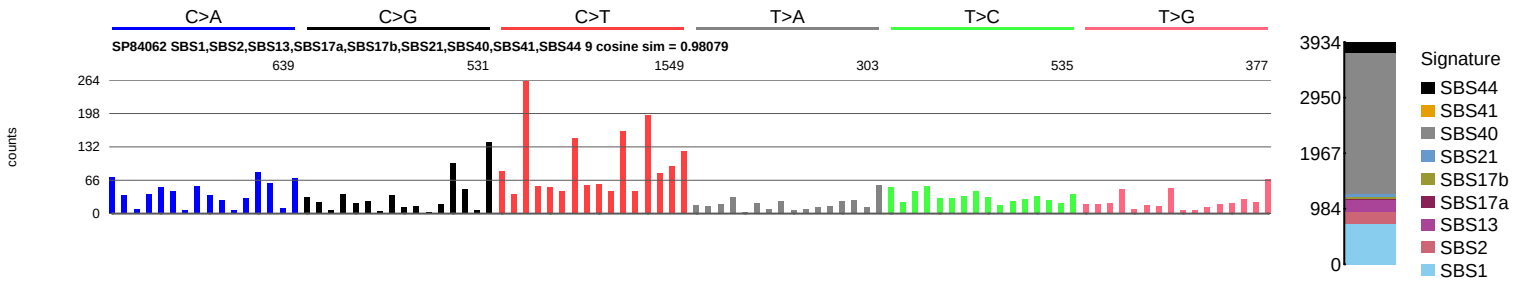

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44,SBS44 10 cosine sim = 0.98088
619
548
1548
311
532
374
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

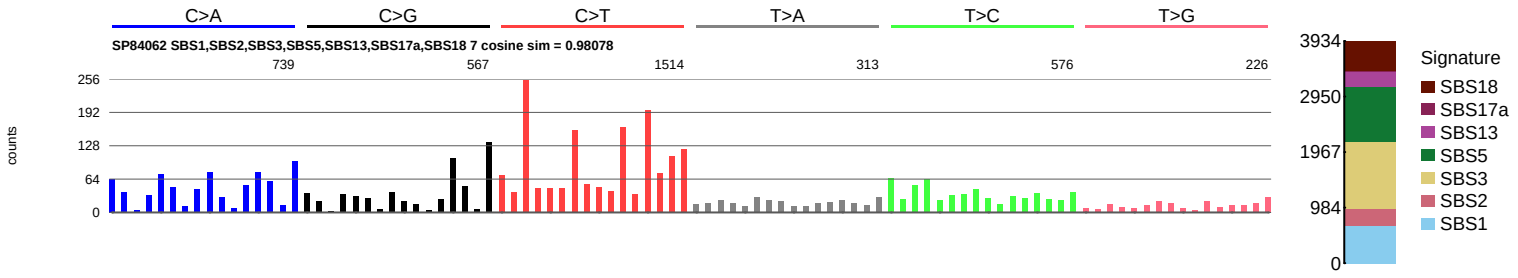

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98088
619
548
1549
311
533
374
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

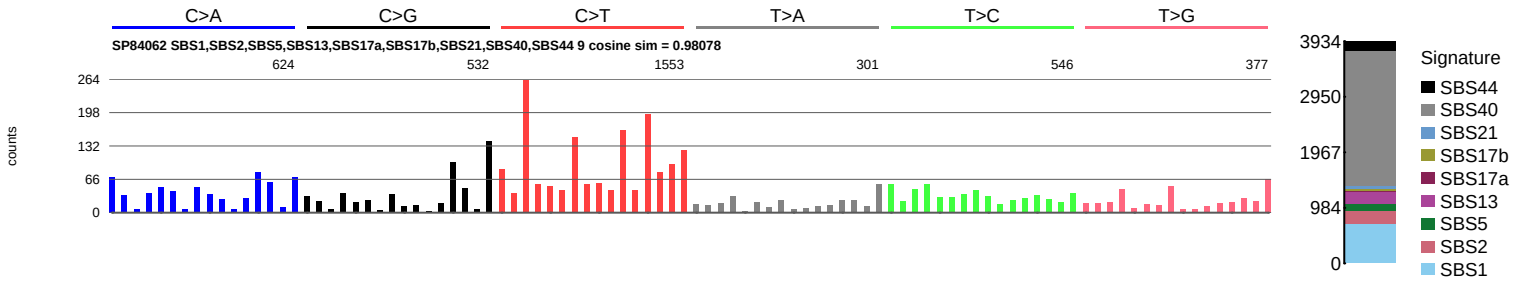

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41 6 cosine sim = 0.98087
760
529
1498
324
458
369
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1

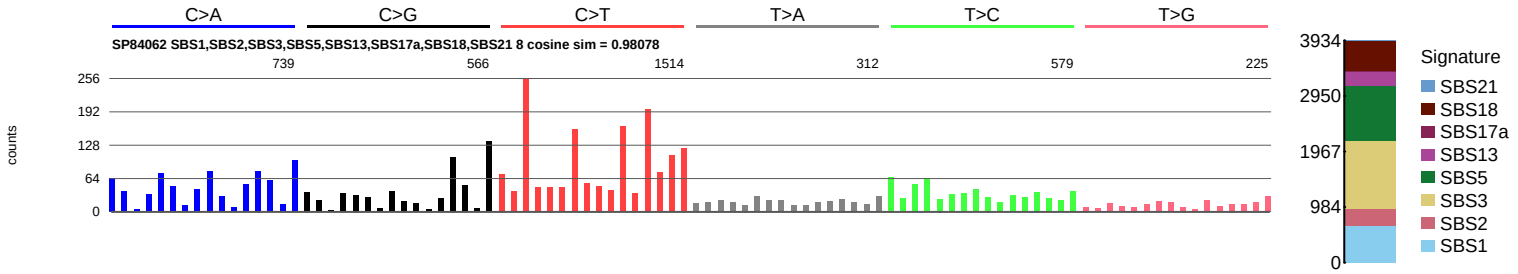

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98085
771
607
1480
349
430
299
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

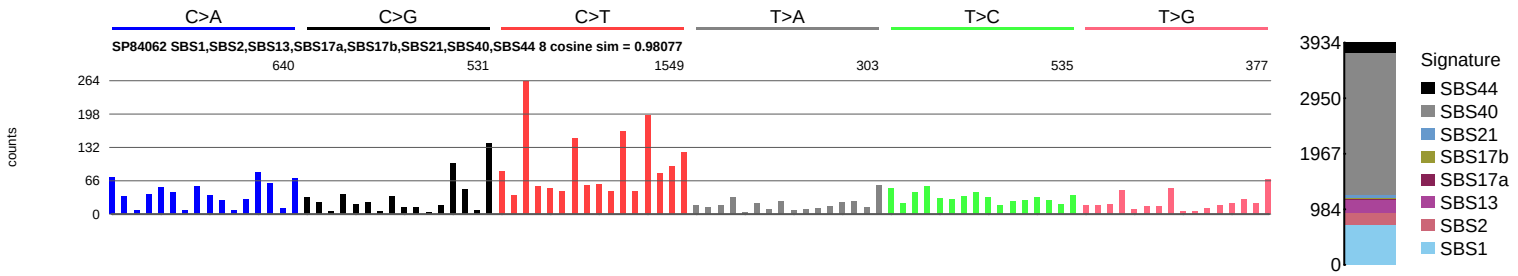

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9808
770
577
1488
336
480
285
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

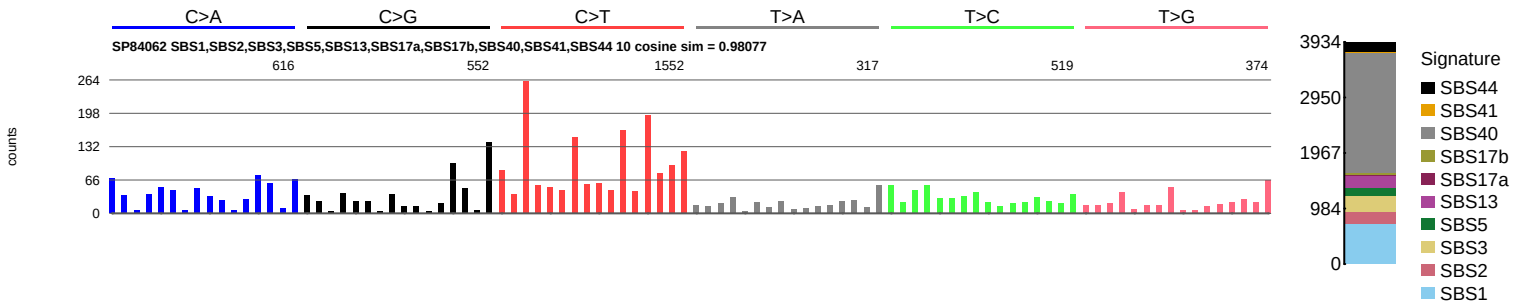

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98079
639
531
1549
303
535
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

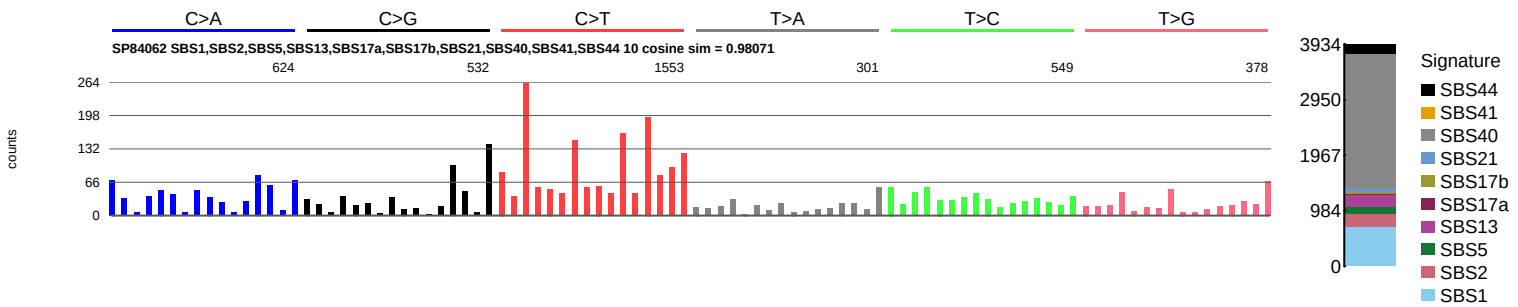

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18 7 cosine sim = 0.98078
739
567
1514
313
576
226
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

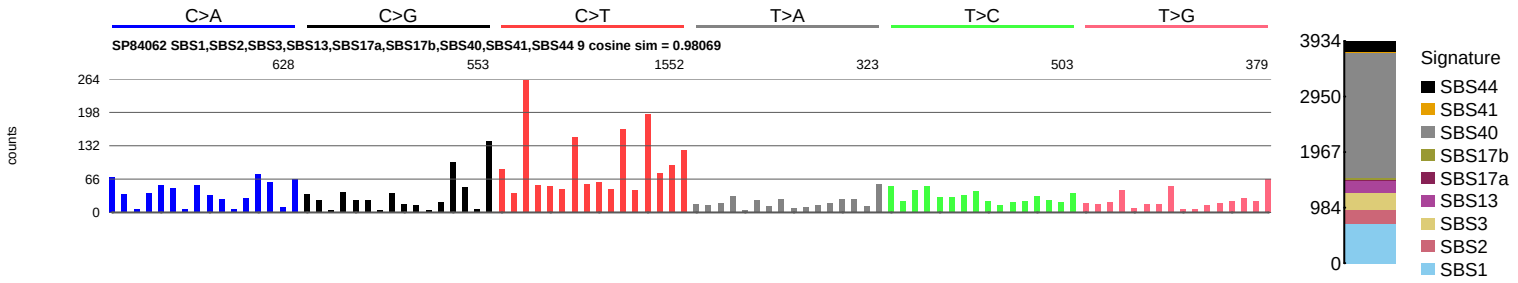

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98078
624
532
1553
301
546
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

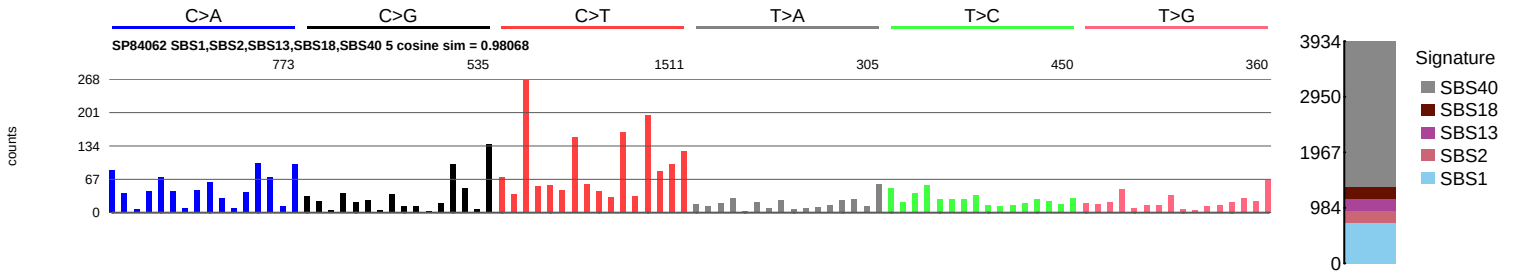

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21 8 cosine sim = 0.98078
739
566
1514
312
579
225
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

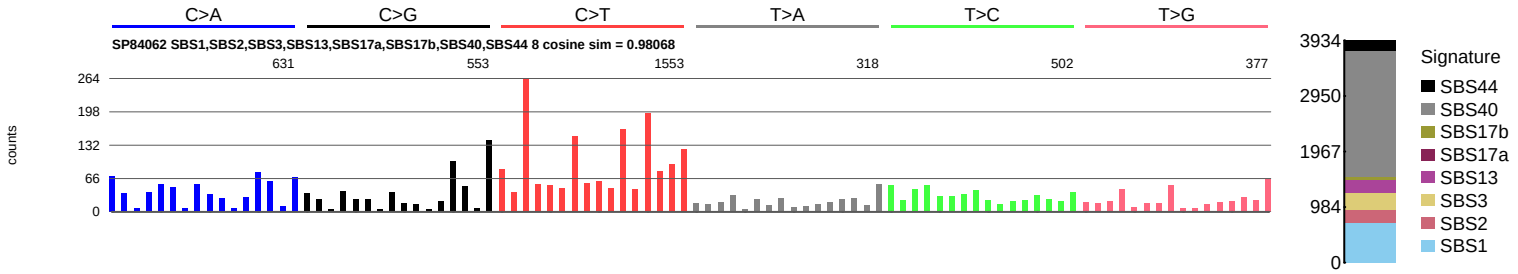

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98077
640
531
1549
303
535
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

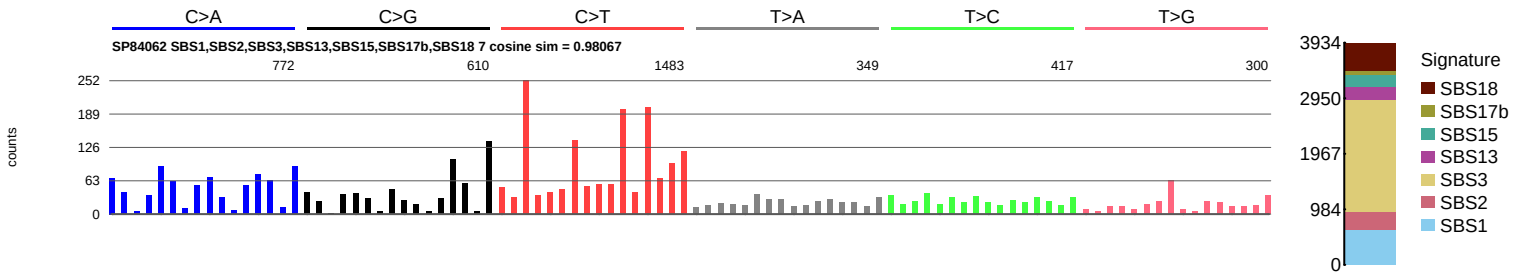

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98077
616
552
1552
317
519
374
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

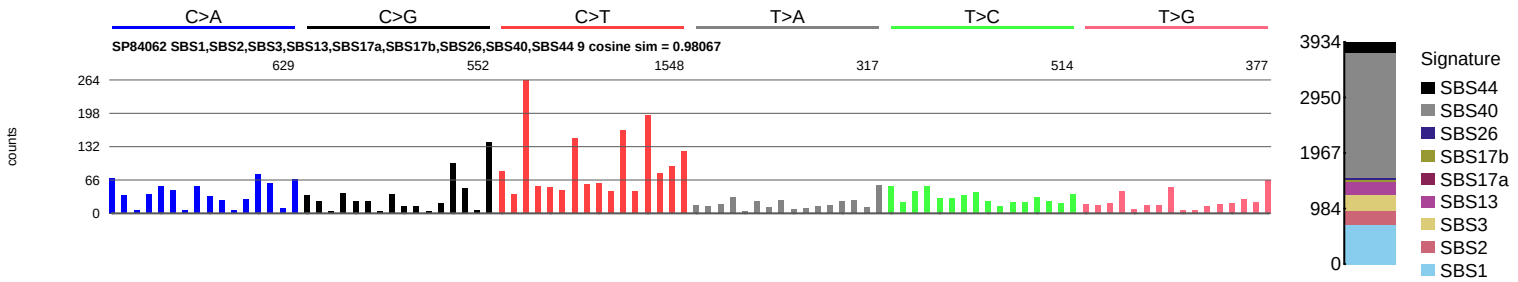

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98071
624
532
1553
301
549
378
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

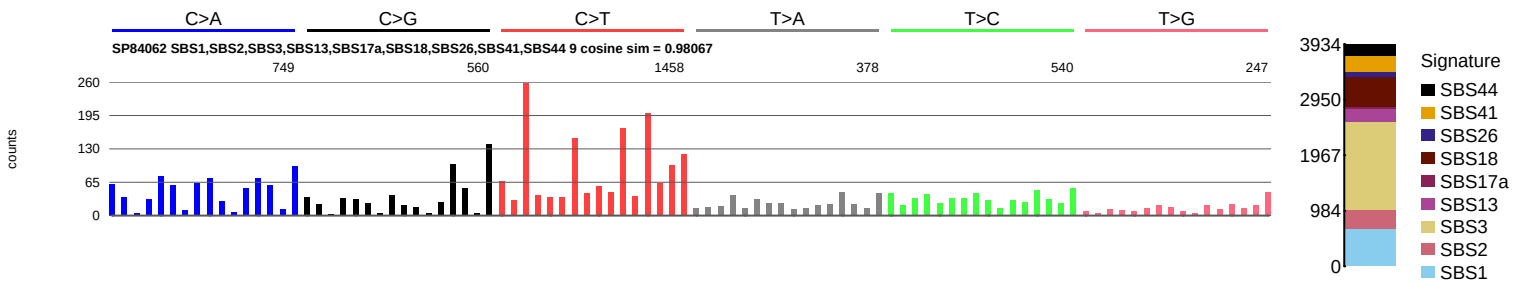

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98069
628
553
1552
323
503
379
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

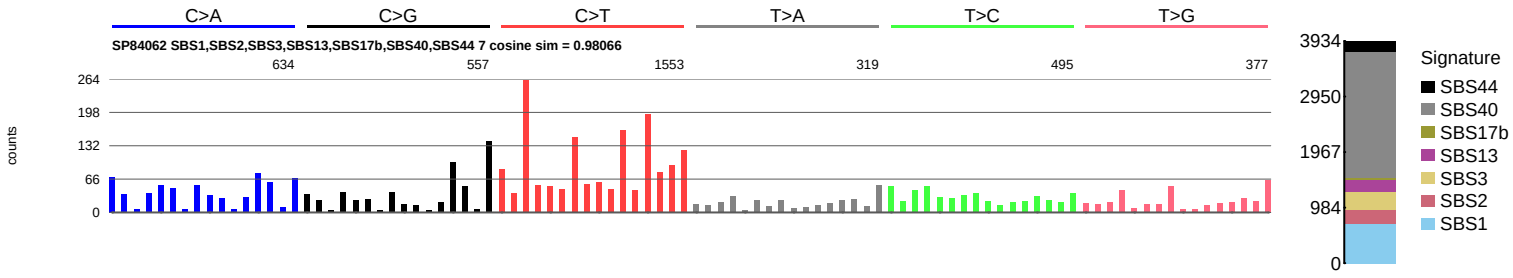

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS18,SBS40 5 cosine sim = 0.98068
773
535
1511
305
450
360
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS13
SBS2
SBS1

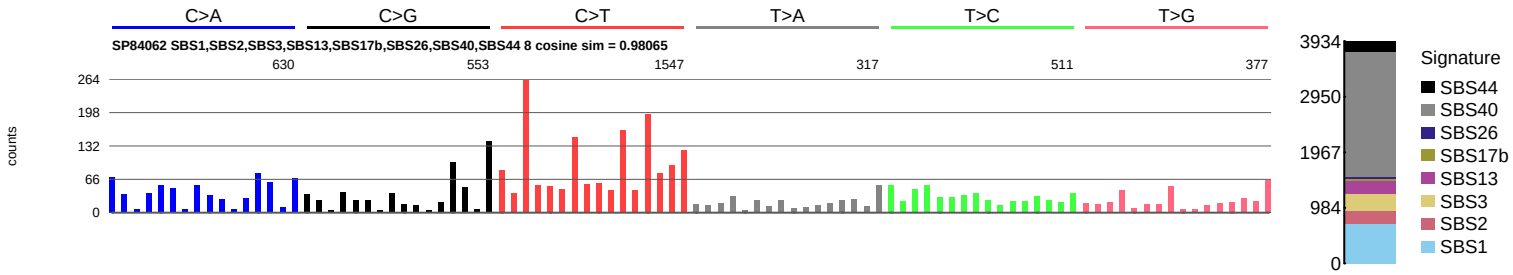

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98068
631
553
1553
318
502
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

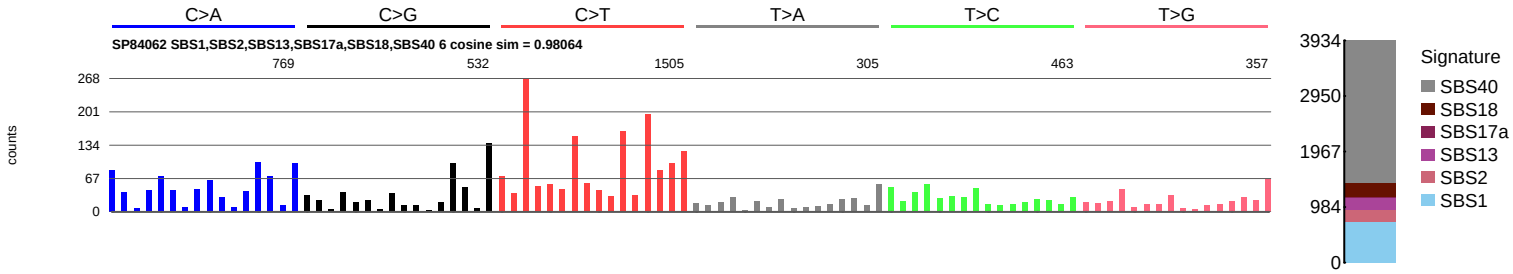

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98067
772
610
1483
349
417
300
counts
252
189
126
63
0
3934
2950
1967
984
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

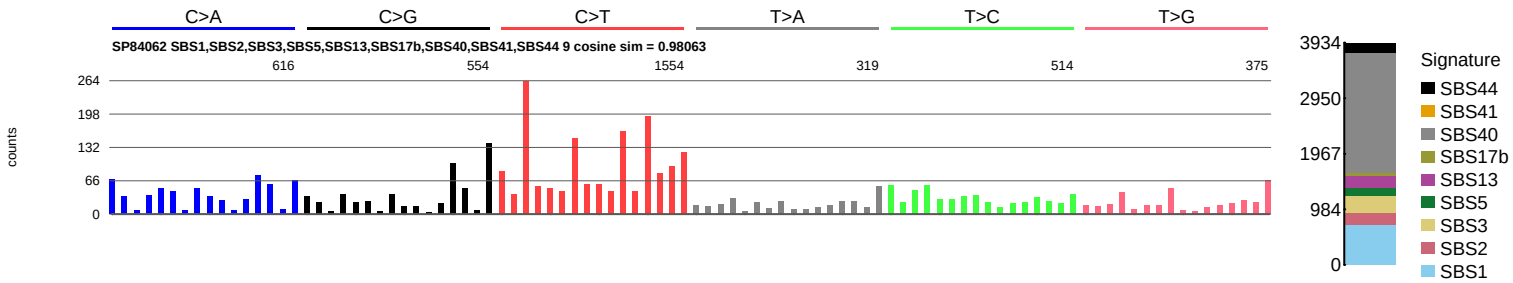

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98067
629
552
1548
317
514
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

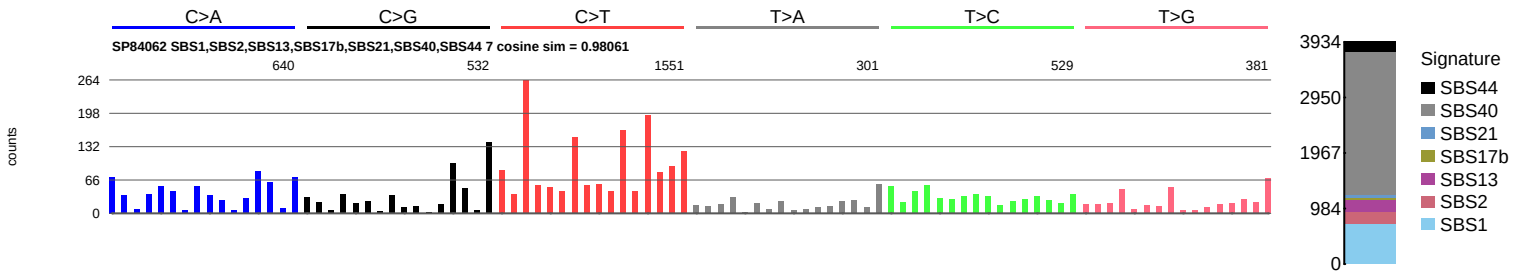

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98067
749
560
1458
378
540
247
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

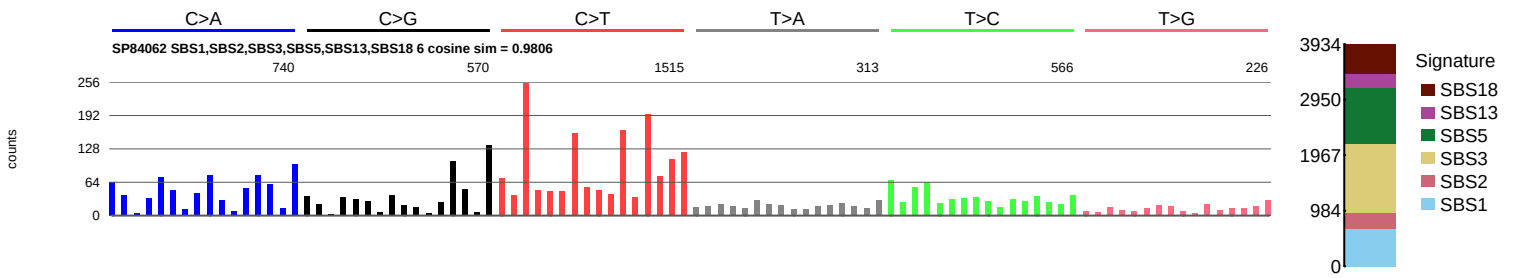

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS40,SBS44 7 cosine sim = 0.98066
634
557
1553
319
495
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS3
SBS2
SBS1

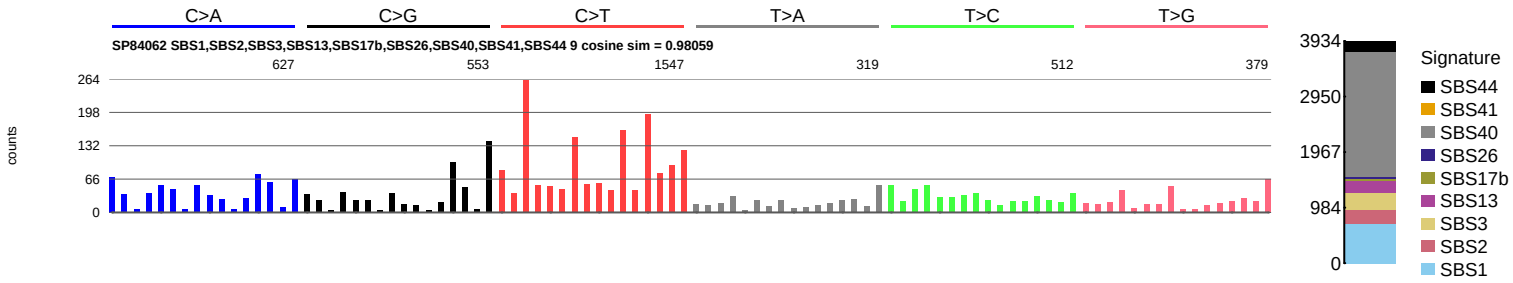

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98065
630
553
1547
317
511
377
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS13
SBS3
SBS2
SBS1

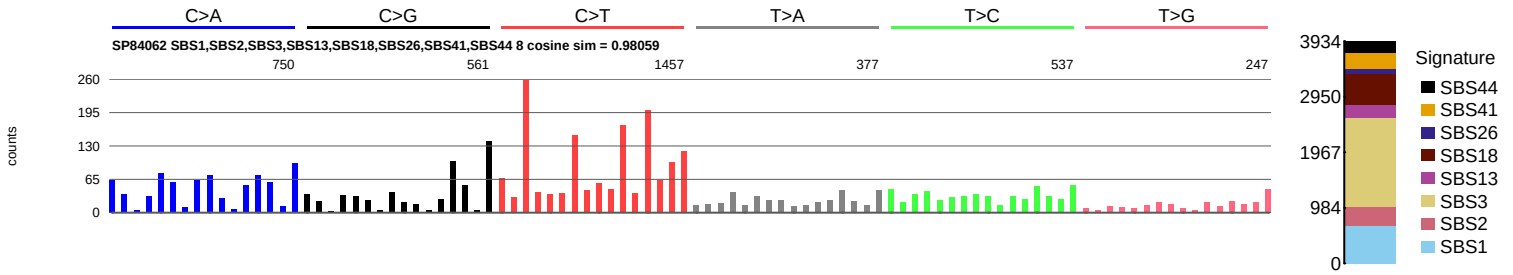

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40 6 cosine sim = 0.98064
769
532
1505
305
463
357
counts
268
201
134
67
0
3934
2950
1967
984
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1

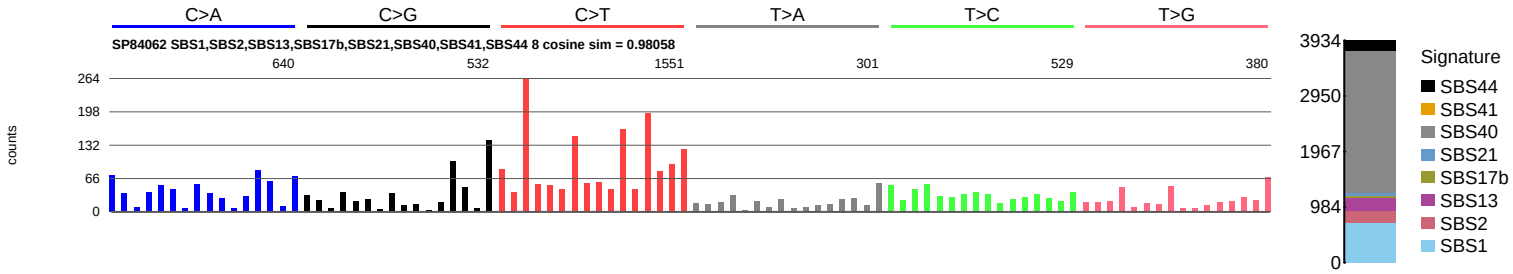

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98063
616
554
1554
319
514
375
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

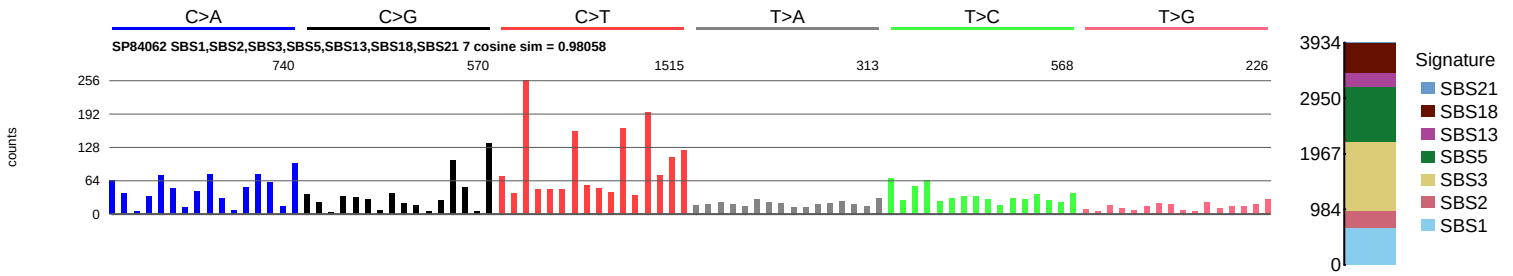

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98061
640
532
1551
301
529
381
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS2
SBS1

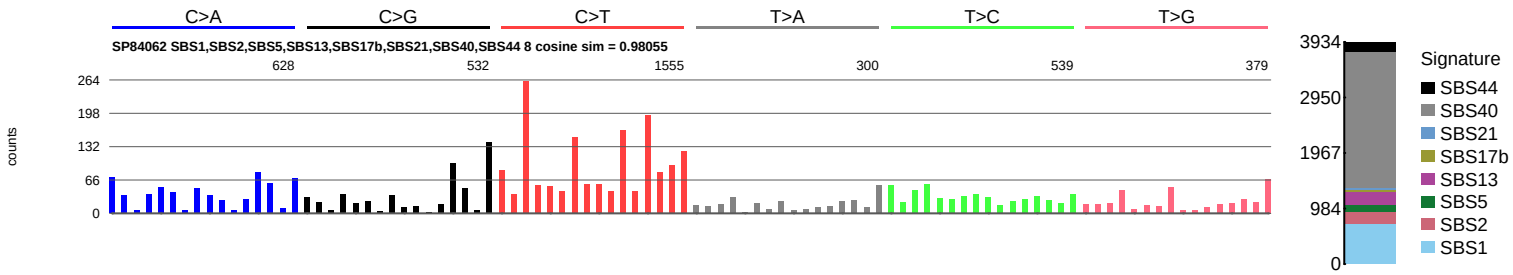

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18 6 cosine sim = 0.9806
740
570
1515
313
566
226
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

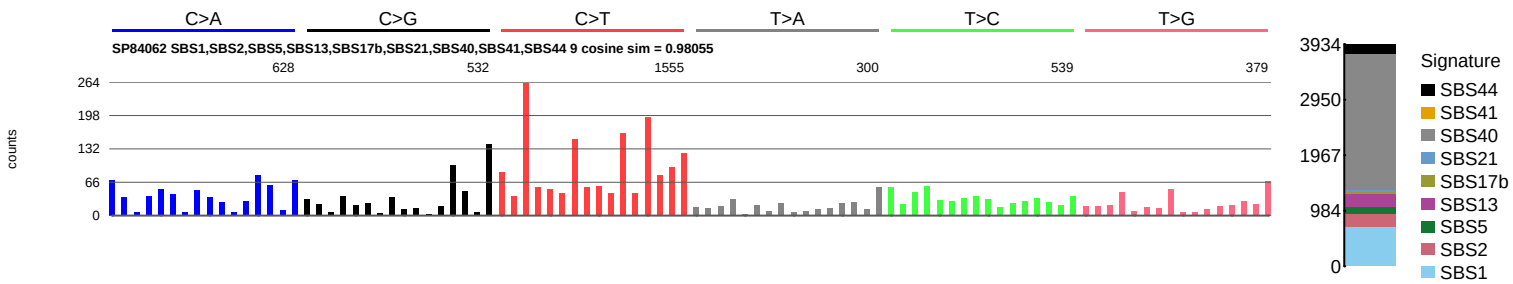

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98059
627
553
1547
319
512
379
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS13
SBS3
SBS2
SBS1

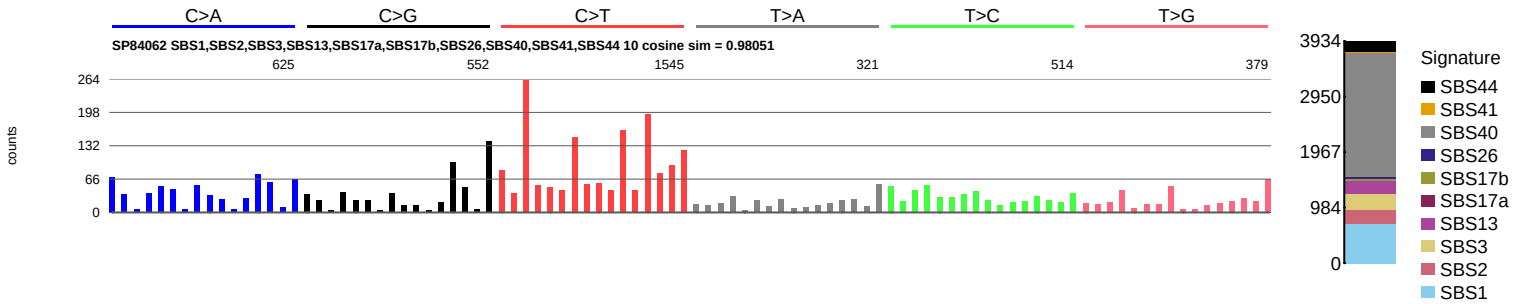

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98059
750
561
1457
377
537
247
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

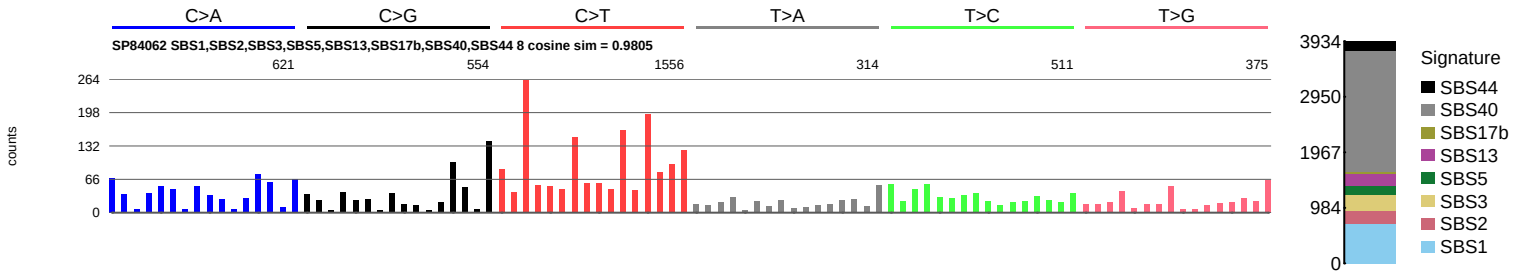

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98058
640
532
1551
301
529
380
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS13
SBS2
SBS1

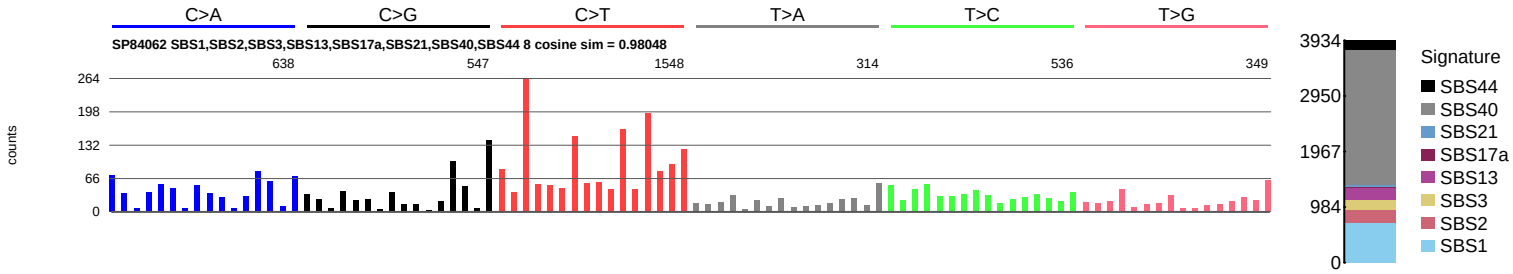

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21 7 cosine sim = 0.98058
740
570
1515
313
568
226
counts
256
192
128
64
0
3934
2950
1967
984
0
Signature
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

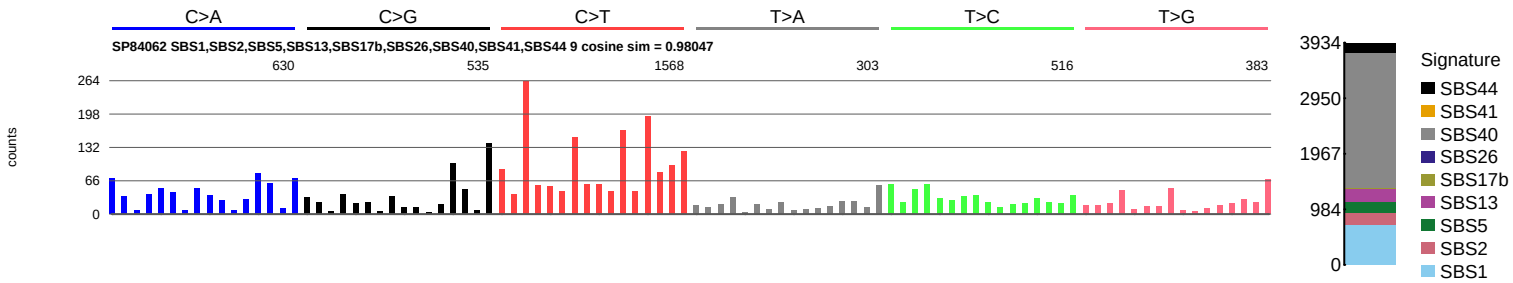

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98055
628
532
1555
300
539
379
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS2
SBS1

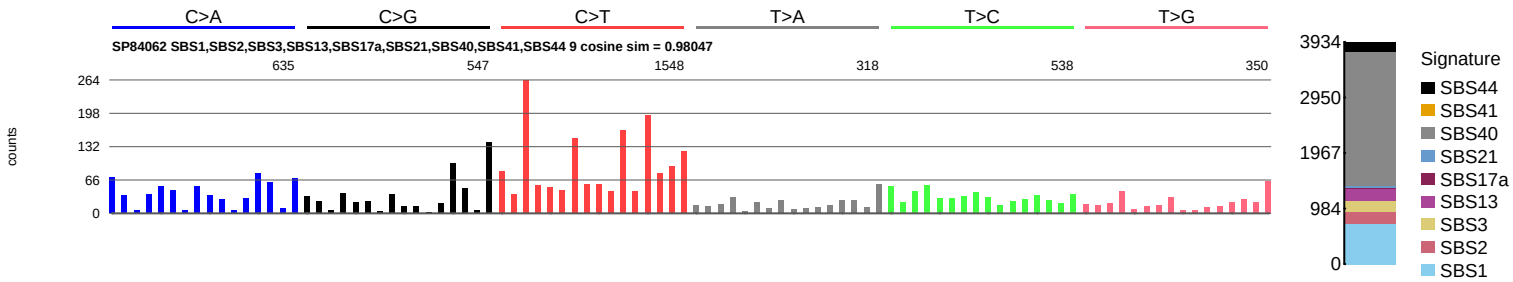

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98055
628
532
1555
300
539
379
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS2
SBS1

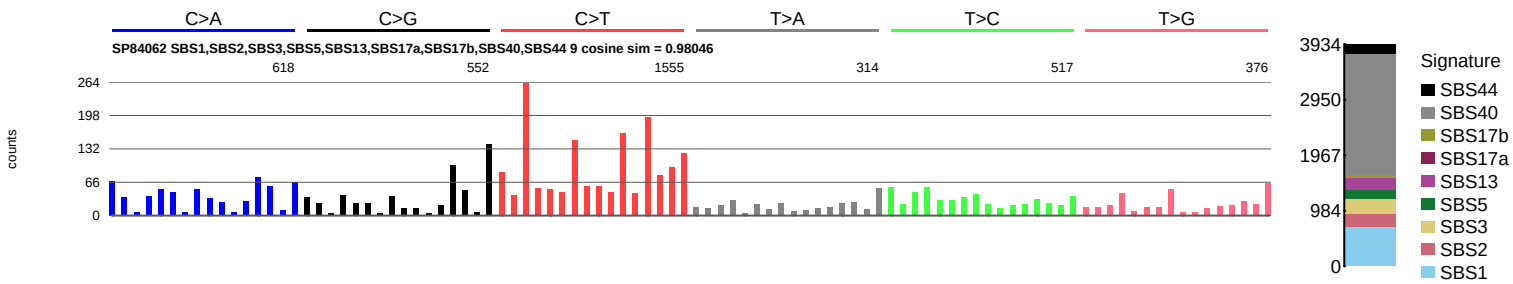

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98051
625
552
1545
321
514
379
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

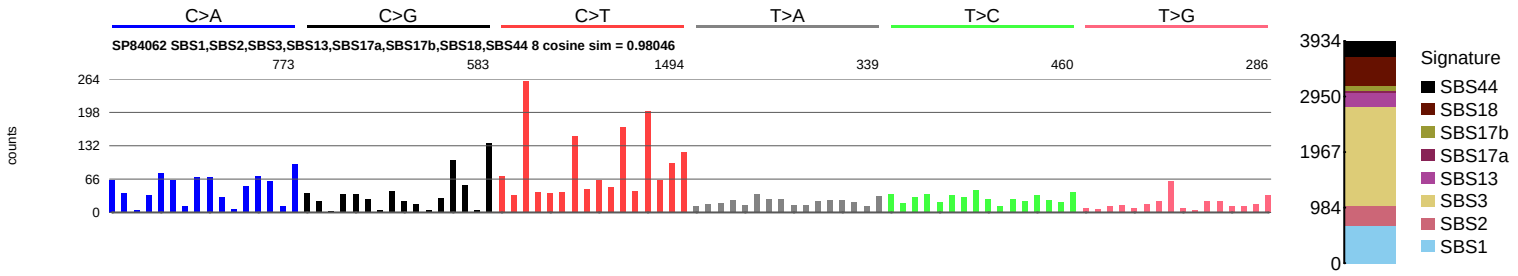

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS44 8 cosine sim = 0.9805
621
554
1556
314
511
375
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

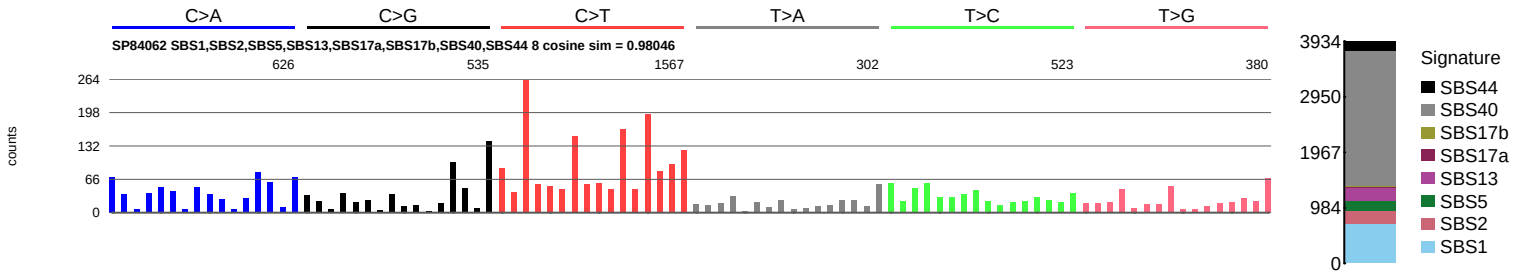

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS21,SBS40,SBS44 8 cosine sim = 0.98048
638
547
1548
314
536
349
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS3
SBS2
SBS1

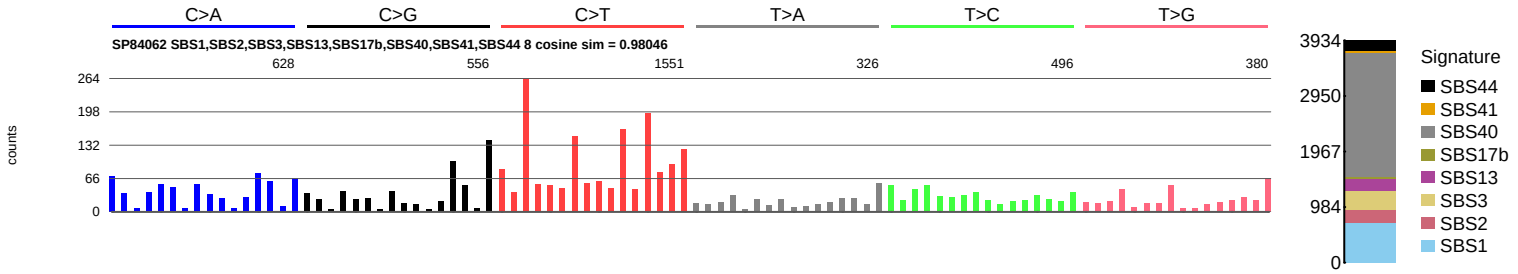

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98047
630
535
1568
303
516
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS13
SBS5
SBS2
SBS1

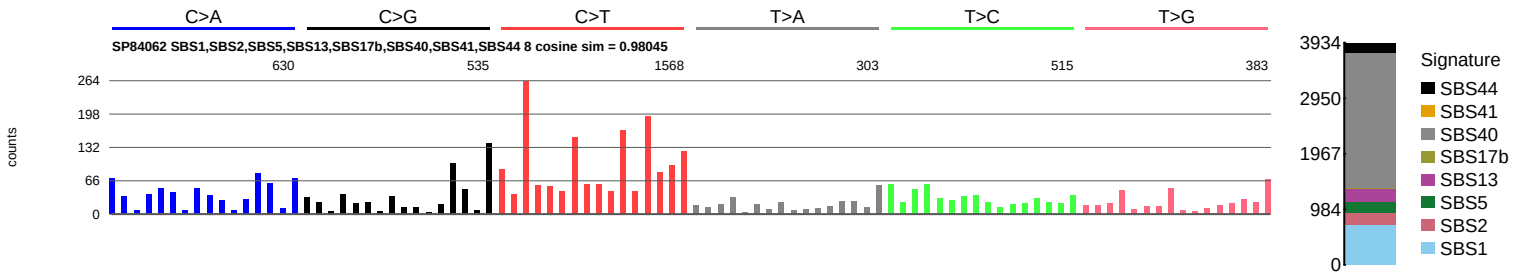

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98047
635
547
1548
318
538
350
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS3
SBS2
SBS1

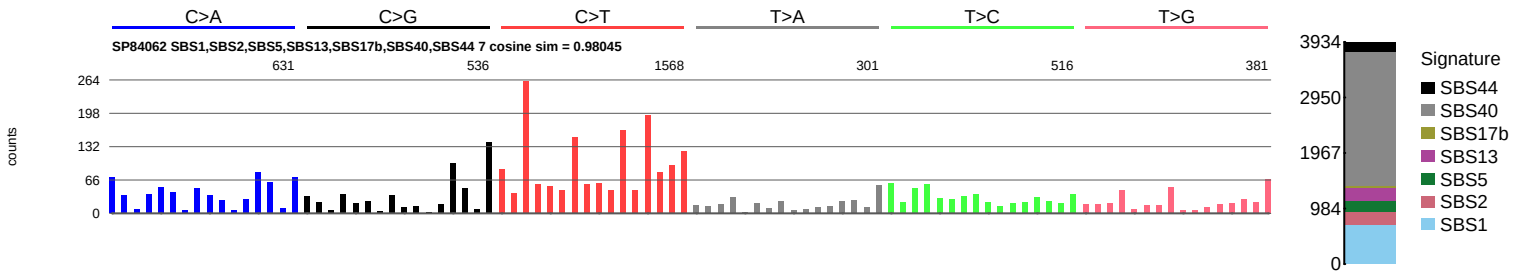

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98046
618
552
1555
314
517
376
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

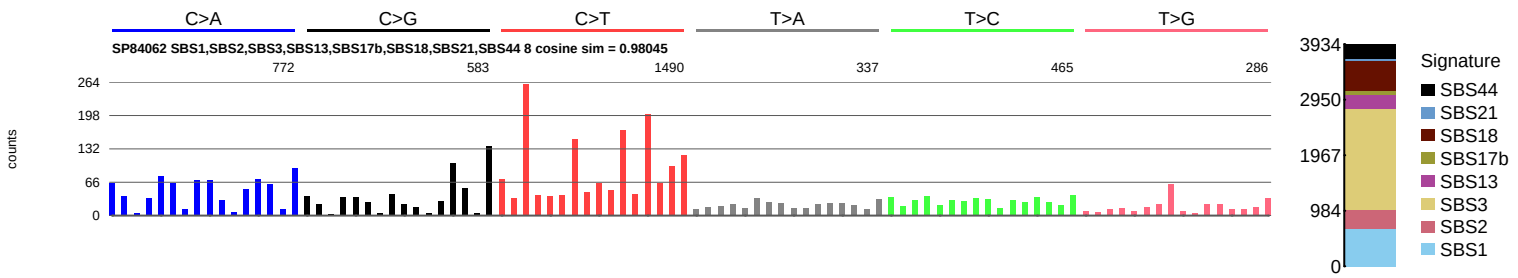

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98046
773
583
1494
339
460
286
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

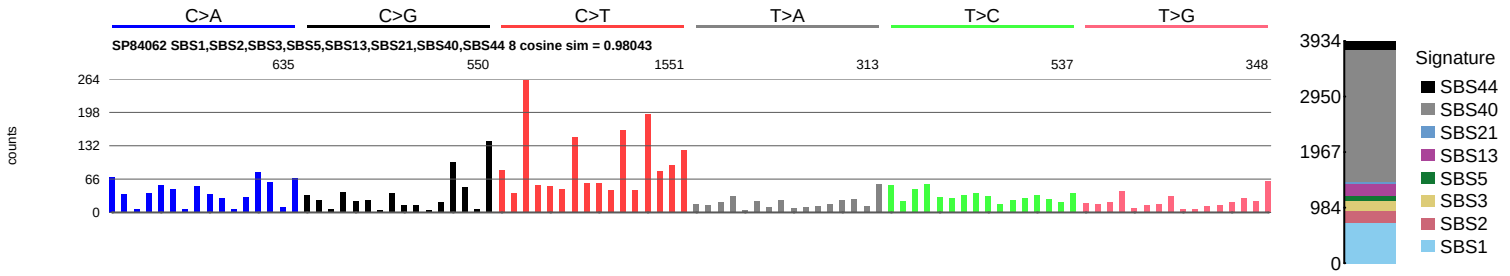

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98046
626
535
1567
302
523
380
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

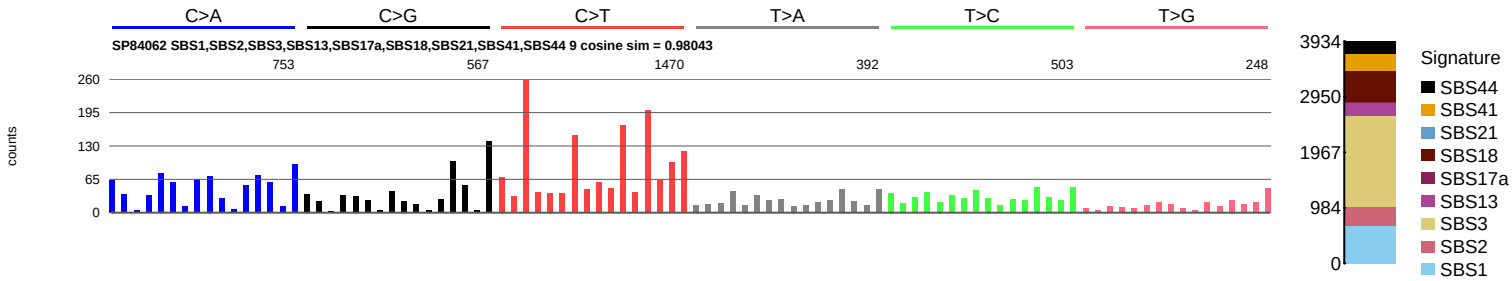

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98046
628
556
1551
326
496
380
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS3
SBS2
SBS1

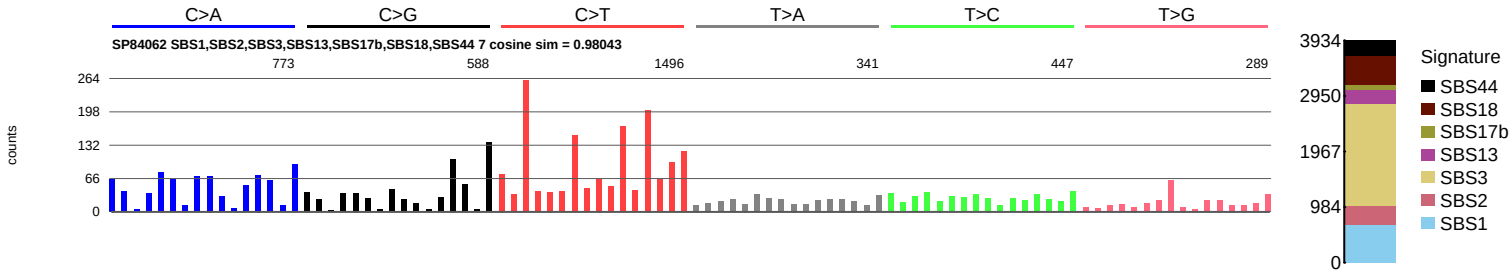

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98045
630
535
1568
303
515
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1

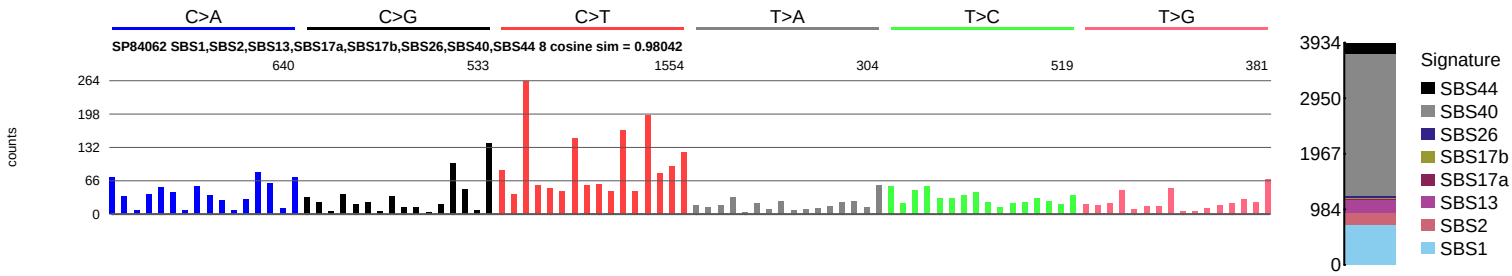

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS44 7 cosine sim = 0.98045
631
536
1568
301
516
381
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1

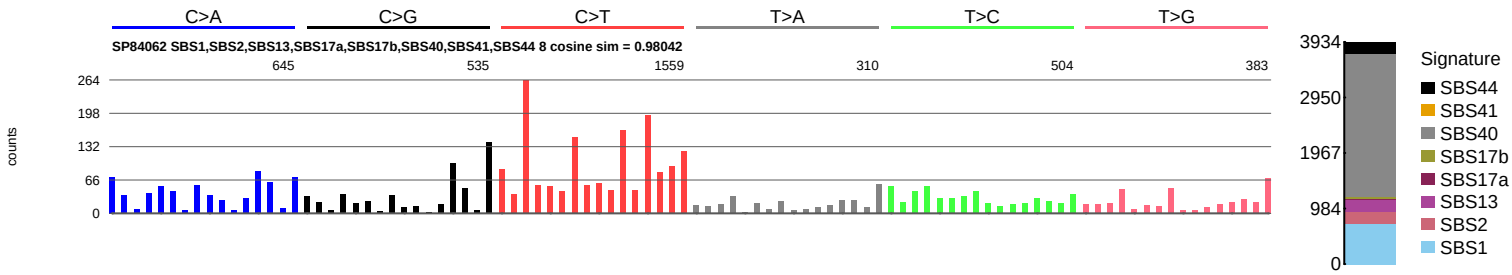

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98045
772
583
1490
337
465
286
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

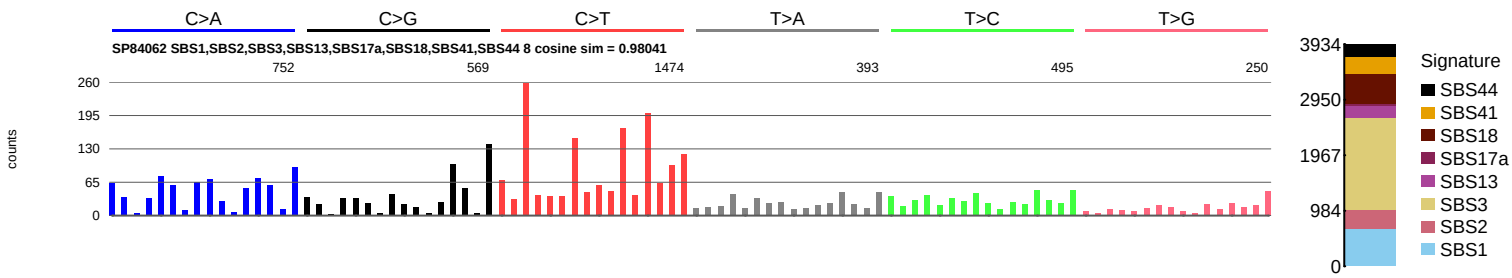

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40,SBS44 8 cosine sim = 0.98043
635
550
1551
313
537
348
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS5
SBS3
SBS2
SBS1

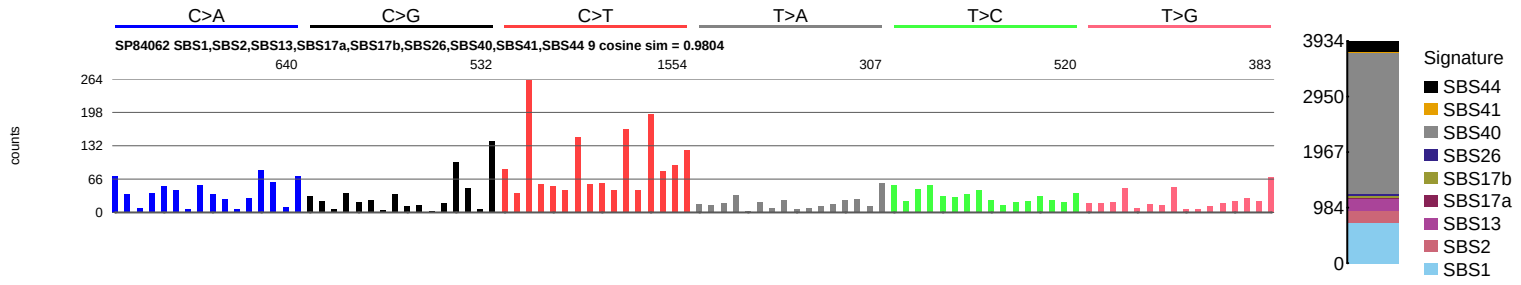

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98043
753
567
1470
392
503
248
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

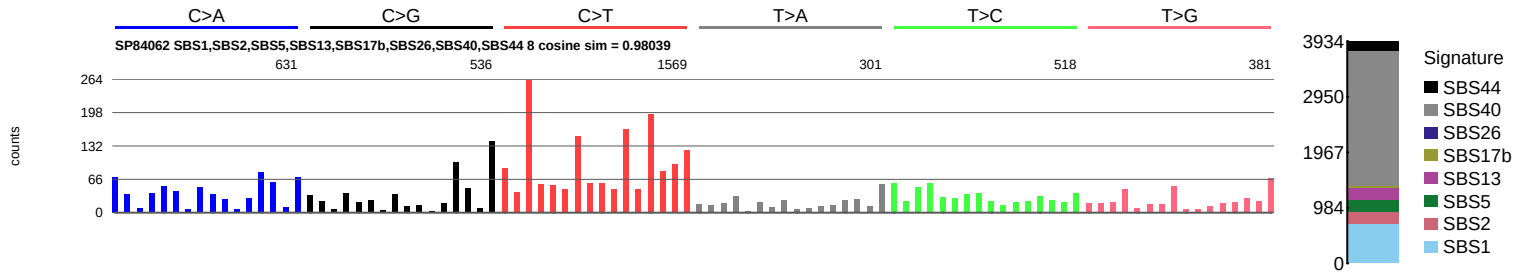

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.98043
773
588
1496
341
447
289
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

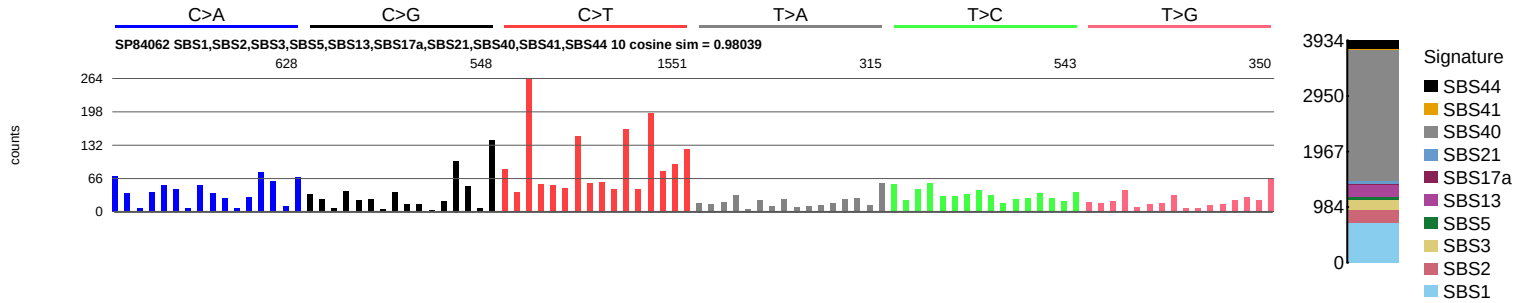

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98042
640
533
1554
304
519
381
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

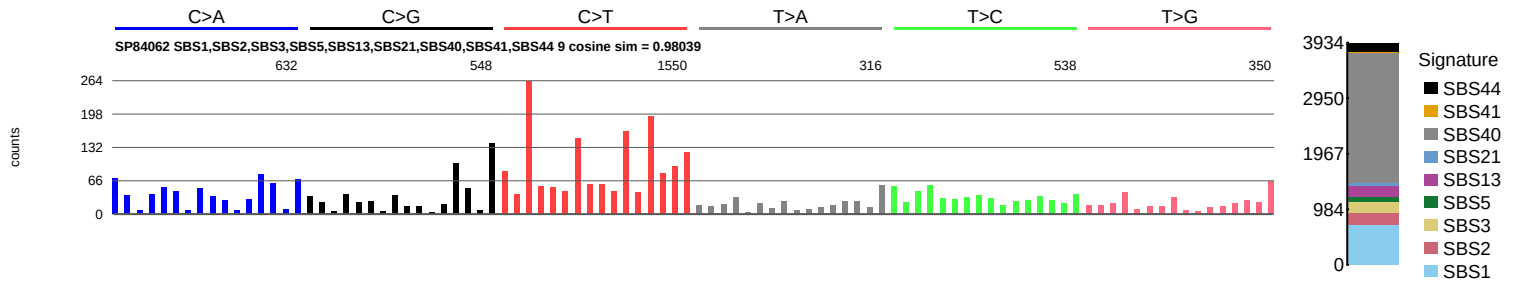

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98042
645
535
1559
310
504
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1

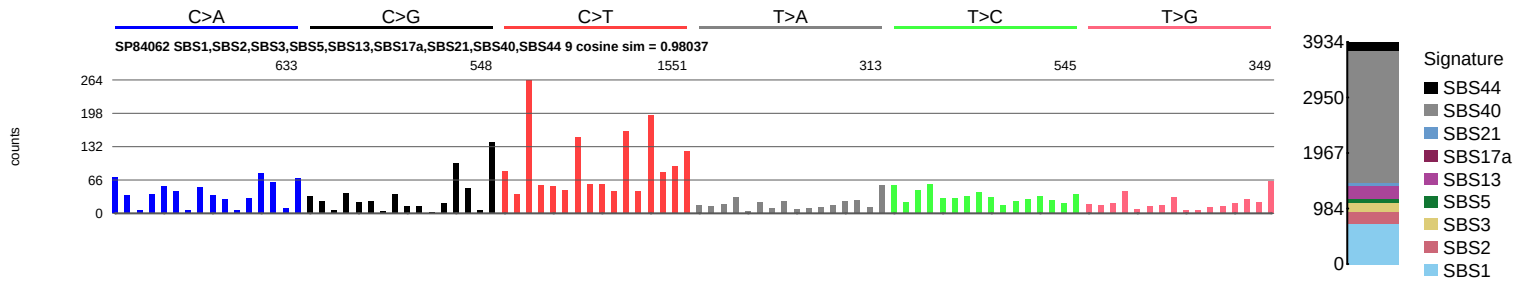

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98041
752
569
1474
393
495
250
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

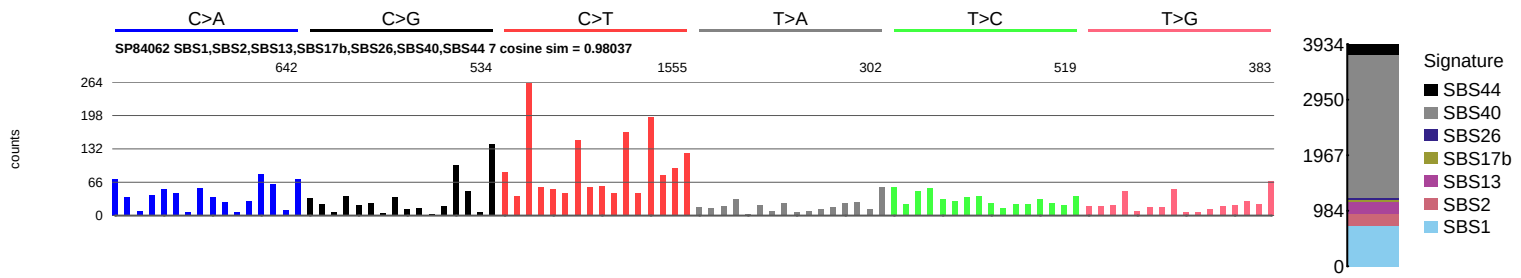

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9804
640
532
1554
307
520
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

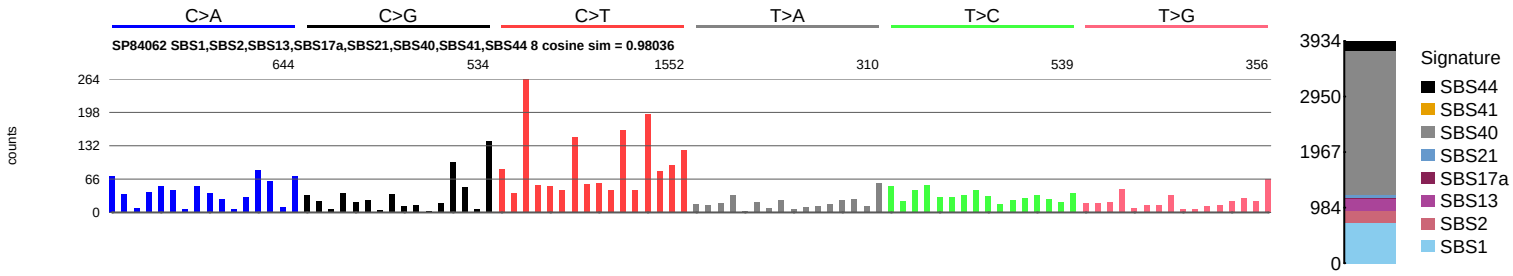

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98039
631
536
1569
301
518
381
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS13
SBS5
SBS2
SBS1

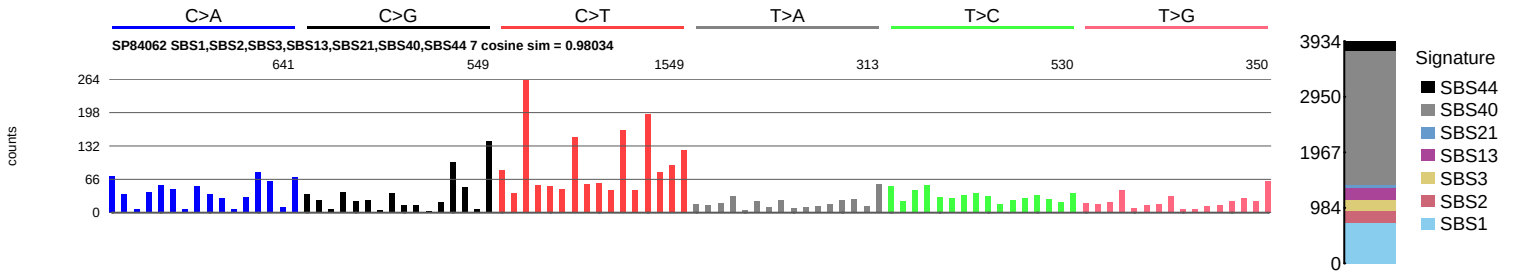

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98039
628
548
1551
315
543
350
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

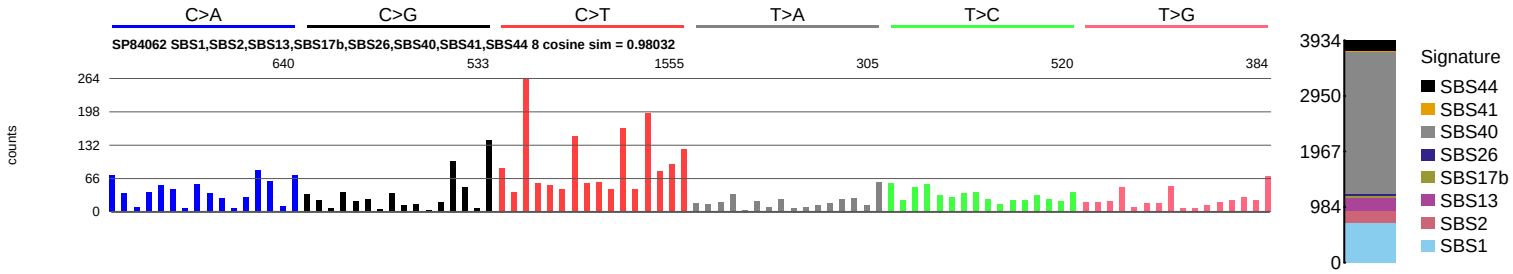

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98039
632
548
1550
316
538
350
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS13
SBS5
SBS3
SBS2
SBS1

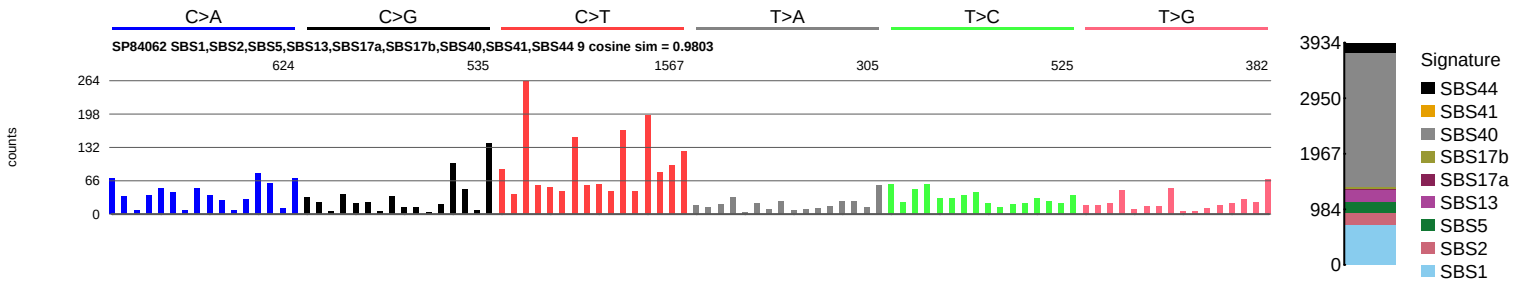

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS44 9 cosine sim = 0.98037
633
548
1551
313
545
349
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

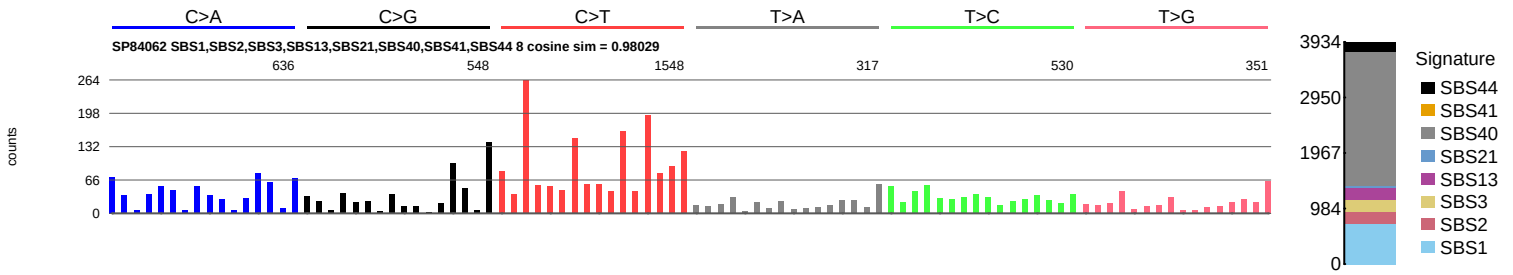

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98037
642
534
1555
302
519
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS13
SBS2
SBS1

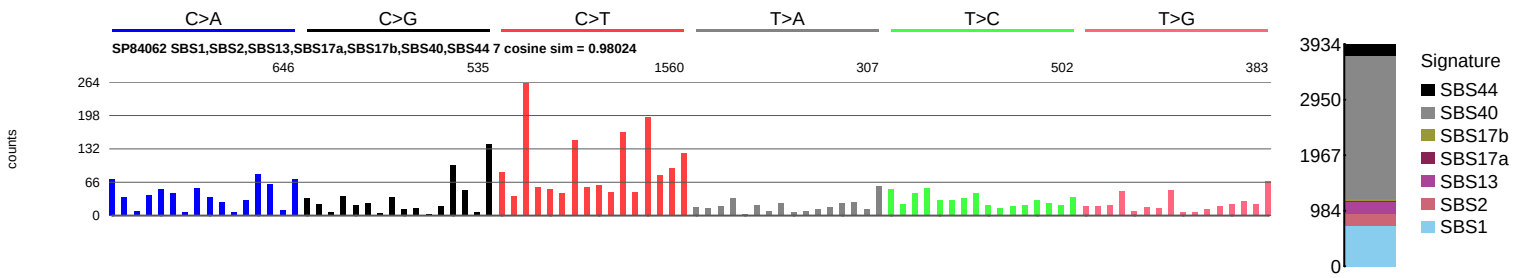

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98036
644
534
1552
310
539
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS2
SBS1

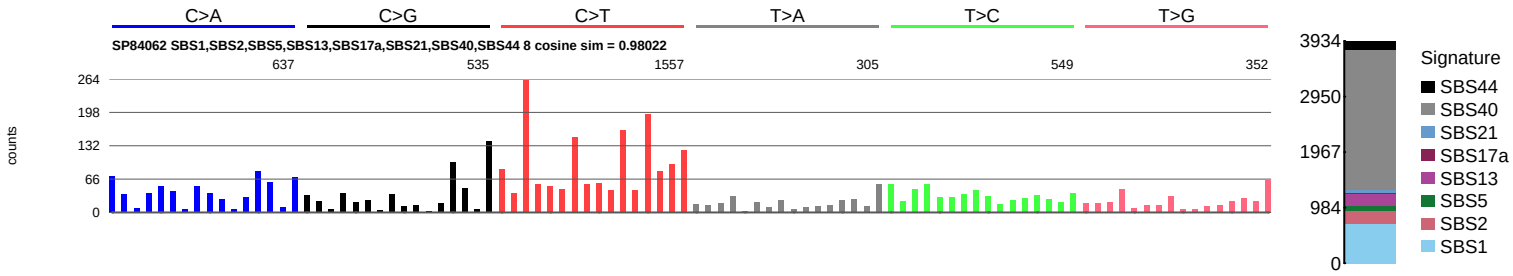

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS21,SBS40,SBS44 7 cosine sim = 0.98034
641
549
1549
313
530
350
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS3
SBS2
SBS1

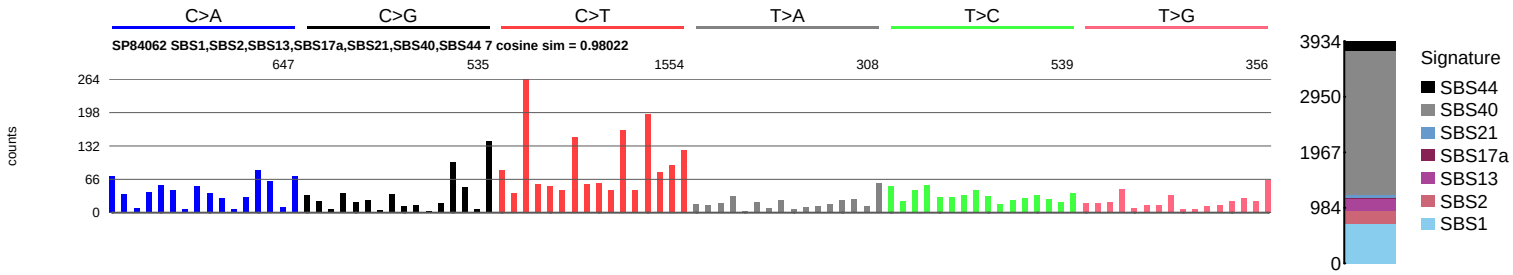

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98032
640
533
1555
305
520
384
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS13
SBS2
SBS1

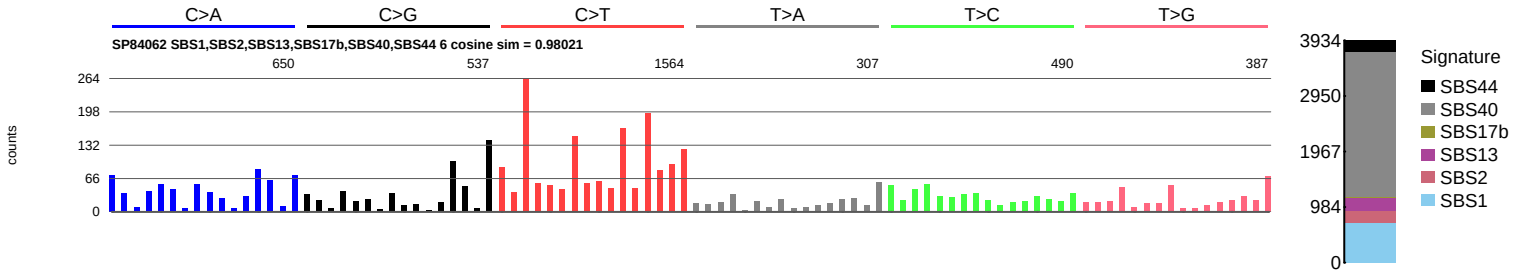

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.9803
624
535
1567
305
525
382
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

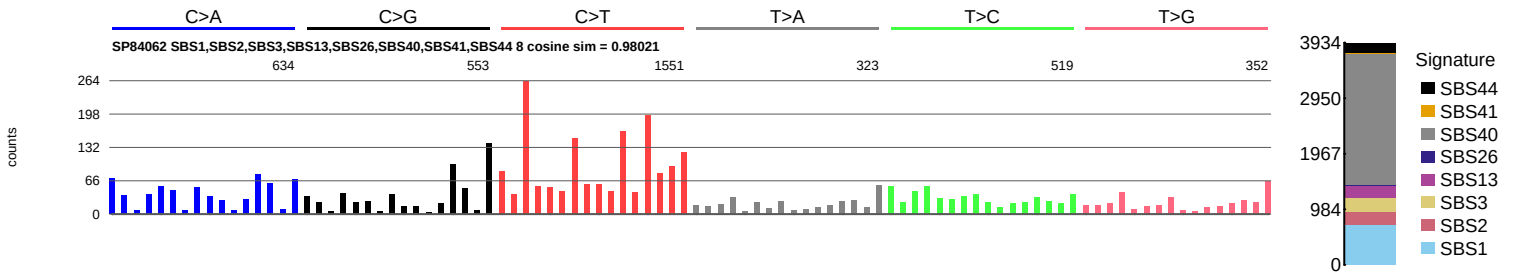

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98029
636
548
1548
317
530
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS13
SBS3
SBS2
SBS1

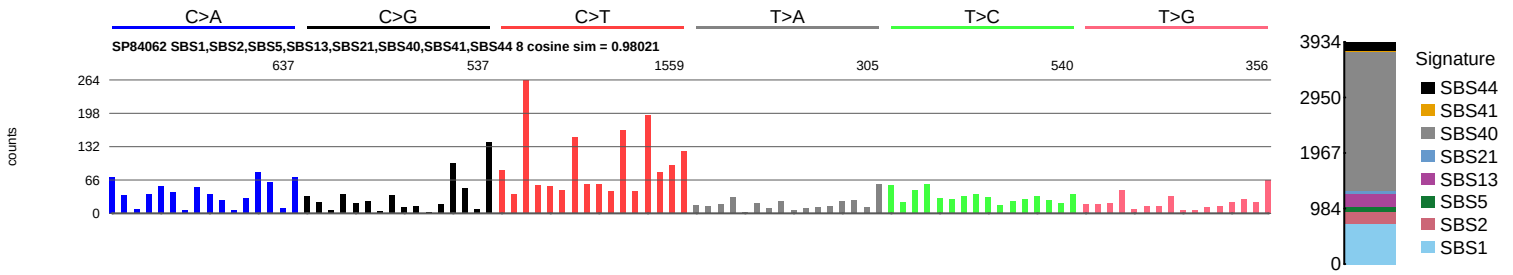

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98024
646
535
1560
307
502
383
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1

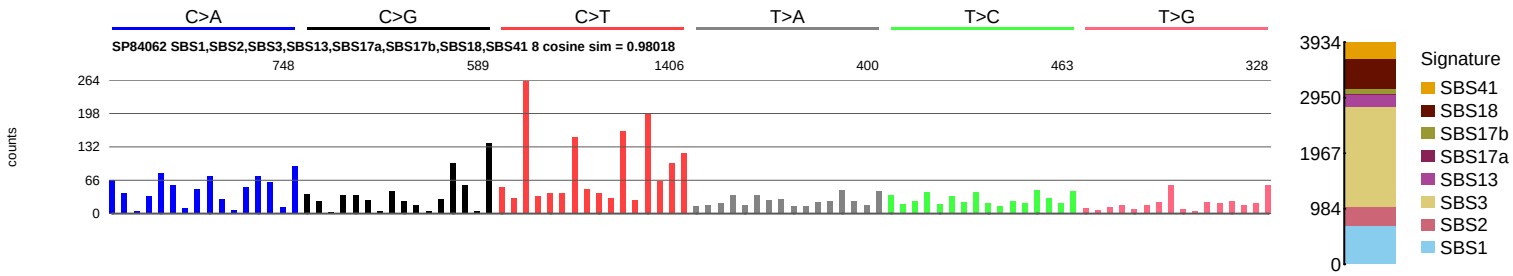

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS44 8 cosine sim = 0.98022
637
535
1557
305
549
352
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1

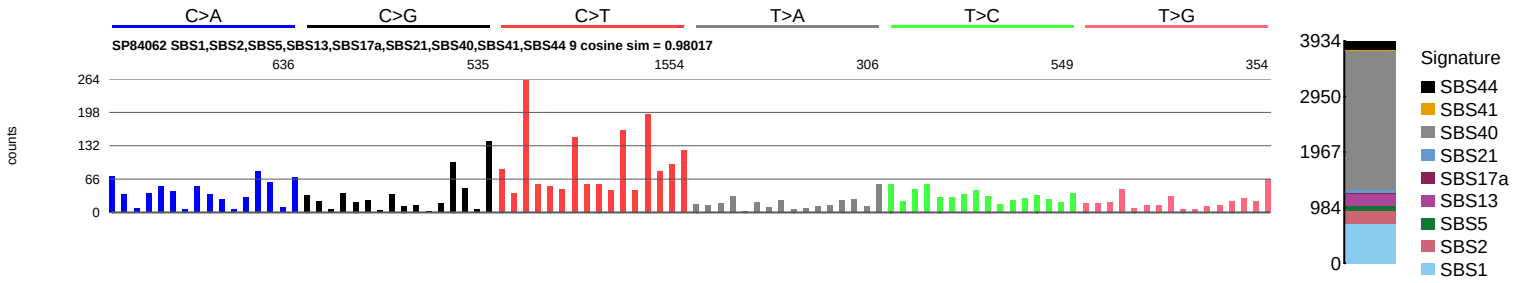

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS21,SBS40,SBS44 7 cosine sim = 0.98022
647
535
1554
308
539
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS2
SBS1

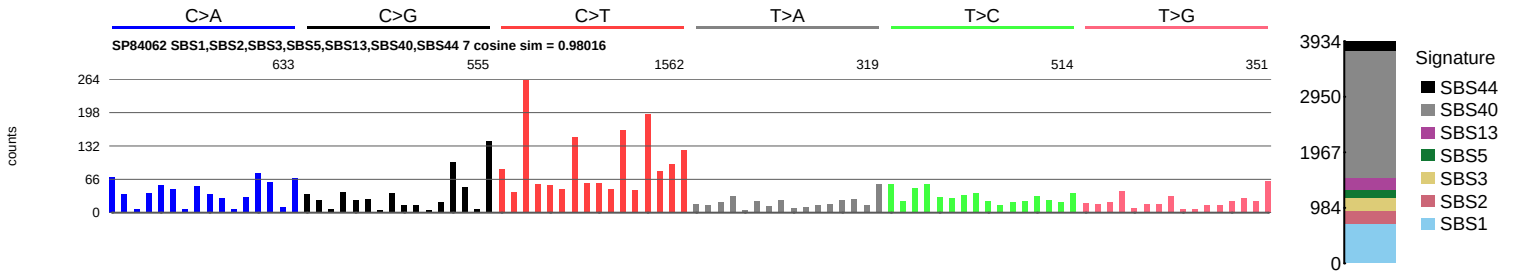

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS40,SBS44 6 cosine sim = 0.98021
650
537
1564
307
490
387
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS2
SBS1

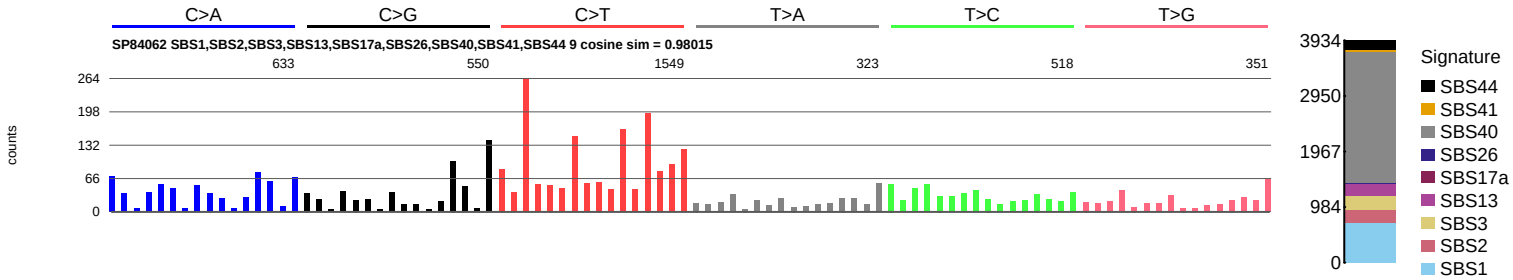

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98021
634
553
1551
323
519
352
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS13
SBS3
SBS2
SBS1

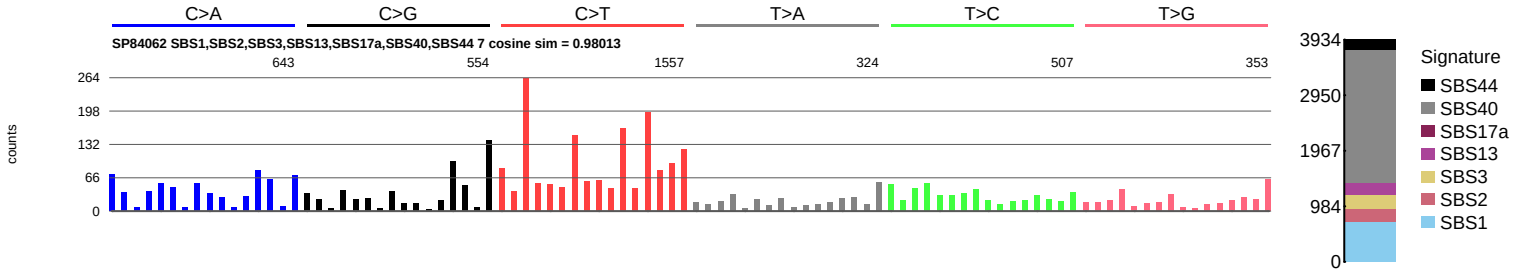

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98021
637
537
1559
305
540
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1

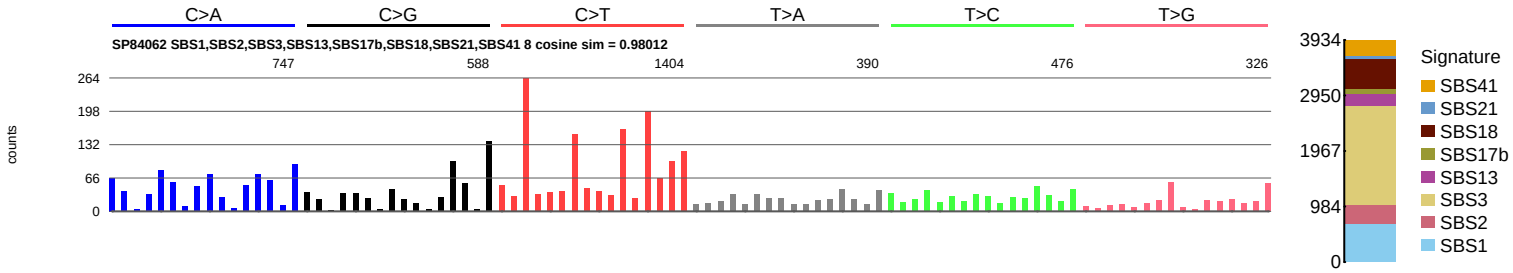

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98018
748
589
1406
400
463
328
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

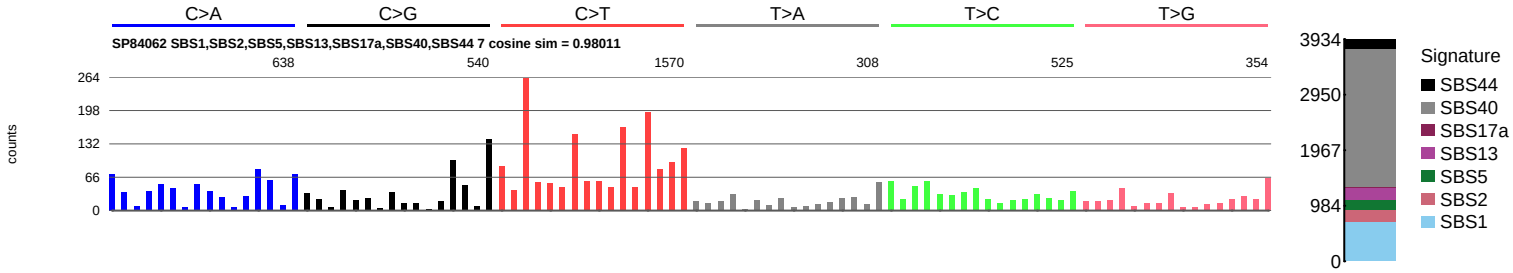

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98017
636
535
1554
306
549
354
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1

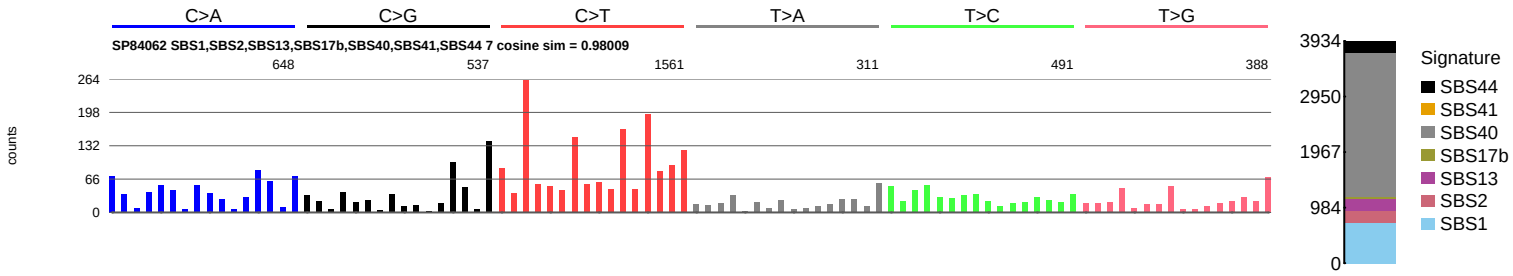

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS44 7 cosine sim = 0.98016
633
555
1562
319
514
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1

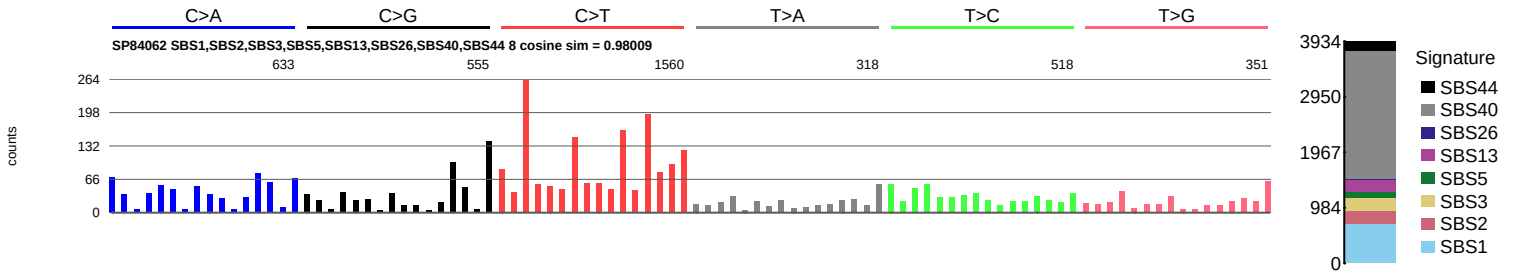

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98015
633
550
1549
323
518
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17a
SBS13
SBS3
SBS2
SBS1

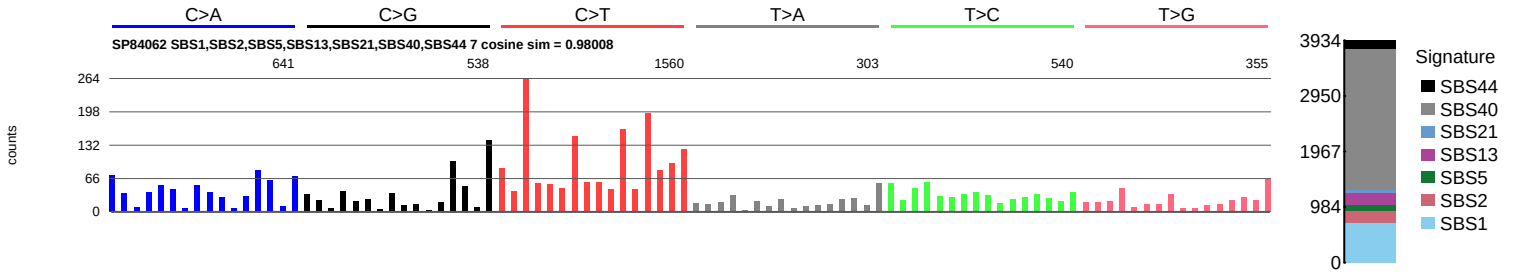

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS40,SBS44 7 cosine sim = 0.98013
643
554
1557
324
507
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS13
SBS3
SBS2
SBS1

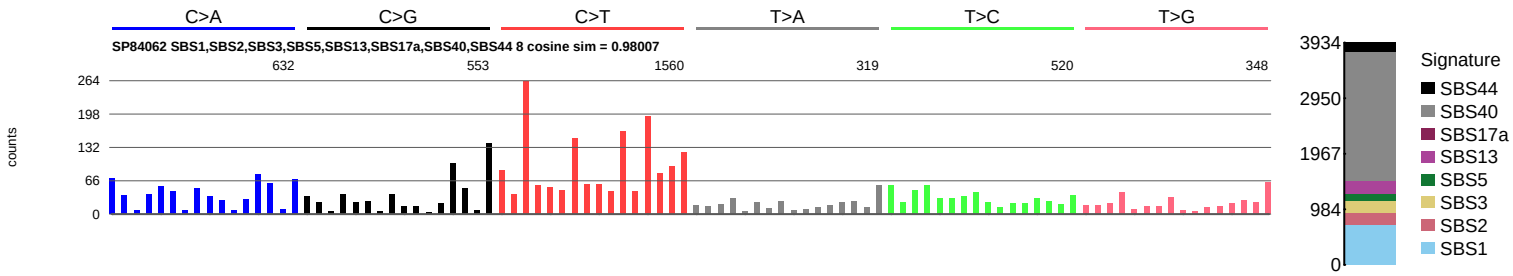

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98012
747
588
1404
390
476
326
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

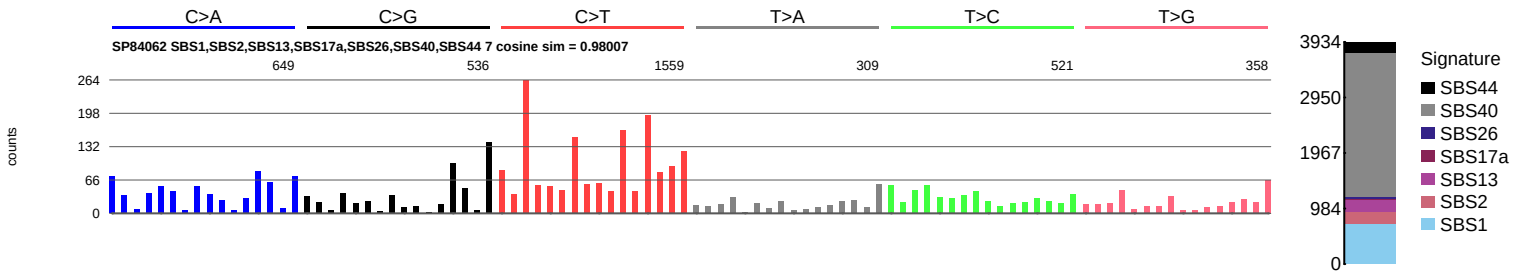

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS44 7 cosine sim = 0.98011
638
540
1570
308
525
354
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1

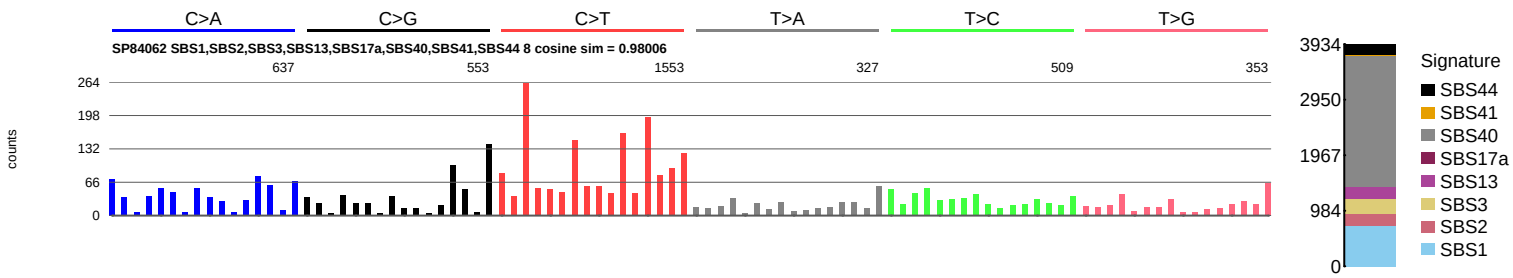

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98009
648
537
1561
311
491
388
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS2
SBS1

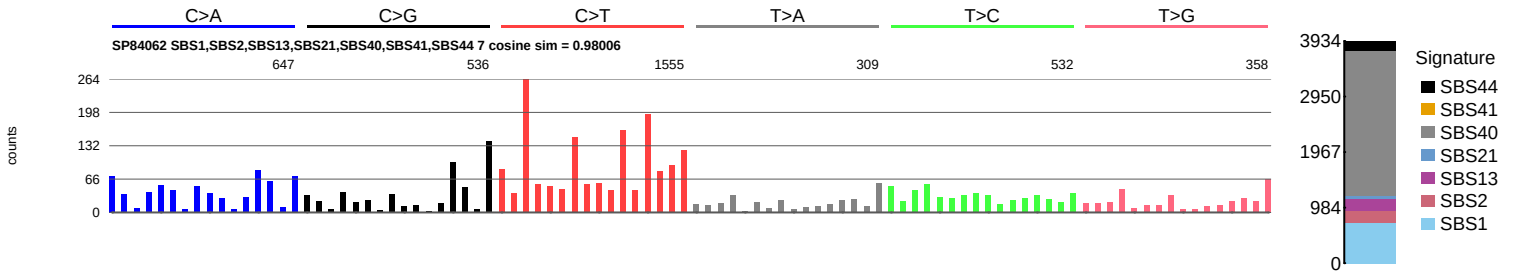

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS26,SBS40,SBS44 8 cosine sim = 0.98009
633
555
1560
318
518
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS13
SBS5
SBS3
SBS2
SBS1

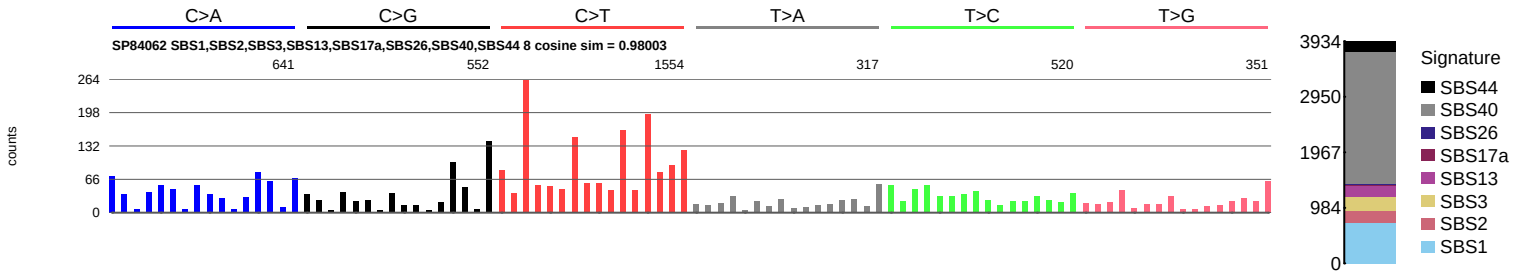

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS44 7 cosine sim = 0.98008
641
538
1560
303
540
355
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1

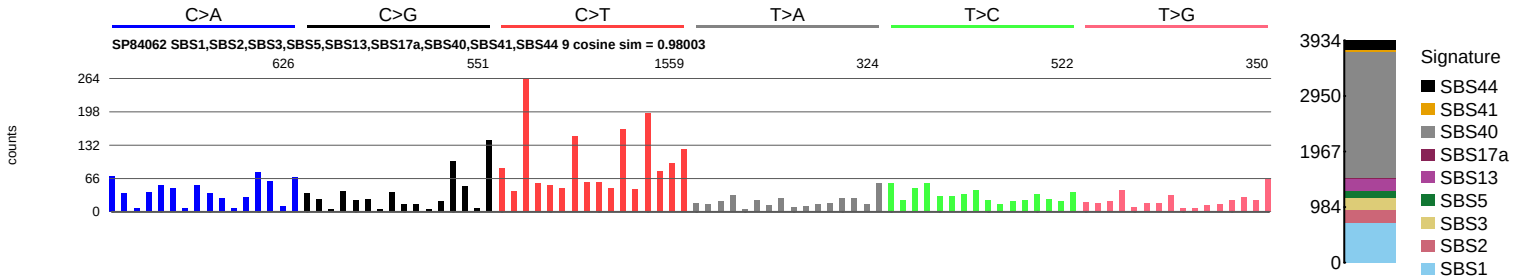

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40,SBS44 8 cosine sim = 0.98007
632
553
1560
319
520
348
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

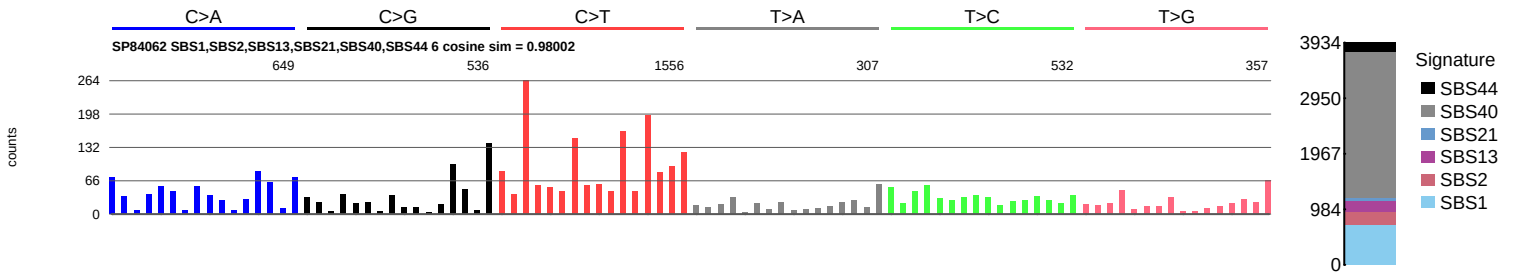

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS17a,SBS26,SBS40,SBS44 7 cosine sim = 0.98007
649
536
1559
309
521
358
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS13
SBS2
SBS1

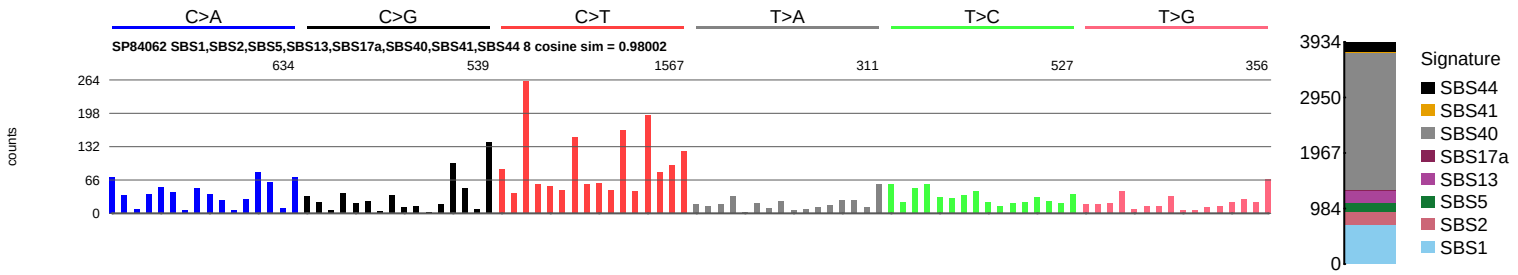

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS40,SBS41,SBS44 8 cosine sim = 0.98006
637
553
1553
327
509
353
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS13
SBS3
SBS2
SBS1

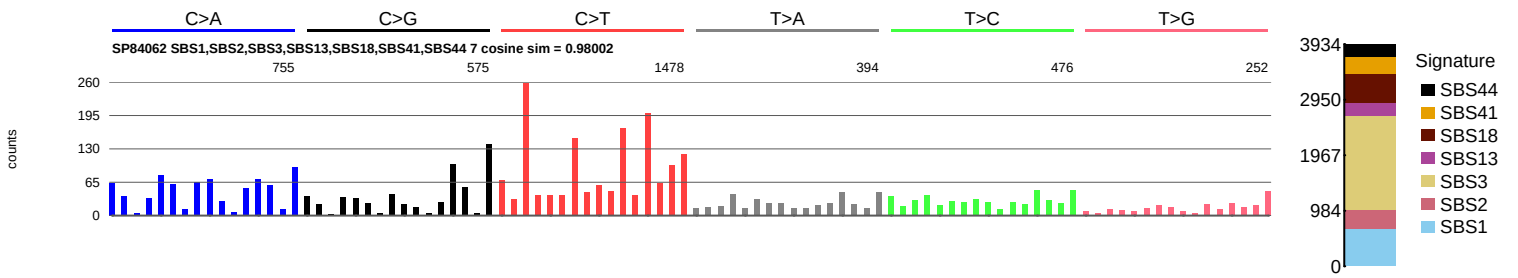

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98006
647
536
1555
309
532
358
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS13
SBS2
SBS1

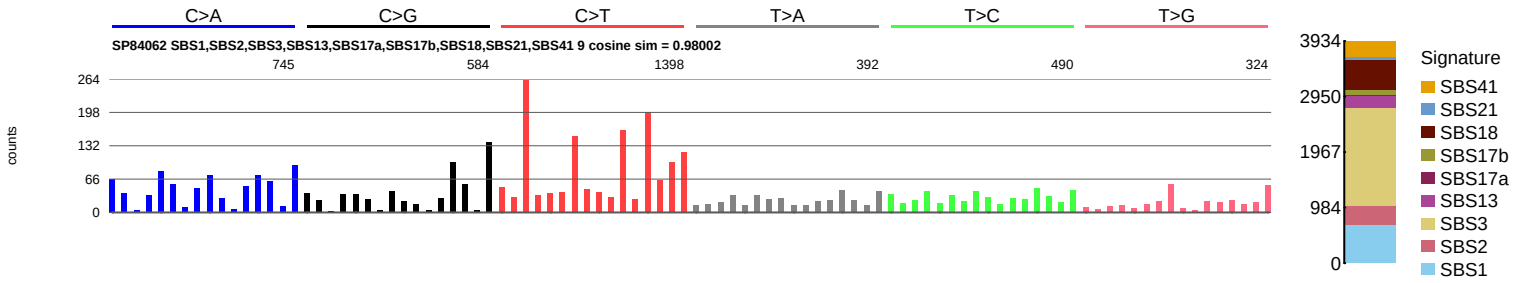

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS26,SBS40,SBS44 8 cosine sim = 0.98003
641
552
1554
317
520
351
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS13
SBS3
SBS2
SBS1

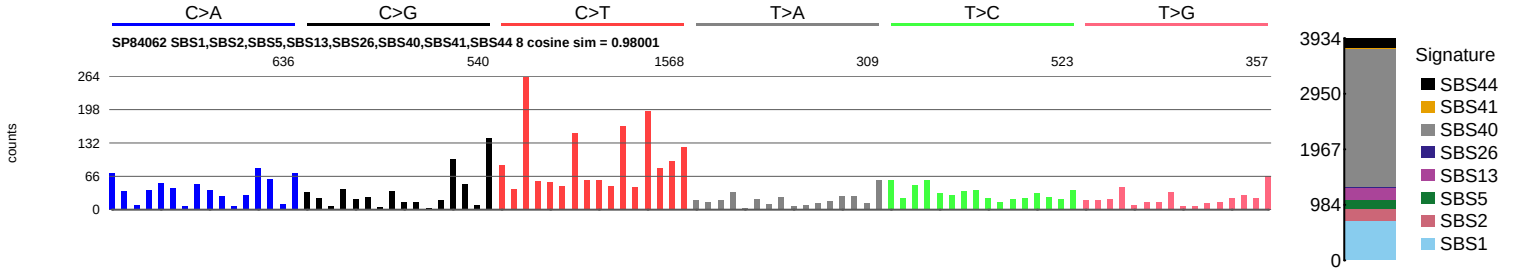

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40,SBS41,SBS44 9 cosine sim = 0.98003
626
551
1559
324
522
350
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

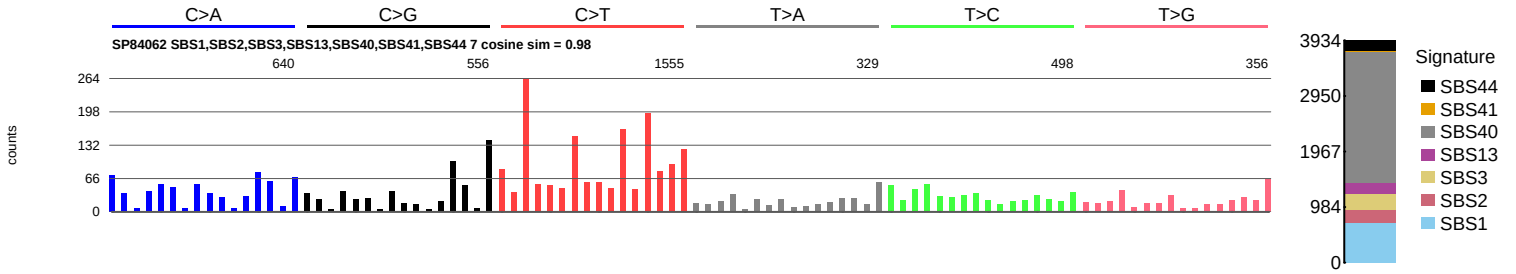

C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS13,SBS21,SBS40,SBS44 6 cosine sim = 0.98002
649
536
1556
307
532
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS41,SBS44 8 cosine sim = 0.98002
634
539
1567
311
527
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS18,SBS41,SBS44 7 cosine sim = 0.98002
755
575
1478
394
476
252
counts
260
195
130
65
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98002
745
584
1398
392
490
324
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS5,SBS13,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98001
636
540
1568
309
523
357
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84062 SBS1,SBS2,SBS3,SBS13,SBS40,SBS41,SBS44 7 cosine sim = 0.98
640
556
1555
329
498
356
counts
264
198
132
66
0
3934
2950
1967
984
0
Signature
SBS44
SBS41
SBS40
SBS13
SBS3
SBS2
SBS1